

EQUITIES

CFA[®] Program Curriculum
2027 • LEVEL I • VOLUME 5

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How to Use the CFA Program Curriculum

The CFA® Program exams measure your mastery of the core knowledge, skills, and abilities required to succeed as an investment professional. These core competencies are the basis for the Candidate Body of Knowledge (CBOK™). The CBOK consists of four components:

- A broad outline that lists the major CFA Program topic areas (www.cfainstitute.org/programs/cfa/curriculum/cbok/cbok)

- Topic area weights that indicate the relative exam weightings of the top-level topic areas (www.cfainstitute.org/en/programs/cfa/curriculum)

- Learning outcome statements (LOS) that tell you the specific knowledge, skills, and abilities you should gain from each curriculum topic area. You will find these statements at the start of each learning module and lesson. We encourage you to review the information about the LOS on our website (www.cfainstitute.org/programs/cfa/curriculum/study-sessions), including the descriptions of LOS “command words” on the candidate resources page at www.cfainstitute.org/-/media/documents/support/programs/cfa-and-cipm-los-command-words.ashx.

- The CFA Program curriculum that candidates receive access to upon exam registration.

Therefore, the key to your success on the CFA exams is studying and understanding the CBOK. You can learn more about the CBOK on our website: www.cfainstitute.org/programs/cfa/curriculum/cbok.

The curriculum, including the practice questions, is the basis for all exam questions. The curriculum is selected/developed specifically to provide candidates with the knowledge, skills, and abilities reflected in the CBOK.

CFA INSTITUTE LEARNING ECOSYSTEM (LES)

Your exam registration fee includes access to the CFA Institute Learning Ecosystem (LES). This digital learning platform provides access to all the curriculum content and practice questions. The LES is organized as a series of learning modules consisting of short online lessons and associated practice questions. This tool is your source for all study materials, including practice questions and mock exams. The LES is the primary method by which CFA Institute delivers your curriculum experience. Here, you will find additional practice questions to test your knowledge, including some interactive questions.

DESIGNING YOUR PERSONAL STUDY PROGRAM

An orderly, systematic approach to exam preparation is critical. You should dedicate a consistent block of time every week to reading and studying. Review the LOS both before and after you study curriculum content to ensure you can demonstrate

the knowledge, skills, and abilities described by the LOS and the assigned learning module. Use the LOS as a self-check to track your progress and highlight areas of weakness for later review.

Successful candidates report an average of more than 300 hours preparing for each exam. Your preparation time will vary based on your prior education and experience, and you will likely spend more time on some topics than on others.

ERRATA

The curriculum development process is rigorous and involves multiple rounds of reviews by content experts. Despite our efforts to produce a curriculum that is free of errors, we must make corrections in some instances. Curriculum errata are periodically updated and posted by exam level and test date on the Curriculum Errata webpage (www.cfainstitute.org/en/programs/submit-errata). If you believe you have found an error in the curriculum, you can submit your concerns through our curriculum errata reporting process found at the bottom of the Curriculum Errata webpage.

OTHER FEEDBACK

Please send any comments or suggestions to info@cfainstitute.org, and we will review your feedback thoughtfully.

Equities

LEARNING MODULE

1

Equity Instrument Features

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	describe basic features and types of equity instruments
☐	contrast features of publicly listed versus private equity securities

INTRODUCTION: EQUITY INSTRUMENT FEATURES

1

Equity instruments, such as common and preferred shares, are securities that represent an investor's fractional ownership in a company. Issuers of equities include publicly listed and private corporations whose owners (i.e., shareholders) receive economic, voting, and legal rights over an indefinite period. Equity investors may receive a proportional share of any distributions made at the issuer's discretion, participate in electing a company's board and decide on important corporate actions, and have a legal claim to remaining company assets upon liquidation once debtholder claims have been satisfied.

Equity securities represent a standardized form of legal ownership and are a core investment for many asset managers, pension funds, and individual investors seeking capital appreciation and income. According to the World Federation of Exchanges, a global industry association for exchanges and clearinghouses, the global market value of publicly listed equity securities exceeds USD100 trillion—nearly 40% of which is US based, followed by the Chinese Mainland, the European Union, and developing or emerging markets, each of which are currently close to 11%. Equities have historically generated the highest long-term returns among major asset classes. Global equity markets are second in size only to fixed-income markets, which are covered in later learning modules. This module introduces the features of equity instruments and compares publicly listed versus private equity investments.

PRE-TEST: EQUITY INSTRUMENT FEATURES

1. Which of the following is the most common characteristic of preferred equity?
- A. Preferred shares represent a claim that is senior to common shares in the company.
 - B. Preferred shares allow holders to participate in the company's shareholder voting process.
 - C. Preferred shares allow holders to only receive dividends at the company's discretion.

Solution:

Choice A is correct. Preferred stock is a more senior claim than common shares.

Choice B is incorrect as preferred shareholders generally do not participate in the company's shareholder voting process.

Choice C is incorrect as preferred shareholders commonly receive non-discretionary, as opposed to discretionary, dividends.

2. The ownership of equity in publicly listed firms is typically characterized by which one of the following?
- A. Firms with publicly held equity typically have few owners with very large ownership stakes.
 - B. Firms with publicly held equity must generally meet legal, regulatory, and compliance requirements of the markets where they trade.
 - C. Firms with publicly held equity are typically much smaller than those with privately held equity.

Solution:

Choice B is correct. Firms with publicly held equity must generally meet legal, regulatory, and compliance requirements of the market where they trade.

Choice A is incorrect as this statement better describes ownership of private equity companies.

Choice C is incorrect as this statement also better describes ownership of private equity companies.

3. Which one of the following statements most correctly reflects a description of preference shares?
- A. Preference shares must be either callable or puttable.
 - B. Preference shares typically have voting rights.
 - C. Preference shares cannot benefit from improved operating performance of the company.

Solution:

Choice C is correct. While preference shares often do not benefit from improved operating performance of the company, these shares may include features that allow for such benefits. Examples include participating preferred or convertible preferred.

Choice A is incorrect. Preference shares do not have to be either callable or puttable.

Choice B is incorrect as preference shares typically do not provide voting rights.

4. Participating preference shares typically entitle shareholders to:

- A. participate in the decision-making process of the company.
- B. convert their shares into a specified number of common shares.
- C. receive an additional dividend if the company's profits exceed a predetermined level.

Solution:

Choice C is correct. Participating preference shares entitle shareholders to receive an additional dividend if the company's profits exceed a predetermined level.

Choice A is incorrect as this statement implies voting rights.

Choice B is incorrect as this statement implies the ability to convert preferred shares into common shares.

5. Which of the following statements about private equity securities is typically correct?

- A. They are commonly sold on secondary markets.
- B. They have public market determined quoted prices.
- C. They are primarily issued to institutional investors.

Solution:

Choice C is correct. Private equity securities are primarily issued to institutional investors.

Choice A is incorrect as private equity securities are not commonly sold on secondary markets.

Choice B is incorrect. Private equity securities do not have public market determined quoted prices.

6. Which one of the following statements most accurately describes a difference between private and publicly listed firms?

- A. Private firms are focused more on short-term results than publicly listed firms.
- B. Private firms are more likely to be in early and growth stages of the company life cycle than publicly listed firms.
- C. Private firms face higher regulatory constraints than publicly listed firms.

Solution:

Choice B is correct. Companies in early life-cycle stages are more likely to be unable to meet public listing requirements, so they are more likely to be private than publicly listed.

Choice A is incorrect as the lack of public financial disclosures and observable share prices allows private equity firms to focus relatively less attention on short-term results as compared to publicly listed firms.

Choice C is incorrect as private equity firms face fewer regulatory constraints than publicly listed firms.

2

EQUITY INSTRUMENT FEATURES



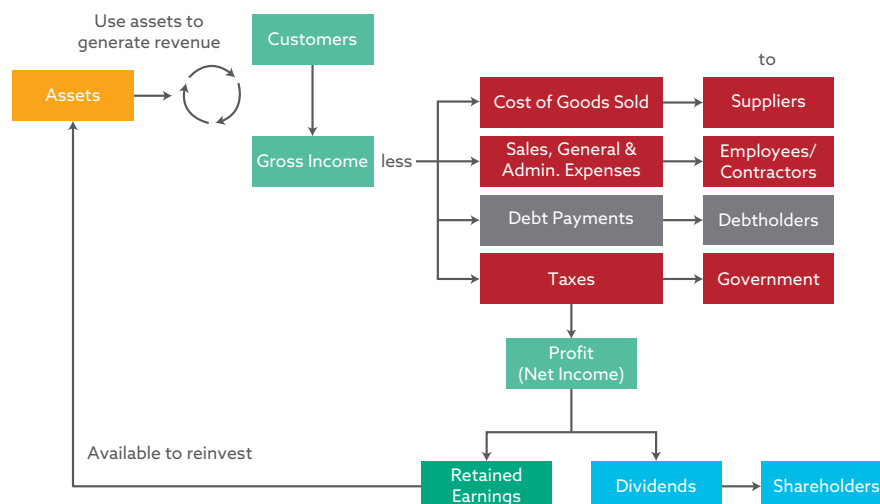
describe basic features and types of equity instruments

Equity instruments are securities that represent a standardized form of company ownership capital. In contrast to sole proprietorships in which the firm's owner supplies equity capital and manages company operations, equity securities in other organizational forms offer several advantages to issuers and investors (as described in earlier Corporate Finance learning modules) as they:

- enable greater transferability of shares,
- expand access to additional external capital, and
- limit the legal liability of owners.

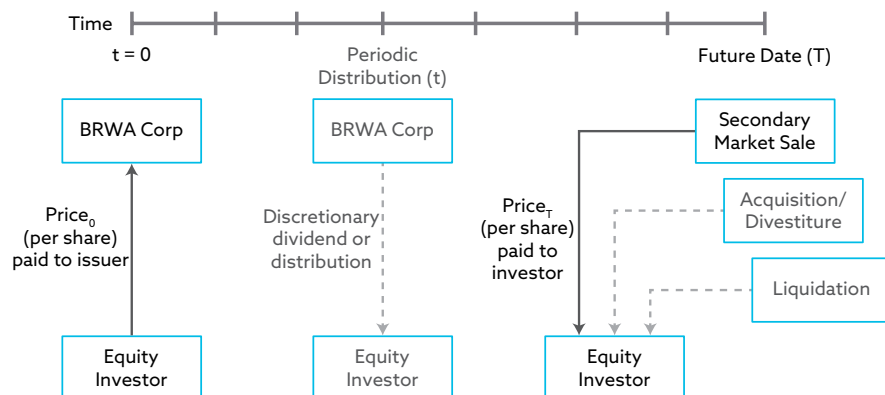
Share ownership represents a fractional stake in an issuer, which is typically a publicly listed or private corporation, granting shareholders specific economic rights and voting rights. Equity investors are owners with the right to receive any distributions as well as net proceeds from the sale of a division or a full dissolution of the company. They may also exercise voting rights on major corporate decisions, such as the election of the board of directors.

Unlike debt securities with a defined maturity, equity securities are outstanding for an indefinite period—that is, until a corporate restructuring event (such as an acquisition or divestiture) occurs, in which shareholders receive either another stock or cash in exchange for their ownership rights, or the firm is liquidated. In contrast to debt, firms issuing equity instruments are neither contractually obligated to repay funds received from shareholders nor to make any periodic payments to shareholders. Instead, common shareholders receive a proportional share of any net income distributed at the company's discretion and have a residual claim on the company's assets upon liquidation once all debts and preferred shareholders have been paid. Exhibit 1 compares the position of equity claims to those of other stakeholders.

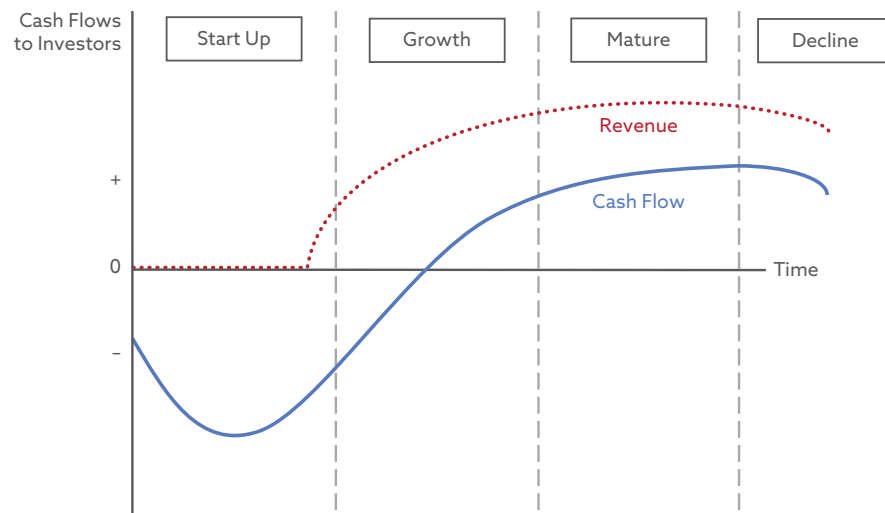
Exhibit 1: Equity versus Other Stakeholder Claims

Once all other stakeholder claims are satisfied, net income is either retained and reinvested in the firm or distributed to its shareholders. Such discretionary net income distributions to shareholders on a periodic basis are **dividends**, which a firm may pay to shareholders regularly, occasionally, or not at all.

Exhibit 2 shows the cash flows for an equity security issued by Bright Wheels Automotive (BRWA) Corporation.

Exhibit 2: BRWA Corporation Equity Cash Flows

If BRWA does not pay periodic dividends and is not subject to restructuring or liquidation, then an investor's return will consist solely of price appreciation or depreciation. The distinction between shares that pay a regular dividend and non-dividend-paying stocks is important for equity investors in assessing the type and expected timing of future cash flows. A key factor affecting a firm's dividend policy is its stage in the company life cycle. The following exhibit shows how cash flows to investors vary over company life-cycle stages.

Exhibit 3: Company Life-Cycle Stages

Start-up and growth companies tend to either generate net losses and therefore no earnings to distribute, or they generate low but growing earnings and cash flows, which are often fully retained and invested internally in expansion and new business opportunities. Equity investors during these life-cycle stages typically seek long-term capital appreciation over current income.

As earning growth slows, mature companies with stable cash flows and fewer new investment opportunities are more likely to regularly distribute a portion of earnings to shareholders. Equity investors seeking recurring income while retaining price appreciation potential often choose mature company stocks, which pay a regular dividend. For example, while around half of the companies in the NASDAQ-100 Index dominated by younger technology companies pay no dividend, over three-quarters of the firms included in the more diversified S&P 500 Index regularly distribute dividends to investors. The following case study highlights the example of an early-stage company's debut on the NASDAQ exchange as a publicly traded stock.

RIVIAN'S INITIAL PUBLIC OFFERING (IPO)

Rivian Automotive, a US-based electric vehicle manufacturer, successfully completed an initial public offering (IPO) of common stock on 10 November 2021. It was the largest US IPO since Alibaba listed its shares in 2014. Rivian sold around 153 million shares of Class A common stock at an IPO price of USD78 per share. The company raised approximately USD12 billion to fund capital to build one million vehicles per year by the end of the decade. Investor demand for Rivian shares showed considerable strength immediately after the firm's shares began trading as investors expected significant price growth.

Rivian's management focus on growth as opposed to shareholder dividends was evidenced by the IPO prospectus, in which the firm states: "We have never declared or paid any cash dividends on our capital stock. We currently intend to retain all available funds and future earnings, if any, for the operation and expansion of our business and do not anticipate declaring or paying any dividends in the foreseeable future."

The **equity prospectus** (or, in the case of private placements, offering memorandum) is distributed to prospective investors coinciding with share issuance. This document outlines the key features of equity securities along with the use of proceeds, details of the issuer's business, risk factors, management discussion and analysis, management and key shareholders, and financial information.

It is important to distinguish between the types of equity instruments being used. The two most widely used are **common shares** (also known as ordinary shares or common stock) and **preference shares** (also known as preferred stock). Common shares involve an ownership interest in a company that entitles an investor to both economic and voting rights, while preference shares involve economic rights but typically no voting rights. Preference shares represent claims that are subordinate to all debt instruments but are senior to common equity, and they may pay periodic dividends. Like common shares, preference shares are outstanding for an indefinite period.

Unlike fixed-income instruments and preference shares that have a specific par value, common shares are issued as distinct units that represent an investor's fractional claim on the issuer's assets less liabilities. This fractional claim may change as the number of shares outstanding changes. The economic and voting rights associated with common stock are as follows:

- *Voting Rights:* Specifies an investor's proportional right to participate in corporate governance decisions.
- *Dividend Rights:* Specifies an investor's proportional right to any regular or extraordinary distributions of net income or assets.
- *Conversion Rights:* Addresses any rights an equity investor may have to exchange one class or type of equity security for another.
- *Liquidation Rights:* Addresses an investor's proportional claim to the distribution of net assets remaining if the corporation is dissolved, assets are sold, and proceeds are distributed based upon the priority of claims.
- *Split, Subdivision, or Combination Rights:* Addresses any changes to an investor's proportional claim if shares are split, subdivided, or combined.
- *Preemptive Rights:* Describes any preferential claims for existing investors to purchase newly issued equity securities or securities convertible into equity.

For example, Exhibit 4 summarizes the terms of BRWA's Class A common shares.

Exhibit 4: BRWA Corporation Common Stock – Brief Summary of Terms

[20,000,000 Shares Bright Wheels Automotive Corporation] Class A Common Stock

Issuer:	Bright Wheels Automotive Corporation
Number of shares issued:	20,000,000
Voting rights:	Holders of Class A common stock are entitled to one vote per share.
Dividends:	Stockholders of Class A common stock will share proportionally in any cash dividend declared by the Board of Directors, subject to any preferential rights of any outstanding preferred stock.

**[20,000,000 Shares Bright Wheels Automotive Corporation]
Class A Common Stock**

Liquidation rights:	In the event of liquidation, dissolution, or winding-up of the issuer, all stockholders of common stock, regardless of class, will be entitled to share ratably in any assets available for distributions to stockholders of shares of the issuer's common stock subject to the preferential rights of any outstanding preferred stock.
Preemptive rights:	Shares of Class A common stock do not entitle a stockholder to any preemptive rights enabling a stockholder to subscribe for or receive shares of stock of any class or any other securities convertible into shares of stock of any class.

Common Shares

Common shares are the predominant type of equity security. These shares offer an investor both economic and voting rights. Economic rights include a proportional claim to any dividends or distributions as well as a claim to the firm's net assets in the event of liquidation. Voting rights allow investors to participate in the corporate governance process. Votes are granted in proportion to share ownership unless otherwise specified.

Voting rights provide shareholders with the opportunity to participate in corporate governance decisions, typically including such considerations as the election of its board of directors, the decision to merge with or take over another company, and the ratification of auditors. Shareholder voting generally takes place during a company's annual meeting and may be conducted in person or via **proxy vote**, where a designated third party—another shareholder, a shareholder representative, or management—votes on a shareholder's behalf.

Issuer and investor benefits of common stock include the following:

- *Issuer benefits:* Issuers maintain operational control of the business while exchanging proportional and economic voting rights to investors in exchange for capital.
- *Investor benefits:* Common shares provide investors access to fractional economic and voting rights in corporations with a relatively small capital outlay regardless of the corporation's size. These ownership claims allow investors to benefit from any dividends paid and/or growth in a firm's net asset value, as well as freely buy and sell these claims in the secondary market.

The issuance of common shares may stem from the desire to source public equity capital for expansion, sell a company that has been restructured under private ownership, or diversify family wealth from a concentrated ownership position. From an issuer's perspective, new common shares may alter the company's capital structure and its weighted average cost of capital. As shown in the following case, a new equity issuance often is publicly listed on an **equity exchange** or stock exchange, which is a rules-based, organized, and regulated centralized market venue where eligible market participants regularly meet to trade shares.

BIRKENSTOCK'S (BIRK) INITIAL PUBLIC OFFERING (IPO)

On 10 October 2023, Birkenstock Holdings PLC, a 250-year-old shoe manufacturer based in Germany, issued around 32 million shares listed on the New York Stock Exchange priced at USD46 per share. Of the USD1.5 billion of stock sold, only about a third represented new proceeds to the company. The remaining two-thirds of IPO proceeds went to existing shareholders who used the offering to sell ownership stakes in the now-public company. Birkenstock

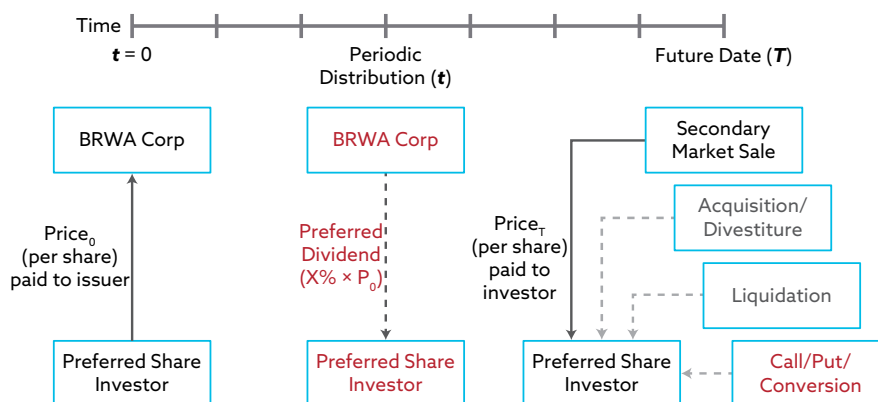
was owned and managed by the namesake family until 2012, when two brothers, Alex and Christian Birkenstock, transferred senior management responsibilities to a long-time company employee along with a company outsider. In 2021, the Birkenstock brothers sold a majority ownership stake to a private equity firm. The 2023 IPO was the final stage of a process in which the family was able to diversify its wealth from the renowned footwear brand.

Preference Shares

Preference shares (or preferred stock) have characteristics of both debt securities and common shares. Preference shareholder claims are senior to those of common shareholders. Like fixed-debt coupons, preference share dividends are a fixed percentage of the **preference share par value** (or original asking price) of each preferred share.

Unlike the failure to make timely scheduled debt payments, deferral of preference dividend payments does not constitute a default on a contractual obligation of the company. As in the case of common shares, preference shares may be perpetual (i.e., have no fixed maturity date) and may pay dividends indefinitely. For example, Exhibit 5 shows the cash flows for preferred shares issued by BRWA Corporation.

Exhibit 5: BRWA Corporation Preferred Share Cash Flows



Preference shares rank above common shares with respect to dividend payments and the distribution of the company's net assets upon liquidation. However, preference shareholders generally do not share in the operating performance of the company and lack voting rights, unless explicitly allowed for upon issuance. Preferred dividends provide investors with income that typically exceeds common stock dividends. However, preferred shares do not offer the same capital appreciation potential as common stock.

Fixed preferred dividends do not typically rise or fall with company performance, as is generally the case with common shares, and they are often **cumulative preferred shares**—that is, unpaid preferred dividends accrue and must be paid before common dividends can be issued.

Issuers frequently offer beneficial contingency provisions to investors, such as participation in improved performance and downside protection, to attract preferred shareholders. For example, **participating preferred shares** provide shareholders the opportunity to receive an additional dividend if company profits exceed a prespecified level. Alternatively, preferred shareholders may seek protection against weaker firm performance by negotiating the right to receive a minimum price for their shares or a fixed payout if the firm is dissolved.

In contrast to these standard features of preference shares, the following ownership stakes sometimes incorporate additional contingency provisions that may benefit the issuer or investor. For example, in the case of **callable preference shares**, the issuer has the right, but not the obligation, to repurchase preferred shares at a predetermined price. **Puttable preference shares** grant investors the right, but not the obligation, to sell preferred shares back to the issuer at a predetermined price referred to as the **put contingency price**. In the case of liquidation, a predefined distribution of cash proceeds to preferred shareholders in advance of any common shareholder payments is referred to as a **liquidation preference payout**.

CASE STUDY



BRWA Preference Share Issuance

BRWA Corporation issues 50,000 puttable preference shares at par value of USD100 per share. Dividends are payable annually at a rate of 5% of par value. BRWA's preferred dividend is cumulative, and the put contingency price is set at 90% of par value. In the year following the issuance of these shares, BRWA's operating performance declines such that the company faces difficulties in paying the preferred dividend.

1. Which one of the following statements is most correct about BRWA's preferred dividend?
 - A. BRWA must borrow sufficient funds to ensure that the preferred dividend cannot fall below USD4.50 per share annually.
 - B. BRWA can skip a preferred dividend payment but must make the payment in the future before it may pay common dividends.
 - C. The put contingency allows BRWA to sell additional preferred shares to fund the preferred dividend of USD5.00 per share.

Solution:

Choice B is correct. The cumulative feature of BRWA's preference shares allows dividend payments to be deferred. However, preferred dividends paid must include any missed dividend payments from prior years before the company is allowed to pay a common stock dividend.

Choice A is incorrect because the statement incorrectly implies that the dividend cannot fall below 90% of the dividend rate of 5%. This is an incorrect interpretation of the put contingency.

Choice C is incorrect as this statement also provides an incorrect statement regarding the put contingency. The put contingency provides the investor with the right to sell BRWA preference shares to BRWA at a price of USD90 per share.

In some situations, it can be advantageous for issuers and investors to consider preferred shares that include the contingent right to change into common shares in the future. For example, closely held, early-stage business ventures with financing needs can lower costs and avoid relinquishing current ownership control by issuing preferred instead of common shares. Debt financing is rarely available to companies with little to no revenue, no assets that can be used as loan collateral, and limited cash flow from operations. However, investors may be willing to purchase a debt-like security in exchange for the right to participate in ownership and future capital appreciation if the business performs well. **Convertible preference shares** entitle shareholders to exchange preference shares into a specified number of common shares (and in some cases at a predetermined price) agreed upon issuance in the future. By granting this

future right to investors, issuers can raise capital at a lower cash cost (that is, at a lower dividend than a standard preferred share) than equity today prior to issuing public equity securities in an initial public offering (IPO).

CASE STUDY



BRWA and Convertible Preference Shares

In assessing the possibility of issuing preference stock, BRWA considered the use of convertible preference shares.

1. Which statement below most correctly reflects the benefit to BRWA and the benefit to investors if BRWA had issued convertible preference shares as opposed to preference shares with no contingency features?
 - A. BRWA benefits by not selling an ownership stake in the company, and investors benefit by receiving downside risk protection.
 - B. BRWA benefits by selling a contingent ownership stake in the company, and investors benefit by receiving a higher dividend rate.
 - C. BRWA benefits by paying a lower dividend rate, and investors benefit by receiving the option to convert the preference shares into common shares if BRWA's stock performs well.

Solution:

Choice C is correct. The convertible contingency provides preference share investors with the option to convert these shares into common stock, thus providing investors with some upside gain potential if BRWA performs well. Investors effectively pay for this option by agreeing to receive a lower dividend, which creates a benefit for BRWA.

Choice A is incorrect because investors do not receive downside risk protection. Additionally, BRWA may not have sold a current ownership stake, but they have sold a contingent ownership stake if investors convert the preference shares into common shares.

Choice B is incorrect because investors receive a lower, not higher, dividend rate in exchange for buying convertible preference shares.

Key features of common versus preference shares as compared to debt are summarized below in Exhibit 6.

Exhibit 6: Features of Common Shares, Preference Shares, and Debt

Features	Common Shares	Preference Shares	Debt
Contractual obligation	No	No	Yes
Economic rights	Residual claimant; Highest uncertainty of payoffs	Senior to common shares; Less uncertainty of payoff than common shares; Dividends are more debt-like	Senior to preference shares; Least uncertainty of payoff than common or preferred
Voting rights	Yes	Rarely	No
Time to maturity	Indefinite life	Indefinite life	Finite maturity

QUESTION SET

1. Which one of the following most correctly describes the features of common shares?
- A. Contractual obligation with finite maturity and rights to vote on corporate decisions
 - B. Non-contractual obligation with indefinite maturity and rights to vote on corporate decisions
 - C. Non-contractual obligation with finite maturity and rights to vote on corporate decisions

Solution:

Choice B is correct. Common stock is not a contractual obligation. It has no defined maturity date and provides holders with voting rights on corporate decisions.

Choice A is incorrect because common stock is not a contractual obligation with finite maturity.

Choice C is incorrect because common stock does not have finite maturity.

2. Which one of the following contingency features of preference shares do *not* provide investors with downside risk protection?
- A. Participating
 - B. Puttable
 - C. Liquidation preference

Solution:

Choice A is correct. Participating preference shares provide investors with the opportunity to receive an additional dividend if company profits exceed a prespecified level. Thus, this contingency provides upside gain potential.

Choices B and C are incorrect as both provide protection against downside risk.

3. Which one of the following statements does *not* typically reflect a primary reason why a company may issue publicly listed shares?
- A. To fund company expansion
 - B. To diversify founders' wealth from a concentrated ownership position
 - C. To take advantage of perceived market mispricing of company stock

Solution:

Choice C is correct as this is the least correct response provided. Perceived market mispricing assumes market inefficiency in that markets will not take into consideration the company's decision to issue shares in their price bids for new shares. While market inefficiency may exist, the other two statements better reflect company rationales for issuing shares.

Choice A is incorrect as the funding of expansion is a common rationale for issuing public shares.

Choice B is incorrect as IPOs are frequently used as a way for the early-stage owners, such as founders and/or family members, to be able to sell their previously illiquid shares.

PUBLICLY LISTED VERSUS PRIVATE EQUITY SECURITIES

3

- | contrast features of publicly listed versus private equity securities

Corporations issue publicly traded equity securities to raise capital from equity investors, reduce the ownership concentration of existing private shareholders, or transfer ownership from government to investors in the case of a state-owned corporation. While most small and medium-size firms are privately held by entrepreneurs and unavailable to outside investors, a sizable proportion of the largest firms are publicly listed. Among those equity investments available to investors, the market value of publicly traded firms far exceeds that of the private equity market, although the *number* of private companies exceeds that of public companies, as described below.

SIZE AND SCOPE OF PUBLICLY LISTED VERSUS PRIVATE EQUITY MARKETS

The vast majority of corporations in both developing and developed economies are privately held. For example, of the over 5.7 million corporations in the United States reported by the Census Bureau in 2020, just 0.1% had publicly traded shares. However, publicly issued shares are far more prevalent among larger companies. For example, Forbes lists twelve privately held firms in the United States with revenues greater than USD25 billion in 2023, while over ten times as many publicly listed corporations exist above this revenue threshold.

The rapid growth in capital raised from institutional investors by private equity funds has drawn significant attention to this market, although it remains much smaller than the public market. For example, while the global market for publicly issued equities is approximately USD100 trillion as noted earlier, the global private equity fund market constitutes less than 10% of the overall equity market in the early 2020s, according to the consulting company McKinsey.

The accessibility of exchange-traded markets allows for public company ownership to be widely distributed among individual and institutional shareholders. As publicly listed companies, firms must meet and maintain the equity listing requirements of an exchange. These may involve achieving a specified level of market capitalization, sales, or profits; a minimum number of shareholders; and adherence to standardized financial and non-financial disclosure requirements, such as the periodic distribution of audited financial statements. While publicly listed equities remain dominant in terms of market value, several market forces are key determinants in the number of publicly listed companies, as described below.

PUBLIC EQUITY MARKET LISTING TRENDS

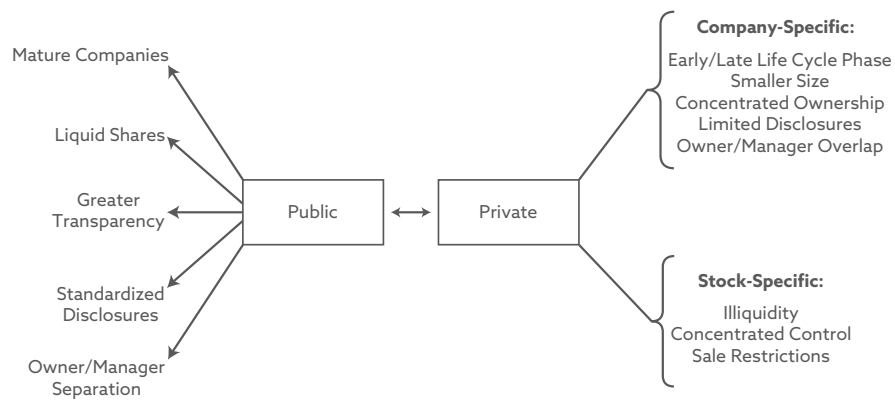
The number and size of equity listings are subject to constant change as firms issue IPOs or delist their shares. For example, the number of companies publicly listed on US stock exchanges reached its peak in 1997. In the following two decades, the number of listed companies fell by over 50%. This decline is in sharp contrast to the approximately 50% growth in US publicly listed companies observed between 1975 and 1997. While this change in public equity listings has been most notable in the United States, other developed markets have also

experienced smaller declines in later years. Private equity investment among institutional investors, on the other hand, has seen significant growth in recent decades from a much smaller base.

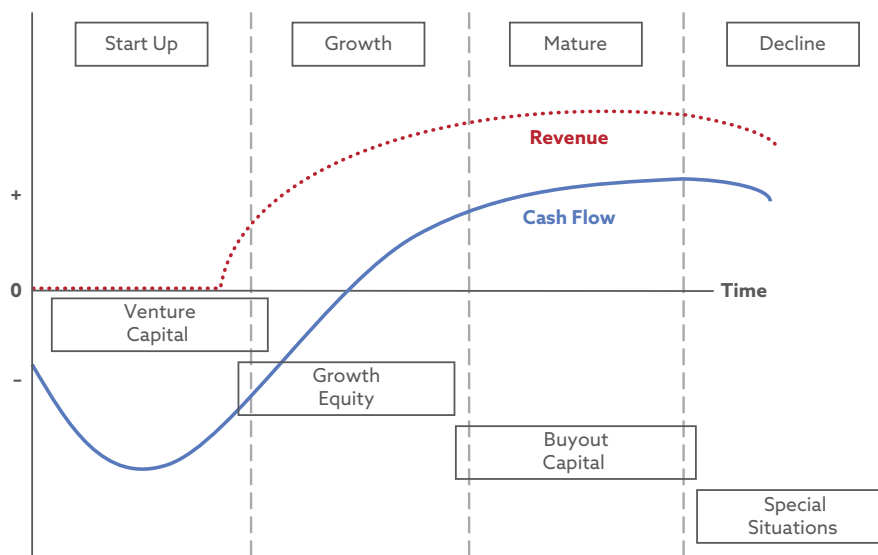
In addition to mergers and acquisitions, more stringent regulations in developed markets have made a public listing less attractive by increasing the costs compared to the benefits. Also, an increase in available capital from private equity investment funds allows for companies to grow larger while remaining private, or for larger publicly listed firms to be acquired by private equity firms.

Private companies range from start-up companies to established single owner operated firms and multigenerational family businesses. Except for entrepreneurs or partners directly contributing ownership capital to an owner-operated firm, private equity investments are far less accessible to individual investors than publicly traded shares. Many large and well-known firms, such as IKEA and the LEGO Group, remain privately held because of their desire to benefit from concentrated ownership and control, which for owner-managed firms includes limiting public disclosure requirements of financial condition. Exhibit 7 contrasts the features of publicly listed versus private companies.

Exhibit 7: Publicly Listed versus Private Company Features



Publicly listed equity securities are usually liquid because they are traded in open and accessible markets, allowing investors to readily buy or sell positions and easily observe current and historical security prices as well as those for relevant benchmarks, such as indexes. Illiquid private equity investments, in contrast, often involve non-standardized terms and trade infrequently on a negotiated basis. Most private companies are relatively small, but those that become sufficiently large to become eligible for an exchange listing often strategically opt to forego access to public equity markets to maintain control or avoid the greater cost and scrutiny associated with public financial disclosures. These private companies usually have few equity investors, with either majority controlling or large minority stakes. Private equity use spans the company life cycle, as shown in Exhibit 8.

Exhibit 8: Private Equity Use over the Company Life Cycle

As shown in Exhibit 8, these private equity fund investments include **venture capital (VC)** for initial development of start-ups as well as buyout equity used to acquire, control, and transform mature companies later sold to private or public investors. In contrast to public equity investors who may buy or sell positions with greater frequency, these private equity fund investments are typically held over multiyear periods with returns realized only upon their sale to a new group of public or private investors.

For example, firms in an early development phase with negative cash flow and little to no revenue are ineligible for a public equity listing and must rely on private capital sources willing to invest in new companies with a high failure rate. Start-up firms often rely on founders, friends and family, and venture capital firms to provide private ownership capital to establish a product and market to reach the next stage of development. Early phase private investment is typically concentrated, with investors who take greater risk and often exercise greater control, as described in the following case study.

RIVIAN'S EARLY-STAGE PRIVATE EQUITY FINANCING

An earlier example described Rivian Automotive's 2021 IPO. The company was founded as Mainstream Motors in 2009 (changing to Rivian Automotive in 2011) by Robert Scaringe, as the sole employee and initial investor, shortly after he earned advanced degrees in mechanical engineering. A car enthusiast with a passion to develop a more sustainable transportation system, Scaringe's ideas for Rivian's electric vehicle models were based on converting standard technology to harness a more environmentally friendly energy source.

Rivian's early capital funding of USD1.25 million consisted of debt and warrants until 2017, when it received an undisclosed investment from the Japanese firm Sumitomo Corporation. In 2019, Rivian raised USD1.5 billion of equity investments from investors, such as Amazon.com, Inc., and Ford Motor Company. Amazon's investment in Rivian was part of the Climate Pledge Fund, a company effort to fund projects seeking net-zero carbon emissions by 2040. Prior to the

IPO, Rivian had raised a total of USD11 billion in several funding stages from a variety of private investors as well as from fund companies, including T. Rowe Price and Third Point.

As described in the above case study, start-up firms often receive multiple rounds of early financing, with investors able to reduce or exit holdings via an initial public offering or acquisition by another company. Alternatively, if a company remains private as it seeks more capital for profitable expansion, a separate group of equity investors in so-called **growth equity** may be used to reduce the concentration of initial investors.

In the case of mature companies, private equity investors may acquire a firm to take control and rapidly transform the company to increase returns. For example, a **leveraged buyout (LBO)** occurs when one or more investors purchase all the outstanding common shares of a publicly traded company. The transaction is referred to as a **management buyout (MBO)** if the group of private equity investors involves a company's existing management. Once fully purchased, often using significant debt financing, the shares cease to trade on an exchange and the investor group takes full control of the company in what is referred to as a **take-private transaction**. These private investors target firms whose assets they consider to be undervalued or may benefit from restructuring or other changes to increase efficiency and generate higher cash flows and profitability once debt costs are included. The objective of buyout equity is to restructure and sell the acquired company, by either issuing new public shares in the primary market or selling existing shares to a publicly listed or private firm. As shown in the following case study, the target of buyout equity may be a private rather than a publicly held company.

THE SALE OF BIRKENSTOCK TO A PRIVATE EQUITY FIRM

The 2023 IPO of Birkenstock Holdings, a 250-year-old German footwear company, was described in an earlier example. Two and a half years earlier, the private family-owned company was acquired for EUR4 billion by L Catterton, a private equity firm funded by the world's largest luxury goods firm LVMH Moët Hennessy Louis Vuitton. LVMH's founder, chair, and CEO, Bernard Arnault, also invested in the acquisition through his family office, Financière Agache. LVMH controls and manages over 70 global luxury brands.

The sale of Birkenstock represented an intermediate step in the family's reduction in ownership and management control of the company. In 2012, the two Birkenstock brothers, Christian and Alex, turned over control of the firm's operations to two co-CEOs, neither of whom were family members (although one was a long-time company employee). The private equity acquisition reduced the family's ownership stake to a relatively small minority position in the firm.

Many companies enter a period of decline as competition or technological advancement causes revenues and cash flow to fall, leading to financial distress. Private equity may enter these **special situations**, which are defined as investments in firms facing financial distress or event-related changes. For example, the shares of a publicly listed company in bankruptcy may be delisted from an exchange if it no longer meets requirements, such as a minimum share price. Alternatively, if a firm facing restructuring or reorganization must quickly raise equity capital to continue operations, it may consider a **PIPE (private investment in public equity) transaction**. Under a PIPE transaction, firms can raise private capital more quickly and with fewer transaction costs than via public equity issuance. This selected group of private investors usually requires an incentive, such as a discounted share price, to participate. This transaction may be subject to existing shareholder approval and may give rise to conflicts between existing and new shareholders given the dilutive nature of such transactions.

CASE STUDY**BRWA PIPE Issuance**

As described in an earlier example, BRWA Corporation issued 20 million public common shares in an IPO. BRWA shares currently trade at USD8. Since its public market debut, BRWA's performance has lagged competitors, and it has lost market share in the traditional gasoline-powered vehicle segment. In response to weaker cash flows in its core business, BRWA management seeks to invest in new models using alternative fuel sources. To pursue this growth initiative, BRWA needs additional equity capital and seeks to complete the financing quickly. Rather than accessing public capital, which would be more time-consuming, BRWA raises new equity capital by selling 2 million shares of stock to several private investors with expertise in alternative energy at a discounted price of USD7 per share through a PIPE transaction.

1. Which one of the following best characterizes BRWA's common equity capital after the PIPE transaction?
 - A. 22 million shares outstanding at USD8 per share
 - B. 22 million shares outstanding at USD7 per share
 - C. 22 million shares outstanding at USD7.91 per share

Solution:

Choice C is correct. By issuing additional common shares at a discounted price, BRWA has diluted the value of their common stock. The company received USD14 million ($= \text{USD7} \times 2 \text{ million shares}$) in addition to the existing common share capital of USD160 million ($= \text{USD8} \times 20 \text{ million shares}$). Dividing the sum of these two numbers by 22 million shares gives a diluted value per share of USD7.91 ($= \text{USD174 million} \div 22 \text{ million shares}$).

As shown in the above examples, companies often take advantage of both private and public equity markets during different life-cycle phases.

QUESTION SET

1. Which one of the following statements comparing publicly listed and private firms in the United States is most correct?
 - A. Private firms outnumber publicly listed firms by approximately ten times.
 - B. Private firms outnumber publicly listed firms by approximately one hundred times.
 - C. Private firms outnumber publicly listed firms by approximately one thousand times.

Solution:

Choice C is correct. About 0.1% of US corporations are publicly traded, thus implying that private firms outnumber publicly listed firms in the United States by about one thousand times.

2. Which one of the following statements provides the most correct comparison between publicly listed and private equity securities?

- A. The total global market value of publicly listed equity securities is significantly greater than that of private equity securities.
- B. Individual investors have similar abilities to own publicly listed and private equity securities.
- C. Mature corporate equities are more likely to be private equity than publicly listed equity.

Solution:

Choice A is correct. Global public equity market values are approximately ten times greater than the market value of private equity.

Choice B is incorrect as private equity investments are far less accessible to individual investors.

Choice C is incorrect as mature firms are more likely to have publicly listed equity securities, although the buyout equity segment of the private equity market consists of mature firms.

3. Which one of the following statements is most correct about companies with private equity securities?

- A. Private companies consist almost entirely of early-stage companies.
- B. Private companies typically are characterized by a separation of ownership and management control.
- C. Private companies face limited disclosure requirements.

Solution:

Choice C is correct. Private companies are not required to provide the same level of public disclosure as publicly listed companies.

Choice A is incorrect as early-stage companies constitute firms that may attract private venture capital, which only comprises a fraction of overall private equity investment.

Choice B is incorrect as private equity owners typically exercise a high degree of management control.

SUMMARY: EQUITY INSTRUMENT FEATURES

- Equity instruments represent an ownership stake in a publicly listed or private issuer that grants investors certain economic and legal rights, such as dividend, conversion, and liquidation rights, as well as voting rights.
- Common shares represent a residual ownership interest that entitles investors to economic and voting rights. Preferred (or preference) shares are senior to common shares in the capital structure, and investors typically receive non-discretionary dividends but generally lack voting rights.
- Publicly listed equities are shares that meet legal, regulatory, and compliance requirements of the exchange or over the counter market where they are traded. Private equity often involves controlling or large minority stakes in firms across the entire company life cycle, which are illiquid and trade on a negotiated basis.
- Private equity is prevalent among smaller firms as well as larger firms controlled by fewer investors seeking to avoid the greater cost and scrutiny associated with public disclosures.
- Over the company life cycle, uses of private equity range from venture capital for start-ups to buyout equity for mature companies. It can also be used for private investments in publicly listed companies.

PRACTICE PROBLEMS

The following information relates to questions 1-4

Audrey Lake is the Chief Investment Officer of Mid States University Management Corporation (MSUMC), an endowment fund overseeing the university's investment portfolio. MSUMC invests over half of its assets in equity securities, both public and private. As part of the equity review process, Lake has assigned the analyst team questions to consider. First, MSUMC's Investment Office is more interested in being able to benefit from improved company performance than the seniority of claims when they invest. Second, MSUMC's Investment Office seeks a "voice" in management decisions of the companies in which they invest. Third, MSUMC targets broad exposure across the equity universe, buying both publicly listed and private equities across company life-cycle stages. Finally, MSUMC has a very long investment horizon.

1. Which of the following securities has the lowest seniority on claiming cash flows from a portfolio company?
 - A. Debt
 - B. Preference shares
 - C. Common shares
2. Which of the following provides the most correct response as to why MSUMC prefers investing in common shares rather than preference shares?
 - A. Common shares typically provide higher dividends than are received from holding preference shares.
 - B. Common shares provide the opportunity to vote on corporate decisions in addition to the possibility for improved dividends or higher stock price as company performance improves.
 - C. Dividends on common shares are less uncertain because of the higher seniority of common shares' payoffs.
3. In which company life-cycle stage is MSUMC most likely to find both publicly listed and private company equities as investments?
 - A. Start-up
 - B. Growth
 - C. Mature
4. MSUMC's long investment horizon is most relevant to which one of the following equity market features in their decision to allocate between publicly listed and private equities?
 - A. Owner–manager separation versus overlap
 - B. Standardized disclosures versus limited disclosures

C. Liquidity versus illiquidity of shares

SOLUTIONS

1. Choice C is correct. Holders of common shares have the residual claim on a company's cash flows and assets. Common shareholders are paid once all other claims are settled.
Choice A is incorrect as debtholders have the highest seniority in receiving cash flows relative to preference and common shareholders.
Choice B is incorrect as preference shareholders are senior to common shareholders.
2. Choice B is correct. MSUMC's stated goals of being able to capitalize on improvements and having a "voice" on corporate decisions best match the features of common shares.
Choice A is incorrect as common shares may have lower or higher dividends than preference shares.
Choice C is incorrect as common shares are less senior than preference shares.
3. Choice C is correct. In the mature stage of the company life cycle, MSUMC is likely to find both publicly listed equity and private equity securities. Publicly listed companies are often mature stage companies, while buyout equity companies in private equity are often mature stage companies.
Choice A is incorrect as start-up companies are unlikely to qualify for public listing because of insufficient revenue and profitability.
Choice B is incorrect given that growth stage companies may not have achieved the profitability or other characteristics needed for public listing.
4. Choice C is correct. MSUMC's long investment horizon allows them to consider holding illiquid private shares if they earn, on average, a higher return than public share holdings.
Choice A is incorrect as this feature is more relevant to MSUMC if they want to be able to exert greater management control through equity ownership, and it is independent of MSUMC's investment horizon.
Choice B is incorrect as the disclosure requirements are also independent of MSUMC's investment horizon.

LEARNING MODULE

2

Equity Jurisdictions, Classes, and the Voting Process

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	describe differences in economic and voting rights and other characteristics among different jurisdictions and equity classes
☐	describe the voting process and roles of the issuer's management, board, proxy advisors, asset managers, and asset owners in the process

INTRODUCTION: EQUITY JURISDICTIONS, CLASSES, AND THE VOTING PROCESS

1

Issuer and investor decisions related to equity securities go well beyond determining whether to issue or invest in publicly listed common or preferred shares. From an issuer perspective, the choice of listing venue and jurisdiction and the distribution of economic and voting rights among shareholders are important considerations in addressing the interests of existing shareholders as well as attracting new equity investors. Investors seeking global diversification who face the decision of how and where to invest must consider factors such as market practices, trading hours, and currency exposures when selecting the best approach to owning individual equity positions. This module addresses shareholder rights across equity classes and jurisdictions as well as the shareholder voting process and the role of issuers, investors, and advisors.

PRE-TEST: EQUITY JURISDICTIONS, CLASSES, AND THE VOTING PROCESS



1. When investing in unsponsored depositary receipts, the voting rights to the shares in the trust belong to:
 - A. the depository bank.
 - B. the investors in the depositary receipts.

C. the issuer of the shares held in the trust.

Solution:

Choice A is correct. In the case of an unsponsored depositary receipt, the depositary bank owns the voting rights to the shares. The bank purchases the shares, places them into a trust, and then sells shares in the trust, not the underlying shares, in other markets.

Choice B is incorrect. Investors in sponsored depositary receipts have voting rights, while investors in unsponsored depositary receipts do not.

Choice C is incorrect. A corporate issuer does not own voting rights to shares issued regardless of whether these are placed in depositary receipts or not.

2. Which one of the following statements about depositary receipts (DRs) is most correct?

- A.** Unsponsored DRs are generally subject to greater reporting requirements than sponsored DRs.
- B.** DRs allow investors to avoid additional costs of directly investing in non-domestic equities.
- C.** American Depositary Receipts facilitate the trading of US equity securities in non-US markets.

Solution:

Choice B is correct. DRs provide a mechanism through which investors can gain access to non-domestic equities while avoiding the costly features of setting up access to directly buy shares in other jurisdictions.

Choice A is incorrect as sponsored DRs are subject to greater reporting requirements compared to unsponsored DRs.

Choice C is incorrect as ADRs are designed to facilitate trading of non-US equities in US markets.

3. Dual listing of shares refers to:

- A.** issuing a class of shares with differing economic rights.
- B.** issuing a class of shares with differing voting rights.
- C.** issuing shares for trading on two or more markets.

Solution:

Choice C is correct. Shares may be listed on multiple exchanges across a variety of jurisdictions, including outside a company's home market, which is referred to as a dual listing.

Choices A and B are incorrect as both responses refer to the practice of creating a dual class structure.

4. Which one of the following statements is most correct about shareholder voting rights?

- A.** Shareholders eligible to vote on corporate actions must be the beneficial owners of the corporation's stock.
- B.** The record date for voting is typically about 3 days in advance of the shareholder meeting.

- C. Shareholders voting by proxy typically must cast their votes at an earlier date than those who attend the annual shareholder meeting to cast votes.

Solution:

Choice C is correct. To run an orderly voting process, votes cast by proxy must be received earlier than those cast by shareholders attending the annual shareholder meeting.

Choice A is incorrect because a portion of eligible voters may have sold their shares after the record date but before the voting deadline.

Choice B is incorrect as the record date for voting is typically about 30 days to allow shareholders time to review and analyze necessary information to make informed voting decisions.

5. Which one of the following statements is most correct about proxy advisors?

- A. Proxy advisors provide voting instructions to shareholders and custodians in advance of shareholder meetings.
- B. Proxy advisors provide institutional investors with voting recommendations in advance of shareholder meetings.
- C. Proxy advisors count the results of voting from votes cast in advance of shareholder meetings.

Solution:

Choice B is correct. Proxy advisors serve many roles in facilitating the shareholder voting process, and one of the most important is to analyze shareholder proposals and provide voting recommendations to institutional investors.

Choice A is incorrect because shareholder voting instructions are provided by voting services, not proxy advisors.

Choice C is incorrect because the vote count is handled by a party known as tabulators or registrars, not proxy advisors.

6. Which one of the following statements most correctly explains to shareholders the importance of an effective corporate voting process?

- A. Corporate voting should serve to protect the board and management from shareholder lawsuits.
- B. Corporate voting should serve to help the board protect and enhance the value of investor capital.
- C. Corporate voting should serve to ensure that the company addresses all ESG-related risks identified by shareholder proposals.

Solution:

Choice B is correct. Corporate voting allows shareholders to have a voice in ensuring that the board is focused on value creation.

Choices A and C are incorrect because both reflect that boards should only be focused on ensuring that the company protects against downside risks. Such strategies are unlikely to create value for the firm's shareholders.

2

JURISDICTIONS AND EQUITY CLASSES

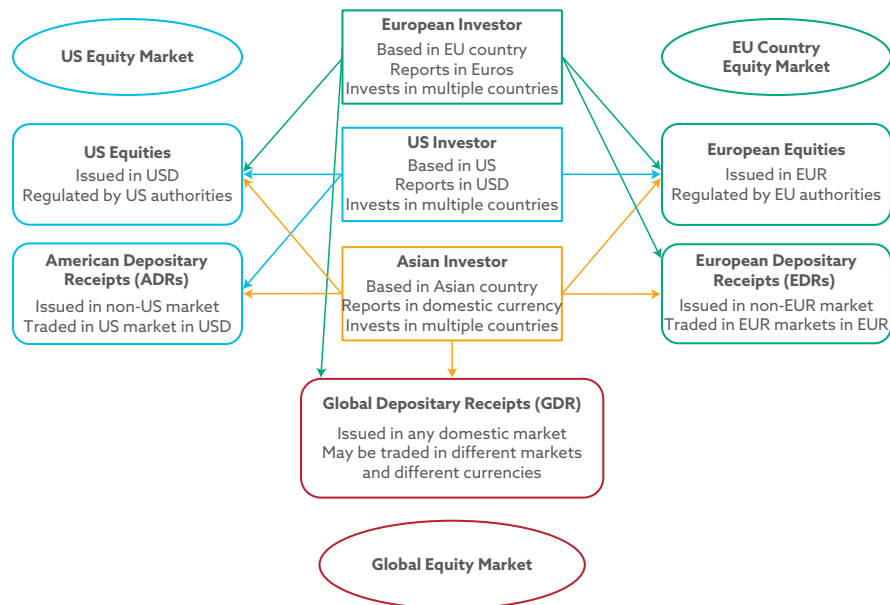


describe differences in economic and voting rights and other characteristics among different jurisdictions and equity classes

An earlier Equities learning module described the use of publicly listed equity issuance among corporations to gain broader access to investors, such as individual and institutional shareholders. While the prior learning module distinguished between publicly traded and privately held equity within the same jurisdiction, we now turn our attention to investments in publicly listed equity securities other than shares listed on a domestic exchange by a domestic company.

Investors seeking to diversify their investments beyond domestic equity securities may be looking to offset a **home asset bias**, a well-known preference for investors to allocate excessive proportions of their investment portfolio in securities issued by companies in their home country. When considering investments in equities issued by foreign companies, investors have several alternatives, as outlined in Exhibit 1.

Exhibit 1: Domestic, Foreign, and Global Equity Markets – Investor Perspective



The individual equity investment alternatives shown in Exhibit 1 may be characterized as direct or indirect as outlined below.

Direct Investing

One approach for an investor to gain non-domestic country equity exposure is to directly purchase or sell the equity securities of a foreign issuer in the issuer's domestic market. When investing directly in foreign equity markets, investors incur setup and transaction costs associated with separate local securities and cash accounts to accommodate foreign equity security buying and selling, dividend disbursements, and

tax payments or rebates. They must become familiar with trading, clearing, settlement, tax rules, and other market procedures and practices, and they must accommodate trading hours in different time zones. Direct investment may also involve language barriers and different degrees of financial disclosure and market liquidity. All related transactions, including share purchases and sales, dividend payments, and other distributions, are conducted in local currency, not the investor's domestic currency, which may involve foreign exchange transaction risk or capital restrictions and different tax implications. While large institutional investors are far more likely to have the expertise, scale, and resources to invest directly in non-domestic shares, smaller and individual investors tend to hold fewer individual foreign shares.

Firms are generally neither restricted to issuing shares in their domestic market nor within a single market. For example, many firms choose to list shares in a single market outside of their home jurisdiction, while others choose a **dual listing**, with shares simultaneously listed and traded in two or more markets. For example, the British oil and gas company Shell plc simultaneously lists its shares on the London Stock Exchange and on the Euronext exchange in Amsterdam.

Indirect Investing

A common means of indirect investment in an individual equity security involves the use of **depository receipts (DRs)**. A depository receipt is created when the equity shares of a foreign company are deposited with a domestic bank (i.e., a depository), which issues receipts that represent the deposited shares.

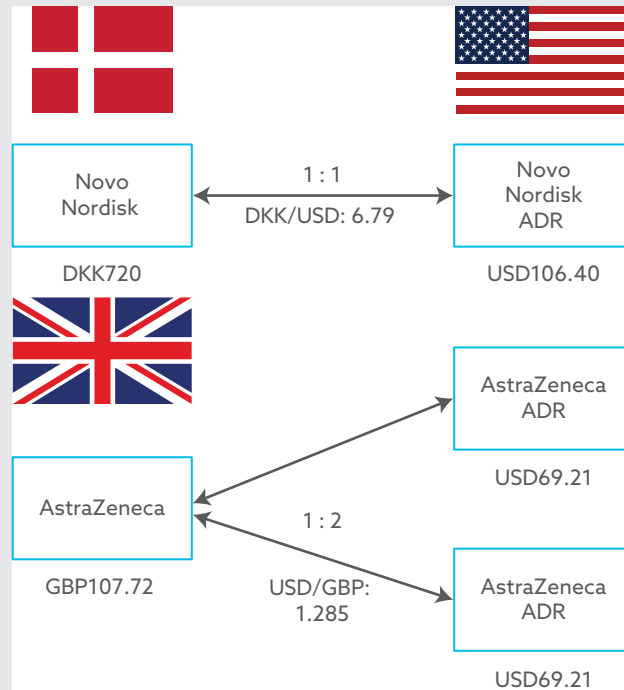
Depository receipts were established to match issuer needs to attract capital from investors outside of the domestic equity market with the desire of many (especially individual) investors to avoid the additional costs of direct investing described earlier. The number of receipts issued, and the price of each DR, is based on a ratio representing the number of depository receipts to shares as well as the exchange rate between the foreign and domestic currencies.

DRs trade like ordinary shares of a local exchange denominated in the domestic currency and represent a proportional economic interest in a foreign company, allowing the publicly listed shares of a foreign company to be traded on an exchange outside its domestic market. DRs are usually named by the market in which they trade, such as American Depositary Receipts (ADRs), the oldest and largest DR market that facilitates trading of non-US equity securities in US markets.

TWO EUROPEAN HEALTHCARE COMPANY ADRS

Novo Nordisk based in Denmark and AstraZeneca, a UK-based company, are publicly listed global pharmaceuticals firms whose shares trade in the US through ADRs. Novo Nordisk's Class B shares trade primarily in Copenhagen in Danish kroner (DKK), with ADRs at a ratio of 1:1, while AstraZeneca shares trade in London in British pounds (GBP). AstraZeneca ADRs reflect a 1:2 ratio, or one share for every two ADRs. The relationship between the respective domestic share prices and ADRs is shown in Exhibit 2.

Exhibit 2: Novo Nordisk and AstraZeneca Share and ADR Prices



On 8 January 2024, the closing price of Novo Nordisk Class B in Copenhagen was DKK720 per share, while later that same day, the company's ADRs closed in New York at a price per share of USD106.40. We may examine the price relationship between domestic shares and ADRs by dividing Novo Nordisk's DKK share price by the USD ADR price:

$$\frac{\text{Novo Nordisk DKK share}}{\text{Novo Nordisk USD ADR}} = \frac{\text{DKK720}}{\text{USD106.40}} = \frac{\text{DKK6.79}}{\text{USD1}}$$

This relationship implies a DKK/USD exchange rate of 6.79, which closely approximates the foreign exchange rate between the DKK and USD on that day.

AstraZeneca shares closed in London at a price of GBP107.72 per share, and their ADRs closed at USD69.21 per share. Given that two ADRs represent one AstraZeneca share, a slightly modified calculation may be used to derive the implied USD/GBP exchange rate as follows:

$$\frac{\text{AstraZeneca USD ADR} \times 2}{\text{AstraZeneca GBP share}} = \frac{\text{USD138.42}}{\text{GBP107.72}} = \frac{\text{USD1.285}}{\text{GBP1}}$$

Note the ADR price (in this case two units) is in the numerator rather than the denominator, as USD serves as the price currency per unit of the base currency GBP in the currency exchange rate term. These two examples demonstrate that depositary receipts are likely to trade at very similar prices to shares issued in their domestic markets. While slight variations may occur due to different market closing times and intraday exchange rate changes, instances of significant pricing differences may arise. Analysts must consider the ratio of depositary receipts to shares as well as the exchange rate to evaluate the degree of mispricing.

European depositary receipts (EDRs) involve non-European equities traded in European markets that are typically denominated in EUR. As more firms outside of the United States and Europe issue shares on exchanges outside of their domestic market, the number of domestic depositary receipt schemes continues to grow, as described in the following case study.

DEPOSITORY RECEIPTS IN INDIA AND CHINA

Beyond the United States and Europe, other large countries have made attempts to create depository receipt markets. As one example, Standard Chartered PLC, the British multinational bank, issued Indian Depository Receipts (IDRs) in May 2010, hoping that these instruments would provide opportunities for Indian investors to participate in the company's growth. Standard Chartered PLC is headquartered in the United Kingdom, but 90% of its profits come from Asia, Africa, and the Middle East. Approximately a year later, hedge funds and foreign institutions sold IDRs heavily after the Securities and Exchange Board of India (SEBI) announced that IDRs could not be converted into shares of common stock. Traders had been buying the IDRs at discounts to the prices available for Standard Chartered common stock traded in Hong Kong SAR and London, and SEBI's ruling effectively negated the arbitrage strategy being employed. The IDR market never recovered from the Standard Chartered experience; no additional IDRs were listed. Standard Chartered delisted their IDRs in mid-2020.

China has focused on creating a depository receipt market to make Chinese Mainland companies that originally listed overseas directly available to domestic investors through Chinese Depository Receipts (CDRs). In October 2020, the Beijing-based electric scooter company Segway-Ninebot offered an IPO of CDRs on the Shanghai STAR market. Despite being based in the Chinese Mainland, Ninebot registered in the Cayman Islands. Several Chinese Mainland technology companies had chosen to register in foreign jurisdictions, and analysts viewed the creation of the CDR market as a way for companies such as Ninebot to return to the Chinese Mainland equity markets.

Global depository receipts (GDRs) are issued outside of the company's home country and outside of the United States. While most GDRs are denominated in USD, the depository bank that issues GDRs is generally located (or has branches) in countries on whose exchanges the shares are traded. Note that although GDRs cannot be listed on US exchanges, they can be privately placed with institutional investors based in the United States.

A key advantage of GDRs is that they are not subject to the foreign ownership and capital flow restrictions that may be imposed by the issuing company's home country because they are sold outside of that country. The issuing company selects the exchange where the GDR is to be traded based on investor familiarity with the issuer company and/or a large international investor base. The London and Luxembourg exchanges were the first to trade GDRs, while others include the Dubai International Financial Exchange and the Singapore Stock Exchange.

As for any secondary market, attracting DR market participants requires that buyers and sellers enjoy sufficient market liquidity and other domestic market benefits that outweigh the cost of trading in foreign markets.

Although investors in theory should be able to align DR prices to foreign share prices through arbitrage, there are instances where this does not occur. DRs trade at a discount to shares in an issuer's home market and in some cases may face restrictions or delisting, as described in the following case study.

TRADING SUSPENSION IN RUSSIAN GDRS DURING THE UKRAINE WAR

Investors in GDRs may face geopolitical risks associated with international sanctions imposed on a specific country. For example, the London Stock Exchange (LSE) suspended trading in the GDRs of Russian-based companies in early 2022. This action followed numerous international sanctions imposed against Russia

in reaction to events in Ukraine. By halting trading, the LSE effectively made the Russian GDRs worthless, ensuring that Russian firms were not able to raise capital by issuing additional GDRs on the LSE.

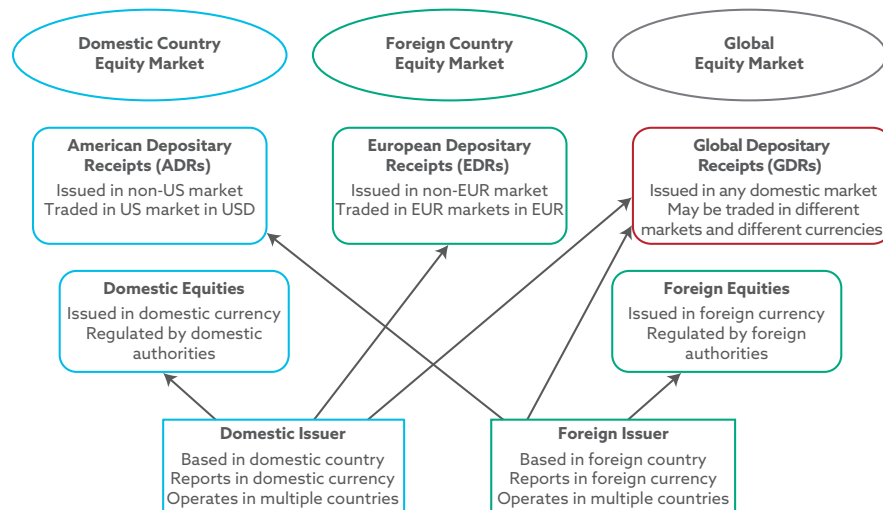
DRs may be sponsored or unsponsored. In a sponsored DR, the foreign company whose shares are held by the depository has a direct involvement in the issuance of the receipts. Investors in sponsored DRs have the same rights as the direct owners of the common shares (e.g., the right to vote and the right to receive dividends).

For unsponsored DRs, the underlying foreign company is not involved with the issuance of the receipts. Instead, the depository purchases the foreign company's shares in its domestic market and then issues the receipts through brokerage firms in the depository's local market. In this case, the depository banks, not DR investors, retain the voting rights.

Sponsored DRs are generally subject to greater reporting requirements than unsponsored DRs. In the United States, for example, sponsored DRs must be registered (meet the reporting requirements) with the US Securities and Exchange Commission (SEC).

Equity issuers must weigh the costs versus benefits between issuing in their own domestic country equity market, issuing in a foreign country's equity market (which may perhaps represent a more important market in terms of sales or available investors), or issuing in a global equity market (engaging in multiple listings on exchanges using DRs as the best approach to accessing investors without diluting the liquidity of shares in their primary trading venue). A breakdown of these domestic, foreign, and global equity market alternatives available from an *issuer* perspective is shown in Exhibit 3.

Exhibit 3: Domestic, Foreign, and Global Equity Markets – Issuer Perspective



Dual Class Shares

Differences in the distribution of economic and/or voting rights arises under a so-called **dual class share structure**, which in many cases involves a widely held, publicly traded share class with one vote per share, while a separate non-traded share class held exclusively by company founders, family members, or insiders has several votes

per share. This structure allows stakeholders who may retain a minority economic ownership to effectively control the company through voting powers, as described in the following case study.

META (FORMERLY FACEBOOK) DUAL CLASS EQUITY OWNERSHIP STRUCTURE

Meta Platforms, formerly known as Facebook, completed its initial public offering (IPO) on 18 May 2012. As part of the IPO process, Facebook issued 180 million shares of Class A stock at USD38.00 per share. Several early-stage investors were able to reduce their concentrated positions by selling Class A shares to the public as well. Following the IPO, Facebook had a total of 635,881,796 Class A common shares outstanding. Each Class A common share was entitled to one vote per share. Facebook's common equity also included over 1,500 million shares of Class B common stock, entitled to 10 votes per share. The company's founder, chair, and CEO, Mark Zuckerberg, owned less than 1% of Class A shares following the IPO and approximately 60% of Class B shares. As a result of this dual class structure, Zuckerberg effectively controlled well over half of Facebook voting rights following the IPO.

His majority voting control has continued despite a reduction in the proportion of Class B to Class A shares. As of early 2024, Zuckerberg's voting control remained above 60%. The primary implication of Meta's dual class structure is that the company's strategic direction over time has reflected Zuckerberg's priorities but not necessarily those of most public investors.

As shown in the Meta (formerly Facebook) example, dual class share structures with an unequal distribution of voting rights allow entrepreneurs to maintain control. While both economic rights and voting rights are often distributed to shareholders on a proportional basis, several jurisdictions allow the uneven distribution of rights across different share classes. For example, the following case involves a firm whose founders have retained *all* the company's voting rights within one share class while granting equal economic rights to a separate widely held share class that has *no* voting powers.

NEWS CORP EXAMPLE

News Corp is a global media and information company operating primarily in the US, Australia, and the UK. The company's products include the *Wall Street Journal* and Realtor.com, and the Australian operations include many leading news outlets in the country. News Corp is incorporated in the United States, and the stock trades on both US and Australian exchanges. The company has two classes of publicly traded shares as of August 2023: approximately 380 million shares of Class A and approximately 192 million shares of Class B. Both classes of News Corp shares have equal rights to receive dividends, in addition to equal liquidation rights. However, Class B shares carry the right to one vote per share, while Class A shares possess no voting rights. The company's Class A shares are widely held, with approximately 15,000 shareholders of record; Class B shares are held across about 400 shareholders, including the Murdoch Family Trust, which owns about 40% of the Class B shares.

In other cases, dual shareholder classes may be subject to both disproportionate voting rights and economic rights. For example, in many cases, an equity issuer seeking to maintain preferential voting rights for one class of shares held by founders or family members may offer additional economic rights to another class in exchange for less voting power. This case is described in the following example.

TATA MOTORS' DVR AND ORDINARY SHARES – ECONOMIC AND VOTING RIGHTS

Tata Motors is the automotive subsidiary of Tata Group, India's largest conglomerate. It is controlled by the family of the same name through the Tata Sons holding company. While the company issued its first ordinary (or common) publicly listed shares decades earlier, in 2008 the firm conducted a follow-on equity offering in two classes to fund its USD2.3 billion purchase of Jaguar-Land Rover. In addition to ordinary shares, the company issued a new class of differential voting rights (DVR) shares, which carry one-tenth the voting rights of ordinary shares while receiving a 5% higher dividend than ordinary shares if one is declared. The company raised INR15 billion in DVR shares, which were offered at a 10% price discount to ordinary shares.

While Tata Motors was able to raise additional equity capital without diluting ownership, individual investors less focused on exercising voting rights were able to acquire newly issued shares at a discount with a potential dividend benefit over ordinary shareholders, as well as benefit from any narrowing of the spread differential between the two share class prices.

Despite the initial positive market reception of DVR shares and follow-on Tata Motor DVR offerings in 2010 and 2015, the lack of institutional investor interest in the shares and added complexity made the DVRs less liquid. Also, as the company ceased paying a regular dividend for several years, DVR shareholders failed to realize any economic benefit from foregoing voting rights; the DVR shares traded at a discount as low as 40% to 50% to ordinary shares.

Given a desire to simplify its capital structure and promote the liquidity of ordinary shares, Tata Motors decided to retire its DVR shares 15 years after they were first issued. The company offered DVR shareholders a 20% premium to the market price, exchanging seven ordinary shares for each 10 DVR shares.

QUESTION SET



1. Which one of the following statements most correctly states an advantage of investing in a foreign country stock using a depositary receipt compared to a direct investment?
 - A. Reduction of home asset bias
 - B. Avoidance of currency exchange rate risk
 - C. Ability to execute trades during domestic market hours

Solution:

Choice C is correct. Indirect investing using a depositary receipt issued on behalf of a foreign company issuer allows an investor to avoid many complications inherent in direct foreign investing, including different market trading hours.

Choices A and B are both incorrect because both direct and indirect investing can reduce an investor's home asset bias; however, both direct and indirect investing may create potential currency exchange rate risk for the investor.

2. Which one of the following statements is most correct for a depositary receipt (DR) with a ratio of one share to five DRs?
 - A. Each DR trades at approximately 5 times the price of each share.

- B. Each DR trades at approximately 5 times the price of each share after accounting for the exchange rate.
- C. Each DR trades at approximately 0.2 times the price of each share after accounting for the exchange rate.

Solution:

Choice C is correct. The DR ratio of five depositary receipts representing one share is equivalent to one DR representing 0.2 shares.

Choice A is incorrect because it reflects a ratio of five shares to one DR and does not mention the exchange rate.

Choice B is incorrect because it also reflects a ratio of five shares to one DR.

3. Which one of the following decisions is most likely for a start-up company founder who wants to eventually take the company public while retaining corporate voting control?
- A. Creating a dual listing of company shares
 - B. Creating a dual class share structure
 - C. Issuing global depositary receipts

Solution:

Choice B is correct. A dual class structure allows the founder to create a class of closely held shares with greater voting rights than the widely held class of shares issued to public investors through an IPO.

Choice A is incorrect as dual listing refers to registering shares for trading in multiple markets.

Choice C is incorrect as depositary receipts are used to access trading in global markets other than the domestic market, but they do not necessarily provide any voting advantages to the founder.

EQUITY VOTING PROCESS

3



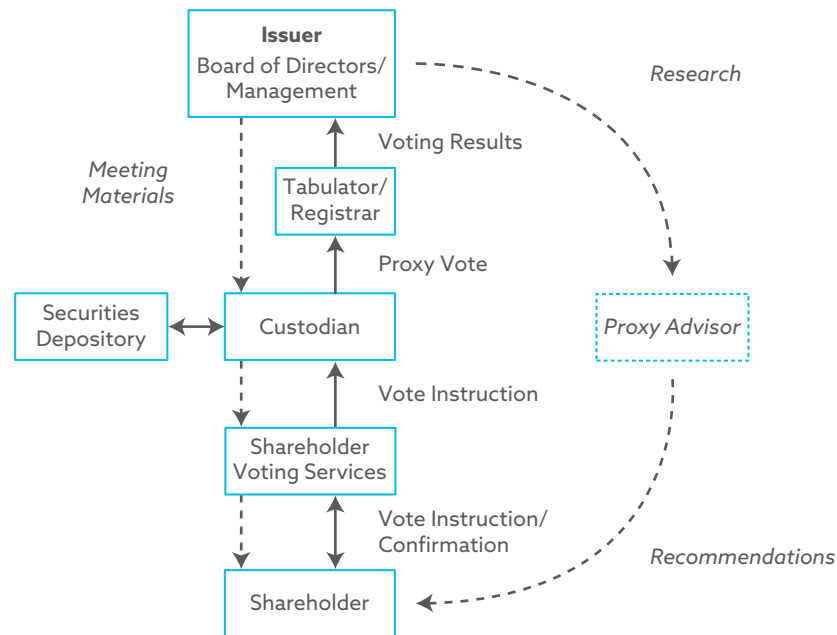
describe the voting process and roles of the issuer's management, board, proxy advisors, asset managers, and asset owners in the process

Equity Voting Process

Although corporate law varies widely across jurisdictions, in most cases the shareholders of a publicly listed corporation have the right to vote on important company matters at the firm's annual general meeting or a special shareholder meeting. In addition to standard proposals put forward by company management—such as the election of the board of directors, resolutions on executive compensation and dividends, and the ratification of an independent auditor—investors can also submit **shareholder proposals** for consideration and an eventual vote. A shareholder proposal involves a recommendation or requirement that the company or its board take a specific action.

The equity voting process is used to ensure sufficient time to collect and vet proposals, identify existing shareholders, confirm voting eligibility, distribute voting materials, and gather the votes of investors who are not physically present at the meeting (or those who delegate their votes to a third party in a **vote by proxy**) in advance in time for tabulation at the meeting. Exhibit 4 outlines the primary steps in the voting process.

Exhibit 4: Shareholder Voting Process



Preparation for the shareholder vote begins well in advance of the annual meeting, as firms weigh procedural matters that require investor approval and solicit shareholder proposals. In the United States, for example, all shareholder proposals must be received at least 120 days prior to the planned release of meeting materials, which must be vetted by the firm with their acceptance or rejection subject to review by regulatory authorities.

Meeting materials are compiled in what is referred to in the United States as a **proxy statement**, which includes management and shareholder proposals along with a detailed agenda for the annual shareholder meeting and voting requirements and deadlines. Proxy statements typically include background information such as board candidate credentials, company performance, policies, and procedures as well as board recommendations for or against all decisions, with management's rationale for supporting or rejecting all proposals.

Proxy statements are usually distributed to shareholders indirectly via a **custodian** in whose name shares are typically held on behalf of an investor. Custodial services provided by a bank or broker involve maintaining ownership records with a securities depository as well as managing the safekeeping, settlement, and reporting for securities held by investors who are the true or so-called **beneficial owner** of shares. In rare instances, beneficial owners themselves are the registered holders of equity securities, with their name directly reflected in company records as a shareholder.

Voting eligibility is limited to **shareholders of record** or those shareholders whose identity is confirmed by a securities depository or custodian as of the **record date** or predetermined cutoff date for voting eligibility. Shareholders of record may or may not be the current beneficial owners of the stock. For example, a broker may serve as custodian for shares owned by a client; the custodian's name is listed on the stock certificates while the client is the beneficial owner of the shares.

Since the record date is often 30 days or more in advance of the shareholder meeting, investors who have sold the stock after the record date remain eligible to vote, while those who purchase shares after the record date but before the shareholder meeting are not.

Once shareholders are identified and receive voting information, they must analyze this information, decide, and cast their votes via a shareholder voting service prior to the voting deadline or cutoff date, which is typically earlier for those shareholders who vote by proxy than those who attend the shareholder meeting in person.

Given the vast amount of information that portfolio managers must evaluate when exercising voting rights for large, well-diversified equity portfolios comprised of many issuers across multiple markets, institutional investors often engage **proxy advisors** to assist in the voting process. Proxy advisors play an important role in gathering information, benchmarking important corporate governance principles across issuers, making voting recommendations, and keeping records associated with exercising shareholder votes across investment portfolios, as described below.

THE ROLE OF PROXY ADVISORS

Two proxy advisory firms, Institutional Shareholder Services (ISS) and Glass Lewis, are dominant players in the US proxy advisory market with a combined 97% market share. Both firms work with institutional investors and publicly listed companies around the world, providing coverage for tens of thousands of corporate shareholder meetings annually across over 100 markets. Given the variety of jurisdictional corporate governance practices, proxy advisors assist institutional investors and asset managers in making informed voting decisions on their public shareholdings. For example, in its 2024 Shareholder Proposal and ESG-Related Issues Benchmark Policy Guidelines, Glass Lewis recommends investor support of proposals that clearly link the subject of the proposal to firm value enhancement or risk mitigation in cases where the company's management and board have failed to address the issue. On the other hand, Glass Lewis tends to recommend against shareholder proposals that are likely to encroach on company management's day-to-day business operations without materially enhancing value or mitigating risk.

In seeking to focus on resolutions designed to enhance firm value or mitigate risk where management has fallen short, proxy advisors seek to create consistent governance standards across companies and industries. While management proposals typically involve legally prescribed matters, shareholder proposals raised by investors may sometimes suggest that boards take action that would encroach on management's day-to-day operation of the company. In their voting decisions, investors should consider whether such proposals have the potential to materially improve corporate value or mitigate risk. These types of proposals often involve measures seeking to strengthen a company's environmental or social policies, as described below.

SHAREHOLDER PROPOSALS TO ADDRESS CLIMATE RISK

Following the adoption by global and US regulators of more stringent rules that prevent companies from excluding investor proposals—particularly those related to environmental, social, or ethical concerns—equity issuers have faced a growing number of shareholder proposals in these areas from socially responsible investment funds and public policy groups. These and other shareholders are actively seeking corporate governance changes via the equity voting process.

As a result, 2023 saw a large increase in climate-related shareholder proposals among oil and gas producers directly involved in greenhouse gas emissions, as well as issuers indirectly involved in the industry.

For example, ExxonMobil faced several climate-related shareholder proposals at its annual meeting, which included proposals on adjusted greenhouse gas emissions reporting, establishment of a Paris Climate Agreement (Scope 3) target, direct methane emissions measurement, and the establishment of a new board committee on decarbonization risk. In each case, the board recommended against the proposals, which failed to pass.

Separately, the City of New York's retirement plans introduced a shareholder proposal at JPMorgan Chase's 2023 annual meeting seeking company disclosure within a year of the bank's plan to align practices for lending and underwriting to oil and gas and power generation companies with 2030 absolute greenhouse gas (GHG) emissions reduction targets. JPMorgan management recommended against voting for the proposal, citing existing company initiatives in support of the goal to provide "secure, reliable, affordable energy while promoting clean energy solutions and strategies to reduce the world's carbon footprint." The proposal came to a vote but was rejected by investors, receiving the support of just 12% of shareholders who cast their ballots.

Despite the frequent rejection of individual shareholder proposals by investors, the equity voting process provides a forum for investors to raise specific matters and exert influence on firms to meet desired standards. While shareholder proposals are typically non-binding, those that are approved, or fail despite the support of a significant fraction of shareholders, cause investors to expect the firm to address the issue raised to avoid reputational harm or signal a lack of responsiveness to investor concerns.

Market regulators often take measures to increase transparency and shareholder protection in the voting process. For example, the European Union changed how proxy voting is conducted via the Shareholder Rights Directive (SRD). These rules were established to improve voting processes and information flows among institutional investors. The SRD improved the voting process by:

- ensuring shareholders may receive immediate electronic confirmation that their votes have been received and further confirmation of how they were recorded and counted;
- enabling the remote exercise of voting rights, with standardized voting-related information available electronically; and
- enhancing duties of care and disclosure applicable to intermediaries across the voting chain (including proxy advisors).

Beyond the shareholder proposals described in the earlier case study, timely and active engagement in the voting process are extremely important aspects of share ownership for many asset managers and asset owners. In some cases, investors delegate their votes to asset managers, while in others, they are actively engaged in the voting process as part of their investment policy. Investor voting should serve to help the board protect and enhance the value of investor capital. As demonstrated in the

following example, the voting process is a critical way for these investors to demonstrate their stewardship and governance principles to stakeholders through concrete voting policies focused on effective boards and shareholder protections at firms held in their investment portfolio.

NORGES BANK INVESTMENT MANAGEMENT (NBIM) VOTING GUIDELINES

Norges Bank Investment Management (NBIM) oversees the investment of the Government Pension Fund Global, Norway's sovereign wealth fund tasked with managing approximately NOK12,340 billion in assets at the end of 2022. NBIM outlines their voting guidelines on minority shareholdings in publicly listed companies around two fundamental pillars: effective boards and shareholder protection.

In their voting guidelines, NBIM views effective boards as characterized by three elements:

1. Independence relates not only to the overall proportion of independent directors but also to the membership of key board committees, such as the remuneration committee, as well as separation of board chair and chief executive officer positions.
2. Board composition areas of focus ensure that board members can commit sufficient time to company board responsibilities; ensure that independent board members bring appropriate industry expertise to their board positions; and finally, ensure that board composition reflects appropriate diversity.
3. Boards should be accountable to shareholders with respect to the board nomination and election process, board decisions and conduct, and recruiting and paying the right CEOs appropriately.

In their voting guidelines, NBIM outlines shareholder protection with the following three components:

1. NBIM has approval guidelines related to voting rights, shareholder meetings, and governing documents. Notably, NBIM does not support companies with unequal voting rights across differing share classes.
2. Reporting guidelines focus on issues associated with companies' external auditors, quality and timeliness of company financial statements, and sustainability disclosures.
3. NBIM's capital guidelines provide value creation expectations for corporate share issuance or M&A plans as well as any related party transactions.

Overall, investors expect the companies whose shares they hold to protect and enhance the value of their investment by mitigating and managing risks through pursuit and execution of business strategies that increase value of shareholdings over time. The equity voting process is designed to ensure that these governance goals are attained.

QUESTION SET



1. Which one of the following statements about shareholder proposals is most accurate?
 - A. Failed proposals receiving the support of many investors are often addressed to protect an issuer's reputation and respond to investor concerns.

- B.** Since shareholder proposals are typically binding, firms must take specific action when they are approved by a majority of investors.
- C.** Shareholder proposals are frequently dismissed by a firm's management without providing a rationale for their rejection.

Solution:

Choice A is correct. Despite their failure to be adopted, shareholder proposals that receive significant investor support are often addressed by the issuer after such a vote.

Choice B is incorrect as shareholder proposals are typically non-binding.

Choice C is incorrect as market regulators typically require some justification for the rejection of shareholder proposals.

2. The primary goal of the corporate shareholder voting process should be to:

- A.** ensure that the corporation's management is focused on value creation.
- B.** ensure that the corporation's board is independent.
- C.** ensure that the corporation's board is accountable to shareholders.

Solution:

Choice A is correct. While corporate shareholder voting may include multiple objectives, the goal is to ensure that the corporation's management is able to focus on improving the firm's value for shareholders.

Choices B and C are incorrect as both responses are only elements of creating an environment in which management can pursue value creation for shareholders.

3. A broker buys Apple shares on behalf of a client, and the stock certificates are listed in the broker's name as of the record date for Apple's next shareholder meeting. The broker then sells the shares after the record date but before the proxy voting deadline. Who has voting rights to the shares?

- A.** The broker
- B.** The broker's client
- C.** The new owner(s) of the shares

Solution:

Choice B is correct. Although the Apple shares are registered in the broker's name, the broker is the custodian of the shares while the client has voting rights of the shares because these are owned as of the record date.

Choice A is incorrect because custody of the shares does not provide the broker with voting rights.

Choice C is incorrect because the new owner did not buy the shares before the record date.

SUMMARY: EQUITY JURISDICTIONS, CLASSES, AND THE VOTING PROCESS

- To access equity markets outside of their domestic jurisdiction, issuers may choose a dual listing, while investors may purchase depositary receipts representing a position in foreign shares denominated in domestic currency.
- Dual class share structures are used to address the interests of a firm's existing shareholders prior to wider ownership versus new shareholders via differences in the distribution of economic and voting rights.
- The equity voting process involves the identification and solicitation of shareholder votes on management proposals, such as electing directors or selecting an auditor, as well as on shareholder proposals for consideration.
- Institutional investors may use the voting process to exert their influence over issuers and to demonstrate investor stewardship and governance principles in accordance with their investment policies, often engaging proxy advisors to conduct research and provide recommendations.

PRACTICE PROBLEMS

The following information relates to questions 1-4

Audrey Lake is the Chief Investment Officer (CIO) of Mid States University Management Corporation (MSUMC), an endowment fund overseeing the university's investment portfolio. MSUMC invests a significant portion of the portfolio in publicly traded equities, and their public holdings are often scrutinized by university stakeholders with respect to companies' track records on managing environmental, social, and governance (ESG) issues. One of Lake's early actions as CIO was to create a set of shareholder voting policies for MSUMC to follow with respect to conversations about their portfolio companies. These guidelines were written after consulting with one of the dominant proxy advisory firms. The following points highlight some statements included within MSUMC's voting policy guidelines.

- MSUMC invests in foreign equities through the purchase of depositary receipts (DRs).
 - MSUMC has implemented a policy in which they will not purchase shares in a company with a dual class structure.
 - MSUMC has instituted voting guidelines that include a statement about voting on shareholder proposals that, if passed, would likely encroach on management day-to-day operations of companies.
 - MSUMC carefully considers their votes on all ESG-related shareholder proposals to ensure that they can effectively communicate their rationales for "no" votes to stakeholders when necessary.
1. Which type of DR would MSUMC most likely own?
 - A. Unsponsored DR
 - B. Sponsored DR
 - C. Either type of DR
 2. Which one of the following statements provides the most likely reason why MSUMC would place a ban on owning shares of companies with dual class structures?
 - A. Dual class structures typically provide multiple classes of traded shares with unequal voting rights.
 - B. Dual class structures typically provide unequal cash flow rights to shareholders.
 - C. Dual class structures typically provide one class of untraded shares with superior voting rights to those of traded shares.
 3. Which one of the following conditions would most likely be necessary for MSUMC to vote "yes" on a shareholder proposal that would cause the board to

encroach on day-to-day management?

- A. Management presents arguments in the proxy statement that the risk mitigation of the proposal is not material.
 - B. Management is already working on the issue underlying the proposal.
 - C. Management following the proposal would materially improve firm value.
4. Which of the following are typically correct about management and shareholder proposals?
- A. Both proposal types are shown in the company's proxy statement.
 - B. Shareholder proposals are binding if passed.
 - C. Shareholder proposals are frequently passed.
-

SOLUTIONS

1. Choice B is correct. MSUMC would prefer to own sponsored DRs because the holder of sponsored receipts retains shareholder voting rights.
Choices A and C are incorrect because the depository bank retains the voting rights on the company's shares in an unsponsored DR.
2. Choice C is correct. MSUMC is concerned with being able to vote meaningfully, so publicly traded shares that are unlikely to have sufficient voting power are not attractive for the endowment.
Choice A is incorrect because dual class structures rarely allow both classes to be traded.
Choice B is incorrect because MSUMC is more concerned with voting rights.
3. Choice C is correct. In general, a shareholder proposal that, if passed, would cause the board to encroach on management's day-to-day operations is not appropriate unless the actions would have material and positive effects on firm value and/or risk mitigation.
Choice A is incorrect because MSUMC is unlikely to base their vote on management's arguments.
Choice B is incorrect as this is typically likely to result in a "no" vote by MSUMC.
4. Choice A is correct. A company's proxy statement includes both management and shareholder proposals, including the company's voting recommendations to shareholders along with the rationales for the recommendations.
Choice B is incorrect because shareholder proposals, if passed, are typically not binding on the company.
Choice C is incorrect because shareholder proposals are not frequently passed.

LEARNING MODULE

3

Equity Issuance and Trading

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	describe primary and secondary public equity markets and their functions
☐	compare exchange, off-exchange, and over-the-counter equities trading
☐	describe liquidity measures for a publicly listed security and calculate float and average daily volume
☐	describe types of equity indexes

INTRODUCTION: EQUITY ISSUANCE AND TRADING

1

Public equity markets play an important role as a source of new ownership capital for growing or established companies as well as a venue for buying or selling existing equity securities. Like other financial markets, public equity markets are composed of primary markets in which issuers raise financing and secondary markets in which previously issued shares are traded. In addition to providing liquidity for investors to sell or buy outstanding shares, secondary equity markets play a key role in offering price discovery to investors as well as performance feedback to management and stakeholders.

The relative ability to trade securities is affected by whether execution occurs via an order-driven exchange or is managed using a broker network on an over-the-counter basis. Investors assess market liquidity based upon transaction costs as well as execution speed and trade volume. Similar to other asset classes, equity market indexes are used to gauge market risk and return versus individual securities, create new investment products, and evaluate fund manager performance. These and other aspects of equity issuance and trading are covered in this learning module.

PRE-TEST: EQUITY ISSUANCE AND TRADING

1. Which of the following is *least* likely to be a reason for a company to issue equity securities on the primary market?

- A. To raise capital
- B. To increase liquidity
- C. To transfer shares from public to private owners

Solution

C is correct. Issuing shares in the primary market often gives existing *private* owners the opportunity to sell shares to *public* investors rather than the opposite.

A is incorrect as issuance of equity securities is likely to raise capital.

B is incorrect as issuance of equity securities is likely to increase the liquidity of the company's stock.

2. Which of the following statements best distinguishes secondary equity markets from primary equity markets?

- A. Primary markets include only first-time equity offerings, while follow-on offerings of new shares are considered secondary market transactions.
- B. Pre-existing private shareholders may take advantage of primary markets to sell shares, while secondary markets are only available to public shareholders who purchase shares after an initial public offering (IPO) or direct listing.
- C. Primary markets are most active when market prices are rising, while secondary markets are active in both rising and falling markets.

Solution

C is correct. Primary issuance of new shares is most active when prices are rising, while secondary markets are active under all market conditions.

A is incorrect since follow-on offerings of new shares are considered primary issuance.

B is incorrect since pre-existing private shareholders may sell shares in the primary or secondary market.

3. Which one of the following trade types is most likely executed on an off-exchange basis?

- A. Limit orders
- B. Market orders
- C. Block trades

Solution

C is correct. Block trades are a major source of off-exchange activity.

A is incorrect as limit orders are typically executed on an exchange.

B is incorrect because market orders are executed on an exchange at the best currently available price.

4. Which one of the following indirect costs of trading can be observed from a limit order book?

- A. Market impact costs
- B. Bid–ask spread costs
- C. Delay costs

Solution

B is correct. The limit order book for a stock provides quoted bid and ask prices from which the bid–ask spread can be observed.

A and C are both incorrect, because neither of these indirect costs can be observed from the limit order book.

5. Which one of the following measures most correctly describes the proportion of a company's shares exchanged on a given trading day?

- A. Average daily turnover divided by market float
- B. Average daily turnover divided by total shares outstanding
- C. Total shares outstanding divided by average daily turnover

Solution

A is correct. Average daily turnover reflects the number of shares traded daily over a specified period, and market float reflects the number of shares available to trade. Thus, average daily turnover divided by market float most correctly describes the proportion of shares exchanged per day.

B is incorrect because not all shares outstanding may be freely traded, thus market float is utilized instead.

C is incorrect because this ratio would describe the number of days required to turn over all shares given average daily trading volume.

6. Which weighting approach is typically used for broad-based equity market indexes?

- A. Price-based
- B. Market capitalization-based
- C. Based on fundamental factors

Solution

B is correct. Typically, broad-based equity market indexes are weighted by each stock's market capitalization within the index. Hence, they offer a more accurate representation of the market's performance than price-based or fundamental factor-based weighting.

A and C are incorrect because neither price-weighting nor fundamental factor weighting are typical for broad-based indexes. These are more likely for more narrow indexes covering smaller subsets of a market.

2

PRIMARY AND SECONDARY PUBLIC EQUITY MARKETS



describe primary and secondary public equity markets and their functions

Primary Equity Markets

Primary equity markets are markets in which an issuer often lists and sells a new equity security to investors to raise ownership capital. In some cases, primary equity markets may be used to transform existing private ownership stakes into public equity capital. On the other hand, **secondary equity markets** are markets in which existing publicly listed shares are subsequently traded.

As described in an earlier lesson on corporate issuers, firms may go from private to public ownership via an initial public offering (IPO) or, less commonly, via a special purpose acquisition company or a **direct listing**. A direct listing involves the listing of already issued and existing shares on an exchange at a price determined by the market without the use of an underwriter.

For example, an IPO, described in earlier learning modules, involves a privately held firm entering the public equity market for the first time. These firms usually engage financial intermediaries known as investment banks to underwrite the sale of new shares to public equity investors. Equities may be sold via a **public offering** or a **private placement**, in which only a selected investor, or group of investors, may purchase the securities.

Primary equity issuance usually involves a transfer from private to public ownership and may occur under the following circumstances:

- An early-stage firm seeks to expand beyond private founder and venture capital financing to attract public equity investment.
- A new public company is created as a result of the divestiture or spin-off of a division or business line from an existing public company.
- A firm whose shares were previously listed returns to the public equity market after a multi-year period over which private owners have restructured the business.

In other cases, a national or regional government accesses the primary equity market in a full or partial **privatization** to transfer part or all of an existing government-held corporation to public investor ownership, as in the following example.

HIDROELECTRICA'S IPO

Hidroelectrica, a Romanian hydropower company, successfully completed an IPO in July 2023, raising over 9 billion Romanian leu (RON). While Hidroelectrica continues to be primarily owned by the Romanian government with a stake of approximately 80%, the IPO was initiated by the Bucharest Stock Exchange listing of a minority shareholder stake by Fondul Proprietatea SA (FP), a publicly traded investment fund with stakes in several government-owned companies. FP was established in 2005 to compensate expropriated property owners during

the Communist era in Romania. Following the IPO, Hidroelectrica became the fourth-largest weighting on the Bucharest Stock Exchange as well as the largest Romanian company traded on the exchange.

Regardless of a prospective equity issuer's circumstances, size, or targeted investor base, the primary issuance process involves fulfilling listing and other regulatory requirements as well as soliciting potential investors for new publicly listed shares. Exhibit 1 captures key elements of the IPO timeline.

Exhibit 1: IPO Regulatory and Marketing Timeline

Timing	T – 6 months	T – 3 months	Issuance Date (T)
Regulatory	Issuer and advisers present proposed listing and timeline to market authorities and regulators	Confidential prospectus filing with market authorities and regulators	Prospectus approved by market authorities and released by issuer
Marketing	Issuer and advisers present company to equity analysts and prospective investors	Analysts publish reports; issuer and advisers speak to equity investors	Initial IPO price range established and book building phase begins

Documentation: The creation and approval of an equity prospectus is a key IPO requirement. This offering memorandum describes the company and its business model, key industry and business risks, audited financial statements, and such details as the number of shares to be issued, expected price range, timetable, and use of issuance proceeds.

Marketing: The issuer, its management, and advisers conduct a parallel process in which they introduce the company to equity analysts and prospective investors, gauging market acceptance and establishing demand as well as a price range for the prospective issuance.

In contrast to the fixed-income book building process that often occurs on the day of issuance, equity issuers and their advisers usually spend much more time educating prospective investors about their business model, business risks, and the use of proceeds as well as establishing an offering price.

Equity underwriters tend to have IPO pricing incentives that conflict with those of firm owners seeking to maximize the value of their business. Although underwriters may generate higher fees with a higher offer price, they often benefit even more from placing underpriced shares with preferred institutional investors identified as part of the book building process. Underpriced shares in IPOs provide incentives for investors to repeatedly seek allocations as new issuers come to the IPO market in hopes of securing attractive returns. As a result, underwriters are less likely to seek the highest offer price for issuers as this pricing behavior will reduce investor demand. This mispricing typically results in a commonly observed first trading day jump in the price of newly traded IPOs, as described below.

IPO UNDERPRICING AND ITS IMPACT ON ISSUERS AND INVESTORS

American academic Jay Ritter has conducted extensive studies on US IPOs. Using data from 1980 to 2023, Ritter documents the average underpricing (as measured by the percentage stock price increase on the first day of trading compared to the offering price) of over 9,000 IPOs at approximately 20%. Across all these IPOs,

this underpricing translates into USD233 billion of additional proceeds that could have hypothetically been raised by corporate issuers based upon the first trading day's closing price. While underpricing has also been observed globally, it is not consistent across all issuers. For example, in 2020, average underpricing in US IPOs was over 40%, but almost 22% of IPO issuers in 2020 experienced negative first-day returns, indicating that the IPOs were overpriced.

Why are IPOs underpriced? First, Ritter argues that underwriters have an incentive to underprice IPOs to generate demand among clients given the attractive first-day returns, especially for so-called “hot” IPOs, which are expected to generate significant investor demand. IPO underwriters also run full-service brokerages that charge fees and commissions. Investor clients are more willing to accept these fees and commissions if they receive preferential access to IPO allocations in return.

Why are corporate issuers willing to accept lower IPO proceeds? Ritter suggests that company managers view first-day returns as a tangible increase in their own wealth based on their holdings of existing company stock. At the same time, the net proceeds not realized due to underpricing are an opportunity cost to the overall firm. These managers typically place a greater value on the positive first-day return versus the opportunity loss from lower IPO proceeds—another example of the principal and agent problem, where the principal is the shareholder and the agent is the manager.

A less common form of primary equity issuance involves the use of a **special purpose acquisition company (SPAC)**. SPACs are created and exist solely for the purpose of raising capital to acquire an unspecified private company within a pre-agreed time frame, typically two years. These shell companies raise equity via an IPO and immediately place net proceeds in a trust account. The IPO proceeds may only be disbursed to complete an acquisition of a private company approved by shareholders (known as a **de-SPAC transaction**) within the contractual time frame or must be returned to investors.

THE RISE AND FALL OF SPAC ISSUANCE

While blank check companies or SPACs offer public market investors the opportunity to invest together with celebrities, successful entrepreneurs, or private equity companies in an unspecified company, SPAC issuance is highly episodic. According to Jay Ritter, of the nearly USD300 billion raised by SPACs in the United States since 1990, over 70% of these proceeds were raised in 2020 or 2021 in just over 860 IPOs. Far fewer such deals were completed in Europe over the same period given restrictions on sales to non-professional investors.

At the end of the two-year contractual time frame, SPACs must either complete a business combination approved by shareholders, extend the contractual deadline, or dissolve the company and return net proceeds to investors. As of mid-2023, nearly a third of SPACs created over this period had been liquidated. Others entered business combinations to convert early-stage or speculative private companies into public companies.

One example of a successfully completed de-SPAC transaction involves Diamond Eagle Acquisition Corporation, a SPAC created by media executive Jeff Sagansky and former MGM CEO Harry Sloan. Within a year of its USD400 million IPO in May 2019, Diamond Eagle partnered with Europe-based SBTech to acquire DraftKings, Inc., a private US-based fantasy sports and betting company established in 2012 to capitalize on the growing legalization of sports gambling, which is now listed on NASDAQ as DKNG.

Other less common forms of primary equity issuance include a **back door listing** or direct listing. A back door listing involves the acquisition of a public company by a private firm to take advantage of an existing listing. Direct listings, described earlier, are usually limited to a few large private firms already familiar to investors, as in the following example.

ROBLOX CORPORATION'S (RBLX) DIRECT LISTING

Roblox, a gaming platform popular with children, experienced rapid growth in users as the COVID-19 pandemic began in 2020. The company capitalized on this growth with plans to go public through a traditional IPO in late 2020. However, Roblox decided against this plan given the significant amount of underpricing observed in other high-demand IPOs. Roblox's CEO sought to take the company's shares public through a process that would improve outcomes for existing investors, including company employees looking to sell their shares in the public offering.

In January 2021, Roblox received USD29.5 billion in new financing from two private investment firms at USD45 per share. The company then went public through a direct listing about two months later, and shares were offered to existing shareholders. Roblox's shares began trading on the New York Stock Exchange on 10 March 2021 at a price of USD64.50 per share, and the stock closed at USD69.50 on its first day of public trading.

In contrast to the first-time offerings described above, a **seasoned equity offering** (or follow-on offering) involves the sale of additional units of a previously issued equity security. While this sale of new shares also requires a prospectus, it is typically an accelerated issuance process given greater market familiarity with the existing issuer. A follow-on offering may be dilutive or non-dilutive to existing shareholders, as shown in the following example.

FACEBOOK'S (META PLATFORMS) 2013 SEASONED EQUITY OFFERING

Following the IPO in May 2012 at an offer price of USD38 per share, Facebook offered an additional 70 million shares to the public at USD55.05 per share, of which:

- 27 million shares were newly issued, and
- 43 million shares were from existing shareholders.

As a result, Facebook's seasoned offering included both dilutive and non-dilutive components. That is, the 27 million new shares issued by Facebook were a source of dilution for existing shareholders.

As shown in the example, a dilutive follow-on offering, also referred to as a **primary follow-on offering**, involves the sale of *new* shares that decreases each existing share's proportional claim to the firm's net income and assets. This approach is used by firms to raise additional equity capital for investment purposes and/or to reduce debt. A non-dilutive or secondary follow-on offering, in contrast, involves the sale of *existing* shares privately held in the public market for the first time, which does not change an existing shareholder's proportional claim to the firm's net income and assets under liquidation. Non-dilutive offerings allow firm founders and private investors to reduce their concentrated share ownership positions and diversify their equity holdings.

Secondary Equity Markets

Secondary equity markets facilitate the trading of previously issued public equity securities. In contrast to primary markets, which are most active when prices and valuations are rising, the ability to buy and sell outstanding equity shares is equally important in both rising and falling markets, as described in the following case study.

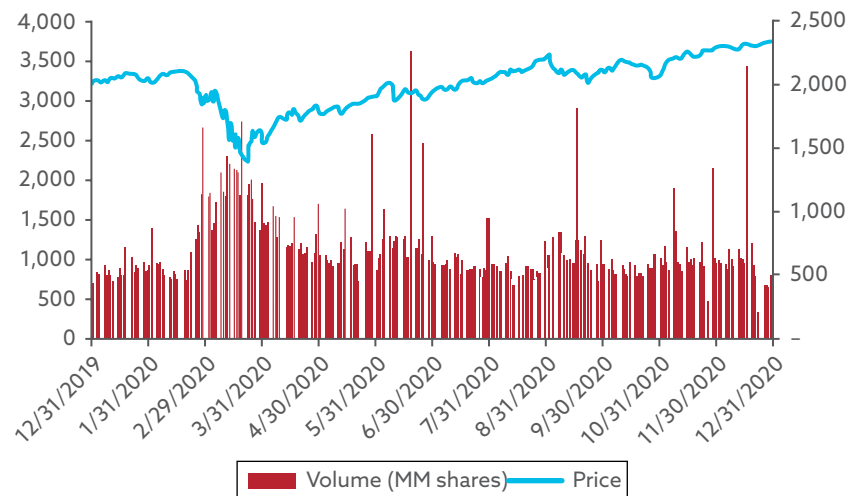
CASE STUDY



Secondary Equity Market Resilience under Volatile Conditions

2020 was a tumultuous year in financial markets due to the global COVID-19 pandemic and the economic impact of social and government policy changes throughout the year. Using the S&P 500 Index as a measure of equity market performance, analysts may consider the 18.4% increase in 2020 as a relatively strong year for equities, as shown in Exhibit 2.

Exhibit 2: S&P 500 Index Price and Trading Volume, 2020



Source: Bloomberg.

For example, as the economic impact of travel restrictions and shutdowns during the COVID-19 pandemic became apparent, the S&P 500 lost over a third of its value from late February to late March of that year. As seen in Exhibit 2, it was also a period of above average secondary market trading volume. Following the initial bear market, government policies in the form of both fiscal stimulus and monetary policy easing provided a significant boost to equity markets, which ended the year over two-thirds higher in price from the March 2020 low.

The S&P 500 Index volatility during 2020, shown in Exhibit 3, is based upon the **VIX Index**, a financial measure that estimates the expected volatility of the S&P 500 Index.

Exhibit 3: S&P 500 Index Volatility (VIX Price), 2020

Source: Bloomberg.

Strong S&P 500 performance in 2020 was not equally distributed across sectors. Remote work policies, online shopping, and the closure of public spaces benefited shares in the Information Technology and Consumer Discretionary sectors, the two strongest sectors with 2020 returns of more than 47% and 34%, respectively. The worst-performing sectors were Energy and Passenger Airlines. While reduced demand drove a sharp drop in oil prices, travel restrictions and lockdowns caused consumer travel demand to plummet, leading to share price declines in both sectors of roughly a third through 2020.

Beyond the ability to readily trade shares, secondary markets offer critical price discovery and other benefits to investors, issuers, company stakeholders, and other market participants. For example, observed market prices allow investors to assess periodic share return, and they serve as a basis for comparison for financial analysts seeking to estimate share value using equity valuation techniques (described in detail later). Issuers and firm stakeholders use secondary market prices as an important gauge of company performance.

Investor access to current and historical price data for individual equities and equity indexes supports a variety of public market investment approaches. For example, investors can easily measure returns, volatilities, and correlations across time and construct equity portfolios with an efficient risk–return tradeoff. Public market data is a critical input in comparing an individual share’s performance over time to firms of a similar size, those in the same sector, or those with other common features as well as to the overall equity market using a broad-based index.

In the case of issuers, secondary equity market prices provide valuable feedback to management and other stakeholders of the firm that supports decision making, potential financing, and compensation activity. For example, a publicly traded firm may consider a corporate restructuring plan to increase operating efficiency. These plans are often tied to expected share price improvements to demonstrate their anticipated value to shareholders.

Firms seeking to either raise additional equity capital or distribute capital to shareholders are likely to factor secondary market pricing and trading as a consideration into their decision-making process. For example, a firm whose secondary market price is rising sharply is more likely to pursue a follow-on offering, while one with a depressed trading price may consider share repurchases in lieu of dividends as a form of shareholder distribution.

Firms that employ **restricted stock units (RSUs)** and other forms of share-based incentive compensation for directors and management use secondary market price changes to gauge performance. RSUs are shares rewarded to employees as part of a compensation package that are restricted from sale during a vesting period that may last several years. Once the vesting period has lapsed and the employee remains in good standing or has met certain requirements after leaving the employer, RSUs may be sold or retained as unrestricted shares.

SHARE-BASED COMPENSATION

Share-based compensation, including restricted stock units and options, has become a more important component of corporate pay packages over the years. According to Morgan Stanley, a US-based financial services firm, share-based compensation has grown from 0.2% to 1.3% of sales among companies in the Russell 3000 Index between 2006 and 2022. The current estimate of share-based remuneration as a fraction of total compensation among these firms is approximately 6%–8%. Share-based compensation typically comprises a greater proportion of total compensation for smaller publicly listed firms as well as firms in the Information Technology sector. A significant rationale for this approach is to better align employee incentives with those of their employer. In addition, this approach replaces a portion of cash compensation and reduces the need to raise additional capital.

Company shares used for compensation may either be issued as new shares or involve previously repurchased shares. Issuance of new shares is dilutive for existing shareholders, while repurchasing existing shares for compensation is less likely to reduce existing shareholder ownership. However, many smaller firms using share-based compensation do so to conserve cash, suggesting these firms lack the ability to pay cash to buy back shares. The executives and employees receiving such compensation are willing to place some of their pay at risk in exchange for expected share appreciation.

Beyond the direct effects on equity investors (the issuer and employees with share-based compensation), changes in a company's market value based on secondary market share price movements are an important gauge of a company's financial health for other stakeholders as well. These include suppliers, competitors, customers, salaried employees, and government policymakers. Given the large swings in the market's perception of a company's value over time, as shown in the case of GameStop in an earlier learning module, it is important for financial analysts to establish a view on a company's value using equity valuation techniques. Equity valuation is covered in detail in later learning modules.

QUESTION SET 1



1. Which one of the following best characterizes an IPO?

- A. Primary market transaction
- B. Secondary market transaction
- C. Direct listing

Solution

A is correct. An IPO is best characterized by raising new capital through share issuance and is generally a primary market transaction.

B is likely incorrect as an IPO is not usually used to trade existing shares, so an IPO is not typically characterized as a secondary market transaction.

C is incorrect because a direct listing is simply a specific mechanism for a company to issue new shares to public investors as an alternative to the IPO process.

2. Which of the following public equity offerings is likely most dilutive?

- A. An IPO in which proceeds are used to buy new assets
- B. An IPO in which all offered shares are currently owned by early-stage investors
- C. A seasoned offering in which all offered shares are currently owned by the company founder

Solution

A is correct. An offering in which new shares are offered to raise capital to accomplish an objective, such as buying new assets, is dilutive to existing shareholders.

B and C are incorrect as both offerings involve no new shares, so these are likely non-dilutive offerings.

3. Which one of the following statements most correctly describes IPO underpricing?

- A. IPOs are typically priced below the stock's intrinsic value.
- B. IPOs typically generate substantial long-run returns.
- C. IPOs typically experience a jump in price on their first trading day.

Solution

C is correct. On average, IPO offerings experience a significant one-day return equal to the difference between the offering price and the publicly traded price at the close of the stock's first day of trading. This suggests the shares were sold at a discount to their first-day closing price, with this difference referred to as IPO underpricing.

A is incorrect as intrinsic value must be estimated and cannot be directly observed to compare to the IPO price.

B is incorrect because underpricing cannot be assessed based on favorable long-run returns because these reflect new information not knowable at the IPO date. Although long-run returns are variable, they cannot reliably predict underpricing.

EQUITY TRADING VENUES

3

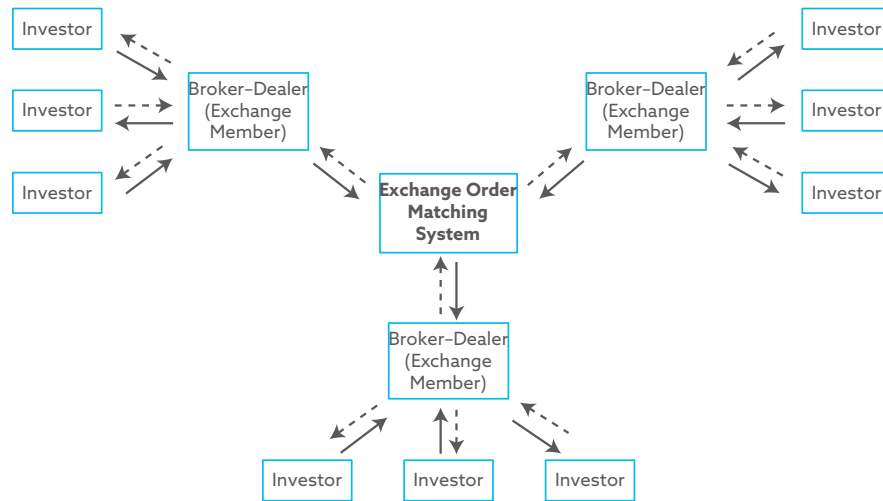


compare exchange, off-exchange, and over-the-counter equities trading

Exchange Trading of Equities

An **equity exchange** is a rules-based venue where market participants regularly trade equity securities. As in the case of other exchanges, trading rules designed to promote liquidity and transparency include the establishment of regularly scheduled trading hours each business day, during which an order matching system is used to link share purchase and sale orders, as shown in Exhibit 4.

Exhibit 4: Exchange-Traded Equity Market Structure



Exchange-based trading activity is executed through firms that are members of the exchange and are registered as **broker-dealers**. These financial intermediaries either act on behalf of clients as an agent or serve as principal counterparty to a financial transaction with a client. In the case of exchange-traded equities, broker-dealers act as an agent for investors by bringing orders to the market via the exchange's order management system. Equity trading orders of all types are typically entered in standardized share lots of a minimum 100 shares each. Stock exchanges also commonly designate a single dealer (or market maker) to ensure the orderly, continuous trading of each listed security. The role of this **designated market maker** involves purchasing, selling, or holding shares to promote liquidity and price discovery when the market opens and closes as well as at times of market imbalance or high volatility.

These exchange rules, centralized systems, and standardization seek to promote a liquid trading environment for company shares, as measured by the **explicit costs of trading**, or out-of-pocket fees and commissions, as well as by implicit costs in the form of a **bid-ask spread** (or bid-offer spread):

$$\text{Bid-ask spread} = \text{Lowest ask price} - \text{Highest bid price.} \quad (1)$$

The bid-ask spread is the asking or offer price (price at which a trader will *sell* a specified quantity of a security) minus the bid price (price at which a trader will *buy* a specified quantity). This may be expressed in absolute currency terms, as in Equation 1, or as a percentage of the lowest ask price, as in Equation 2:

$$\text{Bid-ask spread (\%)} = \frac{(\text{Lowest ask price} - \text{Highest bid price})}{\text{Lowest ask price}}. \quad (2)$$

This spread is a common measure of the cost of purchasing and selling shares in the current market.

Exchange rules also facilitate the execution of different investor order types, such as **market orders** and **limit orders**. While a market order involves directing a broker to buy or sell a specific number of shares at the best available current market price, a limit order imposes restrictions on the sale or purchase of shares. That is, a buy limit order imposes the constraint of a maximum share price to be paid, while a sell limit order specifies a minimum price to be received. If these respective prices are not reached, the order is not executed. Exhibit 5 provides a graphical representation of a limit order book showing these relationships.

Exhibit 5: Equity Limit Order Book



Exhibit 5 provides a broader view of trading interest by comparing **market depth**, or the relative size of exchange orders posted at or very near the best currently available bid and offer prices, to the current market price. The symmetry between buy limit orders at lower prices and sell limit orders at higher prices shown reflects a relative balance between supply of (sell orders) and demand for (buy orders) shares available to trade in the secondary market.

The current market price is often represented by the so-called **mid-market price**, or simply the average of the lowest ask and highest bid price:

$$\text{Mid-market price} = \frac{\text{Lowest ask price} + \text{Highest bid price}}{2} \quad (3)$$

This midpoint between bid and ask prices is often used as a basis to estimate the cost of executing a purchase or sale of stock. The following knowledge check illustrates these concepts as well as how an exchange-based order system captures trade execution.

KNOWLEDGE CHECK

Merlewis Corporation's shares trade on the Worldwide Stock Exchange. Assume the following limit orders are currently open on Merlewis:



Number of Shares Bid	Bid Price
250	99.90
300	99.80
200	99.70
500	99.60
Number of Shares Offered	Ask Price
400	100.10
250	100.20
450	100.30
300	100.40

An investor submits a market order to buy Merlewis shares.

1. Identify the bid–ask spread of Merlewis stock on both an absolute and percentage basis as well as the corresponding mid-market price.

Solution

Using Equation 1, we may solve for the bid–ask spread:

$$0.20 = 100.10 - 99.90.$$

Merlewis is trading at a bid–ask spread of 0.20 in currency terms, which on a percentage basis is 0.001998, or 0.1998% as calculated using Equation 2:

$$0.001998 = \frac{100.10 - 99.90}{100.10}.$$

The mid-market price is the average of the lowest ask and highest bid price:

$$100.00 = \frac{100.10 + 99.90}{2}.$$

2. Determine the weighted average price at which the investor buys Merlewis shares for a market order of 500 shares as well as the difference between this price and the mid-market price from Question 1.

Solution

The market order to buy 500 shares may be executed as follows:

- 400 shares purchased at 100.10, and
- 100 shares purchased at 100.20.

Thus, the investor buys 500 shares at a weighted average price of 100.12:

$$100.12 = \frac{400}{500} \times 100.10 + \frac{100}{500} \times 100.20.$$

The difference between the 100.12 purchase price and the 100.00 mid-market price is 0.12 per share, or 0.02 per share above this difference at the lowest asking or offer price.

In addition to the weighted average price of a *single* order shown above, market participants often track intraday share performance using a **volume-weighted average price (VWAP)**, which is an average price measure derived by weighting the traded price by the volume of shares traded over a given period:

$$\text{VWAP} = \frac{\sum_{i=1}^n (\text{Price}_i \times \text{Volume}_i)}{\sum_{i=1}^n (\text{Volume}_i)}. \quad (4)$$

This dynamic measure of price changes is commonly used by traders to gauge market momentum over the course of a trading day.

KNOWLEDGE CHECK



A trader observes the following Merlewis Corporation share trades following the market open:

Number of Shares Traded	Trading Price
300	98.50
750	100.00
200	99.75

1. Calculate the VWAP for Merlewis shares.

Solution

Using Equation 4, trade-weighted volume in the numerator is as follows:

$$\Sigma(\text{Price}_i \times \text{Volume}_i) = (300 \times 98.5) + (750 \times 100) + (200 \times 99.75) = 124,500.$$

$$\Sigma(\text{Volume}_i) = 300 + 750 + 200 = 1,250.$$

$$\text{VWAP} = 99.60 = (124,500/1,250).$$

For a stock to trade on an exchange, it must be listed and continue to meet exchange listing requirements. Equity listing requirements typically include a minimum number of shareholders, minimum numbers each shareholder may hold, minimum book equity value or net worth and/or profitability levels, and a minimum percentage of stock that is publicly traded. A growing number of equity exchanges exist to accommodate smaller firms at an earlier stage of development, as described in the following case study.

TORONTO STOCK EXCHANGE (TSX) VERSUS VENTURE EXCHANGE (TSXV) LISTINGS

TMX Group, a Canadian-based financial company, includes two stock exchanges for different types of corporate issuers. For larger, more mature issuer companies, the Toronto Stock Exchange (TSX) is an appropriate venue for listing equity offerings and secondary market trading. In contrast, the TSX Venture Exchange (TSXV) is for early-stage and smaller issuer companies, many of which are in the development stage; while they meet alternative listing criteria, they often have yet to earn a profit. TSXV also has a much higher concentration of firms in fewer sectors, such as mineral exploration and technology, while TSX has a more balanced composition across sectors.

Not surprisingly, TSX represents most of the market capitalization across the two exchanges. The total quoted market value of TSX companies is CAD3,792 billion in 2022 compared to CAD70 billion across over 1,700 companies listed on each of the respective exchanges. However, for companies raising capital, the differences are less extreme. The average size of equity capital raised is CAD50 million for TSX companies versus CAD4.4 million for TSXV companies. TMX Group has a streamlined process through which eligible issuers meeting more stringent TSX exchange requirements can “graduate” from TSXV to TSX, 16 of which completed this in 2022.

Off-Exchange Trading of Equities

While standard trades of a small to moderate size are easily digested by the exchange, others may achieve better execution when directly executed with a broker–dealer. These equity trades typically involve illiquid shares, very low volume trades (often in low price shares) initiated by retail investors, or very large trade order sizes known as **block trades**.

Block trades are a major source of off-exchange activity and are usually defined as an order of 10,000 shares or more for a specific issuer. In many cases, block trades are sent to a broker–dealer that either breaks the block into smaller individual orders executed over time to minimize the adverse price impact or sends them simultaneously in smaller pieces to multiple parties to achieve the same effect. In other cases, such as mutual funds, asset managers may agree to bilateral or multilateral exchanges of shares at a market closing price separate from an exchange order system.

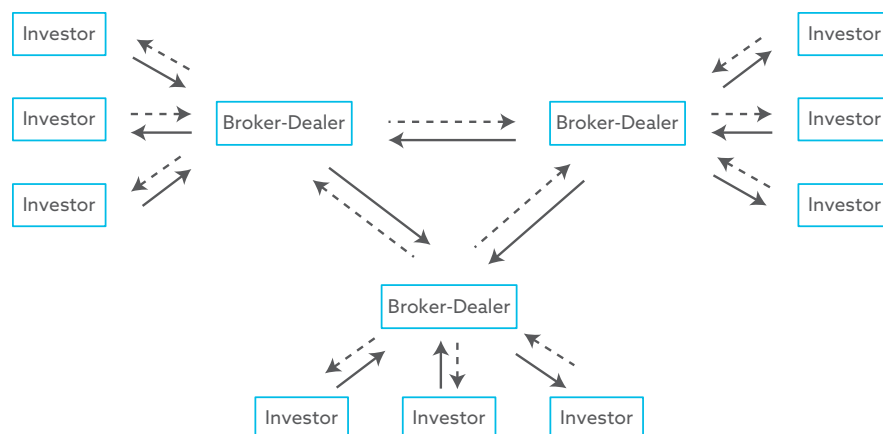
In most cases, off-exchange trading seeks to minimize the **implicit costs of trading**, which are the indirect and generally unobservable costs resulting from the market impact of trading. Implicit costs apart from the bid–ask spread arise from the following sources:

- **Market impact** (or price impact) is the cost of immediately executing a trade on transaction prices. Traders who want to fill large orders often must move prices to encourage others to trade with them.
- **Delay costs** (also called slippage) arise from the inability to complete the desired trade immediately. Traders fail to profit when they fill their orders after prices move as they expect.
- **Opportunity costs** (or unrealized profit/loss) arise from the failure to execute a trade promptly. Traders fail to profit when their orders are not executed, and prices move as they expect.

Over-the-Counter Trading of Equities

In contrast to exchange-traded equity securities, over-the-counter (OTC) equity trading involves a network of broker–dealers who buy and sell shares among themselves rather than via a centralized trading system, as shown in Exhibit 6.

Exhibit 6: Over-the-Counter Equity Markets



The OTC market is sometimes referred to as a quote-driven market. Shares traded on an OTC basis are typically those of firms that are smaller in size that either decide to forego the costs of registration on a larger domestic equity exchange or fail to meet the minimum listing requirements.

In contrast to an electronic order-driven system in the case of exchange trading with a designated market maker as described earlier, a quote-driven trading system consists of multiple market makers, which are typically broker-dealers, that continuously show bid and ask prices to investors seeking to buy or sell shares. Rather than posting an order on a centralized trading platform, these market makers will typically either fill the investor's order from their inventory of shares or seek to match an investor's trading instruction with an offsetting order from another market maker or investor. Quote-driven market systems tend to offer the highest likelihood of trade execution under most market conditions, as a broker-dealer or market maker posts bids and offers and must execute at these prices when receiving orders. On the other hand, OTC markets offer less transparency than exchange-traded markets, as individual trades and orders are not readily visible to market participants. While trading among market makers tends to cause bid and offer prices to be similar across broker-dealers, buyers and sellers have no assurance that they are executing at the best available market price at a given time. OTC broker-dealers may also decide to provide less liquidity in times of market stress.

QUESTION SET 2



1. Which one of the following is most likely a minimum size of an equity trading order submitted on an exchange?

- A. 10 shares
- B. 100 shares
- C. 1,000 shares

Solution

B is correct. Equity trading orders submitted for exchange trading are typically entered in standardized share lots of a minimum 100 shares each.

2. Which one of the following is least likely to be an important part of an equity exchange's minimum listing requirements?

- A. Number of shareholders must be greater than some minimum level
- B. Percentage of shares available to be traded publicly must be greater than some minimum level
- C. Company sales must be greater than some minimum level

Solution

C is correct. Company sales levels are least likely to be employed as a listing requirement as other fundamentals are likely more important (such as profitability).

A is incorrect because exchanges are likely to require enough possible sellers of shares as represented by the number of shareholders.

B is incorrect because exchange trading requires sufficient shares to be available for trading.

3. Which one of the following statements is most likely true of block trades?

- A. Block trades are typically traded on exchanges in their full order size.

- B. Block trades are broken down into smaller sizes to minimize the bid–ask spread for the stock.
- C. Block trades are broken down into smaller sizes to minimize the price impact of the trade.

Solution

C is correct. In many cases, block trades are sent to a broker–dealer that breaks the block into smaller individual orders executed over time to minimize the price impact.

A is incorrect because the larger size of block trades typically makes it impractical to trade all the shares in one order.

B is incorrect as the bid–ask spread reflects both buy and sell orders. A block trade will only reflect a buy or sell order. The bid–ask spread is an explicit cost of trading, while block trades are typically broken into smaller orders to minimize the implicit costs of trading.

4

EQUITY LIQUIDITY MEASURES



describe liquidity measures for a publicly listed security and calculate float and average daily volume

As in the case of other financial markets, equity market liquidity involves the relative ease with which market participants can buy or sell shares in terms of both cost and timing. The explicit and implicit costs associated with trading described earlier can vary widely based upon the size and timing of trade orders as well as the release of other new information affecting share prices, among other factors.

An important quantitative measure of liquidity in terms of market turnover involves the relationship between **market float** and **average daily volume**. While market float is sometimes used to refer to the total market value of shares freely traded, we refer to the number of issuer shares that are available to freely trade in the public market:

Market float

= Total shares outstanding – Restricted shares or shares not available for trading.

(5)

The average daily volume reflects the average number of issuer shares traded each business day over a pre-determined period, shown below as n days:

$$\text{Average daily volume} = \frac{\sum_{i=1}^n (\text{Shares traded per day})_i}{n}. \quad (6)$$

The *ratio* of average daily volume to market float gives an overall indication of the proportion of shares outstanding to be exchanged on a given trading day. This ratio is referred to as the **turnover ratio**:

$$\text{Turnover ratio} = \frac{\text{Average daily volume}}{\text{Market float}}. \quad (7)$$

These liquidity metrics allow equity investors to compare their position in an issuer's stock with the relative ability to liquidate a position based upon overall market activity. This comparison is particularly important for large positions in smaller publicly traded companies.

KNOWLEDGE CHECK

Eastern States Pension Plan owns 300,000 shares of Merlewis Corporation. The number of Merlewis shares outstanding is 1,250 million shares, with 80% of the shares representing the company's market float. The average daily volume in Merlewis over the last month has been 1,500,000 shares. Eastern States internal trading guidelines prohibit the fund from buying or selling more than 8% of a stock's average daily trading volume on any given day. The Chief Investment Officer of Eastern States has instructed the plan's trading staff to implement a complete liquidation of the plan's holdings in Merlewis shares.

1. Calculate the number of days necessary to complete the liquidation of Eastern States' position in Merlewis shares.

Solution

Eastern States' current position in Merlewis represents 20% of average daily trading volume ($= 300,000 \div 1,500,000$). A complete liquidation of their holdings given the 8% constraint implies that Eastern States must sell their Merlewis holdings over at least three trading days. Dividing 20% by 8% gives an answer of 2.5. Thus, they cannot fully liquidate Merlewis shares in two days, but may complete the sale within three days.

Secondary equity market liquidity encompasses several measures beyond the transaction costs mentioned earlier and the proportion of shares traded per day, namely market depth, **market breadth**, and **market resiliency**, or a market's relative speed of price adjustment.

As described earlier, market depth refers to the relative size of orders posted that are at or very near the best currently available bid and offer prices, while market breadth is a measure of the number of orders at different price levels both above and below, but very close to the best bid and offer prices. The combination of market depth and breadth reflects the number, size, and dispersion of orders within a trading book. A high degree of depth and breadth ensures the ability to execute trades of a manageable size quickly and without significant market price moves under relatively stable market conditions. Using algorithms effectively to split large orders into smaller parts hinges on the market's depth and breadth. This strategy facilitates smooth execution of trades without causing a significant adverse price impact.

Market resiliency refers to the relative ability of an issuer's shares to maintain a stable **market-clearing price**, the price level at which all buy and sell orders for a security can be executed, leaving no excess demand or supply, and subsequently revert quickly to such a price following a market shock. Shocks affecting share prices cause an initial change, the impact of which may be dampened by greater market depth and breadth, as well as a period of convergence to a new market-clearing price. The speed of adjustment depends not only on existing orders, but also those not revealed by brokers, including orders with special conditions, block trades executed in smaller portions, or computer-based trades triggered by market moves. These unobservable forces can contribute to significant intraday share price volatility.

QUESTION SET 3

1. An equity market's depth most correctly refers to:
 - A. the market's ability to absorb numerous trade orders.

- B. the typical bid–ask spread observed for stocks trading in the market.
- C. the market’s ratio of market orders to limit orders.

Solution

A is correct. A market with greater depth has many orders at or near the best available bid and offer prices, providing traders with confidence that there are sufficient buyers and sellers to absorb numerous orders.

B is incorrect. Market depth implies more about the limit orders below the highest bid price and above the lowest ask price. The size of the bid–ask spread is not indicative of market depth.

C is incorrect as market depth does not refer to any specific ratio regarding market to limit orders. For an individual stock, a significant number of limit orders might be indicative of the market’s ability to absorb market orders. However, if all limit orders are close to the prices associated with the bid–ask spread, the market would have little depth.

2. Which of the following provides the most accurate estimate of the proportion of shares outstanding exchanged per day given the following information: A company’s number of shares outstanding is 1,250 million shares with 80% of the shares representing their market float. The average daily trading volume in the company’s stock over the last month has been 1,500,000 shares.

- A. 0.60%
- B. 0.12%
- C. 0.15%

Solution

C is correct. The ratio of average daily turnover to market float gives an overall indication of the proportion of shares outstanding exchanged on a given trading day.

Daily turnover is given as 1,500,000 shares, and market float is 1,000 million shares ($= 80\% \times 1,250$ million); so, the ratio is 0.15% ($= 1,500,000 \div 1,000,000,000$).

B fails to adjust for the fact that just 80% of shares outstanding represent market float by dividing average daily volume by *total* shares outstanding rather than those available to trade.

A makes the error of dividing average daily volume by the 20% of shares outstanding that are restricted or privately held and therefore *not* included in market float.

3. Which one of the following statements most correctly describes the factors associated with share prices and market shocks?

- A. A stock with greater market resiliency is more likely to be able to avoid market shocks.
- B. Greater market depth and market breadth can dampen the initial price impact of a market shock.
- C. The speed of market price adjustment following a market shock is determined by the pre-existing market order book.

Solution

B is correct. Greater market depth and market breadth reduce the initial price change following a market shock.

A is incorrect because market resiliency does not help shares *avoid* price shocks; rather, it involves the ability of market prices to adjust to shocks.

C is incorrect because unobserved trades, such as those orders not revealed by brokers and block trades, may also affect the speed of price adjustment. Average trading volume is likely to be a very small fraction of overall market float because of large numbers of shares outstanding for most publicly traded equities.

EQUITY INDEXES

5

☐ describe types of equity indexes

Market indexes support the tracking of risk and return, performance evaluation, and benchmarking investments and investment managers. They also form the basis for indexed investment strategies. Several features distinguish their composition and use in equity markets.

Equity indexes are commonly comprised of a fixed number of constituents, each represented by the issuer's common share outstanding. Each issuer's market share price is used to determine the index value based upon the respective weighting methodology, which is typically based upon market capitalization, price, or other factors described below. The following case study illustrates the composition and determination of a major equity market index.

THE NIFTY 50 EQUITY MARKET INDEX

Established in 1997, the NIFTY 50 equity market index is comprised of 50 of the largest firms traded on the National Stock Exchange (NSE) of India. The index is calculated using a market capitalization-based weighting methodology that is float-adjusted, with the index value representing the total market value of all the stocks in the index relative to a base year value of 1,000 set in 1995.

As of 2024, the largest component of the index was HDFC Bank, with over 13% weightage in the index; the smallest was UPL, a chemical company with an index weight of less than 1%. Financial services firms comprise over 30% of the NIFTY 50's market value, followed by information technology (14%) and oil and gas (11%). NIFTY 50 constituents are subject to review and replacement on a semi-annual basis, with more frequent changes occurring because of a constituent delisting, merger, or other corporate action.

With around two-thirds of total NSE market value and just over half of NSE trading volume comprised by its constituents, the NIFTY 50 is a broad-based reflection of overall Indian stock market performance and serves as the underlying for the world's most actively traded equity derivative contract.

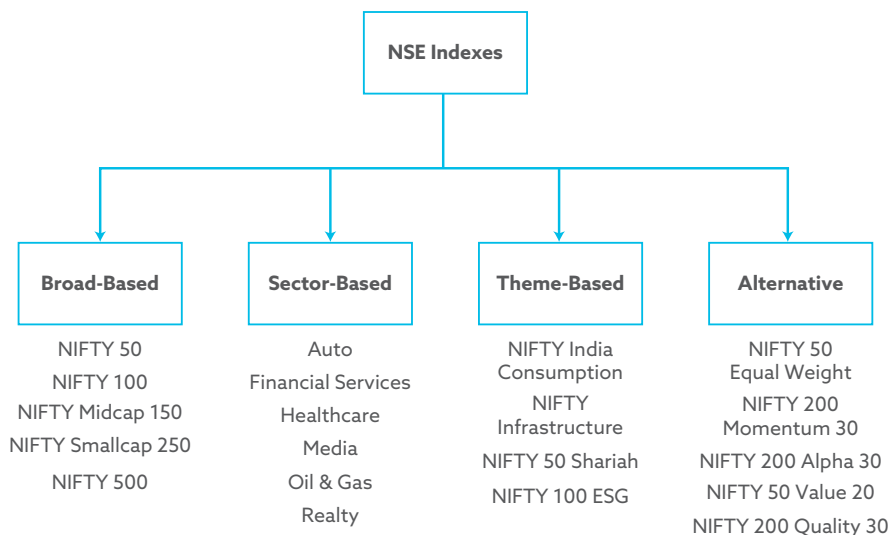
As in the case of the NIFTY 50 index described above, most broad-based equity indexes for a given market include the largest listed firms adjusted by market float and calculated on a market capitalization-weighted basis. Equity market participants commonly track and gauge equity performance across markets using these indexes, as shown in Exhibit 7 using the example of a Bloomberg World Equity Indexes (WEI) summary page.

Exhibit 7: Bloomberg Equity World Indices Example Summary Page

Index	2Day	Value	Net Chg	%Chg
1) Americas				
11) DOW JONES		37919.93	+113.54	+0.30%
12) S&P 500		4890.65 d	+22.10	+0.45%
13) S&P/TSX Comp		21069.29 d	+43.51	+0.21%
14) IBOVESPA		128206.32 d	+390.62	+0.31%
2) EMEA				
21) EURO Stoxx 50		4581.36 d	+17.25	+0.38%
22) FTSE 100		7535.85 d	+8.18	+0.11%
23) CAC 40		7459.97 d	+4.33	+0.06%
24) DAX		16902.59 d	+12.67	+0.08%
3) Asia/Pacific				
31) NIKKEI		36236.47 d	+9.99	+0.03%
32) HANG SENG		16211.96 d	+312.09	+1.96%
33) CSI 300		3342.92 d	+65.81	+2.01%
34) S&P/ASX 200		7555.36	+36.17	+0.48%
35) KOSPI		2470.34	+0.65	+0.03%
36) SPCNX NIFTY		21352.60 d	-101.35	-0.47%
37) TAIEX		18002.62 d	+126.79	+0.71%
38) JAKARTA COMP		7178.04	-49.78	-0.69%
39) FTSE/BM KLCI		1504.10 d	+7.99	-0.53%

Source: Bloomberg.

While the NIFTY 50 (shown as SPCNX NIFTY in Exhibit 7) is the most recognized benchmark within the Indian equity market, as in other jurisdictions, it is by no means the only index available to Indian investors and market participants, as shown in Exhibit 8.

Exhibit 8: Selected National Stock Exchange (NSE) Indexes

As Exhibit 8 demonstrates, numerous equity index types exist to meet the needs of market participants, including broad-based indexes, indexes with an industry sector focus, indexes based upon a specific investment theme, and indexes based upon alternative weightings using so-called **fundamental factors** (i.e., measures other than market capitalization that may incorporate an issuer's revenue, earnings, dividends, or book value).

Broad-based indexes are typically tiered based upon issuer size and often include specific indexes that track the performance of firms of different sizes. The most comprehensive equity market index is typically referred to as a **composite equity index**, which is typically one that includes securities representing more than 90% of the selected market's value. Composite equity indexes are commonly broken down into indexes with fewer constituents based upon size to distinguish the performance of small-, medium-, and large-cap companies. For example, the constituents of the NIFTY 500 composite index are included in three smaller indexes: The NIFTY 100 is comprised of the largest 100 firms in the NIFTY 500, with approximately 75% of the composite index's market capitalization. The NIFTY Midcap 150 includes those firms ranked 101–250, comprising 15% of composite index market value. The NIFTY Smallcap 250 includes the remaining small firms, which total 10% of index market capitalization.

Despite the popularity of equity indexes based upon company size, several considerations give rise to a need for narrower, industry-based indexes. Broad-based capitalization-weighted index performance is heavily influenced by the fortunes of its largest firms (i.e., those whose shares have appreciated) and its firms in concentrated industries. Sectoral indexes facilitate a comparison of firms in the same business, allowing investor and issuer benchmarking across competitors, and supporting the creation of index-based investment products to allow investors to establish equity exposure to a specific industry rather than an individual company or the entire market.

Investment themes, on the other hand, involve indexes created based on investor trends beyond selecting a certain industry or company size. Examples of NSE indexes based upon investment themes include the following areas:

- Firms whose activities are primarily focused on domestic sales in India (NIFTY India Consumption)
- Capital-intensive, long-lived assets used to provide essential services (NIFTY Infrastructure)
- Issuers screened to ensure compliance with Islamic law (NIFTY 50 Shariah)
- Issuers screened to ensure compliance with specific environmental, social and governance (ESG) principles (NIFTY 100 ESG)

The following case study describes the screening process for an ESG index.

S&P EUROPE ESG 350 INDEX

The S&P Europe ESG 350 Index is constructed using the constituents of the underlying S&P Europe 350 Index, a market capitalization-weighted index that seeks to maintain a balanced representation across sixteen developed European markets as well as across global industries.

First, several criteria for exclusion are applied to the constituents. For example, companies engaged in certain business activities deemed not in compliance with ESG principles, such as certain weapons, thermal coal, tobacco, oil sands, small arms, and military contracting, are excluded. Firms considered non-compliant with UN Global Compact Principles are also removed from consideration.

The remaining eligible constituents are ranked by their respective S&P DJI ESG score, a scoring methodology that seeks to measure ESG risk and performance factors for corporations, with the bottom 25% of firms excluded from the index, with a focus on financial materiality. This index is float-adjusted market capitalization weighted.

Indexes based upon alternative weighting methodologies have enjoyed greater popularity due to the perceived biases and shortcomings of standard market capitalization-weighted indexes. The simplest form involves an equally weighted index based upon the same constituents as a broad-based index based on market capitalization. For example, under the NIFTY 50 Equal Weight index, both the largest (HDFC Bank) and smallest (UPL) NIFTY 50 constituents would have a weight of 2%. This approach seeks to remove perceived biases of a greater weighting given to the largest companies and those whose prices have appreciated under a standard market capitalization-weighted method.

Investor interest in so-called **factor-based investing**, or investment selection based upon fundamental factors believed to be underlying drivers of equity performance such as earnings growth and well-capitalized balance sheets, has been the catalyst for the creation of other indexes based upon alternative weighting as described in the following case study.

MSCI QUALITY INDEXES

MSCI, a global provider of financial indexes, data, and analytics, offers a family of fundamental factor indexes known as the MSCI Quality Indexes. MSCI weights these indexes based on a quality score used to determine a revised weighting for index constituents. The quality score is computed using a statistical process based upon standardized measures of three financial ratios for each constituent in a referenced index measuring profitability (return on equity), stability of earnings (earnings variability), and balance sheet strength (debt to equity). That is, firms with higher-than-average return on equity, less earnings variability, and a lower debt-to-equity ratio receive a higher quality score.

MSCI uses the quality score to adjust the market capitalization weighting of the stock. If the score is below zero for a specific stock in the parent index, the stock will be excluded from the quality index. Alternatively, some issuers receiving sufficiently high scores will receive a higher weighting in the quality index versus the parent index. Overall, the quality indexes contain about 20% of the number of stocks in the parent index and account for about 30% of the market capitalization of the parent index.

Multi-market indexes usually comprise indexes from different groups of countries designed to represent the performance of multiple security markets. Multi-market indexes may represent multiple national markets, regions, levels of economic development, or, in some cases, the entire world. As in the case of individual shares as described in an earlier learning module, investment products based upon multi-market equity indexes provide investors the opportunity to offset a home asset bias with an index-based instrument traded on a domestic exchange. While usually denominated in the investor's domestic currency, sources of return on these investments include gains and losses due to exchange rate changes as well as foreign equity index returns. One example for developing or emerging markets is highlighted in the following case study.

MSCI EMERGING MARKETS INDEX AS A BENCHMARK

The MSCI Emerging Markets Index captures large- and mid-cap issuers in representation across 24 developing markets across Asia, Africa, Europe, South America, and the Middle East. With over 1,400 constituents, the index encompasses roughly 85% of the free float-adjusted market capitalization in each jurisdiction.

Asset managers create pooled investments such as mutual funds that use the periodic returns of indexes such as the MSCI Emerging Markets Index as a gauge to measure the relative performance of funds with a similar risk and return profile. For example, BlackRock's Emerging Markets Fund (MADCX) for institutional investors uses the MSCI Emerging Markets Index to benchmark its periodic performance. The principal investment strategy of BlackRock's fund is to invest at least 80% of its assets in countries with developing capital markets. The markets in which it invests may include, but are not limited to, those included in the MSCI Emerging Markets Index.

QUESTION SET 4



1. Which one of the following best describes a broad-based equity index?
 - A. Broad-based indexes are one of the most actively traded shares on most equity exchanges.
 - B. Broad-based indexes include the largest listed firms adjusted by market float and calculated on a market capitalization-weighted basis.
 - C. Broad-based indexes are the benchmark for all index-based investment strategies.

Solution

B is correct. Broad-based indexes include the largest listed firms adjusted by market float and calculated on a market capitalization-weighted basis.

Choice A is incorrect because equity indexes form the basis for products such as index-based shares but are not directly traded as shares.

Choice C is incorrect as many indexes other than broad-based indexes are used as the basis for index-based investment strategies.

2. A composite index is best characterized as:

- A. a large-cap index.
- B. a multi-market index.
- C. a comprehensive index.

Solution

C is correct. A composite index generally includes securities representing more than 90% of the selected market's value, thus making it a comprehensive index covering close to the entirety of a traded equity market.

A is incorrect because a comprehensive index includes not just large-cap equities but also smaller-capitalization equities.

B is incorrect because multi-market indexes are designed to represent multiple security markets but not necessarily the capitalization necessary for each market to consider them composite indexes.

3. Which one of the following most correctly describes an equity index weighted by fundamental factors?

- A. Weightings are never related to market capitalization.
- B. Weightings are based on only one fundamental factor.
- C. Weightings may be based on a statistical process.

Solution

C is correct. As in the case of the MSCI Quality Indexes, the index weightings of each stock may be calculated using a statistical process. Weightings in fundamental factor indexes are derived through statistical analysis of various company fundamentals. Their weighting is distinctly different from market capitalization-weighted indexes.

A is incorrect because fundamental factor weightings may be adjusted from market capitalization by a score, as in the MSCI Quality Indexes.

B is incorrect because fundamental factor weightings may be based on multiple fundamental factors, such as profitability, earnings stability, and balance sheet strength, as shown for the MSCI Quality Indexes.

SUMMARY: EQUITY ISSUANCE AND TRADING

- Primary equity markets involve an issuer selling a new security or additional shares of a previously issued security to investors, while secondary markets are those in which existing shares are traded among investors.
- While exchange-based trading during regular business hours generally offers the highest liquidity, very small or very large trades are often executed off exchange via a broker (often called a block trade). Quote-driven over-the-counter trading is more common among smaller companies and usually offers less liquidity.
- Secondary equity market liquidity is measured in terms of transaction costs such as bid–offer spreads as well as other market features. A common way to assess the market liquidity of a company's shares is to compare the proportion of shares outstanding and available to trade with the average daily number of shares traded.
- Aggregate or broad-based equity indexes allow market participants and observers to gauge the risk and return of the equity market as a whole or in part and allow for relative comparisons to the performance of individual equity securities and investor portfolios.
- Broad-based and narrower indexes, such as those based upon company size, sector, or other features, form the basis of investment products and strategies and are also used to track investment manager performance.

PRACTICE PROBLEMS

The following information relates to questions 1-4

Eastern States is a US public pension plan with an active allocation to publicly listed equities. Eastern States allocates some of their public equity investments to fund managers following specific strategies. With the remainder of their public equity allocation, Eastern States actively manages a portfolio of 30–50 individual stocks. The following points provide some highlights of Eastern States' investment policy:

- As much as 10% of the total equity portfolio may be invested in allocations to IPOs.
- For stocks trading on secondary markets, a minimum of two years of public trading history is required for any new equities to be considered for purchase.
- The average daily turnover of all stock positions must meet minimum thresholds.
- Eastern States follows a globally diversified investment policy focused on large-cap stocks of companies that incorporate ESG principles.

1. Which one of the following provides the most likely argument as to why Eastern States seeks IPO allocations?
 - A. Because of favorable long-run returns from IPOs, on average
 - B. Because IPOs are, on average, underpriced
 - C. Because IPOs are typically non-dilutive to existing shareholders
2. Which one of the following conditions may best reflect Eastern States rationale for requiring two years of trading history for secondary market purchases?
 - A. Trading history provides the ability to assess company management quality.
 - B. Trading history provides the ability to forecast future company stock returns.
 - C. Trading history provides the ability to compare return and risk performance across similar publicly traded companies.
3. Which one of the following observable costs of trading is a likely concern for Eastern States with respect to the policy on average daily trading volume?
 - A. Bid–ask spread costs
 - B. Opportunity costs
 - C. Market impact costs
4. Which one of the following equity index types would be most appropriate for

Eastern States to use as a benchmark for their equity portfolio performance?

- A. A US large-cap ESG index
 - B. A global ESG composite index
 - C. A global large-cap ESG index
-

SOLUTIONS

1. B is correct. IPOs, on average, experience a first-day increase in price from the price at which investors receiving an allocation in the IPO buy the shares and the closing price after the first day of public trading. Often, the level of underpricing may be 20% or greater.
A is incorrect because IPO companies are often likely to underperform over long time horizons.
C is incorrect because Eastern States is unlikely to be affected by the dilutive or non-dilutive effects of the IPO given that they are a new investor. As a new investor, the company is focused on the potential for underpricing benefits rather than dilution effects, unlike existing shareholders.
2. C is correct. Public market data are a critical input in comparing an individual share's performance over time to similar firms. Furthermore, investors can easily measure returns, volatilities, and correlations across time.
A is incorrect because management quality is unlikely to be assessed solely by historical stock prices but may need more examination of financial statements and other sources of company fundamental research.
B is incorrect as historical stock prices do not provide information about future returns. However, they can be valuable for assessing risk and volatility.
3. A is correct. Bid–ask spreads are observable for stocks and are likely higher for stocks with relatively low daily turnover. As such, Eastern States may be looking to avoid owning stocks with high bid–ask spreads.
B and C are incorrect as both opportunity and market impact costs are unobservable.
4. C is correct. Eastern States has a globally diversified portfolio of large-cap stocks in companies incorporating ESG principles. Therefore, they should benchmark with an equity index that matches those criteria.
A is incorrect. Despite being a US investor, Eastern States owns a globally diversified portfolio. Thus, a US large-cap ESG index does not match the global scale.
B is incorrect. The composite index includes not just large-cap stocks but also smaller stocks. Therefore, this index does not match the large-cap characteristic of Eastern States' portfolio.

LEARNING MODULE

4

Sources of Equity Returns

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	describe features and uses of dividends, share repurchases, stock splits, and reverse stock splits
☐	describe dividend payment chronology
☐	calculate price and total return for equity securities

INTRODUCTION: SOURCES OF EQUITY RETURNS

1

The economic rights associated with equity ownership, described in earlier learning modules, drive investor returns in the form of capital appreciation and, in some cases, capital distribution. Over the long term, higher historical and expected future returns relative to other asset classes, such as bonds, make equities an attractive core holding for both individual and institutional investors. Unlike fixed-income investments, which usually have a fixed maturity and contractual cash payments, equities involve discretionary distributions over an indefinite life. In evaluating the sources of equity return, we first compare corporate actions including dividends, share buybacks, stock splits, and reverse stock splits. Next, we focus our attention on the frequency and timing of cash dividends, which are more common among mature companies. Finally, we evaluate the price and total return on equities over a holding period, which may or may not include cash dividends.

PRE-TEST: SOURCES OF EQUITY RETURNS



1. Which one of the following statements about regular cash dividends is most correct?
 - A. A company is legally required to pay a regular cash dividend to shareholders if the company is profitable.
 - B. A company with significant growth potential is likely to pay a large portion of its earnings as regular cash dividends.

- C.** A company's management team and its board of directors make discretionary choices about regular cash dividend payments.

Solution

Choice C is correct. Regular cash dividend payments reflect discretionary decision making by a company's management team and board as to whether regular cash dividend payments are an appropriate use of corporate cash.

Choice A is incorrect because cash dividend payments to shareholders are not a legal obligation regardless of a company's profitability.

Choice B is incorrect because a company with significant growth potential is likely to reinvest more of its profits, leaving a smaller portion for potential cash dividend payments to shareholders.

2. Investors may prefer a company to repurchase shares instead of paying a cash dividend of similar size if:

- A.** tax rates on capital gains income are lower than those on dividend income.
- B.** repurchasing shares reduces the company's shares outstanding.
- C.** the company has relatively little variation in investment opportunities.

Solution

Choice A is correct. Investors facing higher tax rates on dividend income may prefer to receive cash distributions by selling shares and paying taxes at lower capital gains tax rates.

Choice B is incorrect because a company automatically reduces shares outstanding by repurchasing shares, which should have no effect on whether investors would prefer repurchases over dividends.

Choice C is incorrect because a company with little variation in investment opportunities does not require the greater flexibility associated with repurchases versus cash dividends.

3. Identify the most accurate effect of a reverse stock split on a company's stock.

- A.** A reverse stock split increases the number of shares outstanding.
- B.** A reverse stock split is often used by a company that has successfully restructured its operations.
- C.** A reverse stock split increases a company's stock price per share.

Solution

Choice C is correct. A reverse stock split reduces the number of shares outstanding while increasing the stock price per share such that the market value of the company's equity does not change. Each shareholder maintains the same proportional ownership following the reverse split.

Choice A is incorrect because the number of shares outstanding increases in a stock split, not a reverse split.

Choice B is incorrect because a reverse stock split is more likely with a company that is initiating a restructuring rather than with one that has just completed a successful restructuring.

4. On which one of the following dates would a company's stock price be expected to decline, holding all else equal?

- A.** Dividend declaration date

- B. Ex-dividend date
- C. Dividend payment date

Solution

Choice B is correct. On the ex-dividend date, the stock no longer provides the owner with the right to the upcoming dividend payment. As a result, the stock price declines on this date by approximately the amount of the cash dividend, assuming no other causes for the stock to move.

Choice A is incorrect because the declaration date usually precedes the ex-dividend date by a few months.

Choice C is incorrect because the payment date usually follows the ex-dividend date by about two weeks.

5. Two company stocks, Vikaas and Laabhaansh, both trade at INR20 per share. Both stocks generate a 20% total return over a 1-year holding period. Vikaas pays no dividend while Laabhaansh shareholders received a dividend constituting 40% of the stock's total return at the end of the holding period. Which one of the following is correct about the stock prices of Vikaas and Laabhaansh at the end of the holding period?

- A. Vikaas price is INR24.80; Laabhaansh price is INR24.00
- B. Vikaas price is INR24.00; Laabhaansh price is INR22.40
- C. Vikaas price is INR24.00; Laabhaansh price is INR21.60

Solution

Choice B is correct. A 20% total return on a stock priced at INR20 per share implies that the numerator of the return equation is equal to INR4 per share over the holding period ($= 20\% \times \text{INR}20$). Given that Vikaas pays no dividend, this stock must increase to INR24.00 at the end of the holding period. Laabhaansh's holding period return reflects both a dividend payment and a stock price change. Given that 40% of the total return is attributable to the dividend, 60% of the return reflects stock price appreciation. Multiplying 60% by INR4 per share equals INR2.40 per share, so Laabhaansh's stock price is INR22.40 at the end of the holding period.

Choice A is incorrect because it involves a 24% return $[(24.80 - 20)/20]$ on Vikaas shares and mistakenly attributes the full 20% return on Laabhaansh shares to price appreciation.

Choice C is incorrect. While the Vikaas price is correct, the Laabhaansh price mistakenly multiplies the 20% expected total return by the 40% attributable to the dividend to arrive at a dividend of INR1.60 ($= 8\% \times \text{INR}20$).

6. An analyst places a two-year price target of EUR40.00 per share on a stock that is expected to pay an annual EUR1.00 per share dividend at the end of both of the next two years. The analyst believes that a 10% annual required rate of return is appropriate for the stock. Which one of the following is closest to the analyst's estimate of today's fair price for this stock?

- A. EUR34.70
- B. EUR34.80
- C. EUR34.90

Solution

Choice B is correct. The analyst solves for the price by incorporating the expected dividend cash flow of EUR1.00 at the end of each of the next two

years with the target price of EUR40.00 at the end of two years with the following equation:

$$\text{Today's price} = \frac{\text{EUR}1.00}{1.10} + \frac{\text{EUR}41.00}{(1.10)^2}.$$

The solution equals EUR34.79 per share.

Choice A is incorrect because it assumes that both annual dividends are received at the end of two years.

Choice C is incorrect because it fails to incorporate the present value of the dividend received in one year.

2

DIVIDENDS, SHARE REPURCHASES, AND STOCK SPLITS



describe features and uses of dividends, share repurchases, stock splits, and reverse stock splits

As owners of the company, shareholders are entitled to any discretionary capital distributions made by the company on a proportional basis. These distributions are referred to as dividends and vary in terms of timing and type. For example, while many corporate issuers regularly distribute cash to investors, others may make a one-time distribution or pay no dividends at all. These distributions are typically authorized by a firm's board of directors and may be subject to consent by a shareholder vote in some jurisdictions.

Cash Dividends

Cash distributions paid at known intervals are referred to as **regular cash dividends**. These distributions represent a cash return of capital to shareholders. Companies in an early stage of their life cycle with significant growth potential are more likely to reinvest cash flows and as a result less likely to distribute cash to shareholders. As companies mature and generate higher cash flows, they may have fewer growth opportunities, so they are more likely to initiate and pay cash dividends. However, each company's management team and board must assess whether to distribute a portion of company earnings as dividends. In some cases, highly profitable companies choose to pay no dividends.

Those firms that pay cash dividends usually do so with consistent frequency (e.g., quarterly in the United States and Canada; usually semiannually in Europe and Japan; and usually annually in Scandinavia, China, and elsewhere). Many companies attempt to maintain a relatively consistent and constant dividend per share, seeking to avoid dividend changes that investors may interpret as an improvement or deterioration in the issuer's future earnings prospects. Dividend payments have a signal and informational value; maintaining consistent dividends signals management confidence that a cash distribution does not meaningfully reduce a company's future growth and profitability. Alternatively, expected weaker company performance or improved growth prospects may lead management to reduce or eliminate dividends to preserve cash.

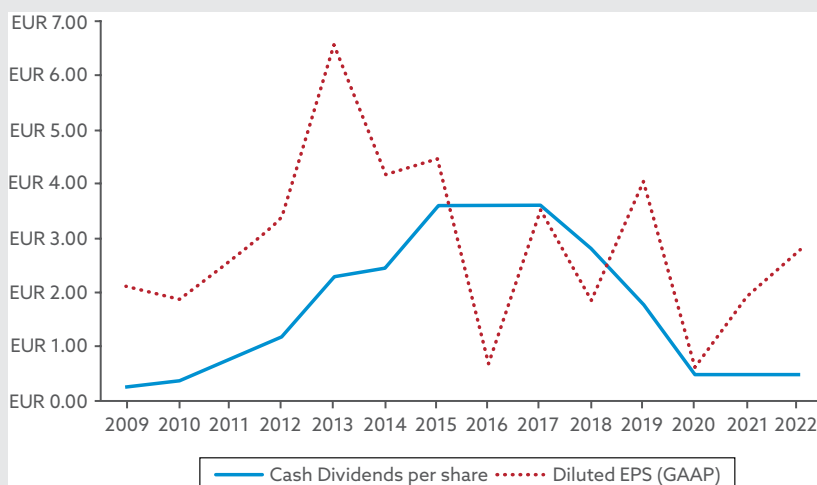
Similarly, companies often forego a dividend increase if a rise in earnings or cash flow generation is unlikely to be sustained. In cases where a company experiences significant restructuring, strategic disruption, or earnings changes over the economic cycle, cash dividends may be forced to fluctuate, as described in the next example.

EXAMPLE 1

Anheuser-Busch InBev SA/NV Dividend Changes

Anheuser-Busch InBev SA/NV (ABI), the world's largest brewer, was formed by combining Brazilian beverages company AmBev, the Belgian-based drinks firm InBev, and Anheuser Busch, the largest US brewing company. Over the 15 years following the acquisition of Anheuser-Busch in 2008, ABI's fortunes fluctuated as reflected in both its earnings per share (EPS) and dividends per share (DPS), as shown in Exhibit 1.

Exhibit 1: Anheuser-Busch InBev EPS and DPS, 2009 to 2022



Source: Bloomberg.

The early years of the ABI merger were met with investor enthusiasm. During the seven years following the Anheuser-Busch acquisition, ABI's stock price grew by nearly 10 times from a low of EUR12.89 at the end of November 2008 to a high of EUR121.95 at the end of November 2015, with its dividend rising from less than EUR1 to over EUR6.00 per share. In late 2015, ABI announced its acquisition of SABMiller, its primary rival in the global beer market, a deal that was finalized in October 2016 at a price of over USD100 billion.

Unfortunately, the acquisition coincided with a slowdown in beer demand and subsequent production decline. Due to ABI's dominant position as a newly merged company in multiple parts of the world, regulatory approval of the SABMiller deal required ABI to sell key brands in several countries under less favorable market conditions. Earnings suffered as a result, leading ABI to cut its dividend in each of the years 2018, 2019, and 2020, reaching a low of EUR0.50 per share. It wasn't until 2023 that ABI was able to once again increase dividends given its slow post-COVID earnings recovery.

While tax policies vary by jurisdiction, investors are typically subject to income tax on cash dividends in the tax period they are received. In some jurisdictions, corporations are legally required to withhold taxes on gross dividends and only distribute net dividends.

An *extraordinary or special cash dividend* is paid by a company that either does not pay dividends on a regular schedule or pays as a one-time cash payment that supplements a regular dividend. Companies may use special dividends to reward shareholders with a one-time distribution of unexpectedly high earnings or pay out cash proceeds from the liquidation of an investment, as described in the following example.

FORD MOTOR COMPANY'S SPECIAL DIVIDEND

Ford Motor Company, a US-based multinational manufacturer of commercial and passenger vehicles, missed earnings expectations in 4Q and full year 2022 due to several factors, including “execution issues” and limited profitability of its electric vehicle segment.

Despite reporting a net loss in 2022, Ford announced a special dividend in February 2023 of USD0.65 per share. The dividend involved a distribution related to Ford’s sale of around 90% of its ownership stake in Rivian, a new electric vehicle company in which it held a strategic equity stake. This sale of more than 90 million shares over the course of 2022 resulted in a gain to Ford of USD1.8 billion. Shareholders received the supplemental dividend of USD0.65 per share in addition to Ford’s regular cash dividend of USD0.15 per share.

As in the above example, special dividends are typically larger than those regularly paid to shareholders. Another form of extraordinary dividend is a **liquidating dividend**, which involves a return of capital rather than an earnings distribution. Liquidating dividends are used when a company ceases operations and distributes its net assets or sells a portion of its business for cash and distributes the proceeds. The use of a liquidating dividend may also involve a distribution that exceeds a firm’s accumulated retained earnings.

Share Repurchases

Share repurchases provide an alternative way to distribute cash in contrast to cash dividend payments. A **share repurchase** (or buyback) is a transaction in which a company uses available cash to buy back its own shares. In contrast to cash receipts from dividend payments that are received by all shareholders of a company, cash receipts from share repurchases are only received by shareholders who voluntarily choose to sell.

Shares that have been repurchased are not considered for dividends, voting, or computing earnings per share. A share repurchase is viewed as equivalent to the payment of cash dividends of equal value in terms of the effect on *total* shareholder wealth, although differences in *realized* returns among shareholders occur, as illustrated in the following example.

KNOWLEDGE CHECK: ZHAO CORPORATION DIVIDENDS VERSUS SHARE REPURCHASE



Zhao Corporation has 10 million shares outstanding at a price of CNY20 per share and CNY50 million in cash. Zhao is considering two possible ways to distribute this CNY50 million in cash to its shareholders:

- One-time CNY5 cash dividend per share paid to all shareholders
- Use of CNY50 million to repurchase shares at the current market price

1. Ignoring any tax effects and assuming no share price changes, calculate the realized return to shareholders if the cash dividend is paid.

Solution

Each shareholder receives a CNY5 cash dividend for a share priced at CNY20, resulting in a realized return of 25%:

$$\text{Realized return} = \frac{\text{Distribution}}{\text{Share Price}} = \frac{\text{CNY5}}{\text{CNY20}}.$$

2. Assuming no share price changes, calculate the remaining Zhao shares outstanding following the repurchase.

Solution

The use of CNY50 million to purchase shares priced at CNY20 results in a decline of 2.5 million shares outstanding, with 7.5 million shares remaining:

$$\text{Shares repurchased} = \frac{\text{Total Cash Distribution}}{\text{Share Price}} ;$$

$$2.5 \text{ million} = \frac{\text{CNY50 million}}{\text{CNY20}}.$$

The two mechanisms for distributing cash in the example have a similar aggregate effect, although the impact on individual shareholders may differ. Company managers cite several reasons for engaging in share repurchases:

- Signals a view that shares are undervalued, or, more generally, to support share prices and/or change an issuer's debt-to-equity ratio
- Offers an issuer greater flexibility in the amount and timing of cash distribution to shareholders versus regular cash dividends because of the ability to reduce cash payouts when the issuer has relatively more investment opportunities
- Offers investors greater flexibility in managing investment returns, providing tax efficiency in markets where tax rates on dividends paid to investors exceed tax rates on capital gains, and avoids double taxation
- Enables firms to manage the number of outstanding shares due to the exercise of employee stock options to manage the impact on earnings per share
- Enables firms to increase earnings per share by reducing shares outstanding during a reporting period

While many dividend-paying companies make efforts to maintain consistency in their distribution of regular cash dividends, stock repurchases may be announced in advance on a multi-year basis but conducted opportunistically based on market conditions, as described next.

ADIDAS AG DIVIDEND AND STOCK REPURCHASE HISTORY

Adidas, the German athletic footwear and clothing marketer, historically limited its shareholder distributions to dividends. Starting in 2014, the company announced a EUR1.5 billion, three-year share buyback program. The first repurchase tranche of EUR300 million was executed in November 2014 at an average of EUR61, or well below Adidas' EUR75 average share price for the first nine months of the year. However, cash distributions through buybacks remained consistently below cash distributions through dividend payments to shareholders.

In March 2018, Adidas announced a new three-year plan to repurchase EUR3 billion of its stock. From 2018 to 2020, Adidas spent nearly double the cash on share repurchases that it paid in dividends, marking a major shift in its shareholder distribution strategy. The company continued its buyback plans with over EUR1 billion in 2021 and over EUR2.5 billion in 2022, which included proceeds from its divestiture of Reebok. In 2021, management announced a plan to continue to distribute most of its free cash flow to shareholders, with 30% to 50% of net income in the form of dividends and expected buybacks of over EUR8 billion through 2025.

Some companies choose to distribute returns to shareholders entirely through stock repurchases. For example, many large US technology firms, such as Alphabet, have never paid dividends but consistently spend billions of dollars each quarter to buy back stock.

Stock Dividends, Stock Splits, and Reverse Splits

A **stock dividend** (or bonus issue of shares) is a type of dividend in which a company distributes newly issued shares instead of cash to shareholders. In some cases, this may involve the distribution of the company's existing common shares, while in others, a company may spin off an existing division as a new company. For example, when the US conglomerate General Electric embarked on a plan to shed businesses and narrow its strategic focus to the aerospace industry, it distributed common shares of new stand-alone companies from existing divisions, including health care (GE HealthCare Technologies) and electric power generation (GE Vernova Inc.), to shareholders.

In contrast to a cash dividend, a stock dividend of a company's own shares divides the firm's market value of shareholders' equity into smaller pieces without a change in the value or any shareholder's proportional ownership in the company. Thus, stock dividends are neither a source of income nor relevant for valuation purposes.

Specific forms of stock dividends include stock splits and reverse stock splits. A **stock split**, sometimes referred to as a forward split, involves an *increase* in the number of shares outstanding and a proportionate *decrease* in the price per share such that the total market value of equity as well as investors' proportional ownership claim in the company are unchanged. A common reason for issuers to consider a stock split includes maintaining a unit price per share within an optimal range for trading purposes.

A very low or high unit price per share can affect trading dynamics via a stock's bid–offer spread. For example, a trader showing a bid–ask spread of USD0.03 per share for a company stock trading at USD4.00 may discourage investors due to the relatively high 1.5% cost ($= 0.03 \times 2 \div 4$) to acquire and subsequently sell shares. The same USD0.03 spread for a USD1,000 share price results in just a 0.006% ($= 0.03 \times 2 \div 1,000$) charge, which may not be sufficient to attract market makers to provide liquidity.

Equity issuer approaches to stock splits vary widely, as described in the following example.

STOCK SPLIT POLICIES OF AMAZON.COM VERSUS BERKSHIRE HATHAWAY

Two major US corporations, Amazon.com and Berkshire Hathaway, offer an interesting contrast in stock split behavior. Amazon.com went public as an online book retailer in June 1997, and the company's share price rose to approximately 50 times the initial public offering (IPO) price during the internet boom of the late 1990s. Amazon.com shares split three separate times over this period, with an original share equivalent to 12 shares by September 1999. At the beginning

of 2000, Amazon's price per share was near USD80 per share, and by the end of 2021, each share traded at well over USD3,000. In 2022, as stock prices declined, Amazon.com split its stock 20:1, lowering the price per share to closer to USD100.

Berkshire Hathaway has taken a very different approach to its rising share price. The company's Class A shares went public in March 1980 at USD290 per share. By the end of 2022, each Class A share of Berkshire Hathaway traded at around USD542,000 per share, a 186,900% price increase with no splits during the life of the Class A shares. However, relenting to pressure from shareholders seeking to directly own individual shares at a lower price, Berkshire Hathaway issued Class B shares in 1996 at 1/30 of the price of Class A shares. In 2010, the Class B shares split on a 50:1 basis and now trade at approximately 1/1,500 of the Class A share price.

A **reverse stock split** involves a *decrease* in the number of shares outstanding and a proportionate *increase* in the price per share such that the total market value of equity and investors' proportional ownership claim in the company do not change.

Common reasons for issuers to consider a reverse stock split include maintaining a minimum price per share to meet exchange requirements as well as bringing the current price into a range more optimal for trading purposes at a time when a firm takes measures to turn around or restructure its business.

BOMBARDIER'S 2022 SHARE CONSOLIDATION

In 1942, Joseph-Armand Bombardier founded a company bearing his name to build and sell his snowmobile invention. In the following decades, Bombardier went public and expanded globally as a manufacturer of recreational equipment, rail transport equipment, and commercial aircraft. After narrowly averting bankruptcy, the company initiated a multiyear plan to shed most of its divisions, pay down debt, and focus solely on corporate jet production. Bombardier's share price declined by almost 95% from a peak in July 2018 to a low of CAD0.28 in November 2020. While the price recovered to well over CAD1.00 per share in 2021, early the following year, Bombardier shareholders voted to consolidate both Class A shares (with multiple votes) and Class B (single vote shares) on a 25-for-1 basis as of June 2022.

The reverse split reduced the number of shares outstanding by a factor of 25 while increasing Bombardier's price per share to CAD26.86 on the split date. Despite an initial negative market reaction, market sentiment improved as the company successfully executed its focused strategy, with the price per share exceeding CAD50 by the end of 2023.

QUESTION SET 1



1. Investors in Kettlewood Corporation have received cash dividends per share of GBP10 for the past three years. Kettlewood decides to halve its dividend and use the remaining cash to repurchase shares. Which one of the following statements best describes the impact of this change on the *realized* returns of Kettlewood's shareholders?
 - A. All Kettlewood shareholders will experience a 50% reduction in their realized return after the change.
 - B. Since Kettlewood distributes the same amount of cash as in the past, all Kettlewood shareholders will have the same realized return after the change.

- C. All Kettlewood shareholders will experience a 50% reduction in their realized return, except for those who choose to sell their shares.

Solution

Choice C is correct. All shareholders will experience a halving of the cash dividend, with only those who choose to sell receiving additional cash and any realized earnings.

Choice A is incorrect because it excludes those shareholders who choose to sell their shares back to the company for cash.

Choice B is incorrect because not all shareholders will choose to sell shares back to the company in exchange for cash, and only those who do will realize the return.

2. A share repurchase differs from cash dividend payments because:

- A. repurchases decrease shareholder proportional ownership stakes while dividends do not.
- B. repurchases are not direct cash distributions to shareholders, unlike cash dividends.
- C. repurchases may provide signals to the market about an issuer's perception of share value, while dividends do not.

Solution

Choice C is correct. When an issuer repurchases its stock, investors may consider this to be a signal that management believes its stock is undervalued. Cash dividends provide no signal about the stock's value (although an increase may signal management's expectation of higher future earnings, and this signal requires more than simply the payment of dividends).

Choice A is incorrect because the repurchase increases the proportional ownership of all shareholders who do not participate in the stock repurchase. Only shareholders who choose to sell might reduce their proportional ownership.

Choice B is incorrect because repurchases are a cash distribution to shareholders. The difference is that only selling shareholders receive the distribution from repurchases.

3. Baixo Corporation's stock recently increased from EUR800 to EUR1,600 per share. Baixo's chief financial officer decides to split the company's stock to achieve a price between EUR70 and EUR90 per share. Which one of the following is the most appropriate split adjustment for Baixo?

- A. 10:1
- B. 20:1
- C. 25:1

Solution

Choice B is correct. At a price of EUR1,600 per share, a 20:1 split would decrease Baixo's price per share to approximately EUR80 per share. That is, an $X:1$ split results in a new price (P_{new}) equal to P_{old}/X , or $\text{EUR}1,600 \div 20$, which is in the middle of the desired stock price range.

Choice A is incorrect because this split adjustment would only decrease the stock price to approximately EUR160 per share ($= \text{EUR}1,600 \div 10$), which is well above the desired range.

Choice C is incorrect because this split adjustment would lower the stock price to approximately EUR64 per share ($= \text{EUR}1,600 \div 25$), which is below the desired range.

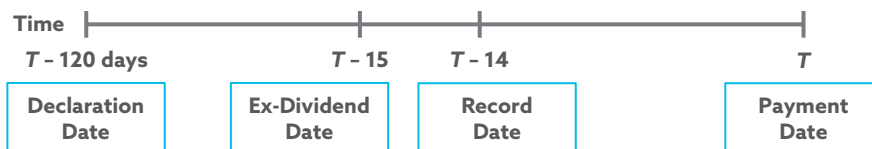
DIVIDEND PAYMENT CHRONOLOGY

3

☐ describe dividend payment chronology

The payout of regular cash dividends to common shareholders generally follows a standard timeline, as shown in the hypothetical example in Exhibit 2.

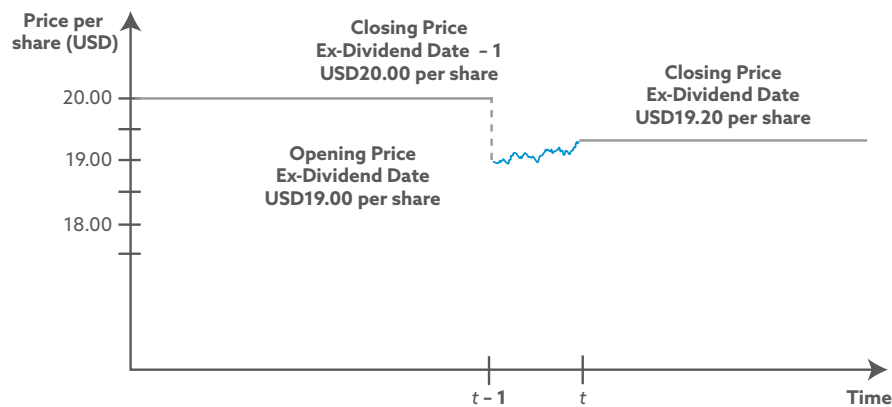
Exhibit 2: Dividend Payment Chronology



While timing varies, the dividend payout process is set into motion once the company's board of directors (and shareholders, as necessary) votes to pay the dividend.

- First is the **declaration date**, the day that the company issues a statement declaring a specific dividend.
- This is followed by the **ex-dividend date** (or ex-date), the first date that a share trades without (or "ex") the dividend. The ex-dividend date is followed closely (one or two business days later) by the **record date**, or the date that a shareholder registered with a securities depository is deemed to own the shares for purposes of receiving the upcoming dividend.
- The difference between the ex-dividend date and the record date is linked to equity market trade settlement. **Settlement** refers to the official transfer of securities to the buyer's account and cash to the seller's account. If an equity trade settlement takes two days, usually there is a two-day gap between the ex-dividend date and the record date, ensuring all transactions made before the ex-dividend date are settled by the record date.
- The final step in the process is the **payment date**, which is the day that the company electronically transfers a dividend payment to shareholders.

Exhibit 3 shows the impact of a USD1.00 per share dividend from the prior day's closing price to the closing price on the ex-dividend date.

Exhibit 3: Hypothetical Stock Price Change on Ex-Dividend Date

As buyers of a company's shares on or after the ex-dividend date are ineligible to receive the pending dividend, all else being equal, the company's share price immediately falls by the amount of the foregone dividend on that day. However, this decline in a company's share price is obscured by the effects of intraday equity price changes, as shown in Exhibit 3 and summarized in Equation 1:

Closing Price Change

$= -\text{Dividend per share} + \text{Overall equity market change} + \text{Company-specific price change.}$

(1)

In addition to dividends, these observed price changes reflect changes in equity prices overall as well as any idiosyncratic or company-specific price changes. In this simplified approach, we importantly assume that the risk of the company's stock is *identical* to that of the market, approximating daily stock price changes due to the market using a broad-based equity index on the company's ex-dividend date, as shown in the following example.

KNOWLEDGE CHECK: TECH MAHINDRA EX-DIVIDEND DATE STOCK RETURN



Tech Mahindra, an information technology services and solutions provider based in India, declared a dividend in the amount of INR32 per share with an ex-dividend date of 21 July 2023. On 20 July 2023, Tech Mahindra's stock price closed at INR1,252.95, and its stock price declined to INR1,195.10 at the close on 21 July 2023. Tech Mahindra's stock is a component of the broad-based Indian NIFTY 50 Index, the price of which fell from INR19,979.15 to INR19,745 between 20 July and 21 July 2023.

1. Calculate Tech Mahindra's percentage price decline on the ex-dividend date, and show how much of the percentage decline is comprised of:

- the expected dividend;
- the daily stock price change due to the market; and
- the impact of company-specific price changes.

Solution

Tech Mahindra's stock declined by 4.62% on the ex-dividend date of 21 July:

$$-4.62\% = \frac{1,195.10 - 1,252.95}{1,252.95}.$$

Recall from Equation 1 that the *overall* price decline is the sum of the dividend, overall market changes, and company-specific price movements. Given the information above, we may estimate the first two of these effects as follows:

- Dividend of INR32 results in a price change of -2.55% by dividing the expected dividend by the share price on the day before the ex-dividend date:

$$-2.55\% = \frac{-32}{1,252.95}.$$

- The one-day equity market change is -1.17% based upon the NIFTY 50 Index percentage price change:

$$-1.17\% = \frac{19,745 - 19,979.15}{19,979.15}.$$

Solve for company-specific changes by substituting the results into Equation 1:

$$-4.62\% = -2.55\% - 1.17\% + \text{Company-specific changes}$$

$$\text{Company-specific changes} = 0.90\%.$$

In the absence of company-specific price changes, we would expect a smaller stock price decline of 3.72% ($= -2.55\% + -1.17\%$).

QUESTION SET 2



1. Wiley Creek Limited declared a quarterly dividend of AUD1.25 per share on its common stock on Monday, 7 January. The dividend was payable on 1 February to shareholders of record on 10 January. On which one of the following business days would a decline in Wiley Creek's stock price of the dividend amount, holding all else equal, be expected?

- A. 9 January
- B. 10 January
- C. 1 February

Solution

Choice A is correct. The ex-dividend date is the first date at which the stock trades without rights to the upcoming dividend payment. The ex-dividend date typically occurs one business day before the record date. In this case, 10 January is the record date, so 9 January is the most likely ex-dividend date.

Choice B is incorrect because the record date occurs after the ex-dividend date.

Choice C is incorrect because the payment date occurs several weeks later than the ex-dividend date.

2. A junior analyst proposes the following short-term trading strategy: Buy stocks on the date before their ex-dividend dates, and then sell the stocks following the dividend payment date. Which one of the following statements is most correct about this trading strategy?

- A. This trading strategy will likely be profitable, on average, because of returns from collecting dividends.

- B.** This trading strategy will likely generate negative profits, on average, because of the combined effect of market movements and company-specific price changes.
- C.** This trading strategy will likely earn profits, on average, proportional to market movements.

Solution

Choice C is correct. Buying before the ex-dividend date and selling after collecting the dividend, on average, should generate zero profits because the stock price should decline by the amount of the dividend. The effect of company-specific price changes should be canceled by trading in many different stocks. The only profits (positive or negative) will be based on market movements between the purchase and sale dates for each stock.

Choice A is incorrect because the dividends collected simply offset the price declines on the ex-dividend dates.

Choice B is incorrect because profits could be positive or negative, depending on market movements. Company-specific price changes should be eliminated because of the diversified trading strategy.

3. Verizon Communications declared a quarterly dividend of USD0.665 per share on its common stock on 7 December 2023. The dividend was payable on 1 February 2024 to shareholders of record on 10 January 2024. On 8 January, Verizon stock closed at USD40.10 per share and the S&P 500 Index closed at USD4,763.54. On 9 January, Verizon stock closed at USD39.04 while the S&P 500 Index closed at USD4,756.50. If the S&P 500 Index accurately reflects how Verizon share prices change due to market price changes, which one of the following most correctly shows the percentage of Verizon's decline on the ex-dividend date that reflects company-specific price changes?

- A.** -0.14%
- B.** -0.85%
- C.** -1.66%

Solution

Choice B is correct. Verizon's decline to USD39.04 reflected a total daily return of -2.65% combining dividend, market, and company-specific price changes:

$$-2.65\% = \frac{39.04 - 40.10}{40.10}.$$

The expected price change from the ex-dividend day effect is equal to the USD0.665 dividend divided by the prior day's closing price, or -1.66%:

$$-1.66\% = \frac{-0.665}{40.10}.$$

The market-based price change on the ex-dividend date was -0.14%:

$$-0.14\% = \frac{4756.5 - 4763.54}{4763.54}.$$

Overall, we would expect a decline of 1.80% (= 1.66% + 0.14%) in Verizon stock on the ex-dividend date.

Therefore, company-specific changes contributed an additional -0.85% [= -2.65% - (-1.80%)] return on the ex-dividend date.

Choice A is incorrect as this is solely the percentage change from overall market movements.

Choice C is incorrect because this percentage is the expected amount of decline due to the size of the dividend.

EQUITY TOTAL RETURN

4

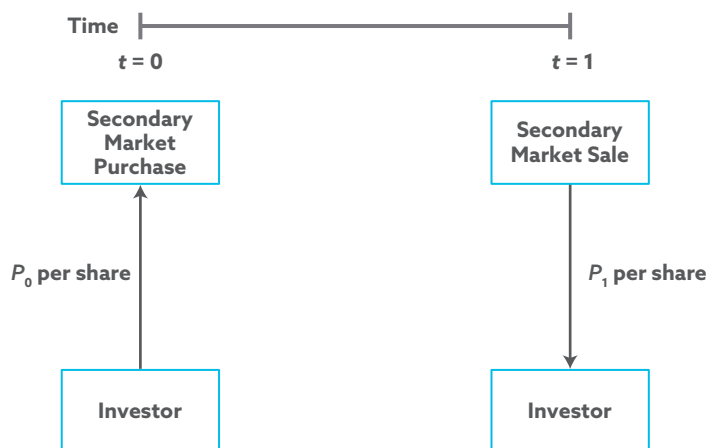
- ☐ calculate price and total return for equity securities

Two sources of equity return exist, namely the distribution of discretionary cash flows in the form of cash dividends and the per share price appreciation or depreciation (capital gain or capital loss) over a specific period. Investors use current and historical security prices (adjusted for stock splits or reverse stock splits) to measure returns over a specific holding period, as well as equity returns on a daily, weekly, or annual basis. The following analysis ignores the impact of taxes on returns.

Equity Return with No Dividends

Non-dividend-paying share returns are solely based upon price increases or decreases. Exhibit 4 shows an example of the return on a share of non-dividend-paying stock over a holding period.

Exhibit 4: Holding Period Return for a Non-Dividend-Paying Stock



In general, assuming that an equity issuer is subject to neither a restructuring nor a liquidation event over the holding period, then for an investor who owns a share with a market price of P_t at time t and observes a per share market price of P_{t+1} at time $t + 1$, the total realized equity rate of return r_E over the period t to $t + 1$ consists of price appreciation only, as shown in Equation 2:

$$\text{Realized rate of return} = r_E = \frac{P_{t+1} - P_t}{P_t}. \quad (2)$$

KNOWLEDGE CHECK

1. Consider an investor who purchases a stock in the secondary market at USD20.00 per share. Calculate the holding period rate of return if the investor sells the stock for USD22.50 per share at the end of the period.

Solution

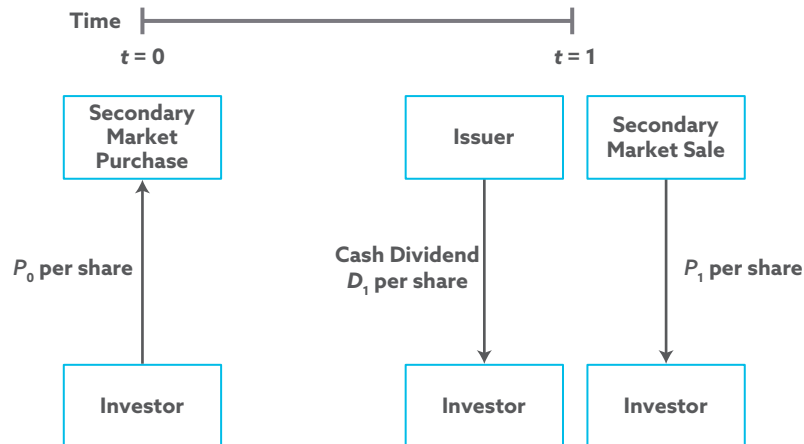
Solve this using Equation 2, with P_t of USD20 and P_{t+1} equal to USD22.50:

$$r_E = 12.50\% = \frac{22.50 - 20.00}{20.00}.$$

Equity Return with Dividends

Any cash dividends to which an investor is entitled must be included in the return calculation for a given holding period. Exhibit 5 shows an example of the return on a share of stock over a holding period that ends on a dividend payment date.

Exhibit 5: Holding Period Return for Dividend-Paying Stock



As shown in Exhibit 5, for an investor who owns a share priced at P_t at time t , receives a regular cash dividend D_{t+1} , and observes a per share market price of P_{t+1} at time $t + 1$, we may calculate the equity security's total rate of return r_E over the holding period from t to $t + 1$:

$$\text{Realized rate of return} = r_E = \frac{P_{t+1} - P_t + D_{t+1}}{P_t}. \quad (3)$$

KNOWLEDGE CHECK

Consider an investor who purchases a stock for USD20.00 per share. The investor receives a cash dividend of USD1.00 per share at the end of the holding period, at which time the price per share is USD22.50.

1. Calculate the holding period rate of return.

Solution

Solve this problem using Equation 3, with P_t of USD20, D_{t+1} of USD1.00, and P_{t+1} equal to USD22.50:

$$r_E = 17.50\% = \frac{22.50 - 20.00 + 1.00}{20.00}.$$

Note that in the prior example, the cash dividend (D_1) is received at the same time as the shares are valued (at P_1) at time $t + 1$. For non-dividend-paying stocks, $D_{t+1} = 0$ and the total rate of return consists of price appreciation only, as shown in Equation 2.

Recall from an earlier lesson that the *number* of shares associated with a specific proportion of ownership claim may change over time. This may occur, for example, when a firm issues a stock dividend or declares a stock split or reverse stock split. When calculating the total return on a share position, we must therefore adjust the number of shares outstanding as well as any dividends paid per share for the entire holding period to consistently calculate total return.

KNOWLEDGE CHECK



Recall the prior example in which an investor buys a stock for USD20.00 per share and receives a cash dividend of USD1.00 at the end of the holding period, at which time the share price is USD22.50.

1. Calculate the holding period return on a percentage basis if the issuer conducts a 1-for-5 reverse share split of the stock at the end of the period, with the cash dividend paid to shareholders at the same time.

Solution

Note that the *change* in price and dividend are proportionally the same before and after the reverse split. Using Equation 3, we may calculate the total return based upon the new share price of USD112.50 ($= \text{USD}22.50 \times 5$) with a USD5 dividend ($= \text{USD}1 \times 5$) per share as follows:

$$r_E = \frac{P_{t+1} - P_t + D_{t+1}}{P_t}$$

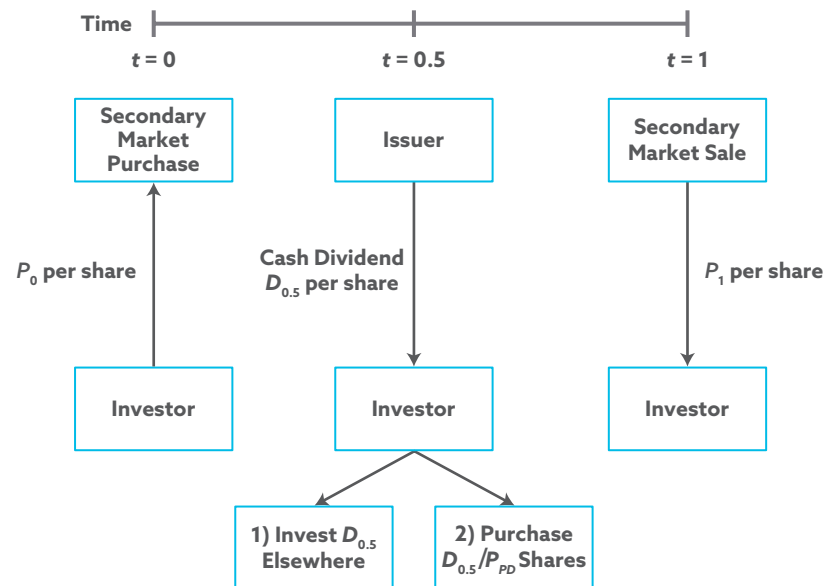
$$17.50\% = \frac{112.50 - 100 + 5}{100}.$$

The reverse share split has no effect on the holding period return.

Equity Return with Reinvested Dividends

While an investor could choose to invest cash dividends in many ways, **reinvested dividends** are cash dividends that the investor receives and uses to purchase additional shares issued by the same company. Since dividend reinvestment is a common investor approach, we consider its impact on holding period returns (ignoring taxes) when dividends are received during a given holding period.

Exhibit 6 shows the case of an investor who receives a cash dividend in the middle of a holding period.

Exhibit 6: Holding Period Return for Reinvested Dividends

As shown in Exhibit 6, a shareholder who immediately reinvests a dividend D will purchase D/P_{PD} additional shares, where P_{PD} is the prevailing market price on the dividend payment date. The calculation of holding period returns in the case where dividends are reinvested is shown in Equation 4:

$$r_E = \frac{(P_{t+1} \times (1 + \frac{D}{P_{PD}})) - P_t}{P_t} \quad (4)$$

KNOWLEDGE CHECK



Recall the earlier examples in which an investor buys a stock for USD20.00 per share, receives a cash dividend of USD1.00, and observes an end of period share price of USD22.50. Assume that instead of receiving the dividend at the end of the holding period, the investor receives the dividend halfway through the period at a time at which the share price is USD21.

1. Calculate the holding period return if the dividend is reinvested at a risk-free rate of 1.00% to the end of the holding period.

Solution

To solve for holding period return using Equation 4, we must first calculate the future value of the dividend. A shareholder investing the USD1.00 dividend at a risk-free rate of 1% for the remainder of the holding period will receive USD1.005 $[= 1.00 \times (1.01)^{0.5}]$ at time $t + 1$. Substituting this into Equation 4 as the future value of the dividend D_{t+1} , solve for the holding period rate of return as:

$$r_E = 17.525\% = \frac{22.50 - 20 + 1.005}{20}$$

2. Calculate the holding period rate of return if the dividend is reinvested in shares at the prevailing market price upon receipt.

Solution

Given the cash dividend of USD1.00 and the prevailing market price S_{PD} of USD 21, we may determine that the investor purchases $1/21$ (or 0.04762) additional shares. Substituting this into Equation 4, we may solve for a holding period rate of return of 17.857%:

$$r_E = \frac{(P_{t+1} \times (1 + \frac{D}{P_{PD}})) - P_t}{P_t}$$

$$r_E = 17.857\% = \frac{(22.50 \times (1 + \frac{1}{21})) - 20}{20}.$$

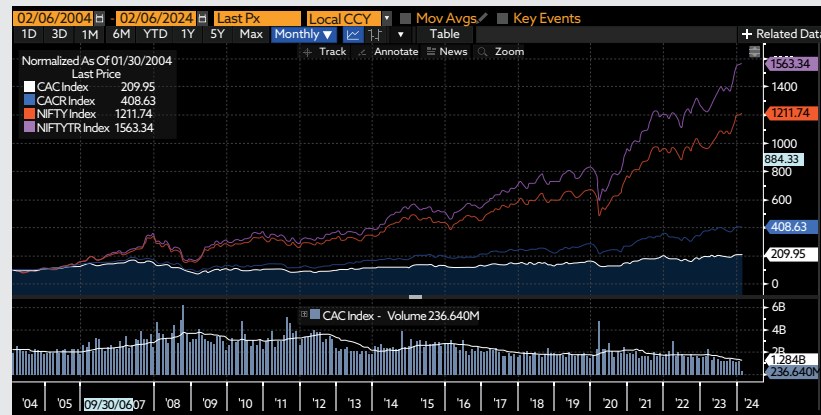
The reinvested dividend increases the investor's ownership from one share at time t to $(1 + D/P_{PD})$ shares at time $t + 1$. Note that the reinvestment of dividends (as well as a reverse share split) may give rise to **fractional share ownership**—that is, a situation in which an investor owns a fraction of a share with a similar proportional claim to distributions and net assets. In this case, the investor has acquired an additional 0.04762 shares over the holding period at a price of USD21. The appreciation of the fractional share purchased in this case increases the investor's total return. The effect of reinvested dividends can provide significant improvements in the overall returns experienced by investors in equity markets, as illustrated in the following example of two equity indexes.

TOTAL EQUITY RETURNS VERSUS PRICE-ONLY RETURNS: A GLOBAL COMPARISON

The effect of reinvested dividends can have a significant impact over 20 years and can vary greatly by equity market.

From February 2004 through February 2024, the French CAC 40 Index roughly doubled while the Indian NSE NIFTY 50 Index rose by over 1,200%. These equity indexes only reflect price changes, however, not total returns for the index. An examination of the total returns on these two indexes shows that the CAC 40 increased over 400% while the NIFTY 50 increased by well over 1,500%. Exhibit 7 illustrates the trajectories of these two indexes (CACR and NIFTYTR, respectively) from both a price and a total return perspective.

Exhibit 7: NSE NIFTY 50 versus CAC 40 Index Performance (Total Return and Price)



Source: Bloomberg.

By dividing the price index amount by the total return index amounts shown in Exhibit 7, we can isolate the pure price appreciation component of the holding period return. Doing so for the NIFTY 50 shows that 77.5% ($= 1,211.74 \div 1,563.34$) of the 20-year holding return is strictly from price appreciation. The remaining 22.5% of returns reflect the effects of reinvested dividends on the total returns of the NIFTY 50.

By contrast, the CAC 40 has generated only 51.4% ($= 209.95 \div 408.63$) of its returns from pure price appreciation. Reinvested dividends provide the other 48.6% of the CAC 40 total returns. Note that in contrast to unrealized capital gains, reinvested dividend income is generally subject to taxation, which has the effect of dampening total returns.

The developing Indian equity market is relatively more reliant on capital gains in driving higher long-run returns compared to the more developed French market. Furthermore, the sector classifications of the two indexes are quite different: The NIFTY 50 Index is dominated by financial, information technology, and energy firms, and the CAC 40 Index has industrials and consumer classifications as the largest sectors.

Equity Price

Using the dividend and price appreciation components of an equity security's return over a holding period, we now turn our attention to the *current* price of an equity security in relation to these components. In general, the price of a stock at time t should be equal to the *present value* of *expected* future cash flows from cash distributions and its anticipated future price at the end of a holding period.

For example, if we assume an issuer pays no dividends, we can rearrange Equation 2 for the case of a non-dividend-paying stock to show that today's share price should equal the present value of the expected future price at the end of a holding period:

$$P_t = \frac{E(P_{t+1})}{1 + r_e}. \quad (5)$$

Note that we have replaced P_{t+1} with $E(P_{t+1})$ given that the future price of a share is an *expected* quantity at time t , which is not known until $t + 1$. The uncertainty of future stock prices ensures that investors are less able to predict future stock returns. In this case, r_e may be interpreted as the appropriate periodic discount rate that shareholders apply to future cash flows and reflects the required return on equity.

For a firm that pays a cash dividend at time $t + 1$, the current share price P_t is equal to the sum of the present value of the expected dividend $E(D_{t+1})$ plus the present value of the expected future price $E(P_{t+1})$, as shown in Equation 6:

$$P_t = \frac{E(P_{t+1}) + E(D_{t+1})}{1 + r_e}. \quad (6)$$

KNOWLEDGE CHECK



An investor observes that an analyst has established a share price target for Volksbein AG of EUR55 in one year's time. The investor considers 12% to be the appropriate annual discount rate.

1. Calculate the Volksbein AG share price the investor would expect to pay today assuming the issuer pays no dividends.

Solution

Using Equation 5 with $E(P_{t+1})$ equal to EUR55 and r_e of 12%, solve for P_t :

$$\text{EUR}49.11 = \frac{E(P_{t+1})}{(1 + r_e)} = \frac{\text{EUR}55}{1.12}.$$

2. Assume that Volksbein AG announces a year-end dividend of EUR2.50, and the analyst maintains the same EUR55 year-end price target. Calculate the revised share price the investor would expect to observe today.

Solution

Using Equation 6 with $E(P_{t+1})$ equal to EUR55, $E(D_{t+1})$ equal to EUR2.50, and r_e of 12%, solve for P_t :

$$\text{EUR}51.34 = \frac{\text{EUR}2.50 + \text{EUR}55}{1.12}.$$

Note that the difference in today's share price between 1. and 2. is equal to the present value of the EUR2.50 dividend equal to EUR2.23 ($= 2.50/1.12$).

Investors and financial analysts commonly develop expectations for share prices several periods into the future based upon financial statement analysis and forecasting among other equity valuation approaches, detailed coverage of which begins in the next Equities learning module. These future share price projections are used as the *expected* end of holding share price to estimate today's share price. If we consider the expected share price $E(P_{t+n})$ at the end of an investment horizon of n periods where a constant dividend D is paid at the end of each period, we may solve for P_t by discounting expected future cash flows at the annualized expected rate of return of r_e , as shown in Equation 7:

$$P_t = \sum_{i=1}^n \frac{D}{(1 + r_e)^i} + \frac{E(P_{t+n})}{(1 + r_e)^n}. \quad (7)$$

Equation 7 allows us to compare the relative impact of these cash flow components, which are explored in greater detail in later equities learning modules.

KNOWLEDGE CHECK



The same analyst report referenced in the prior example had a long-term price target for Volksbein AG of EUR65 in three years' time.

1. Calculate today's Volksbein AG share price based upon the three-year price target if the investor expects the recently announced EUR2.50 dividends per share to remain constant over the next three years and the discount rate to remain unchanged at 12%.

Solution

Using Equation 7 with $E(P_{t+1})$ equal to EUR65, D of EUR2.50 each period, and r_e of 12%, solve for P_t :

$$\text{EUR}52.27 = \frac{\text{EUR}2.50}{1.12} + \frac{\text{EUR}2.50}{(1.12)^2} + \frac{\text{EUR}67.50}{(1.12)^3}.$$

2. Describe the relative impact of expected dividends versus expected future share price appreciation over the next three years on Volksbein AG's stock price.

Solution

Derive the relative impact of dividends versus share price appreciation by comparing the first component of Equation 7 as a fraction of total share price with its second component. From Question 1 of this Knowledge Check:

$$\text{Dividend cash flows: EUR}6.00 = \frac{\text{EUR}2.50}{1.12} + \frac{\text{EUR}2.50}{(1.12)^2} + \frac{\text{EUR}2.50}{(1.12)^3}$$

$$\text{Expected future share price: EUR}46.27 = \frac{\text{EUR}65}{(1.12)^3}$$

Expected dividend cash flows represent 11.5% (or $6 \div 52.27$) of today's Volksbein AG share price, while the remaining 88.5% ($= 46.27 \div 52.27$) is based upon the expected future share price.

If an investor intends to hold a share with an expected constant dividend D indefinitely, the present value of future dividend cash flows will be the predominant factor in today's share price. On the other hand, if the firm is expected to pay no dividends ($D = 0$) during the investment time horizon, then firm cash flows that contribute to net asset growth will be most important for today's stock price. The contribution of the present value of the terminal value estimate generally exceeds the PV of the interim cash flows, unless the investment time horizon is very long or the discount rate is very high.

QUESTION SET 3



1. An investor buys a stock at a price of CAD200 per share. After one year, the issuer of the stock conducts a 4-for-1 split of the shares. Immediately following the split, the issuer pays a CAD3.00 per share cash dividend, and the stock price is CAD58 per share. Which one of the following is closest to the one-year holding period return on the stock?
 - A. 17.5%
 - B. 22.0%
 - C. 40.0%

Solution

Choice B is correct. To properly account for the split, the purchase price of the stock, P_p , may be adjusted to CAD50:

$$\text{CAD50} = \frac{\text{CAD200}}{4}.$$

The one-year return is equal to 22.0%:

$$22.0\% = \frac{58 + 3 - 50}{50}.$$

Choice A is incorrect, as 17.5% is the result if we were to incorrectly adjust the end-of-year dividend *downward*:

$$17.5\% = \frac{58 + 0.75 - 50}{50}.$$

Choice C is incorrect, as 40% is the result if we were to incorrectly adjust the end-of-year dividend upward:

$$40\% = \frac{58 + 12 - 50}{50}.$$

2. An investor buys a stock at a price of USD75 per share. One year later, the stock's price is USD85 per share and pays a dividend of USD3.00 per share. The investor reinvests this dividend. After the second year, the stock's price is USD80 per share, and the stock pays a dividend of USD3.00 per share. Which one of the following is closest to the two-year holding period return of this investor?

- A. 14.57%
- B. 14.67%
- C. 14.98%

Solution

Choice A is correct. This may be confirmed by two different solution processes. First, the approach shown earlier in which we explicitly calculate the fractional share purchased with the reinvested dividend is:

$$r_E = 14.57\% = \frac{[(80 + 3) \times (1 + \frac{3}{85}) - 75]}{75}.$$

Note that the USD3 dividend at the end of the second year is added to the USD80 share price as this dividend does not compound. Rather, only the USD3 dividend received at the end of the first year is used to buy a fractional share at a price of USD85.

An alternative approach to finding the correct answer uses the concept of compounding periodic returns:

$$\text{Year 1 return} = 17.33\% = (85 + 3 - 75)/75$$

$$\text{Year 2 return} = -2.35\% = (80 + 3 - 85)/85$$

$$r = 14.57\% = (1 + 0.1733) \times (1 - 0.0235) - 1$$

$$r = 14.57\% = (1.1733) \times (0.9765) - 1$$

Choice B is incorrect because 14.67% is derived from simply adding the two dividends to the second-year price [= (80 + 3 + 3 - 75)/75]. Thus, this approach does not properly account for the reinvested dividend at the end of the first year.

Choice C is incorrect because 14.98% is simply the sum of the Year 1 and Year 2 returns.

3. An analyst has placed a five-year target price of EUR75.00 per share on a stock. The stock recently paid a EUR3.00 dividend, and this dividend is expected to be paid annually over the next five years with a 4% growth rate each year. The analyst believes that a discount rate of 8% is appropriate for the stock. Which one of the following most closely estimates the current price of this stock?

- A. EUR61.98
- B. EUR64.46
- C. EUR78.00

Solution

Choice B is correct. Using Equation 7 presents the basic structure of the problem:

$$\begin{aligned} \text{EUR64.46} = & \frac{\text{EUR3} \times (1.04)}{(1.08)^1} + \frac{\text{EUR3} \times (1.04)^2}{(1.08)^2} + \frac{\text{EUR3} \times (1.04)^3}{(1.08)^3} \\ & + \frac{\text{EUR3} \times (1.04)^4}{(1.08)^4} + \frac{\text{EUR3} \times (1.04)^5 + \text{EUR75}}{(1.08)^5}. \end{aligned}$$

The resulting cash flow stream consists of dividends and Year 5 price of the following:

- Year 1 dividend: EUR3.12
- Year 2 dividend: EUR3.24
- Year 3 dividend: EUR3.37
- Year 4 dividend: EUR3.51
- Year 5 dividend: EUR3.65
- Year 5 price: EUR75.00

Choice A is incorrect because EUR61.98 results if the first-year dividend is assumed as EUR3.00 with 4% growth following.

Choice C is incorrect because a result of EUR78.00 assumes 4% growth *for-ever* in the dividend, disregarding the EUR75.00 target price in Year 5.

SUMMARY: SOURCES OF EQUITY RETURNS

- Regular cash dividends are a source of periodic income and subject to tax. While share repurchases have a similar wealth effect for owners, they also support share prices and offer flexibility and tax efficiency to some investors.
- Stock splits and reverse splits change the number of shares outstanding, with a proportional change in share price, but do not affect shareholder wealth, a firm's equity market value, or an investor's ownership claim.
- Once a cash dividend is *declared*, a date is specified upon which shares are traded *without* receiving the future dividend, at which time the share price typically drops by the amount of the future dividend.
- Shareholders entitled to receive dividends are typically identified on a date *before* a dividend payment is made. Dividend payments are made based upon the number of shares held as of that date.
- Sources of equity holding period return include any cash distributions and share appreciation or depreciation over the time horizon.
- A stock's price should equal the present value of expected future cash distributions and the expected future sale price at the end of a holding period.

PRACTICE PROBLEMS

1. Which one of the following is typically a more predictable part of an expected holding period return?
 - A. Dividend portion of total return
 - B. Capital gains portion of total return
 - C. Dividend and capital gains portions are equally unpredictable.

The following information relates to questions 2-6

An investor is evaluating three different stocks as candidates for purchase. The following statements provide basic descriptions of each of the three stocks.

- Addison Company's stock is trading at a price of GBP54 per share and recently paid a quarterly dividend of GBP0.50 per share. On average, analysts have set a one-year target price of GBP60. Addison's dividends in the past have typically been a small portion of its ability to pay shareholders, and Addison has commonly repurchased significant amounts of its stock in the past.
 - Bryce Limited's stock traded two years ago at a price of GBP76 per share. One year ago, the company paid a GBP5.00 per share dividend when its stock price was GBP96 per share; it then completed a 2-for-1 stock split immediately after. Today, Bryce paid a GBP5.00 per share dividend, and its stock trades at GBP82 per share.
 - Caddell Corporation operates in a commodity-based industry with high current cash flows and negative long-run growth. Caddell's stock is currently trading at GBP80.76 per share and is expected to pay dividends of GBP10 per share for the next five years. The expectation is that Caddell's share price will be GBP60 per share in five years. Analysts believe that 8% is an appropriate discount rate for Caddell stock.
2. Which one of the following is closest to Addison's expected stock return over the next one-year holding period if the dividend stays constant, dividends are reinvested at 1% per quarter, and analysts' target price estimate is correct?
 - A. 11.1%
 - B. 14.8%
 - C. 14.9%
 3. Which one of the following statements is likely most correct as to how the investor may assess Addison's policy regarding dividend payments compared to money spent on repurchasing stock?
 - A. The investor may value Addison's policy if tax rates on capital gains are lower than tax rates on dividend income.

- B.** The investor may value Addison's policy if the company has stable investment opportunities.
 - C.** The investor likely is not in favor of Addison's policy because cash flows from repurchase are not of equivalent value to cash flows from dividends.
- 4.** Which one of the following statements provides the most correct rationale to justify Bryce Limited's decision to complete a stock split?
 - A.** Bryce is attempting to attract more institutional investors to buy its stock.
 - B.** Bryce wants a higher per share price on its stock to improve its reputation in the equity market.
 - C.** Bryce is attempting to keep its stock price within a reasonable price range to attract individual investors.
- 5.** Which one of the following is closest to the two-year holding return on Bryce's stock if the dividend from one year ago is reinvested?
 - A.** 20.4%
 - B.** 127%
 - C.** 141%
- 6.** Which one of the following most correctly states the relative percentage impact of Caddell Corporation's future share price of USD60 on its current price?
 - A.** -25.7%
 - B.** 50.6%
 - C.** 74.3%

SOLUTIONS

- Choice A is correct. Issuers typically attempt to maintain some consistency in their dividends, which allows investors to be able to view recent dividends as a somewhat predictable source of expected returns.
Choice B is incorrect because future prices are highly uncertain. Although analysts will build target prices into their expected return models, the fact that future target prices are very unpredictable implies that the capital gains proportion is much less predictable.
Choice C is incorrect following from the above arguments.
- Choice C is correct. The expected return combines the price increase from GBP54 to GBP60, the dividends paid, and the return generated from the return on reinvested dividends as follows:
Price return = 11.11% [= (60 – 54) ÷ 54]
Dividend return incorporating reinvestment = 3.76% (= 2.03 ÷ 54)
Compounded value of dividends = GBP2.03
[= 0.50(1.01)³ + 0.50(1.01)² + 0.50(1.01)¹ + 0.50]
Total return = 14.87% (= 11.11% + 3.76%), which rounds to 14.9%.
Choice A is incorrect because 11.11% reflects only the price change.
Choice B is incorrect because 14.8% reflects only the dividends paid without incorporating the return earned on dividend reinvestment.
- Choice A is correct. Payout to shareholders through share repurchases generates a possible capital gain for shareholders who sell shares back to the company. Thus, repurchases may be favored over dividends when capital gains tax rates are lower than those on dividend income.
Choice B is incorrect because repurchases provide flexibility for the company to pay less to shareholders when investment opportunities fluctuate higher. This statement implies that this flexibility is not necessary.
Choice C is incorrect. Assuming all else equal, a cash flow from repurchasing stock is equivalent in value to a cash flow from payment of dividends.
- Choice C is correct. A stock split lowers the per share price by increasing the number of shares outstanding to keep the price per share at a level low enough to attract interest from retail investors.
Choice A is incorrect because institutional investors are less likely to respond to the price per share changes created by stock splits.
Choice B is incorrect because this statement better describes a reverse stock split as an attempt to improve the stock price optics by increasing the price per share.
- Choice C is correct. The two-year holding period return on Bryce Limited shares is closest to 141%. The following table shows the share price and dividend details over the two-year holding period:

Bryce Limited Share Price and Dividend Details

Measure / Time	$t = -2$	$t = -1$	$t = 0$
Price	76	96	82
Dividend	—	5	5
Split factor	—	2	—
Split-adjusted price	38	48	—
Split-adjusted dividend	—	2.50	—

Computing the two-period return from the split-adjusted prices and dividends yields the following:

$$r = 140.87\% = \frac{[(82 + 5) \times (1 + \frac{2.50}{48}) - 38]}{38}.$$

Choice A is incorrect because 20.4% is only obtained if the computation fails to account for the split from a year earlier.

Choice B is incorrect because 127% is obtained if the computation fails to account for the dividend just received.

6. Choice B is correct. Calculating the present value of the expected price in five years and dividing this by Caddell's current price of GBP80.76 yields the following:

$$50.6\% = \frac{\frac{60}{(1.08)^5}}{80.76}.$$

Choice A is incorrect because $-25.7\% [= (60 - 80.76) \div 80.76]$ is simply the *price* return of Company C stock over a five-year holding period.

Choice C is incorrect because 74.3% simply divides 60 by 80.76, thus ignoring any time value of money considerations.

LEARNING MODULE

5

Introduction to Equity Valuation

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	contrast price and value for an equity security
☐	contrast the use of book value of equity, market capitalization, and enterprise value as indicators of intrinsic value
☐	describe major categories of equity valuation models and explain their advantages and disadvantages

INTRODUCTION: INTRODUCTION TO EQUITY VALUATION

1

Earlier Equities learning modules addressed the economic and voting rights of shareholders as well as equity issuance, trading, and sources of return. We now turn our attention to how investors and financial analysts assess the value of equity securities.

If equity markets are informationally efficient, then the current observed market price per share should reflect all available information and the aggregated views of market participants. Investors can attempt to evaluate the intrinsic value of a stock using different approaches. These approaches are meant to generate indicators or estimates of a stock's true intrinsic value. We refer to these estimates generated by investors as perceived value indicators of the stock. As investors may use different approaches and information, with varying levels of risk tolerance and investment time horizons, these perceived values may differ, sometimes greatly, across investors. Differences between investor perceived value and current market price are a key driver of issuer and investor decisions.

Equity valuation typically involves a financial analyst's assessment of whether today's market price for a share accurately reflects the present value of expected future cash flows over a specific period. These assessment methods rely on information about the company at a point in time. As new information about the company and its prospects becomes available, an analyst's assessment and even the true intrinsic share value may change.

In this learning module, we first contrast price and value, assessing market-based and accounting-based measures often used in the equity valuation process. We then introduce equity valuation approaches that assess company performance on an absolute or relative basis and their advantages and disadvantages. The details of valuation models applying these approaches are the subject of subsequent learning modules.

PRE-TEST: INTRODUCTION TO EQUITY VALUATION



1. Which one of the following statements is most correct about price and value of an equity security?
 - A. Equity market price must reflect its intrinsic value.
 - B. Equity market price is observed while its intrinsic value is unobserved.
 - C. Equity market price reflects investor agreement on the security's expected value.

Solution:

B is correct. Choice B is correct because a firm's equity market price is observed, while intrinsic value reflects the collective (and differing) expectations of market participants. Each market participant has their own unobserved value of an equity security.

Choice A is incorrect because this statement implies that a stock is always fairly valued, which does not allow for periods of overvaluation or undervaluation.

Choice C is incorrect because a stock's price reflects differing expectations of a stock's value among investors.

2. An analyst estimates the intrinsic value of a stock to be in the range of EUR17.85 to EUR21.45. Recent stock trading has been in a market price range from EUR21.15 to EUR24.35. Assuming the analyst's intrinsic value range for the stock is accurate, this stock is most likely:
 - A. overvalued.
 - B. undervalued.
 - C. fairly valued.

Solution:

A is correct. Choice A is correct because the range of intrinsic values is almost entirely below the range of recent trading prices. Thus, the price is likely greater than value.

Choice B is incorrect because the range of intrinsic values needs to be greater than the range of traded prices for the stock to likely be undervalued.

Choice C is incorrect because the overlap in the two ranges is very small from EUR21.15 to EUR21.45. Almost all of the price range lies above the intrinsic value range.

3. Which one of the following assets is most likely to be deducted in an analyst's adjustment of book value of equity?
 - A. Short-term investments
 - B. Inventories

C. Goodwill**Solution:**

C is correct. Choice C is correct because goodwill is an intangible asset and financial analysts typically exclude intangible assets when calculating the book value of equity.

Choice A is incorrect because short-term investments are part of a company's tangible assets and thus included in book value calculations.

Choice B is incorrect because inventories are part of a company's tangible assets and thus included in book value calculations.

4. The market value of equity for a company can be calculated as enterprise value:

- A. minus market value of debt, preferred stock, and cash and short-term investments.
- B. plus market value of debt and preferred stock minus cash and short-term investments.
- C. minus market value of debt and preferred stock plus cash and short-term investments.

Solution:

C is correct. Choice C is correct because enterprise value is calculated as the market value of equity plus the market value of debt and preferred stock minus cash and short-term investments. Therefore, the market value of equity is enterprise value minus the market value of debt and preferred stock plus cash and short-term investments.

Choice A is incorrect because this description states that cash and short-term investments are subtracted instead of added.

Choice B is incorrect because this description reverses the correct arithmetic operations on both debt plus preferred stock and cash and short-term investments.

5. Which of the following is most likely considered a weakness of present value models?

- A. Present value models cannot be used for companies that do not pay dividends.
- B. Changes in model assumptions and inputs can result in large changes in the computed intrinsic value of the security.
- C. The value of the security depends on the investor's holding period; thus, comparing valuations of different companies for different investors is difficult.

Solution:

B is correct. Choice B is correct because computed values can be sensitive to changes in inputs, such as required rate of return or growth rates.

Choice A is incorrect because other present value models, such as free cash flow to equity (FCFE) or residual income models, can be used to value companies that do not pay dividends.

Choice C is incorrect because the intrinsic value computed from a present value model should be independent of the investor's holding period.

6. An analyst is attempting to calculate the intrinsic value of a company and has gathered the following data for similar companies as well as the company being valued: EBITDA, total market value, and market value of cash and short-term investments, debt, and preferred shares. The analyst is most likely to use:

- A. a multiplier model.
- B. a present value model.
- C. an asset-based valuation model.

Solution:

A is correct. Choice A is correct because the analyst has gathered sufficient information to calculate the enterprise value to EBITDA multiple for comparable firms and to compare this multiple to that of the company of interest.

Choice B is incorrect because the analyst requires dividend, FCFE or FCFF (free cash flow to the firm), growth rate expectations, and a required return estimate to use a present value model.

Choice C is incorrect because the analyst requires market value estimates of the company's specific assets and liabilities individually.

2

EQUITY PRICE AND VALUE



contrast price and value for an equity security

Market Price, Intrinsic Value, and Investor Perceived Value

While sometimes used interchangeably, the price and value of a security can have very different meanings for financial analysts who work with issuers or investors. The observed *market price* of a firm's equity at which it may be freely bought and sold in the secondary market reflects the *collective* (and differing) expectations of market participants regarding the amount, timing, and uncertainty of the firm's future cash flows available to its owners.

The intrinsic value of a firm's share is generally an unobserved and unobservable quantity. **Intrinsic value (or fundamental value)** is based on key characteristics that affect a firm's future cash flows. Financial analysts often view a firm's intrinsic value as based on the "true" present value of a share's expected future cash flows using an appropriate discount rate.

While this approach to intrinsic value is common for firms whose equity is traded in primary or secondary markets, cases exist in which a firm's intrinsic value and future cash flows depend greatly on the transaction and the parties involved. For example, in a merger, a firm's intrinsic value relies heavily on how well the acquired firm may be integrated into the acquiring firm to generate additional value. For this learning module, we focus on the true present value of a share's discounted expected future cash flows when considering intrinsic value.

While an equity security has only one market price at any given time, each market participant has their own **perceived value** of the equity security. These individual assessments of how a security *should* be priced may vary widely and differ significantly

from the observed market price. These differences can influence both investor decisions, such as the purchase or sale of shares, as well as issuer decisions, including the issuance of new shares or repurchase of outstanding stock. These decisions are predicated on the assumption that as market participants digest new information and take actions to capitalize on differences between current market prices and perceived value, these forces will cause the share price to converge over time to its true intrinsic value.

Investor Approaches to Determining Perceived Value

Market participants use three major approaches to develop perceived values that they use as guides to a firm's intrinsic value:

1. Naive or story-based valuation
2. Relative valuation
3. Absolute or fundamental valuation

Each approach is described briefly below, and the latter two are addressed later in more detail in this and later learning modules.

The first approach is naive or story-based valuation. A naive valuation approach focuses on a narrative about a company's value that is not directly linked to an in-depth economic or financial evaluation. This approach is often used for companies known as **meme stocks** or **cult companies**. Meme stocks are company shares for which investors make investment decisions heavily influenced by social media posts, while cult companies are those for which investor decisions may be based upon a loyal or fanatical product following rather than expected company performance. Naive valuation may also feature in market "bubbles" to justify an excessively high or low valuation of a company, a sector, or even an entire market. While a naive evaluation may prove correct in the end, it is not based on any type of deep analysis of the share value.

Relative valuation uses observed financial or market information to develop indicators of value that are compared to a relevant peer group. A company or sector is then said to be over- or undervalued relative to the indicators of the chosen peer group. An example of a relative value indicator is the **price-to-earnings ratio (P/E)** or price-to-book ratio (P/B). The chosen indicators for a company and its peer group must be appropriate and calculated consistently to serve as a useful estimate of intrinsic value.

Absolute or fundamental valuation is an approach to estimating share prices using expected cash flows introduced in the prior learning module. For example, recall that we calculated the present value of expected dividends per share plus the future expected share price at the end of n periods with an annualized equity discount rate of r_e . Using actual cash distributions to shareholders is one approach, while others include assessing expected cash flows *available* to investors as well as relative comparisons to other equity issuers or equity rate of return-based estimates. In each case, we seek to establish an intrinsic value (IV_t) assessment at time t . We may show this using the example of a generalized discounted cash flow approach as in Equation 1:

$$IV_t = \sum_{i=1}^n \frac{E(CF_{t+i})}{(1+r_e)^i} + \frac{E(P_{t+n})}{(1+r_e)^n}, \quad (1)$$

where the expected share price in n periods is denoted $E(P_{t+n})$ and the expected periodic cash flow available to investors is $E(CF_{t+i})$. This assessment requires us to estimate the expected future price using today's information. As we will see in later lessons, an important value distinction involves whether the intrinsic value of the firm is measured from an **enterprise value** perspective, that is, from the point of view of both debtholders and shareholders, or the assessment focuses solely on equityholders based on share price.

A well-constructed equity valuation model seeks to estimate share value while eliminating the individual biases incorporated by market participants. The result of such an unbiased model is an estimate of intrinsic value (or fundamental value) that is distinct from the current market share price. Intrinsic value is based on analyzing key characteristics believed to be essential elements affecting a firm's future cash flows. An analyst estimating the intrinsic value of an equity security often uses more than one valuation approach to arrive at a value estimate.

The rise during the COVID-19 pandemic of meme stocks is an extreme example of the divergence between a share's market price and value, as described in the following case study.

CASE STUDY



The Meteoric Rise and Fall of GameStop Corporation's (GME) Share Price

Based in the United States, GameStop is the world's largest video game retailer with over 4,400 stores in North America, Europe, and Australia. The rise in online shopping caused the company's fortunes to decline as it closed stores and ceased dividend payments in 2019, a trend that accelerated during the initial months of the COVID-19 pandemic.

Institutional investors who believed that the GME share price continued to exceed its intrinsic value borrowed and sold shares short to gain from continued price declines. In mid-2020, the aggregate short position in GME shares was equivalent to over 25 times the average daily trading volume, making it one of the most widely shorted US stocks during that time.

Exhibit 1: GME Share Price (USD), July 2020 to December 2021



Source: Bloomberg.

Individual investors viewing social media sites such as the WallStreetBets Reddit forum were swayed by a narrative that GME would survive as a company and that share purchases would punish professional short sellers. To force short sellers to cover losses, retail investors aggressively purchased GME shares, driving up trading volumes and prices sharply in early 2021, as shown in Exhibit 1.

GameStop's shares jumped from a low of under USD1 in mid-2020 to nearly USD87 at its peak in early 2021, adjusting for the effects of a 4-for-1 stock split in July 2022. The company's management took advantage of its elevated share price to issue new shares *twice* in the first half of 2021. The company initially sold 3.5 million shares for gross proceeds of over USD550 million and used

almost half of the proceeds to retire all its long-term debt. It then raised an additional USD1.13 billion by selling 20 million new shares at a price of roughly USD56 in June of 2021 to invest in new growth initiatives.

Investor Perceived Value, Market Prices, and Market Dynamics

The example of GameStop highlights how the actions of both investors and issuers often interact. Differences in perceived value of a company's shares across investor types, such as institutional and individual investors, as well as company management can generate price dynamics as the market seeks to find a market-clearing share price that reflects these conflicting valuations.

For example, when considering a firm's intrinsic value from the perspective of shareholders, comparison of the market share price P_t to the estimated value IV_t leads a financial analyst to one of the following conclusions:

Share is currently undervalued: $P_t < IV_t$

Share is currently overvalued: $P_t > IV_t$

Share is fairly valued: $P_t \approx IV_t$

Investors will either increase (for an undervalued security) or decrease share exposure to earn a portfolio return greater than r_e under the assumption that the market price-to-value difference will narrow over the holding period. Issuers, on the other hand, often view undervaluation as a share repurchase opportunity and overvaluation as leading to additional share issuance.

In the GameStop example, institutional investors considered the market price for the company's shares to be overvalued, presumably based on a relative or absolute valuation analysis. This perception led the institutional investors to reduce GME exposure and even take a short position in GME shares. In contrast, individual investors swayed by social media posts considered GME to be undervalued. Their purchases pushed market prices higher, contrary to what institutional investors expected. This push led to short-run losses for investors who evaluated GameStop based on relative or absolute valuation analyses.

Finally, GameStop's management, seeing the sharp rise in their share price, acted to raise capital through share issuance. This action allowed management to raise far more capital than at a lower market price, providing GME with an unexpected source of funding to address its business challenges. This example demonstrates the complex market dynamics associated with differences in perceived values across investors and even the issuer itself.

Exhibit 2 provides a high-level summary of how actions of both investors and issuers depend on their perceived values versus the market price.

Exhibit 2: Issuer and Investor Signals When Price and Value Diverge

Market Situation	Investor Action	Issuer Action
Undervalued Value > Price	Purchase Shares Increase Share Exposure	Repurchase Outstanding Shares
Overvalued Price > Value	Sell Shares Decrease Share Exposure	Issue New Shares

While market prices are readily observable, true intrinsic value may be more difficult to ascertain with a high degree of certainty. For example, stocks of issuers with stable, predictable growth that commonly distribute regular cash dividends are more

straightforward to value than those experiencing rapid growth, cyclical, or substantial change. Valuation estimates are more likely to diverge across market participants for “harder to value” stocks. The level of conviction surrounding value estimates combined with their divergence from market prices influences whether market participants act based on the results of an equity valuation exercise.

KNOWLEDGE CHECK



An investor analyzes a stock and concludes that it is currently under-valued. The investor values the stock at EUR26.59 and buys shares at the current market price of EUR21.65. The stock is not expected to pay any cash dividends.

1. Calculate an investor's expected holding period rate of return from buying this stock under the assumption that shares are purchased at the current market price and the price-to-value difference goes to zero.

Solution:

$$\text{Expected rate of return} = \frac{\text{EUR}26.59 - \text{EUR}21.65}{\text{EUR}21.65} = 22.82\%.$$

2. Identify two reasons why the investor may not achieve a positive result from buying this stock during the holding period.

Solution:

1. The investor's valuation model incorrectly interpreted information about the stock or incorporated individual biases that caused the valuation to be higher than it should have been.
2. Negative information about the company's expected future performance emerged during the investor's holding period.

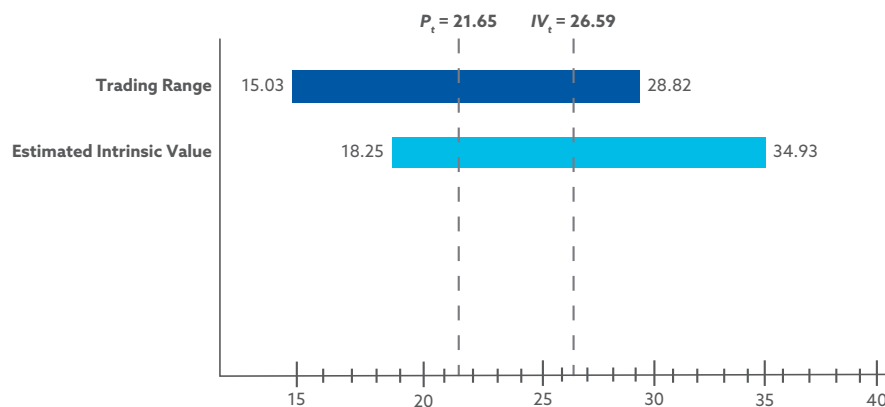
Market changes and additional information affect both share prices and valuation drivers, leading analysts to typically consider ranges rather than point estimates for both P_t and IV_t . In the case of share prices, analysts frequently reference a share's price range as defined by its recent high and low price. A value estimate range may apply to more than one valuation method and/or change the key model inputs used under a single valuation method based on **scenario analysis**. Scenario analysis combines a base case valuation with alternative outcomes, allowing the use of more favorable (bullish) or adverse (bearish) forecasts in the valuation process:

$$IV_t = \sum_{i=1}^n \text{Probability}(i) \times P_{t,i}, \quad (2)$$

where $P_{t,i}$ is the expected share price at time t under scenario i .

There are various approaches available for estimating a security's intrinsic value, and within each approach there are often multiple different ways to estimate value. It is important for an analyst to be able to compare these various estimates against one another and against recent experience with the security's market price. This synthesis-of-valuation approach seeks to identify ranges for intrinsic value produced by a variety of approaches and compare that to the range of the stock's price over the recent past.

Exhibit 3 is a side-by-side comparison of a stock's actual trading range and one or more estimated value ranges referred to as a **floating bar valuation chart**, sometimes also called a football field valuation chart.

Exhibit 3: Estimated Value versus Market Value**KNOWLEDGE CHECK**

An investor analyzes a non-dividend-paying stock and buys shares at the current trading price of EUR21.65. The investor estimates intrinsic value associated with three potential scenarios at the end of an investment horizon:

- Base case: Stock price rises to 26.59
- Best case: Stock price rises to 34.93
- Worst case: Stock price falls to 18.25

1. If the outcomes are equally probable, calculate the stock's expected price and return.

Solution:

The analyst's expected price is EUR26.59, as calculated below using Equation 2.

$$IV_t = (1 \div 3) \times 26.59 + (1 \div 3) \times 34.93 + (1 \div 3) \times 18.25$$

$$IV_t = 26.59$$

$$R_e = \frac{\text{EUR}26.59 - \text{EUR}21.65}{\text{EUR}21.65} = 22.82\%.$$

2. Suppose that the analyst changes the probabilities on the scenarios to the following:

- Probability of base case = 50%
- Probability of best case = 20%
- Probability of worst case = 30%

Calculate the stock's updated expected price and return given the change in probabilities.

Solution:

The stock's expected price and return with the updated probabilities is EUR25.76 and 18.97%, as shown below:

$$IV_t = 25.76[(0.5 \times 26.59) + (0.2 \times 34.93) + (0.3 \times 18.25)]$$

$$r_e = \frac{\text{EUR}25.76 - \text{EUR}21.65}{\text{EUR}21.65} = 18.97\%.$$

The prior knowledge check illustrates that an analyst often assigns probabilities to different equity valuation scenarios based upon judgment. The building blocks used to estimate firm value as well as the most common valuation approaches are the subject of the following lessons.

QUESTION SET 1



1. Which one of the following statements provides the most correct characterization of meme stocks?
 - A. Meme stocks typically exhibit low trading volumes and high volatility.
 - B. Meme stocks can show extreme divergence between price and value.
 - C. Meme stocks typically are undervalued.

Solution:

B is correct. Choice B is correct because high trading volumes from strong individual investor buying behavior cause price to temporarily diverge from intrinsic value.

Choice A is incorrect because meme stocks typically exhibit high, not low, trading volumes.

Choice C is incorrect because prices of meme stocks typically increase above intrinsic value and thus are likely to be overvalued.

2. A corporate issuer's chief financial officer values its shares between USD50 and USD60; these shares have recently traded at prices between USD44 and USD51. Which one of the following most correctly describes an appropriate action by this corporate issuer?
 - A. The issuer should issue new shares of company stock.
 - B. The issuer should repurchase shares of company stock.
 - C. The issuer should neither issue shares nor repurchase shares of company stock.

Solution:

B is correct. Choice B is correct because the company's chief financial officer believes the stock to be undervalued. Repurchasing undervalued shares is an appropriate action for a corporate issuer.

Choice A is incorrect because an issuer is more likely to issue new shares when they are overvalued, not undervalued.

Choice C is incorrect because this statement more accurately describes the appropriate action if the company's shares are fairly valued.

3. Which one of the following is most likely to be true for a stock that is easier to value compared to a stock that is harder to value?
 - A. Less divergence between price and intrinsic value
 - B. Higher potential future rates of return

C. More cyclical in profits and dividends

Solution:

A is correct. Choice A is correct because valuation estimates are more likely to diverge across market participants for harder-to-value stocks.

Choice B is incorrect because the greater divergence for harder-to-value stocks increases the potential for high returns, although this divergence also increases the potential for negative returns on these stocks.

Choice C is incorrect because stocks of companies with more cyclical in profits and dividends may be harder to value.

OBSERVABLE MEASURES OF COMPANY VALUE

3



contrast the use of book value of equity, market capitalization, and enterprise value as indicators of intrinsic value

In establishing a firm's estimated value using an equity valuation approach, financial analysts often employ observable measures based upon financial statements or market data to gain further insight into the intrinsic value of an equity security, namely the book value of equity, market capitalization, and enterprise value.

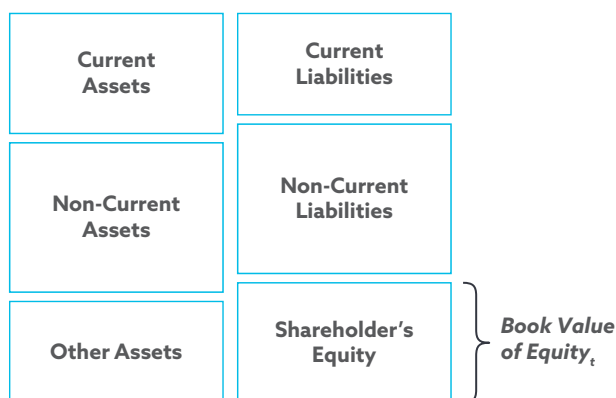
Book Value of Equity

The **book value of equity** (BVE_t) represents the difference between the carrying value of assets and liabilities in a firm's audited financial statements:

$$BVE_t = \text{Book value (Assets)}_t - \text{Book value (Liabilities)}_t \quad (3)$$

Exhibit 4: Book Value of Equity

Balance Sheet



Book value is often evaluated on a per share basis $(BVE/\text{Share})_t$ as a measure of the value of a firm's net assets per share of ownership capital:

$$\left(\frac{\text{BVE}}{\text{Share}} \right)_t = \left(\frac{\text{Shareholder's equity}}{\text{Shares outstanding}} \right)_t \quad (4)$$

Shareholder's equity includes closely held shares but excludes minority interest as it represents a non-controlling stake with limited control over ownership decisions. As illustrated in Exhibit 4, the measurement of a firm's net assets is based on balance sheet items. As discussed in earlier Financial Statement Analysis learning modules, balance sheet items may reflect historical cost or market values. Furthermore, depreciation and amortization methods affect the reported balance sheet amounts of fixed assets.

In the case of tangible assets, differences in historical cost and depreciation methods can distort book value comparisons across firms. For example, newer fixed assets are less depreciated and therefore reflected in financial statements at a higher carrying value than older assets.

Intangible assets—or non-monetary, non-physical items of value owned or controlled by a company and that provide economic benefits—may not be consistently recorded in accounting statements. For example, the treatment of intellectual property, such as patents, licenses, or trademarks, differs depending upon whether they are internally generated or purchased from a third party. If identifiable, intangible assets generated internally are usually expensed in a single period. In contrast, intangible assets that are purchased are amortized over multiple periods if their useful life is finite or subject to impairment tests if they are of use for an indefinite period. Balance sheet assets also include any portion of the purchase price of companies acquired that exceeds the fair value of identifiable assets and liabilities, referred to as **goodwill**. For these reasons, the assets of acquisitive companies may overstate a firm's underlying capacity to generate future cash flow versus firms with organic growth. As a result, financial analysts typically exclude intangible assets when calculating book value.

Book value has greater relevance as a value measure for firms with significant tangible assets and similar asset structures and is less useful for asset-light companies or early-stage firms where the balance sheet may not reflect future earnings potential. A firm's book value of equity reflects the past historical operating and financing decisions of its management as well as abrupt changes, such as the write-down of assets highlighted in the following case study.

ØRSTED A/S ASSET WRITE-DOWNS

Ørsted A/S, a Danish company, operates in the renewable energy sector and is one of the world's largest offshore wind farm developers. One of Ørsted's largest projects was a 2.2 GW (gigawatt) offshore wind farm to be located off the coast of New Jersey. While the project's initial phase received US government approval for construction in July 2023, the company canceled the project in October 2023 due to supplier delays, tax credit limitations, and unforeseen cost increases.

The project cancellation resulted in Ørsted reporting over DKK19 billion in asset write-downs in its 2023 income statement and contributed to a decline in the company's book value of equity from DKK95.5 billion at year-end 2022 to DKK77.8 billion at year-end 2023. Ørsted's stock price initially declined almost 25% on 30 August as the company first announced the pending write-downs of wind farm assets, and its market capitalization fell over 40% between the second and third quarters of 2023.

The book value of liabilities typically includes long-term debt and other liabilities carried at face value but may also underestimate any *contingent* liabilities arising from sources such as product liability, environmental, or legal risks not fully captured on a firm's balance sheet in accounting terms.

Aside from the difference between assets and liabilities, the book value of equity may be separately calculated as the sum of shareholder's equity line items in a company's audited financial statements shown as follows:

$$\begin{aligned} \text{Shareholder's equity} &= \text{Shares capital} + \text{Additional paid-in capital} \\ &+ \text{Retained earnings} - \text{Treasury stock}. \end{aligned} \quad (5)$$

When an issuer has a negative book value of equity, it may signal a greater risk of financial distress or bankruptcy given a company's hypothetical inability to meet outstanding obligations with existing assets based upon accounting measures. That said, firms experiencing rapid growth or asset impairment may continue to add value for investors despite a negative book value of equity. In other cases, a company's distribution policy choices, such as share repurchases, combined with the distortions associated with measurement of intangible assets may contribute to sustained negative equity for well-performing companies as described in the following case study.

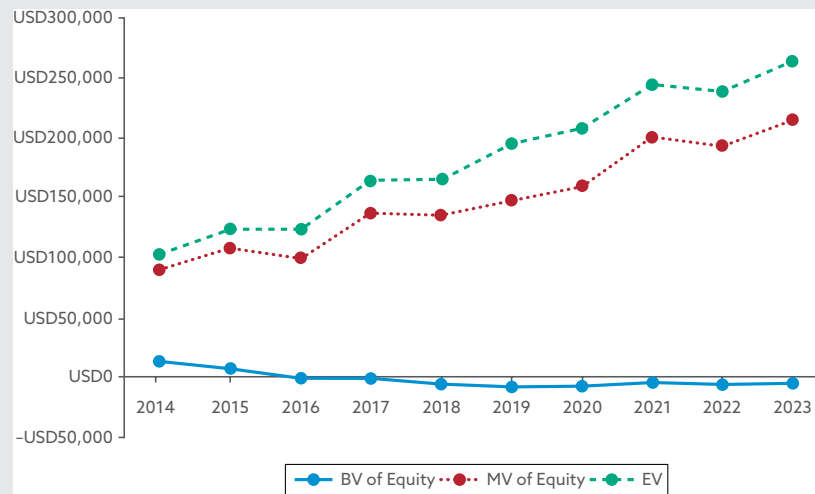
MCDONALD'S CORPORATION AND NEGATIVE BOOK VALUE OF EQUITY

As of year-end 2013, McDonald's Corporation (MCD) reported an all-time high book value of equity of over USD16 billion. However, by the end of 2016, the company's book value of equity had fallen below zero. MCD's book value of equity continued its decline until year-end 2019, reaching –USD8.2 billion and has continued to operate with negative book value of equity.

While some companies experience negative equity book value due to net losses, MCD has enjoyed consistent profitability as its book value of equity fell from an all-time high to negative values. In fact, the company reported *cumulative* net income of over USD63 billion in net income between 2013 and 2023. However, the company paid USD39 billion in dividends and spent USD42 billion to repurchase shares over the same period. These cash distributions well above MCD's net income drove its book value of equity downward.

During this same period, MCD restructured its business by further increasing its focus on franchise operations and away from company-owned stores. The market value of MCD's equity rose from USD96 billion at year-end 2013 to USD214 billion at year-end 2023, more than doubling despite the equity book value decline. Exhibit 5 illustrates the evolution of MCD's book and market values of equity as well as the company's enterprise value (EV).

Exhibit 5: MCD Book Value of Equity, Market Capitalization, and Enterprise Value (in millions of USD), 2014–2023



Source: Bloomberg.

The large positive difference in MCD's market capitalization compared to its book value of equity reflects the value of its highly regarded global brand, expected future franchise fees and royalties, and its real estate value. In addition to franchising well over 90% of its stores to operators, MCD typically owns the real estate utilized by these franchisees. The company's real estate has an estimated value of over USD40 billion and generates a significant portion of MCD's revenues.

Market Capitalization

In contrast to historical balance sheet entries that comprise the book value of equity, a share's market price reflects forward-looking investor expectations about future company cash flows generated by positive net present value investment opportunities. The product of the price per share and number of shares outstanding is a company's market capitalization.

$$(\text{Market capitalization})_t = P_t \times (\text{Shares outstanding})_t \quad (6)$$

Unlike the *book* value of equity, changes in which are only observable when periodic financial statements are issued, this measure of the *market* value of a firm's equity changes daily as share prices fluctuate. The number of shares outstanding may change within a financial reporting period due to issuance or buybacks and is available to market participants only once financial statements or other filings are released. Note that total shares outstanding include both those publicly held and available to trade as well as shares held privately or by management, such as restricted stock.

KNOWLEDGE CHECK

Eckwald AG's (ECKW) finance department analysts are discussing the effect of a recent stock repurchase on the company's book value and market value of equity. Eckwald repurchased four million shares at an average price of EUR125 on the last day of its most recent financial reporting period.



1. Describe the effect of ECKW's stock repurchase on its book value of equity as well as on the market value of equity.

Solution:

The repurchase reduces ECKW's equity in terms of both book value and market value. The book value of Eckwald's equity declines by EUR500 million (= 4 million × EUR125) when the repurchase occurs, as this reflects the amount of capital distributed to shareholders.

If the company repurchases shares at market value, this action should technically not affect the market price. However, the actual change in ECKW's market value will depend upon the market's interpretation of ECKW's *motivation* to repurchase shares. On the one hand, it may signal management's confidence and belief that shares are undervalued, which may lead to a share price increase. Alternatively, the buyback may indicate a lack of investment opportunities, which may have a negative effect on the share price.

2. Explain the effect of ECKW's stock repurchase on its market value and book value of equity as observed by an *external* equity analyst.

Solution:

Eckwald's share repurchase on the final day of the reporting period is not transparent to investors as would be the case with dividends. The reduction in shares outstanding is only disclosed once Eckwald releases this information publicly. Thus, reported market capitalization will be based upon the prior period shares outstanding until that date. Unlike the decline in share prices on the ex-dividend date, share repurchases generally occur at market prices and do not coincide with observable share price changes. Once financial statements are released, an analyst should be able to isolate the effect of stock repurchases made during the prior financial reporting period. The time lag associated with the release of financial reports delays the ability of public analysts to observe the effects of ECKW's repurchase activity.

The effect of share repurchases on market capitalization is more nuanced than the one-time ex-dividend day price change outlined in the prior learning module. As shown in the above exercise, share repurchases may occur on an opportunistic basis and are only disclosed to market participants after the fact. Changes in share price are not typically directly attributable to the cash outflow of the repurchase but rather by subsequent changes in market perception and financial statement disclosures.

Beyond the timing differences and historical versus forward-looking nature of the book value of equity and market capitalization, investors often compare the two to draw initial conclusions about price versus value. Exhibit 6 and Exhibit 7 demonstrate the conceptual relationship between market capitalization and the book value of equity for a given proportion of liabilities and equity to assets.

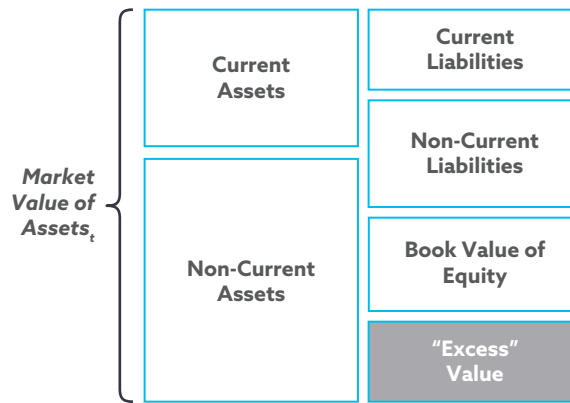
Exhibit 6: Market Value of Assets Exceeds Book Value

Exhibit 6 shows a scenario in which investors attribute a greater hypothetical market value to the firm's assets than their carrying value in its most recent financial statements, leading to a market capitalization that exceeds the book value of equity. Ignoring the potential book value distortions cited earlier, a common investor interpretation of this premium over net balance sheet assets is that the firm creates value for its investors.

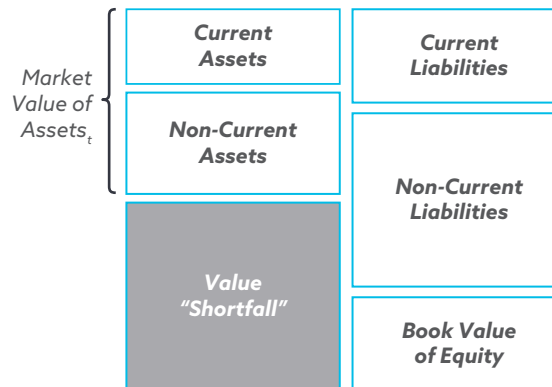
Exhibit 7: Book Value of Assets Exceeds Market Value of Equity

Exhibit 7 shows a case in which the market value of firm assets is below the book value, leading to a shortfall that partially or fully offsets the book value of equity. This case implies that if sold individually and separately from the business at book value, the firm's assets would command a materially higher price than if sold together as a going concern. Shares trading at a discount to their book value may signal an undervalued share price. However, interpreting the relationship between market and book value requires an analyst to look beyond the actual numbers. These shares may be fairly priced, or even overvalued, if the firm is uncompetitive, owns substandard or obsolete assets, or has ineffective management. If a firm's performance is far inferior to its peers, the *market* value of liabilities may be considerably below the recorded book value due to a greater likelihood of financial distress. Despite a positive share price, if the issuer cannot generate sufficient cash flow to meet creditor obligations as a going concern, it may eventually result in the liquidation of assets to settle creditor claims and a total loss to shareholders.

The **price-to-book ratio (P/B)** reflects the relationship between the market value and book value of equity per share:

$$\left(\frac{P}{B}\right)_t = \left(\frac{\frac{P}{\text{BVE}}}{\text{Share}}\right)_t \quad (7)$$

This and other ratios used to compare measures of value are also known as **multiples**, and these will be covered extensively in a later learning module. The lower the multiple, the more attractive a firm's price appears to be based on a specific chosen metric. Like the expected convergence of price and value described earlier, this scenario assumes that a measure of market to book value (shown here using the P/B) will normalize or rise in line with comparable companies in the future. On the other hand, the higher the price-to-book ratio among firms with similar features, the more favorably investors value a company's future cash flow generation potential versus existing net assets. P/B is a common metric for firms within an industry with a similar reliance on debt versus equity financing and none of the significant book value distortions mentioned earlier.

CASE STUDY



Logistics Industry Competitors: Comparison of Price-to-Book Ratios

Following a series of bilateral agreements designed to strengthen trade and investment between the United States, India, Australia, and Japan, an analyst seeking to capitalize on expected growth compares logistics firms based upon multiples. Advani Cargo Limited (ACL) based in India shows INR800 million of assets and INR200 million of shareholders' equity on its balance sheet. Bairnsworth (BRW) is an Australian-based industry peer of ACL with a balance sheet including AUD20 million of assets and AUD5 million of shareholders' equity. Advani Cargo has 10 million shares outstanding, and its shares currently trade at INR95.00. Bairnsworth has 400,000 shares outstanding, and its stock is priced at AUD50 per share. The two companies have similar sales-to-asset ratios and generate similar profit margins.

1. Based solely on a comparison of P/Bs, which one of these two companies would an analyst likely view as being less expensive?

Solution:

Use Equation 6 and Equation 7 to solve for the respective P/Bs:

Advani Cargo:

Market capitalization = INR950 million(= INR95 × 10 million shares)

$P/B = 4.75$ (= INR95 million ÷ INR20 million)

Bairnsworth:

Market capitalization = AUD20 million(= AUD50 × 400,000 shares)

$P/B = 4.00$ (= AUD20 million ÷ AUD5 million)

BRW has the lower P/B, allowing the analyst to conclude the stock is cheaper based on this multiple given that the two companies are similar with respect to leverage, profit margin, and asset intensity.

2. The analyst finds a third industry competitor, Chidori Express Corporation (CEC), based in Japan. CEC shows JPY1,500 million in assets on its balance sheet with a book value of equity equal to JPY100 million. CEC's market

capitalization is JPY900 million. Evaluate whether the analyst can adequately compare CEC's P/B to those of ACL and BRW.

Solution:

No, the P/Bs of Advani Cargo and Bairnsworth are not comparable to the Chidori Express P/B due to a significant difference in financial leverage. Chidori Express has assets equal to 15 times its book value of equity, far above the leverage of Advani Cargo and Bairnsworth, with assets only 4 times their book values of equity.

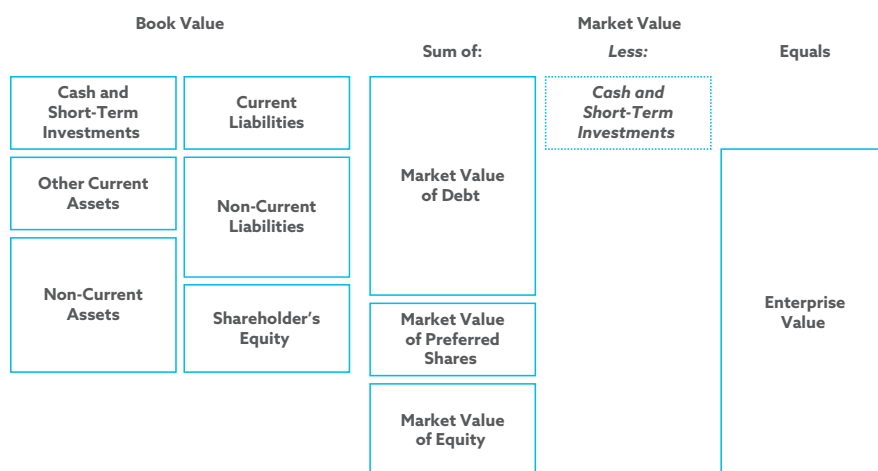
Attribute	Advani Cargo Limited (ACL) in INR	Bairnsworth (BRW) in AUD	Chidori Express Corporation (CEC) in JPY
Assets	800 million	20 million	1,500 million
Liabilities	600 million	15 million	1,400 million
Shareholders' Equity	200 million	5 million	100 million
Shares Outstanding	10 million	400,000	Not Provided
Share Price	INR95	AUD50	Not Provided
Market Capitalization	950 million	20 million	900 million
P/B	4.75	4.00	Not Directly Comparable

The use of multiples as a basis for comparing a company's value to other companies considered to have similar characteristics, as shown in the above case study, is known as the **method of comparables approach** to valuation. As shown in the prior knowledge check, P/Bs may be less desirable in such an approach because of differences in leverage across seemingly similar companies.

While company actions such as growing earnings or increasing leverage via greater borrowing or share buybacks directly affect a firm's book value, they indirectly affect the market value of its equity. To reduce the impact of balance sheet changes or differences among firms on valuation estimates, financial analysts often turn to a broader measure of company value.

Enterprise Value

An alternative to estimating the market value of equity is to consider the value of the *entire* firm (or enterprise) from the perspective of a potential buyer. In the case of a buyout, an acquiring company not only assumes the target company's debt but also receives its cash. Since a buyer may use cash and short-term investments easily convertible into cash to pay the acquisition price or reduce debt, firm value considers only **net debt**, or debt minus cash and cash-like investments. A firm's enterprise value (EV) is comprised of the sum of the market value of a firm's equity, preferred shares, and net debt, as illustrated in Exhibit 8.

Exhibit 8: Enterprise Value

Enterprise value(EV) = Market value of debt + Market value of equity

+Market value of preferred shares – Cash and short-term investments. (8)

As shown in Equation 8, the broader EV measure of company value incorporates the claims of both debtholders and preferred shareholders as well as common shareholders. As a result, enterprise value is often a more useful measure than share price if comparing companies with varying degrees of reliance on debt financing. Note that EV is measured in the company's reporting currency. As a result, it is not possible to directly compare the EV of companies reporting in different currencies. To make direct comparisons, all company EVs must be translated into a single common currency.

CASE STUDY**Logistics Industry Competitors: Enterprise Value**

An earlier case study provided information to assess P/Bs of three companies in the logistics industry. Suppose we add the following information regarding the debt and cash balances of the three companies and assume that none of the three have preferred debt outstanding:

Firm	Market value of debt	Cash & short-term investments
Advani Cargo Limited (ACL)	INR500 million	INR100 million
Bairnsworth (BRW)	AUD12 million	AUD3 million
Chidori Express Corporation (CEC)	JPY1,100 million	JPY200 million

1. Calculate the enterprise values of the three companies.

Solution:

Use Equation 8 to solve for the respective enterprise values, noting that none of the three firms has preferred shares:

$$EV = \text{Market value of debt} + \text{Market value of equity}$$

– Cash and short-term investments

EV(ACL):

INR1,350 million(= INR500 million + INR950 million – INR100 million)

EV(BRW):

AUD29 million(= AUD12 million + AUD20 million – AUD3 million)

EV(CEC):

JPY1,800 million(= JPY1,100 million + JPY900 million – JPY200 million)

As enterprise value (EV) incorporates additional forms of capital, such as debt, it is an appropriate comparison to cash flow or operating profit measures available to *all* capital providers. These include funds available for reinvestment, distribution to shareholders via dividends or share buybacks, or debt repayment. When used as the numerator in a multiple, EV may be interpreted as a firm's value per unit of sales or operating profit among industry peers, as shown below:

$$\text{EV multiple} = \frac{\text{Enterprise value}}{\text{Revenue or Operating profit measure}} \quad (9)$$

In contrast to absolute measures of intrinsic value, such as enterprise value, multiples are not a currency-based measure and are therefore directly comparable regardless of the currency of measurement.

CASE STUDY



Logistics Industry Competitors: Enterprise Value Multiple

In an earlier case study, the enterprise values were calculated for three logistics industry competitors. Suppose we incorporate the following information about revenues and operating profit as measured by EBITDA:

Firm	Revenue	EBITDA
Advani Cargo	INR1,200 million	INR150 million
Bairnsworth	AUD30 million	AUD4 million
Chidori Express	JPY2,250 million	JPY300 million

An analyst uses these data in addition to the enterprise values provided earlier to calculate enterprise value multiples for comparative purposes.

1. Which one of the following stocks appears most likely undervalued relative to the other competitors based upon enterprise value multiples?

- A. Advani Cargo
- B. Bairnsworth
- C. Chidori Express

Solution:

C is correct. Choice C is correct because Chidori Express exhibits the lowest enterprise value multiples (EV/Revenue and EV/EBITDA) as shown below using Equation 9:

ACL:

EV/Revenue: 1.125($= 1,350 \div 1,200$)

EV/EBITDA: 9.00($= 1,350 \div 150$)

BRW:

EV/Revenue: 0.967($= 29 \div 30$)

EV/EBITDA: 7.25($= 29 \div 4$)

CEC:

EV/Revenue: 0.800($= 1,800 \div 2,250$)

EV/EBITDA: 6.00($= 1,800 \div 300$)

As a result, CEC appears to be undervalued based on both enterprise value-based ratios.

Choice A is incorrect because ACL's EV multiples are the highest of the three competitors.

Choice B is incorrect because BRW's EV multiples are higher than those of CEC.

As in the case of the price-to-book ratio, a higher EV-based multiple indicates a greater market value assigned per unit of cash flow or operating profit generated by one firm versus another or the industry average.

In practice, it may be difficult to precisely measure EV if the market value of a firm's debt is not observable. EV measures therefore often substitute the par or book value for market value. Expected debt values that differ significantly from par are estimated from current quotations for bonds with similar maturity, sector, and credit characteristics.

QUESTION SET 2



- Two companies, Arthurdale Corp. and Chapmanville Inc., exhibit similar book values of assets, liabilities, and capital structure. The primary difference between the two is that Arthurdale has grown significantly by acquiring other companies while Chapmanville has grown organically without the use of acquisitions. The book value of equity is more likely overstated for:

- A. Arthurdale.
- B. Chapmanville.
- C. neither Arthurdale nor Chapmanville.

Solution:

A is correct. Choice A is correct because the assets of acquisitive companies include goodwill, which is more likely to be deducted to estimate book value of equity.

Choice B is incorrect because Chapmanville is unlikely to have goodwill that must be deducted in estimating book value of equity.

Choice C is incorrect because the differences in intangible assets between the two companies imply that book value of equity may be different after adjustments.

2. Two comparable industry peers, Macintosh Corporation and Nesbitt Industries, are otherwise similar but have different P/B multiples. Macintosh trades at a P/B multiple of 4.0, while Nesbitt trades at a P/B multiple of 5.0. Which one of the following statements is most correct?
- A. Investors should buy Macintosh stock because this company has more expected growth potential than Nesbitt.
 - B. Investors should buy Macintosh stock because its price is more attractive relative to book value versus that of Nesbitt's stock.
 - C. Investors should buy Macintosh stock because it should provide higher returns than Nesbitt due to higher leverage.

Solution:

B is correct. Choice B is the correct response because Macintosh and Nesbitt are similar companies operating in the same industry. As a result, we are more likely to be able to interpret Macintosh as more attractively priced than Nesbitt based upon the P/B.

Choice A is incorrect because Nesbitt's higher P/B might reflect that investors believe that it has more growth potential, not Macintosh.

Choice C is incorrect because higher leverage implies a higher P/B. If investors are making a buy recommendation because of higher leverage, then they would prefer Nesbitt stock.

3. Two companies, Rodgers Inc. and Stafford Co., are of similar size in terms of the book and market values of their assets, but Stafford has far more debt than Rodgers. Which one of the following statements is most correct?
- A. Rodgers' market capitalization is likely larger than that of Stafford.
 - B. Rodgers' enterprise value is likely larger than that of Stafford.
 - C. Stafford's market capitalization is likely larger than that of Rodgers.

Solution:

A is correct. Choice A is correct because market capitalization is the market value of a company's equity. Because both companies have a similar market value of assets but Rodgers has less debt, Rodgers should have a higher market value of equity.

Choice B is incorrect because the statement that the two companies have similar market value of assets implies that their enterprise values are similar.

Choice C is incorrect because Stafford's market capitalization is likely lower than that of Rodgers because a greater proportion of Stafford's balance sheet consists of debt.

4

EQUITY VALUATION MODELS



describe major categories of equity valuation models and explain their advantages and disadvantages

Discounted cash flow and multiple-based comparisons are among the approaches underlying equity valuation models commonly used by industry practitioners. While all such models aim to establish an unbiased estimate of intrinsic value, the choice of

model(s) depends upon whether expected returns are based on capital appreciation versus capital income, the applicability for a firm or industry, and the availability and reliability of inputs. While absolute valuation approaches seek to quantify actual cash flows or asset prices that drive intrinsic share value, relative value approaches rely on price- or enterprise value-based ratio comparisons.

Absolute Valuation Approaches

Present Value Models

Present value models estimate the intrinsic value of expected future capital income and capital appreciation as of today using an appropriate discount rate. While universally applied across investments, in contrast to fixed-income models with a finite maturity, equity cash flows continue indefinitely or until a firm is restructured, acquired, or liquidated. Analysts applying present value models to equity therefore usually project a series of *forecasted* cash flows and estimate a so-called **terminal value** that seeks to capture the value of future cash flows beyond the forecast period into the indefinite future.

Equation 10 is a generalized form of the discounted cash flow valuation model used for equity investments:

$$IV_t = \sum_{i=t+1}^n \frac{E(CF_i)}{(1 + \text{Required rate of return})^i} + \frac{E(P_{t+n})}{(1 + \text{Required rate of return})^n}, \quad (10)$$

where $E(CF_i)$ represents the estimated cash flow per share measure in period i . Key features that distinguish present value models include the following:

- Cash flow metric used to establish value, $E(CF)$
- Required rate of return
- Length of the forecast period, n
- Terminal value, $E(P_{t+n})$

Cash flow measures used in the numerator of the first term in Equation 10 may be limited to *actual* distributions or include funds *available* that may be reinvested in the company to fund operations. The former case includes dividends per share using a so-called **dividend discount model**. For firms not expected to pay consistent cash dividends in the future, **free cash flow** is a more appropriate measure. Free cash flow (FCF) is the excess of operating cash flow over capital expenditures, with two primary forms:

$$\begin{aligned} \text{Free cash flow to the firm (FCFF)} &= \text{Cash flow from operations} + \text{Interest expense} \\ &\times (1 - \text{Tax rate}) - \text{Capital expenditures}. \end{aligned} \quad (11)$$

$$\begin{aligned} \text{Free cash flow to equity (FCFE)} \\ &= \text{Cash flow from operations} - \text{Capital expenditures} \\ &+ \text{Net debt issuance}, \end{aligned} \quad (12)$$

where $\text{Net debt issuance} = \text{Additional borrowing} - \text{Debt repayment}$.

Free cash flow to the firm (FCFF) is a measure of cash flow available to *both* debtholders and equityholders, while **free cash flow to equity (FCFE)** is the cash flow available to the company's common stockholders. Both free cash flow measures exclude a firm's necessary capital expenditures, and the two metrics differ based on net debt repayment and the after-tax effect of interest expense.

The choice of cash flow metric and its respective discount rate varies based upon the characteristics and expected policies of the firm. For example, a dividend cash flow measure is appropriate for mature firms with stable, consistent cash distributions, while FCFE is appropriate for non-dividend-paying stocks. Both dividends and FCFE measure shareholder returns and are discounted at the required equity rate of return. FCFE, a broader measure of cash flows available to both debtholders and shareholders, is applicable when a firm has a high reliance on debt and either a negative FCFE or a changing capital structure over the forecast period. Here the appropriate discount rate is the weighted average cost of capital combining the cost of debt and equity.

KNOWLEDGE CHECK



An analyst is following three companies: Gisborne, Hastings, and Picton. Each of these three companies generates positive free cash flow to pay to shareholders. Each company has attributes that may require different cash flow metrics used to value each stock. These attributes are summarized as follows:

Company	Dividend Pay-out Ratio	Stock Repurchases	Funding Source
Gisborne	25% to 40%	Significant	Cash flow from operations, new debt issuance
Hastings	5% to 10% (very low)	Significant	Cash flow from operations, stable debt ratio
Picton	70% to 80% (substantial)	None	Slow growth of revenues and net income

1. Which one of the three companies listed above is likely to be most appropriately valued using FCFF?

- A. Gisborne
- B. Hastings
- C. Picton

Solution:

A is correct. Choice A is correct for the following reasons: First, Gisborne distributes cash flow using both dividends and repurchases, so a free cash flow metric is better suited as compared to a dividend model. Second, Gisborne primarily funds share buybacks with debt. The potential for changing leverage makes FCFF a more appropriate metric than FCFE for valuing Gisborne.

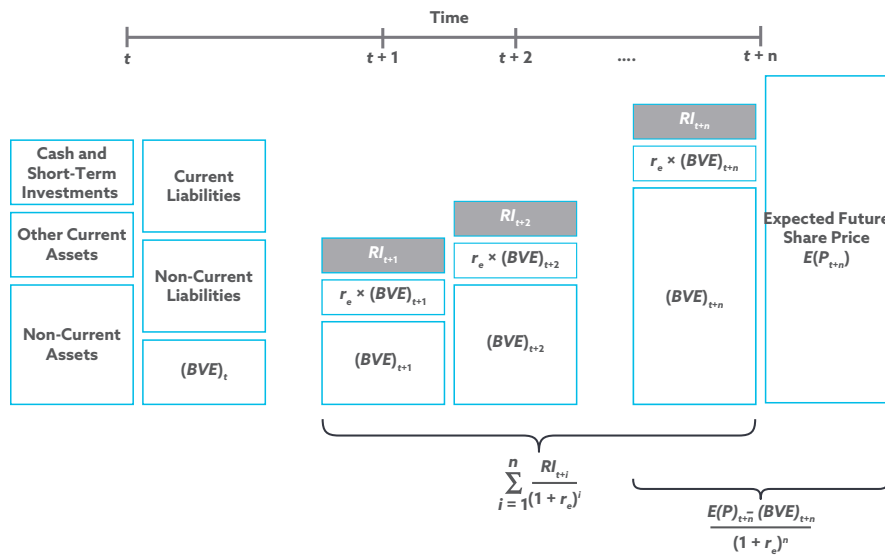
Choice B is incorrect because Hastings may be more likely to be valued using FCFE. As was the case for Gisborne, a free cash flow approach is more useful for Hastings because it distributes a relatively small portion of its cash flow as dividends. Hastings' stable debt ratio suggests that FCFE is an appropriate cash flow metric for valuation.

Choice C is incorrect because Picton's high dividend payout and stable net income make this company a good candidate for a dividend-based valuation model.

The length of the forecasted time horizon varies based upon company circumstances. The period should extend as long as an analyst has a sufficient degree of confidence and available inputs to quantify a narrative associated with value changes. For example, a two-stage model may involve a phase of above-trend growth followed by a terminal value based on slower constant growth or an observed multiple relationship.

A **residual income model** in contrast seeks to estimate the periodic return to shareholders above the required equity rate of return r_e , known as residual income. As a measure of *excess* return, residual income is smaller than earnings, which measures *all* funds available for reinvestment or distribution via dividends or buybacks, and FCFE, which also includes net debt issuance. Exhibit 9 provides an illustration of the estimation of intrinsic value using a residual income approach.

Exhibit 9: Residual Income Approach



The intrinsic value based on the residual income approach is shown in Equation 13:

$$IV_t = BVE_t + \sum_{i=1}^n \frac{RI_{t+i}}{(1+r_e)^i} + \frac{E(P)_{t+n} - (BVE)_{t+n}}{(1+r_e)^n}. \quad (13)$$

The residual income approach estimates a company's value by calculating the present value of expected future residual incomes, which are the profits remaining after deducting the equity cost of capital from net income. Given their reliance on accounting measures, such as book value, residual income models are most appropriate for companies that have no book value distortions, do not pay consistent dividends, or do not even have negative free cash flow over the forecast horizon. Use of book value-based excess returns reduces the emphasis on terminal value versus other present value approaches. Residual income is a measure of incremental return over the required rate. If a firm is expected to generate residual income over time, we would expect the price-to-book ratio to be greater than one.

Exhibit 10 summarizes the different cash flow measures, required rates of return, and applications for major discounted cash flow valuation approaches.

Exhibit 10: Present Value Models and Their Uses

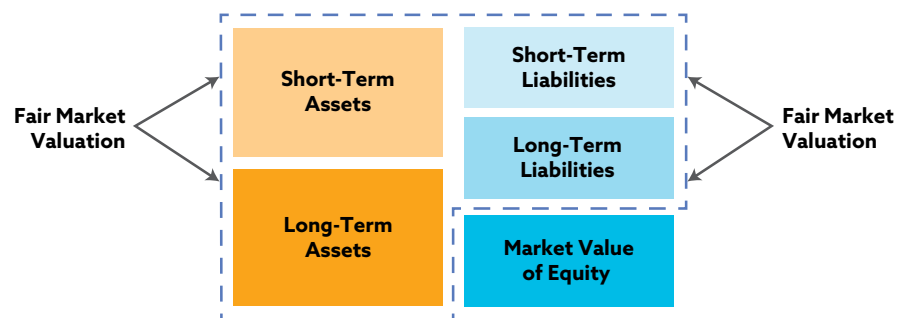
Model	Cash Flow Metric	Discount Rate	Applications
Dividend discount	Dividends	Shareholder required rate of return (r_e)	Value the equity of mature firms with stable dividends or consistent dividend policy
Free cash flow to the firm (FCFF)	FCFF	Weighted average cost of capital (WACC)	Value the business enterprise for firms with higher leverage or changing capital structure
Free cash flow to equity (FCFE)	FCFE	Shareholder required rate of return (r_e)	Value equity of non-dividend-paying firms or those with less consistent distribution policy with expected stable capital structure
Residual income	Residual income per share	Shareholder required rate of return (r_e)	Value equity of firms without book value distortions or that have either an inconsistent dividend policy or negative cash flow and less emphasis on terminal value

Advantages of absolute valuation models based on present value include their rigorous approach in using financial statement measures to directly estimate the size and timing of actual or potential cash flows to shareholders. The different cash flow metrics used allow for flexibility in valuing firms with distinct characteristics and also allow for the incorporation of changes in revenue, margins, or capital investments as well as distribution policies.

The level of detail associated with present value-based models is also one of the major drawbacks. Use of different models and forecast horizon choices may cloud comparisons, and present value models are often quite sensitive to uncertain input variables. By focusing on cash flows derived solely from company performance, they may also fail to adequately incorporate equity market dynamics or industry changes, such as the rise of competitors or technological change. For example, forecasting the cash flows of a traditional retail company using historical measures may ignore the rise of online sales and new industry entrants.

Asset-Based Valuation Models

This category of absolute valuation model seeks to estimate the intrinsic value of a common share by subtracting the estimated value of a firm's liabilities from its assets, as shown in Exhibit 11.

Exhibit 11: Asset-Based Valuation Models


In contrast to present value models focused on the *use* of assets to generate cash flows, the theory underlying the **asset-based valuation model** approach is that a firm's market value depends primarily on the price of the resources it owns, or controls, as

opposed to net cash flows generated. For example, the market value of the mineral resources a firm holds or the oil reserves it controls will fluctuate based upon commodity price changes.

When a company's core activity involves the extraction, refining, and sale of these assets, as for commodity producers, market price changes are often reflected in a company's share price as described in the following example.

ALBEMARLE CORPORATION (ALB) SHARES AND THE PRICE OF LITHIUM

Albemarle Corporation, a US-based company, is one of the world's top producers of lithium. Lithium production comprises over 70% of Albemarle's revenues. Demand for lithium soared in the early 2020s as the demand for this critical component of electric vehicle (EV) batteries sharply rose. Exhibit 12 illustrates the price of lithium along with Albemarle's stock price from 2021 through early 2024. While Albemarle's stock price appeared to behave independently of lithium prices for much of 2021, strong demand drove the commodity's price significantly higher, and Albemarle's stock was a beneficiary of high lithium prices into 2022. However, the market price for lithium has been mostly in decline since mid-2022, and Albemarle's stock price has followed.

Exhibit 12: Albemarle Corporation and Lithium Price, 2021–24



Source: Bloomberg.

Intangible assets often involve forms of intellectual property, such as the value of a brand or trademark.

The valuation of assets that may be more incidental to a firm's business, such as real estate ownership, are also often factored into the valuation process separately from their cash flow generation potential in the company's business.

WALMART'S REAL ESTATE ASSETS AND MARKET VALUE

Walmart Inc., a US-based discount retailer, operates over 10,000 stores in approximately 20 countries globally with an enterprise value of over USD500 billion. While Walmart continues to expand its e-commerce capabilities, the company's physical stores continue to generate significant amounts of value. Walmart owns, instead of leases, a majority of the real estate used in its businesses, especially in the United States where it owns approximately 85% of its property. Walmart operates three models of stores including its large Supercenters, which sell a wide array of both general merchandise and grocery items and average 178,000 square feet (or over 16,500 square meters) in size. The company's smaller store models are designed for more urban and suburban neighborhood locations, while still using sizable portions of real estate. Estimates suggest that Walmart's current retail operations cover 1.1 billion square feet (over 102 million square

meters), which is larger than the island of Manhattan in New York City. This square footage does not include the very large parking areas and surrounding real estate parcels of land owned along with the stores. Overall, a large proportion of the value of Walmart is likely implied by the value of its real estate holdings.

Advantages of an asset-based approach include the relatively straightforward process of valuing assets, which does not depend on the many underlying assumptions or forecasted cash flows associated with present value models. Given the use of market-based prices for valuation, this approach is useful for separately valuing a division or individual assets within a company or in the event of firm liquidation.

Given the focus on asset valuation, drawbacks of this approach include inadequate consideration of firm value as a going concern, including profit margins, the efficient use of assets, and the competitive landscape. It also may ignore the effect of taxes and financial and other liabilities that may impact intrinsic value.

Relative Valuation Approaches

Comparables models establish ratios (or multiples) of market-based measures of value to standardized income statement or balance sheet items to establish price comparisons per unit of revenue, cash flow, or profit between companies. The numerator of a multiple is typically based upon a firm's share price or enterprise value, while the denominator is a measure of earnings, free cash flow, or book value.

Multiples that use a company's share price in the numerator focus solely on a firm's value to shareholders. In this case, an analyst uses comparable company ratios of share price per unit to a historical or projected fundamental variable such as revenue, earnings, cash flow, or book value to estimate a company's intrinsic value. For example, an analyst may consider the share price per unit of earnings, or price-to-earnings ratio (P/E):

$$\text{P/E ratio} = \frac{P_t}{\text{EPS}_t} \quad (14)$$

The earnings per share (EPS) is either based on current or forecast earnings, that is, on a forward basis (i.e., forecasted EPS for the next year) or a trailing basis (i.e., EPS for the past year) consistently applied across comparable companies. Like other multiples, this measure relies on a consistently positive denominator to ensure meaningful results. P/E comparisons therefore only apply to firms and industries that are profitable over the measurement period and should exclude those with very low or negative earnings. Also, to ensure an accurate comparison of earnings potential, comparisons should exclude extraordinary, non-recurring items and other inconsistencies.

CASE STUDY



Logistics Industry Competitors and the Price-to-Earnings Ratio

In addition to key financial measures provided or calculated for the three logistics industry competitors introduced in earlier knowledge checks, the most recent annual net income for each is included in the following table:

Firm	Share Price	Shares Outstanding	EBITDA	Net Income
Advani Cargo Limited (ACL)	INR95	10 million	INR150 million	INR100 million
Bairnsworth (BRW)	AUD50	400,000	AUD4 million	AUD2.75 million
Chidori Express Corporation (CEC)	JPY450	2 million	JPY300 million	JPY120 million

The analyst also learns that half of Bairnsworth's net income in the past year resulted from a one-time gain on the sale of used equipment.

1. Which one of the following stocks appears most likely overvalued relative to the other competitors solely based on trailing P/E?

- A. Advani Cargo
- B. Bairnsworth
- C. Chidori Express

Solution:

B is correct. B is correct as BRW has the highest trailing P/E (14.55) among the three competitors, suggesting it is overvalued. BRW's P/E is computed based on net income of AUD1.375 million to exclude the effects of the one-time gain.

A is incorrect as ACL has a lower P/E ($9.5 < 14.55$) than BRW.

C is incorrect as CEC has the lowest P/E (7.5), making it the most *undervalued* stock based upon this measure.

Use Equation 14 to calculate the trailing P/E for each of the firms, using AUD1.375 million ($= 2.75 \div 2$) for adjusted Bairnsworth net income as half involves extraordinary, non-recurring income:

$$\text{ACL: P/E} = 9.5 [= \text{INR}95 \div (\text{INR}10 \text{ million} \div 100 \text{ million})]$$

$$\text{BRW: P/E} = 14.55 [= \text{AUD}50 \div (\text{AUD}1.375 \text{ million} \div 0.4 \text{ million})]$$

$$\text{CEC: P/E} = 7.5 [= \text{JPY}450 \div (\text{JPY}120 \text{ million} \div 2 \text{ million})]$$

Enterprise value (EV) multiples, in contrast, seek to value an entire firm from the perspective of both debtholders and shareholders. Rather than dividends or earnings available only to shareholders in the denominator, EV multiples typically include revenue, operating income (EBITDA), or income after operating expenses (EBIT) available to all capital providers.

Based on Equation 9, an analyst might use an industry composite or average $[(\text{EV}/\text{EBITDA})_{\text{Composite}}]$ versus individual company $[(\text{EV}/\text{EBITDA})_{\text{Firm}}]$ multiple comparison to draw a conclusion about relative value versus share price. A higher enterprise value to EBITDA multiple suggests a firm has a greater enterprise value per unit of operating income. As a result:

$(\text{EV}/\text{EBITDA})_{\text{Composite}} > (\text{EV}/\text{EBITDA})_{\text{Firm}}$: Suggests the firm is *undervalued* relative to comparable companies based on this measure

$(\text{EV}/\text{EBITDA})_{\text{Firm}} > (\text{EV}/\text{EBITDA})_{\text{Composite}}$: Suggests the firm is *overvalued* relative to comparable companies based on this measure

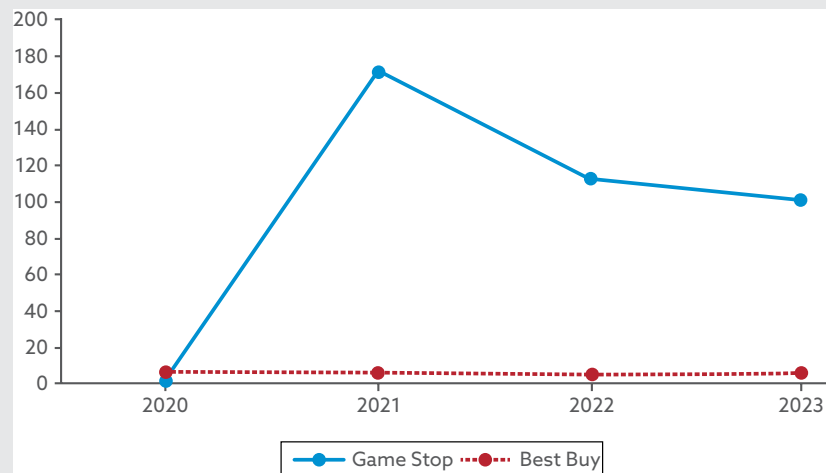
As a broad measure of operating income, EBITDA approximates cash flows available to all capital providers and is less likely to be negative, unlike EPS. As a broader measure, EV/EBITDA multiples may be applicable in situations where P/E multiples are more difficult to use or interpret due to earnings fluctuations.

Earlier in this learning module, we introduced GameStop to discuss a situation in which price diverged from likely value. We illustrate the multiple comparison in the following example.

GAMESTOP (GME) RELATIVE VALUE MULTIPLE COMPARISON

As GameStop's stock price soared in early 2021 on an absolute basis, its relative valuation behaved in a similar fashion. GameStop had generated negative EPS, a trend that analysts expected to continue, rendering its forward P/E multiple a meaningless statistic in 2021. On the other hand, GameStop's forecasted EBITDA was consistently positive, allowing the firm's EV/EBITDA multiple to be compared to industry peers. For example, if we consider consumer electronics and software retailer Best Buy (BBY) for comparative purposes, Exhibit 13 provides a comparison of multiples that suggests clear evidence of GameStop's overvaluation versus Best Buy based upon these measures over this period.

Exhibit 13: GME versus BBY Multiples, 2020–2023



Source: Bloomberg.

The example of GameStop discussed above implies comparison of a company's multiple to a composite average and we illustrate such a comparison explicitly in the following case study.

CASE STUDY



Logistics Industry Competitors and EV/EBITDA Multiples

A final step in the relative valuation process of the three logistics competitors introduced in earlier case studies used comparisons of EV/EBITDA multiples relative to a composite average. The EV/EBITDA multiples for the three industry competitors are as follows:

- Advani Cargo Limited (ACL): 9.00
- Bairnsworth (BRW): 7.25

- Chidori Express Corporation (CEC): 6.00

Furthermore, the analyst calculates the P/E multiples based upon most recent reported earnings for each as follows:

- Advani Cargo Limited (ACL): 24.0
- Bairnsworth (BRW): 17.0
- Chidori Express Corporation (CEC): 12.0

The composite industry averages for EV/EBITDA and P/E are 8.0 and 25.0, respectively. These averages are created from the set of all industry competitors with economically meaningful multiples.

Research shows that the market views ACL as having better growth potential relative to the other two competitors shown, but its growth potential appears comparable to the overall industry. CEC appears to exhibit considerably more debt in its capital structure as compared to the industry.

1. Which of the following statements is likely to be true for the comparable sets of multiples used in relative valuations of the logistics industry competitors?

- A. The comparable set of companies with P/E multiples is likely larger than that with EV/EBITDA multiples.
- B. The comparable set of companies with P/E multiples is likely smaller than that with EV/EBITDA multiples.
- C. The comparable sets of companies likely consist of the same number of firms for both P/E and EV/EBITDA multiples.

Solution:

B is correct. Choice B is correct because as a broader measure, EBITDA is positive while negative EPS causes P/E calculations to be economically meaningless, so those observations are likely excluded from the comparison. Because EBITDA is more likely to be positive than EPS for any given firm, the EV/EBITDA set of comparable firms is likely larger than the set of P/E comparables.

Choice A is incorrect because this statement implies that EV/EBITDA is more likely to be economically meaningless than P/E for any given firm.

Choice C is incorrect because this statement implies that EPS and EBITDA are equally likely to be economically meaningless.

2. Evaluate which one of the three companies above is likely overvalued.

Solution:

Advani Cargo Limited (ACL) is likely overvalued. Using the EV/EBITDA multiple of 8.0, ACL's multiple of 9.0 is greater. The analyst's research suggests that ACL's growth potential is no better than that of the overall industry, thus providing less justification for ACL to trade at a higher multiple.

Relative valuation models are appealing due to their simplicity and straightforward application. Rather than forecasting cash flows over a specific time horizon, a multiple approach facilitates comparisons among similar companies and can include expectations of future earnings or other variables from analyst forecasts. In contrast to establishing an intrinsic value, a **justified price multiple** for the stock may be estimated to compare intrinsic value and price. The justified price multiple is the estimated fair value of that multiple, which can be supported using the method of comparables or the method of forecasted fundamentals. Limitations of a multiple-based approach

include the challenge of choosing comparable companies, which may significantly influence results. Company multiples may vary based on differences in fundamentals. These and other issues related to the relative value equity valuation approach will be explored in a later learning module.

Reconciling Multiple Equity Valuation Approaches

Most analysts use more than one type of model to estimate value, recognizing that each involves a simplification with uncertainty due to model appropriateness, specification, and inputs used. For example, Exhibit 14 illustrates the comparison of several model results to the recent share price range on a side-by-side basis in a floating bar valuation chart:

Exhibit 14: Equity Valuation Process

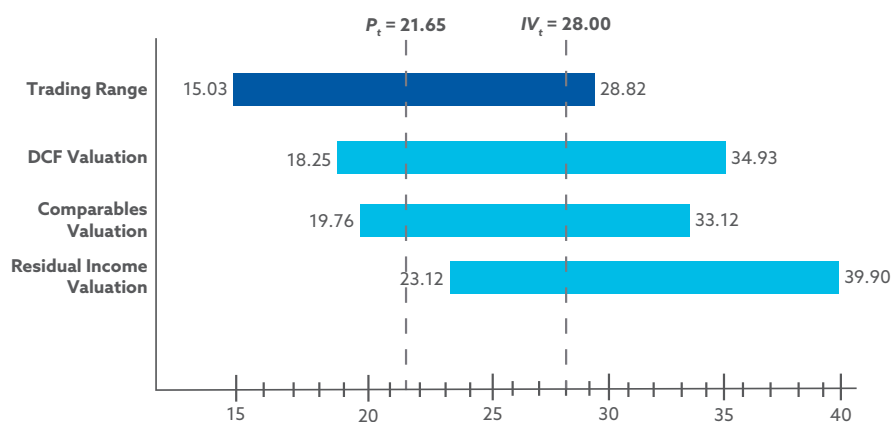
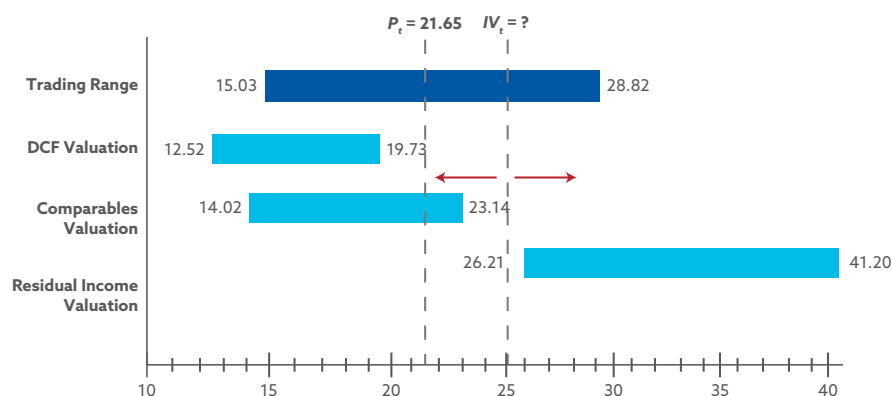


Exhibit 14 shows an example in which each of the three valuation estimates results in a higher estimated intrinsic value than the current share price, suggesting that the stock is undervalued ($P_t < IV_t$).

In other cases, the combination of multiple equity valuation methods may provide less conclusive or even contradictory results due to the wide dispersion of outcomes. For example, Exhibit 15 shows a case in which different valuation approaches suggest that the stock may be undervalued or overvalued.

Exhibit 15: Equity Valuation Process with High Dispersion

A high degree of dispersion among valuation estimates often causes analysts to dig deeper into model assumptions, examine industry peers and market dynamics more closely, and possibly change the weights associated with different approaches to reflect the relative confidence in model results.

KNOWLEDGE CHECK



An analyst has valued a stock using three different methodologies: discounted cash flows, relative valuation, and residual income. The stock currently trades at EUR21.65 with a recent trading range from EUR15.03 to EUR28.82. Each valuation methodology was conducted with two scenarios as well as each methodology's midpoint:

Valuation Method	Worst Case	Midpoint	Best Case
Discounted cash flow	EUR12.52	EUR16.13	EUR19.73
Comparables	EUR14.02	EUR18.58	EUR23.14
Residual income	EUR26.21	EUR33.71	EUR41.20

1. Calculate the expected percentage difference between this stock's value and price if the analyst uses the midpoints from each methodology and weights each methodology equally.

Solution:

$$\text{Expected value} = (1 \div 3) \times 16.13 + (1 \div 3) \times 18.58 + (1 \div 3) \times 33.71 = 22.81.$$

$$\text{Difference between estimated value and price} = \frac{\text{EUR}22.81 - \text{EUR}21.65}{\text{EUR}21.65} = 5.36\%$$

Thus, the analyst might conclude that this stock is moderately undervalued.

2. Analyze the effect on the analyst's conclusion if the residual income approach is weighted at 10% while the other two approaches are equally weighted at 45% each.

Solution:

$$\text{Expected value} = 0.45 \times 16.13 + 0.45 \times 18.58 + 0.10 \times 33.71 = 18.99.$$

$$\begin{aligned}\text{Expected difference between value and price} &= \frac{\text{EUR}18.99 - \text{EUR}21.65}{\text{EUR}21.65} \\ &= -12.29\%.\end{aligned}$$

Because the residual income approach generates a much higher estimate of value on this stock compared to the other approaches, reducing its weighting also reduces the expected value of this stock. The proposed weighting change increases the possibility that the analyst concludes that this stock is overvalued.

The relative importance of a given model's results in making an investment decision will depend upon the analyst's confidence in the inputs and outputs and whether the valuation approach is appropriate for a specific firm, industry, and/or market environment. These valuation approaches and associated models will be the subject of subsequent learning modules.

QUESTION SET 3



1. An analyst plans to value the equity of Buttonwood Inc. and calculates the firm's FCFF and FCFE for the most recent year. He finds that FCFE is greater than FCFF. Which one of the following statements must be true for the analyst to observe this difference?
 - A. Additional borrowing minus debt repayment is greater than interest expense multiplied by one minus tax rate.
 - B. Debt repayment minus additional borrowing is greater than interest expense multiplied by one minus tax rate.
 - C. Additional borrowing is greater than debt repayment.

Solution:

A is correct. Choice A is correct because $\text{FCFE} > \text{FCFF}$, which implies that $\text{FCFE} - \text{FCFF} > 0$. Substituting Equation 11 and Equation 12 for FCFE and FCFF into this equation reduces to the following expression:

$$(\text{Additional borrowing} - \text{Debt repayment})$$

$$-[\text{Interest expense} \times (1 - \text{Tax rate})] > 0.$$

The "cash flow from operations" and "capital expenditure" terms appear in both FCFE and FCFF, so these terms are eliminated. The inequality above is only true if additional borrowing minus debt repayment is greater than interest expense multiplied by 1 minus tax rate.

Choice B is incorrect because this statement implies that the term "(Additional borrowing - Debt repayment)" is negative and the overall term becomes more negative when subtracting "[Interest expense $\times$ (1 - Tax rate)]."

Choice C is incorrect because this statement does not include whether the additional borrowing minus the debt repayment is greater than the after-tax interest expense term.

2. Which one of the following is both an advantage and a drawback of many present value models?
 - A. Their simplicity and straightforward application
 - B. Their ability to model equity market dynamics

C. The level of detail required to create the models

Solution:

C is correct. Choice C is correct because many present value models require significant amounts of detailed inputs that can incorporate specific value drivers, such as changes in revenue, margins, and capital investments—a definite advantage of these types of models. The level of detail associated with present value-based models is also one of their major drawbacks. While analysts can use different models and forecast horizons, they may face challenges in establishing both model assumptions and company forecasts.

Choice A is incorrect because the simplicity and straightforward application is cited as an advantage of relative valuation models.

Choice B is incorrect because the inability to model equity market dynamics is only a drawback of present value models.

3. Cheshire Inc. exhibits a lower EV/EBITDA multiple compared to the average multiple of a group chosen from a set of the fastest-growing companies operating in its industry. Which one of the following statements most correctly reflects how an analyst might interpret this comparison?

- A.** Cheshire is most likely undervalued because the comparison group is drawn from companies in its industry.
- B.** Cheshire may be undervalued only if it is also one of the fastest-growing companies in its industry.
- C.** Cheshire is unlikely to be undervalued because the comparison group is likely inappropriate.

Solution:

B is correct. Choice B is correct because this statement reflects the fact that Cheshire's multiple is less than the comparison group while also incorporating the condition that Cheshire should also be among the fastest-growing companies in the industry for this to be an appropriate comparison group.

Choice A is incorrect because the comparison group appears to only include a subset of the industry. As a result, it is possible that choosing a more comprehensive set of comparison companies would result in a different multiple that might be lower.

Choice C is incorrect because the comparison group is from the same industry as Cheshire. However, without assessing whether Cheshire shares similarities with the comparison group, it is not possible to determine whether the comparison group is appropriate.

SUMMARY: INTRODUCTION TO EQUITY VALUATION

- An equity security's observed market price incorporates the perceived value of all active market participants, while a security's intrinsic value is unobservable and reflects the valuation of its expected future cash flows.
- A divergence between a share's market price and its perceived value may give rise to issuer and/or investor actions. For example, investor perception that shares are undervalued may lead to an increase in exposure, while an issuer with the same perception may be motivated to repurchase shares.
- Analysts often compare market capitalization, or the market value of a firm's equity, to book value, an accounting-based measure, to assess cash flow generation potential versus existing net assets.
- Enterprise value estimates an entire firm's potential acquisition price and therefore includes both debt and equity capital. Relative to market value of equity, this measure allows for better comparison across firms because it is less sensitive to variations in capital structure.
- The absolute valuation approach used in present value or asset-based models aims to estimate a firm's future cash flows or asset prices that drive intrinsic share value. Relative valuation methods use comparisons of observable measures of value to balance sheet or income statement units to draw conclusions about a firm's value versus market price.
- Absolute valuation methods seek to directly assess intrinsic value using financial projections but are more complex and subject to analyst assumptions. Relative valuation approaches are simpler to use and easier to compare across industry peers but may not fully account for company-specific factors that affect intrinsic value.

PRACTICE PROBLEMS

The following information relates to questions 1-6

A financial adviser meets with two high-net-worth clients, Sophie Cho and Cynthia Lee, to discuss three different stocks that the adviser believes may be appropriate investments. Financial details of the three companies are below (in thousands of USD):

	Rivercreek Corp	Lough Co	Arkanoid Inc
Revenue	50,000	12,000	30,000
EBITDA	9,000	2,500	5,000
Net income	3,000	1,750	4,000
Total assets	45,000	24,000	25,000
Book value of liabilities	32,000	15,000	30,000
Shares outstanding	1,000	500	1,200
Market cap	180,000	25,000	100,000
EV	190,000	26,000	105,000

All three are retail companies within the Consumer Discretionary sector. Rivercreek owns most of its physical store locations, while the other two companies lease their space. The table below shows composite multiples:

	P/B	P/E	EV/Revenue	EV/EBITDA
Retail composite	8.0	50.0	3.0	16.0

- The financial adviser finds that the two clients have differing opinions about the future of Rivercreek. Sophie Cho believes that Rivercreek has tremendous potential to grow its product lines and increase its profitability and cash flows. Cynthia Lee believes that Rivercreek is likely to remain a low-margin business and faces potential for disruption from competitors. Which statement below most correctly describes investor perceived values of Rivercreek?
 - Sophie Cho's perceived value of Rivercreek stock suggests that she is likely to be a buyer of this stock.
 - Sophie Cho's perceived value of Rivercreek stock is likely to be closer to the stock price compared to Cynthia Lee's perceived value.
 - Cynthia Lee's perceived value of Rivercreek stock implies that she is unlikely to take a position in the stock.
- The composite industry P/E multiple has risen considerably over the last year from about 20 to the current level of 50, and several companies in the industry have been issuing new public shares. Which of the following statements provides a likely explanation for these observations?
 - Retail companies may be issuing shares to pay higher dividends.

SOLUTIONS

1. B is correct. Choice B is correct because Rivercreek is priced at multiples that are higher than the composite industry multiples, as shown below:

$$\frac{P}{B} = \frac{180,000}{45,000 - 32,000} = 13.85 > 8.0$$

$$\frac{P}{E} = \frac{180,000}{3,000} = 60.00 > 50.00$$

$$\frac{EV}{\text{Revenue}} = \frac{190,000}{50,000} = 3.80 > 3.0$$

$$\frac{EV}{EBITDA} = \frac{190,000}{9,000} = 21.11 > 16.0$$

As a result, Sophie Cho's perceived value of Rivercreek stock better matches the optimistic view associated with Rivercreek's stock price as shown by its multiples. Choice A is incorrect because Sophie Cho and the market both have an optimistic view of Rivercreek's future as reflected by the market price and Sophie Cho's likely high perceived value. However, Sophie Cho may believe that Rivercreek's price reflects more optimism than she has about the company's future. As a result, we cannot conclude that Sophie Cho believes that Rivercreek stock is undervalued.

Choice C is incorrect because Cynthia Lee may choose to take a short position by short selling Rivercreek stock given her pessimistic view of the company and the current optimism priced by the market.

2. C is correct. Choice C is correct because a rapid increase in P/E accompanied by issuers deciding to sell new shares of stock to the public implies that the industry's stocks have become overvalued, on average.

Choice A is incorrect because issuing shares to pay more dividends does not explain the rise in P/E. Furthermore, this action would increase equity and then reduce it with extra dividend payments.

Choice B is incorrect because a rising EBITDA is not a sufficient explanation for a rising P/E.

3. B is correct. Choice B is correct because P/B creates problems in comparison for at least one reason. Arkanoid's book value of equity is negative because its book value of liabilities is greater than its book value of assets. Thus, Arkanoid's P/B is economically meaningless. Additionally, there appears to be the potential for differences in leverage across the three companies.

Choice A is incorrect because net income is positive and, for all three companies, does not appear to be too close to zero to create economically meaningless P/E multiples.

Choice C is incorrect because EBITDA shows higher values than net income, thus EV/EBITDA is even less likely to provide economically meaningless results than P/E.

4. B is correct. Choice B is correct because Lough's net debt to assets is only 4.2%. Net debt is equal to EV minus equity market capitalization, so Lough's net debt is 1,000 (= 26,000 – 25,000). Dividing this by Lough's book value of assets of 24,000 gives a result of 4.2% (= 1,000 ÷ 24,000).

Choice A is incorrect because Rivercreek's net debt to assets is 22.2%. Its net debt is 10,000 (= 200,000 – 190,000), and its book value of assets is 45,000.

Choice C is incorrect because Arkanoid's net debt to assets is 20%. Its net debt is 5,000 ($= 105,000 - 100,000$), and its book value of assets is 25,000.

5. B is correct. Choice B is correct because Lough's EV/EBITDA multiple of 10.4 ($= 26,000 \div 2,500$) is lower than the composite multiple of 16.0.
Choice A is incorrect because Rivercreek's EV/EBITDA multiple of 21.11 ($= 190,000 \div 9,000$) is higher than the composite multiple of 16.0.
Choice C is incorrect because Arkanoid's EV/EBITDA multiple of 21.0 ($= 105,000 \div 5,000$) is greater than the composite multiple of 16.0.
6. A is correct. Choice A is correct because Rivercreek's owned real estate may provide substantial amounts of the company's overall value. As such, it may be useful to value Rivercreek using an asset-based approach.
Choice B is incorrect. While Lough does have a lower sales-to-assets ratio compared to its two competitors, the higher capital intensity does not imply that an asset-based valuation may be useful. Rather, we would need to know that Lough's assets are an integral aspect of understanding the value of the business. It was stated earlier that Lough does not own its real estate so is unable to capitalize on increases in market value of leased retail space.
Choice C is incorrect. Firm size is not a factor as to whether an asset-based valuation approach is useful.

LEARNING MODULE

6

Discounted Cash Flow (DCF) and Growth Models

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	contrast the inputs used in dividend discount, free cash flow to equity (FCFE), free cash flow to the firm (FCFF), and residual income models, and explain the process for using present value models to value equity
☐	calculate and interpret the intrinsic value of an equity security based on present value models that assume constant cash flow growth or multistage cash flow growth
☐	explain the shortcomings of constant and multistage cash flow growth assumptions in present value models
☐	calculate the intrinsic value of a non-callable, non-convertible preferred stock and describe how contingency features affect intrinsic value

INTRODUCTION: DISCOUNTED CASH FLOW (DCF) AND GROWTH MODELS

1

Financial asset values depend on future cash flows and claims adjusted for their expected timing and likelihood. Unlike bond cash flows, which are finite and contractual, shareholder distributions are unknown and at the discretion of management over an indefinite period. As described in earlier learning modules, some equity issuers consistently distribute capital income as cash dividends or repurchase shares, while others distribute no cash at all.

Discounted cash flow (DCF) models incorporate forecasted measures of future cash flow available to investors discounted at their required rate of return. This learning module first contrasts the available cash flow measures used in discounted cash flow models and spells out the detailed process of applying this approach to estimate intrinsic value. The choice of a growth pattern is a critical element in valuing an indefinite series of cash flows. The use of constant or multistage growth rate assumptions

is explained, along with the benefits and drawbacks of these alternatives. The learning module concludes with the DCF valuation of preferred shares and considers the effect of contingency features on intrinsic value.

PRE-TEST: DISCOUNTED CASH FLOW (DCF) AND GROWTH MODELS



1. Which one of the following most correctly describes a reason for using a mature company's dividends as the cash flow metric to value the company's equity?
 - A. Dividends are a cash flow metric favored by a controlling investor.
 - B. Dividends represent a complete measure of cash flow to shareholders.
 - C. Dividends are received by all shareholders of the company.

Solution

C is correct because dividends are a common cash flow received by all shareholders. Because a financial asset should be valued based on cash flows expected to be received, dividends are a rational choice as a cash flow metric.

A is incorrect because dividends can be changed by a controlling investor; thus dividends are more typically used to value a company's equity from a non-controlling investor's perspective.

B is incorrect because dividends are a complete measure of cash flow to shareholders only if the company does not engage in stock repurchases or stock issuances.

2. Which one of the following characteristics is most likely for a company with negative free cash flow to equity?
 - A. Growing cash balances
 - B. Large stock issuances
 - C. Large stock repurchases

Solution

B is correct because stock issuance reflects a cash flow from, rather than to, shareholders. Thus, stock issuances reduce FCFE, all else equal, and large stock issuances may be more likely observed for companies with negative FCFE.

A is incorrect because growing cash balances reflect higher cash flow to equity. The improvement in cash balances is the portion of FCFE not paid to shareholders.

C is incorrect because stock repurchases are cash flows to equity, so large stock repurchases are a likely characteristic of a company with positive FCFE.

3. Which one of the following statements provides the most accurate argument for why an analyst may prefer a shorter cash flow forecast time horizon when valuing a company's stock?
 - A. A shorter forecast horizon places less weight on the terminal value.
 - B. A shorter forecast horizon reflects an analyst's degree of confidence in the forecast.

- C. A shorter forecast horizon reflects the analyst's expected holding period for the stock.

Solution

B is correct because analysts typically have a greater degree of confidence in their thesis of a company over a shorter future period, so they prefer to limit the forecast length to shorter horizons.

A is incorrect because the terminal value estimate in an equity valuation carries more weight with shorter forecast horizons.

C is incorrect because the expected holding period should have no bearing on the forecast time horizon used in the present value model.

4. Kruoth Corporation stock currently pays a dividend of AUD3.20 per share. An analyst assumes that Kruoth's dividend will grow at a rate of 12% annually for the next three years. In later years, the analyst assumes 4% annual constant growth. If Kruoth's cost of equity is 10%, which one of the following is closest to the correct value estimate for Kruoth stock?

- A. AUD66.25 per share
B. AUD68.50 per share
C. AUD73.00 per share

Solution

B is correct because the value estimate can correctly be estimated as follows:

$$\begin{aligned} \text{Kruoth value per share} &= \frac{3.20(1.12)^1}{1.10^1} + \frac{3.20(1.12)^2}{1.10^2} \\ &+ \frac{3.20(1.12)^3}{1.10^3} + \frac{3.20(1.12)^3(1.04)}{0.10 - 0.04} \div 1.10^3 = 3.26 + 3.32 + 3.38 + 58.55 = 68.51. \end{aligned}$$

A is incorrect because an estimate of 66.25 results if the terminal value is incorrectly calculated using the Year 3 expected dividend, rather than the Year 4 dividend, in the numerator of the constant growth equation.

C is incorrect because an estimate of 73.00 results if the terminal value incorrectly assumes 12% growth in the computation of the Year 4 dividend.

5. Which one of the following statements most correctly identifies a potential limitation of both FCFE and FCFF growth models?

- A. FCFE and FCFF growth models may be prone to value estimation bias because of base effects.
B. FCFE and FCFF growth models may be prone to value estimation bias because shares outstanding may fluctuate.
C. FCFE and FCFF growth models may be prone to value estimation bias when debt ratios fluctuate.

Solution

A is correct because both FCFE and FCFF often exhibit historical volatility, which creates the potential for a base effect that may bias an analyst's forecast of expected future cash flows.

B is incorrect because fluctuation in shares outstanding affects dividend growth models in which an analyst uses per share amounts. Fluctuation in shares outstanding does not create any potential limitations for either FCFE or FCFF models.

C is incorrect because fluctuations in debt ratios create potential limitations for FCFE models but not for FCFF models.

6. Which one of the following statements best describes a common feature of preferred and common equity valuation?

- A. Both preferred and common equity may be valued using dividend-based models.
- B. Both preferred and common equity valuation processes include a terminal value estimate.
- C. Both preferred and common equity valuation processes use the same required return.

Solution

A is correct because both preferred and common shares may be valued using dividend-based models, although common shares can be modeled using other cash flow metrics as well.

B is incorrect because preferred equity valuation does not incorporate a terminal value estimate as part of its valuation process.

C is incorrect because preferred equity has a priority claim over that of common equity, so preferred equity is less risky and has a lower required return compared to common equity.

2

PRESENT VALUE MODELS FOR EQUITY INSTRUMENTS



contrast the inputs used in dividend discount, free cash flow to equity (FCFE), free cash flow to the firm (FCFF), and residual income models, and explain the process for using present value models to value equity

The use of present value models to value equity is based on the idea that a financial asset's price should equal its promised or expected future cash flows adjusted for their timing and risk. Cash flows to equity holders are discretionary over an indefinite period. Some firms regularly pay cash dividends, while other valuation approaches may involve expected future cash flows *available*—but not distributed—to owners. The generalized discounted cash flow valuation model introduced in an earlier learning module may be broadly applied to establish the intrinsic value of a firm's shares:

$$IV_t = \sum_{i=1}^n \frac{E(CF_{t+i})}{(1 + \text{Required rate of return})^i} + \frac{E(P_{t+n})}{(1 + \text{Required rate of return})^n} \quad (1)$$

In Equation 1, IV_t denotes intrinsic value, CF_t is periodic cash flow, and $E(P_{t+n})$ is expected price. Since they are based on financial statement measures, present value-based valuation approaches have substantial flexibility to incorporate changes in sales, costs, investments in fixed assets, and working capital, as well as a firm's use of debt versus equity and shareholder distribution policies. This flexibility makes model outputs very sensitive to both the type of model used and input assumptions. The application of present value models should therefore follow a well-structured process, the first step of which involves selecting an appropriate cash flow measure (CF_{t+i}), as described in the following section.

Present Value Model Metrics

The metric best suited for a specific issuer in a present value model depends on a company's characteristics and circumstances, as well as assumptions underlying the analysis and the perspective of an investor. When using discounted cash flow equity valuation models, analysts often choose from one of the four following metrics introduced in an earlier learning module:

- dividends,
- free cash flow to equity,
- free cash flow to the firm, or
- residual income (RI).

In contrast to the other three measures, RI is not a cash flow metric but, rather, a function of net income, book value of equity, and required return on equity.

Dividends offer an intuitively appealing choice for valuing firms because these cash flows are distributed proportionally to all shareholders of record. The timing and size of dividends paid for publicly listed equities are widely reported, allowing analysts to directly assess historical growth and develop expectations of future growth, as in the following example.

COCA-COLA'S DIVIDEND GROWTH AND SHARE PRICE

The Coca-Cola Company, a US-based multinational corporation in the beverage industry, has a long history of paying dividends to shareholders, with a dividend per share that has steadily risen over six decades. Exhibit 1 shows the company's dividend per common share along with its share price during a 20-year period, over which dividends per share and the share price roughly tripled.

Exhibit 1: Coca-Cola Share Price and Dividend Growth, 2004–2023



Source: Bloomberg.

Note that the company maintained consistent payouts despite share price fluctuations during the COVID-19 pandemic. As a mature company, Coca-Cola distributes most of its net income in the form of cash dividends. Over the final five years shown in Exhibit 1, the company distributed an average of nearly 80% of net income in cash dividends. For example, in 2023, Coca-Cola paid out nearly USD8 billion in dividends from over USD10.7 billion in net income.

Dividends are the most appropriate measure of shareholder return for use in a DCF model when a firm distributes a consistently high proportion of earnings as cash to shareholders with little change in the number of shares outstanding.

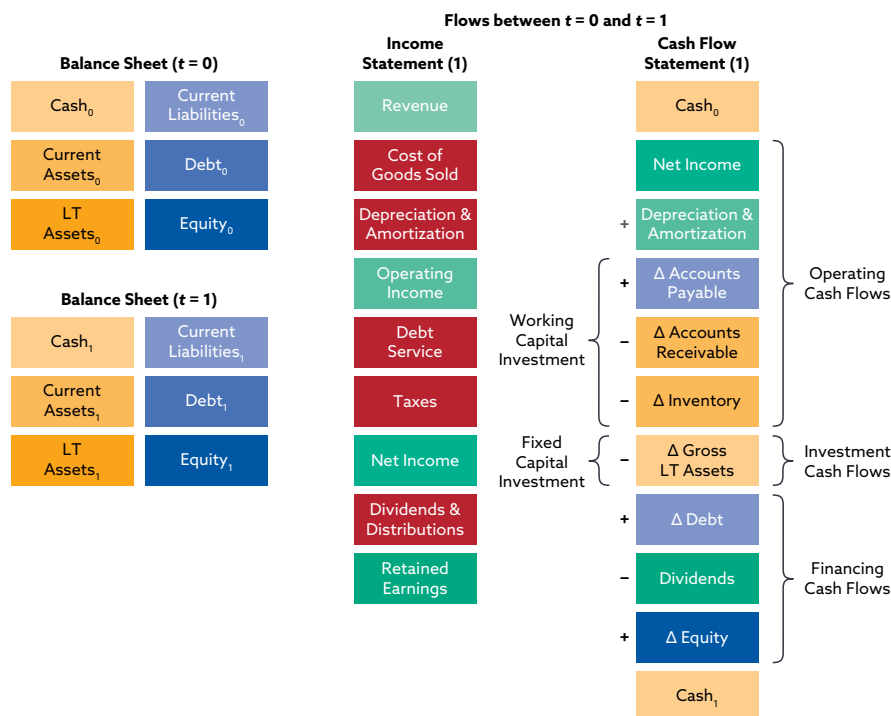
Free cash flow to equity provides a direct alternative to dividends as a cash flow measure to shareholders. FCFE includes dividends and other forms of cash flow available to equity owners. For example, share buybacks are a form of shareholder return excluded from the dividend measure but are included in FCFE as a *use* of equity capital, which is a cash flow to shareholders. Similarly, share issuance is a *source* of equity capital that *reduces* FCFE, while leaving dividends unchanged. Increases or decreases in cash retained beyond that needed to fund operations that remain available to shareholders will also affect FCFE but not dividends. As a result of the inclusion of available cash and share capital changes, FCFE can be negative, whereas dividends can never be less than zero.

Similarly, free cash flow to the firm is a measure of cash flow available to investors but includes debtholders, as well as shareholders. As a result, FCFF is often used to value entire companies rather than only a firm's equity. A key difference between the two involves the cash flows to and from debtholders, interest payments, and changes in debt outstanding, as described in detail later.

Residual income differs from the other three available present value metrics in that it is a return-based measure rather than a cash flow estimate. This accounting-based measure estimates company earnings after deducting a charge for the cost of equity capital used to generate the earnings. An RI-based valuation model begins with an assumption that a company's value is partially determined by its book value of equity, which is adjusted based on the company's ability earn a return above the cost of equity over time.

In what follows, we consider each one of the four present value metrics in greater detail to provide additional clarity in understanding important similarities and differences. As a first step, it is helpful to review the underlying economics of a company's financial statements. A simple three statement model from earlier Financial Statement Analysis learning modules based on US GAAP is shown in Exhibit 2.

Exhibit 2: Three Statement Model

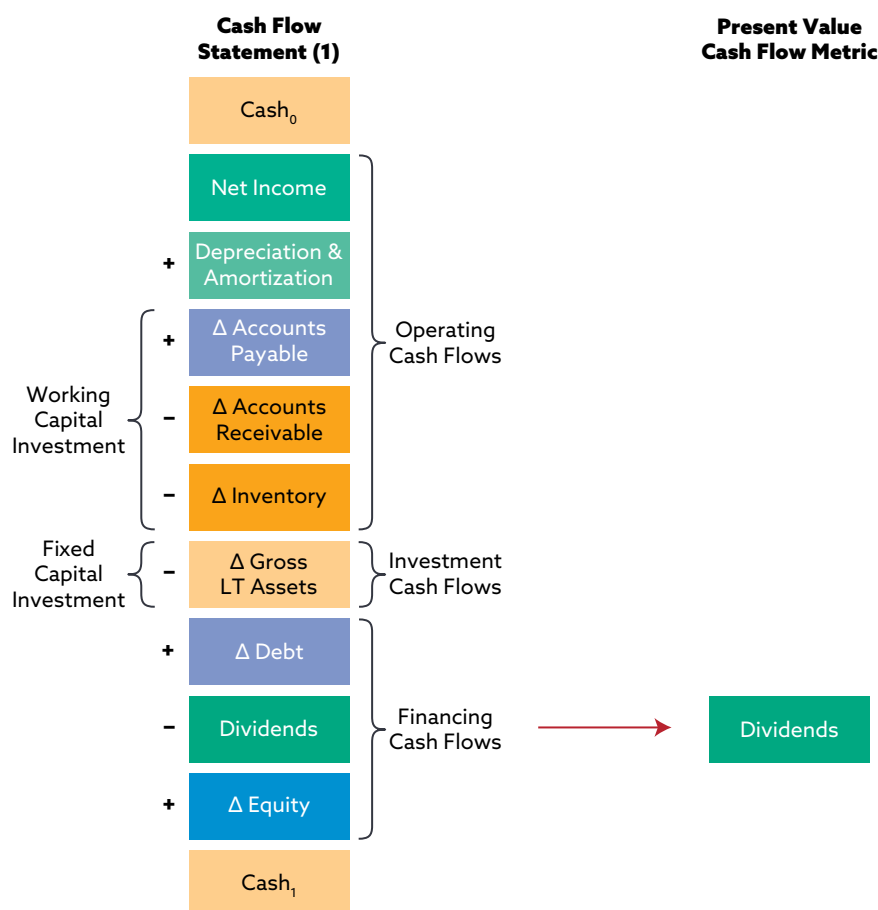


Three of the four metrics are based on cash flow statements, which incorporate all information from both the income statement and changes in balance sheets between time periods. Since depreciation is shown as an operating cash flow, we refer to changes in *gross* long-term assets as an investment cash flow in Exhibit 2.

Dividends

The narrowest cash flow metric used in equity present value models involves cash dividends, or the portion of net income which is neither retained and reinvested nor distributed in the form of share repurchases. A **dividend discount model** (DDM) assumes a shareholder's sole access to firm value is through capital income in the form of cash distributions. This model is most appropriate for valuing firms with positive and stable earnings that distribute a large and consistent proportion of their net income as dividends. Under these circumstances, expected future dividends may be forecasted with reasonably high levels of confidence. This valuation approach is from the perspective of a non-controlling shareholder—that is, an investor with no influence over future dividend policy. Exhibit 3 shows dividends in the context of the cash flow statement.

Exhibit 3: Dividends as a Present Value Cash Flow Metric



It is important to note not only what is included in the dividend discount model but also what is *excluded*, as shown in Exhibit 3. For example, financing cash flow changes, such as greater reliance on debt or new equity issuance, are ignored, leading to distorted results if they significantly impact equity value.

A simple dividend discount model involves a firm paying a dividend D that grows at a constant rate g into the indefinite future, as in Equation 2:

$$IV_t = \frac{D_t(1+g)}{r_e - g}, \quad (2)$$

where the required equity rate of return is r_e , also sometimes referred to as the cost of equity. Analysts may assume a growth rate based on most recent dividends or directly forecast future dividends (D_{t+1}) to solve for intrinsic value, as shown in Equation 3:

$$IV_t = \frac{D_{t+1}}{r_e - g}. \quad (3)$$

The numerator of Equation 3 may be expressed as total dividends paid or as dividends per share (DPS), in which case we solve for the intrinsic value as a share price. The inclusion of a constant growth rate is a useful element that will be expanded to other present value metrics later in this learning module.

KNOWLEDGE CHECK: VALUING EXCELITATE BASED ON DIVIDENDS



Excelitate Corporation currently pays a dividend of USD1.33 per share. Analysts at Broadrange Inc. and Stratospheric Securities have decided to estimate the intrinsic value of Excelitate shares using a dividend discount model. While both firms estimate that the stock's required return is 8.5%, the Broadrange analyst believes the company has reached a mature phase and investors should therefore expect a steady dividend stream. Stratospheric's analyst considers the company well positioned for long-term expansion, with 5% annual dividend growth into the future as the most likely scenario.

1. Which one of the following is closest to Broadrange's estimate of Excelitate's intrinsic value?

- A. USD6.40
- B. USD11.30
- C. USD15.65

Solution

C is correct because there is no expected dividend growth, so we use Equation 2 with $g = 0$ as follows:

$$IV_t = \frac{D_t(1+g)}{r_e - g} = \frac{1.33(1+0)}{0.085 - 0} = 15.65.$$

A is incorrect because it results from dividing 8.5 by 1.33.

B is incorrect because it results from multiplying 8.5 by 1.33.

2. Which one of the following is closest to Stratospheric's estimate of Excelitate's intrinsic value?

- A. USD10.34
- B. USD26.60

C. USD39.90

Solution

C is correct because using Equation 2 with D_t of 1.33, r_e of 8.5%, and a growth rate g of 5%, we obtain the following:

$$IV_t = \frac{D_t(1+g)}{r_e - g} = \frac{1.33(1+0.05)}{0.085 - 0.05} = 39.90.$$

A is incorrect because it results from incorrectly adding the growth rate to the required return on equity in the denominator:

$$\frac{1.33(1+0.05)}{0.085+0.05} = 10.34.$$

B is incorrect because it results from incorrectly dividing the current dividend by the growth rate, g :

$$\frac{1.33}{0.05} = 26.60.$$

3. Which statement is most correct if Stratospheric's analyst mandates an increase in Excelitate's required return on equity?

- A. The estimated intrinsic value will fall.
- B. The estimated intrinsic value will rise.
- C. The estimated intrinsic value could rise or fall.

Solution

A is correct because an increase in the required return in Equation 2, all else equal, results in a decline in the value estimate:

$$IV_t = \frac{D_t(1+g)}{r_e - g}.$$

The required return on equity, r_e , is a positive term in the denominator of the equation. Accordingly, a higher r_e leads to a lower IV_t .

Dividend discount models are also applicable in industries operating under specific constraints that cause dividends to closely track earnings. For example, **real estate investment trusts (REITs)** own a portfolio of income-producing properties, as described in detail later in the curriculum. Under this structure, business activities are limited to real estate, and a REIT is legally required to distribute almost all of its annual taxable income to shareholders to avoid corporate taxes. Another sector in which dividend-based models are commonly applied is regulated utilities, as described in the following case study.

DIVIDENDS IN REGULATED INDUSTRIES: ENEL SPA

Utilities are firms that provide essential services, such as electricity, water, or natural gas, to residential and commercial customers. Given the importance of continuous power at predictable prices, utilities are heavily regulated by government, especially with respect to residential accounts. For example, utilities are restricted from competition, with transparent pricing structures and rates subject to regulatory approval. Such regulations constrain utility profit margins and growth potential, limiting the potential upside for equity investors. As a result, it is not uncommon for investors in utility shares to earn most of their return in the form of dividends.

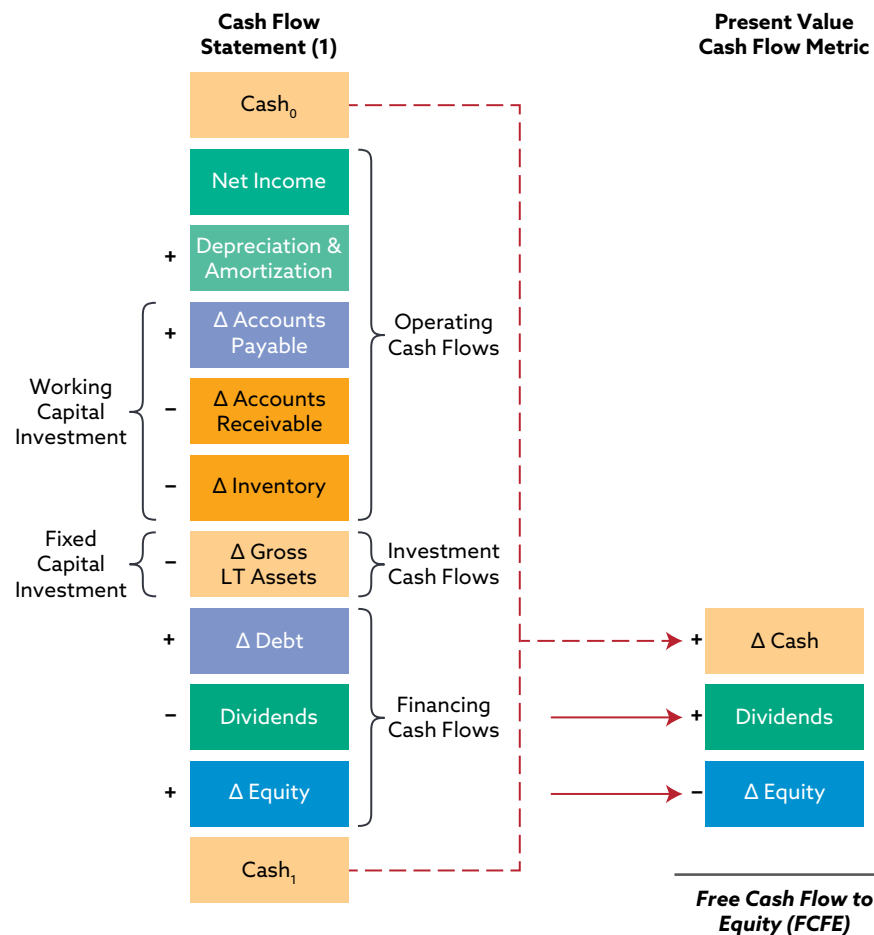
Enel SpA is an Italy-based utility operating in 30 countries globally with its primary revenue sources in Italy and the Iberian Peninsula. The company generates, transmits, and distributes electricity, in addition to distributing natural gas. Enel is also a leader in energy generation from renewable sources. The company paid a semiannual dividend of EUR0.215 per share in early 2024 at a time when Enel shares traded at approximately EUR6.60 per share. An investor in Enel shares therefore would expect to earn approximately 6.5% on an annualized basis from dividends. In comparison, Enel's total annual return compounded annually over the last 20 years is approximately 7.5%.

In the case of Enel SpA and other utilities, the price appreciation potential is often limited. As for other companies or industries, growth may be a more important component of dividend discount models, as shown in the next lesson.

While dividend-based models are a valid cash flow metric for firms whose earnings are predominantly paid out in cash, for other firms, they often provide an incomplete picture of total cash flow available to shareholders. Companies may pay few or no cash dividends to preserve investment flexibility or because of investor tax preferences. They may also increase debt outstanding. In these cases, dividends may not fully reflect future expected cash flows and therefore may possibly underestimate equity value. As a result, broader cash flow metrics are often more useful than dividends in valuing the equity of most companies.

Free Cash Flow to Equity

A broader measure of cash flow for valuing equity combines dividends with other forms of cash flow available to equity investors that are not necessarily distributed. Free cash flow to equity, introduced earlier, captures these additional cash flow elements, as shown in Exhibit 4.

Exhibit 4: FCFE as a Present Value Cash Flow Metric

FCFE is the remaining cash flow available to shareholders once all operating expenses, including taxes, are covered and all necessary investments for continuing operations and projected growth, both working capital and long-term fixed assets, are made. Recall from the prior learning module that FCFE is defined as follows:

$$\begin{aligned} &\text{Free cash flow to equity} \\ &= \text{Cash flow from operations} - \text{Capital expenditures} + \text{Net debt issuance}, \end{aligned} \quad (4)$$

where

$$\text{Cash flow from operations} = \text{Net income} + \text{Depreciation} - \Delta \text{Working capital}$$

and

$$\text{Net debt issuance } (\Delta \text{Debt}) = \text{Additional borrowing} - \text{Debt repayment}.$$

As shown in Exhibit 4, another way to express what remains available to shareholders is as follows:

$$\text{Free cash flow to equity (FCFE)} = \Delta \text{Cash} + \text{Dividends} - \Delta \text{Equity}, \quad (5)$$

where ΔEquity is equal to share issuance less repurchases. As a measure of shareholder return, note that FCFE includes cash flows after cash operating expenses, interest, and taxes.

KNOWLEDGE CHECK: VALUING EXCELITATE USING FCFE

Excelitate Corporation recently posted its annual financial results. An analyst has compiled the following selected data in US dollars:

Position	Value
Net income	4,800,000
Depreciation	700,000
Inventory change	+2,000,000
Accounts receivable change	+1,000,000
Accounts payable change	-1,000,000
Capital expenditures	2,000,000
New debt issuance	600,000

1. Which of the following is the closest estimate of Excelitate's free cash flow to equity?

- A. -500,000
- B. 100,000
- C. 1,500,000

Solution

B is correct because this result incorporates cash flow from operations as defined previously:

$$\begin{aligned}\text{Cash flow from operations} &= \text{Net income} + \text{Depreciation} - \Delta \text{Working capital} \\ &= 4,800,000 + 700,000 - [2,000,000 + 1,000,000 - (-1,000,000)] = 1,500,000.\end{aligned}$$

Following this, we can estimate FCFE using cash flow from operations, capital expenditures, and change in debt as shown in Equation 4:

$$\text{FCFE} = 1,500,000 - 2,000,000 + 600,000 = 100,000.$$

A is incorrect because this result neglects to add back the change in debt.

C is incorrect because this result includes only cash flow from operations.

2. The analyst notes that Excelitate paid USD2,000,000 in dividends and repurchased USD5,000,000 of its stock. Additionally, the company issued USD600,000 in new stock. Calculate the change in the company's cash position given the FCFE calculated in the prior question.

Solution

Since FCFE is USD100,000, we can apply Equation 5 to solve for the change in cash of USD6,300,000 as follows:

$$\text{Free cash flow to equity (FCFE)} = \Delta \text{Cash} + \text{Dividends} - \Delta \text{Equity}.$$

$$100,000 = \Delta \text{Cash} + 2,000,000 + 5,000,000 - \Delta \text{Equity}.$$

$$\Delta \text{Cash} = 2,000,000 + 5,000,000 - 600,000 - 100,000 = 6,300,000.$$

While changes in cash flow statement items have a clear impact on FCFE, its use in present value models to project *future* cash flows requires some assumptions to maintain consistency over time. For example, to isolate cash flows to shareholders, FCFE forecasts often assume that future company financing follows a “target” debt ratio (DR):

$$DR = \frac{\text{Total debt}}{(\text{Total debt} + \text{Total equity})} \quad (6)$$

The assumption of a target debt ratio simplifies the FCFE calculation by eliminating the need to forecast net borrowing to finance both capital expenditures that exceed depreciation and working capital investment. Assuming depreciation as the only non-cash charge, we may show FCFE by distinguishing between maintaining going-concern investments (depreciation) and capital growth investment:

$$FCFE = \text{Net income} - \text{Net investment} + \Delta\text{Debt}, \quad (7)$$

where

$$\text{Net investment} = (\Delta\text{Gross LT assets} - \text{Depreciation}) + \Delta\text{Working capital}.$$

A target debt ratio (DR) implies that net debt issuance (ΔDebt) rises proportionally to equity when financing business expansion and working capital increases:

$$\Delta\text{Debt} = DR \times \text{Net investment}. \quad (8)$$

Substituting ΔDebt into Equation 7 and solving for FCFE, we see that FCFE is net income minus the *proportion* of net increase in long-term assets and working capital investment financed by equity, since $(1 - DR) = \frac{\text{Total equity}}{(\text{Total debt} + \text{Total equity})}$:

$$FCFE = \text{Net income} - (1 - DR)(\text{Net investment}). \quad (9)$$

A company’s FCFE is equivalent to its dividends when the company pays out all available equity cash flows as dividends. However, the FCFE approach is also appropriate for firms with an inconsistent relationship between dividends and earnings or to those that pay no dividends at all. We can make this more explicit by restating Equation 5 as follows:

$$FCFE = \Delta\text{Cash} + \text{Dividends} + \text{Stock repurchases} - \text{Stock issuance}. \quad (10)$$

In addition to the effect of dividends, changes in a company’s equity are also affected by a company’s repurchase and issuance of new shares. Furthermore, changes in a company’s cash balance after covering necessary investments reflect cash that is theoretically available to shareholders but not distributed. Beyond dividends, firms with positive FCFE may repurchase stock or choose to increase cash balances for later use. In contrast, FCFE may be negative for a firm that issues stock.

Assuming a positive cash flow from operations, Exhibit 5 illustrates conditions under which FCFE is more likely to be positive or negative.

Exhibit 5: Features of Positive vs. Negative FCFE Companies

Feature	FCFE > 0	FCFE < 0
Capital expenditures	Lower capital expenditures and/or debt-financed capital expenditures	Large equity-financed capital expenditures relative to cash flow from operations
Debt policy	Growing debt balance	Large drawdowns in debt balance
Cash management	Growing cash balances	Large drawdowns in cash balances

Feature	FCFE > 0	FCFE < 0
Payout policy	Positive payout of either dividends or stock repurchases	No dividends or stock repurchases
Share issuance	Little to no share stock issuance	Large share issuance

Observations of a company's financing and capital expenditure activities provide simple clues to analysts seeking to predict whether such actions will result in an increase or decrease in FCFE in the future.

KNOWLEDGE CHECK: CHANGES TO EXCELITATE'S FCFE



An analyst examines the Excelitate Corporation in greater detail to determine how the FCFE picture may change given changes in certain inputs, as shown in the following "Revised" column:

Position	Original	Revised
Net income	4,800,000	---
Depreciation	700,000	---
Inventory change	2,000,000	---
Accounts receivable change	1,000,000	---
Accounts payable change	-1,000,000	1,000,000
Capital expenditures	2,000,000	1,500,000
New debt issuance	600,000	500,000

1. Which of the following is closest to the change in FCFE between the original and revised scenarios?

- A. +USD1,400,000
- B. +USD2,400,000
- C. +USD2,600,000

Solution

B is correct because the components of FCFE show the following three changes between the two scenarios:

- +2,000,000 in accounts payable and a positive change in a liability *increase* FCFE,
- -500,000 in capital expenditures and a decline in capital expenditures *increase* FCFE, and
- -100,000 in new debt issuance and a negative change in debt *decrease* FCFE.

Change in FCFE = 2,000,000 + 500,000 - 100,000.

A is incorrect because this result deducts, rather than adds, the change in capital expenditures.

C is incorrect because this result subtracts, rather than adds, the change in new debt issuance.

2. The analyst decides to alter the assumption about new debt issuance in the revised scenario. Based on statements about target debt usage from Exceli-

tate corporate management, the analyst assumes a 40% debt ratio. Calculate the FCFE under this new scenario.

Solution

Under this change to the debt assumption, Excelitate's FCFE is USD3,120,000 in the revised scenario, which may be calculated using Equation 9:

$$\text{FCFE} = \text{Net income} - (1 - \text{DR})(\Delta \text{Gross LT assets} - \text{Depreciation}) - (1 - \text{DR}) \Delta \text{Working capital}.$$

$$\begin{aligned} \text{FCFE} &= 4,800,000 - 0.6 \times (1,500,000 - 700,000) - 0.6 \times 2,000,000 \\ &= 3,120,000. \end{aligned}$$

3. The analyst believes that Excelitate could potentially support a 60% target debt ratio. Calculate an updated estimate of Excelitate's FCFE under this revised assumption.

Solution

Under this change to the debt assumption, Excelitate's FCFE is USD3,680,000 in the revised scenario. This may be calculated using Equation 9:

$$\text{FCFE} = \text{Net income} - (1 - \text{DR})(\Delta \text{Gross LT assets} - \text{Depreciation}) - (1 - \text{DR}) \Delta \text{Working capital}.$$

$$\begin{aligned} \text{FCFE} &= 4,800,000 - 0.4 \times (1,500,000 - 700,000) - 0.4 \times 2,000,000 \\ &= 3,680,000. \end{aligned}$$

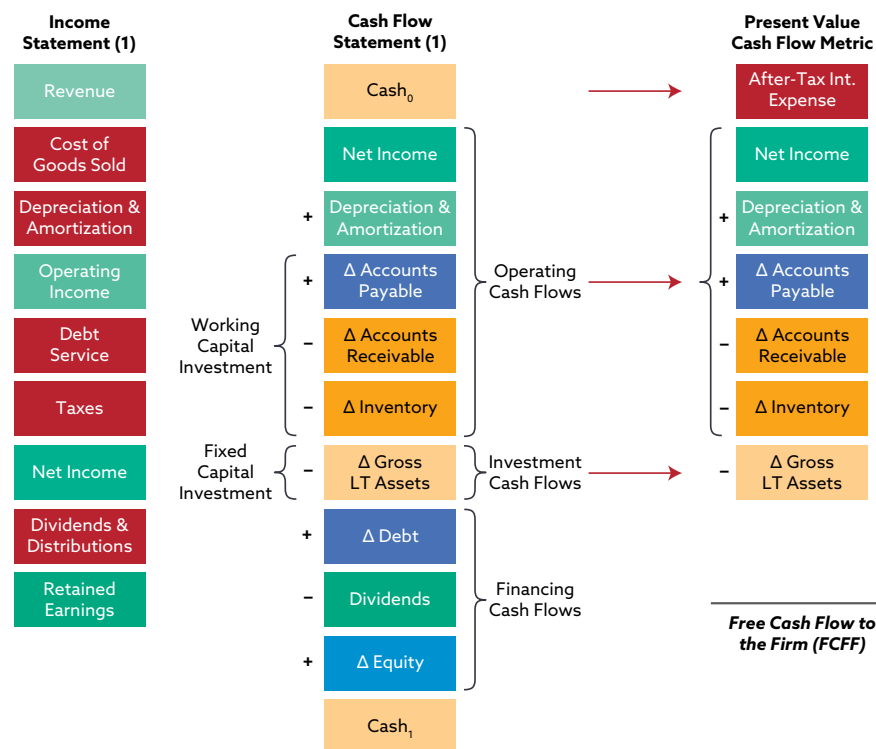
Because the analyst assumes greater debt issuance while holding investment expenditures for long-term and working capital assets constant, we observe that assumed FCFE increases.

The prior exercise highlights the important roles of both investment expenditures and financing choices in estimating a company's FCFE. Lower investment expenditure and higher proportions of debt used to finance investment lead to higher FCFE, holding net income constant.

The use of FCFE for valuing a company's equity is appropriate for controlling investors who can influence corporate policies, such as those shown in Exhibit 5. However, the FCFE model is less useful for companies with significant future capital requirements because of the possibility of negative FCFE. Additionally, companies with significant capital structure changes exhibit fluctuations in FCFE. For these types of companies, equity valuation may benefit from using a cash flow metric that does not rely on a target debt ratio as illustrated earlier.

Free Cash Flow to the Firm

Free cash flow to the firm is a broader measure than FCFE; it explicitly incorporates cash flows to both debt and equity investors. The components of FCFF are shown in Exhibit 6.

Exhibit 6: FCFF as a Present Value Cash Flow Metric

FCFF combines cash flow from operations (CFO) with capital expenditures and interest paid to debtholders:

$$\text{FCFF} = \text{Cash flow from operations} + \text{Interest expense} \times (1 - \text{Tax rate}) - \text{Capital expenditures.} \quad (11)$$

It is useful to draw a direct comparison between FCFF and FCFE to highlight differences between these two measures:

$$\text{FCFF} = \text{FCFE} + \text{Interest expense} \times (1 - \text{Tax rate}) - \Delta \text{Debt}, \quad (12)$$

where $\Delta \text{Debt} = \text{Debt issuance} - \text{Debt repayment}$.

For firms that rely solely on equity financing, FCFF and FCFE are equal. When debt financing is used, a firm's FCFF will exceed FCFE as long as debt outstanding remains unchanged. Changes in debt outstanding, however, can cause FCFF to fall below FCFE, as shown in the following exercise.

KNOWLEDGE CHECK: VALUING EXCELITATE USING FCFF

The analyst assessing Excelitate's cash flow metrics now turns her attention to the company's free cash flow to the firm. She begins by continuing to assume that the company's cash flow from operations is USD3,500,000 and capital expenditures are USD1,500,000. Excelitate's corporate income tax rate is 20%.

- Which one of the following is closest to Excelitate's FCFF if before-tax interest expense is USD840,000?

A. USD1.33 million

B. USD2.67 million

C. USD2.84 million

Solution

B is correct because, as shown in Equation 11, FCFF is equal to CFO *plus* after-tax interest expense minus capital expenditures. Since *before*-tax interest expense is USD840,000, we must first solve for after-tax interest expense:

$$\text{After-tax interest expense} = 840,000 \times (1 - 0.2) = 672,000.$$

Applying Equation 11, we solve for FCFF:

$$\text{FCFF} = 3,500,000 + 672,000 - 1,500,000 = 2,672,000.$$

A is incorrect because this result is computed by subtracting, rather than adding, after-tax interest expense, using Equation 11.

C is incorrect because this result is computed by adding before-tax interest expense instead of after-tax interest expense.

2. The analyst notes that Excelitate's FCFE from the scenario above is USD3,120,000 because she is assuming a 40% target debt ratio. Calculate Excelitate's change in debt given the FCFF estimate from the prior question.

Solution

Excelitate's change in debt in this scenario is USD1,120,000, as shown below using Equation 12.

$$\text{FCFF} = \text{FCFE} + \text{After-tax interest expense} - \Delta\text{Debt}.$$

$$2,672,000 = 3,120,000 + 672,000 - \Delta\text{Debt}.$$

$$\Delta\text{Debt} = 1,120,000.$$

Note that a change in debt of USD1,120,000 with a 40% target debt ratio implies net investment of 2,800,000 ($= 1,120,000 \div 0.4$), and this result is also consistent with the details of capital expenditures, depreciation, and working capital change used in earlier scenarios.

3. The analyst returns to an assumption of a 60% Excelitate debt ratio. Under this assumption, Excelitate's FCFE was USD3,680,000, as shown earlier. In this new scenario, Excelitate's before-tax interest expense is USD875,000. Calculate Excelitate's FCFF if these are the only changes from the earlier scenario.

Solution

In this scenario, Excelitate's FCFF is USD2,700,000, as shown below using Equation 11.

$$\text{FCFF} = \text{CFO} + \text{After-tax interest expense} - \text{Capital expenditures}.$$

$$\text{FCFF} = 3,500,000 + 875,000 \times (1 - 0.2) - 1,500,000 = 2,700,000.$$

Alternatively, we can derive this result using the USD2,800,000 in total investment and the target debt ratio to find ΔDebt of USD1,680,000 ($= 0.6 \times 2,800,000$). Then, we can apply Equation 12, as follows:

$$\text{FCFF} = 3,680,000 + 875,000 \times (1 - 0.2) - 1,680,000 = 2,700,000.$$

As shown in the previous exercise, shareholders initially benefit from increased debt due to higher leverage, potentially resulting in a higher FCFE than FCFF, although higher interest expense will increase future FCFF. The exercise also illustrates that FCFF estimates are much more stable compared to FCFE estimates as assumptions about debt usage are changed.

While FCFE-based forecasts often rely on a target debt ratio, FCFF may be directly applied to firms with varying capital structures that change over time. The flexibility of this cash flow metric also makes it the most suitable choice for controlling investors with influence over investment and debt policies.

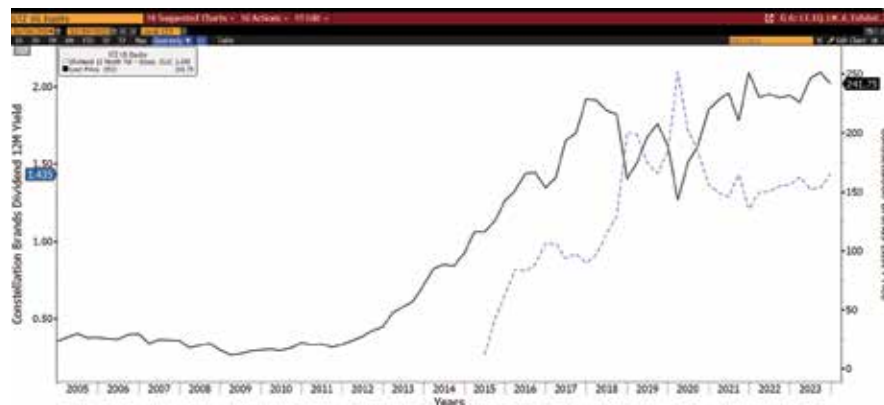
CASE STUDY



Constellation Brands (STZ) and Free Cash Flow to the Firm

Constellation Brands, Inc., is a major beer, wine, and spirits company based in North America. In addition to being the world's largest premium wine producer, in 2013, the company acquired from AB Inbev SA the license to produce, import, and sell Grupo Modelo beer brands in the United States. In contrast to the earlier example of Coca-Cola, following a period of growth through acquisitions, the company only recently began paying dividends, which have reflected a high degree of variability, as shown in Exhibit 7.

Exhibit 7: Constellation Brands Price and Dividend Growth, 2004–2023

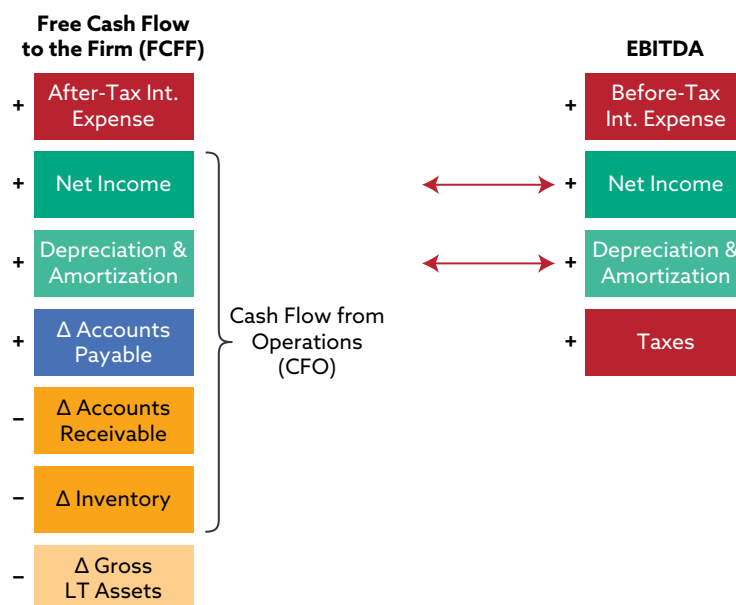


Source: Bloomberg.

In addition to its changing dividends, Constellation also significantly increased its reliance on debt financing since the 2013 Grupo Modelo acquisition and increased its share repurchase program. Constellation's changing leverage and distribution policy to shareholders make FCFF a more appropriate cash flow metric to consider for discounted cash flow valuation than dividends.

EBITDA vs. Free Cash Flow Measures

Earnings before interest, tax, depreciation, and amortization (EBITDA) is a broad measure that *includes* revenues minus operating expenses but *excludes* payments to debtholders, non-cash expenses, and taxes. Exhibit 8 compares the components of EBITDA and FCFF.

Exhibit 8: EBITDA vs. FCFF

Equation 13 shows how EBITDA differs from FCFF:

$$\text{FCFF} = \text{EBITDA}(1 - \text{Tax rate}) + \text{Depreciation}(\text{Tax rate}) - \Delta \text{Gross LT assets} - \Delta \text{Working capital.} \quad (13)$$

EBITDA considers cash flows solely related to operating performance and excludes capital expenditure and investment in working capital, as well as the effect of taxes. Advantages of EBITDA usage as a cash flow metric include its applicability to companies with net losses but positive operating income, as well as its focus on a firm's cash generation prior to any discretionary outlays. This allows for broader comparison between firms with different levels of financial leverage, tax rates, and investment expenditures.

Although frequently disclosed by companies and commonly used by analysts in equity valuation, EBITDA and adjusted EBITDA are not recognized earnings measures under GAAP or IFRS accounting standards. **Adjusted EBITDA** is defined as EBITDA plus one-time, non-recurring expenses or minus non-operating or non-recurring income. The lack of consensus surrounding what may be included in or excluded from these adjustments allows those firms seeking to overstate financial performance to make arbitrary adjustments to these figures, as shown in the following example.

WEWORK'S "COMMUNITY ADJUSTED EBITDA" MEASURE

WeWork Inc., a US-based company founded in 2010 that provides co-working spaces for firms and individuals, funded its initial rapid expansion into office leasing using private equity capital. Despite an asset-light business model and negative cash flow, it was able to issue public bonds in April 2018 and soon thereafter sought to publicly issue equity via an IPO. Based on the preliminary IPO prospectus filed in late 2018, WeWork's 2017 financial statements showed revenue of USD866 million and a net loss of USD933.5 million, with the following adjusted EBITDA disclosure.

**WeWork Companies Inc. 2018 Form S-1 Registration Statement
(Excerpt)**

Amounts in USD thousands, as of December 31st	2016	2017
Net loss	(429,690)	(933,494)
Income tax (benefit) provision	16	(5,727)
Interest and other (income) expense net	33,400	7,387
Depreciation and amortization	88,952	162,892
Adjustments for impact of straight-lining of rent	188,746	272,927
Stock-based compensation expense	22,660	295,362
Stock-based payments for services rendered by consultants	1,594	7,326
Adjusted EBITDA	(94,322)	(193,327)
Other revenue	(1,744)	(19,106)
Other operating expenses	---	1,322
Sales and marketing	42,653	139,180
Growth and new market development	33,245	98,336
Pre-opening community expenses	21,223	23,039
Adjusted EBITDA before growth investments	1,055	49,444
General and administrative expenses	94,888	183,703
Community adjusted EBITDA	95,943	233,147

Source: SEC Form S-1, WeWork Companies Inc., 28 December 2018. www.sec.gov/Archives/edgar/data/1533523/000162827918000370/filename1.htm.

By adding back a wide range of costs beyond depreciation and amortization, including rent holidays, stock-based compensation, growth and market development, and general and administrative expenses, which the company claimed were unrelated to the ongoing operation of its co-working office space, WeWork was able to shift its “community adjusted EBITDA” measure to positive growth despite mounting losses.

Using adjusted EBITDA as a cash flow metric, value estimates were as high as USD50 billion, but WeWork’s widely anticipated IPO was postponed in 2019 due to market perception of inflated growth and earnings expectations. The company made subsequent management changes, cut costs, and benefited from the trend toward hybrid office space usage during the COVID-19 pandemic. WeWork subsequently went public via an IPO two years later but at a much lower market value, USD9 billion.

Cases such as the previous example have led many prominent investors, such as Warren Buffett of Berkshire Hathaway, to express skepticism over the use of EBITDA and adjusted EBITDA as measures of financial performance, considering them to be incomplete and in some cases misleading gauges of cash flow. Academic studies suggest that firms using EBITDA in financial reporting tend to be smaller, use more financial leverage, have more fixed assets and longer operating cycles, and tend to be less profitable than those that solely use standardized earnings measures. In cases where EBITDA is an available alternative cash flow metric, these findings support closer scrutiny of this metric versus standardized measures of earnings, as well as any adjustments.

Depreciation charges as a percentage of EBITDA can also vary among companies and industries. It is also important to note that EBITDA is a *before-tax* measure, so unlike the after-tax weighted average cost of capital (WACC), any discount rate applied to EBITDA must be before taxes.

KNOWLEDGE CHECK: EXCELITATE'S EBITDA



Given that its earnings disclosures include EBITDA, the analyst reviewing Excelitate is asked about using this measure as a potential alternative to measuring the company's FCFF. To address the question, she shows the following financial information for Excelitate Corporation in US dollars, also noting Excelitate's 20% corporate income tax rate.

Position	Value
Net income	4,800,000
Depreciation	700,000
Interest expense	840,000
Capital expenditures	1,500,000
FCFF	2,672,000

1. Which of the following is closest to Excelitate's EBITDA?

- A. USD6,340,000
- B. USD7,300,000
- C. USD7,540,000

Solution

C is correct because a computation of EBITDA adds the effect of taxes, interest, and depreciation to the company's net income, as follows:

$$\text{EBITDA} = \text{Pretax income} + \text{Interest} + \text{Depreciation}.$$

$$\text{EBITDA} = 4,800,000 / (1 - 0.2) + 840,000 + 700,000 = 7,540,000.$$

Excelitate generates USD6,000,000 of pre-tax income, as shown by the first term, and pays 1,200,000 ($= 0.2 \times 6,000,000$) in taxes. The final two terms reflect interest and depreciation.

A is incorrect because it is the result if we fail to incorporate the tax rate.

B is incorrect because it is the result if we use net income multiplied by 1 plus the tax rate to get the pretax income.

2. Which of the following statements is most correct about Excelitate's EBITDA and FCFF if the analyst assumes that capital expenditures increase by USD2,000,000 (from 1,500,000 to 3,500,000) while everything else remains unchanged.

- A. Both EBITDA and FCFF decrease by 2,000,000.
- B. EBITDA decreases by 2,000,000, and FCFF does not change.
- C. EBITDA does not change, and FCFF decreases by 2,000,000.

Solution

C is correct because capital expenditures do not affect EBITDA calculations. As a result, a change in capital expenditure will not affect EBITDA. However, capital expenditures are a direct component of the FCFF calculation.

Holding all else equal (including depreciation), FCFF declines with each extra unit of currency invested in capital expenditures.

As the prior exercise demonstrates, EBITDA does not provide a complete cash flow measure because it does not incorporate information about corporate investment.

Residual Income

Present value cash flow metrics are typically derived from cash flow statement entries. However, residual income measures income rather than cash flow. The residual income metric aims to quantify the profit a firm generates *in excess of* the opportunity cost of its equity capital, which is also referred to as **economic profit**. That is, while a company may have positive net income, it may not add value for shareholders unless it earns a return on equity that exceeds the cost of its equity (r_e).

As shown in Exhibit 9, residual income represents net income less a charge (deduction) for common shareholders' opportunity cost in generating net income. The per-share residual income in period t , RI_p , is defined as

$$RI \text{ per share}_t = EPS_t - r_e \left(\frac{BVE}{\text{Share}} \right)_{t-1}, \quad (14)$$

where EPS_t represents earnings per share for period t , r_e is the required rate of return on equity, and BVE/Share is the book value per share at the beginning of the period, or at time $t - 1$.

Exhibit 9: Residual Income as a Present Value Cash Flow Measure

Income Statement (1)		Cash Flow Statement (1)		Present Value Cash Flow Metric
Revenue		Cash ₀		
Cost of Goods Sold		Net Income	→	$\frac{NI_t}{r_e \times (BVE)_t}$
Depreciation & Amortization	+	Depreciation & Amortization		
Operating Income	+	Δ Accounts Payable		
Debt Service	-	Δ Accounts Receivable		
Taxes	-	Δ Inventory		
Net Income	-	Δ Gross LT Assets		
Dividends & Distributions	+	Δ Debt		
Retained Earnings	-	Dividends		
	+	Δ Equity		
		Cash ₁		

The RI valuation method relies on an assumption of clean surplus accounting, in which book value of equity includes only the effects of income statement items and cash flows to and from shareholders (i.e., dividends, stock repurchases, and issuances). Any other surplus items that flow directly into equity, such as foreign currency translation gains and losses, require an adjustment to the book value of equity in the calculation.

While RI is not a cash flow metric, its reliance on the clean surplus relationship and the measurement of BVE as reflecting elements of FCFE implies that the result of an RI valuation should in theory be similar to an FCFE valuation.

Residual income may be a useful valuation metric in a variety of circumstances because, like FCFE and FCFF models, RI models do not require a firm to distribute cash flows. Unlike FCFE and FCFF models, RI models may be useful for companies who are currently generating negative free cash flows.

KNOWLEDGE CHECK: VALUING EXCELITATE USING RESIDUAL INCOME


The analyst studying Excelitate decides to consider a residual income measure as a possible equity valuation tool. Recall from an earlier knowledge check that Excelitate's net income is 4,800,000 and the analyst finds that Excelitate's book value of equity (BVE) is USD14,000,000. Excelitate currently has 1,000,000 shares outstanding.

1. If Excelitate's cost of equity, r_e , is 8.5%, which of the following is closest to its residual income per share?

- A. USD4.70
B. USD3.60
C. USD4.40

Solution

B is correct because we can derive residual income per share from Equation 13 as follows:

$$RI = \frac{4,800,000}{1,000,000} - 0.085 \times \left(\frac{14,000,000}{1,000,000} \right) = 3.610.$$

A is incorrect because it is the approximate result if we simply subtract the cost of equity from EPS.

C is incorrect because it is the approximate result if we multiply $(1 - 0.085)$ by the company's EPS.

Exhibit 10 summarizes basic features of each valuation metric discussed in this lesson and provides some key contrasts.

Exhibit 10: Features and Contrasts of Valuation Metrics

Present value metrics	Description	Unique features
Dividends	Cash flow distributed to shareholders only Reflects cash flows received by all shareholders	Easiest measure to observe Single measure of shareholder payout Value can only be zero or positive
FCFE	Total cash flow <i>available</i> to shareholders Includes share buybacks and cash accumulation Incorporates stock issuances (i.e., negative cash flows) and changes in cash balances	More comprehensive than dividends Does not necessarily reflect cash flows paid to all shareholders Sensitive to changes in debt Value may be positive or negative
FCFF	Extends FCFE to include cash flows to debtholders and preferred shareholders Measures firm value, rather than only equity value	Most comprehensive metric of company cash flows Invariant to changes in debt Suitable for controlling investors
EBITDA	Cash flows from operations excluding capital expenditure, investment in working capital, and the effect of taxes Measure of cash generation prior to discretionary outlays	Applies to companies with net loss, positive operating income Allows comparison across firms with different leverage, taxes, and investment expenditure Non-standardized earnings measure, with no universal agreement on adjustments
RI	Metric reflecting excess earnings (i.e., economic profits)	Not a cash flow metric Incorporates book values and costs of equity

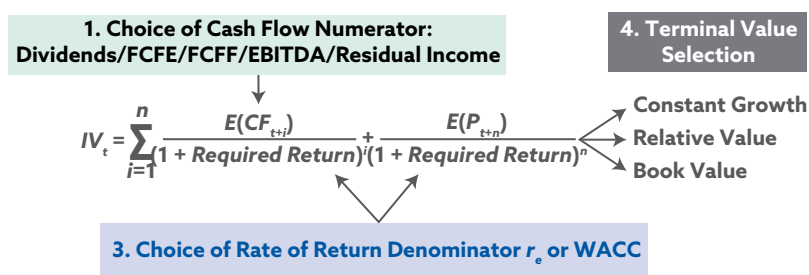
Valuation Process

A well-executed process used to estimate intrinsic value based on discounting expected future cash flows (or residual income) over a finite time horizon is structured as follows:

1. Select an appropriate cash flow measure.
2. Forecast the cash flow measure over an appropriate time horizon.
3. Determine an appropriate required rate of return for all cash flows.
4. Establish a terminal value.
5. Establish an intrinsic value (IV_t) *point estimate* (based on a single set of assumptions) or *range* (using scenario analysis) for comparison with the current share price and other value estimates.

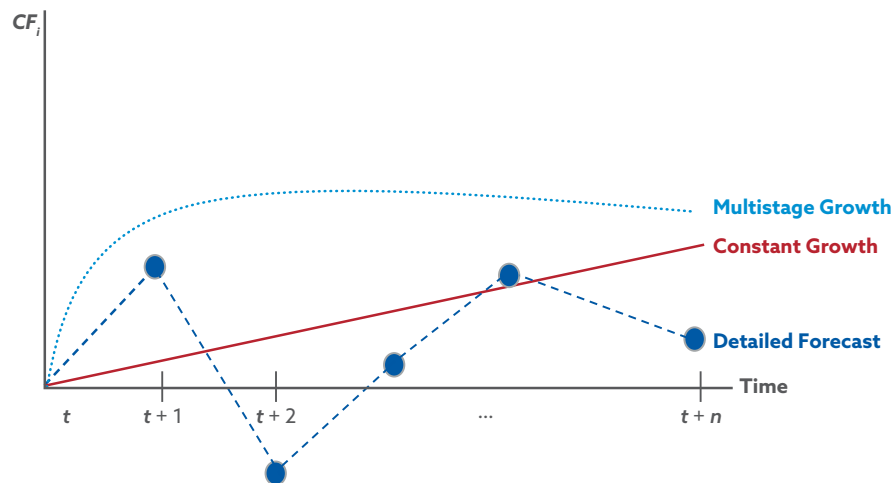
Exhibit 11 outlines the first step of choosing a cash flow metric.

Exhibit 11: Choosing a Cash Flow Metric



The choice of an appropriate cash flow metric in the prior section corresponds to the selection of a numerator in Equation 1 for the forecast period. An analyst must review the valuation process for each cash flow measure selected.

Once an appropriate numerator is selected, as a second step, an analyst must select a method to project the cash flow measure over the forecast time horizon, as shown in Exhibit 12.

Exhibit 12: Cash Flow Forecast Method

In this learning module, we consider the choice of either a constant growth rate or a changing (or so-called multistage) growth rate to forecast expected future cash flows over a specific time horizon. This simple approach of using a cash flow metric rate of change is extended in a later learning module to a detailed forecast approach, where we allow for changing relationships between revenues, expenses, and assets by forecasting the underlying drivers of cash flow in the financial statements. For example, a mature firm in a cyclical industry may face cash flow growth fluctuations over a business cycle, as shown in Exhibit 12.

Once a cash flow forecast method is established, Steps 3 and 4 of the process involve specifying the appropriate discount rate over a forecasted time horizon and a terminal value capturing all remaining future cash flows to complete the intrinsic value calculation.

Exhibit 13: Discount Rate and Terminal Value Selection

4. Terminal Value Selection

$$IV_t = \sum_{i=1}^n \frac{E(CF_{t+i})}{(1 + \text{Required Return})^i} + \frac{E(P_{t+n})}{(1 + \text{Required Return})^n}$$

Constant Growth
 Relative Value
 Book Value

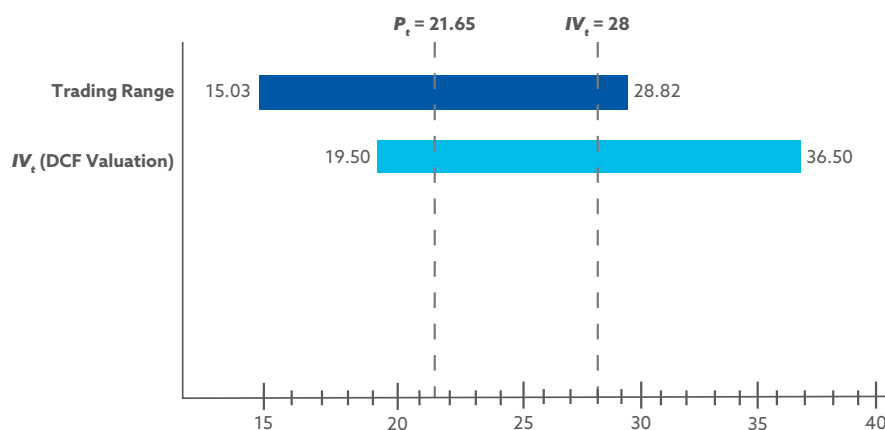
3. Choice of Rate of Return Denominator r_e or WACC

The third step of the process involves choosing an appropriate discount rate for the cash flow in the numerator of the intrinsic value calculation, as shown in Exhibit 13. As discussed earlier, cash flows that are distributed or available exclusively to shareholders are discounted by the required return on equity, r_e , while those available to both debtholders and equityholders are discounted by the weighted average cost of capital, WACC. In a later learning module, we consider detailed methodologies for estimating the required return on equity.

The fourth step in the process involves estimating a terminal value at the end of the forecast horizon. In this learning module, we focus on present value methodology and use a constant growth assumption to estimate terminal values. In the next Equities learning module, which addresses relative valuation, we extend the method of comparables using market multiples to estimate terminal values. Once the components of an intrinsic value calculation have been specified, we can solve for IV_t .

In the fifth and final stage of the valuation process, we compare value estimates to the recent trading range by either using different assumptions associated with the same valuation metric or using different valuation metrics, as shown in Exhibit 14.

Exhibit 14: Establishing a Forecast Valuation Range



QUESTION SET 1



1. Steinschlager AG pays stable dividends of EUR1.50 on an annual basis. The consensus among equity analysts is that the share price will reach EUR20 at the end of five years. Which of the following is closest to today's intrinsic value if the required return on equity is 10.0%?

- A. EUR17.17
- B. EUR18.10
- C. EUR25.68

Solution

B is correct. Using Equation 1, where $E(CF_i)$ is equal to EUR1.50 for five years and $E(P_{t+5})$ is equal to EUR 20,

$$IV_t = \sum_{i=1}^5 \frac{1.5}{(1.1)^i} + \frac{20}{(1.1)^5} = \text{EUR}18.10.$$

A is incorrect because it ignores the final dividend payment.

C is incorrect because it fails to properly discount the terminal value to solve for its present value.

2. The analyst is told that Steinschlager's terminal value may vary from EUR15 to EUR28 and that the required return on equity can range from 7.5% to 15%. Steinschlager AG pays stable dividends of EUR1.50 on an annual basis.

Which of the following best defines the range of possibilities for intrinsic values?

- A. EUR12.49 to EUR25.57
- B. EUR14.97 to EUR18.95
- C. EUR16.52 to EUR18.95

Solution

A is correct. It is derived by first entering 15% for r_e and EUR15 for $E(P_{t+5})$, the terminal value, into Equation 1 to find the lower bound of the range and r_e of 7.5% and $E(P_{t+5})$ of EUR28 for the upper bound, as follows:

$$IV_L(\text{Lower bound}) = \sum_{i=1}^5 \frac{1.5}{(1.15)^i} + \frac{15}{(1.15)^5} = \text{EUR}12.49.$$

$$IV_U(\text{Upper bound}) = \sum_{i=1}^5 \frac{1.5}{(1.075)^i} + \frac{28}{(1.075)^5} = \text{EUR}25.57.$$

B is incorrect because it uses r_e of 15% and $E(P_{t+5})$ of EUR20 to derive the lower bound and r_e of 15% and $E(P_{t+5})$ of EUR28 to derive the upper bound.

C is incorrect because it uses r_e of 7.5% and $E(P_{t+5})$ of EUR15 for the terminal value to derive the lower bound and r_e of 15% and $E(P_{t+5})$ of EUR28 to derive the upper bound.

3. Suppose an analyst is valuing a company using FCFE and uses the company's required return on equity as the discount rate. If the analyst is directed to use WACC instead of required return on equity and the company makes use of both debt and equity financing, which of the following statements is *most correct*?

- A. The updated estimated intrinsic value will decrease, and this value estimate is appropriate.
- B. The updated estimated intrinsic value will increase, and this value estimate is not appropriate.
- C. The updated estimated intrinsic value will neither rise nor fall, and this value estimate is not appropriate.

Solution

B is correct because we should expect WACC to be less than the required return on equity, r_e , because WACC is a blended rate of return for equity and debt. Given that debt has lower risk than equity, the debt component would make WACC less than the required return on equity. We therefore discount at a lower rate, and the valuation would increase. However, FCFE reflects cash flow to shareholders, so the discount rate change to WACC from required return to equity creates an inconsistency between the numerator and the denominator of the valuation. So, this updated estimate is not an appropriate one.

A is incorrect because it would result if WACC were to exceed r_e . Additionally, the updated value estimate is not appropriate, as explained above.

C is incorrect because the question states that the company uses debt and equity. WACC will equal r_e only in a case where a company uses no debt and therefore would have an unchanged discount rate and valuation.

GROWTH ASSUMPTIONS IN PRESENT VALUE MODELS

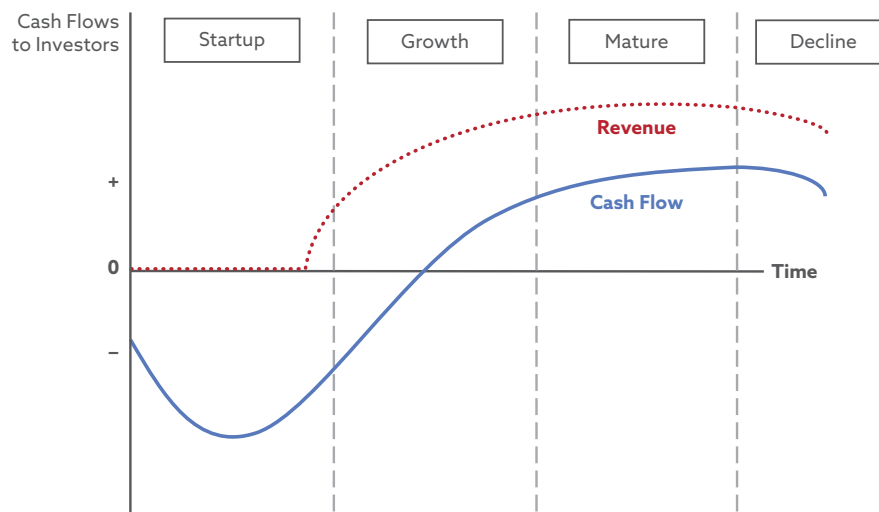
3

- calculate and interpret the intrinsic value of an equity security based on present value models that assume constant cash flow growth or multistage cash flow growth

Once a cash flow input is selected, the next step in the valuation process involves forecasting this cash flow metric into the indefinite future. Here, we will focus on the use of constant or changing rate of growth assumptions, addressing detailed financial statement forecasts in a later learning module. These forecasted cash flows are discounted at an appropriate rate of return and combined with a discounted terminal value capturing cash flows beyond the forecast period to provide point estimates and ranges of company value.

In the first Equities learning module, we introduced the company life cycle shown in Exhibit 15. The use of present value models varies across different stages of the cycle.

Exhibit 15: Company Life-Cycle Stages



For example, startup firms often have little or no revenue and net losses, making it challenging to identify an appropriate cash flow metric. While a few are highly successful and enjoy meteoric growth, many startups fail. As a result, standard present value model techniques based on past cash flow measures are generally not useful in valuing such firms. Once a firm enjoys initial success in establishing its products, services, and customers, it enters a high-growth phase as the market expands. Over time, this growth levels off as firms reach a mature phase with more stable cash flows and profitability, with some firms eventually entering a phase of decline.

In what follows, we consider both constant and multistage growth assumptions in estimating a firm's intrinsic value, as well as the choice of a discount rate and terminal value.

Constant Growth Models

Mature, well-established firms that exhibit stable cash flows are those for which an assumption of a constant future growth rate (g) is most appropriate. Stable-growth firms typically represent average risk relative to the market, a debt ratio approaching the industry average, and a return on capital close to their weighted average cost of capital. Given fewer new investment opportunities, these firms also tend to have lower capital expenditures and higher dividend payout ratios than firms in a high-growth phase.

As shown in the prior lesson in Equation 3 using a dividend discount approach, the present value of cash flows that are assumed to grow at a fixed rate forever into the future may be expressed as a perpetuity. Since constant growth may be applied to any of the valuation metrics, Equation 15 shows a generalized version of the constant growth model:

$$IV_t = \frac{E(CF)_{t+1}}{\text{Required rate of return} - g} \quad (15)$$

As in the case of the dividend discount model, an analyst may use the most recent measure, CF_t , and apply the growth rate, g , as follows:

$$E(CF)_{t+1} = CF_t \times (1 + g).$$

Alternatively, an analyst may use a current estimate of the next period's cash flow measure, as in the following example.

KNOWLEDGE CHECK: EXCELITATE'S INTRINSIC VALUE BASED ON CONSTANT GROWTH



The Excelitate analyst begins to estimate an intrinsic value of the company's share price using expected free cash flow to equity.

- Excelitate's next-period FCFE is expected to be USD4.2 million, and the expected growth rate is 4.3%. If the required return on equity is 8.5%, which of the following is closest to the intrinsic value of Excelitate's equity?

- A. USD32.8 million
- B. USD49.4 million
- C. USD100.0 million

Solution

C is correct because applying Equation 15 with an expected next-period cash flow $E(CF_{t+1})$ of USD4,200,000, a required return r_e of 8.5%, and an expected long-run growth rate g of 4.3% provides a solution of USD100,000,000:

$$IV_t = \frac{E(CF)_{t+1}}{\text{Required rate of return} - g}$$

$$100\text{million} = \frac{4.2\text{ million}}{0.085 - 0.043}$$

A is incorrect because it is the result if we incorrectly divide the FCFE by the sum of the required return on equity and the growth rate.

B is incorrect because this result happens if we divide the FCFE by the required return on equity.

2. If Excelitate has 1,500,000 shares of stock outstanding and USD25,000,000 of debt, which of the following is closest to the company's intrinsic value per share?

- A. USD50.00
- B. USD67.00
- C. USD83.00

Solution

B is correct because the intrinsic value per share calculated in the prior question reflects the intrinsic value of Excelitate's equity in terms of market capitalization. To translate this value into a per share value, we must divide this measure by the number of shares outstanding.

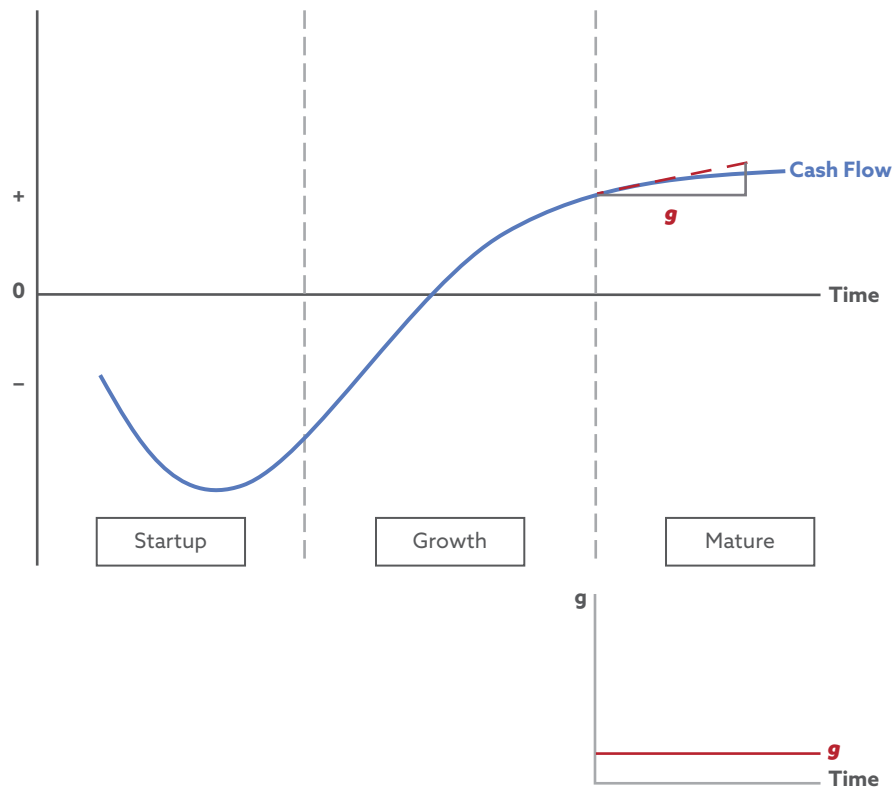
$$\text{Intrinsic value of equity per share} = \frac{100,000,000}{1,500,000} = 66.67.$$

A is incorrect because this result assumes that debt should be deducted from the intrinsic value of equity.

C is incorrect because this result assumes that debt should be added to the intrinsic value.

When a constant growth assumption is applied to the most recent or next period's cash flow metric, the numerator of Equation 15 also serves as the terminal value.

The simplicity and lack of a forecast time horizon makes the use of a constant growth assumption appealing. However, analysts must exercise caution in using this approach because the intrinsic value estimate is highly sensitive to the choice of g . For example, not only must the constant growth rate be below the required rate of return (WACC or r_e) used to discount cash flows for the intrinsic value to be a positive number, it should also not exceed the growth rate of the underlying economy. While some firms can maintain higher rates of growth for extended periods, all firms will approach stable growth at some future date. Exhibit 16 illustrates the constant growth rate in the context of the company life cycle.

Exhibit 16: Constant Growth Assumption for Mature Firms

From a macro perspective, when overall economic activity comes from both high-growth and stable-growth industries and firms, a realistic growth rate for a stable-growth firm should be below that of the economy. Firms in a phase of decline may have a negative growth rate. Also, since both cash flow measures and discount rates are typically estimated using nominal values, it is important to maintain consistency by applying a nominal constant growth rate denominated in the same currency as the valuation. The risk-free rate of return can be used to approximate the nominal growth rate in an economy.

An alternative application of constant growth approach involves using the observed market share price and estimated required rate of return to solve for an **implied constant growth rate**, or the hypothetical rate of steady growth into the indefinite future necessary for the intrinsic value to equal the current market value. Here, we substitute P_t for IV_t in Equation 15 and solve for the implied constant growth rate ($g_{implied}$) instead:

$$P_t = \frac{E(CF)_{t+1}}{\text{Required rate of return} - g_{implied}} \quad (16)$$

KNOWLEDGE CHECK: IMPLIED GROWTH FOR EXCELITATE'S COMPETITOR



The Excelitate analyst shifts her attention to a competitor, Lotormate. She expects Lotormate's FCFE to reach USD8.40 per share next year, with a required cost of equity of 8.5%. Lotormate's market share price is USD200.

1. Which of the following is closest to Lotormate's implied FCFE growth rate?

- A. 4.3%
- B. 8.5%
- C. 12.7%

Solution

A is correct because a 4.3% implied growth rate can be shown to be consistent with the current share price using Equation 16. We can rearrange Equation 16 to solve directly for the implied growth rate as follows:

$$P_t = \frac{E(CF)_{t+1}}{\text{Required rate of return} - g_{\text{implied}}}$$

$$g_{\text{implied}} = \text{Required rate of return} - \frac{E(CF)_{t+1}}{P_t}$$

$$0.043 = 0.085 - \frac{8.4}{200}$$

B is incorrect because this response is the required return on equity, r_e .

C is incorrect because this result occurs if we add the required return on equity to the ratio of expected cash flow over the intrinsic value.

As in the case of constant growth, we can solve for implied growth using any of the cash flow metrics introduced earlier. In reducing the growth expectations embedded in a current market price to a single growth factor, analysts can more easily compare these measures across companies within an industry, among industries, or to overall economic growth.

Most companies and industries exhibit changing growth dynamics due to competitive and market forces. Under these circumstances, assuming constant growth based solely on a cash flow measure from the previous year or next year's forecast is an oversimplification that can lead to incorrect or misleading valuation results.

A more realistic modeling approach is to quantify these cash flow dynamics over an expected time horizon (n periods, as shown in Equation 1) that can represent a significant proportion of a firm's value before it reaches steady growth. To incorporate these variations over time, we turn our attention next to *changing* rates of growth.

Multistage Growth Models

Multistage growth models are applicable when a firm's expected cash flow growth varies over time. These changes may be associated with a growth phase in which a company expands rapidly as it enters new, growing markets, eventually slowing to a more stable, constant rate of growth. The following example illustrates the growth trajectory of Starbucks Corporation.

CASE STUDY

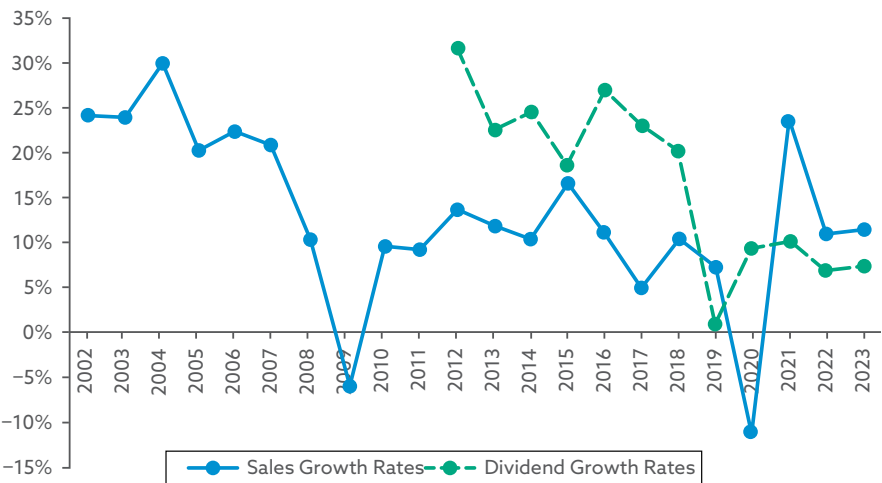


Starbucks: Historical Revenue Growth vs. Cash Flow Metrics

Starbucks Corporation, the global coffee retailer, began its growth trajectory in the late 1980s, ending 1989 with 55 stores operating exclusively in the United States. Starbucks' stock went public in June 1992, initiating an expansion strategy to end the decade with approximately 2,500 stores. This rapid store and revenue growth trend continued until the Global Financial Crisis of 2008–2009 affected consumer demand. Starbucks hit a temporary peak in the number of

stores at just under 17,000 in 2008–2010. Annual revenue growth rates fell from over 20% before the crisis to roughly half the earlier pace over the following decade. As growth slowed, Starbucks began paying regular dividends to its shareholders, with dividend growth exceeding revenue growth rates before leveling off in 2019. Exhibit 17 shows annual growth rates in Starbucks' revenue and dividends through 2023.

Exhibit 17: Starbucks' Revenue and Dividend Growth, 2002–2023

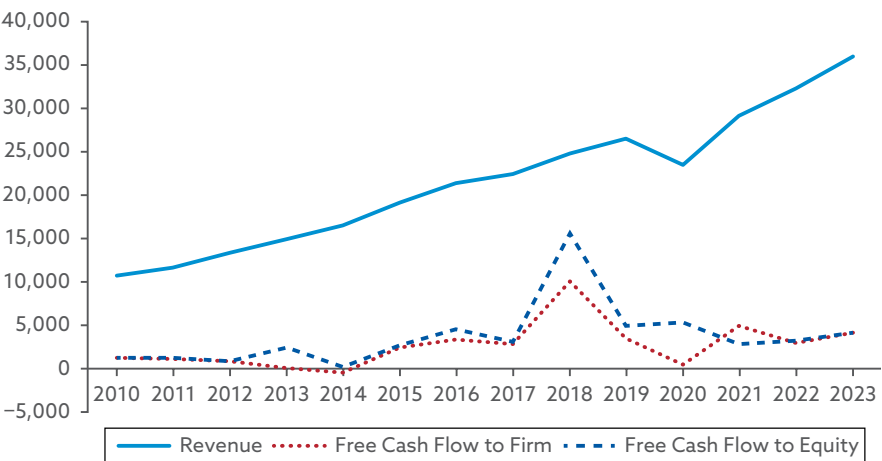


Source: Bloomberg.

Starbucks also changed its capital structure by greatly increasing its reliance on debt financing starting in 2018 (which rose from under USD4 billion in 2017 to over USD20 billion in 2020) and increasing share buybacks over the same period.

When valuing a company such as Starbucks using a present value model, it is important to avoid confusing revenue growth with cash flow growth. Exhibit 18 compares these measures over the slower-growth period starting in 2010.

Exhibit 18: Starbucks Revenue and Cash Flow, 2010–2023 (USD millions)



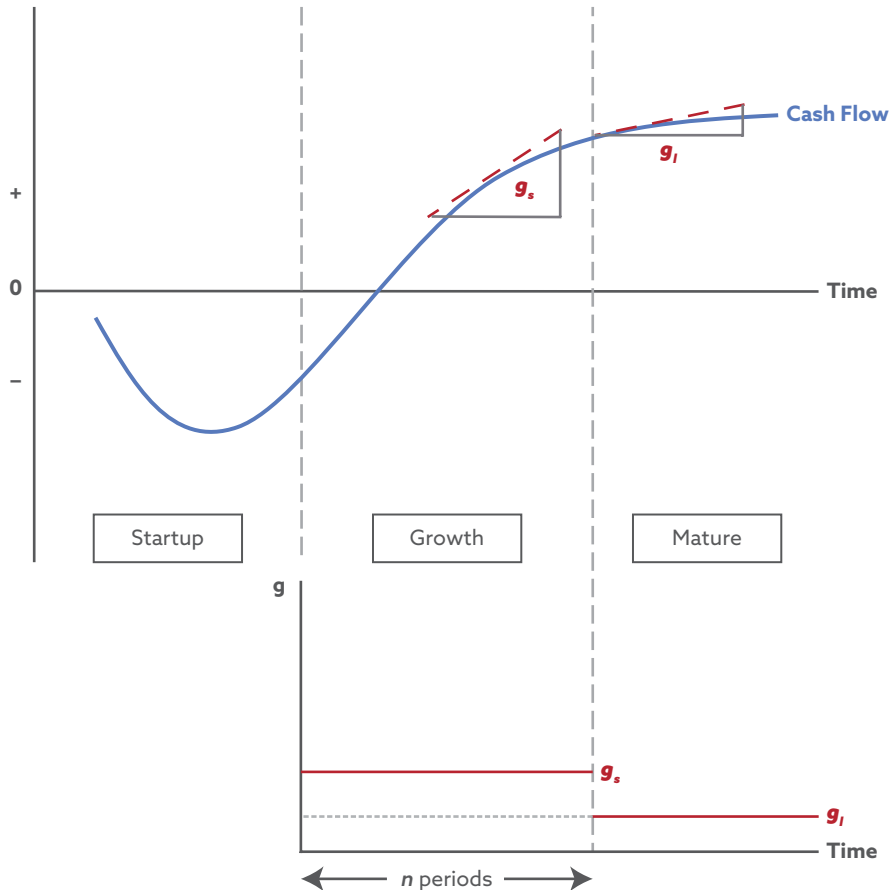
Source: Bloomberg.

A company's growth phase may assume a *single* rate of higher growth for n periods, as shown next in the two-stage case, or as many as n different growth rates—one for each period—depending on the expected trajectory of cash flow growth. Once a firm reaches stable growth, a terminal value is established to capture remaining cash flows. The rationale and mechanics associated with multistage growth are addressed in this lesson, but analysts must validate its use based on detailed company and industry analysis covered in a later learning module.

Two-Stage Growth Models

The simplest form of multistage growth involves an initial phase of constant (above- or below-trend) growth followed by a different long-term stable rate. For example, if a firm has started to generate positive cash flow as it enters a growth stage, an analyst may assume an initial higher short-term cash flow based on the flow growth rate (g_s) for a finite period followed by an indefinite period of lower sustained growth at a long-term rate (g_l). Exhibit 19 illustrates the structure of a two-stage growth model.

Exhibit 19: Two-Stage Growth Model Structure



The assumption of initial-stage growth at a higher rate of g_s for n periods followed by a second stage of slower growth at g_l results in a modified version of the constant growth model shown earlier. Using FCFE as the cash flow measure, we can solve for intrinsic value using Equation 17:

$$IV_t = \sum_{i=1}^n \frac{FCFE_t(1+g_s)^i}{(1+r_e)^i} + \frac{\text{Terminal value}}{(1+r_e)^n}. \quad (17)$$

The second expression in Equation 17 represents the present value of a terminal value that assumes constant growth of g_l starting in n periods, for which we can substitute the geometric series simplification:

$$IV_t = \sum_{i=1}^n \frac{FCFE_t(1+g_s)^i}{(1+r_e)^i} + \frac{IV_{t+n}}{(1+r_e)^n}, \quad (18)$$

where the intrinsic value of the stock in n periods (IV_{t+n}) equals

$$IV_{t+n} = \frac{FCFE_{t+n+1}}{r_e - g_l}. \quad (19)$$

KNOWLEDGE CHECK: TWO-STAGE GROWTH ASSUMPTION FOR EXCELITATE



More detailed research on company growth prospects leads the Excelitate analyst to the conclusion that she should use a two-stage growth model. Specifically, she assumes an initial FCFE growth rate of 12% annually for the first three years, followed by stable 4% growth thereafter. Excelitate's most recent FCFE is USD3,500,000, and the company's required return on equity is 8.5%. The company has 1,500,000 shares outstanding.

- Which of the following is closest to the intrinsic value per share under the above two-stage growth assumption?

- A. USD62.13
- B. USD66.78
- C. USD83.22

Solution

B is correct, based on the following process. First, we must determine FCFE for the next four years:

	Year 1	Year 2	Year 3	Year 4
FCFE	3,920,000	4,390,400	4,917,248	5,113,938

For Years 1–3, we simply calculate it as

$$FCFE_t = 3,500,000 \times (1 + 0.12)^t.$$

For Year 4, the first year with the long-term growth rate, we have

$$FCFE_4 = 3,500,000 \times (1 + 0.12)^3 \times (1 + 0.04)^1 = 5,113,938.$$

Use the Year 4 value to estimate the terminal value as follows:

$$\text{Terminal value} = \frac{5,113,938}{0.085 - 0.04} = 113,643,067.$$

As a future cash flow, we must convert the terminal value into the present value by discounting at the required return on equity over the first-stage horizon of three years.

$$PV(\text{Terminal value}) = \frac{113,643,067}{(1 + 0.085)^3} = 88,972,078.$$

Similarly, we need to add the present values of the FCFE amounts from Years 1–3, as illustrated in the following table.

	Year 1	Year 2	Year 3
FCFE	3,920,000	4,390,400	4,917,248
PV	3,612,900	3,729,448	3,849,753

Finally, we add all the present values plus the terminal value.

$$IV = 3,612,900 + 3,729,448 + 3,849,753 + 88,972,078 = 100,164,179.$$

And with 1,500,000 shares outstanding, we have

$$\text{Share price} = 100,164,179 \div 1,500,000 = 66.78.$$

A is incorrect because it results from an error in discounting the terminal value to the present (using 4 instead of 3 as the exponent in the denominator).

C is incorrect because it is the result of incorrectly failing to discount the terminal value.

2. If the analyst increases the initial growth period to five years and reduces the stable growth rate to 3.25%, how will this change the intrinsic value per share?

- A. Intrinsic value decreases by over USD0.25.
- B. Intrinsic value remains within USD0.25.
- C. Intrinsic value increases by over USD0.25.

Solution

B is correct. To arrive at this answer, we first extend the table from Question 1 to include Years 5 and 6.

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
FCFE	3,920,000	4,390,400	4,917,250	5,507,320	6,168,198	6,368,660
PV	3,621,900	3,729,450	3,849,750	3,973,940	4,102,130	---

As before, the Year 6 FCFE is obtained by taking the Year 5 value and applying the 3.25% stable growth rate:

$$FCFE_6 = 3,500,000 \times (1 + 0.12)^5 \times (1 + 0.0325)^1 = 6,368,660.$$

Solve for the terminal value:

$$\text{Terminal value} = \frac{6,368,660}{0.085 - 0.0325} = 121,307,900.$$

And solve for the present value:

$$PV(\text{Terminal value}) = \frac{121,307,900}{(1 + 0.085)^5} = 80,675,230.$$

Adding the FCFE present values to the discounted terminal value gives us an intrinsic value of USD99,943,410:

$$IV = 3,621,900 + 3,729,450 + 3,849,750 + 3,973,940 + 4,102,130 + 80,675,230.$$

Finally, we divide this result by the number of shares outstanding to obtain the new intrinsic value:

$$66.63 = \frac{\text{USD}99,943,410}{1,500,000 \text{ shares outstanding}}$$

The new estimated intrinsic value of USD66.63 represents a change of USD0.15.

3. The analyst reverts to her original assumptions in Question 1 but revises her required return on equity higher by 1%, from 8.5% to 9.5%. Which of the following is closest to the magnitude of the change to Excelitate's intrinsic value?

- A. 12%
- B. 18%
- C. 29%

Solution

B is correct because the valuation, especially the effect on the terminal value, must be recalculated with the updated assumption about the cost of equity.

	Year 1	Year 2	Year 3	Year 4
FCFE	3,920,000	4,390,400	4,917,248	5,113,938
PV at 8.5%	3,612,900	3,729,448	3,849,753	---
PV at 9.5%	3,579,910	3,661,640	3,745,240	---

If the required return on equity rises to 9.5%, the terminal value declines:

$$\text{Terminal value} = \frac{5,113,938}{0.095 - 0.04} = 92,980,700.$$

Solve for the revised value:

$$\text{PV(Terminal value)} = \frac{92,980,700}{(1 + 0.095)^3} = 70,819,100.$$

Adding the FCFE present values to the discounted terminal value once again gives us a revised intrinsic value of USD81,805,890, which is 18.3% lower than the original value:

$$\text{IV} = 3,579,910 + 3,661,640 + 3,745,240 + 70,819,100.$$

Growth Rate Determinants

Detailed forecasts of financial statement measures that affect cash flow are addressed in a later learning module, but here, we begin the valuation process with a top-down base-year cash flow measure and make assumptions about its future growth potential. These cash flow growth forecasts are typically based on historical growth rates or fundamental growth determinants.

Firms directly report historical dividends, while FCFE and FCFF are calculated from several variables. An analyst's first task to measure historical growth is to either obtain historical data on dividends or inputs necessary to calculate FCFE and/or FCFF.

The next step involves assessing a company's cash flow volatility over time. The **base effect**, or impact of the reference point selected to determine historical growth rates of cash flow measures, may be small, large, or not meaningful, depending on the years from which a historical average is determined, as shown in the following example.

KNOWLEDGE CHECK: SELECTION OF A BASE YEAR FOR STARBUCKS

An analyst seeks to assess long-term growth rates of Starbucks' cash flow metrics from the data underlying the following chart (in USD millions).

	2017	2018	2019	2020	2021	2022	2023
FCFF	2,794.2	10,094.5	3,506.8	461.2	4,887.5	2,930.7	4,095.1
FCFE	3,082.6	15,545.5	4,886.4	5,280.7	2,837.9	3,229.1	4,143.7

Source: Bloomberg.

1. Explain why an analyst should likely avoid calculating a compound annual growth rate of FCFE or FCFF from 2018 to 2023 for Starbucks.

Solution

Both cash flow metrics showed a one-year surge in 2018 that quickly reversed in 2019. Thus, a compound annual growth rate calculated using 2018 data for either FCFE or FCFF will be biased downward because of a negative base effect associated with the choice of 2018 as the reference point.

An alternative means of estimating a firm's ability to generate cash flow growth is based on fundamental factors, including (1) the level of investment and (2) the productivity of funds invested. We refer to this rate as a **sustainable growth rate (SGR)**, which is the product of the percentage of earnings retained and the return on equity, as shown in Equation 20.

$$\text{Sustainable growth rate}_t = \left(1 - \frac{\text{Dividends}_t}{\text{Net income}_t}\right) \times \left(\frac{\text{Net income}_t}{\text{Book value of equity}_{t-1}}\right). \quad (20)$$

The first term in Equation 20 measures the relative amount of a company's investment, while the second term measures a company's return on equity (ROE), which is a measure of a company's productivity of equity capital invested. The use of sustainable growth is appropriate for firms where these ratios are expected to remain stable.

KNOWLEDGE CHECK: SUSTAINABLE GROWTH FOR BHARDWAJ SUPPLY

The Bhardwaj Supply Company, based in Mumbai, has a dividend payout ratio of 70% and return on equity of 15%. The required return on equity is 10%, and the risk-free INR interest rate is 4.0%, with expected Indian GDP growth of 10.5%.

1. Which of the following is the best estimate of the sustainable growth rate for Bhardwaj Supply?

- A. 4.5%
- B. 7.0%
- C. 10.5%

Solution

A is correct. We can apply Equation 18 given the company's dividend payout ratio and return on equity as follows:

$$\text{Sustainable growth rate}_t = \left(1 - \frac{\text{Dividends}_t}{\text{Net income}_t}\right) \times \left(\frac{\text{Net income}_t}{\text{Book value of equity}_{t-1}}\right).$$

Sustainable growth rate = $(1 - 0.70) \times 0.15 = 0.045$.

B is incorrect because it comes from taking the product of the dividend payout rate with the required return on equity.

C is incorrect because it comes from taking the product of the dividend payout rate and the return on equity.

2. If Bhardwaj Supply's most recent dividend per share is INR5.00 and its sustainable growth rate is 4.5%, which of the following is closest to the company's intrinsic value per share?

A. INR86.67

B. INR90.91

C. INR95.00

Solution

C is correct because the sustainable growth rate is a likely assumption for the long-run growth rate of dividends when using the constant growth dividend discount model.

$$\frac{\text{DPS}_0 \times (1 + \text{SGR})}{\text{Required return on equity} - \text{SGR}} = \frac{5 \times (1.045)}{0.1 - 0.045} = 95.$$

A is incorrect because it uses the risk-free rate of 4% instead of the sustainable growth rate.

B is incorrect because this result occurs if we do not multiply dividends per share by $(1 + \text{SGR})$.

The SGR framework is applicable to both FCFE and FCFF. For FCFE, the dividend term in Equation 21 substitutes FCFE_t for Dividends_t to estimate FCFE SGR_t:

$$\text{FCFE SGR}_t = \left(1 - \frac{\text{FCFE}_t}{\text{Net income}_t}\right) \times \left(\frac{\text{Net income}_t}{\text{Book value of equity}_{t-1}}\right). \quad (21)$$

If a company consistently pays dividends and repurchases stock, FCFE will exceed dividends, with a smaller percentage of net income retained and a lower sustainable growth rate compared to that based on dividends.

For FCFF, the SGR framework is similar but incorporates a denominator consistent with cash flows available to all capital providers.

$$\text{FCFF SGR}_t = \left(1 - \frac{\text{FCFF}_t}{\text{NOPAT}_t}\right) \times \left(\frac{\text{NOPAT}_t}{\text{Book value of total capital}_{t-1}}\right). \quad (22)$$

Equation 22 defines a company's reinvestment by the ratio of the FCFF cash flow measure divided by the company's **net operating profit after tax (NOPAT)**. NOPAT was introduced earlier in the curriculum as a measure of income to both debt- and equityholders, as shown in Equation 23:

$$\text{NOPAT} = \text{Operating income} \times (1 - \text{Tax rate}). \quad (23)$$

With this term in the numerator and the book value of both debt and equity representing the book value of total capital in the denominator, the second term reflects a company's return on capital (ROC) ratio.

Analysts must be aware that different cash flow metrics may result in different growth rates, even when assuming a long-run constant growth rate.

KNOWLEDGE CHECK: SWISSERV AG'S SGR USING DIFFERENT CASH FLOW METRICS


An investor is considering a significant investment in Swisserv AG, a growing business consultancy and support services firm. As he considers the best cash flow metric for valuation purposes, the investor takes a closer look at sustainable growth rates implied by such measures as dividends, FCFE, and FCFF using the following company data (in CHF):

Dividends	2,000,000
FCFE	3,120,000
FCFF	2,672,000
Net income	4,800,000
EBIT × (1 – Tax rate)	5,472,000
Book value of equity	14,000,000
Book value of debt	9,333,000

1. Select the best answer to complete the sentence: “The SGR calculated using _____ is greater than when using the other two measures.”

- A. Dividends
- B. FCFE
- C. FCFF

Solution

A is correct because the SGR calculated using dividends as a cash flow metric is the highest of the three. Using Equation 20, we can calculate SGR based on dividends:

$$SGR_t = \left(1 - \frac{\text{Dividends}_t}{\text{Net income}_t}\right) \times \left(\frac{\text{Net income}_t}{\text{Book value of equity}_{t-1}}\right).$$

From the first expression, the dividend payout ratio is equal to 41.67%:

$$\frac{\text{Dividends}}{\text{Net income}} = \frac{2,000,000}{4,800,000} = 41.67\%.$$

The second expression, return on equity, is 34.3%:

$$\frac{\text{Net income}}{\text{Book value of equity}} = \frac{4,800,000}{14,000,000} = 34.3\%.$$

Solve for SGR in Equation 20 by substituting these values:

$$SGR(\text{Dividends}) = (1 - 0.4167) \times 0.343 = 20.0\%.$$

B is incorrect because the FCFE-based SGR is below that of the same measure using dividends. The FCFE-based SGR is determined using Equation 21:

$$FCFE\ SGR_t = \left(1 - \frac{FCFE_t}{\text{Net income}_t}\right) \times \left(\frac{\text{Net income}_t}{\text{Book value of equity}_{t-1}}\right).$$

From the first expression, we substitute the FCFE to net income ratio for the dividend payout ratio, which is 65%:

$$\frac{\text{FCFE}}{\text{Net Income}} = \frac{3,120,000}{4,800,000} = 65\%.$$

Substituting this expression and ROE of 34.3% from the prior calculation, we can solve for FCFE-based SGR of 12%:

$$\text{SGR}(\text{FCFE}) = (1 - 0.65) \times 0.343 = 12.0\%.$$

C is incorrect because the FCFF-based SGR is lower than that of the same measure using dividends. The FCFF-based SGR is calculated using Equation 22, where we include FCFE, NOPAT, and return on capital, as follows:

$$\text{FCFF SGR}_t = \left(1 - \frac{\text{FCFF}_t}{\text{NOPAT}_t}\right) \times \left(\frac{\text{NOPAT}_t}{\text{Book value of total capital}_{t-1}}\right).$$

First, calculate the FCFF to NOPAT and NOPAT to return on capital ratios:

$$\frac{\text{FCFE}}{\text{NOPAT}} = \frac{2,672,000}{5,472,000} = 48.8\%.$$

$$\frac{\text{NOPAT}}{\text{Book value of capital}} = \frac{5,472,000}{14,000,000 + 9,333,000} = 23.5\%.$$

Substitute these two expressions into Equation 22, and solve for SGR of 12%:

$$\text{SGR}(\text{FCFF}) = (1 - 0.488) \times 0.235 = 12.0\%.$$

Present Value Models: Discount Rates and Terminal Value

Using estimates of the base-year cash flow measure and growth rates over a forecast time horizon, recall that the present value calculation shown earlier in Equation 1 requires that these cash flows be discounted at an appropriate required rate of return:

$$\text{IV}_t = \sum_{i=1}^n \frac{E(\text{CF}_{t+i})}{(1 + \text{Required rate of return})^i} + \frac{E(P_{t+n})}{(1 + \text{Required rate of return})^n}.$$

The discounted terminal value in n periods captures all remaining cash flows beyond the forecast period.

Present Value Model Discount Rates

The applicable discount rate must be consistent with the type of investor claim reflected in the cash flow measure in the numerator of the present value model. For example, a company's weighted average cost of capital introduced in an earlier learning module on corporate issuers is a single required return for all capital providers to the firm. As shown in Equation 24, WACC provides a market value weighted average of debt (adjusted for the tax deductibility of interest) versus equity cost:

$$\text{WACC} = \frac{\text{MV}_{\text{Debt}}}{\text{MV}_{\text{Debt}} + \text{MV}_{\text{Equity}}} r_d (1 - \text{Tax rate}) + \frac{\text{MV}_{\text{Equity}}}{\text{MV}_{\text{Debt}} + \text{MV}_{\text{Equity}}} r_e, \quad (24)$$

where MV_{Debt} and $\text{MV}_{\text{Equity}}$ are the current market values of the firm's debt and equity and $r_d(1 - \text{Tax rate})$ and r_e are the after-tax required return of debt and the required rate of return on equity. Detailed estimation of these required returns, also referred to as the costs of debt and equity, is addressed in later learning modules.

A company's WACC is an appropriate discount rate for calculating a firm's value from the perspective of all investors (FCFF) rather than only equity investors (FCFE). As shown in Equation 25, however, we must subtract the market value of debt from the resulting *enterprise* value (shown in brackets) to derive a firm's equity value (IV_t):

$$\text{IV}_t = \left[\sum_{i=1}^n \frac{\text{FCFF}_{t+i}}{(1 + \text{WACC})^i} + \frac{E(P_{t+n})}{(1 + \text{WACC})^n} \right] - \text{MV}_{\text{Debt}} \quad (25)$$

The cost of equity (r_e) is the appropriate discount rate for cash flows available to equityholders, such as dividends, FCFE, and residual income. This relationship is shown in Equation 26 based on FCFE.

$$IV_t = \sum_{i=1}^n \frac{FCFE_{t+i}}{(1+r_e)^i} + \frac{E(P_{t+n})}{(1+r_e)^n}. \quad (26)$$

Since contractual debt claims involve less risk to investors relative to equity, the cost of debt is lower than the cost of equity. This relationship implies the inequality shown in Equation 27:

$$WACC \leq r_e. \quad (27)$$

While WACC and r_e will be the same for firms using no debt, the degree of reliance on debt and the difference between debt and equity costs dictate the difference between WACC and the cost of equity. Valuing a company with either FCFE or FCFF should yield similar equity valuation results if the analyst maintains consistency between the two sets of model assumptions, as reflected in the following example.

KNOWLEDGE CHECK: BELCHAMP CORPORATION'S COST OF CAPITAL



Belchamp Corporation is a Canadian company with a market value of equity that is four times larger than its market value of debt. The Canadian dollar risk-free rate is 2.5%, and the Canadian corporate tax rate is 20%.

1. If the required return on equity is 12% and the firm's cost of debt is 6%, which of the following most correctly reflects Belchamp's weighted average cost of capital?

- A. 10.00%
- B. 10.24%
- C. 10.56%

Solution

C is correct because 10.56% is the WACC determined by using Equation 24, as follows: With equity value four times the company's debt value, 80% of Belchamp's value is equity and 20% is debt. We can substitute the values provided into Equation 24 to solve for WACC:

$$WACC = \frac{MV_{Debt}}{MV_{Debt} + MV_{Equity}} r_d (1 - \text{Tax rate}) + \frac{MV_{Equity}}{MV_{Debt} + MV_{Equity}} r_e.$$

$$WACC = 0.2 \times 0.06(1 - 0.2) + 0.8 \times 0.12 = 10.56\%$$

A is incorrect because this result incorrectly uses the risk-free rate, 2.5%, in place of the cost of debt of 6% in Equation 24.

B is incorrect because this result uses incorrect weights of 25% on debt and 75% on equity in Equation 24.

2. Belchamp's FCFE for the base year was CAD40 million, and its FCFF was CAD40.8 million. The expected FCFE growth rate is 2.00%. Calculate the intrinsic value of Belchamp's equity.

Solution

Recall that Belchamp's required return on equity is 12%. The base-year FCFE is CAD40 million, so applying the constant growth model in Equation 15 using FCFE produces an intrinsic equity value of CAD408 million:

$$IV_t = \frac{E(CF)_{t+1}}{\text{Required rate of return} - g}.$$

$$IV(FCFE) = \frac{FCFE_0(1 + g_{FCFE})}{r_e - g_{FCFE}} = \frac{40(1.02)}{0.12 - 0.02} = 408.$$

3. The market value of Belchamp's debt as of the base year is estimated to be CAD104 million. Calculate the implied FCFF growth rate under the assumption that a constant growth version of the FCFF model must produce an equity value of CAD408 million. Assume that Belchamp's WACC is 10.56%.

Solution

We can solve for firm value by adding the value of Belchamp's debt to the intrinsic value of CAD408 million to get a firm value of CAD512 million:

$$\text{Firm value} = MV_{Debt} + MV_{Equity} = 104 + 408 = 512.$$

For the FCFE and FCFF models to result in the same intrinsic value of equity, we must solve for g_{FCFF} , or the implied growth rate of Belchamp's FCFF, using Equation 16:

$$P_t = \frac{E(CF)_{t+1}}{\text{Required rate of return} - g_{implied}}.$$

Substituting $FCFF_0(1 + g_{FCFF})$ for $E(CF)_{t+1}$, WACC for the required rate of return, and g_{FCFF} for $g_{implied}$, we get

$$\frac{FCFF_0(1 + g_{FCFF})}{\text{WACC} - g_{FCFF}} = 512 = \frac{40.8(1 + g_{FCFF})}{0.1056 - g_{FCFF}}.$$

$$512(0.1056 - g_{FCFF}) = 40.8(1 + g_{FCFF}).$$

$$g_{FCFF} = 2.40\%.$$

The previous example yields identical results, but this outcome is far more difficult to achieve in practice due to challenges in maintaining internal consistency among FCFE and FCFF models. If equity valuation results diverge significantly when using both cash flow metrics, it suggests an analyst should revisit the assumptions used to address these differences.

Present Value Model Terminal Values

The terminal value in any valuation is an expected future share price at the end of an n period forecast horizon over which an analyst has projected future cash flows. The terminal value represents a greater proportion of a value estimate when an analyst uses shorter forecast horizons or when there are smaller differences between forecast horizon growth rates and the long-run constant growth rate. Given the impact of terminal values on intrinsic value estimates, analysts seek to minimize biases by following standardized approaches, which include

- present value estimation and
- market multiples approaches.

In this learning module, we focus on present value estimation, and we address the use of market multiples in the next learning module.

As shown earlier, the discount rate used for terminal growth consisting of required return on equity minus a growth rate is a primary driver of intrinsic value. The sustainable growth rate calculation based on the reinvestment rate and return on capital

shown earlier is one approach to estimating the constant growth rate, shown here as g_t . This approach may lead to estimation of positive growth rates, which may be close to zero or even negative for firms in a declining phase of their company life cycle. Other approaches include observed historical growth and one that considers nominal economic growth when selecting a stable growth rate.

Analysts often assume the constant growth rate used to compute a terminal value estimate should lie in the range of the expected inflation rate and nominal economic growth rate. Thus, constant growth rates may vary when examining companies across countries with variation in inflation and GDP growth. As shown in the following example, analysts must consider long-term averages and markets relevant to a specific firm when establishing terminal value growth assumptions.

KNOWLEDGE CHECK: TERMINAL VALUE GROWTH ASSUMPTIONS FOR SWISSERV AG



More detailed analysis of the Swiss business consultancy firm Swisserv AG introduced earlier leads to the conclusion that a two-stage growth model is a better choice for valuing the firm. As stated in an earlier knowledge check, Swisserv had a base-year FCFE of CHF3,120,000 and an estimated initial FCFE growth rate of 12% annually. We assume a required return on equity of 10%.

At the end of three years, the analyst must solve for a terminal value with a stable long-term growth rate. Observing long-term historical Swiss nominal GDP growth of 2% and inflation of 0.5%, the analyst decides that the 1.25% average of these two measures is an appropriate growth rate to consider.

1. Which of the following is closest to Swisserv's intrinsic value using an FCFE model and the two-stage growth assumptions provided?

- A. CHF43.1 million
- B. CHF47.8 million
- C. CHF60.4 million

Solution

B is correct based on the following process. First, we must determine FCFE for the next four years:

	Year 1	Year 2	Year 3	Year 4
FCFE	3,494,400	3,913,728	4,383,375	4,438,168

For Years 1–3, we simply calculate this as

$$\text{FCFE}_t = 3,120,000 \times (1 + 0.12)^t.$$

For Year 4, the first year with the long-term growth rate, we have

$$\text{FCFE}_4 = 3,120,000 \times (1 + 0.12)^3 \times (1 + 0.0125)^1 = 4,438,167.55.$$

Use the Year 4 value to estimate the terminal value:

$$\text{Terminal value} = \frac{4,438,167.55}{0.10 - 0.0125} = 50,721,915.$$

As a future cash flow, we must convert the terminal value into the present value by discounting at the required return on equity over the first-stage horizon of three years.

$$PV(\text{Terminal value}) = \frac{50,721,915}{(1 + 0.1)^3} = 38,108,125.$$

Similarly, we need to add the present values of the FCFE amounts from Years 1–3 as illustrated in the following table.

	Year 1	Year 2	Year 3
FCFE	3,494,400	3,913,728	4,383,375
PV @ 10%	3,176,727	3,234,486	3,293,295

Finally, we add all the present values plus the terminal value to arrive at an equity value of CHF47.8 million:

$$IV = 3,176,727 + 3,234,486 + 3,293,295 + 38,108,125 = 47,812,633.$$

A is incorrect because it results from mistakenly solving for the terminal value by dividing Year 4 results by the required return on equity without subtracting the stable growth rate.

C is incorrect because it results from incorrectly failing to discount the terminal value.

2. Swisserv's stated business strategy is to grow by offering outsourcing solutions in emerging Eastern Europe. Based on this stated strategy, an analyst at another firm disputes the original valuation given that these economies have long-term average nominal GDP growth of 7% and inflation of 4%, suggesting a 5.25% stable growth rate in these markets.

Assuming that roughly half of Swisserv's future business is expected from slower-growth, developed markets, such as Switzerland, and the other half will be derived from more rapidly growing Eastern European markets following an initial growth phase, the analyst revises the stable growth rate assumption to 3.25%. Based on this change, which of the following is closest to the increase in intrinsic value versus the original value?

- A. 26%
- B. 30%
- C. 60%

Solution

A is correct since the revised terminal value based on the higher stable growth rate of 3.25% is 26% greater than the original value.

Recall that the original intrinsic value from Question 1 was CHF47.8 million:

$$IV = 3,176,727 + 3,234,486 + 3,293,295 + 38,108,125 = 47,812,633.$$

Using the 3.25% stable growth rate for Year 4, we recalculate $FCFE_4$ as follows:

$$FCFE_4 = 3,120,000 \times (1 + 0.12)^3 \times (1 + 0.0325)^1 = 4,525,835.$$

Use the Year 4 value to estimate the terminal value:

$$\text{Terminal value} = \frac{4,525,835}{0.10 - 0.0325} = 67,049,408.$$

As a future cash flow, we must convert the terminal value into the present value by discounting at 10% over the first-stage horizon of three years.

$$PV(\text{Terminal value}) = \frac{67,049,408}{(1 + 0.1)^3} = 50,375,213.$$

The revised intrinsic value using 3.25% stable growth is CHF65.5 million, or a 40% increase in intrinsic value over the original:

$$IV = 3,176,727 + 3,234,486 + 3,293,295 + 50,375,213 = 60,079,721.$$

This result is 26% greater $[= (60,079,721 \div 47,812,633) - 1]$ than the original intrinsic value. One could interpret this difference as the incremental present value associated with the higher-growth opportunity.

B is incorrect because it uses the long-term growth rate of 1.25% to solve for the terminal value and fails to discount the value to the present.

C is incorrect because even though it uses the correct 3.25% stable growth rate to solve for the terminal value, it fails to discount the value to the present.

As shown in the previous knowledge check, it is preferable to base terminal value growth assumptions on long-run averages rather than current rates, which may reflect unsustainable factors. Also, rather than solely considering growth rates for a country where a firm is based, an analyst may need to take economic variables into consideration in other markets where the firm operates. Analysts must also be aware that a terminal value estimate may require a constant growth rate assumption below the inflation rate or, in some cases, even a negative rate, as in the following example.

KNOWLEDGE CHECK: MACARTHUR COMPANY'S GROWTH DECLINE



MacArthur Company is a well-established Australian office supply retailer. Because the firm faces continued competitive pressures of digitization and the rise of low-cost regional substitutes for its offering, an equity analyst believes that the most recent year's FCFE is an appropriate basis for estimating future cash flows but assigns a negative stable growth rate to the firm.

1. If MacArthur's most recent FCFE was AUD250 million, the required return on equity is 10%, and the stable long-term growth rate of FCFE is -3.5%, which of the following is closest to the company's intrinsic value?

- A. AUD1,800 million
- B. AUD2,500 million
- C. AUD3,700 million

Solution

A is correct. We can solve for this approximate value by applying an FCFE constant growth model with negative growth, as follows:

$$IV_t = \frac{FCFE_t(1+g)}{r_e - g}.$$

$$IV_t = \text{AUD1,787 million} = \frac{250(1 - 0.035)}{0.10 - (-0.035)}.$$

B is incorrect because this result occurs if we incorrectly assume the long-term growth rate is equal to zero.

C is incorrect because this result occurs by mistakenly subtracting 3.5% in the denominator, which would be the case if the growth rate were positive.

A stable and negative growth rate implies that a firm is likely to face a slow decline and cease to operate in the future.

The first four steps of the valuation process shown earlier in this lesson establish a *point estimate* of a company's intrinsic value based on a single set of expected cash flows. Because the underlying assumptions that are the inputs to present value models often involve an estimated range, key assumptions such as the length of the growth period, the initial higher growth rate, and the stable growth rate used for the terminal value may vary in alternative scenarios of the valuation model. Depending on the assumptions, the point estimate deemed most likely under expected assumptions may be considered a base case, with one or more alternatives used to establish a range or distribution of outcomes, as shown in the following example.

KNOWLEDGE CHECK: VARYING SPEEDS OF DECLINE FOR MACARTHUR COMPANY



Analysts at the firm estimating MacArthur's intrinsic value have starkly different views on the pace of the company's decline because of technological change to both the company's products and business model, as well as the rise of competitors in the Asia-Pacific region. One group believes that the firm will be able to maintain growth of 1% in the long term, while another believes the FCFE decline will accelerate to 5% per annum going forward. The mean consensus is -3.5%, and the required return on equity remains 10%.

1. Which of the following provides the best range of intrinsic values to capture these divergent analyst views of MacArthur?

- A. AUD1.58 billion to AUD2.81 billion
- B. AUD1.79 billion to AUD2.50 billion
- C. AUD2.25 billion to AUD5.25 billion

Solution

A is correct. It results from using the additional growth rates (as well as the growth rate from the prior knowledge check) in the FCFE constant growth model from Equation 2, as follows:

$$IV = \frac{FCFE_0(1+g)}{r_e - g}$$

$$IV(\text{Lower bound}) = \text{AUD1.58 billion} = \frac{250(1 - 0.05)}{0.10 + 0.05}$$

$$IV(\text{Upper bound}) = \text{AUD2.81 billion} = \frac{250(1 + 0.01)}{0.10 - 0.01}$$

B is incorrect because it uses a growth rate g of -3.5% for the lower bound and 0 for the upper bound.

C is incorrect because it uses a growth rate g of -1% for the lower bound and +5% for the upper bound.

QUESTION SET 2

1. Which of the following statements best characterizes the relationship between the weighted average cost of capital and the cost of equity?
 - A. Since it takes tax deductibility into account, the WACC is always lower than the cost of equity.
 - B. Higher tax rates will cause the WACC to rise in comparison to the cost of equity, which is not tax deductible.
 - C. The difference between WACC and the cost of equity depends on the degree of reliance on debt and the difference between debt and equity costs.

Solution

C is correct because all else equal, a firm with greater reliance on debt versus equity or with a much lower debt versus equity cost will have a greater difference between WACC and the cost of equity.

A is incorrect because it ignores the case in which a company has no debt, for which the WACC equals the cost of equity.

B is incorrect because a higher tax rate will reduce the after-tax cost of debt, causing the WACC to fall in comparison to the cost of equity.

2. Which of the following statements regarding an equity valuation using FCFE versus FCFF as a cash flow metric is most accurate?
 - A. Equity valuation using FCFE is typically higher than when using FCFF due to the higher discount rate used for equity investors versus all capital providers.
 - B. Equity valuation results using FCFE or FCFF should be similar provided an analyst maintains consistency between model assumptions.
 - C. Equity valuation using FCFF as a cash flow metric is typically higher since the cash flows to all capital providers are higher than those to equity investors.

Solution

B is correct because equity valuation models using FCFE or FCFF have differing discount rates that adjust for the different cash flow measures and should provide similar equity valuation results if an analyst maintains consistent assumptions.

A is incorrect because although the return on equity used as an FCFE discount rate is usually greater than the WACC used for FCFF, this does not result in a higher equity valuation for FCFE-based models versus FCFF-based models.

C is incorrect because although the FCFF cash flow metric is in fact usually higher than FCFE, the intrinsic value of equity is determined by subtracting the market value of debt from the resulting enterprise value.

3. Which of the following statements about the constant growth rate used to compute a terminal value estimate is most accurate?
 - A. Analysts most often assume a constant growth rate between the expected inflation rate and nominal growth rate of the economy.

- B.** Constant growth rate assumptions cannot be below zero, because this would result in a negative share price in the future.
- C.** A firm's constant growth rate should be primarily determined based on growth rates for the country where a firm is based.

Solution

A is correct because analysts commonly assume a constant growth rate will reach or exceed expected inflation.

B is incorrect because constant growth rates may be negative for firms in decline, which does not imply a negative future share price but, rather, implies that a firm will cease to exist at some point in the future.

C is incorrect because a constant growth rate assumption should take the geographic distribution of a firm's operations or sales into consideration rather than simply where it is headquartered.

4

LIMITATIONS OF DISCOUNTED CASH FLOW GROWTH MODELS



explain the shortcomings of constant and multistage cash flow growth assumptions in present value models

As described in earlier lessons, the use of DCF models should involve a well-structured process based on logically consistent assumptions—for example, matching a cash flow metric in the numerator with an appropriate required rate of return for investors with access to those cash flows in the denominator. Despite this approach, analysts must understand the drawbacks of such models when using projections based on cash flow growth rates related to both company-specific or bottom-up factors and top-down, macroeconomic or market considerations affecting value. For example, company-specific disadvantages of the DCF approach may involve limitations of the cash flow metric itself, incorrect assumptions or models used, and company dynamics that are poorly or incorrectly captured in a present value approach.

The choice of a single cash flow metric may lead to biased results if it fails to fully capture the underlying value dynamics of a firm over a forecast horizon. For example, dividends are an intuitively appealing shareholder return metric but provide an incomplete measure of cash flow to stockholders. For example, a firm that *increases* shareholder distributions via buybacks while maintaining a *consistent* dividend payout ratio may have an understated equity value using a dividend-based DCF model versus one based on broader cash flow measures. On the other hand, declining or flat EPS growth that coincides with higher dividends may produce divergent results using different cash flow measures, as shown in the following example.

KNOWLEDGE CHECK: DIVIDENDS AND FCFE FOR MACARTHUR COMPANY VALUATION



A member of the analyst team for MacArthur Company, an Australian office supply retailer, believes the firm can survive as a going concern and continue to grow FCFE at 1% indefinitely despite competitive pressures. Citing

the 5% base-year increase in MacArthur's dividends per share, a second team member suggests the use of dividends as an alternative cash flow measure, arguing for this higher growth rate to be applied to MacArthur's valuation instead.

1. If MacArthur's base-year FCFE was AUD250 million, its total base-year dividends were AUD150 million, and the required return on equity is 10%, what is the approximate valuation discrepancy between the two analyst estimates using a constant growth model?

- A. AUD340 million
- B. AUD900 million
- C. AUD1,235 million

Solution

A is correct. We can compare the results of the following FCFE- and dividend-based constant growth models. The results of the FCFE-based model were shown in a prior knowledge check using Equation 2 with an FCFE growth rate of 1%:

$$IV_t = \frac{FCFE_t(1+g)}{r_e - g}$$

$$IV_t = \text{AUD2.81 billion} = \frac{250(1+0.01)}{0.10-0.01}.$$

This result is compared with a similar dividend-based constant growth model with a dividend growth rate of 5%:

$$IV_t = \frac{\text{Dividends}(1+g)}{r_e - g}$$

$$IV_t = \text{AUD3.15 billion} = \frac{150(1+0.05)}{0.10-0.05}.$$

The dividend-based model provides a higher intrinsic value, with a discrepancy of AUD340 million (= AUD3.15 billion – AUD2.81 billion).

B is incorrect because this result occurs if we incorrectly subtract rather than add the long-term FCFE growth rate in the numerator and add rather than subtract it in the denominator.

C is incorrect because this result occurs by mistakenly omitting the dividend growth rate from the denominator of the dividend-based constant growth model.

2. Which of the following statements best explains the discrepancy between the analyst valuations of MacArthur Company?

- A. The discrepancy between the FCFE- and dividend-based valuations may be explained by possible changes in MacArthur's dividend payout ratio.
- B. The FCFE- and dividend-based valuation discrepancy may be explained by possible changes in the number of MacArthur shares outstanding.
- C. The FCFE- and dividend-based valuation discrepancy may be a result of changes in the dividend payout ratio and/or changes in the number of shares outstanding.

Solution

C is the correct answer because the discrepancy may arise from either a higher dividend payout ratio or a decline in shares outstanding due to a

share buyback. Note that the analyst using the dividend-based approach cites dividends *per share* as the basis for a 5% growth assumption, while ignoring any possible changes in shares outstanding by using *total* dividends as a cash flow valuation metric.

A is incorrect because it cites only one possible explanation for the discrepancy based on the dividend payout ratio.

B is incorrect because it cites only one possible explanation for the discrepancy based on the number of shares outstanding.

Broader cash flow measures, such as FCFE and FCFF, face similar model-based or estimation uncertainties, as described previously in the case of dividends. For example, in the case of FCFE, base-year effects may bias value estimates when an analyst must specify a constant debt ratio over the forecast period. For both FCFE and FCFF, operating profits, margins, and investments in long-term assets and/or working capital may be difficult to define.

Present value models based on a growth rate are most appropriate for firms in the mature life-cycle stage for which a base-year cash flow metric can serve as an accurate basis for estimating intrinsic value. The use of a higher growth forecast horizon followed by a terminal value based on stable growth allows for changes over time within predefined bounds. However, challenges arise when firms experience unexpected patterns of growth, transformation, or decline over indefinite periods. These situations are often incompatible with present value model assumptions, as shown in the following case study.

CASE STUDY



Amazon's Transformational Growth

Founded in 1994 by Jeff Bezos as an online book seller, Amazon.com was among the world's largest companies, with sales of over a half trillion US dollars nearly three decades later, in 2023. While the company expanded rapidly across products and regions during its first two decades, its third decade was characterized by transformational growth, including outsized growth in its Amazon Web Services (AWS) cloud storage business; the 2017 acquisition of supermarket chain Whole Foods Market, with physical retail stores; and over USD100 billion in assets to support the firm's fulfillment and delivery network.

As these changes transformed the asset side of Amazon's balance sheet, the firm increased its reliance on debt while largely reinvesting earnings, forgoing major share repurchases and dividends initiated by other technology companies.

The use of a base-year cash flow metric and a cash flow growth rate to value Amazon's equity provides an incomplete picture of Amazon's value drivers, given the profound changes in leverage, profitability, and investments during this transformational growth period.

In contrast to firms in a growth phase, companies in decline face a greater risk of financial distress, such as bankruptcy, restructuring, and liquidation. Present value models that discount future cash flows back to the present implicitly assume that the firm will remain a going concern over the forecast horizon. While negative growth rates may be used to model a firm in slow decline, financial distress is often sudden, and the likelihood of a firm's failure may change rapidly based on market and competitive forces. It is often preferable to explicitly incorporate this likelihood into a valuation exercise using scenario analysis, as shown next.

KNOWLEDGE CHECK: MACARTHUR COMPANY'S LIKELIHOOD OF FINANCIAL DISTRESS

Recall that analysts estimating MacArthur's value disagree on its future viability as a going concern. While the base case involves slow FCFE growth of 1% in the long term, a downside scenario involves financial distress. Rather than the slow decline implied by a negative growth rate, the analysts observe that financial distress among traditional retailers usually involves liquidation.

1. If we assume a 1/3 chance of a financial distress scenario (implying a 2/3 chance of survival) under which MacArthur is liquidated with a total loss to shareholders and a cost of equity of 10%, which of the following is closest to MacArthur Company's intrinsic value?

- A. AUD2.81 billion
- B. AUD2.295 billion
- C. AUD1.87 billion

Solution

C is correct. It is calculated using the FCFE constant growth model from Equation 2 weighted by the 2/3 probability of survival:

$$IV = \frac{FCFE_0(1+g)}{r_e - g}.$$

$$IV(\text{Survival}) = \text{AUD2.81 billion} = \frac{250(1+0.01)}{0.10-0.01}.$$

$$IV = \frac{2}{3} \times \text{AUD2.81 billion} + \frac{1}{3} \times \text{AUD0} = \text{AUD1.87 billion}.$$

A is incorrect because it fails to adjust the intrinsic value given slow growth by the likelihood of survival.

B is incorrect because it incorrectly adds the growth rate to the denominator and fails to adjust for the likelihood of survival.

Macroeconomic or market considerations affecting a firm's intrinsic value can have as large a positive or negative impact on valuation as the company-specific factors outlined earlier. These top-down variables commonly affect estimates of the cost of capital (WACC and required return on equity) in the denominator and are subject to several estimation challenges. For example, the risk-free interest rate embedded in both the costs of debt and equity changes over time based on macroeconomic conditions. The following knowledge check illustrates the impact of these changes on intrinsic value.

KNOWLEDGE CHECK: BELCHAMP'S CHANGING COST OF CAPITAL AND INTRINSIC VALUE

Belchamp was introduced earlier as a Canadian company with a market value of equity four times the size of the market value of its debt. Using a Canadian dollar risk-free rate of 2.5% and a Canadian corporate tax rate of 20%, an earlier knowledge check showed a WACC of 10.56% based on a required return on equity of 12% and the firm's cost of debt of 6%.

An investor reviewing the analyst's assumptions and valuation points out that while the Canadian dollar benchmark interest rate used to approximate the risk-free rate is *currently* 2.5%, the rate is forecast to rise to 5% over the next 6–12 months to a level close to the long-term historical average.

1. Which of the following is closest to the *change* in Belchamp's WACC as a result of using the long-term historical average benchmark interest rate rather than the original assumption?

- A. 2.00%
- B. 2.50%
- C. 2.40%

Solution

C is correct because the revised WACC of 12.96% is 2.40% greater than the original 10.56% WACC using Equation 24. Note that both debt and equity costs increase by 2.5%.

$$\text{WACC} = \frac{\text{MV}_{\text{Debt}}}{\text{MV}_{\text{Debt}} + \text{MV}_{\text{Equity}}} r_d (1 - \text{Tax rate}) + \frac{\text{MV}_{\text{Equity}}}{\text{MV}_{\text{Debt}} + \text{MV}_{\text{Equity}}} r_e$$

$$\text{WACC} = 0.2 \times 0.085(1 - 0.2) + 0.8 \times 0.145 = 12.96\%$$

A is incorrect because it results from incorrectly multiplying the change in the risk-free rate, 2.5%, by the proportion of equity.

B is incorrect because it results from incorrectly adding the 2.5% increase in the risk-free rate to the current WACC.

2. Belchamp's FCFF for the base year was CAD40.8 million, and the expected FCFF growth rate is 2.40%. Calculate the revised intrinsic value of Belchamp's equity using the new WACC from Question 1.

Solution

Recall that Belchamp's required return on equity is 12%. Using a base-year FCFF of CAD40.8 million, g_{FCFF} of 2.40%, and a revised WACC of 12.96%, applying the constant growth model in Equation 16 using FCFF produces an intrinsic equity value of CAD395.6 million:

$$\text{IV}_t = \frac{\text{FCFF}_0(1 + g_{\text{FCFF}})}{\text{WACC} - g_{\text{FCFF}}} = \frac{40.8(1 + 0.024)}{0.1296 - 0.024} = \text{CAD395.6 million.}$$

Note that this revised intrinsic value of CAD395.6 million is 23% lower than the original CAD512 million estimate using the lower original WACC.

Changing cost of capital assumptions due to evolving market conditions can have profound effects on valuation. While the cost of capital also depends on the timing of cash flows and can vary according to the time to maturity, analysts often simply match the term of the risk-free rate with that of the forecast horizon and use long-term averages. Developing markets pose additional challenges when government debt is not considered risk free or long-term sovereign rates are not available.

The cumulative effect of distortions due to company-specific or macroeconomic factors on present value models may create misleading results. While most appropriate for stable firms with consistent cash dividends and cash flow measures, they are less applicable to non-dividend-paying firms, those with negative cash flow, or firms at an early- or late-stage inflection point of the company life cycle. Technological change, competitive pressures, or other company- or industry-based changes can lead to a violation of the static model assumptions that underlie constant growth models. For these

and other reasons, analysts typically use more than one type of model and approach to estimate equity value, recognizing that each has limitations and involves simplification with uncertainty due to model appropriateness, specification, and inputs used.

QUESTION SET 3



1. Which of the following statements most accurately describes the valuation discrepancies that may arise when using dividends versus FCFE as a cash flow metric for present value models?
 - A. A declining dividend payout ratio over time will result in a lower valuation when using an FCFE model than when using a dividend-based model.
 - B. Rising share buybacks over time will result in a lower valuation when using a dividend-based model than when using an FCFE model.
 - C. Increasing reliance on debt over time will result in a lower valuation when using a dividend-based model than when using an FCFE model.

Solution

B is correct because an increase in share buybacks will result in an FCFE-based valuation that captures these distributions, while a cash dividend metric will not.

A is incorrect because a lower dividend payout ratio will result in a higher, not lower, valuation when using an FCFE model than when using a dividend-based model.

C is incorrect because changes in reliance on debt financing will result in a valuation discrepancy not between dividend- and FCFE-based valuation but, rather, between FCFE- and FCFF-based valuation.

2. Which of the following statements best describes the applicability of the present value model to a firm in slow decline?
 - A. The present value model forecast horizon should be adjusted to match the end of the firm's expected life.
 - B. The present value model should incorporate a negative growth rate.
 - C. The present value model is inapplicable to firms in slow decline.

Solution

B is correct because a present value model that incorporates a negative growth rate is consistent with the narrative that the firm will enter a phase of slow decline and may cease to exist in the future.

A is incorrect because a finite forecast horizon is not consistent with a firm in slow decline.

C is incorrect because a present value model can accommodate slow decline using a negative growth rate.

3. Which of the following approaches best characterizes how to overcome the challenges associated with changing interest rates when determining the cost of capital for a present value model?
 - A. To estimate the current intrinsic value, it is important to use an updated short-term benchmark rate in determining the cost of capital for all cash flows.

- B.** To estimate the current intrinsic value, it is best to apply the current long-term benchmark rate in determining the cost of capital for a valuation.
- C.** To estimate the current intrinsic value, it is best to apply a long-term average benchmark rate that matches the term of the forecast horizon.

Solution

C is correct because long-term average rates capture macroeconomic changes over longer periods.

A is incorrect because it incorrectly uses the current short-term rate for cash flows that occur over the long term.

B is incorrect because the current long-term rate does not capture changing market conditions in a present value valuation.

5

PREFERRED STOCK VALUATION



calculate the intrinsic value of a non-callable, non-convertible preferred stock and describe how contingency features affect intrinsic value

An earlier Equities learning module stated that like debt, preferred stock pays periodic dividends equal to a fixed percentage of the preference share's par value (or original asking price). Preferred shares generally represent a very small fraction of total equity securities outstanding and are typically limited to large public issuers in regulated sectors, such as banking, insurance, and utilities, as well as private companies and venture capital. Like common shares, preference dividends are not contractual obligations, but preferred shareholders have a priority claim over common stock in the receipt of dividends and on the issuer's assets in the event of liquidation. While preferred shares may include such features as cumulative and/or participating dividends, as well as contingency provisions, such as call, put, or conversion features, our focus here is on non-callable, non-convertible preferred shares with constant, fixed dividends.

The similarity of preferred stock cash flows based on these assumptions to those of a dividend-paying common stock allows us to use a dividend discount model to calculate the intrinsic value (IV_t) of preferred stock with three key modifications, as shown in Equation 28:

- Because preferred shares pay a *fixed* periodic dividend, we denote this as D rather than D_i for each period in the DDM.
- For preferred shares with a finite maturity at time n , investors receive the face or par value (FV) rather than an expected terminal value $E(P_{t+n})$.
- The priority of preferred over common share claims is reflected in the preferred share discount rate (r_p), with $r_p < r_e$ representing required return for common equity.

$$IV_t = \sum_{i=1}^n \frac{D}{(1+r_p)^i} + \frac{FV}{(1+r_p)^n} \quad (28)$$

For an otherwise identical non-callable, non-convertible *perpetual* preferred share, the intrinsic value is a simple perpetuity calculation:

$$IV_t = \frac{D}{r_p} \quad (29)$$

**KNOWLEDGE CHECK: MONTE CABRIO AND MONTE CORSA
PREFERRED SHARES**


A high-net-worth European investor is considering an investment in non-callable, non-convertible preferred shares issued by one of two regional banks. Monte Cabrio Bank has outstanding preferred shares with a EUR1,000 face value and an annual dividend of 6.75% that mature in five years, while Monte Corsa Bank has a new issuance of perpetual shares with a EUR1,000 face value.

1. If the high-net-worth investor has a required return on preferred equity of 7.5%, which of the following is closest to the intrinsic value per share of Monte Cabrio Bank's preferred offering?

- A. EUR724
- B. EUR970
- C. EUR1200

Solution

B is correct. The intrinsic value using Equation 28 with D of 67.50 (or 6.75% $\times$ 1,000), FV of 1,000, and r_p of 7.5% for $n = 5$ is

$$IV_t = \sum_{i=1}^5 \frac{67.5}{(1 + 0.075)^i} + \frac{1,000}{(1 + 0.075)^5} = \text{EUR}969.66.$$

We can solve this problem using a financial calculator or the Excel PV function [=PV(rate,nper,pmt,fv,type)] by entering PV(7.5%,5,67.5,1000,0).

A is incorrect because it mistakenly uses $D = 6.75$ as the numerator in the first expression.

C is incorrect because it calculates the present value of principal without the exponent of $n = 5$ in the second expression.

2. If the high-net-worth investor observes that Monte Corsa's newly issued perpetual shares are trading at a market price of EUR1,020 per share with an 8% annual dividend, which of the following is closest to the market's implied rate of return for this issuance?

- A. 8.5%
- B. 8.0%
- C. 7.8%

Solution

C is correct because it correctly applies Equation 29 to solve for r_p using IV_t of EUR1,010 and $D = \text{EUR}80$ ($= 8\% \times 1,000$), as follows:

$$IV_t = \frac{D}{r_p}.$$

$$\text{EUR}1,020 = \frac{80}{r_p}; r_p = 7.84\%.$$

A is incorrect because it incorrectly divides the dividend of 80 by the market price of 1,020 minus the dividend of 80 to solve for r_p .

B is incorrect because it uses the dividend rate as the market's implied rate of return.

Contingency features involve the right—but not the obligation—to either redeem preferred shares for a predetermined fixed price or convert preferred shares to common shares under certain conditions. Each of these features affects the intrinsic value of preferred shares because these rights have economic value to either issuers or investors. This value should be reflected by a difference in price relative to preferred shares with otherwise similar characteristics but without contingency features.

For example, **callable preferred shares** with a finite maturity are preferred shares that may be redeemed by an *issuer* prior to maturity, often at par value or at a predetermined premium to par that declines to par as the maturity date approaches. Issuers must typically offer a higher dividend than for a non-callable preferred security with similar features. Stated differently, the option to call preferred shares typically reduces the value of a preferred share to an investor versus a non-callable preferred share with the same dividend rate because the right to redeem will be exercised by the issuer when it is in the issuer's favor and allowed to expire when it is not.

Conversely, **putable preferred shares** allow an investor to sell shares back to an issuer at either par or a predetermined price, which generally involves a lower dividend rate versus similar non-putable shares.

Convertible preferred shares are a form of preferred stock that may both pay a dividend and be converted into common shares at a fixed **conversion ratio**, or number of common shares received for each preferred share, after a specific period. Investors in early-stage companies often contribute capital in the form of convertible preferred shares rather than ordinary shares due to several attractive features. For example, these shares have both debt- and equity-like characteristics. Like debt, they represent claims senior to those of common shareholders and may receive a fixed dividend or a cash distribution to preferred shareholders made in full before any payment to common shareholders. The conversion right to exchange preferred shares for ordinary shares in the future represents an equity-like return. The valuation of these contingency features will be addressed in detail later in the curriculum.

KNOWLEDGE CHECK: CONTINGENCY FEATURES FOR MONTE CORSA BANK PREFERRED



Recall that a high-net-worth European investor is considering an investment in non-callable, non-convertible perpetual preferred shares issued by Monte Corsa Bank with a EUR1,000 face value. At the time of Monte Corsa Bank's non-callable, non-convertible perpetual share issuance, the investor's wealth manager raises several comparable preferred share investments with contingency features for comparative purposes.

1. Which of the following choices properly ranks the expected dividend rates for otherwise identical preferred shares in order from highest to lowest?
 - A. Convertible preferred/non-callable, non-convertible preferred/callable preferred
 - B. Callable preferred/putable preferred/non-callable, non-convertible preferred
 - C. Callable preferred/non-callable, non-convertible preferred/putable preferred

Solution

C is correct because it correctly ranks preferred shares from the highest to lowest expected dividend. Investors expect to receive a premium in the form of a higher dividend in exchange for an issuer's right to redeem shares and expect to receive a lower dividend when acquiring the right to sell shares back to an issuer at a predetermined price.

A is incorrect because convertible preferred dividends would be expected to be lower than non-convertible preferred dividends.

B is incorrect because putable preferred dividends are expected to be below non-callable, non-convertible preferred dividends.

QUESTION SET 4



1. Which of the following statements most accurately describes the difference between the dividend discount model for equity valuation and the valuation of non-callable, non-convertible preferred shares with a finite maturity?

- A. Preferred share valuation involves a higher discount rate, a higher dividend, and a lower terminal value than the DDM.
- B. Preferred share valuation involves a lower discount rate than the DDM, a fixed dividend, and a par value rather than terminal value.
- C. Preferred share valuation involves a lower discount rate and a lower dividend than the DDM with a par value rather than terminal value.

Solution

B is correct because it correctly describes the valuation of non-callable, non-convertible preferred shares with a finite maturity as having a lower discount rate than common equity, a fixed dividend, and a return of par at maturity.

A is incorrect because valuation of non-callable, non-convertible preferred does not involve a terminal value.

C is incorrect because preferred shares do not necessarily have a lower dividend than common equities under the DDM.

2. An institutional investor has the choice between two non-callable, non-convertible preferred securities with the following characteristics:

- Millstone Financial: Three years to maturity with a 7.75% dividend
- Millennial Securities: Perpetual with a 7.75% dividend

If the required rate of return for both preferred shares is 8%, which of the following is closest to the expected price difference for the Millstone and Millennial preferred shares based on a par value of USD1,000?

- A. USD25
- B. USD5
- C. USD155

Solution

A is correct because it represents the difference between Millstone and Millennial preferred share prices based on Equation 28 and Equation 29, where D is 77.50 (or $7.75\% \times 1,000$), FV is 1,000, and r_p is 8% for $n = 3$:

$$\text{Millstone IV}_t = \sum_{i=1}^3 \frac{77.5}{(1 + 0.08)^i} + \frac{1,000}{(1 + 0.08)^3} = \text{USD}993.56.$$

$$\text{Millennial IV}_t = \frac{77.5}{0.08} = \text{USD}968.75.$$

We can solve for Millstone's value using a financial calculator or the Excel PV function [=PV(rate,nper,pmt,fv,type)] by entering PV(8%,3,77.5,1000,0). The expected share price difference is USD24.81 (= 993.56 – 968.75).

B is incorrect because it mistakenly uses the dividend rate for both the numerator and denominator to calculate Millennial's expected price.

C is incorrect because it calculates the present value of principal for Millstone without the exponent of $n = 3$ in the second expression.

3. An institutional investor is considering several alternatives to the non-callable, non-convertible Millennial preferred shares described in Question 2. Which of the following statements represents the most accurate comparison of contingency features?

- A. A convertible preferred contingency feature would be expected to result in a higher dividend than for a comparable non-callable, non-convertible share.
- B. A callable preferred contingency feature would be expected to result in a lower dividend than for a comparable non-callable, non-convertible share.
- C. A putable preferred contingency feature would be expected to result in a lower dividend than for a comparable non-callable, non-convertible share.

Solution

C is correct because a put feature involves an economic right for the investor that would result in a lower expected dividend rate.

A is incorrect because the convertibility feature would result in a lower, not higher, dividend rate given its economic value to the investor.

B is incorrect because the callability feature would result in a higher, not lower, dividend rate as an economic right for the issuer.

SUMMARY: DISCOUNTED CASH FLOW (DCF) AND GROWTH MODELS

- Discounted cash flow valuation models assume a share's value should equal the present value of expected future income and capital appreciation.
- The DCF equity valuation process involves selecting and forecasting a suitable cash flow measure, establishing a terminal value, and using an appropriate discount rate to determine the present value.
- Present value models based on dividends apply to firms with value derived from future capital income.
- Free cash flow–based metrics involving cash flows available for distribution to shareholders (FCFE) or to all capital providers (FCFF) are broadly applicable to firms with varying distribution policies and capital structures.
- Discounted cash flow models that use residual income as a measure seek to determine the value a firm generates *above* its expected return on equity.
- Drawbacks of present value models based on company-specific or bottom-up factors include both challenges of selecting a base year and static assumptions of underlying value drivers over a forecast horizon.
- Limitations of cash flow growth models based on top-down factors include estimating inputs used to determine the cost of capital given changing macroeconomic conditions.
- Non-callable, non-convertible preferred share values are estimated by discounting preferred dividends and share par value to the present with a growth rate of zero.
- Preferred share contingency features that benefit issuers will decrease the intrinsic value of otherwise similar preferred shares, while features of value to investors increase intrinsic share value.

PRACTICE PROBLEMS

1. Which of the following statements best describes the difference between the FCFE (free cash flow to equity) metric and dividends?
 - A. FCFE involves cash flows to shareholders in the form of both share buy-backs and cash dividends.
 - B. FCFE includes dividends, retained earnings, and changes in cash.
 - C. FCFE includes changes in cash plus dividends minus changes in equity.
2. Which of the following statements best describes the relationship between FCFE and FCFF?
 - A. FCFF and FCFE are equal as long as debt outstanding does not change.
 - B. When debt financing is used, a firm's FCFF will exceed FCFE as long as debt outstanding remains unchanged.
 - C. FCFF is always either greater than or equal to FCFE.
3. Storeham plc is expected to reach an FCFE of GBP15 per share next year with a cost of equity of 12.5%. Storeham's market price is GBP150 per share. Which of the following is closest to Storeham's implied FCFE growth rate?
 - A. 1.875%
 - B. 2.5%
 - C. 10%
4. Which of the following statements best describes the sustainable growth rate for a firm with required return on equity of 10%, a dividend payout ratio of 70%, and return on equity of 15%?
 - A. 4.5%
 - B. 7.0%
 - C. 10.5%
5. Which of the following statements best explains why preferred shares have a lower required rate of return than common shares for a given issuer?
 - A. Preferred shareholders have a priority claim over common stock in the receipt of dividends and on the issuer's assets in the event of liquidation.
 - B. Preferred shareholders receive fixed dividends, while common shares receive discretionary dividends.
 - C. Preferred shares generally have contingency features that represent an economic benefit to the issuer, resulting in a lower expected return.

SOLUTIONS

1. C is correct because it incorporates net cash flows distributed and available to shareholders once other obligations are satisfied.
B is incorrect because it ignores changes in equity from new issuance and share buybacks.
A is incorrect because it omits cash available to shareholders.
2. B is correct because FCFF is equal to FCFE plus after-tax interest expense less debt changes.
C is incorrect because FCFF can fall below FCFE if debt outstanding changes.
A is incorrect because FCFF and FCFE cannot be equal if debt is outstanding; they can be equal only if no debt is used by the firm.
3. B is correct because a 2.5% implied growth rate is consistent with the current share price using Equation 16. We can rearrange Equation 16 to solve directly for the implied growth rate as follows:

$$P_t = \frac{E(CF)_{t+1}}{\text{Required rate of return} - g_{\text{implied}}}$$

$$g_{\text{implied}} = \text{Required rate of return} - \frac{E(CF)_{t+1}}{P_t}$$

$$0.025 = 0.125 - \frac{15}{150}$$

C is incorrect because this response is the ratio of expected FCFE to the share price.

A is incorrect because this response is the product of the required rate of return and the FCFE per share.

4. A is correct. We can apply Equation 18 given the company's dividend payout ratio and return on equity, as follows:

$$\text{Sustainable growth rate}_t = \left(1 - \frac{\text{Dividends}_t}{\text{Net income}_t}\right) \times \left(\frac{\text{Net income}_t}{\text{Book value of equity}_{t-1}}\right)$$

$$\text{Sustainable growth rate} = (1 - 0.70) \times 0.15 = 0.045$$

B is incorrect because it results from taking the product of the dividend payout rate and the required return on equity.

C is incorrect because it results from taking the product of the dividend payout rate and the return on equity.

5. A is correct because the priority claim over common stock in the receipt of dividends and on assets under liquidation is the primary driver of the lower expected rate of return.
B is incorrect because the type or size of the dividend does not affect the discount rate applied based on the required rate of return.
C is incorrect because contingency features associated with preferred shares may represent an economic benefit to either an issuer or an investor.

LEARNING MODULE

7

Relative Value Equity Valuation Approaches

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	explain the process for using multiples to value equity based on the method of comparables and the method of forecasted fundamentals
☐	calculate and interpret price and enterprise value multiples, and describe their appropriate uses
☐	determine a potential peer group for multiples-based equity valuation based on a company's industry classification or specific drivers of expected return
☐	describe the use of multiples based on past, current, and projected future values in estimating equity value

INTRODUCTION: RELATIVE VALUE EQUITY VALUATION APPROACHES

1

Relative valuation compares market-based measures to income statement, balance sheet, or cash flow metrics to draw conclusions about a firm's value versus its current share price. In contrast to absolute valuation approaches, such as discounted cash flow models, which directly estimate share price or firm value, multiples or ratios are used as a basis of comparison under the assumption that similar assets should trade at similar prices. In addition, forecasts of the fundamental drivers underlying a multiple may be used to determine expected price or firm value. While simple to use and straightforward to compare, relative valuation requires a rigorous, well-constructed approach to maximize the level of confidence surrounding its application in selecting an equity investment.

In this learning module, we first discuss the process associated with applying multiples in relative valuation, which includes defining, describing, and analyzing the multiple. We then show how to calculate and interpret multiples and assess which are most appropriate under different circumstances. While we initially assume that the group of comparable companies is given, identification and selection of appropriate comparables is a critical step in the relative valuation process that can have a significant

impact on results. Thus, we later address the approaches used in determining a proper peer group. Finally, we describe how analysts consider historical, current, and projected future multiples to identify potential opportunities.

PRE-TEST: RELATIVE VALUE EQUITY VALUATION APPROACHES



1. A justified multiple for a company may be estimated using:

- A. only the method of comparables.
- B. only the method of forecasted fundamentals.
- C. either the method of comparables or the method of forecasted fundamentals.

Solution:

C is correct because a justified multiple reflects an estimate of a fair value of the multiple that can be compared to its actual multiple. In the method of comparables, a justified multiple is estimated from the multiples of the peer group. A justified multiple using the method of forecasted fundamentals can be calculated using a present value equity valuation model.

A is incorrect because the method of forecasted fundamentals can also be used to estimate a justified multiple.

B is incorrect because the method of comparables can also be used to estimate a justified multiple.

2. Outro Company trades at a price of EUR42 per share. The company's book value of equity per share is EUR16. Outro's return on equity (ROE) is 20%. The company's cost of equity is 10%. The company's expected future growth rate is 5%. Which one of the following statements about Outro's value is most correct?

- A. Outro stock cannot be valued with the given information.
- B. Outro stock trades at a multiple less than its justified multiple.
- C. Outro stock is overvalued.

Solution:

B is correct because Outro's P/B multiple is 2.625 ($= 42 \div 16$) while its justified P/B multiple is 3.0, calculated as follows:

$$\text{Justified } \frac{P}{B} = \frac{\text{ROE} - g}{r_e - g} = \frac{0.20 - 0.05}{0.10 - 0.05} = 3.0.$$

A is incorrect because there is sufficient information to assess the relative valuation by comparing Outro's actual P/B to a justified multiple, as shown above. Additionally, the information provided is sufficient to calculate a value using a residual income valuation.

C is incorrect because Outro stock is more likely to be undervalued given that the actual P/B is lower than the justified P/B.

3. Price-to-earnings ratios (P/Es):

- A. are higher for stocks with greater cost of equity.
- B. are higher for stocks with higher expected growth.

C. can be increased for a stock by increasing dividends.

Solution:

B is correct because P/Es are positively related to growth rates, holding all else equal, as shown in the following equation:

$$\frac{P_t}{\text{EPS}_t} = \frac{(\text{Payout ratio}) \times (1 + g)}{r_e - g}.$$

The growth rate, g , is positive in the numerator and negative in the denominator, thus leading to a higher P/E when g is higher.

A is incorrect because the cost of equity, r_e , is positive in the denominator, thus leading to a lower P/E when the cost of equity is greater.

C is incorrect because the equation above implies that for two or more firms with the same cost of equity and growth rate, the firm with the higher dividend payout should have a higher P/E. However, the equation does not imply that a company can increase its own P/E by paying a higher dividend, because that will reduce reinvestment and negatively affect its growth rate, g .

4. Two similar companies trade at different EV/EBITDA multiples. Which one of the following statements most likely describes features of the stock with the lower EV/EBITDA?

- A. Lower cost of capital, higher capital expenditures, and lower growth
- B. Higher cost of capital, higher capital expenditures, and lower growth
- C. Higher cost of capital, higher capital expenditures, and higher growth

Solution:

B is correct is because the constant growth enterprise value present value model reduces to the following justified multiple:

$$\frac{\text{EV}}{\text{EBITDA}} = \frac{(1 - t)}{\text{WACC} - g} + \frac{\text{Depreciation}(t)/\text{EBITDA}}{\text{WACC} - g} - \frac{\text{CapEx}/\text{EBITDA}}{\text{WACC} - g} - \frac{\Delta \text{Working capital}/\text{EBITDA}}{\text{WACC} - g}.$$

Cost of capital, WACC, is positive in the denominator, so it is inversely related to the multiple. Capital expenditures (CapEx) are subtracted in the numerator, so they also reduce the multiple. Growth, however, has a positive effect on the multiple because this term is subtracted in the denominator. Thus, higher cost of capital, higher capital expenditures, and lower growth are all factors implying a lower EV/EBITDA multiple.

A is incorrect because the lower cost of capital implies a higher EV/EBITDA multiple.

C is incorrect because higher growth implies a higher EV/EBITDA multiple.

5. When conducting a relative valuation for a target company stock, the most important criteria for selecting a peer group of companies is to define a subset that most closely matches the target's:

- A. industry classification.
- B. risk, growth, and cash flow characteristics.
- C. market capitalization.

Solution:

B is correct because the underlying criteria for peer group selection is to define a group that best matches the target company on the characteristics of

risk, growth, and cash flow. These criteria are difficult to satisfy in practice because no two firms are identical. However, the selection of an appropriate peer group should be cognizant of these criteria.

A is incorrect. Although peer groups commonly consist of industry peers, a final selection process should eliminate companies that have different risk, growth, and cash flow characteristics. Thus, industry classification is not the most important criterion.

C is incorrect. While company size as measured by market capitalization might be a factor *related* to risk, growth, and/or cash flow characteristics when considering for peer firms, it is unlikely to be the most important criterion.

6. An analyst completes a present value model with a five-year forecast horizon to estimate the value for a company. Which one of the following statements is most correct as to how the analyst could incorporate relative valuation into the value estimate?

- A. The analyst could multiply forecast future free cash flow to equity at the end of five years by a median ratio of price-to-FCFE from a peer group.
- B. The analyst could multiply the forecast base-year free cash flow to equity by a median ratio of price-to-FCFE from a peer group.
- C. The analyst could multiply the forecasted future free cash flow to equity at the end of five years by an average ratio of the company's historical price-to-FCFE.

Solution:

A is correct because the analyst has forecasted free cash flow to equity (FCFE) as part of the present value model process. By multiplying an end-of-forecast FCFE by the median peer group multiple, the analyst has an estimate of terminal value of the company's equity at the end of the forecast horizon.

B is incorrect because this statement reflects a simple application of the method of comparables. In this case, the analyst ignores the forecast of FCFE and simply applies the peer group multiple to current FCFE to estimate a current equity value.

C is incorrect because this statement reflects using the company's own historical multiples to estimate terminal value. This approach does not use relative valuation, because of the lack of any information from comparable company multiples.

2

PRICE MULTIPLES AND EQUITY VALUATION



explain the process for using multiples to value equity based on the method of comparables and the method of forecasted fundamentals

In an earlier learning module, candidates were exposed to the basics of using multiples as part of an equity valuation. In this lesson, we expand on earlier lessons by explaining the overall relative valuation process as was done for the present value approach in the prior learning module.

A firm's multiple is commonly compared to those of other companies considered to have similar characteristics to draw conclusions about the relationship between intrinsic value and market price using what is referred to as the **method of comparables approach**. No two firms are identical. However, applying the principle of no arbitrage, we might expect a firm's multiple to approach the multiples of similar firms that investors may consider to be close substitutes. If a firm's multiple is well *above* those of similar firms, then it may be considered *overvalued*, leading investors to reduce exposure or sell shares short. A multiple far *below* those of similar firms may lead an investor to conclude that a company is *undervalued*, representing an attractive investment under the assumption that the gap between estimated intrinsic value and price will converge. Alternatively, an issuer might decide to sell shares deemed to be overvalued and repurchase those believed to be undervalued using a relative valuation approach. While multiple differences may point to mispricing, there may also be valid reasons for why a multiple is higher or lower that warrant further analysis.

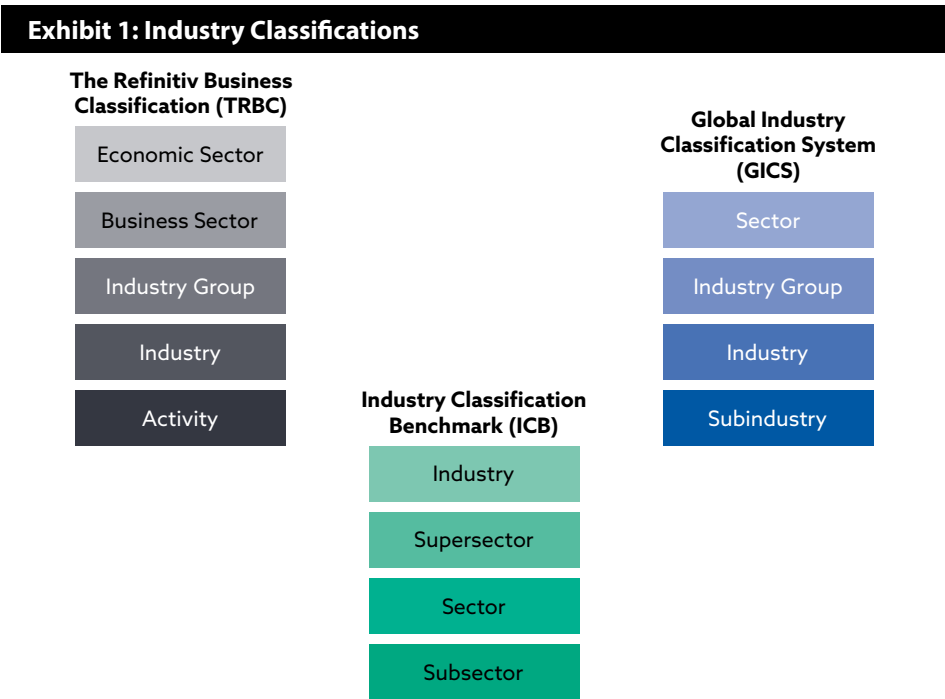
The Process of Relative Valuation Using the Method of Comparables

A well-organized process used to estimate relative value based on the method of comparables is structured as follows:

1. Choose a group of companies with similar characteristics for comparative purposes.
2. Select an appropriate multiple.
3. Calculate, compare, and interpret multiples for a company and its peers.
4. Establish whether a firm is overvalued, undervalued, or fairly valued relative to its peers using past, current, and/or expected future measures.
5. Use one or more multiples to substantiate DCF valuations and/or establish relative value comparisons to support an estimated valuation range versus a firm's actual trading range.

Peer Group Selection

The first step of a relative valuation process is to select an appropriate **peer group** for the company being valued. A peer group involves a group of comparable companies selected by a financial analyst as a benchmark for relative value comparisons. These comparable companies ideally should have similar cash flow, risk, and growth characteristics to the target firm being valued. Analysts often select a peer group using firms operating in the same industry or choose firms with similar statistical characteristics or factors that may operate in different businesses. Exhibit 1 shows common classification frameworks used to define industries.

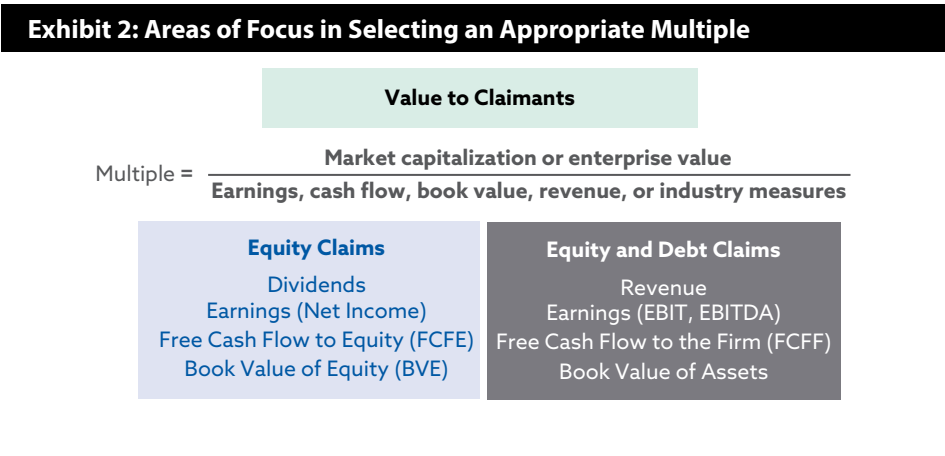


A key reason why comparable groups are commonly aligned with industry peers is that these firms typically face similar competitive forces and value drivers. These and other aspects of company and industry analysis will be covered in detail in a later learning module.

The importance of properly constructing an appropriate peer group is paramount to an analyst’s effective use of relative valuation. As a result, we devote a later lesson in this learning module to the construction of peer groups.

Multiple Selection

The next step of an analyst’s relative valuation process is to select one or more appropriate multiples to use to compare across the peer group. Exhibit 2 outlines this step in the process. As shown below, this ratio comprises a numerator that measures market value and a denominator that is a gauge of company fundamentals, such as earnings.



Analysts must ensure that multiples are consistently defined and computed for a company and its peers, which includes the standardization of measures across the numerator and denominator. For example, when a multiple has a company's stock price or market capitalization in the numerator, it should also use a denominator that reflects an equity claim. As shown later in detail, multiples considered for selection should incorporate the most appropriate value drivers for a firm and comparable companies. For example, a dividend-based multiple is applicable to firms with stable cash dividends and a constant payout ratio. Using such a multiple to compare non-dividend-paying firms with dividend-paying firms would reflect poor practice.

As shown in Exhibit 3, multiples may be categorized by the desired investor's perspective represented by the numerator (columns) and the source of fundamental information in the denominator (rows).

Exhibit 3: Overview of Commonly Applied Multiples

	Shareholders	All Capital Providers
Income Statement	P/E P/Sales	EV/EBITDA EV/EBIT EV/Sales
Balance Sheet	P/B	Firm Value/Book Value of Capital EV/Book Value of Capital less Cash
Cash Flow Statement	Dividend Yield Mkt. Cap/FCFE	EV/FCFF

To properly apply the method of comparables, once selected, a multiple must be defined and calculated consistently across all firms used for comparative purposes. Inconsistencies that may arise include differences in accounting methodology, reporting frequencies, and one-time, non-recurring items or timing differences, as shown in the following example.

KNOWLEDGE CHECK: COMPARING FIRMS WITH DIFFERENT REPORTING PERIODS



An analyst gathers the market price per share and book value of equity per share (as of the end of the most recent reporting period) of three companies operating in the same industry. The quarterly reporting periods for the three firms end on 31 December, 30 November, and 31 October, respectively.

1. Identify the most important timing issue associated with comparing the price-to-book ratios (P/Bs) of the three companies.

Solution:

The prices of the three stocks are measured at different points in time, thus distorting comparisons of P/Bs based on changing market conditions.

2. Recommend a possible solution to the timing issue identified in Question 1.

Solution:

The analyst should recalculate the P/Bs for each company using prices for each stock measured at the same point in time. For example, the analyst could use the prices on 31 December.

3. Describe any additional timing issues that may be problematic in comparing the three P/Bs.

Solution:

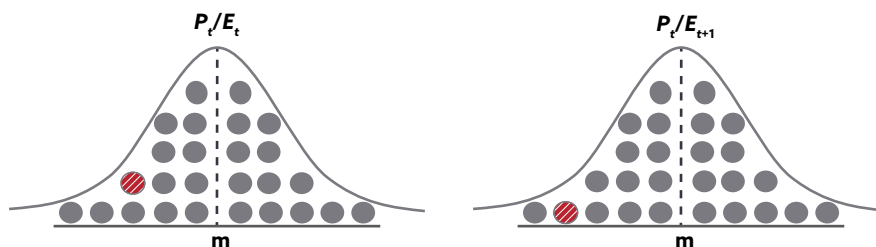
Because of the differences in reporting periods, the book values of equity (i.e., the denominators of the P/Bs) are not measured at the same point in time. For the two companies with earlier reporting dates, management may have taken actions (such as a stock repurchase) between the reporting date and December that affect the book value of equity and possibly the market value of equity. Unfortunately, an analyst may not be able to adequately adjust for this type of timing difference.

Calculate, Compare, and Interpret Multiples

Once a multiple has been consistently defined for a firm and its peer group, analysts calculate the multiple across the peer group. By comparing the multiple(s) of the company being valued against its peer group, an analyst can interpret the potential mispricing of the company being analyzed. As a first step in this portion of the process, analysts determine where the company's multiple lies within the distribution of peer group multiples. A measure of central tendency, such as the peer average (mean) or median, is commonly assumed to reflect the multiple's representative "fair value" for the peer group, with outliers more likely to be mispriced or incorrectly included in the sample.

For example, Exhibit 4 shows a firm's price-to-earnings multiple in relation to its peer group using current and estimated future earnings.

Exhibit 4: Firm Multiples Relative to a Peer Group



Based on the firm's present and expected future P/E relative to its peers shown in Exhibit 4, the firm appears to be undervalued. The use of multiples over different time periods is addressed in detail in a later lesson.

Beyond the mean value of the multiple, a thorough analysis involves a more complete assessment including such measures as dispersion and skewness. For example, a positively skewed distribution of a multiple may be associated with an average multiple that is well above its median value. In such cases, the median multiple is likely a better reflection of central tendency. The effect of extreme values on the overall distribution should be evaluated, and where possible, a consistent strategy should be developed for including or excluding outliers. A related issue arises from biases associated with peer firms for which a chosen multiple is not meaningful, such as P/E multiples that are negative when a firm has net losses. Both issues arise in the following example.

TREATMENT OF OUTLIERS IN RELATIVE VALUATION

The presence of outliers or observations for which a ratio is not meaningful due to a negative earnings denominator commonly arise when analysts seek to compare a set of multiples across peer firms. Consider the case of large European pharmaceutical companies as of 2024 shown in Exhibit 5.

Exhibit 5: Selected Multiples of Large European Pharmaceutical Firms

Company	Market Cap (EUR millions)	EV (EUR millions)	EV/EBITDA	P/E
Median	196,701	222,631	9.60	22.17
Bayer AG	28,033	65,536	3.77	N/A
GSK plc	85,974	101,263	8.18	16.06
Roche Holding AG	196,701	222,631	9.60	16.48
Novartis AG	215,470	231,678	11.25	23.66
Sanofi	112,922	121,589	9.18	20.68
AstraZeneca PLC	219,376	244,034	16.45	37.61
Novo Nordisk A/S-B	540,534	542,916	34.36	45.24

Source: Bloomberg.

Based on the information in Exhibit 5, Bayer AG is an outlier on the downside in the case of EV/EBITDA, while its P/E is not meaningful due to a net loss. Novo Nordisk, in contrast, is an outlier on the upside and appears to be over-valued relative to these industry peers. Assuming the firms are comparable, an analyst must decide whether to exclude Bayer and/or Novo Nordisk for these reasons, which will affect the median or mean for the remaining firms in the comparable set.

At first glance, the method of comparables may appear to be an attractive approach compared to DCF valuation because multiples are simple to calculate and require fewer assumptions. However, each multiple incorporates implicit assumptions. To properly apply multiples in a valuation, one must analyze these assumptions, assessing whether the company's fundamentals imply that it should trade at higher or lower multiples relative to the peer group, as shown in the following example.

EXAMPLE 1

Interpreting Multiples within a Peer Group

As mentioned earlier, variations among multiples may result from mispricing but can also reflect significant differences in expected risk and return associated with firms that may at first appear to operate in the same businesses.

For example, while Bayer AG has a large pharmaceutical division, nearly half of the firm's sales are derived from its Crop Sciences segment. Declining prices for its agricultural products were a key driver of Bayer's nearly EUR3 billion net loss in 2023. Also, the firm's 2016 acquisition of US-based Monsanto Corporation led to over USD10 billion in legal settlements due to allegations that its best-selling herbicide caused cancer, with ongoing legal battles that continue to weigh on the company's share price and earnings. Just over five years after

the Monsanto acquisition, Bayer's market capitalization was less than half of the USD63 billion purchase price of the company. The assumption of Monsanto's liabilities and the shift to underlying drivers of profitability that differ from the pharmaceutical business are factors that may explain Bayer's lower multiples.

In contrast to Bayer, Novo Nordisk focuses entirely on pharmaceutical products. The risk and return profiles of pharmaceutical companies vary widely across industry activities. For example, on the one hand, low-risk, low-growth companies focused on generic or over-the-counter medications typically produce and distribute products in markets with more than one supplier and few barriers to entry, leading to a competitive market for relatively low-cost drugs. On the other hand, firms actively focused on research to develop, test, and market new treatments incur significant risk over the product development cycle. If successful, these firms obtain regulatory approval and patent protection to sell unique products that generate considerably higher returns.

Novo Nordisk's activities are concentrated in the areas of diabetes and obesity care, as well as rare diseases. The company has enjoyed rapid growth in sales and profitability, both of which roughly doubled over a five-year period, due largely to soaring demand for Ozempic, developed to treat diabetes but approved for weight loss.

Relative Valuation Using the Method of Forecasted Fundamentals

As described in the prior example, a first step in analyzing a multiple is to understand underlying firm dynamics, such as profitability, growth, and investment, which may cause differences in multiples across firms.

A **justified multiple** for a company's equity reflects an estimate of a fair value of the multiple that can be compared to its actual multiple. In the method of comparables, the justified multiple is assumed as an average or median of peer group multiples.

Analysis of multiples can be extended to account for differences in key characteristics that likely cause variation in the multiples. To do this, we integrate basic principles from discounted cash flow valuation. The **method of forecasted fundamentals** refers to the use of multiples derived from drivers of value, including profitability, growth, and the reinvestment of earnings. These multiples include both past multiples and projected future multiples. A **trailing multiple** is based on the prior 12-month period. A **forward multiple** is based on the next 12-month period.

In the next lesson, we use the method of forecasted fundamentals in conjunction with several multiples. For now, we introduce this method using the dividend constant growth model used in the prior Equities learning module as a starting point:

$$P_t = \frac{\text{Div}_{t+1}}{r_e - g}, \quad (1)$$

where

P_t = a stock's price per share

Div_{t+1} = next period's expected dividend per share

g = assumed long-run constant dividend growth rate

r_e = stock's required rate of return

We can convert this formula into a justified multiple by dividing both sides of the expression by an appropriate denominator, such as the book value of equity per share, to solve for the justified price to book value of equity (BVE) per share:

$$\frac{P_t}{\text{BVE}_t} = \frac{\frac{\text{Div}_{t+1}}{\text{BVE}_t}}{r_e - g}. \quad (2)$$

Next, we modify Equation 2 to show the justified multiple in terms of underlying fundamentals. We do so by restating next period's dividend, D_{t+1} , as the product of expected earnings per share (EPS_{t+1}) and the expected payout ratio and recognizing that earnings per share (EPS) divided by book value of equity equals return on equity:

$$\text{Div}_{t+1} = \text{EPS}_{t+1} \times \text{Dividend payout ratio}.$$

$$\text{ROE} = \frac{\text{EPS}}{\text{BVE}}.$$

Substituting both fundamental relationships into Equation 2, we express the price-to-book multiple in terms of ROE and the payout ratio in Equation 3.

$$\frac{P_t}{\text{BVE}_t} = \frac{\text{ROE} \times \text{Dividend payout ratio}}{r_e - g}. \quad (3)$$

Equation 3 is an example of the method of forecasted fundamentals. In this case, for a firm with stable long-run characteristics, we can simply substitute *estimates* of the four inputs on the right-hand side of Equation 3 to solve for a company's justified P/B multiple. In this case, a justified multiple reflects characteristics of the underlying stock in terms of its return on equity, dividend payout ratio, cost of equity, and dividend growth rate.

To further simplify Equation 3, we recall the sustainable growth rate equation presented in a prior Equities learning module.

$$g = (1 - \text{Dividend payout ratio}) \times \text{ROE}. \quad (4)$$

After rearranging the sustainable growth rate equation to solve for the product of ROE and dividend payout ratio, we can rewrite Equation 3 as follows.

$$\frac{P_t}{\text{BVE}_t} = \frac{\text{ROE} - g}{r_e - g}. \quad (5)$$

As shown in Equation 5, the justified multiple using the method of forecasted fundamentals demonstrates that a P/B multiple may be simply expressed as a function of the *spread* between a company's ROE and its cost of equity (since the growth rate g is subtracted from both). The relative difference of the spread determines valuation: A more rapidly growing company should have a higher multiple.

KNOWLEDGE CHECK: PRICE-TO-BOOK MULTIPLE AND FUNDAMENTALS



An analyst gathers or estimates data on actual P/Bs, return on equity, expected constant growth rate (g), and cost of equity (r_e) for three firms.

Firm	P/B	ROE	g	r_e
Ardfield	2.42	13%	2%	8.20%
Bantry	1.32	10%	3%	8.00%
Cloyne	4.56	18%	5%	8.30%

1. Which one of the three firms is most likely undervalued based on its P/B multiple?

- A. Ardfield
- B. Bantry

C. Cloyne

Solution:

B is correct because Bantry's actual P/B is only 1.32, while its justified P/B is 1.40, as shown below using Equation 5:

$$\frac{P_t}{\text{BVE}_t} = \frac{\text{ROE} - g}{r_e - g}.$$

$$\text{Justified P/B (Bantry)} = \frac{0.10 - 0.03}{0.08 - 0.03} = 1.40 > 1.32.$$

Thus, Bantry's stock price appears to be lower than its fair value using the method of forecasted fundamentals.

A is incorrect because Ardfield's actual P/B is 2.42 while its justified P/B is 1.77, as shown below.

$$\text{Justified P/B (Ardfield)} = \frac{0.13 - 0.02}{0.082 - 0.02} = 1.77 < 2.42.$$

Thus, Ardfield's stock price appears to be higher than its fair value using the method of forecasted fundamentals.

C is incorrect because Cloyne's actual P/B is 4.56 while its justified P/B is 3.94, as shown below.

$$\text{Justified P/B (Cloyne)} = \frac{0.18 - 0.05}{0.083 - 0.05} = 3.94 < 4.56.$$

Thus, Cloyne's stock price appears to be higher than its fair value using the method of forecasted fundamentals.

While the prior example uses a constant growth dividend discount model, justified multiples using the method of forecasted fundamentals can be calculated using other, more complex equity valuation models. In each case, the critical element of the method of forecasted fundamentals is to identify the key value driver(s) associated with a specific multiple for comparison across companies when applying multiples. In the next lesson, we apply constant growth models to determine the key value drivers of other multiples.

Once a multiple has been defined, calculated, and analyzed for a firm and a set of comparable companies, the next step in the relative valuation is to *combine* the methods of comparables and forecasted fundamentals to not only compare multiples but also adjust for the differences in characteristics that drive potential variation in multiples across companies. Two common approaches applied to achieve this result are shown below.

1. Adjustment approach: Modify a multiple based on a fundamental variable to make it more directly comparable to others.
2. Regression approach: Create a statistical estimate of the multiple after controlling for one or more key company characteristics using statistical analysis, such as a multivariate regression.

A common application of the adjustment approach involves the application of different growth rates. In a later lesson, we show that expected growth rates are a primary characteristic that should drive differences in P/E multiples across similar firms. The **price/earnings-to-growth (PEG) ratio** seeks to adjust P/E multiples to account for differing growth rate expectations across comparable firms.

$$\text{PEG ratio} = \frac{\frac{P}{E} \text{ multiple}}{\text{Expected EPS growth rate}}. \quad (6)$$

Like P/E, the higher the PEG ratio, the more likely a stock is overvalued. The following example demonstrates how the PEG ratio is applied across a set of comparable firms to standardize P/E multiples among peers with different growth rate expectations.

EXAMPLE 2

P/E Multiples and PEG Ratios for the “Magnificent Seven”

In 2023, seven firms dominated the S&P 500's performance: Alphabet, Amazon.com, Apple, Meta, Microsoft, Nvidia and Tesla. Shares of these companies rose over 100% and constituted roughly two-thirds of the S&P 500 Index appreciation in that year. While often broadly referred to as technology companies, these firms are in different industries, with products ranging from online retail (Amazon.com) to graphics processing units and chips used in artificial intelligence and other applications (Nvidia) to electric vehicles (Tesla). A key common characteristic of these seven firms is their high rate of implied growth, which explains their high trailing P/Es, as shown in Exhibit 6.

Exhibit 6: Magnificent Seven P/E and PEG Ratios as of 2024

Company	Trailing P/E	Implied growth (%)	PEG	P/E rank	PEG rank
Nvidia	77.50	30.9	2.51	1	2
Amazon.com	51.87	25.3	2.05	2	5
Tesla	45.35	7.3	6.21	3	1
Microsoft	36.41	16.3	2.23	4	4
Meta	29.53	12.1	2.44	5	3
Apple	27.19	17.7	1.54	6	6
Alphabet	27.11	23.1	1.17	7	7

Source: FactSet.

The differences shown in the rankings of multiples after converting P/E multiples to PEG ratios implies some reinterpretations across the Magnificent Seven stocks. For example, Amazon.com showed the second-highest P/E, but after adjusting for growth expectations, its PEG ratio becomes more moderate.

As shown in Equation 6, the PEG ratio is based on the simple assumption that P/E multiples are linearly related to the growth rate. While the adjustment of a single variable may suffice in some cases, other approaches are preferable when the relationship between multiples and fundamental factors is more complex.

Regression analysis offers advantages over the adjustment approach when more than one fundamental factor is involved. For example, Equation 7 shows the relationship between P/B multiples and ROE, the growth rate, and the cost of equity. An analyst analyzing the P/B multiple for a firm and a comparable set of companies may gather data on ROE, growth rates, and the cost of equity for each firm to estimate the following cross-sectional regression:

$$\frac{P}{B} \text{ Multiple}_i = a_i + b_{1,i} \times \text{ROE}_i + b_{2,i} \times g_i + b_{3,i} \times r_{e,i} + e_i, \quad (7)$$

where the a_i and b_i terms represent the intercept and the coefficient estimates for the i th firm estimated from a cross-sectional regression of P/B multiples on ROE; growth rate, g_i , and cost of equity, $r_{e,i}$, respectively. The error term, e_i , for each firm provides

the unexplained difference between the firm's justified multiple predicted from the regression and the firm's actual multiple. Firms exhibiting larger error terms are more likely to be mispriced, as demonstrated in the following example.

KNOWLEDGE CHECK: A REGRESSION-BASED APPROACH



An analyst gathers the following data for 10 comparable companies.

Firm	P/B	ROE	Growth rate	Cost of equity
1	1.98	13%	2.0%	8.2%
2	1.49	10%	2.5%	8.0%
3	4.56	18%	5.0%	8.3%
4	3.74	15%	4.0%	7.5%
5	2.04	12%	2.5%	7.5%
6	2.48	14%	2.0%	7.8%
7	4.88	17%	4.0%	8.0%
8	4.80	20%	3.5%	7.8%
9	2.22	11%	3.5%	7.0%
10	3.27	16%	3.0%	8.5%

The analyst runs a regression of the P/Bs on the ROE, growth rate, and cost of equity variables. The resulting regression results suggest that the coefficient on the cost of equity variable is not significantly different from zero. The analyst runs the regression again with only ROE and growth rates as the explanatory variables and obtains the following regression output.

	Coefficient	Standard error	t-Stat.	P-value
Intercept	-2.58	0.49	-5.24	0.0012
ROE	29.02	3.99	7.27	0.0002
g	46.53	13.09	3.55	0.0093

1. Calculate Firm 3's justified P/B multiple based on a constant growth model.

Solution:

Using Equation 5 presented earlier,

$$\frac{P_t}{\text{BVE}_t} = \frac{\text{ROE} - g}{r_e - g} = \frac{18\% - 5\%}{8.3\% - 5\%} = 3.94.$$

2. Calculate the justified P/B multiple for Firm 3 predicted by the regression results.

Solution:

$$\text{Justified Firm 3 } \frac{P}{B} = -2.58 + (29.02 \times 18\%) + (46.53 \times 5\%) = 4.97.$$

3. Evaluate whether Firm 3 is fairly valued based on the regression results.

Solution:

The measured P/B for Firm 3 is given as 4.56. So, Firm 3's actual P/B is below the justified P/B of 4.97 estimated from the regression results. This difference implies that Firm 3 may be undervalued.

As demonstrated in the previous knowledge check, the regression approach to estimating justified multiples uses the same fundamental variables as in the development of the method of forecasted fundamentals assuming constant growth shown earlier. By incorporating data from the rest of the comparable multiples, the justified multiple predicted from regression results also incorporates the method of comparables without relying on the implications of the constant growth model.

Like the adjustment approach demonstrated using the PEG ratio, the regression approach to estimating justified multiples assumes a linear relationship between a multiple and its explanatory variables. Since these relationships may not be linear, analysts may need to explore other, non-linear approaches in estimating the impact of these fundamental factors.

The final step in the relative valuation process involves applying one or more multiples to estimate the intrinsic value per share. The use of multiples to establish intrinsic value occurs either through direct use of a multiple to infer a current price that reflects a share's fair value or indirectly as an input in a discounted cash flow model to establish the terminal value.

Recall from earlier Equities learning modules that the present value approach involves a forecast of a series of expected future cash flows plus a terminal value used to capture the value of future cash flows beyond the forecast period, as shown in Equation 8.

$$IV_0 = \sum_{t=1}^n \frac{E(CF_t)}{(1 + \text{Required rate of return})^t} + \frac{E(P_{t+n})}{(1 + \text{Required rate of return})^n} \quad (8)$$

The terminal value, $E(P_{t+n})$, in the numerator of the second term can be estimated using an appropriate multiple.

KNOWLEDGE CHECK: CALASTRA'S ESTIMATED MARKET CAPITALIZATION AND FCFE BASED ON PEER MULTIPLES



In late February, an analyst is estimating the intrinsic value of Calastra Group's equity as of December of the previous year. The company has numerous publicly listed competitors that serve as a comparable peer group of companies. The analyst decides to use trailing P/E multiples as her preferred relative valuation measure and finds that the median P/E in the comparable set is 16.0.

1. Calastra generated USD700 million in earnings in the prior four quarters. Calculate the analyst's likely intrinsic value estimate if she uses the peer group P/E to directly estimate Calastra's market-cap value.

Solution:

In this case, the analyst simply multiplies Calastra's trailing earnings by the peer group multiple to obtain a market-cap value estimate of USD11,200 million.

$$\text{Estimated market cap} = 700 \text{ million} \times 16.0 = 11,200 \text{ million.}$$

2. Using an alternative approach, the analyst forecasts the next five years of Calastra's net income and free cash flow to equity as follows:

Year	1	2	3	4	5
Net income	779	857	934	1009	1080
FCFE	580	661	745	830	920

The analyst then estimates Calastra's terminal value at the end of Year 5 using the peer group P/E multiple of 16.0. Calculate the analyst's likely revised estimate of Calastra's market-cap value if she assumes that Calastra has a 10% cost of equity.

Solution:

This process reflects a mixture of a present value approach and a relative value approach. In one step of the valuation process, the analyst calculates the present value of the five years of future free cash flow estimates.

$$\text{PV of FCFE} = \frac{580}{(1.10)^1} + \frac{661}{(1.10)^2} + \frac{745}{(1.10)^3} + \frac{830}{(1.10)^4} + \frac{920}{(1.10)^5} = 2,771.$$

The analyst calculates the terminal value at the end of the forecast horizon as the net income expected in Year 5 of the forecast by the peer group multiple.

$$\text{Terminal value} = 1,080 \times 16.0 = 17,280.$$

The terminal value is a future amount as of the end of five years, so the analyst discounts the terminal value using the cost of equity.

$$\text{PV of Terminal value} = \frac{17,280}{(1.10)^5} = 10,730.$$

Thus, the analyst's estimate of Calastra's market capitalization under these assumptions is USD13,501 million, or the sum of the present values of the FCFE and terminal value:

$$\text{USD13,501 million} = 2,771 \text{ million} + 10,730 \text{ million}.$$

Later in this learning module, we further explore the issues involved with using multiples to estimate a terminal value as part of an equity valuation that combines present value and relative valuation approaches. In particular, the peer group may need to be adjusted to reflect that today's peer group multiple may not be an appropriate comparison point for estimating a firm's future value.

QUESTION SET 1



1. An analyst argues that relative valuation is a poor tool for valuing companies and makes the following statements:

Statement 1 "No two firms are identical."

Statement 2 "It is pointless to compare P/Es across companies because of the differences in fundamental characteristics."

Which one of the following is correct regarding the analyst's two statements?

- A. Both Statement 1 and Statement 2 are true.

B. Statement 1 is true, and Statement 2 is false.

C. Statement 1 is false, and Statement 2 is true.

Solution:

B is correct because no two firms are truly identical since there are always differences among even very similar businesses. However, differences in fundamental characteristics among firms may be addressed by using adjustment approaches, such as those used in the PEG ratio or by conducting a regression of the P/E on fundamental characteristics.

A is incorrect because Statement 2 is not true.

C is incorrect because this response is not true for either statement.

2. Ferrillton Corporation generates a stable return on equity of 12%, and the company's cost of equity is 8%. The company pays a significant dividend that is expected to grow by 2% per year indefinitely. Which one of the following options most closely approximates Ferrillton's P/B and the company's dividend payout ratio?

A. P/B is 1.67, and the dividend payout ratio is 83%.

B. P/B is 1.50, and the dividend payout ratio is 75%.

C. P/B is 4.0, and the dividend payout ratio is 67%.

Solution:

A is correct because we can use Equation 5 to calculate Ferrillton's justified P/B as follows:

$$P/B = \frac{12\% - 2\%}{8\% - 2\%} = 1.67.$$

The dividend payout ratio can be found using either Equation 3 or Equation

4. Recall Equation 3:

$$\frac{P_t}{BVE_t} = \frac{ROE \times \text{Dividend payout ratio}}{r_e - g}.$$

$$1.67 = \frac{ROE \times \text{Dividend payout ratio}}{r_e - g} = \frac{12\% \times \text{Dividend payout ratio}}{8\% - 2\%}.$$

Using Equation 4,

$$g = (1 - \text{Dividend payout ratio}) \times ROE.$$

$$2\% = 12\% \times (1 - \text{Dividend payout ratio}).$$

Solving for the dividend payout ratio provides a result of 83.33%.

B is incorrect because P/B is calculated by dividing ROE by the cost of equity and the dividend payout ratio is calculated by subtracting the ratio of growth to cost of equity from 1.

C is incorrect because P/B is calculated by dividing the cost of equity by growth. The dividend payout ratio is calculated by dividing the cost of equity by ROE.

3. An analyst gathers data for P/Bs for each company among a set of 20 similar companies. She also gathers ROE and growth rates and runs a cross-sectional regression of P/B on ROE and growth rate. After doing so, she concludes that the stock with the lowest P/B among the companies is undervalued.

Which one of the following statements is most consistent with the analyst's conclusion?

- A. The lowest-P/B stock's predicted P/B is lower than its actual P/B.
- B. The lowest-P/B stock's predicted P/B is higher than its actual P/B.
- C. The lowest-P/B stock's predicted P/B is equal to its actual P/B.

Solution:

B is correct because a stock with a predicted multiple from a regression on the multiple's fundamental characteristics that is greater than its actual multiple is more likely to be undervalued. The regression controls for the fundamental characteristics (ROE and growth rate) of the P/B multiple, so the company's ROE and growth rates suggest that P/B should be higher than observed.

A is incorrect because a predicted multiple below the actual value of the multiple implies that the multiple is too high given its fundamental characteristics.

C is incorrect because equality of the predicted and actual multiples implies that the stock is fairly valued. In other words, it has a low P/B because of its poor fundamentals.

4. The application of multiples in a relative valuation may refer to:

- A. using comparable multiples as a direct approach to estimating the current intrinsic value of a company's equity.
- B. using comparable multiples to infer a company's future cash flow from its terminal value.
- C. using comparable multiples as an indirect approach to estimate a company's next expected dividend payment.

Solution:

A is correct because the process of applying multiples in a valuation often refers to estimating a multiple, such as a P/E, from comparable companies and then multiplying the multiple by an appropriate company fundamental, such as EPS, to obtain a current estimate of intrinsic value for the company.

B is incorrect because this statement reverses the order of a common application of using multiples in a terminal value estimation. A correct statement would include the fact that a multiple can be applied by multiplying it by a forecasted future fundamental variable, such as EPS, to obtain a terminal value estimate. Then, this terminal value may be used as part of estimating the company's value.

C is incorrect because a company's dividend payment is a possible fundamental variable that might be used as part of a multiple. However, the dividend is not estimated from the multiple as part of applying the multiple.

3

PRICE AND ENTERPRISE VALUE MULTIPLES



calculate and interpret price and enterprise value multiples, and describe their appropriate uses

Now that we have established the relative valuation process using the method of comparables and the method of forecasted fundamentals, we proceed to calculating, interpreting, and applying the most common multiples used to compare a company's share price or enterprise value to underlying income statement, balance sheet, or cash flow metrics, as shown earlier in Exhibit 1. Key areas of focus in calculating and interpreting multiples include the market value to claimants in the numerator and the relevant measure of earnings, cash flow, assets, or other resources available to satisfy these claims captured in the denominator.

Market Value as a Multiple Numerator

The value of a company that represents a multiple numerator may be measured from either the perspective of equityholders or all claimants to a firm's cash flows and net assets. While the former involves market value of equity, enterprise value (EV) is commonly used to measure a firm's value to all claimants:

$$\begin{aligned} \text{Enterprise value} &= \text{Market value of debt} + \text{Market value of equity} \\ &+ \text{Market value of preferred shares} - \text{Cash and short-term investments.} \end{aligned} \quad (9)$$

The book value of debt and preferred shares are commonly used as a proxy to market value in Equation 9. Cash and short-term investments are subtracted because they are non-operating assets.

Recall that multiples should reflect a value to claimants in the numerator aligned to cash flows or assets available to those same claimants in the denominator. For example, the market value of equity is divided by dividends or free cash flow to equity, while EV may be divided by free cash flow to the firm (FCFF), which includes interest payments to debtholders. While share prices are readily available each trading day, in some cases, price averages over a specific period are used. Other variables that make up the numerator, which may include the number of shares, debt outstanding, and cash, are typically based on the most recently reported figures in financial statements.

KNOWLEDGE CHECK: SCAROLA S.P.A.'S ENTERPRISE VALUE AND DEBT



An analyst is estimating the enterprise value for Scarola S.p.A., a publicly listed Italian corporation, five months after the company's fiscal year-end. The company reports financial results semiannually. As of the end of the fiscal year, Scarola's common equity included 2.5 billion shares outstanding, which closed at EUR30 per share as of year-end. Scarola's most recent financial statements reported EUR25 billion in debt and EUR8 billion of cash.

1. Calculate Scarola's EV as of the end of its fiscal year.

Solution:

Scarola's EV at the end of the fiscal year is EUR90 billion, calculated using Equation 9:

$$\text{Enterprise value} = \text{Debt} + \text{MV of equity} - \text{Cash} = 25 + (2.5 \times 30) - 8 = 92.$$

The analyst uses the book value of debt of EUR25 billion as a proxy for the company's market value of debt in the calculation.

2. Using large language models, the analyst incorporates the impact of news announcements during the last five months to estimate changes to the company's EV during the current year, noting the following news releases:

- At the beginning of the year, Scarola announced a EUR10 billion debt offering with a stated use of proceeds for share repurchases over the next several months. The company's stock price did not change materially at the time of the announcement.
- Scarola's share price rose from EUR30 to EUR33 over the five-month period following the debt issuance.

Calculate an appropriate EV estimate for the analyst.

Solution:

An appropriate estimate of EV would be EUR98.5 billion. The rationale for this estimate is shown below.

$$35 + (65 \times 1.10) - 8 = 98.5.$$

The analyst's mid-year estimate of EV should account for the following:

- Debt increased by EUR10 billion from the debt issue.
- The debt is used to decrease the company's equity by EUR10 billion, to EUR65 billion.
- The market price of the company's shares did not change upon the announcement, and the equity increased by 10% following the capital structure change.

Thus, the company's EV at the beginning of the year consisted of EUR35 billion in debt and EUR65 billion of equity, less EUR8 billion of cash. The change in the value of the corporation's equity between the beginning of the year and after five months was 10% [= (33 ÷ 30) – 1]. So, the increase in enterprise value reflects the 10% increase in equity value of EUR6.5 billion (= 65 × 0.10).

3. Calculate an alternative answer to the prior question if the company's debt offering were to be used to purchase new operating assets.

Solution:

If the company uses the debt offering to purchase operating assets, the EV estimate would be EUR109.5 billion.

$$\text{Enterprise value} = 35 + (75 \times 1.10) - 8 = 109.5.$$

In this case, the debt is not used to reduce the company's equity, so the equity term increases by EUR11 billion (= 82.5 – 71.5) from the prior answer.

Standardized Measures in a Multiple Denominator

The fundamental variables underlying the denominator of multiples used in relative valuation involve measures that are critical components of a standard discounted cash flow calculation, including elements of the three-statement financial model described earlier in the curriculum. We address multiples calculated based on each of these statements, in addition to measures that are applied on an industry-specific basis.

Income Statement Measures

Multiples using income-based measures in the denominator are among the most frequently used because earnings are a primary driver of firm value. Exhibit 7 shows how these variables relate to the income statement.

Exhibit 7: Income Statement Measures

Income Statement	Earnings	Sales	EBIT	EBITDA
Revenue	Net Income	Revenue	Before-Tax Int. Expense	Before-Tax Int. Expense
Cost of Goods Sold			Net Income	Net Income
Depreciation & Amortization			Taxes	Depreciation & Amortization
Operating Income				Taxes
Debt Service				
Taxes				
Net Income				

In Exhibit 7, net income is shown as the most frequently used measure of shareholder return; it incorporates all costs although analysts may adjust net income as suggested in earlier Financial Statement Analysis learning modules. Sales or revenue measures are common for early-stage, unprofitable companies and fully ignore a firm's cost structure. Both EBIT and EBITDA are based on operating income. EBIT excludes depreciation and amortization to measure operating income available to both debt and equity investors, whereas EBITDA is a broader, non-standardized measure that excludes depreciation, amortization, and other non-cash expenses. We examine these measures in greater detail below.

Earnings-Based Multiples

Net income, often referred to as earnings, reflects accounting income after deducting all expenses, including debt service and taxes. While earnings are not a cash flow measure, they constitute a theoretical amount of income available for distribution to shareholders or retained for reinvestment. The **price-to-earnings ratio (P/E)**, or the market share price per unit of earnings, introduced earlier uses net income in the denominator divided by the weighted average number of shares outstanding during the period to derive earnings per share.

$$P/E = \frac{P_t}{EPS}. \quad (10)$$

Equation 10 omits a time subscript on the EPS term because P/E multiples may make use of consensus forward earnings estimates from analysts covering a specific stock (EPS_{t+1}) to calculate a forward P/E or the last four quarters (EPS_t) to calculate a trailing P/E.

This widely cited multiple is often the primary focus of security analysts, with its inverse, E/P , referred to as **earnings yield**, representing a commonly used measure of equity return. In fact, a company's earnings yield computed based on expected *future*

EPS (referred to as forward EPS) can be used to gauge the market's assessment of a company stock's price growth opportunities. As shown in an earlier learning module, the stock of a company with no growth opportunities may be valued as follows:

$$P_t = \frac{\text{Div}_{t+1}}{r_e}. \quad (11)$$

Because this firm has zero growth opportunities, the firm should pay 100% of its earnings as dividends, and we can restate Equation 11 as

$$P_t = \frac{\text{EPS}_{t+1}}{r_e}. \quad (12)$$

As Equation 12 represents the equity value of a firm with zero growth opportunities, we can state a general expression of the equity value of a firm with growth opportunities:

$$P_t = \frac{\text{EPS}_{t+1}}{r_e} + \text{PVGO}, \quad (13)$$

where PVGO stands for **present value of growth opportunities**. By dividing the above expression by P_t and rearranging terms, we can show the following relationship between a firm's forward earnings yield and its cost of equity and growth opportunities:

$$\frac{\text{EPS}_{t+1}}{P_t} = r_e \left(1 - \frac{\text{PVGO}_t}{P_t} \right). \quad (14)$$

The forward earnings yield is simple to observe for a publicly listed equity by inverting the stock's forward P/E. Analysts commonly estimate the cost of equity, r_e , when conducting a valuation exercise. Using these two elements of Equation 14, the ratio of PVGO to a company's stock price can be estimated as follows:

$$\frac{\text{PVGO}_t}{P_t} = 1 - \frac{\frac{\text{EPS}_{t+1}}{P_t}}{r_e}. \quad (15)$$

KNOWLEDGE CHECK: PRESENT VALUE OF GROWTH OPPORTUNITIES



An analyst compiles the following information for four different firms.

	Firm 1	Firm 2	Firm 3	Firm 4
Price	50.00	120.00	35.00	80.00
Cost of equity	10%	9%	12%	8%
Forward EPS estimate	3.00	8.75	0.84	7.20

1. Calculate Firm 1's earnings yield and the proportion of its stock price that reflects the present value of its growth opportunities.

Solution:

Firm 1's earnings yield is 6% ($= 3.00 \div 50.00$).

Using Equation 15, we solve for the present value of growth opportunities as a proportion of Firm 1's stock price as follows:

$$\frac{\text{PVGO}_t}{P_t} = 1 - \frac{\frac{3.00}{50}}{0.10} = 40\%.$$

Firm 1's stock price of 50 per share consists of 40% PVGO, which reflects 20 per share ($= 40\% \times 50$). The additional 30 per share reflects the present value of Firm 1's forward earnings ($= 3.00 \div 0.10$).

2. Which one of the following companies has the lowest ratio of PVGO to stock price?

- A. Firm 2
- B. Firm 3
- C. Firm 4

Solution:

C is correct because the PVGO to stock price for Firm 4 is –12.5%, calculated using Equation 15:

$$\frac{\text{PVGO}_t}{P_t} = 1 - \frac{\frac{7.20}{80}}{0.08} = -12.5\%.$$

This proportion is the lowest among the three companies, as shown below.

A is incorrect because the PVGO to stock price for Firm 2 is 19.0%:

$$\frac{\text{PVGO}_t}{P_t} = 1 - \frac{\frac{8.75}{120}}{0.09} = 19.0\%.$$

B is incorrect because the PVGO to stock price for Firm 3 is 80.0%:

$$\frac{\text{PVGO}_t}{P_t} = 1 - \frac{\frac{0.84}{35}}{0.12} = 80.0\%.$$

A negative PVGO value as shown in the prior example may be interpreted as a sign that reinvestment in the company will diminish its value, suggesting that earnings should instead be distributed to shareholders.

The prior exercise highlights the inverse relationship between earnings yields and investor perceptions of growth opportunities reflected in share prices. However, an analyst who compares similar stocks based on earnings yield should adjust this measure by any cost of equity differences to make appropriate comparisons. In so doing, the analyst can compare growth expectations already priced into each stock and draw other comparisons as necessary to assess whether expectations are too high or too low across firms. Because PVGO reflects what investors are willing to pay for expected long-term growth, we return to the Magnificent Seven stocks to consider the growth expectations embedded in current prices.

EXAMPLE 3

The Magnificent Seven and the Present Value of Growth Opportunities

We can demonstrate the present value of growth opportunities as related to price (PVGO/P) using the seven rapidly growing firms from an earlier example (see Exhibit 8).

Exhibit 8: PVGO and Related Data for the Magnificent Seven

Company	Forward P/E	Forward Earnings Yield	Cost of Equity	PVGO/P
Tesla	72.89	1.37%	11.90%	88.47%
Nvidia	36.78	2.72%	12.27%	77.84%
Amazon.com	40.62	2.46%	10.27%	76.03%
Microsoft	35.60	2.81%	9.46%	70.31%

Company	Forward P/E	Forward Earnings Yield	Cost of Equity	PVGO/P
Apple	28.82	3.47%	9.61%	63.89%
Meta	23.34	4.28%	9.96%	56.98%
Alphabet	23.57	4.24%	9.66%	56.08%
S&P 500	19.16	5.22%	9.02%	42.14%

Source: FactSet.

The higher PVGO/P of the Magnificent Seven versus the S&P 500 suggests that investors in these seven companies are paying a higher proportion of the share price based on future investment opportunities than the rest of the market. Exhibit 8 highlights that a higher forward P/E multiple does not perfectly correlate with expectations of growth priced into a stock after controlling for differences in the cost of equity. For example, Amazon.com shows the second highest forward P/E, but because of its lower cost of equity, its PVGO/P falls slightly below that of Nvidia.

Empirical research suggests that stock P/E differences are related to long-term average investment returns. As in the case of the P/B multiple shown in the prior lesson, we can show the fundamental relationship between the P/E multiple and growth, risk as estimated by the required rate of return, and a firm's payout policy by starting with the constant growth dividend discount model from Equation 1:

$$P_t = \frac{D_{t+1}}{r_e - g},$$

where D is the dividend, r_e is the required return on equity, and g is the growth rate. We can divide both sides of the expression by earnings per share and incorporate the fact that the payout ratio is equal to dividends per share (DPS) divided by earnings per share:

$$\text{Payout ratio} = \frac{D_{t+1}}{\text{EPS}_{t+1}} = \frac{\text{DPS}}{\text{EPS}}.$$

These steps allow us to convert Equation 1 into a justified multiple using the method of forecasted fundamentals for a company's trailing P/E:

$$\frac{P_t}{\text{EPS}_t} = \frac{(\text{Payout ratio}) \times (1 + g)}{r_e - g}. \quad (16)$$

This approach is illustrated in the following example.

KNOWLEDGE CHECK: P/E AND FUNDAMENTALS



An analyst is preparing a presentation to demonstrate the relationship between P/E multiples and company fundamentals. She prepares a sensitivity analysis to show justified trailing P/E multiples based on varying two fundamental characteristics while holding the third characteristic fixed.

1. First, the analyst assumes a fixed payout ratio of 80%. Calculate the justified trailing P/E multiple for a company with 3% expected growth and 8% cost of equity.

Solution:

The justified P/E multiple is 16.48, and the calculation is shown below using Equation 16:

$$\frac{P_t}{\text{EPS}_t} = \frac{(\text{Payout ratio}) \times (1 + g)}{r_e - g} = \frac{0.8 \times (1 + 0.03)}{0.08 - 0.03} = 16.48.$$

Next, the analyst completes the sensitivity analysis of justified trailing P/E multiples assuming a fixed dividend payout ratio of 80%, with varying cost of equity from 8% to 12% (shown in the rows in Exhibit 9) and varying growth from 0% to 5% (in the columns).

2. Explain the change in P/E multiples as cost of equity increases.

Solution:

P/E multiples decline as cost of equity increases. Observing Equation 16 from the prior solution, cost of equity (r_e) is a positive term in the denominator, so increasing this term results in a lower P/E. To provide a direct comparison to the scenario shown in the prior question, the analyst uses 12% instead of 8% for the cost of equity in Equation 16 while keeping the growth rate at 3%.

$$\frac{P_t}{\text{EPS}_t} = \frac{(\text{Payout ratio}) \times (1 + g)}{r_e - g} = \frac{0.8 \times (1 + 0.03)}{0.12 - 0.03} = 9.16.$$

An increase in cost of equity from 8% to 12% results in a decrease in the justified multiple from 16.48 to 9.16.

3. Explain the change in P/E multiples as the growth rate increases.

Solution:

P/E multiples increase as the growth rate increases, holding other factors equal. The growth rate (g) appears in both the numerator and the denominator of Equation 16. Because the numerator term shows growth as a positive and the denominator shows growth as a negative, a positive relation between the growth rate and P/E multiples is implied. To provide a direct comparison to the scenario shown in the prior question, the analyst uses 5% instead of 3% for the growth rate in Equation 16 while keeping cost of equity at 8%.

$$\frac{P_t}{\text{EPS}_t} = \frac{(\text{Payout ratio}) \times (1 + g)}{r_e - g} = \frac{0.8 \times (1 + 0.05)}{0.08 - 0.05} = 28.00.$$

An increase in growth rate from 3% to 5% results in a decrease in the justified multiple from 16.48 to 28.00.

As shown in the prior exercise, firms with a higher g and lower r_e should have higher P/Es, holding other factors constant. Exhibit 9 shows the complete sensitivity analysis of justified P/E multiples described previously.

Exhibit 9: Justified Trailing P/E Multiples at an 80% Dividend Payout Ratio

r_e/g	0%	1%	2%	3%	4%	5%
8%	10.00	11.54	13.60	16.48	20.80	28.00
9%	8.89	10.10	11.66	13.73	16.64	21.00
10%	8.00	8.98	10.20	11.77	13.87	16.80
11%	7.27	8.08	9.07	10.30	11.89	14.00
12%	6.67	7.35	8.16	9.16	10.40	12.00

Alternatively, an analyst could conduct a sensitivity analysis using a fixed cost of equity or a fixed growth rate to illustrate how P/E multiples should differ across firms as the other two variables change. Although not explicitly demonstrated here, a higher payout ratio (that is, a higher fraction of earnings distributed to shareholders) should also result in a higher P/E, holding the other two variables constant. However, higher growth rates may be less likely with higher payout ratios as implied by the sustainable growth rate equation presented in an earlier learning module. Higher growth rates often implicitly assume a firm is earning a higher return on equity.

Drawbacks of P/E as a relative value measure include that EPS can be zero, negative, or insignificantly small relative to price, in which case the P/E measure may be either misleading or not meaningful. Also, while the P/E numerator is quoted on either the current market price or the average price over a period, the earnings denominator is based on complex accrual accounting rules and presents significant interpretation issues. For example, recurring earnings components that are most important in establishing intrinsic value can be difficult to distinguish from non-operating and non-recurring items, and differences in accounting choices and estimates may distort EPS, adversely affecting the comparability of P/Es among companies.

GAMESTOP P/E AND THE SOURCE OF A COMPANY'S EARNINGS

An earlier learning module introduced GameStop (GME), the world's largest video game retailer, with over 4,400 stores, and examined its competitive challenges before, during, and following the COVID-19 pandemic. GameStop's business problems were highlighted by large operating losses from 2019 to 2023, in the range of USD200 million to USD300 million per year.

In its fiscal 2024 annual results, GameStop reported an operating loss of just USD34.5 million, which represented a significant improvement compared to prior years. Combining this operating loss with USD49.5 million in interest income resulting from higher short-term interest rates on its large cash balances, GameStop reported positive net income of USD6.7 million (USD0.02 per share). With a stock price hovering around USD14 per share, the stock's P/E multiple was about 700. However, despite having a positive denominator, the economic interpretation of GameStop's P/E was difficult to determine given the combination of continuing operating losses and the offsetting non-operating income generated solely from the firm's large cash balances.

Non-recurring items, such as gains and losses from asset sales, write-downs, goodwill impairment, provisions for future losses, changes in accounting estimates, or income/loss from discontinued operations, should be removed from the earnings denominator. The P/E calculation should target what are defined as **core earnings**, or those that exclude these one-time items that do not reflect ongoing operating performance.

Analysts use alternative P/E definitions to capture different time horizons over which earnings are measured, to address cyclicity over the economic cycle, or to filter out non-recurring items that distort earnings over time. In the case of cyclical companies or industries, the most recent earnings or a near-term forecast may not accurately reflect the long-term earning power of a business that is highly sensitive to the economic cycle. Trailing EPS for such stocks is often low or negative at the bottom of a cycle and unusually high at the peak. Analysts of cyclical firms often use **normalized EPS**, or an estimate of EPS under mid-cycle market conditions. This approach uses either the historical average EPS over the most recent full cycle or the method of average return on equity, in which normalized EPS is calculated as the average return on equity from the most recent full cycle, multiplied by current book value per share.

KNOWLEDGE CHECK: ZENSIA'S TRAILING, NORMALIZED, AND ADJUSTED P/ES

Zensia SA, a European chemical company, faces fluctuating demand for its products that mirrors the economic cycle. The following table shows Zensia's EPS, book value of equity per share (BVPS), and return on equity.

	Year -7	Year -6	Year -5	Year -4	Year -3	Year -2	Year -1
EPS	0.62	0.53	0.24	0.63	1.49	1.02	0.89
BVPS	3.00	3.43	3.57	4.10	5.49	6.41	7.20
ROE	20.7%	15.5%	6.7%	15.4%	27.1%	15.9%	12.4%

Zensia's stock price is EUR18.21. Dividing this number by Zensia's most recent EPS, an analyst finds that the stock's P/E multiple is 20.46. However, she is concerned that Zensia's current P/E may be biased due to the company's cyclicity.

1. Calculate a revised P/E for Zensia using historical average EPS.

Solution:

Zensia's P/E multiple using the average of its EPS is 23.52:

$$\text{Average EPS} = \frac{0.62 + 0.53 + 0.24 + 0.63 + 1.49 + 1.02 + 0.89}{7} = 0.7743.$$

$$\text{P/E} = \frac{18.21}{0.7743} = 23.52.$$

2. Calculate a revised P/E for Zensia using average return on equity.

Solution:

Zensia's P/E multiple using the average return on equity is 15.6:

$$\text{Average ROE} = \frac{20.7\% + 15.5\% + 6.7\% + 15.4\% + 27.1\% + 15.9\% + 12.4\%}{7} = 16.2\%.$$

$$\text{Implied EPS} = \text{BVPS}_{\text{Year-1}} \times \text{Average ROE} = 7.20 \times 16.2\% = 1.1664.$$

$$\text{P/E} = \frac{18.21}{1.1664} = 15.6.$$

The prior exercise demonstrates that application of these two methods can lead to significant differences in the calculated EPS. In cases in which a company's size changes significantly, the approach based on average ROE is typically preferred. The average EPS approach does not adequately control for change in company size over time.

An additional factor to consider in calculating EPS for the P/E is the effect of dilution—that is, the case in which all outstanding executive stock options, equity warrants, and convertible bonds were to be exercised, resulting in an increase in shares outstanding and a lower EPS.

Revenue-Based Multiples

Revenue, or sales, is the broadest income statement measure used in equity valuation. If revenue recognition policies across firms are similar, multiples using revenue in the denominator are less subject to distortion or manipulation than net income and are usually positive even when other income-based measures are negative.

As sales typically fluctuate less than earnings, sales-based multiples are usually more stable than those based on net income or operating income and are considered appropriate for valuing the stocks of mature, cyclical, and negative-income companies. However, these multiples may be misinterpreted when businesses show high sales growth but do not operate as profitably or efficiently as other firms based on other income and cash flow measures.

Revenue-based multiples are commonly applied using either the share price or enterprise value in the numerator. Although sales revenue is available to all capital providers, it is commonly used in multiples combined with the share price for firms with little or no leverage.

The **price-to-sales ratio** (P/S) is the market price per share divided by annual net sales per share, where net sales are total sales minus returns and customer discounts. We can show the fundamental relationship between P/S and P/E by dividing both sides of the constant growth dividend discount model from Equation 1 by sales and noting that dividends divided by sales may be expressed as the product of the net income margin and the payout ratio:

$$P_t = \frac{\text{Div}_{t+1}}{r_e - g}$$

$$\frac{\text{Dividends}}{\text{Sales}} = \frac{\text{Net income}}{\text{Sales}} \times \frac{\text{Dividends}}{\text{Net income}} = \text{Net profit margin} \times \text{Payout ratio}.$$

$$\frac{P_t}{\text{Sales}_t} = \frac{(\text{Net profit margin}) \times (\text{Payout ratio}) \times (1 + g)}{r_e - g} \quad (17)$$

The fundamental factors shown to drive P/S are those for P/E (Equation 16) with the inclusion of the net profit margin in the numerator.

Like other multiples observed so far, P/S should be negatively related to cost of equity, holding other factors constant. The effect of the payout ratio on the multiple should be positive, although payout decisions may affect future growth rates. As a result, it may be difficult to separate the effects of the payout ratio and growth rates.

KNOWLEDGE CHECK: P/S AND FUNDAMENTALS



In an earlier knowledge check, an analyst considered a firm with a dividend payout ratio of 80%, an expected growth rate of 3%, and cost of equity of 8%. Suppose this company generates a stable net profit margin of 15%.

1. Calculate the justified P/S for the company.

Solution:

Using Equation 17, the justified P/S is 2.47:

$$\frac{P_t}{\text{Sales}_t} = \frac{(0.15) \times (0.80) \times (1.03)}{0.08 - 0.03} = 2.472$$

2. Using the justified P/S multiple, calculate the firm's justified P/E multiple.

Solution:

As mentioned above, the justified P/S is equal to the justified P/E multiplied by the net profit margin (since net profit margin is equal to earnings divided by sales). Solving for the justified P/E, we get the following:

$$\text{Justified } \frac{P}{E} = \frac{\text{Justified } \frac{P}{S}}{\text{Net profit margin}}$$

$$\text{Justified } \frac{P}{E} = \frac{2.472}{0.15} = 16.48.$$

This calculation verifies the result shown in an earlier knowledge check.

3. The analyst finds another company with the same payout ratio, expected growth, and cost of equity. However, the second company generates a stable net profit margin of 20% rather than 15%. Calculate the percentage difference between the second firm's justified P/S multiple and the 2.47 multiple of the first company.

Solution:

Since the other three terms are identical for the two firms, the only source of difference between the two is the 5% net profit difference in the numerator terms of the calculation. Thus, the percentage difference is simply 33.33% [= (0.2 ÷ 0.15) – 1]. Alternatively, the percentage difference may be determined by calculating the justified P/S multiple using the second firm's net profit margin of 20% and then finding the percentage difference between the two justified multiples.

4. Suppose the first firm decreases its payout ratio from 80% to 70% and can increase its growth rate from 3% to 3.5%. Calculate the first firm's new justified P/S, and interpret the change in the ratio compared to its original value.

Solution:

The updated value of the first firm's justified P/S is 2.415. The calculation is shown below.

$$\frac{P_t}{\text{Sales}_t} = \frac{(0.15) \times (0.70) \times (1.035)}{0.08 - 0.035} = 2.415.$$

The justified P/S declines slightly from 2.472 to 2.415. In this example, the current dividend decline associated with the change in payout ratio from 80% to 70% had more of a negative effect on the ratio than the positive influence from the improvement in expected growth rate of future dividends from 3.0% to 3.5%.

The prior exercises not only illustrate calculation of the justified multiple but also demonstrate a linear relationship between justified P/E and justified P/S. A firm cannot mechanically improve its multiples with higher growth, because the additional growth likely requires lower payouts to shareholders, other things held equal.

Despite their advantages, price-to-sales multiples have some drawbacks, most notably the logical mismatch between shareholder claims in the numerator and a measure available to both equityholders and debtholders in the denominator. While it is less of an issue for early-stage companies, which typically rely more on equity financing, many analysts prefer to use an enterprise value (EV) to sales multiple when valuing firms with significant debt or complex capital structures. Comparisons of price-to-sales ratios also require analysts to adjust for cost structure differences across companies and are subject to differing revenue recognition practices, which may distort comparisons.

Multiples Based on Operating Income

Since operating earnings are available to both debtholders and equityholders, these income statement measures are usually considered in combination with enterprise value or **firm value**, allowing for a useful comparison among firms that have different financial structures. Firm value is simply the sum of the market value of debt and the market value of equity, while the enterprise value deducts cash and short-term

marketable investments from firm value. As shown earlier for price-based multiples, we can connect measures based on firm value to fundamentals using a constant growth discounted cash flow model for the enterprise value of the firm using free cash flow to the firm in the numerator and the weighted average cost of capital (WACC) in the denominator:

$$EV_t = \frac{FCFF_{t+1}}{WACC - g} \quad (18)$$

An earlier learning module defined FCFF as

$$FCFF = EBITDA(1 - t) + \text{Depreciation}(t) - \text{CapEx} - \Delta \text{Working capital} \quad (19)$$

Substituting this into Equation 18 gives us the enterprise value as a function of fundamental factors, as shown in Equation 20:

$$\text{Enterprise value} = \frac{EBITDA(1 - t) + \text{Depreciation}(t) - \text{CapEx} - \Delta \text{Working capital}}{WACC - g} \quad (20)$$

We can divide both sides of Equation 20 by EBITDA to establish the drivers of an EV-based EBITDA multiple:

$$\begin{aligned} \frac{EV}{EBITDA} &= \frac{(1 - t)}{WACC - g} + \frac{\text{Depreciation}(t)/EBITDA}{WACC - g} \\ &\quad - \frac{\text{CapEx}/EBITDA}{WACC - g} - \frac{\Delta \text{Working capital}/EBITDA}{WACC - g} \end{aligned} \quad (21)$$

The following knowledge check explores these disaggregated drivers of value in greater detail.

KNOWLEDGE CHECK: DRAYTON'S EV/EBITDA MULTIPLE AND FUNDAMENTALS



An analyst is analyzing the observed EV/EBITDA multiple of 3.52 for Drayton Company stock. The analyst gathers a variety of fundamental data for the company to estimate a justified multiple assuming constant growth.

She estimates Drayton's WACC as 7.1% and long-run growth of FCFF as 2.0%. Drayton's base-year ($t = 0$) EBITDA is equal to EUR300 million, and its depreciation expense is EUR150 million. Drayton's capital expenditures total EUR190 million, and the firm operates with zero net working capital. Drayton's tax rate is 30%.

1. Calculate a justified EV/EBITDA multiple for Drayton given the fundamental information provided.

Solution:

Using Equation 21, we can calculate Drayton's justified EV/EBITDA multiple.

$$\begin{aligned} \frac{EV}{EBITDA} &= \frac{(1 - t)}{WACC - g} + \frac{\text{Depreciation}(t)/EBITDA}{WACC - g} \\ &\quad - \frac{\text{CapEx}/EBITDA}{WACC - g} - \frac{\Delta \text{Working capital}/EBITDA}{WACC - g} \\ \frac{EV}{EBITDA} &= \frac{1 - 0.30}{0.071 - 0.02} + \frac{\frac{150 \times 0.3}{300}}{0.071 - 0.02} - \frac{\frac{190}{300}}{0.071 - 0.02} = 4.25. \end{aligned}$$

2. After further research, the analyst believes that Drayton's stable tax rate is 25% and the company can grow at a long-run rate of 2.5%. To achieve this higher level of growth, the analyst assumes capital expenditures should be higher, at EUR230 million, along with depreciation of EUR180 million.

Calculate the analyst's updated estimate of Drayton's justified EV/EBITDA multiple.

Solution:

Applying the analyst's updated assumptions to Equation 21 leads to an updated EV/EBITDA multiple of 2.90:

$$\frac{\text{EV}}{\text{EBITDA}} = \frac{1 - 0.25}{0.071 - 0.025} + \frac{\frac{180 \times 0.25}{300}}{0.071 - 0.025} - \frac{\frac{230}{300}}{0.071 - 0.025} = 2.90.$$

3. The decline in Drayton's justified EV/EBITDA multiple following the change in assumptions surprised the analyst because she thought that each of these changes should positively affect the multiple. Which one variable change primarily accounted for the decline in the multiple?

- A. Tax rate
- B. Depreciation
- C. Capital expenditures

Solution:

C is correct because capital expenditures have a negative relationship with the justified EV/EBITDA multiple, as shown by the composition of Equation 21. Specifically, there is a negative sign on the capital expenditures term of the equation; thus, higher capital expenditures are predicted to have a negative effect on the justified multiple.

A is incorrect because the decline in tax rate has a positive effect on the first term in the equation. The decline in tax rate also creates a small negative effect on the second term of the equation. Overall, the tax rate has an ambiguous effect on the justified multiple.

B is incorrect because the increase in depreciation has a positive effect on the second term of the equation.

Other factors held constant, a lower cost of capital, lower capital expenditures, a higher growth rate, and lower taxes all contribute to a higher EV/EBITDA multiple. The complex interaction among these factors is best addressed by modeling a firm's financial statements for equity valuation purposes, which will be addressed in detail in the next learning module.

EV/EBITDA controls for differences in depreciation and amortization practices and policies among businesses and is therefore frequently used in the valuation of capital-intensive businesses with substantial depreciation and amortization expenses.

Alternative earnings measures to using EBITDA in the denominator of EV multiples include EBITA (earnings before interest, taxes, and amortization) and EBIT. EBITA is used if amortization (associated with intangibles) but not depreciation (associated with tangibles) is a major expense for companies, while EBIT is often chosen if neither depreciation nor amortization are major items.

PRICE-BASED VS. EV-BASED MULTIPLES: RYANAIR VS. EASYJET

In addition to measuring the relative level of single multiples across companies, comparisons across multiples involving different capital providers in the numerator and measures of income in the denominator can also shed light on variations among firms that warrant further investigation.

For example, European budget passenger airlines Ryanair Holdings plc and EasyJet plc faced similar challenges in recovering from losses during the COVID-19 pandemic due to travel restrictions that decimated consumer demand for low-cost travel. Key market data and multiples for the two industry peers as of 2024 are shown in Exhibit 10.

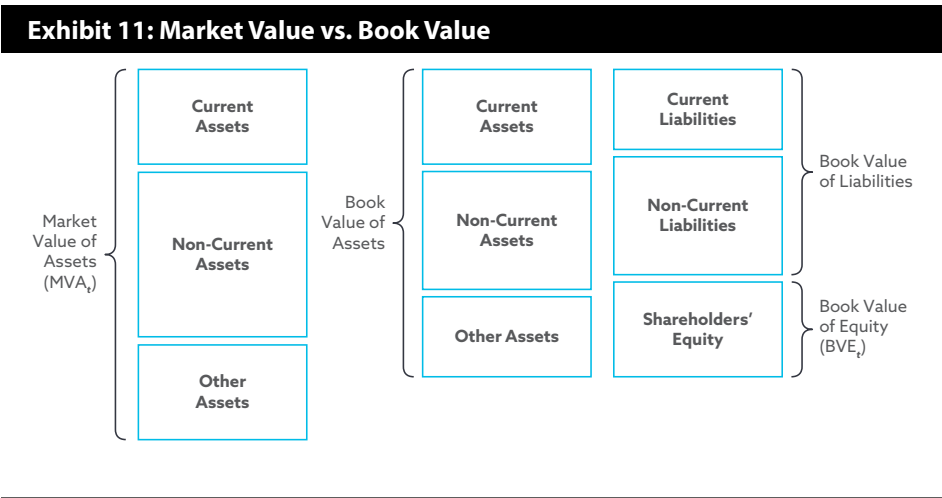
Exhibit 10: Ryanair vs. EasyJet: Key Data and Multiples as of 2024				
Company	Market Cap (EUR billions)	EV (EUR billions)	EV/EBITDA	P/E
Ryanair	20.53	19.27	6.18	10.75
EasyJet	4.13	3.93	2.93	10.89

Source: Bloomberg.

While a cursory comparison of P/Es suggests that these firms are closely aligned from a relative value perspective, the large discrepancy in EV/EBITDA values tells a different story. Areas that differ in the numerator and denominator of these ratios warrant more detailed analysis, including the relative use of debt versus equity and the difference between operating income and earnings as measured by net income. We revisit the comparison of Ryanair and EasyJet in a later example.

Balance Sheet Measures

Unlike multiples based on income statement items that involve a measure of market value per unit of income or sales, multiples based on balance sheet items seek to capture the price or market value per unit of *assets* available to generate returns. The balance sheet metrics used to derive these multiples are shown in Exhibit 11.



The most common equity-based multiple compares the market value of equity to the book value of shareholders' equity in the firm's balance sheet, or the price to book value of equity, covered in an earlier lesson in this learning module.

As discussed earlier, a higher P/B is associated with both a higher return on equity versus equity rate of return (r_e) and a larger payout ratio. In an earlier Equities learning module, it was noted that book value comparisons may be distorted by variations in the

treatment of intangible assets, as well as differences in historical cost and depreciation methods for tangible assets. Additionally, P/B multiples may be difficult to interpret across firms if some comparable companies engage in significant share buybacks.

For firms with similar leverage and asset profiles and consistent accounting practices, price-to-book multiples can be a valuable gauge of a firm's expected future returns versus existing net assets. For example, global banks hold few fixed assets; maintain balance sheet positions including deposits, loans, and securities with similar characteristics; and operate with a high degree of financial leverage constrained by regulatory equity capital requirements. These balance sheet features make P/B and the related **price to tangible book value** multiple, which excludes intangible items from the book value of assets such as goodwill, common metrics for the banking industry, as shown next.

COMPARISON OF PRICE-TO-BOOK RATIOS AMONG GLOBAL BANKS

As described earlier, P/B multiples are commonly used as a relative valuation measure for banks given their underlying business and balance sheet characteristics. Exhibit 12 presents P/Bs for selected major global banks.

Exhibit 12: Price-to-Book Multiples for Global Banks, 2024

Company	Market Cap (USD bn)	Price to Tangible Book Value	P/B
J.P. Morgan	561.64	2.24	1.83
Bank of America	303.59	1.57	1.15
Deutsche Bank AG	33.77	0.54	0.48
Barclays PLC	41.20	0.65	0.56

Source: Bloomberg.

Although J.P. Morgan and Bank of America may appear overvalued relative to European counterparts based on the P/B multiples in Exhibit 12, differences in the business composition and market positions of these firms may justify these multiple discrepancies.

For example, while traditional commercial banking activities predominantly involve lending and accepting deposits, both J.P. Morgan and Bank of America are larger players than their European peers in less balance sheet-intensive but high-margin fee-based businesses, such as investment banking underwriting and advisory, as well as wealth management.

In cases where comparable companies have differences in financial leverage or a firm may even have negative equity, a broader **value-to-book multiple** comparing the firm's value (V_f) to the book value of *capital* is a more appropriate measure than price to book, where the book value of capital is the sum of book value of equity and book value of debt. The value-to-book measure is the ratio of the market value to book value of both debt and equity, while a similar EV-based multiple subtracts cash from the numerator and denominator:

$$\text{Value to book} = \frac{\text{Market value of equity} + \text{Market value of debt}}{\text{Book value of equity} + \text{Book value of debt}} \quad (22)$$

$$\text{Enterprise value to book} = \frac{\text{Market value of equity} + \text{Market value of debt} - \text{Cash}}{\text{Book value of equity} + \text{Book value of debt} - \text{Cash}} \quad (23)$$

Using the value of the firm from Equation 18 as a starting point, we can determine the underlying factors affecting enterprise value to book:

$$EV_t = \frac{FCFF_{t+1}}{WACC - g}.$$

FCFF was shown earlier to be a function of EBITDA, depreciation, and investment in long-term assets and working capital. We may alternatively express it as the proportion of after-tax EBIT that is *not* reinvested and is therefore available to debtholders and shareholders:

$$FCFF_{t+1} = EBIT_{t+1} \times (1-t) \times (1 - \text{Reinvestment rate}). \quad (24)$$

Note that “reinvestment” combines depreciation, capital expenditures, and changes in working capital. Substituting Equation 24 into Equation 18, we get

$$EV_t = \frac{EBIT_{t+1} \times (1-t) \times (1 - \text{Reinvestment rate})}{WACC - g}. \quad (25)$$

Dividing both sides of Equation 25 by the book value of capital and recognizing that, in contrast to ROE, **return on capital (ROC)** is the ratio of after-tax EBIT to capital,

$$ROC = \frac{EBIT_{t+1} \times (1-t)}{\text{Book value of capital}},$$

then

$$\frac{\text{Enterprise value}}{\text{Book value of capital}} = \frac{\text{Return on capital} \times (1 - \text{Reinvestment rate})}{WACC - g} \quad (26)$$

A common reason for selecting an approach using enterprise value to book over a price-to-book multiple is to address differences in or changes in leverage. Let us return to an earlier example that compared price-based and EV-based multiples to look more closely at differences between two companies.

A CLOSER LOOK AT RYANAIR AND EASYJET

An earlier comparison of price- and EV-based multiples for two low-cost European airlines showed similar P/Es for Ryanair and EasyJet but a much lower EV/EBITDA (2.93 versus 6.18) for EasyJet than Ryanair. A closer look at the components in the numerator and denominator of the broader EV-based measure sheds further light on key value drivers at both firms.

From a numerator perspective, the difference between market capitalization and enterprise value for both firms is shown as of 2024 in Exhibit 13.

Exhibit 13: Enterprise Value Components as of 2024, in Millions

	EasyJet	Ryanair
Market Cap	GBP3,532.3	EUR20,526.5
+ Total Debt	GBP3,186.0	EUR2,746.8
– Cash and Equivalents	GBP3,363.0	EUR4,133.2
= Enterprise Value	GBP3,355.3	EUR19,140.1

Source: Bloomberg.

Note that both firms have more cash and equivalents than debt, although EasyJet has proportionally more of both than Ryanair. This growth appears to be a result of EasyJet’s expansion into the vacation segment, which represents just over 10% of sales, in which prepaid vacations generate significant balance sheet cash.

As for the denominator, Exhibit 14 shows a breakdown of operating income and earnings for both firms as of the most recent fiscal year:

Exhibit 14: Selected Income Statement Components as of Fiscal Year-End

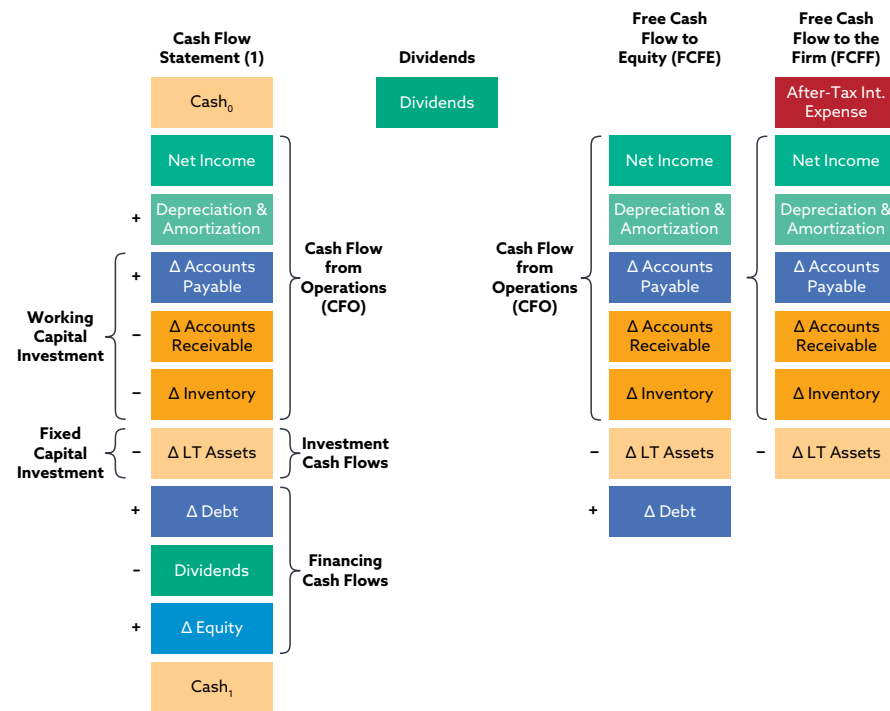
	EasyJet FYE 9/30/23 (GBP millions)	EasyJet % of Revenue	Ryanair FYE 3/31/24 (EUR millions)	Ryanair % of Revenue
Revenue	8,171.0	100%	13,443.8	100%
EBITDA	1,146.0	14.03%	3,120.2	23.21%
Operating Income	473.0	5.79%	2,060.7	15.33%
Non-Operating Income	21.0	0.26%	67.3	0.5%
Net Income	324.0	3.97%	1,917.1	14.26%

Source: Bloomberg.

Exhibit 14 shows that Ryanair has higher EBITDA margins (23.2% versus 14.0%) than EasyJet, while Ryanair's operating and profit margins are roughly triple those of EasyJet. While the price-based P/E shows similar multiples across firms, equity investors benefit from EasyJet's higher leverage and larger proportional cash balances due to the growing vacation segment. However, EasyJet's much weaker EV-based EV/EBITDA measure shows the effect of both subtracting cash and focusing on operating earnings.

Cash Flow Statement Measures

Cash flow-based multiples have some advantages over income statement-based measures. Since cash flows are also the direct inputs in DCF valuations, they tend to have a more direct link to relative value drivers. Cash flow also tends to be less susceptible to manipulation and quality differences than earnings. Common cash flow metrics used in multiples include those shown in Exhibit 15.

Exhibit 15: Key Cash Flow Metrics

Dividend-Based Multiples

Cash dividends paid to shareholders are the narrowest cash flow measure used in relative valuation. The price-to-dividend multiple represents the share price divided by dividends per share. Like the earnings yield, mentioned earlier, the inverse of this multiple (Div_t/P_t), known as the **dividend yield**, is a preferred way to present this multiple. Dividend yield is used as a valuation indicator because it is often a more predictable component of total return than capital appreciation. When a company pays no dividends, its price-to-dividend multiple is undefined, with zero in the denominator.

Trailing dividend yield, $\frac{Div_{t-1}}{P_t}$, is calculated by using the dividend rate divided by the current market price per share, with the annualized amount of the most recent periodic dividend referred to as the **dividend rate**. The **leading dividend yield**, $\frac{Div_{t+1}}{P_t}$, is calculated as forecasted dividends per share over the next year divided by the current share price.

Dividend yield reflects a firm's fundamental characteristics. For example, one way to express a company's dividend yield is as a simple function of the firm's dividend payout ratio and its earnings yield:

$$\frac{Div_t}{EPS_t} \times \frac{EPS_t}{P_t} = \frac{Div_t}{P_t} \quad (27)$$

Alternatively, we can once again use the dividend constant growth model in Equation 1 as a starting point:

$$P_t = \frac{Div_{t+1}}{r_e - g}.$$

The relationship between trailing dividends (Div_t) and expected future dividends (Div_{t+1}) is a function of the growth rate g :

$$\text{Div}_{t+1} = \text{Div}_t \times (1 + g).$$

Therefore, we can substitute this term into the dividend constant growth model and solve for the trailing dividend yield $\left(\frac{\text{Div}_t}{P_t}\right)$:

$$\frac{\text{Div}_t}{P_t} = \frac{r_e - g}{1 + g}. \quad (28)$$

Equation 28 demonstrates that the fundamental drivers of dividend yield are the expected growth rate in dividends and the required rate of return. If the numerator of Equation 28 reflected the expected future dividend, Div_{t+1} , we would find that the justified leading dividend yield is simply equal to cost of equity, r_e , minus the growth rate, g . Dividend yields are higher for firms with a higher cost of equity but lower for those with higher growth expectations.

KNOWLEDGE CHECK: P/E VS. DIVIDEND YIELD



In an earlier knowledge check, an analyst showed that a company with a dividend payout ratio of 80%, cost of equity of 8%, and a growth rate of 3% has a justified trailing P/E multiple of 16.48 based on a constant growth dividend discount model.

1. Calculate the company's justified trailing dividend yield using the information above.

Solution:

The company's justified dividend yield is equal to 4.85%. This amount can be calculated using either Equation 27 or Equation 28, as shown below.

$$\text{Equation 27: } \frac{\text{Div}_t}{\text{EPS}_t} \times \frac{\text{EPS}_t}{P_t} = 80\% \times \frac{1}{16.48} = 4.85\%.$$

$$\text{Equation 28: } \frac{\text{Div}_t}{P_t} = \frac{r_e - g}{1 + g} = \frac{0.08 - 0.03}{1.03} = 4.85\%.$$

2. Suppose the company has a 15% return on equity and lowers its payout ratio from 80% to 70%. Calculate the updated justified dividend yield for the company.

Solution:

The justified dividend yield in this scenario is 3.35%. This amount can be calculated most easily using Equation 28. However, the analyst must first update the expected growth rate, g , to be consistent with this scenario by using the sustainable growth rate in Equation 4.

$$g = (1 - \text{Payout ratio}) \times \text{ROE}.$$

$$g = (1 - 0.70) \times 15\% = 4.5\%.$$

$$\frac{\text{Div}_t}{P_t} = \frac{r_e - g}{1 + g} = \frac{0.08 - 0.045}{1.045} = 3.35\%.$$

The prior knowledge check shows that the dividend yield reflects not only the dividend payout ratio but also the firm's P/E multiple. As is true for any multiple, dividend yields should be compared only after controlling for the multiple's fundamental drivers. The justified dividend yield computation can provide a way to assess whether a firm's actual dividend yield is higher or lower than it should be. A drawback of using the dividend yield as a return measure is that it is only one component of total return.

The dividend yield is zero for all firms that choose not to pay a dividend and may be very low for others that pay out only a small portion of their free cash flow to equity. Therefore, dividend yield comparisons are most appropriate among comparable firms with clear and consistent dividend policies.

Practitioners and financial data vendors often use simple cash flow approximations that may ignore some adjustments and simply combine net income with depreciation and amortization charges. These less technically accurate measures based on earnings plus non-cash charges often simply use the abbreviation “CF.” More technically precise measures shown in Exhibit 15 that are commonly used in multiples include the following:

- Cash flow from operations (CFO) is found in the statement of cash flows and includes changes in working capital. Adjustments for components not expected to persist or for different accounting standards (IFRS versus GAAP) may be required.
- Free cash flow to equity is closely aligned with firm value to shareholders, as shown in an earlier Equities learning module. FCFE is sensitive to the timing of capital expenditure among companies being compared, as well as changes in debt outstanding.
- Free cash flow to the firm is also affected by capital expenditures and incorporates after-tax interest expense to capture returns to debtholders.

The fundamental drivers of a value-to-FCFF multiple include the cost of capital and the growth rate, g . Dividing the value of the firm from Equation 18 by FCFF, we get

$$\frac{V_t}{\text{FCFF}_{t+1}} = \frac{1}{\text{WACC} - g}. \quad (29)$$

Similarly, an equity valuation using the constant growth FCFE model produces the following:

$$\frac{P_t}{\text{FCFE}_{t+1}} = \frac{1}{r_e - g}. \quad (30)$$

In both cases, the multiples are negatively related to the firm’s required return on capital or equity (r_e) and positively related to the expected growth rate (g) of the firm’s future cash flows.

Cash flow–based multiples have the disadvantage of being dependent on short-term or cyclical fluctuations in capital expenditures and working capital investments, as well as short-term volatility in debt usage for FCFE-based multiples.

Industry Specific Measures

Analysts often apply industry-specific measures of relative value that use specific assets or reserves, output, or operational metrics in the denominator rather than standardized income statement, balance sheet, or cash flow measures from audited financial statements. Advantages include a more direct link with sector-based value drivers and a lack of accounting distortions or differences that may lead to valuation biases. These measures are prevalent in both capital-intensive industries, such as mineral extraction and transportation, and asset-light sectors, such as technology.

For example, analysts valuing firms in the energy, metals, or mining industries often consider asset- or output-based multiples tied directly to an underlying commodity, as in the following example. These firms are capital intensive, with generally long lead times between investment and production, a high level of operational leverage, and substantive assets in place.

USE OF RESERVES AND DAILY PRODUCTION TO VALUE OIL PRODUCERS

Oil producers are often measured not only in terms of current production but also based on reserves developed and available for immediate production, as well as those that have been discovered but not yet developed for near-term production. Industry-specific valuations therefore often include multiples that use reserves or daily production figures in the denominator of a multiple, as shown in Exhibit 16.

Exhibit 16: Selected Multiples for European Oil Producers, 2024

Company	Market Cap (EUR billions)	P/E	EV/ Production	EV/ Developed Reserves
Median	42.29	7.7	58.65	26.81
Shell plc	208.99	8.6	85	36.86
TotalEnergies SE	158.49	7.87	74.46	26.81
BP p.l.c.	96.16	7.7	58.65	33.19
Equinor ASA	79.28	8.5	31.68	19.82
Eni SpA	47.91	6.69	39.3	17.12
Repsol SA	18.11	4.72	47.36	25.47
OMV AG	15.74	6.88	75.88	34.18

Source: Bloomberg.

Both production and developed reserves are measured in barrel-of-oil equivalents to allow the combination of oil and natural gas production into one multiple. Despite some deviations, the rankings of P/Es and these multiples are closely aligned. For example, Shell plc has both the highest P/E and the highest EV-related multiples among its peers.

The disadvantages of sector-specific multiples and measures include that they lack comparability with other industries and may therefore lead to the mispricing of an entire industry versus the market. They are also difficult to connect to fundamentals and control for other differences among firms when applied.

Exhibit 17 summarizes the major multiples, their fundamental drivers, and applicability across firms and industries.

Exhibit 17: Key Price and Value Multiples

Multiple	Fundamental Drivers	Applications
Price/Earnings	Dividend payout ratio Growth rate	Firms with positive and recurring EPS
Dividend Yield	Dividend size and growth	Mature, dividend-paying firms
Price/Sales	Net profit margin	Zero- or low-leverage firms Firms with cyclical profitability
Value/EBITDA	Tax rate Depreciation Investment	Capital-intensive firms
Price/Book Value of Equity	Return on equity	Firms with similar leverage and asset profiles

Multiple	Fundamental Drivers	Applications
Enterprise Value/Book Value of Capital	Return on capital	Firms with differing leverage Firms with negative book value of equity
Price/Free Cash Flow to Equity	FCFE growth rate	Firms with stability of FCFE
Value/Free Cash Flow to the Firm	FCFF growth rate	Firms with stability of FCFF

QUESTION SET 2



1. An analyst is comparing two stocks, Alpha and Beta, based on P/S multiples. Alpha is a 100% equity-financed company, and Beta is financed with 60% equity and 40% debt. Both companies have costs of equity of 8% and expected growth rates of 3%. Alpha has a net profit margin of 6% and a dividend payout ratio of 70%. Beta has a net profit margin of 5% and a dividend payout ratio of 84%. Which one of the following statements is most correct?

- A. The analyst concludes that Alpha has a higher justified P/S because of its higher net profit margin.
- B. The analyst concludes that Beta has a higher justified P/S because of its higher dividend payout ratio.
- C. The analyst concludes that the P/S multiples of Alpha and Beta cannot be compared because their debt ratios are different.

Solution:

C is correct because the P/S multiple should typically be used only for companies with no debt because the numerator of the multiple (P) reflects an equity claim while sales revenue (S) is applicable to both equity and debt claims. As a result, Alpha and Beta should not be compared based on P/S multiples because of Beta's financial leverage.

A and B are incorrect because the justified P/S multiples for Alpha and Beta are equal, as shown below.

$$\text{Alpha: } \frac{P_t}{\text{Sales}_t} = \frac{(\text{Net profit margin}) \times (\text{Payout ratio}) \times (1 + g)}{r_e - g} = \frac{0.06 \times 70\%}{0.08 - 0.03} = \frac{0.042}{0.05} = 0.84.$$

$$\text{Beta: } \frac{P_t}{\text{Sales}_t} = \frac{(\text{Net profit margin}) \times (\text{Payout ratio}) \times (1 + g)}{r_e - g} = \frac{0.05 \times 84\%}{0.08 - 0.03} = \frac{0.042}{0.05} = 0.84.$$

As shown in the equation for justified P/S, both net profit margin and the dividend payout ratio are positively related to P/S. Thus, Alpha's higher net profit margin is offset by its lower dividend payout ratio when compared to Beta. Nevertheless, as highlighted above, a comparison of P/S between these two companies is not appropriate, because of the inconsistency between the claimants implied by the numerator and denominator.

2. An analyst is comparing two mature companies with stable dividend policies. Firm 1 has a dividend payout ratio of 70%, and its stock has a dividend

yield of 5.5%. Firm 2 has a dividend payout ratio of 85%, and its stock has a dividend yield of 8.0%. The analyst finds that the two stocks have the same cost of equity of 10%. The analyst calculates Firm 1's expected dividend growth rate at 4.0% and Firm 2's growth rate at 2.0%. Which one of the following is the most correct relative valuation conclusion from this analyst?

- A. Firm 1's stock is underpriced and Firm 2's stock is overpriced based on their justified P/E multiples.
- B. Firm 2's stock is underpriced compared to Firm 1's stock price based on their P/E multiples.
- C. Firm 2's stock is underpriced and Firm 1's stock is overpriced based on their justified P/E multiples.

Solution:

C is correct because Firm 2's actual P/E of 10.63 is less than its justified P/E of 10.84 and Firm 1's actual P/E is greater than its justified P/E. The results are shown below.

A firm's actual P/E can be inferred by dividing the firm's dividend payout ratio by its dividend yield:

$$\text{Actual } \frac{P}{E} = \frac{\text{Dividend payout ratio}}{\text{Dividend yield}} = \frac{\left(\frac{D}{E}\right)}{\left(\frac{E}{P}\right)}.$$

Therefore,

- Firm 1 actual $\frac{P}{E} = \frac{0.7}{0.055} = 12.73$.
- Firm 1 justified $\frac{P}{E} = \frac{70\% \times (1 + 0.04)}{0.10 - 0.04} = 12.13 < 12.73 = \text{Actual } \frac{P}{E}$.
- Firm 2 actual $\frac{P}{E} = \frac{0.085}{0.08} = 10.63$.
- Firm 2 justified $\frac{P}{E} = \frac{85\% \times 1.02}{0.10 - 0.02} = 10.84 > 10.63 = \text{Actual } \frac{P}{E}$.

A is incorrect because this statement is the reverse of the correct statement.

B is incorrect because the P/Es cannot be reasonably compared without adjusting for the differences in dividend payout ratios and growth rates.

3. Which of the following variables are necessary to estimate the proportion of a stock's present value of growth opportunities (PVGO) as a percentage of its stock price?

- A. Forward P/E and dividend payout ratio
- B. Forward P/E and cost of equity
- C. P/B and ROE

Solution:

B is correct because the ratio of PVGO to stock price may be computed using the following equation:

$$\frac{\text{PVGO}_t}{P_t} = 1 - \frac{\frac{\text{EPS}_{t+1}}{P_t}}{r_e}.$$

In the second term on the right-hand side of the equation, we see the inverse of the forward P/E (i.e., the expected earnings yield) in the numerator and the cost of equity, r_e , in the denominator.

A is incorrect because the computation of PVGO does not require an assumption of dividend payment.

C is incorrect because this combination of ratios may be used to infer P/E (i.e., $P/BVE \div EPS/BVE$) but does not include the cost of equity.

4. Suppose a firm has a current dividend payout ratio of 75%, a cost of equity of 10.5%, an expected growth rate of dividends of 4.5%, and an ROE of 18%. Which one of the following is closest to the justified trailing P/E of this stock under the assumption that the variables are expected to be stable over time in the future?

- A. 12.5
- B. 13.1
- C. 14.3

Solution:

B is correct because the formula to estimate a justified trailing P/E is given by the following equation:

$$\frac{P_t}{EPS_t} = \frac{\text{Payout ratio} \times (1 + g)}{r_e - g}$$

$$\frac{P_t}{EPS_t} = \frac{75\% \times 1.045}{10.5\% - 4.5\%} = 13.0625.$$

A is incorrect because a result of 12.5 occurs if we omit the growth term in the numerator. This would be correct if we were calculating the justified forward P/E instead of the trailing P/E.

C is incorrect because this result is generated by improperly using 0.82 (i.e., 1 minus the ROE) in the numerator of the equation instead of the dividend payout ratio.

4

PEER GROUPS FOR MULTIPLES VALUATION



determine a potential peer group for multiples-based equity valuation based on a company's industry classification or specific drivers of expected return

Deciding how to select companies to establish a peer group or benchmark for comparing multiples is in many respects as important as the choice of a multiple itself and is often the most challenging decision in the relative valuation process. As demonstrated in the prior lesson, multiples may be defined in terms of a firm's fundamental characteristics, such as the growth rate, risk as measured by the rate of return on equity or capital, and cash flow or earnings metrics, such as net profit margin. While no two firms are exactly alike, ideal candidates for comparable firms are those with very similar fundamental characteristics.

Peer groups for equity valuation based on multiples are usually selected in one of two ways. Companies with the same principal business activity as the target firm are often chosen as comparable companies. Alternatively, companies may be grouped by statistical characteristics or factors that are of significance in explaining equity risk and return.

Industry-Based Peer Groups

A group of companies that offer similar goods or services within a branch of manufacturing or trade are considered to operate within an industry. While firms within an industry may serve different geographic areas or have a different product mix, to the extent that they operate in the same business environment and face similar competitive forces, suppliers, and customers, industry peers are often believed to have similar fundamental characteristics. Investors commonly compare the performance of equities among industry competitors and consider them to be substitutes. In the following case study, we revisit the energy producer from the prior learning module.

PEER ANALYSIS FOR ENEL SPA

Recall that the electric and natural gas utility Enel SpA operates in 30 countries, with its revenue generated primarily in Italy, Spain, and Portugal. An analyst seeking to compare Enel with its European peers on a relative value basis might use the Bloomberg Relative Value (RV) function to obtain an initial set of 11 firms, as shown in Exhibit 18, along with select multiples based on the most recent financial statements.

Exhibit 18: Selected Information and Multiples for Enel SpA and Peers

Company	Market Cap (EUR millions)	Enterprise Value (EUR millions)	EV/ EBITDA	P/E
Median	18,311	27,475	6.60	11.65
Enel SpA	63,033	144,005	6.60	16.41
EnBW Energie Baden-Württemberg	18,311	33,237	5.13	11.65
Fortum Oyj	11,090	12,487	6.38	7.23
A2A S.p.A.	5,869	11,120	6.08	8.95
Iberdrola SA	74,864	139,222	8.93	15.00
Engie SA	39,573	73,779	5.58	17.65
VERBUND AG	24,632	27,475	6.14	10.87
MYTILINEOS S.A.	5,438	7,402	7.24	8.45
SSE plc	21,471	33,753	184.27	31.40
BKW AG	7,406	8,896	9.01	15.11
EVN AG	5,163	6,591	8.56	9.76

Source: Bloomberg.

While Enel shares are priced in line with the median EV-based multiple for this initial group of selected utilities, it appears to be overvalued ($16.41 > 11.65$) based on trailing P/E.

As a next step, an analyst might seek to define the peer universe more precisely by considering the geographic distribution of activities for firms under consideration. She observes that unlike Enel, utilities with a smaller market capitalization also tend to be less geographically diversified. She therefore decides to exclude firms with a market capitalization below EUR10 billion. The revised results are shown in Exhibit 19.

Exhibit 19: Revised Information and Multiples for Enel SpA and Peers

Company	Market Cap (EUR millions)	Enterprise Value (EUR millions)	EV/ EBITDA	P/E
Median	24,632	33,753	6.38	15.00
Enel SpA	63,033	144,005	6.60	16.41
EnBW Energie Baden-Württemberg	18,311	33,237	5.13	11.65
Fortum Oyj	11,090	12,487	6.38	7.23
Iberdrola SA	74,864	139,222	8.93	15.00
Engie SA	39,573	73,779	5.58	17.65
VERBUND AG	24,632	27,475	6.14	10.87
SSE plc	21,471	33,753	184.27	31.40

Source: Bloomberg.

Note that Enel's overvaluation based on trailing P/E has nearly disappeared. An additional area for adjustment raised earlier relates to the treatment of outliers. In this example, a closer investigation reveals that SSE plc had an extraordinary loss in the prior year, which led to sharply lower EBITDA and earnings, increasing the two multiples shown in Exhibit 19. The analyst should consider excluding this firm, using the median rather than the average, or adjusting the income measures for non-recurring items and changing the two multiples accordingly.

As the case study suggests, the selection of comparable firms from within an industry goes well beyond simple classification and extends to weighing and adjusting the peer group based on other criteria affecting multiples. Although utilities are characterized by price controls, limited competition, and barriers to entry in a given area that serve to limit differences, firms may have varying proportions of revenue and operating profit from power generation, power transmission, and trading activities.

Moreover, even those firms with a similar proportion of power generation revenue may have very different cost structures from the target firm due to the use of different underlying *sources* of energy. For example, a firm's composition of traditional (coal, natural gas) versus renewable (wind, solar) energy sources and differences in green transition timing can affect multiples.

Analysts may also need to adjust a peer group due to differences in the demand environment faced by otherwise comparable companies versus the firm being valued when comparing firms operating in separate markets.

The selection of comparable companies must consider such features as market structure, cyclicity, and the life-cycle stage of firms when establishing a peer group to apply the method of comparables. Using industry-based peer groups for relative valuation has several advantages. Companies within an industry often have a business mix very similar to that of the company being analyzed. The approach is also consistent with the idea underlying the method of comparables: Similar firms generating similar cash flows should trade at similar prices.

Analysts may estimate how industry-specific factors affect business characteristics for a target company relative to the profitability, growth, or financial strength for firms with similar suppliers and customers and comparable revenue and cost and value drivers.

The use of price or EV-based multiples as a basis for comparison between industry competitors can also provide insight via best- and worst-in-class performance. Beyond assessing the relative value of an individual firm versus a peer group, this analysis is also useful for active investors who may over- or underweight investments in one or more firms among peers.

Industry-based peer groups are commonly used in many applications of relative valuation. Specific challenges in creating peer groups exist, including the following types of situations:

- Unique firms or those operating in very concentrated industries may have a limited number of companies from which to create a peer group.
- Companies operating in industries in which competitors exhibit a high variation of stages across the company life cycle.
- Selection of companies in a single industry that are based in different jurisdictions may face different regulations and taxes that affect fundamental data and therefore limit comparability.
- Multi-industry companies, or those with significant revenues, income, and assets across different industries, can present challenges, especially if these industries have differing risk, growth, and cash flow characteristics, as in the following example.

AMAZON.COM: RETAIL OR TECHNOLOGY COMPANY?

Amazon.com is one of the world's largest companies by revenue, with just under USD575 billion in 2023. While the company established its reputation in the world of e-commerce and as a disruptor of physical retail stores, much of the company's growth has been powered by its Amazon Web Services (AWS) business segment. AWS provides cloud storage services and is now the largest by market share in this space, surpassing even Microsoft's Azure. Between 2013 and 2023, AWS revenues grew at a compound annual rate of 40% and its share of company revenue grew from 4% to almost 16%.

Apart from the revenue growth, the fundamental characteristics of Amazon.com's business units differ as well. Amazon.com reports three distinct business segments: (1) North America, (2) International, and (3) AWS. The North America and International segments are characterized by operating profit margins below 5% (and that occasionally are negative). Meanwhile, the annual operating profit margins for the AWS segment are typically 25%–30%, with this segment usually accounting for well over half of Amazon.com's overall operating profit.

The challenges noted above may be addressed using the following approaches:

- For companies operating in a concentrated industry, an analyst may place little or no emphasis on relative valuation. In such cases, discounted cash flow models are more likely to be relied on for the valuation estimate.
- For a company operating in an industry in which the comparable firms show significant variation across life-cycle phases, such as the pharmaceutical industry, an analyst may choose to eliminate those firms from the peer group that are in different life-cycle phases than the company being valued. However, this type of approach must be moderated to avoid creating an overly concentrated peer group.
- For a company with multiple industries, an analyst may create peer groups for each industry in which the company operates. The peer group multiple can be estimated as a weighted average of the different industry peer group

multiples. The weightings might be based on the proportion of the company's revenues, operating income, or assets in each industry, as shown in the following exercise.

KNOWLEDGE CHECK: EXINC LABS ACTIVITIES ACROSS SEGMENTS



eXinc Labs Limited is an Indian company engaged in manufacturing and marketing pharmaceuticals globally. eXinc's original line of business was the manufacture of active pharmaceutical ingredients for use in generic medications. The company has built research capabilities over time and now has a portfolio of drugs sold under patent protection; 35% of eXinc's sales last year consisted of patented drug sales, with the remainder attributable to ingredients for generic drugs. An analyst valuing eXinc assesses these two pharmaceutical segments and compiles the following data.

Pharmaceutical Segment	Median EV/EBITDA Multiple
Generic	10.5
Patented	18.3

1. Calculate an appropriate peer group EV/EBITDA multiple to apply to eXinc's valuation.

Solution:

Given the available information on the proportion of eXinc's sales across operating segments, we calculate a weighted average of the two segment EV/EBITDA multiples as follows:

$$\text{Peer group } \frac{\text{EV}}{\text{EBITDA}} \text{ multiple} = 0.65 \times 10.5 + 0.35 \times 18.3 = 13.2.$$

Therefore, the analyst can apply a 13.2 EV/EBITDA multiple to eXinc's EBITDA as part of her relative valuation.

Factor-Based Peer Groups

Factor investing is a process that seeks to identify statistical characteristics important to risk and return used to group companies and make investment decisions. Specific characteristics or factors have been shown to earn a long-term risk premium over time and represent exposure to systematic sources of risk. **Systematic risk** refers to the unavoidable, inherent market risks affecting all equity securities, while **non-systematic risk**, or idiosyncratic risk, is the risk that affects a single security or industry.

Factor-based peer groups classify companies using statistical characteristics to gauge relative value between a target company and a factor-based benchmark. The most prominent single-factor or market model is the **capital asset pricing model (CAPM)**, in which the primary determinant of an equity security's expected return is how well the security correlates with the overall market. This relationship, introduced in an earlier Quantitative Methods learning module is shown in Equation 31:

$$E(R_i) = R_f + \beta_i [E(R_m) - R_f], \quad (31)$$

where

R_i = return on the target company

R_f = risk-free rate

$E(R_m)$ = expected market return

β_i , or **beta** = correlation between the target security and the overall market

Assets with $\beta > 1$ have an expected return above the market return, whereas assets with $\beta < 1$ have an expected return below the market return. Exhibit 20 shows examples of high- and low-beta stocks in the Indian NIFTY 50 stock index.

Exhibit 20: Selected High- and Low-Beta Stocks in the NIFTY 50, 2024

Company	Sector	Forward P/E	Beta
JSW Steel Limited	Steel Producers	12.63	1.72
Asian Paints Limited	Specialty Chemicals	46.88	1.41
Dr. Reddy's Laboratories	Specialty and Generic Pharmaceuticals	16.50	1.00
Apollo Hospitals Enterprise Ltd.	Health Care Facilities	57.03	0.46
Tech Mahindra Limited	Information Technology Services	20.14	0.37

Source: Bloomberg.

Factor-based peer groups establish a more direct relationship between features used to classify stocks and their risk-versus-return profile. If markets are efficient, then factor returns reflect compensation for bearing systematic risk and therefore serve as a reasonable benchmark for company stocks with similar risks, while allowing for different cash flow and growth characteristics. A further advantage of a factor-based approach is that it helps overcome some of the constraints associated with industry-based valuation described earlier. The following example demonstrates how this relationship is translated into an investment strategy used to select stocks.

HIGH-BETA EQUITY FUNDS AND RELATIVE VALUATION

The S&P 500 High Beta Index includes the 100 constituents in the S&P 500 Index with the highest beta. Some exchange-traded funds seek to track the performance of this index. The top 10 constituents in terms of beta as of May 2024 are shown in Exhibit 21.

Exhibit 21: 500 High-Beta Index Top 10 Constituents

Company	Sector	Forward P/E	Beta
Nvidia	Information Technology	36.78	1.71
Comerica	Financials	10.99	1.76
Generac Holdings	Industrials	23.73	1.56
Zebra Technologies	Information Technology	26.28	1.48
KeyCorp	Financials	13.39	1.58
Monolithic Power Systems	Information Technology	55.46	1.50

Company	Sector	Forward P/E	Beta
Norwegian Cruise Line Holdings	Consumer Discretionary	11.53	1.59
Carnival Corp	Consumer Discretionary	14.79	1.74
Citizens Financial Group	Financials	11.78	1.67
Teradyne	Information Technology	42.39	1.43

Sources: S&P and FactSet.

Note that these firms among the ones with the highest beta come from a wide range of industries across a significant range of P/Es. In comparison, the S&P 500 forward P/E average, observed at the same time, is just above 20.

Disadvantages of a factor-based approach include possible model misspecification. That is, factor models capture valuation relationships for a specific stock (or group of stocks) over a particular period, while the predictive power of the regression for different stocks over different periods is unknown. In practice, many factors believed to drive asset returns underperform over long periods and may be subject to cyclicity.

QUESTION SET 3



- Which one of the following statements best describes the preferred approach for estimating a multiple for a company operating in multiple industries?
 - Estimate the multiple as a weighted average of the industry peer group multiple based on the proportion of firm revenues, income, or assets in each industry.
 - Estimate the multiple by compiling a list of the firm's peers across all the industries in which it operates and taking an average of the multiples across all the peers.
 - Estimate the multiple using the median of the peers across all industries in which the firm operates with similar risk, growth, and cash flow characteristics.

Solution:

A is correct because a weighted average of industry peer group multiples best mirrors the risk, growth, and cash flow characteristics of a multi-industry firm.

B is incorrect because an average of multiples across peers fails to take the proportion of firm activity in each industry segment into account.

C is incorrect because this more selective approach also fails to consider the proportion of a firm's activity in each industry in the estimate.

- Which one of the following statements is an appropriate description of the composition of a factor-based final peer group for relative valuation?
 - The final peer group consists of all firms in the same industry classification.
 - The final peer group consists of firms with similar beta coefficients despite industry classification.

- C. The final peer group consists of firms in the same industry and with similar fundamental characteristics.

Solution:

B is correct because this response reflects a factor-based approach to peer group selection. By using firms with similar beta coefficients, the peer group is chosen from a large group of firms with similar systematic risk. Firms with very similar betas should be similar in terms of their costs of equity and should have similar expected returns.

A is incorrect because of the inclusion of “all firms.” A subset of firms is typically removed from the industry peer group before finalizing. For example, the final peer group may remove any firms that fall below (or above) a certain market capitalization or size threshold. In some cases, firms are removed because their earnings include non-recurring items that significantly affect earnings-based multiples.

C is incorrect because differences in fundamental characteristics, such as the dividend payout ratio or growth, can be incorporated into the analysis of multiples through an adjustment or regression methodology. Thus, the final peer group can consist of firms with differing characteristics.

3. Which one of the following is a possible approach to establishing a peer group multiple for a company that has significant proportions of operations in different industries?

- A. Use the peer group multiple from an industry that is the same as the industry of the company’s largest operation.
- B. Use a peer group multiple that reflects the equally weighted average of multiples across the different industries.
- C. Use a peer group multiple that reflects a weighted average of multiples based on the company’s proportional weightings of revenues.

Solution:

C is correct because this statement implies using a peer group multiple that reflects both the company’s different business lines by industry and the proportionality of the company’s business across different industries.

A is incorrect because this statement implies using a multiple that neglects to include the multiple(s) from any other significant operating divisions.

B is incorrect because this statement implies using a multiple that ignores the proportionality of the company’s business across different industries. This statement would only be correct if the firm’s business activities are equally weighted across industries.

USE OF MULTIPLES TO ESTIMATE EQUITY VALUE

5



describe the use of multiples based on past, current, and projected future values in estimating equity value

So far, we have focused on current multiples to establish a foundation for assessing whether a company is undervalued, overvalued, or fairly valued. Before proceeding, we offer a warning for analysts: One must always consider the possibility that the peer group used as part of the relative valuation process may not be fairly valued. Exhibit 22 addresses this phenomenon.

Exhibit 22: True Justified Multiples vs. Observed Multiples

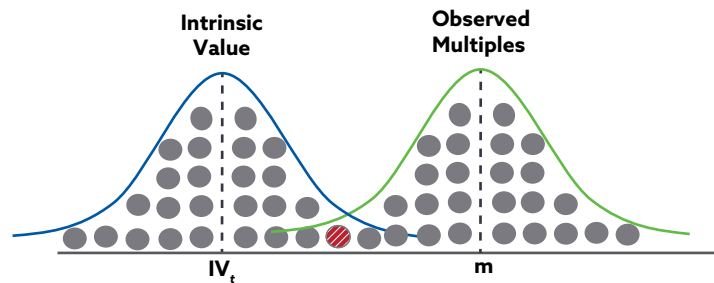


Exhibit 22 shows the example of a firm whose *intrinsic* value, IV_t , may seem high relative to those of its peers but may be at the low end of observed multiples in the current market. This outcome suggests that the entire industry or market may be overvalued.

As a result, analysts must exercise caution about strict conclusions regarding a stock's valuation solely derived from using multiples. Relative valuation is best used in combination with present value methods to better assess valuation recommendations. However, analysts may also use historical multiples and projected future multiples as part of a company's valuation process, as described later.

Multiples Based on Current Values

As shown in prior lessons, current multiples typically involve the ratio of earnings, cash flow, or sales over the most recent quarter to the current share price or enterprise value. These multiples were interpreted as the current share price (or enterprise value) relative to the most recent earnings, cash flows, book values, or sales and were compared to an average benchmark or justified multiple.

Because the data used in the denominator are limited to the most recent period, current multiples face the effect of non-recurring income items, different accounting methods, and/or cyclicalities for a given multiple, which must be taken into consideration when drawing conclusions using these measures. This issue is like the base-year effect that arises in present value models and may be addressed by analyzing past and expected future multiples, as shown in the next two sections.

Multiples Based on Past Values

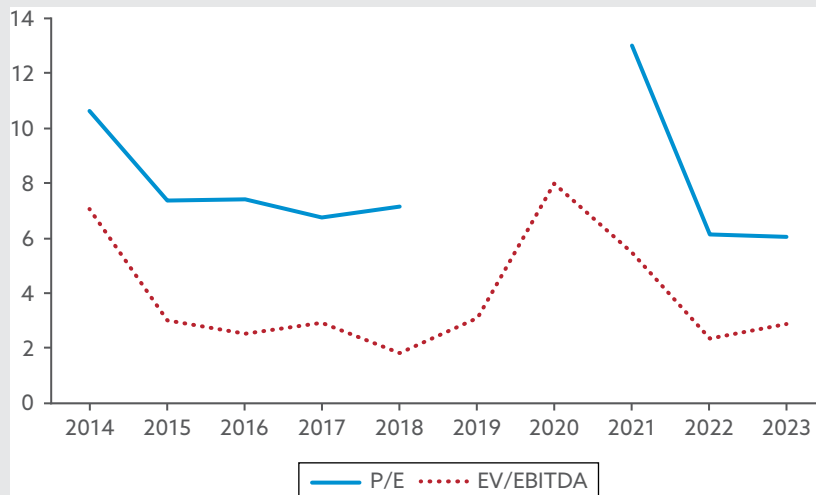
Historical multiples are commonly used by analysts in relative valuation to gauge trends of actual past multiple observations relative to current multiples. Multiples based on past values are used in two different ways to establish the relative value of a company's stock:

1. Past multiples relative to a company's own history: This use of past multiples allows an analyst to closely assess any deterioration or improvements in a company's performance that might be expected to continue into the future.

2. Multiple comparisons for a company relative to peer historical averages: This allows for assessment of a firm's relative performance under similar market and business conditions to a benchmark peer group over time.

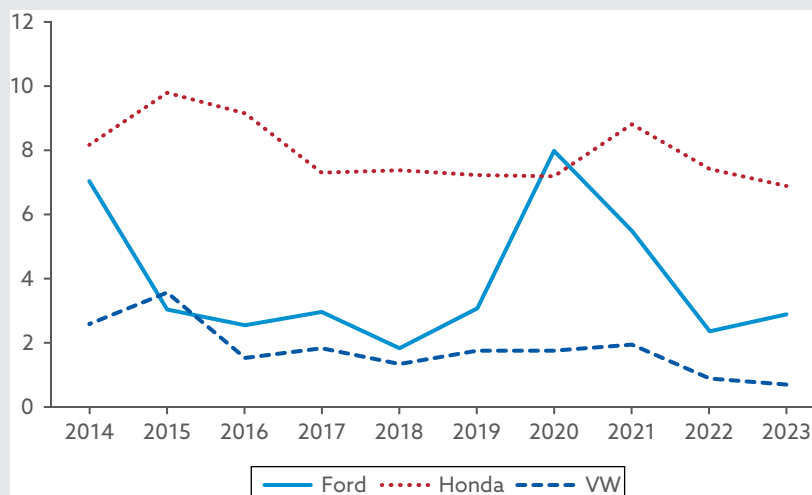
EXAMPLE 4**Ford Motor Company Absolute and Relative Multiples over Time**

An analyst considering an investment in Ford Motor Company, a US-based multinational automobile manufacturer, decides to compare company multiples over time and compare multiples to those of peers over a similar period. As a first step, the analyst compares Ford Motor Company's current P/E and EV/EBITDA multiples of 6.47 and 3.38, respectively, to the 10-year historical averages of 8.06 and 3.92. The analyst considers the firm to be undervalued based on this assessment. Multiple changes over 10 years are shown in Exhibit 23.

Exhibit 23: Ford P/E and EV/EBITDA Multiples, 2014–2023

Source: Bloomberg.

Note that the average P/E for Ford omits the years 2019 and 2020 because the firm had a net loss in both years. Exhibit 23 shows Ford's relative performance over time, but the analyst decides to compare Ford's performance with Japan-based Honda Motor Company and European automaker Volkswagen AG to establish Ford's relative performance over the same period. Exhibit 24 uses EV/EBITDA for comparative purposes for the three companies over the same period.

Exhibit 24: Relative Ford EV/EBITDA Multiples, 2014–2023

Source: Bloomberg.

Based on the 10-year historical averages shown in Exhibit 24, the analyst finds that Honda's EV/EBITDA was roughly double that of Ford, which, in turn, averaged approximately two times that of Volkswagen over the decade in question. Except for 2020–2021, however, Exhibit 24 suggests that Ford was originally valued more like Honda in 2014 but has trended toward VW's lower valuation over time.

The results using historical multiples may be influenced by changes in a company's business operations and strategy or those of its peers. However, we may still draw comparisons using the method of forecasted fundamentals to gauge the effects of these changes over time.

Multiples Based on Projected Future Values

Multiples based on future values include projections or estimates in the denominator (such as forward P/E) that are typically used to compare a target company to a benchmark. For example, consider the projected future multiples for the prior large European pharmaceuticals example, shown in the following example.

PROJECTED FUTURE MULTIPLES FOR LARGE EUROPEAN PHARMACEUTICALS

Consider the case of large European pharmaceutical companies' multiples as of 2024, shown in Exhibit 25, which includes trailing P/E and P/E based on next year's earnings (P/E FY 1) and the following year's earnings (P/E FY 2).

Exhibit 25: Selected Multiples of European Pharmaceutical Firms, 2024

Company	Market Cap (EUR millions)	EV (EUR millions)	Trailing P/E	P/E FY 1	P/E FY 2
Median	196,701	222,631	22.17	13.25	12.04
Bayer AG	28,033	65,536	N/A	5.56	5.19
GSK plc	85,974	101,263	16.06	11.21	10.00
Roche Holding AG	196,701	222,631	16.48	13.25	12.04
Novartis AG	215,470	231,678	23.66	14.23	13.05
Sanofi	112,922	121,589	20.68	11.63	10.22
AstraZeneca PLC	219,376	244,034	37.61	18.92	16.83
Novo Nordisk A/S-B	540,534	542,916	45.24	38.44	31.35

Source: Bloomberg.

Exhibit 25 shows expected positive earnings for Bayer AG, leading to an economically meaningful P/E, while Novo Nordisk's forward P/E shows an expected decline in line with industry peers.

A common application of multiples based on future projected values includes the determination of a terminal value at the end of a forecast period, as shown earlier in Equation 8:

$$IV_0 = \sum_{t=1}^n \frac{E(CF_t)}{(1 + \text{Required rate of return})^t} + \frac{E(P_{t+n})}{(1 + \text{Required rate of return})^n}$$

The numerator of the second term is often estimated using an appropriate multiple, as shown in the following example.

KNOWLEDGE CHECK: TERMINAL VALUE FOR SWISSERV AG USING A MULTIPLE



Recall the valuation from an earlier learning module of the Swiss business consultancy firm Swisserv AG. Swisserv had a base-year FCFE of CHF3,120,000 and an estimated 12% annual FCFE growth rate over a three-year period under the assumption of a 10% required return on equity.

In contrast to the prior terminal value estimation approach using a stable growth assumption, the analyst believes that at the end of the forecast period, the company will reach the price-to-FCFE multiple of 15 observed for more mature business consultancy firms.

- Which of the following is closest to Swisserv's intrinsic value using an FCFE present value model and a terminal value based on a P/FCFE multiple of 15?

- A. CHF59.1 million
- B. CHF649.0 million
- C. CHF75.5 million

Solution:

A is correct. It is calculated as follows:

$$\sum_t^n \frac{FCFE_t}{(1+r)^t} + \frac{FCFE_{t=0} \times (1+g)^n \times \text{Multiple}}{(1+r)^n}.$$

First, we must determine FCFE for the next three years. For Years 1–3, we simply calculate it as follows:

$$FCFE_t = 3,120,000 \times (1 + 0.12)^t.$$

This yields the following results, rounded to the nearest whole unit:

	Year 1	Year 2	Year 3
FCFE	3,494,400	3,913,728	4,383,375

For the terminal value at the end of Year 3, we multiply the Year 3 FCFE by 15 (multiple) to arrive at the *future* terminal value:

$$\text{Terminal value} = 4,383,375 \times 15 = 65,750,630.$$

Because it is a future cash flow, we must convert the terminal value into the present value by discounting at the required return on equity over the first-stage horizon of three years.

$$PV(\text{Terminal Value}) = \frac{65,750,630}{(1 + 0.1)^3} = 49,399,422.$$

Similarly, we need to add the present values of the FCFE amounts from Years 1–3 as follows:

	Year 1	Year 2	Year 3
FCFE	3,494,400	3,913,728	4,383,375
PV @ 10%	3,176,727	3,234,486	3,293,295

Finally, we add all the present values plus the terminal value to arrive at an equity value of CHF59.1 million:

$$IV = 3,176,727 + 3,234,486 + 3,293,295 + 49,399,422 = 59,103,930.$$

B is incorrect because it results from mistakenly solving for the present value of the terminal value by discounting by the required return on equity over two years rather than three.

C is incorrect because the result is arrived at by incorrectly failing to discount the terminal value.

QUESTION SET 4



- Which one of the following statements most accurately describes the use of past multiples in the relative valuation process?
 - Multiple comparisons for a company relative to peer historical averages allow one to gauge deterioration or improvement in company performance.
 - Past multiples relative to a company's own history allow assessment of performance under similar market conditions relative to a peer group.

- C.** Results using historical multiples may be influenced by changes in business operations and strategy and/or those of peers.

Solution:

C is correct because business operations or strategy changes can change a company's risk, growth, and cash flow characteristics, which affect multiples.

A is incorrect because it describes the best use of past multiples relative to a company's own history.

B is incorrect because it describes the best use of multiple comparisons for a company relative to peer historical averages.

2. Company M has experienced improving growth along with higher shareholder payouts over the last four years. Company M trades at a current P/E multiple of 22. An analyst observes that during the last four years, Company M's P/E multiple has ranged from a low of 15 to a high of 30. Its average P/E based on daily observations over each of the last four years is shown below.

	Year -3	Year -2	Year -1	Current year
Average P/E	19.8	19.5	21.1	20.5

Which one of the following statements is most correct about Company M's current P/E relative to its own historical P/E?

- A.** Company M is currently overvalued relative to its historically observed P/E multiples.
- B.** Company M is currently undervalued because of its improving fundamental characteristics.
- C.** Company M cannot be valued relative to its historical multiples without incorporating comparables.

Solution:

C is correct because comparison of a company's current multiple to its own historical multiples does not allow for a conclusion of misvaluation of its stock.

A is incorrect because a comparison of a company's current multiple to its historical multiple does not allow for a conclusion of misvaluation of its stock. In this case, Company M's P/E multiple appears to possibly be trending upward over time, which makes the current multiple consistent with an upward trend.

B is incorrect because more information about the improvement in fundamental characteristics of growth and shareholder payouts is needed to assess whether these improvements are consistent with the increase in multiples. Furthermore, a conclusion of undervaluation would require some incorporation of comparables.

3. Company N is a technology firm with high growth potential expected over the next seven years. Following the high-growth period, the company's growth is expected to decline to a much more sustainable pace. An analyst is preparing a present value model for valuing Company N's stock and will use a peer group P/E multiple to estimate the stock's terminal value at the end of

seven years. Which statement is most correct about the analyst's choice of peer group for the P/E multiple?

- A. The analyst should use the current P/E multiple taken from a peer group of high-growth technology firms.
- B. The analyst should use the current P/E multiple taken from a peer group of larger and slower-growth technology firms.
- C. The analyst should use an average of Company N's historical P/E observations over the last seven years.

Solution:

B is correct because the most appropriate peer group consists of comparables that share the growth rate characteristics expected in Company N's future. In other words, a larger and slower-growth peer group makes more sense.

A is incorrect because the higher-growth peer group is more comparable to Company N's current fundamentals. While this would likely be an appropriate peer group if the analyst were performing a relative valuation of current values, this peer group is likely to exhibit higher P/E multiples than might be expected seven years into the future when Company N is expected to be larger and exhibit slower growth.

C is incorrect because Company N's historical P/E multiples likely incorporated much different fundamentals, such as growth. As a result, Company N's historical P/E multiples are unlikely to be useful in estimating a future multiple used to compute terminal value.

SUMMARY: RELATIVE VALUE EQUITY VALUATION APPROACHES

- The relative valuation process involves selecting an appropriate multiple, establishing a representative peer group, and calculating and interpreting the multiple for a company and its peers. The results are used to estimate intrinsic value versus observed market price.
- Variations in multiples for companies with similar characteristics should reflect differences in fundamental drivers of value, such as profitability, growth, or the reinvestment of earnings.
- Multiples comprise a numerator consisting of equity or enterprise value and a denominator that includes income statement, balance sheet, or cash flow metrics available to the same capital providers.
- Multiples can vary widely across peer group companies due to cash flow, risk, and growth features, with differences among firms often classified by these fundamental characteristics.
- Comparable companies for equity valuation are often selected from firms in the same industry, country, or regional market with similar business activities and size as the target firm.
- Peer groups may be selected using statistical characteristics or factors that explain equity risk and return, such as the relationship between the target firm and the overall equity market.
- Past multiples for a company may be used to understand positive or negative trends over time and to assess the evolution of the company's multiples relative to its industry or market benchmark over time.
- A terminal value estimate reflects an implied or explicit future expected multiple for a company. Relative valuation may be used to estimate a company's terminal value in the context of a present value model.

PRACTICE PROBLEMS

1. Which one of the following statements most correctly describes a positively skewed distribution of P/E multiples?
 - A. The average P/E is greater than the median P/E.
 - B. The distribution includes firms with negative EPS.
 - C. The average P/E more likely reflects the distribution's central tendency.
2. Relative valuation is least likely to be a primary method for valuing equity of:
 - A. firms operating in concentrated industries.
 - B. firms operating in industries with wide variation in company age.
 - C. firms operating in multiple industries.
3. An analyst analyzes StevCo's observed value-to-book multiple of 1.60 based on the following information:
 - StevCo's WACC is 7.5%, and its long-run growth of FCFE is 3.0%.
 - StevCo's book value of equity is EUR400 million, and its book value of debt is EUR900 million.
 - StevCo's base-year ($t = 0$) EBITDA is equal to EUR300 million, and its depreciation expense is EUR150 million. The company spends EUR190 million on capital expenditures. StevCo operates with zero net working capital. StevCo's tax rate is 30%.

Which one of the following is closest to the analyst's estimate of StevCo's justified value-to-book multiple assuming constant growth?

 - A. 0.17
 - B. 1.11
 - C. 2.22
4. An analyst is comparing two mature companies with stable dividend policies. Firm 1 has a dividend payout ratio of 60%, and its stock has a dividend yield of 4.2%. Firm 2 has a dividend payout ratio of 70%, and its stock has a dividend yield of 4.5%. The analyst finds that the two stocks have the same cost of equity of 9%. The analyst calculates Firm 1's expected dividend growth rate at 4.7% and Firm 2's growth rate at 4.2%. Which one of the following is the most correct relative valuation conclusion from this analyst?
 - A. Firm 2's stock is underpriced compared to Firm 1's stock based on the differences in dividend yield.
 - B. Firm 1's stock is underpriced compared to Firm 2's stock price based on their P/E multiples.
 - C. Firm 1's stock is underpriced and Firm 2's stock is overpriced based on their justified dividend yields.
5. Which one of the following statements most correctly describes a weakness of

using relative valuation multiples in trying to identify misvalued stocks?

- A. The peer group used as part of the relative valuation may include firms that are not identical.
- B. The peer group used as part of the relative valuation process may exhibit variation in fundamental characteristics.
- C. The peer group used as part of the relative valuation process may not be fairly valued.

The following information relates to questions 6-8

An analyst gathers the following information for a set of firms from the same industry.

Firm	P/E	Dividend payout ratio	Growth rate	Cost of equity
1	15.80	80.00%	2.0%	8.2%
2	14.20	75.00%	2.5%	8.0%
3	28.10	70.00%	5.0%	8.3%
4	23.80	72.00%	4.0%	7.5%
5	16.50	82.00%	2.5%	7.5%
6	17.50	90.00%	2.0%	7.8%
7	26.40	77.00%	4.0%	8.0%
8	25.10	81.00%	3.5%	7.8%
9	19.30	70.00%	3.5%	7.0%
10	21.60	79.00%	3.0%	8.5%

The average P/E of this peer group is 20.83, with a median of 20.45. Across the distribution, the average payout ratio is 77.6%, the average growth rate is 3.2%, and the average cost of equity is 7.9%.

The analyst runs a linear regression of the P/E multiples on payout ratios and growth rates and the regression generates the following results.

	Coefficient	Standard Error	t-Stat.	P-Value
Intercept	-32.63	9.49	-3.44	0.0108
Payout ratio	42.54	10.16	4.19	0.0041
Growth rate	639.14	64.50	9.91	0.0000

The analyst is curious about Firm 3's stock given the fact that its expected growth rate is the highest among the firms. However, she also notes that Firm 3's stock trades at the highest P/E in the peer group. Additionally, she observes that Firm 7 offers higher payout to shareholders along with relatively high growth expectations.

- 6. The analyst considers Firm 2 stock as possibly undervalued because its P/E of 14.2 is the lowest in the distribution. Which of the following statements about Firm 2's characteristics most correctly implies that its P/E should be low relative

to its peers?

- A. Firm 2's payout ratio and cost of equity are low relative to its peers.
 - B. Firm 2's payout ratio and growth rate are low relative to its peers.
 - C. Firm 2's growth rate and cost of equity are low relative to its peers.
7. Which one of the following is closest to the justified P/E multiple on Firm 3's stock using a constant growth assumption?
- A. 14.0
 - B. 22.3
 - C. 30.3
8. Which of the following is a reasonable conclusion for the analyst from observing Firm 7's P/E multiple compared to its justified multiple computed from the regression results?
- A. The comparison suggests that Firm 7's stock may be undervalued.
 - B. The comparison suggests that Firm 7's stock may be overvalued.
 - C. The comparison suggests Firm 7's stock is probably neither under- or overvalued.
-

SOLUTIONS

- A is correct because the mean (i.e., average) P/E exceeds the median P/E in a positively skewed distribution since the average is affected by the presence of very high P/E multiples that may be outliers within the distribution.

B is incorrect because P/E multiples are not economically meaningful and are typically not included in the distribution.

C is incorrect because the median is a more likely reflection of the distribution's central tendency when the distribution is positively skewed.
- A is correct because the ability of an analyst to construct a peer group is much more limited for firms in which there are fewer competitors. For these types of firms, present value methods of valuing equity may be preferable.

B is incorrect because firms of differing phases in the company life cycle can be included in the analyst's peer group as long as differences in fundamental characteristics can be incorporated into the analysis.

C is incorrect because an analyst can create multiple peer groups across industry classifications and weight the peer group multiples according to the firm's proportion of business activity in each industry.
- B is correct because StevCo's justified value-to-book multiple assuming constant growth is approximately 1.11, as shown below.

$$\begin{aligned}\text{StevCo return on capital} &= \frac{\text{EBIT}_{t+1} \times (1 - t)}{\text{Book value of capital}} = \frac{(300 - 150) \times (1 - 0.30)}{400 + 900} \\ &= 8.077\%.\end{aligned}$$

In the previous equation, StevCo's EBIT is calculated by subtracting depreciation of EUR150 million from its EBITDA of EUR300 million. Then, in the equation below, StevCo's reinvestment rate is computed by subtracting depreciation from capital expenditures of EUR190 million. The resulting EUR40 million is divided by StevCo's after-tax EBIT as shown in the numerator of the return on capital calculation.

$$\text{Justified value to book} = \frac{\text{Return on capital} \times (1 - \text{Reinvestment rate})}{\text{WACC} - g}$$

$$\text{StevCo justified value to book} = \frac{0.08077 \times \left(1 - \frac{190 - 150}{105}\right)}{0.075 - 0.03} = 1.11.$$

A is incorrect because this result occurs if the reinvestment rate is incorrectly calculated by omitting depreciation in the numerator and using after-tax EBITDA in the denominator of the reinvestment rate equation.

C is incorrect because this result occurs if after-tax EBITDA is incorrectly used to calculate return on capital.

- C is correct because Firm 1's actual dividend yield of 4.2% is greater than its justified dividend yield of 4.1% while Firm 2's actual dividend yield is below its justified dividend yield. The results are shown below.

$$\begin{aligned}\blacksquare \text{ Firm 1 justified } \frac{\text{Div}}{P} &= \frac{9\% - 4.7\%}{1.047} = 4.1\% < 4.2\% = \text{Actual } \frac{\text{Div}}{P}. \\ \blacksquare \text{ Firm 2 justified } \frac{\text{Div}}{P} &= \frac{9\% - 4.2\%}{1.042} = 4.6\% > 4.5\% = \text{Actual } \frac{\text{Div}}{P}.\end{aligned}$$

A is incorrect because the dividend yields cannot be reasonably compared between the two companies without adjusting for the difference in growth rates (which are likely related to the differences in dividend payout ratios).

B is incorrect because the P/Es cannot be reasonably compared without adjusting for the differences in dividend payout ratio and growth rates.

5. C is correct because a misvalued peer group creates a situation in which firms may be classified only as undervalued or overvalued relative to the peer group. For example, a stock that is undervalued relative to an overvalued peer group may be overvalued if valuing the stock on an absolute basis using a present value model.

A is incorrect because relative valuation is based on the concept that similar companies should trade at similar multiples after controlling for differences. Peer companies do not need to be identical to be appropriate peers.

B is incorrect because differences in fundamental characteristics among peer group firms may be controlled for by using adjustment or regression based on the method of forecasted fundamentals.

6. B is correct because Firm 2's payout ratio and growth rate are both well below the average of the distribution. Because these two fundamental characteristics should be related to a company's P/E multiple, it should not be surprising that Firm 2 has a low P/E.

A is incorrect because a low cost of equity is associated with a higher, not lower, P/E multiple.

C is incorrect because a low cost of equity is associated with a higher, not lower, P/E multiple.

7. B is correct because applying the justified P/E equation provides a result of 22.27:

$$\frac{P_t}{\text{EPS}_t} = \frac{\text{Payout ratio} \times (1 + g)}{r_e - g} = \frac{0.7 \times (1.05)}{0.083 - 0.05} = 22.27.$$

A is incorrect because this choice occurs if the analyst divides the payout ratio of 70% by the growth rate of 5%.

C is incorrect because this choice occurs if the analyst uses the equation for price to FCFE, dividing 1 by the difference between the cost of equity and the growth rate.

8. B is correct because analysis of Firm 7's payout ratio, growth rate, and cost of equity implies that the stock's justified P/E from the regression should be approximately 25.7:

$$\begin{aligned} \text{Predicted Firm 7 } \frac{P}{E} &= -32.63 + 42.54 \times 77\% + 639.14 \times 4\% \\ &= 32.76 + 25.57 - 32.63 = 25.7 \end{aligned}$$

Because Firm 7's actual P/E of 26.4 is greater than 25.7, this comparison suggests that Firm 7's stock may be overvalued. However, a formal statistical test might conclude that this difference is not significantly different from zero because of the small sample size of only 10 observations.

A is incorrect because this statement would require Firm 7's actual P/E to be less than the justified P/E from the regression.

C is incorrect because this statement would require the actual P/E and justified P/E from the regression to be approximately equal.

LEARNING MODULE

8

Financial Statement Forecasting in Equity Valuation

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	explain the rationale, construction, and uses of financial statement forecast models in equity valuation
☐	evaluate the construction of an equity valuation model based on characteristics of the subject company
☐	estimate and interpret the value of an equity security based on the outputs of a financial statement forecast model

INTRODUCTION: FINANCIAL STATEMENT FORECASTING IN EQUITY VALUATION

1

Prior Equities learning modules focused on valuing firms by either extrapolating or applying base year financial information. Under the present value method, expected future cash flows (such as dividends or free cash flow) are assumed to grow in proportion to base year measures by applying one or more growth rates over a forecast horizon. Relative value multiples include both base year metrics and an appropriate market multiple to estimate value. While simple to compute, these two approaches rely on the assumption of static relationships among underlying value drivers.

In this learning module, we discuss the practice of equity valuation using forecasted financial statements, referred to as disaggregated valuation models. We first address the rationale, use, and construction of forecasts in equity valuation. This approach allows an analyst to generate a more detailed valuation narrative. Rather than simply estimating a cash flow growth rate, analysts can separately model and project the firm's operating, investing, and financing performance—solving for growth as the combined result of these activities. Next, we consider different forecast approaches based on a company's characteristics, such as its stage in the company life cycle. Finally, we show how a firm's financial forecast translates into an equity valuation comparable to the results of other approaches.

While aligned with fundamental equity valuation concepts introduced in earlier learning modules, the disaggregated approach is more consistent with that of professional equity analysts and detailed management forecasts, allowing for adjustments to be made as underlying market factors change.

PRE-TEST: FINANCIAL STATEMENT FORECASTING IN EQUITY VALUATION



1. Disaggregated models are most appropriate for valuing a company when its cash flow components are expected to:

- A. grow at the same rate in the future.
- B. grow at differing rates in the future.
- C. remain constant in the future.

Solution

B is correct. Choice B is correct because differing growth rates in various cash flow components may cause the overall cash flow growth rate to differ from that of the individual components. A disaggregated model is likely to yield different results than a constant growth model.

Choice A is incorrect because this implies that cash flow is expected to grow at the same rate as that of its components and disaggregated and cash flow models will have comparable results.

Choice C is incorrect because with zero growth for each component, cash flow is constant and disaggregated and constant growth models will be the same.

2. Which one of the following sets of disaggregated model assumptions best determines the net investment component of forecasted free cash flow?

- A. Revenue model assumptions
- B. Asset model assumptions
- C. Expense model assumptions

Solution

B is correct. Choice B is correct as asset model assumptions determine working capital and net long-term asset growth, which comprise net investment.

Choice A is incorrect because revenue model assumptions are a partial determinant of forecasted net income.

Choice C is incorrect because expense model assumptions are a partial determinant of forecasted net income.

3. Which of the following company life-cycle phases is least likely to be valued using a disaggregated model?

- A. Start-up
- B. Growth
- C. Decline

Solution

A is correct. Choice A is correct because a company in the start-up phase of the company life cycle is likely to be valued based solely on revenue poten-

tial. Thus, a disaggregated model is not necessary because only the revenue model assumptions affect the valuation.

Choice B is incorrect because a company in the growth phase of the company life cycle is likely to be valued based on not only revenue growth but also profit margin improvements. Thus, both revenues and expense model assumptions may be incorporated into a disaggregated valuation.

Choice C is incorrect because a company in the decline phase of the company life cycle is likely to be valued based on declining profit margins during the forecast horizon. The valuation may incorporate separate revenue and expense forecasts as in the case of a growth phase company.

4. A company in the growth phase of the company life cycle is likely to be valued based on:

- A. revenue growth and asset efficiency improvements.
- B. margin improvements and asset efficiency declines.
- C. revenue growth and margin improvements.

Solution

C is correct. Choice C is correct because the growth phase is usually characterized by significant revenue growth and improvements in profit margins.

Choice A is incorrect because the growth phase does not have a specific prediction about asset efficiency improvements. Nevertheless, high revenue growth may require significant investment in which asset efficiency declines during the forecast horizon.

Choice B is incorrect because this choice neglects to mention revenue growth, which is a key assumption underlying a company in the growth phase of the company life cycle.

5. A disaggregated model values forecasted free cash flows based on:

- A. growth of the free cash flow from the base year.
- B. output of forecasted financial statements.
- C. market multiples of current free cash flow.

Solution

B is correct. Choice B is correct because the disaggregated model creates forecasted financial statements from which forecasted free cash flows are calculated.

Choice A is incorrect because a disaggregated model typically uses multiple growth rates within the forecasted financial statements; thus, forecasted cash flows are not calculated solely based on base year free cash flow.

Choice C is incorrect because this response implies a relative valuation using base year free cash flow. In this case, valuation does not require a disaggregated model to generate forecasted financial statements.

6. Scenario analysis combined with disaggregated models allows analysts to:

- A. include narratives that may affect the market's current valuation of an equity security.
- B. objectively determine the probabilities associated with each scenario.

C. more accurately determine whether an equity security is fairly valued.

Solution

A is correct. Choice A is correct because disaggregated models allow for explicit narratives to be included by the analyst through the assumptions and outputs of the model. The valuation associated with each scenario presents an analyst's picture of possible returns on an equity security, or the scenarios may be probability-weighted to imply the analyst's estimated value of the security.

Choice B is incorrect because the probabilities assigned to each scenario reflect the analyst's subjective view.

Choice C is incorrect because the incorporation of scenarios does not increase accuracy; rather, the scenarios reflect the analyst's subjective view of the company.

2

FINANCIAL STATEMENT MODEL USES IN EQUITY VALUATION



explain the rationale, construction, and uses of financial statement forecast models in equity valuation

An earlier learning module introduced present value models in which base year cash flow metrics were used to value firms based on direct assumptions of expected aggregate cash flow growth.

In this learning module, rather than directly assuming an overall cash flow growth rate, we consider a forecast of financial statements into the future. Financial statement projections allow an analyst to make assumptions about changes in individual cash flow components, such as expected future revenues and expenses, as well as investment and financing cash flows. We refer to valuations of expected future cash flows using financial statement forecasts as **disaggregated valuation models**.

Exhibit 1 illustrates one possible approach to estimating the next period's free cash flow to the firm (FCFF) based on individual income statement and balance sheet component projections.

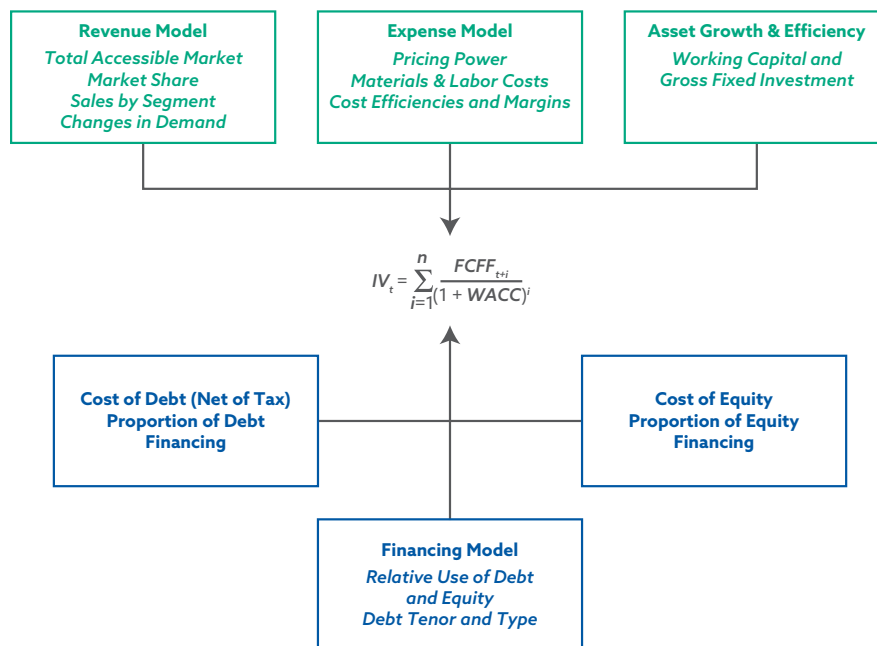
Exhibit 1: Illustration of Disaggregated Model Approach

Exhibit 1 shows areas of detail under a disaggregated approach. For example, an analyst may include a more granular sales forecast, incorporate trends related to operating margins and specific expense drivers, specify working capital changes and investment in long-term assets, and/or build out a financing model that specifies the timing, amounts, and cost of debt and equity needed to support underlying growth. The resulting cash flow measure reflects the *combined* effects of these assumptions.

By extending these projections over a multi-year forecast horizon, the disaggregated model produces a free cash flow forecast that can be used in either a present value model or a relative value multiple based, for example, on enterprise value (EV), as shown in Exhibit 2.

Exhibit 2: Financial Statement Models in Equity Valuation

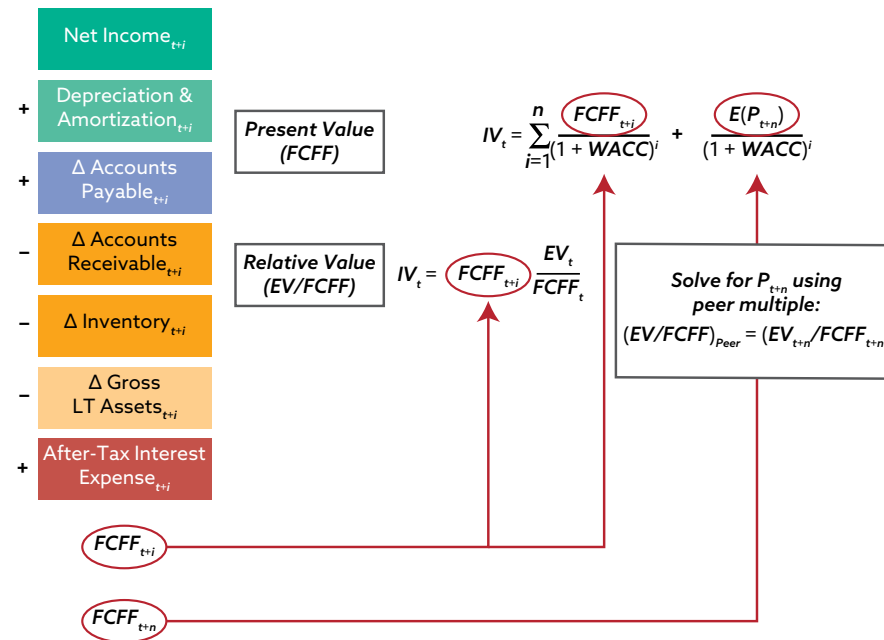


Exhibit 2 shows that projected free cash flow to the firm (FCFF_{t+i}) may be used as an input in the *numerator* of a discounted cash flow (DCF) estimate of intrinsic value (IV_t), also shown in Equation 1:

$$IV_t = \sum_{i=1}^n \frac{FCFF_{t+i}}{(1+WACC)^i} + \frac{E(P_{t+n})}{(1+WACC)^n} \quad (1)$$

Alternatively, FCFF forecasts may be used in the *denominator* of a multiple used to determine relative value (Equation 2), as described in an earlier learning module:

$$IV_t = FCFF_{t+1} \times EV \text{ Multiple}_t \quad (2)$$

where:

$$EV \text{ Multiple}_t = \frac{EV_{t+i}}{FCFF_{t+i}}$$

A disaggregated model usually includes company revenue and/or EBITDA forecasts, allowing an analyst to use the projected income measure combined with an appropriate peer multiple estimate to calculate the terminal value, $E(P_{t+n})$. We discuss terminal value estimation for a disaggregated valuation model later in this learning module.

Rationale for Disaggregated Models

Disaggregated models allow an analyst to translate a narrative about a company's future into an estimate of company and/or equity value using underlying value drivers. This valuation narrative may incorporate industry projections, competitive forecasts, and/or individual company analysis in the form of specific inputs (such as revenue, profit margins, and asset growth) that result in outputs (such as earnings, investment, cash flow, and other measures) used to value a firm's equity. Extending well beyond a simple estimate of base year cash flow and growth rate over a time horizon, disaggregated models allow an analyst to explicitly communicate assumptions underlying a valuation, such as expected profit margin changes or disproportionate investment

needs relative to revenue growth. In general, disaggregated models are appropriate in any valuation exercise in which cash flow components may grow at different rates over time, as in the following case.

EXAMPLE 1

Consensus Analyst Estimates for Amazon.com

Analyst estimates of company performance are available from press releases and financial data vendors. Exhibit 3 shows Bloomberg's Earnings Estimates Overview, or EEO, function, for Amazon.com as of May 2024.

Exhibit 3: Bloomberg EEO Function for Amazon.com, May 2024

Amazon.com Inc		Periodicity		Quarters	Source	Standard	Cur	USD		
More estimates are available		MODL »			Broker	-----	Guidance			
1	Headline	2		Headline Growth	3					Company-Specific
3 Months Ending		Q1 2024 Act	Q2 2024 Est		Q3 2024 Est		Q4 2024 Est		Q1 2025 Est	
		03/31/2024	06/30/2024	#	09/30/2024	#	12/31/2024		03/31/2025	
11)	EPS, Adj	1.590	1.313 ↑	20	1.441	19	1.655 ↑	19	1.541 ↑	
12)	EPS, GAAP+	1.170	1.029 ↑	40	1.155 ↑	39	1.356 ↑	39	1.320 ↑	
13)	Revenue	143.313B	148.874B	50	159.155B	47	187.877B	47	158.989B	
14)	Gross Margin %	49.319	49.508	37	48.804	35	47.014	35	50.194	
15)	Operating Profit	15.307B	13.662B ↑	44	15.505B ↑	45	18.185B ↑	44	18.274B ↑	
16)	EBIT	15.307B	13.805B ↑	38	15.552B ↑	39	18.155B ↑	39	18.355B ↑	
17)	EBITDA	31.952B	33.045B	35	34.646B	34	38.249B	34	37.044B ↑	
18)	Pre-Tax Profit	14.983B	13.687B ↑	31	15.528B ↑	31	18.046B ↑	32	17.799B ↑	
19)	Net Income Adj	16.972B	13.990B	17	15.978B	16	21.286B ↑	16	14.526B ↑	
20)	Net Income, GAAP+	12.011B	11.136B ↑	37	12.496B ↑	37	14.503B ↑	37	14.472B ↑	
21)	Net Debt	68.667B	47.422B ↓	6	66.103B ↓	6	98.709B ↓	6	103.431B ↓	
22)	BPS	20.867	22.924	5	24.572	5	26.636 ↑	5	27.960	
23)	CPS	1.827	2.500 ↑	1	2.680	1	4.680 ↑	1		
24)	DPS	0.000								
25)	Return on Equity %	20.305	19.532 ↓	5	18.870 ↓	5	19.632 ↓	5	16.173 ↓	

Source: Bloomberg.

Users of this summary information should develop an understanding of how it is compiled, as analyst estimates and adjustments can vary widely.

Amazon.com's Q1 2025 consensus revenue forecast is shown as USD158.989 billion, or 10.9% higher than actual Q1 2024 revenue of USD143.313 billion. On average, analysts are even more optimistic about Amazon.com's operating profit growth. The Q1 2025 consensus estimate for operating profit of USD18.274 billion is 19.4% above actual Q1 2024 operating profit of USD15.307 billion.

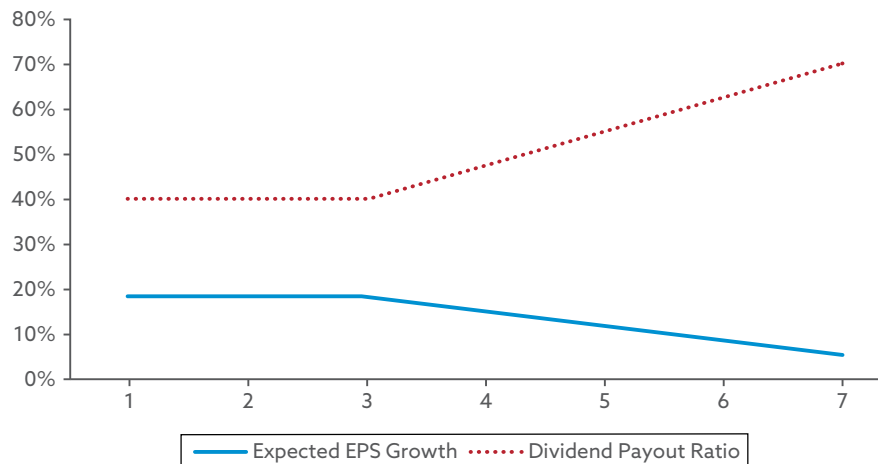
While the above case shows varying growth rates among income statement components, a disaggregated approach may also capture a valuation story that anticipates changes in investment policy, financing, or earnings distribution policies.

For example, consider a mature firm with a stable dividend payout ratio. If an analyst expects earnings growth to decline, they may also expect the firm to increase its dividend payout ratio due to fewer profitable investment opportunities. The following knowledge check explores the effect of this narrative on the firm's dividends per share (DPS) used for valuation purposes.

KNOWLEDGE CHECK: SPINCLIFF PLC'S DIVIDEND PAYOUT POLICY



An analyst seeks to value Spincliff plc shares using a dividend discount model. Last year, Spincliff generated earnings per share (EPS) of GBP1.50. The analyst separates Spincliff's dividend payout ratio and EPS growth to allow for the assumption that as its EPS growth slows, Spincliff will increase the percentage of EPS distributed as dividends. Exhibit 4 shows the analyst's forecast.

Exhibit 4: Spincliff plc Dividend Payout and EPS Assumptions

As shown in Exhibit 4, the forecast dividend payout ratio in Years 4 to 7 rises, while EPS growth slows for the same period. Details are shown in the following table:

Year	Expected EPS Growth	Dividend Payout Ratio
1	18%	40.0%
2	18%	40.0%
3	18%	40.0%
4	14.75%	47.5%
5	11.50%	55.0%
6	8.25%	62.5%
7	5%	70.0%

1. Calculate the expected percentage growth in Spincliff's DPS over the seven-year period and compare it with expected EPS growth.

Solution

The following table shows Spincliff's expected DPS and DPS growth rates, with EPS and DPS results rounded from exact calculations to two decimals.

Year	EPS	DPS	DPS Growth
0	1.50	0.60	---
1	1.77	0.71	18.0%
2	2.09	0.84	18.0%
3	2.46	0.99	18.0%
4	2.83	1.34	36.3%
5	3.15	1.73	29.1%
6	3.41	2.13	23.0%
7	3.58	2.51	17.6%

For each year, EPS may be calculated from assumed growth rates, as shown in the case of Years 3 and 4 (rounded to four decimals):

$$\text{Year 3 EPS} = 1.50 \times (1.18)^3 = 2.4645.$$

$$\text{Year 4 EPS} = 1.50 \times (1.18)^3 \times (1.1475) = 2.8281.$$

Years 3 and 4 DPS are calculated based on their respective EPS and the assumed Years 3 and 4 dividend payout ratios (rounded to four decimals):

$$\text{Year 3 DPS} = 2.4645 \times 0.4 = 0.9858.$$

$$\text{Year 4 DPS} = 2.8281 \times 0.475 = 1.3433.$$

Year 4 DPS growth is projected as follows:

$$\text{Year 4 DPS growth rate} = (1.3433 \div 0.9858) - 1 = 0.3626.$$

Comparing DPS growth rates to the assumed EPS growth rates, we observe that DPS growth rates in Years 4 to 7 are significantly higher than EPS growth rates during the period over which dividend payout ratio increases.

In the prior knowledge check, the assumed rise in the payout ratio amid slower EPS growth results in a forecast in which DPS grows much faster than corresponding EPS. If the analyst had instead assumed a constant 40% dividend payout ratio, then forecasted DPS growth would equal that of EPS. This example illustrates the rationale for disaggregated models, namely that varying assumptions across individual components may result in very different growth patterns in forecasted cash flow outcomes.

Disaggregated Model Construction

Disaggregated model construction first requires an analyst to determine the level of detail to be considered. This decision may vary greatly for different income statement and balance sheet components depending on an analyst's focus and the value drivers involved. As shown in Exhibit 1, an analyst may consider the following areas of focus in a model over the forecast horizon:

- Revenue
- Expense and/or profit margin
- Asset growth and efficiency
- Financing

The first two choices above produce income assumptions used to compute FCFF and/or FCFE (free cash flow to equity) cash flow metrics. The third modeling choice allows for the incorporation of investment assumptions that affect cash flow measures. Finally, the financing model supports the assumption of possible debt and equity combinations available for a given forecast of company income and investment and their effects on cash flow and firm value.

Level of Model Detail and Areas of Focus

Model construction often begins by downloading a company's historical financial statements from an external source. Here we assume an analyst has made any necessary adjustments for different accounting standards or policy choices as discussed in earlier Financial Statement Analysis learning modules.

The original data format may lead an analyst to build a model using the same line items as in the historical financial statements. However, a more effective model often results from either consolidating line items or incorporating additional detail beyond what is included in the financial statements. The degree of detail in a disaggregated model usually follows from the area(s) of focus, including value drivers, as shown in Exhibit 5.

Exhibit 5: Disaggregated FCFF Forecast

$$\begin{aligned}
 & \left[\text{Revenue} - \text{Cost of Goods Sold} - \text{SG \& A} - \text{Dep \& Amort} \right] (1 - \text{Tax Rate}) - \Delta \text{Working Capital} - \Delta \text{Net LT Assets} \\
 & \text{FCFF} = \text{NOPAT} - \text{Net Investment} \\
 & IV_t = \sum_{i=1}^n \frac{\text{FCFF}_{t+i}}{(1 + \text{WACC})^i}
 \end{aligned}$$

The simplest FCFF forecast combines expected future **net operating profit after taxes (NOPAT)** and **net investment**. Exhibit 5 shows that NOPAT equals revenue less cost of goods sold (COGS); selling, general, and administrative expenses (SG&A); and depreciation and amortization (D&A) after taxes. This calculation is also shown in Equation 3:

$$\text{NOPAT} = [\text{Revenue} - \text{COGS} - \text{SG\&A} - \text{D\&A}] (1 - t), \quad (3)$$

where t is the firm's tax rate.

Net investment consists of both long-term assets net of D&A and working capital changes, as shown in Equation 4:

$$\text{Net investment} = \Delta \text{Working capital} + \Delta \text{Net long-term assets}. \quad (4)$$

As shown in Exhibit 5, a more detailed NOPAT projection may include separate forecasts of the following:

- Revenues
- COGS
- Additional operating expenses, such as SG&A and depreciation
- Tax rates

Added details might include revenues based on estimated market share or by product segment. Costs may be broken down by product or type, such as research and development (R&D), salaries, or sales expenses. Net investment may include separate forecasts for different components of working capital in addition to distinct capital expenditure and depreciation forecasts.

While greater model complexity allows for a richer valuation narrative with more flexibility, it can reduce the tractability and interpretability of model results and add a greater potential for model error or noise. Detailed models with more inputs often must be updated with greater frequency to incorporate potential changes in relationships among underlying variables as markets evolve and competitive forces change. An analyst should therefore always weigh these trade-offs when adding model complexity. The following knowledge check provides a simple example to illustrate key issues for analysts to consider.

KNOWLEDGE CHECK: ALMSTUBE AG

An analyst generates a cash flow forecast for Almstube AG, gathering income statements for the most recent year, t , as shown in the following table (in millions of EUR).



Year	<i>t</i>
Revenue	140
COGS	65
SG&A	45
Depreciation	15
Earnings before interest and taxes (EBIT)	15
Interest expense	6
Interest income	1
Earnings before taxes (EBT)	10
Taxes	3
Net income	7

1. The analyst creates a forecast for year $t+1$ in which a growth rate of 7% is applied to the following year t income statement line items: revenue, COGS, SG&A, and depreciation. The analyst expects stable interest rates and therefore holds interest expense and interest income items constant in the forecast. The analyst assumes a tax rate of 30%. Which one of the following statements most correctly describes the resulting forecasted growth of Almstube's net income compared to the 7% growth rate assumption?

- A. The year $t+1$ net income forecast shows growth of exactly 7%.
- B. The year $t+1$ net income forecast shows growth of *less* than 7%.
- C. The year $t+1$ net income forecast shows growth of *greater* than 7%.

Solution

C is correct. Choice C is correct because the combination of 7% growth applied to operating items and zero growth of nonoperating items implies that the *weighted* net income growth rate is greater than 7%. This occurs because net interest expense (i.e., interest expense minus interest income) is greater than zero in year t . To show this, an analyst may begin this forecast with the EBIT line item given that the same growth rate of 7% is applied to all income statement line items above EBIT, as shown below:

$$\begin{aligned} & \text{EBIT}(1 + g) \\ &= \text{Revenue}(1 + g) - \text{COGS}(1 + g) - \text{SGA}(1 + g) - \text{Depr\&Amort}(1 + g). \end{aligned}$$

$$15(1.07) = 140(1.07) - 65(1.07) - 45(1.07) - 15(1.07) = 16.05.$$

Since interest expense of EUR6 and interest income of EUR1 are assumed to be constant:

$$\text{Forecasted EBT} = \text{Forecasted EBIT} - \text{Net interest expense}$$

$$\text{Forecasted EBT} = 16.05 - (6 - 1) = 11.05.$$

Applying the tax rate of 30%:

$$\text{Forecasted net income} = \text{EBT} \times (1 - \text{Tax rate})$$

$$\text{Forecasted net income} = 11.05 \times (1 - 0.30) = 7.735.$$

The year $t + 1$ net income forecast of EUR7.735 is 10.5% greater than year t net income of EUR7 [= $(7.735 \div 7) - 1$], exceeding the 7% used for the operating items.

Choice A is incorrect, as this would only be true if net interest expense were zero. In such a case, the effect of the nonoperating items would be irrelevant to the forecast and only the 7% growth rate would affect the forecast.

Choice B is incorrect, as this would only be the case if net interest expense were negative. If interest income exceeds interest expense, then the analyst would be growing EBIT at 7% and net interest income (a positive number) at 0%. The combined effect would result in a net income growth rate below 7%.

2. Recommend an approach to simplify the above income statement for the year t . Explain why such a simplification should be applied with caution.

Solution

A simplified version of Almstube's year t income statement (in millions of EUR) is shown below:

Year	t
Revenue	140
Operating expense	125
EBIT	15
Net interest expense	5
EBT	10
Taxes	3
Net income	7

If the analyst assumes the same growth rate on all operating expenses (i.e., COGS, SG&A, and depreciation), these may be consolidated into one line item. The same logic applies to consolidating interest expense and interest income into one line item: net interest expense. However, such a simplification should be applied with caution, as this imposes model constraints that will need to be reversed if the analyst later decides that different operating line items have varying growth rates.

Revenue Model Construction

The first step in building an income statement forecast is to project expected future revenue. As revenue is a key driver of cash flow measures and value, analyst revenue assumptions often receive a high degree of scrutiny. In this lesson, we consider several approaches to revenue forecasting:

1. Historical approach
2. Bottom-up approach
3. Top-down approach

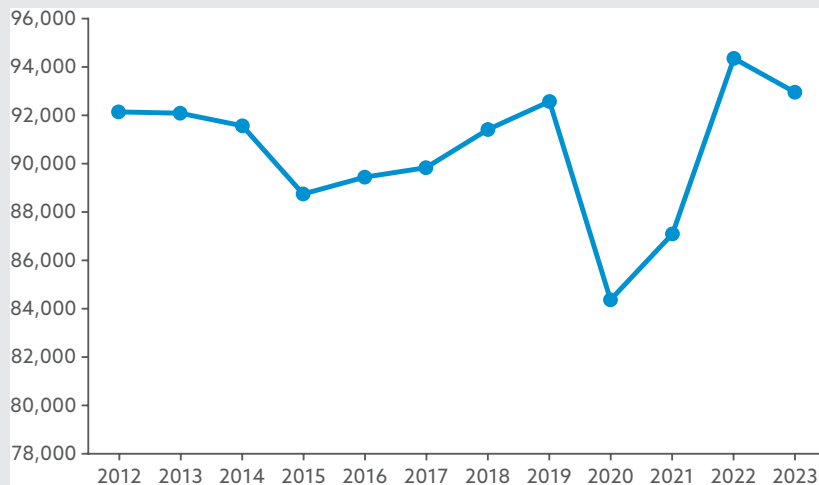
Historical Revenue Model

In the case of mature companies—those with established products, a stable product mix and geographic footprint, and no expected change to the competitive environment—past company revenues are often used to generate estimates of future revenue growth. This approach may be quite reasonable for companies with well-established business models that are unlikely to face significant changes to demand for their products or services during the forecast horizon. A historical approach may be useful for firms with relatively low revenue growth potential, as in the following example.

EXAMPLE 2**Nestlé S.A.'s Limited Growth**

Based in Switzerland, Nestlé, the world's largest food company, has struggled to generate consistent revenue growth. Exhibit 6 shows Nestlé's revenues (in CHF millions) from 2012 to 2023.

Exhibit 6: Nestlé Revenues, 2012 to 2023 (in CHF millions)



Source: FactSet.

Nestlé operates in nearly 190 jurisdictions with over 2,000 brands, including coffee and other beverages; prepared, frozen, and refrigerated foods; and nutrition products. While the company has grown in past years through acquisition, the scale and breadth of Nestlé's revenue in mature segments with stable demand makes past sales a reasonable starting point on which to base projections of future revenue. In 2012, the company generated just over CHF92 billion in revenue. In 2023, just over a decade later and following a decline during the COVID-19 pandemic, the company had approximately CHF93 billion in annual sales. Nestlé's annual revenue growth rate over this period was 0.16%, with a compound annual growth rate (CAGR) of 0.08% and a standard deviation of annual growth rates of 4.22%.

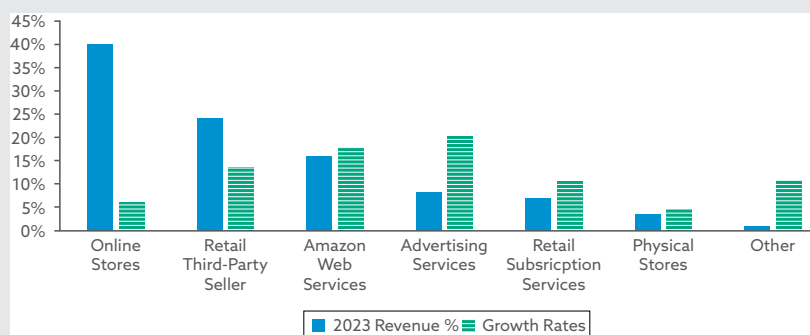
While historical growth may provide a basis for a company's revenue potential, it should also incorporate information about the company's strategy and the market outlook for its products and services. Analysts should also consider a company's industry and competitive environment when assessing its future potential. This approach ensures that the valuation narrative incorporates both expansion opportunities and threats, which may jeopardize future performance over the forecast horizon. Later Equities learning modules explore topics in company and industry analysis.

Historical growth analysis also applies to companies operating in multiple business lines with differing growth prospects. Firms are typically required to provide separate revenue and other financial information on economically material operating segments. In some cases, a company's lines of business may complement one another and face similar revenue growth prospects; in other cases, however, company segments may involve very different revenue growth prospects, as illustrated in the following example.

EXAMPLE 3**Amazon.com: Differing Growth Across Operating Segments**

Amazon.com, the global leader in e-commerce, continues to transform its business model across a variety of segments. While traditional e-commerce, or the Online Stores segment in Exhibit 7 below, remains the largest business line as a percentage of sales, this segment along with Physical Stores are expected to generate the slowest growth among Amazon.com's business lines.

Exhibit 7: Amazon.com Segment Revenues (% of Sales) and Growth Rates, 2023



Source: Bloomberg.

The company continues to capitalize on rapid growth in cloud computing and targeted advertising using artificial intelligence through its Amazon Web Services and Advertising Services business segments. These segments underscore the company's ability to maintain double-digit percentage revenue growth despite its already sizable revenue of nearly USD575 billion in 2023.

Bottom-Up Revenue Model

In some industry settings, analysts may prefer to forecast firm revenue based on more fundamental units of business activity. For example, analysts may initiate a revenue forecast using distinct assumptions about unit increases and revenue per unit across products or services. The airline industry is a case in point. Analysts seeking to forecast passenger airline company revenue will typically generate forward-looking estimates of passenger seat capacity using **available seat kilometers (ASKs)**. ASKs are a measure of capacity used in the airline industry derived by multiplying the number of seats available for scheduled airline passengers by the total number of kilometers in which those seats were flown over a given period, such as a year. An airline industry analyst may gather historical ASKs and incorporate planned changes to an airline's route structure to forecast ASKs. The analyst then forecasts revenue per ASK and multiplies this unit revenue estimate by the ASK estimate to determine a future revenue estimate.

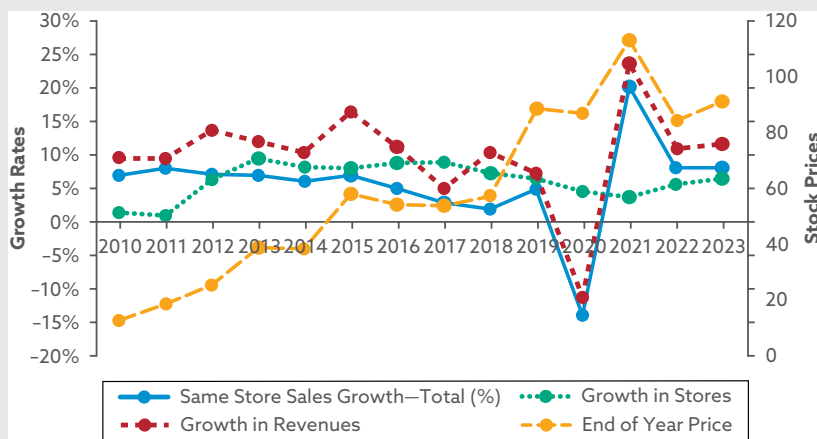
Bottom-up approaches based on capacity analysis are often applied in valuing businesses dependent on existing physical locations, such as retail outlets or restaurant chains. For example, analysts and investors closely track corporate releases of same-store sales growth to revise expectations of future revenues and earnings, as discussed in the following example.

EXAMPLE 4**Starbucks Corporation (SBUX) and the Importance of Same-Store Sales**

In April 2024, shares of Starbucks, the global coffee giant, fell by 12% following news of weaker than expected quarterly earnings and revenues due to lower same-store sales growth. Starbucks management lowered earnings and revenue forecasts for the rest of 2024 and outlined strategies to reverse the revenue decline at its global outlets.

While Starbucks' 2023 global revenue of almost USD36 billion rose in part due to an uptick in new stores, over the longer term, annual revenue growth more closely follows same-store sales growth rates, as shown in Exhibit 8. For example, between 2015 and 2018, Starbucks' stock price languished while same-store sales growth rates declined below 5%.

Exhibit 8: SBUX Growth in Revenues, Stores, Same-Store Sales, and Stock Price, 2010–2023



Source: FactSet.

Top-Down Revenue Model

In other cases, company revenue forecasts may be determined using a firm's market share in an industry. The first step in this process is to project overall market size. For a mature industry in a stable market, this often involves estimating future demand for products or services within the context of a macroeconomic forecast. This forecast may be broken down further into unit sales and expected price per unit. Company revenue estimates may be determined based on the company's market share per segment and the changes in units sold and price per unit, as shown in the following example.

KNOWLEDGE CHECK: ESTIMATING TOYOTA MOTOR COMPANY'S REVENUE GROWTH



An analyst considers a top-down forecast of Toyota Motor Company's FY2025 revenue using 2024 as the base year. The analyst first compiles information from the company's audited financial statements and footnotes as well as a financial data provider, making assumptions due to timing differences and data inconsistencies to arrive at the inputs shown in Exhibit 9:

Exhibit 9: Toyota—Selected Financial Information as of 31 March 2024

Region	Revenue (JPY billions)	Vehicle sales (thousands of units)	Revenue per vehicle (JPY thousands)	Market share estimates (% of total)
North America	17,694	2,816	6,284	14%
Asia	7,742	1,804	4,292	13%
Japan	7,399	1,993	3,713	52%
Other	6,863	1,638	4,190	8%
Europe	5,397	1,192	4,527	7%
Total	45,095	9,443	---	---

Sources: Toyota Motor Company 10-K; Bloomberg.

Based on a survey of 2025 economic forecasts and the global automotive industry outlook, the analyst establishes an expectation for 4% growth in vehicle sales in North America, Asia, and Japan and 3% growth in vehicle sales in Europe and elsewhere, with wholesale auto prices increasing by 5% in Japan and 4% year over year in other regions. Also, while Toyota is expected to maintain a stable market share in other regions, the analyst expects a 1% uptick in Toyota's North American market share given recent model upgrades and the renewed popularity of plug-in hybrid vehicles.

- Using the data in Exhibit 9 and the above assumptions, calculate a top-down 2025 revenue forecast for Toyota Motor Company.

Solution

As described above, a top-down revenue forecast consists of the following steps:

- Determine overall market size/growth for the forecast year or period
- Establish expected firm market share as a percentage of sales
- Solve for revenue as the firm's share of total expected market revenue

First, we calculate total units sold (in thousands) by multiplying prior year units sold by the growth rate if market share is unchanged. For North America, the 4% growth combined with a market share increase of 1% from 14% to 15% is equal to 11.14% growth, or $0.04 + [(15/14) - 1]$:

$$\text{Units sold} = 2,816(1.1114) + 1,804(1.04) + 1,993(1.04) + 1,638(1.03) + 1,192(1.03) = 9,994 \text{ units sold.}$$

Next, we calculate forecasted revenue per unit (in JPY thousands). For example, in the case of North America:

$$\text{Revenue per unit}_{\text{North America}} = \text{JPY}6,535 = \text{JPY}6,284(1.04).$$

We then calculate forecast revenue by finding the sum of the products of the respective units per region and the revenue per unit:

$$\text{Total revenue} = \sum_{i=1}^n \text{Units sold}_i \times \text{Revenue per unit}_i.$$

These calculations are summarized in the following table:

Region	2024 Units Sold (000s)	% Unit Sales Growth	2025 Units Sold (000s)	2024 Revenue per Unit (000s)	% Unit Price Increase	2025 Revenue per Unit (000s)	2025 Total Revenue
North America	2,816	11.14%	3,130	6,284	4.00%	6,535	20,453
Asia	1,804	4.00%	1,876	4,292	4.00%	4,463	8,374
Japan	1,993	4.00%	2,073	3,713	5.00%	3,898	8,080
Other	1,638	3.00%	1,687	4,190	4.00%	4,357	7,352
Europe	1,192	3.00%	1,228	4,527	4.00%	4,708	5,781
Total	9,443	---	9,994	---	---	---	50,040
Based on these calculations, the analyst forecasts 2025 Toyota Motor Company revenue to be JPY50,040 billion, a 10.97% increase over FY 2024.							

Expense or Profit Model Construction

Forecasted expenses and resulting profit margins are key drivers of firm profitability and value. While constant growth models assume expenses grow at the same rate as revenue, disaggregated models allow for changes in gross and operating profit margins relative to the base year of a forecast. An analyst may incorporate a view of rising or falling margins over a forecast horizon based on a company's cost structure and how it may change as sales grow. As in the case of revenue, expense and profit margin forecasts may consider historical as well as other qualitative approaches. We consider the following:

1. Historical approach
2. Itemization of economically significant expenses approach
3. Direct profit margin assumption approach

Historical Expense Model

Analysts may estimate expenses in the same way as revenue by using past growth rates. However, this method ignores the link between revenues and expenses. An alternative method combining historical expense and revenue changes is regression analysis. For example, an analyst might regress a cost line-item change on revenue changes as shown below:

$$\text{Change in expense line item}_t = a + b \times (\text{Change in revenue})_t \quad (5)$$

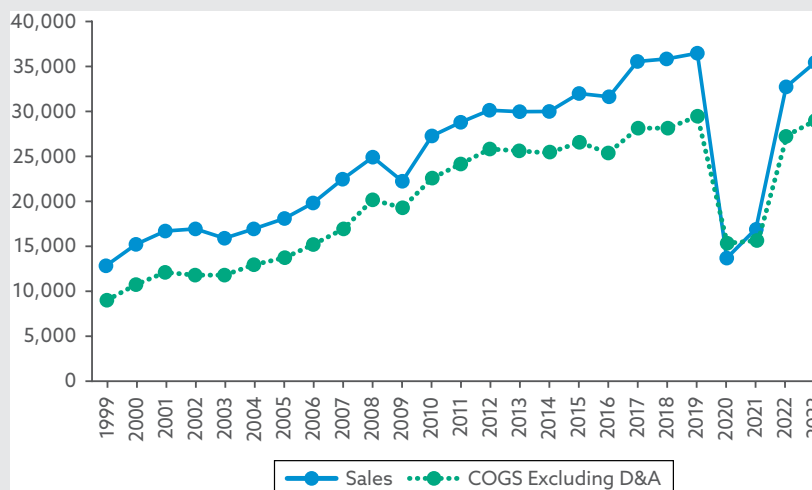
The intercept term, a , estimates the fixed cost component of the expense line item. The slope coefficient term, b , estimates how the expense line item varies with each unit of change in revenue. The regression should use either absolute changes or percentage changes, as the use of expense and revenue *values* creates the potential for incorrect statistical inferences about the relationship between the two variables. The next example illustrates this regression methodology.

EXAMPLE 5

Deutsche Lufthansa AG's Cost Structure

Exhibit 10 illustrates the revenues and COGS excluding D&A for the German airline company, Deutsche Lufthansa.

Exhibit 10: Deutsche Lufthansa AG Revenues and COGS, 1999–2023
(in EUR millions)



Source: FactSet.

Exhibit 10 shows that COGS has grown from approximately 70% of Lufthansa's revenue in the late 1990s to early 2000s to over 80% since 2008. COGS grew at a 24-year CAGR of 5.0% between 1999 and 2023, while sales grew 4.3%.

An analyst seeks to estimate the fixed cost component of Lufthansa's COGS and better understand its relationship to revenue. To remove the effects of the COVID-19 pandemic, the analyst excludes observations after 2019. The data through 2019 show higher compound annual growth of both COGS and sales at 5.4% and 6.1%, respectively. Regression results of the annual percentage change in COGS (y -variable) on the percentage change in sales (x -variable) are shown in the following table:

	Coefficients	Standard Error	t-Statistic	p-Value
Intercept	0.0159	0.0088	1.7944	0.0896
% Δ Sales	0.8468	0.0925	9.1549	0.0000

The analyst's interpretation of the regression coefficient is that Lufthansa's COGS changed by approximately 0.85% per 1.00% change in sales. The intercept term implies that an additional 1.59% of COGS growth reflects fixed costs unrelated to sales growth.

To apply these results, an analyst creating a COGS forecast for the next year would use an assumed sales forecast growth rate. For example, if Lufthansa is expected to grow revenues by 4% over the base year, then the forecasted COGS growth rate is estimated as 4.98%, as shown below:

$$\text{Expected COGS growth rate} = 0.0159 + 0.8468 \times 4\% = 4.98\%.$$

Regression methodology may be useful for forecasting expense growth using revenue growth projections for firms with sufficient historical financial data. However, when using any historical data, one must be confident that the firm's future cost structure remains similar to that of the historical period in question. As a result, a regression-based approach is most appropriate for mature firms and less applicable to early-stage or highly cyclical companies.

Itemization of Expenses

One weakness of reliance on downloaded standardized financial statements to build a disaggregated model is that for certain types of firms, significant expenses may not be explicitly shown. As a result, analysts may need to contrast a company's financial statements as summarized by a financial data provider with the detail shown in the annual report. When economically significant but non-standardized expenses should be modeled with different growth dynamics, analysts may prefer to incorporate more industry or company-specific line items, as illustrated in the example below.

EXAMPLE 6

Deutsche Lufthansa Income Statements: Standardized versus As-Reported

An analyst downloads the following standardized operating income portion of Deutsche Lufthansa's income statement for 2023 (see Exhibit 11) from a major data provider.

Exhibit 11: Deutsche Lufthansa Standardized Format Income Statement (Operating Income only), 2023 (in EUR millions)

<i>GAAP/IFRS Income Statement</i>	2023
Sales	35,442
COGS excluding D&A	31,158
D&A expense	2,228
Depreciation	2,140
Amortization of intangibles	88
Gross income	4,284
SG&A expense	3,062
R&D	56
Other SG&A	3,006
Other operating expense	(295)
EBIT (operating income)	1,517

Source: FactSet.

The analyst finds that the standardized format does not provide line-item disclosures for the two most economically significant expenses faced by airlines: labor and jet fuel. Since jet fuel costs are unlikely to be affected by revenue growth and labor costs are likely more closely aligned with revenue changes, the analyst seeks to model these expenses separately. The analyst then uses the same data provider to download Deutsche Lufthansa's "as-reported" income statement, as shown in Exhibit 12.

Exhibit 12: Deutsche Lufthansa As-Reported Income Statement (Operating Income only), 2023 (in EUR millions)

<i>GAAP/IFRS Income Statement</i>	2023
Total revenue	35,442
Traffic revenue	29,926
Other revenue	5,516

GAAP/IFRS Income Statement	2023
Other operating income	3,140
Changes in inventories and work performed by entity and capitalized	727
Other operating income excluding changes in inventories and work performed by entity and capitalized	2,413
Total operating income	38,582
Operating expenses	(36,126)
Cost of materials and services	(20,378)
Staff costs	(8,344)
Depreciation, amortization, and impairment	(2,242)
Other operating expenses	(5,162)
Profit/loss from operating activities	2,456

Through this exercise, the analyst observes staff costs of EUR8,344 million but is disappointed to find no separate line item for Lufthansa's jet fuel expenses in these financials. The analyst then accesses the company's financial statements directly from the company's investor relations portal. The analyst finds jet fuel costs of EUR7,931 million as part of the company's materials and services of EUR20,378 disclosed only in the Financial Performance section of Lufthansa's Annual Report.

Direct Profit Margin Assumptions

As an alternative to directly assuming disaggregated expense line items within a model, an analyst may directly assume profit margin changes. EBIT margin is an important driver in both FCFF and FCFE valuations, so analysts may choose to create specific forecasts of this key variable. For some industries heavily reliant on direct costs of goods sold, such as the earlier Deutsche Lufthansa example, gross profit margin changes may be assumed directly as part of an analyst's narrative. Nevertheless, direct margin assumptions imply some indirect assumptions about underlying expenses and their implied growth rates, as demonstrated in the following knowledge check.

KNOWLEDGE CHECK: JELENICO S.A.'S PROFIT MARGIN



An analyst is generating a three-year gross profit forecast for Jelenico S.A., a Brazilian construction materials supplier. During its base year (year t), Jelenico reports revenue of BRL6,500 million, COGS of BRL2,600 million, and gross profit of BRL3,900 million for a gross margin in year t of 60% ($= 3,900/6,500$). The analyst's assumed revenue growth and gross margin assumptions are shown below:

Year	$t+1$	$t+2$	$t+3$
Revenue growth rate	12%	11%	10%
Gross margin assumption	61.5%	63.0%	64.5%

1. Calculate the analyst's assumed COGS growth rates implied by the revenue growth and gross margin assumptions.

Solution

The analyst's implied assumption of annual COGS growth rates during the three-year forecast horizon is 7.80%, 6.68%, and 5.54%, respectively. The solution process is shown for year $t+1$ as follows.

- Calculate year $t+1$ revenue forecast:

$$\text{Revenue} = 7,280 \text{ million} = 6,500 \text{ million} \times (1.12)$$

- Calculate year $t+1$ gross profit forecast:

$$\text{Gross profit} = 4,477.2 \text{ million} = 0.615 \times 7,280 \text{ million}$$

- Calculate year $t+1$ COGS:

$$\text{COGS} = 2,802.8 \text{ million} = 7,280 \text{ million} - 4,477.2 \text{ million}$$

Alternatively, the analyst could skip the gross profit step and calculate COGS as follows:

$$\text{COGS} = (1 - 0.615) \times 7,280 \text{ million} = 2,802.8 \text{ million}$$

The process is repeated for years $t+2$ and $t+3$ with complete results below:

Year	$t+1$	$t+2$	$t+3$
Revenues	7,280.000	8,080.800	8,888.880
COGS	2,802.800	2,989.896	3,155.552
Gross profit	4,477.200	5,090.904	5,733.328
Gross profit growth	14.80%	13.71%	12.62%
COGS growth rate	7.80%	6.68%	5.54%

The exercise above illustrates a hypothetical example in which economically meaningful increases in gross margin require expense growth rates to be significantly below revenue growth rates. In this example, COGS represents a significant expense line item. Significant differences between revenue and expense growth rates have a lesser effect on profit margins when expense items are smaller relative to revenue.

An analyst's profit margin forecast into the future is a particularly important aspect of the valuation narrative. Rising profit margin forecasts may reflect optimism about a company's ability to improve operating efficiencies, its ability to downsize or shed less profitable activities, and/or grow those with higher returns. This involves an assumption that the company's revenue growth can exceed expense growth, on average, over the forecasting horizon. Such a valuation narrative should include detailed company and industry analysis that substantiates this optimism. The opposite holds true for pessimistic forecasts with falling profit margins.

While analysts may forecast gross and/or operating profits directly by using forecasted margins, *net* margin forecasts incorporate assumptions that are outside of company operations. Non-operating expenses include asset write-downs, one-off restructuring or legal costs, interest for firms that carry interest-bearing debt, and income taxes. Additionally, firms may generate significant interest income from holdings of cash and marketable securities. Because these non-operating items are unlikely to be related to revenue changes, analyst assumptions may cause net margins to improve or decline due to factors unrelated to company fundamentals.

Whether determined through regression analysis, expense itemization, or in aggregate, margin assumptions are a critical driver of valuation models. Forecasted margin improvement or deterioration should reflect a solid understanding of company and industry dynamics, a topic discussed in detail in later Equities learning modules.

Asset Model Construction

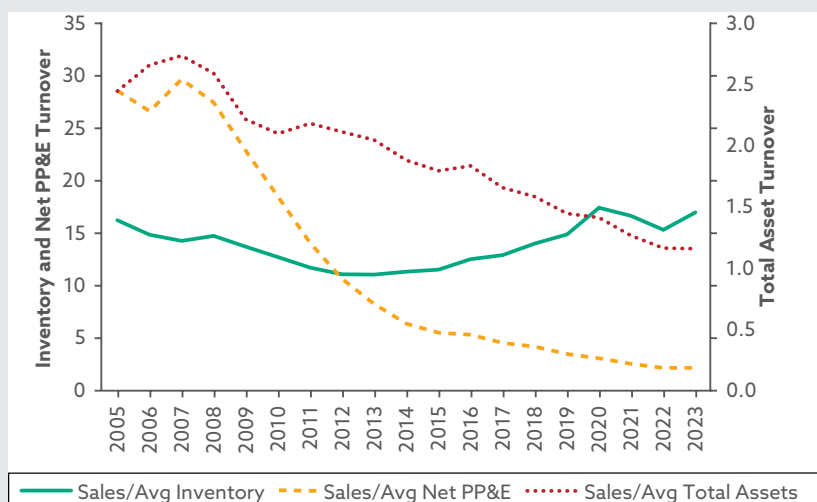
Disaggregated models allow for the use of different asset levels relative to revenue necessary to support a company's current operations as well as future revenue growth. A firm's current and expected future business model is an important determinant of the need for working capital and fixed assets. While some firms can generate revenue with a very small, efficient asset base with less financing requirements, in other cases, an asset-light business model may be challenging to maintain as a company grows.

EXAMPLE 7

Amazon.com's Evolution from an Asset-Light Business Model

Asset-light business models are a common feature of disruptive start-up companies founded in the past three decades. Companies such as Uber Technologies Inc. (passenger transportation) and Airbnb, Inc. (lodging) operate in industries typically characterized by significant asset requirements. These two companies chose business models that focus on operating technology platforms for use by auto and rental property owners. Amazon.com was an asset-light business model pioneer with its e-commerce retail platform founded in the 1990s. Operating almost exclusively online, Amazon.com's market capitalization achieved parity with that of Walmart, a retail chain mostly focused on its physical stores, by 2016 while making just a quarter of the investments made historically by Walmart.

By the 2020s, Amazon.com was harder to classify as an asset-light business. At its year-end 2023, the company's total asset turnover had declined to 1.16 from its all-time high of 2.74 in 2007. More notably, Amazon.com became increasingly reliant on long-term investments. In 2007, the company's fixed asset turnover was 29.67. By 2023, this ratio had declined to 2.17; moreover, net property, plant, and equipment (PP&E) had become Amazon.com's largest asset category as the company increasingly invested in fixed technology assets to support its Amazon Web Services segment (see Exhibit 13).

Exhibit 13: Amazon.com Asset Turnover, 2005 to 2023

Source: Bloomberg.

Asset models provide the foundation for estimating investing cash outflows as part of the free cash flow calculation. As in the case of profit margins, asset model forecasts may either assume a company's assets grow at similar rates to revenue or rely on an itemized forecast.

Historical Asset Models

Regression analysis can be a useful tool in forecasting asset growth. Similar to Equation 1 showing a relation between expense growth and revenue growth, we can formulate the following equation for the change in assets relative to revenues:

$$\% \text{ Change in asset line item}_t = a + b \times \% \text{ Change in revenue}_t \quad (6)$$

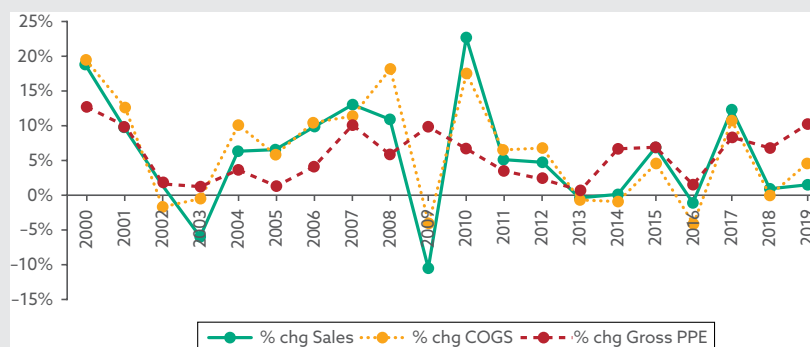
Assuming the same growth rate for revenues and assets implies that a equals zero and b equals one in Equation 5 as part of the company forecasting process. Over a finite time horizon, this assumption implies that the company's asset turnover ratios remain constant. However, some firms may need to grow assets more rapidly to achieve their growth and profitability plans, while others may be able to achieve their goals while investing relatively less. One simple test to assess the degree to which a company has historically grown its assets is to simply compare the compound annual growth rates of key asset metrics compared to revenues. A company with historical asset growth greater than historical revenue growth may be somewhat more dependent on additional investment compared to a similar company whose growth rates show opposite historic patterns.

An analyst seeking to estimate Equation 5 may find the implementation and interpretation more challenging than the expenses versus revenue regression example shown earlier. Annual asset growth is less likely to be as closely related to revenue as are expenses, and time coefficients may be less likely to match, as additional revenues are earned once asset purchases have taken place. We illustrate this idea in the following example.

EXAMPLE 8**Deutsche Lufthansa Changes in Assets (2000–2019)**

Airline companies, such as Deutsche Lufthansa, typically exhibit significant cyclical in revenues and expenses over the business cycle. Exhibit 14 shows annual percentage changes in Deutsche Lufthansa's revenues, COGS (excluding depreciation), and gross PP&E. Between 2000 and 2019, Lufthansa experienced volatile annual revenue and COGS growth rates. The primary assets underlying Lufthansa's income statement are its aircraft and other related tangible assets. Annual percentage changes shown in Exhibit 14 provide insight into Lufthansa's historical investment spending, suggesting that the company's gross PP&E growth is relatively stable compared to its revenue growth.

Exhibit 14: Deutsche Lufthansa Changes in Sales, COGS, and Gross PP&E, 2000–2019



Source: FactSet.

Using the growth rates from Exhibit 14 in Equation 5 produces the following regression output, with percentage change in assets as the dependent variable.

	Coefficients	Standard Error	t-Statistic	p-Value
Intercept	0.0478	0.0097	4.9189	0.0001
% Δ Sales	0.1716	0.1017	1.6884	0.1086

The relationship between the annual growth rates of sales on annual growth rates of PP&E is positive but appears to be statistically insignificant. The coefficients suggest that most of annual gross PP&E growth is related to factors unrelated to annual revenue growth. Nevertheless, application of the regression results implies the following growth rates of gross PP&E associated with differing assumed levels of revenue growth. These results demonstrate the basic notion that Deutsche Lufthansa is likely to pursue a more stable fixed asset investment policy as compared to the volatility in its operating business.

Assumed Sales Growth	Implied Gross PP&E Growth
-5.0%	3.9%
-1.0%	4.6%
3.0%	5.3%
7.0%	6.0%
11.0%	6.7%

The prior example illustrates a situation in which a company's asset spending exhibits greater stability than its income, which can easily be incorporated into a disaggregated forecast model. Alternatively, analysts may sometimes model asset growth for companies in which historical asset growth exhibits more volatility relative to revenue growth. The regression methodology demonstrated above may be useful in providing an analyst with a more smoothed approach to estimating asset growth in a forecast model.

Itemized Asset Model Forecasts

Analysts use balance sheet projections as a mechanism to estimate forecasted investment spending by assuming that balance sheet account changes reflect uses (or sources) of cash and, therefore, decrease or increase forecasted free cash flow estimates. Each company's balance sheet reflects a unique set of assets; analysts should therefore view each asset (and liability) for a company from the following perspectives:

- Does the line item arise as part of the company's operating process?
- Is the item an asset (or liability) that involves cash payments or receipts?

If the answers to both above questions are "yes" for a particular line item, then it should likely be projected to grow over the forecast horizon. In contrast, non-operating assets (such as marketable securities) should typically *not* be forecast to grow in a disaggregated model except to capture changes in cash, an issue addressed later in the context of the financing model.

Only those assets (and liabilities) reasonably assumed to reflect cash payments or receipts should be forecasted to grow in an asset model. Historical changes in assets or liabilities that reflect market value changes should not be modeled unless there is a cash change component embedded in the line item. One example of an asset line item that sometimes reflects market value changes, not cash changes, is the goodwill line item when a company takes an impairment write-down, as described in an earlier learning module.

Financing Model Construction

The final aspect of an analyst's modeling process is to integrate a company's funding needs. Financing requirements are required to accurately estimate FCFE. Recall the two different (although equivalent) forms of expressing forecasted FCFE from an earlier learning module:

$$\text{FCFE} = \text{Net income} - (\Delta \text{Gross LT assets} - \text{Depreciation})$$

$$- \Delta \text{Working capital} + \Delta \text{Debt}. \quad (7)$$

$$\text{FCFE} = \Delta \text{Cash} + \text{Dividends} + \text{Stock repurchases} - \text{Stock issuance}. \quad (8)$$

As may be observed from Equation 6 and Equation 7, a company's expected changes in debt, cash, and equities (as reflected by dividend payments, stock repurchases, and stock issuances) are all components of a company's expected FCFE and must be incorporated into a disaggregated valuation model based on this measure.

Estimating Financing Needs

A simple approach to estimating a company's financing needs for a future forecasted year involves the following process:

1. Forecast a company's operating assets to obtain the expected change in a given year.

2. Forecast a company's non-debt liabilities to obtain the expected change each year.
3. Forecast the change in a company's retained earnings by deducting forecasted dividends from forecasted net income.

The first step in the process involves new asset expenditures, while the second and third components capture any internal financing sources available to pay for these new assets, with the remainder sourced externally. Equation 8 provides an expression for this approach to estimating financing needs:

Financing Needs

$$= \Delta \text{Operating Assets} - \Delta \text{Non Debt Liabilities} - \Delta \text{Retained Earnings.} \quad (9)$$

While a positive financing need must be covered by external debt or equity, a negative result implies that the company generates surplus cash. This situation is illustrated in the following knowledge check.

KNOWLEDGE CHECK: PT INDORINJANI'S FINANCING NEED

An analyst has created an income statement and balance sheet forecast for PT Indorinjani, an Indonesian manufacturer. The following table shows a simplified version of the company's balance sheet (in IDR thousands).

	Year t	Year $t+1$
Non-cash assets	122,000	130,540
Non-debt liabilities	133,000	136,010

The analyst forecasts that PT Indorinjani's net income (in IDR thousands) for the next year ($t+1$) is 6,440 and that the company does not pay dividends or repurchase stock.

1. Which one of the following best describes the analyst's estimate of PT Indorinjani's financing needs for year $t+1$?
 - A. The company needs additional funds to finance its growth.
 - B. The company generates more than enough cash to finance its growth.
 - C. The company generates exactly enough cash to finance its growth.

Solution

B is correct. Choice B is correct because PT Indorinjani's financing needs are negative, as shown using Equation 8 below:

$$\begin{aligned} \text{Financing needs} &= (130,540 - 122,000) - (136,010 - 133,000) - (6,440 - 0) \\ &= -910. \end{aligned}$$

Choice A is incorrect because financing needs must be a positive number to imply that the company needs additional funds to finance its growth.

Choice C is incorrect because this statement implies that financing needs equal exactly zero.

2. The analyst further forecasts that PT Indorinjani adds no debt or new stock to its balance sheet between years t and $t+1$ and that the company adds the

surplus IDR910,000 to its cash balance during year $t+1$. Calculate the analyst's forecast of PT Indorinjani's FCFE for year $t+1$.

Solution

The analyst forecasts PT Indorinjani's FCFE to be IDR910,000 in year $t+1$. This result may be obtained using Equation 7, as shown below:

$$\begin{aligned}\text{FCFE} &= \Delta \text{Cash} + \text{Dividends} + \text{Stock repurchases} - \text{Stock issuance} \\ &= 910,000 + 0 + 0 - 0.\end{aligned}$$

This negative financing need of IDR910,000 becomes a positive addition to the company's cash balance. This amount reflects 100% of the company's forecasted FCFE.

A variation of Equation 6 could also be used to compute this result by recognizing that the investment terms related to long-term assets, depreciation, and working capital should be equivalent to the change in operating assets minus the change in non-debt liabilities. This alternative approach is shown below:

$$\text{FCFE} = 6,440 - [(130,540 - 122,000) - (136,010 - 133,000)] = 910.$$

3. The analyst considers the possibility of assuming that PT Indorinjani uses the IDR910,000 cash surplus to decrease its debt balance. Calculate the analyst's revised estimate of the company's FCFE.

Solution

The revised FCFE forecast is equal to exactly zero. Using Equation 7, all input terms are zero. Alternatively, the Equation 6 framework equals the following:

$$\text{FCFE} = 6,440 - (130,540 - 122,000) + (136,010 - 133,000) - 910 = 0.$$

The final term on the right-hand side of the expression reflects a decline in debt of 910 during year $t+1$.

In the prior example, we assume that the analyst allocates 100% of the financing need/surplus to a single line item such as debt or cash. In the case of the cash surplus allocation, the financing need equals the forecasted FCFE. When the financing surplus is allocated to debt reduction, the expected FCFE falls to zero. These examples provide simple illustrations of how assumed corporate financing choices affect forecasted FCFE.

In many forecasting situations, an analyst may choose to allocate portions of the total financing need amount across debt, cash, and/or equity sources. Exhibit 15 shows a summary of the potential sources and uses associated with these different accounts.

Exhibit 15: Financing Needs and Sources of Financing

	Financing Needed > 0 (Additional Capital Needed)	Financing Needed < 0 (Excess Cash Available)
Cash	May be assumed as a source of capital Analyst should check to ensure that resulting cash balance does not fall below an acceptable level, which varies among companies	May be assumed to be added to cash balance
Debt	May be assumed as a source of capital Analyst should check resulting debt ratio to avoid overly high debt ratio assumption Assumption may have implications for interest expenses	May be assumed to be deducted from debt balance Analyst should check resulting debt ratio to ensure debt ratio falling below appropriate level
Equity	May be assumed as a source of capital Analyst may want to avoid using equity unless company has track record of issuing additional equity	May be assumed to be deducted from equity balance Funds are likely assumed as stock repurchases although may be assumed as additional dividends

A positive financing need implies a company must raise funding from cash, debt, and/or equity. As firms prefer to raise equity less frequently due to cost, potential dilution, and market timing issues, publicly listed firms are likely to use cash balances and/or debt as funding sources rather than equity. However, an analyst must take care to avoid making assumptions that will result in very low cash balances or overly high debt ratios. Negative financing needs may result in a return of capital to debtholders and/or equity holders. Equity capital is likely to be returned by repurchasing stock, although it may also take the form of higher dividends.

Corporate Debt Ratios, Dividend Policies, and Financing Needs

On the one hand, the prior example of forecasting financing needs ignores the effects of corporate target debt ratios and dividend policies. The effect of dividends on financing needs is quite straightforward. Equation 5 shown earlier included a term for change in retained earnings, and this term reflects internal equity available to pay for assets. The change in retained earnings is reduced if dividends are paid, implying a negative relationship between a firm's expected future dividends and its forecasted financing needs.

On the other hand, modeling target debt ratios can create nuanced challenges for an analyst. For example, a higher debt ratio may adversely impact a company's credit rating and increase its cost of debt, as addressed in detail in later learning modules on Fixed Income. For now, we present a simplified target debt ratio example, including a forecasted dividend based on our prior example.

KNOWLEDGE CHECK: PT INDORINJANI FINANCING NEED REVISITED

Recall the earlier example of PT Indorinjani. The following table shows a simplified version of the company's balance sheet (in IDR thousands).

Year	T	$t+1$
Change in non-cash assets	122,000	130,540
Change in non-debt liabilities	133,000	136,010

The analyst forecasts that PT Indorinjani's net income (in thousands) for the next year ($t+1$) is 6,440 and that the company's expected dividend payment to shareholders during year $t+1$ is 2,000. Additionally, the analyst expects Company A to borrow 40% of its forecasted asset investment, net of non-debt liabilities.

1. Explain the separate effects of PT Indorinjani's expected dividend and borrowing policies on its forecasted financing needs.

Solution

On the one hand, the dividend of 2,000 reduces PT Indorinjani's change in retained earnings; so, the analyst's dividend policy assumption increases forecasted financing needs by the amount of the dividend. On the other hand, the analyst's assumption that PT Indorinjani borrows 40% of its net investment spending provides additional assumed financing.

$$\begin{aligned}\text{Planned borrowings} &= 40\% \times [(130,540 - 122,000) - (136,010 - 133,000)] \\ &= 2,212.\end{aligned}$$

The analyst forecast includes planned borrowing for PT Indorinjani in the amount of 2,212.

2. In the earlier example when the analyst assumed zero dividends and no additional borrowings, Indorinjani's forecasted financing need for year $t+1$ was calculated as -910. Calculate the company's financing needs if it pays 2,000 in dividends and borrows 2,212.

Solution

Following from the result of Question 1, the analyst's forecast of PT Indorinjani's financing needs decreases by 212 because the dividends of 2,000 increase financing needs, while the borrowings of 2,212 decrease financing needs. Thus, PT Indorinjani's financing needs decline from -910 to -1,122 ($= -910 + [2,000 - 2,212]$).

3. Explain the effect of the analyst's inclusion of a forecasted dividend and planned borrowings for PT Indorinjani on its FCFE forecast in year $t+1$. Assume that PT Indorinjani's forecasted FCFE was 910 without these inclusions and that the analyst continues to assume that the financing surplus is entirely allocated to cash.

Solution

The company's FCFE increases by the amount of the planned borrowings, as may be seen by observing Equation 3 (shown below):

$$\text{FCFE} = \text{Net income} - (\Delta \text{LT assets} - \text{Depreciation}) - \Delta \text{Working capital} + \Delta \text{Debt}$$

The change in debt term is the only one changed from the application of this equation from the earlier knowledge check. Thus, the analyst forecast of FCFE should increase by 2,212 (i.e., the planned borrowing amount) to 3,122 ($= 910 + 2,212$).

Alternatively, the FCFE may also be considered using Equation 7 (shown below):

$FCFE = \Delta \text{Cash} + \text{Dividends} + \text{Stock repurchases} - \text{Stock issuance}.$

The change in cash term is 1,122, as shown in the solution to the prior question, and dividends are 2,000, as assumed.

4. Explain how the analyst's calculations and interpretations of PT Indorinjani's forecasted financing needs and FCFE change if the analyst updates the year $t+1$ non-operating asset amount to 140,540 (i.e., an increase of 10,000).

Solution

Additional asset requirements for year $t+1$ imply that PT Indorinjani's forecasted financing needs increase by 10,000 relative to prior scenarios.

The change in forecasted FCFE is dependent on how the analyst assumes that Indorinjani raises the extra 10,000. The analyst forecast of FCFE may become negative if cash is used or if equity is raised. However, if the analyst assumes that Indorinjani raises the 10,000 with new debt, then the forecast FCFE is more likely to be similar relative to the original scenarios.

The exercises above highlight the linkage between model assumptions, forecasted financing needs, and forecasted FCFE. Although, because forecasted FCFE depends only on profitability and investment, forecasted FCFE is less sensitive to forecasted financing needs. However, forecasted FCFE may change slightly with differing modeling assumptions of the source of financing needs because of financing costs, which are ignored in the above exercises.

Financial Statement Model Uses in Equity Valuation

Audited financial statements and related industry and company information are important inputs in assessing a company's past performance and projecting future cash flows for valuation purposes.

Recall from an earlier learning module that both investors and issuers apply valuation models to develop perceived values as a guide to a firm's intrinsic value. Investors use this information to make decisions about buying or selling shares, which can also inform issuer decisions, such as new share issuance or the repurchase of outstanding stock.

Disaggregated valuation models with similar or greater detail versus a firm's financial statements can be a powerful tool for estimating intrinsic value, changing an original valuation estimate as new information becomes available, or conducting what-if analysis of potential value changes under different scenarios. Use of financial statement components rather than constant or varying aggregate cash flow growth rates facilitates a direct comparison between changes in observable variables and share prices as demonstrated in the following examples.

Forward-looking revenue models based on top-down or bottom-up approaches described earlier can gauge the near-term share price impact of changes in sales reflected in quarterly financial statements when other factors such as expenses, assets, or financing remain relatively unchanged.

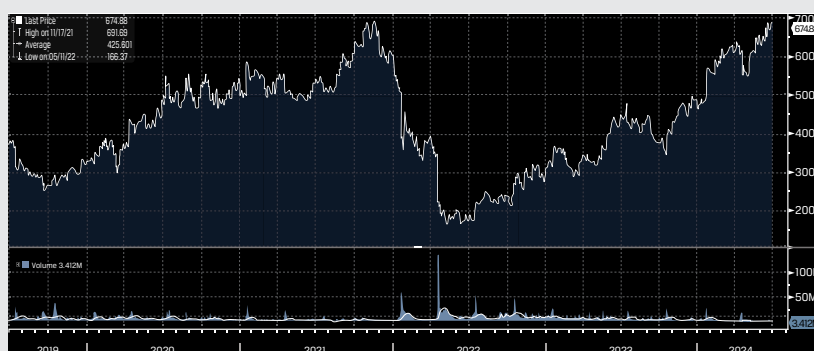
Subscription-based business models in which firms collect a monthly or annual fee from customers using their services represent a different set of parameters that may also be captured in a disaggregated model. As subscribers are a source of recurring periodic revenue, an analyst's focus shifts to forecasting subscriber acquisition and retention, as well as the average revenue per user to project overall revenue. As the following example suggests, subscriber dynamics can play an important role in driving share price changes.

EXAMPLE 9**Netflix's Subscriber Base and Share Price**

Netflix, Inc., is a US-based provider of video streaming services of films and television programming, which it either acquires or produces, that serves over 270 million subscribers in 190 jurisdictions.

The economic shutdowns during the COVID-19 pandemic in 2020 fueled Netflix's rapid growth. It reached over 200 million subscribers by year end, more than doubling its subscriber base from the end of 2016.

Exhibit 16: Netflix, Inc., Share Price (USD), 2019 to 2024



Source: Bloomberg.

As shown in Exhibit 16, Netflix shares reached a peak closing price of nearly USD692 per share in late 2021. This price plunged to under USD200 the following year as the firm's revenue generation potential eroded due to a reset of investor expectations following the COVID-19 pandemic, declining subscriptions, and widespread password sharing on its platform. In response, Netflix cracked down on account sharing and changed subscription pricing policies, introducing a lower cost membership option with advertising. Tighter controls, tiered pricing, as well as expanded depth and variety of content contributed to a rebound in revenue and operating profit growth, with the company's share price approaching the 2021 historical high as of mid-2024.

Unlike other valuation approaches, a disaggregated financial projection model can capture the effects of more complex cost structures. Rather than simply assuming distinct aggregate profit margin changes as described earlier, this technique allows analysts to factor in the effect of fixed versus variable costs under different market scenarios. For example, an industrial firm with high fixed costs may enjoy a sharp rise in profit margins if growth is strong during a cyclical growth phase, while facing losses in a cyclical downturn due to an inability to adjust near-term expenses. We illustrate this phenomenon by extending an earlier knowledge check.

KNOWLEDGE CHECK: JELENICO S.A.'S PROFIT MARGIN UNDER DIFFERENT REVENUE GROWTH SCENARIOS



Recall from an earlier knowledge check that an analyst generated a three-year gross profit forecast for Jelenico S.A., a Brazilian construction materials supplier. In its base year (year t), Jelenico had revenue of BRL6,500 million, COGS of BRL2,600 million, and gross profit of BRL3,900 million for a gross margin in year t of 60% ($= 3,900/6,500$).

While earlier analysis used a single scenario with falling revenue growth and rising gross margins, the analyst refines assumptions to incorporate the cyclical nature of Brazil's construction industry and details of Jelenico's cost structure. The analyst observes that half of the firm's year t COGS are fixed, and the other half are variable and depend on revenue. The analyst assumes three revenue growth scenarios over the next three years:

Scenario	$t+1$	$t+2$	$t+3$
Growth	18%	16.5%	15%
Base	12%	11%	10%
Downturn	-6.0%	0.0%	2.0%

1. Calculate the revised profit margin forecast under the base case scenario if Jelenico has the same cost structure as described above in year t .

Solution

From the discussion above, Jelenico's COGS are 40% of sales and 50% of Jelenico's COGS are fixed. Thus, the variable portion of Jelenico's COGS reflects 20% ($= 40\% \times 50\%$) of sales. We may show year t COGS as a function of sales as follows:

$$\text{COGS} = \text{Fixed costs} + \text{Variable costs}$$

$$\text{BRL}2,600 \text{ million} = \text{BRL}1,300 \text{ million} + (0.2 \times \text{Sales}_t)$$

As a first step, we solve for Jelenico's base case revenue through year $t + 3$:

$$\text{Sales}(\text{Base})_{t+1} = \text{BRL}7,280 \text{ million} = 6,500 \times (1.12)$$

$$\text{Sales}(\text{Base})_{t+2} = \text{BRL}8,080.8 \text{ million} = 6,500 \times (1.12) \times (1.11)$$

$$\text{Sales}(\text{Base})_{t+3} = \text{BRL}8,888.88 \text{ million} = 6,500 \times (1.12) \times (1.11) \times (1.10)$$

Next, we solve for Jelenico's COGS to year $t + 3$ as shown above:

$$\text{COGS}(\text{Base})_{t+1} = \text{BRL}2,756 \text{ million} = \text{BRL}1,300 \text{ million} + (0.2 \times 7,280)$$

$$\text{COGS}(\text{Base})_{t+2} = \text{BRL}2,916.16 \text{ million}$$

$$= \text{BRL}1,300 \text{ million} + (0.2 \times 8,080.8)$$

$$\text{COGS}(\text{Base})_{t+3} = \text{BRL}3,077.78 \text{ million}$$

$$= \text{BRL}1,300 \text{ million} + (0.2 \times 8,888.88)$$

Finally, we solve for expected gross margin through year $t + 3$:

$$\text{Gross margin} = 1 - \left(\frac{\text{COGS}}{\text{Revenue}} \right)$$

$$\text{Gross margin}(\text{Base})_{t+1} = 62.1\% = \left(1 - \frac{2,756}{7,280} \right).$$

$$\text{Gross margin}(\text{Base})_{t+2} = 63.9\% = \left(1 - \frac{2,916.16}{8,080.8} \right).$$

$$\text{Gross margin}(\text{Base})_{t+3} = 65.4\% = \left(1 - \frac{3,077.78}{8,888.88} \right).$$

2. Calculate and interpret the gross margin range across Jelenico's growth and downturn scenarios in year $t + 3$.

Solution

First, we solve for Jelenico's growth scenario gross margin in year $t + 3$:

$$\begin{aligned}\text{Sales (Growth)}_{t+3} &= \text{BRL}10,275.88 \text{ million} \\ &= 6,500 \times (1.18) \times (1.165) \times (1.15)\end{aligned}$$

$$\begin{aligned}\text{COGS (Growth)}_{t+3} &= \text{BRL}3,355.18 \text{ million} \\ &= \text{BRL}1,300 \text{ million} + (0.2 \times 10,275.88)\end{aligned}$$

$$\text{Gross margin (Growth)}_{t+3} = 67.3\% = \left(1 - \frac{3,355.18}{10,275.88}\right).$$

Next, solve for Jelenico's downturn scenario gross margin in year $t + 3$:

$$\text{Sales (Downturn)}_{t+3} = \text{BRL } 6,232.20 \text{ million} = 6,500 \times (0.94) \times (1) \times (1.02)$$

$$\begin{aligned}\text{COGS (Downturn)}_{t+3} &= \text{BRL}2,546.44 \text{ million} \\ &= \text{BRL}1,300 \text{ million} + (0.2 \times 6,232.20)\end{aligned}$$

$$\text{Gross margin (Downturn)}_{t+3} = 59.1\% = \left(1 - \frac{2,546.44}{6,232.20}\right).$$

These results show an 8.2% gross profit range ($= 67.3\% - 59.1\%$) across the optimistic and downside cases, which results from the distribution of fixed costs over higher or lower sales figures, respectively.

Detailed models projecting balance sheet positions, such as assets or financing, are often designed to support a growth forecast based on an existing business model. Growing a company's working capital and fixed assets in proportion to expected sales using external debt or equity preserves the internal consistency of a forecast of expanded operations.

In contrast to organic growth, another important use of such models is for firms entering a transformative phase that involves either acquiring another company, selling a division or significant firm assets, or undertaking other measures to improve performance in what is referred to as a **corporate restructuring**. While companies pursuing corporate restructuring are often publicly listed, in many other cases private investors take control of and restructure mature underperforming public companies. This is also referred to as a **leveraged buyout (LBO)** due to the significant use of incremental debt financing. These applications will be covered in greater detail later in the curriculum.

The more-detailed inputs that characterize financial statement models afford analysts a higher degree of flexibility and granularity than under constant or multistage growth assumptions. For example:

- *Revenue drivers.* Bottom-up forecasts by line of business and/or region as well as market share changes may be explicitly included.
- *Expense drivers.* The effect of specific cost drivers, including labor, fixed assets, and raw materials, may be taken into consideration.
- *Efficiency drivers.* This includes improvements or deterioration in working capital management, such as inventory or receivables.

Forecasts using key financial statement entries affecting revenue, expense, and efficiency drivers, among others, allow analysts to conduct sensitivity analysis—that is, to gauge the impact on estimated equity valuation of changes in one or more key variables while holding the others constant.

QUESTION SET 1



1. A disaggregated valuation model including more line items:

- A. allows for a richer valuation narrative.
- B. decreases potential for model error or noise.
- C. requires fewer updates as new information becomes available.

Solution

A is correct. Choice A is correct because a more complex valuation with more line items addresses more economic aspects of a firm's outlook. In other words, the analyst has built more possible levers into the valuation process.

Choice B is incorrect because the inclusion of more line items creates more model noise; the variability of more line items creates more possible points of error.

Choice C is incorrect because more line items imply more items to be updated as the future unfolds.

2. An analyst creates a revenue forecast of a retail store chain using increases in numbers of stores and same-store sales growth, which is an example of a:

- A. historical revenue model.
- B. bottom-up revenue model.
- C. top-down revenue model.

Solution

B is correct. Choice B is correct because a bottom-up revenue model forecasts firm revenue based on fundamental units of business activity, which for a retail store chain involves numbers of stores and growth in same-store sales.

Choice A is incorrect because a historical revenue model would base the forecast on the chain's past revenue growth.

Choice C is incorrect because top-down revenue models are based on market share as a fraction of total sales, not store numbers or same-store growth.

3. Which one of the following statements most correctly describes historical asset models?

- A. Historical approaches using regression are often more precise for forecasting assets compared to expenses.
- B. Historical approaches based on compound annual growth rates of assets compared to those of revenues are typically not useful.

- C. Historical approaches using regression to create asset models are complicated by timing differences in asset purchases and revenue.

Solution

C is correct. Choice C is correct because asset changes often do not occur in the same year as resulting revenue changes, making it less straightforward to model a regression of asset changes on revenue changes.

Choice A is incorrect because regressions of asset changes on revenue changes are typically less precise than those of expense changes on revenue changes.

Choice B is incorrect because comparisons of compound growth rates of assets versus revenues demonstrate the relative importance of investments in generating revenue. A company with historically higher asset growth rates must invest more than a similar revenue-growth company with lower historical asset growth rates.

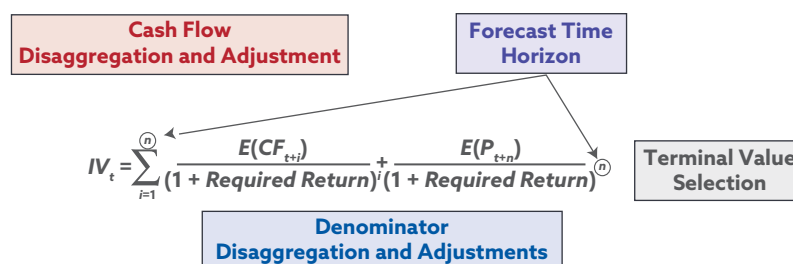
MODEL CONSTRUCTION BASED ON COMPANY CHARACTERISTICS

3

- ☐ evaluate the construction of an equity valuation model based on characteristics of the subject company

Earlier Equities learning modules underscored the importance of selecting an appropriate cash flow metric and discount rate based on a firm's features, which are applied over a forecast horizon over which the company's business model and corporate structure are assumed to remain relatively stable. Under a more flexible financial statement modeling approach, model criteria may be adjusted to reflect circumstances, such as a company's development stage, the availability and quality of financial data as a basis for projections, and the perspective of an analyst conducting the valuation. Key areas of valuation focus when company characteristics vary are shown in Exhibit 17.

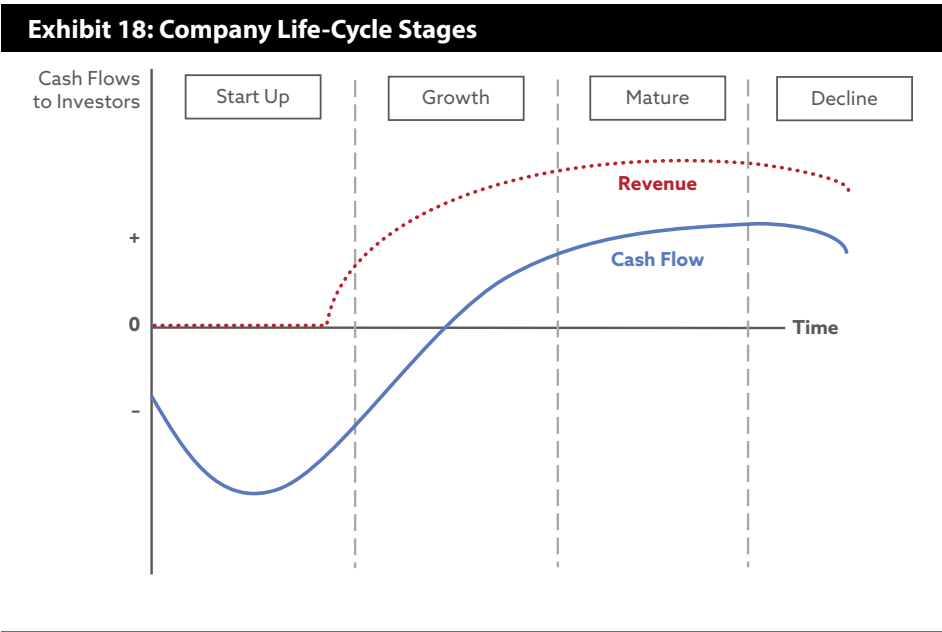
Exhibit 17: Areas of Valuation Focus Based on Company Characteristics



While the cash flow “numerator,” discount rate “denominator,” and terminal value building blocks shown in Exhibit 17 are familiar features of a valuation model, we now focus our attention on how their application changes when considering a company's life-cycle stage and the impact of shareholder control.

Company Life-Cycle Stage and Model Selection

In earlier Corporate Finance learning modules, we addressed how a company evolves from its establishment to a start-up period, to a phase of rapid expansion, with rising cash flow and profits that stabilize over time to a mature phase of its life cycle. These stages are shown in Exhibit 18.



While Exhibit 18 represents a generalized view, the progression and length of each company’s development phase can vary widely. Most start-ups are small private companies. Many will fail in their first few years, some will be acquired, and a small proportion will become publicly listed. From a valuation perspective, we distinguish between key metrics, value drivers, and data available in Exhibit 19.

Exhibit 19: Equity Valuation Metrics, Value Drivers, and Data Availability Across the Company Life Cycle

Phase	Start-Up	Growth	Mature	Decline
Metric	Sales	Cash flow to fund high growth	Earnings and cash flow to fund stable growth	Cash flow to fund return of capital to investors
Value driver	Market potential	Sustainable growth and margins	Sustainable profits	Long-run margins following decline
Historical financial data	Little to none	Limited	Available over longer periods	Available over longer periods

Except for the two-stage growth model, which allowed for an above-trend expansion phase, the valuation approaches introduced so far generally assume a firm is in a mature development phase, as shown in Exhibit 19. Valuing firms in other life-cycle phases require some adjustments, which we turn to next.

Model Construction in the Start-Up Phase

Start-up companies are often in an early product development phase with little to no revenue, negative cash flow, no operating history, and a high likelihood of failure. Valuation approaches based on existing financial statements are therefore of little use. Instead, these companies derive their value from a firm's future potential rather than revenue or earnings growth projections from base year results.

Valuation models for start-up firms typically forego a near-term cash flow forecast and are primarily focused on the firm's expected future terminal value. Combining and modifying Equations 1 and 2 from the prior lesson, we may express intrinsic value as follows:

$$IV_t = \frac{\text{Expected firm revenue} \times \left(\frac{\text{Price}}{\text{Sales}}\right)}{(1 + \text{Required rate of return})^n}, \quad (10)$$

Unlike the case of a mature firm, for which a terminal value may be derived using a constant cash flow growth rate beyond a forecast horizon, the numerator as shown in Equation 9 is often derived using a price-to-sales multiple combined with a firm's expected future revenue potential. The denominator is derived from the investor's expected rate of return over the expected investment horizon. To attract private equity investors to these high-risk ventures, companies must often offer returns on investment (ROIs) of more than 10 times the original investment over five years or more. As shown earlier in the curriculum, returns on investment with no cash flows occurring during the intermediate time periods may be expressed as an annualized internal rate of return (IRR) over an n -period time horizon as follows:

$$ROI = (1 + IRR)^n. \quad (11)$$

Combining Equation 10 and Equation 11, we can express IV_t as a function of ROI:

$$IV_t = \frac{\text{Expected firm revenue} \times \left(\frac{\text{Price}}{\text{Sales}}\right)}{ROI}. \quad (12)$$

Start-up valuations are usually conducted to attract these investors who offer small but in some cases controlling equity stakes known as **venture capital (VC)** to fund a firm's next stage of development.

KNOWLEDGE CHECK: RAJAH CART START-UP COMPANY VALUATION



RajahCart is an early-stage online auction and sales platform for small-scale capital equipment in India. It intends to offer sales, servicing, and financing opportunities to its industrial customers. Over the next six years, RajahCart is expected to reach INR8 billion in sales. A VC analyst determines that firms in similar businesses at a later development stage exhibit a price-to-sales ratio of 3 times. The analyst's firm has a target ROI of 15 times for similar investments over a six-year time horizon.

1. Calculate RajahCart's intrinsic value based on the analyst's assumptions and the VC firm's target return.

Solution

We may use Equation 11 to solve for intrinsic value. First, solve for the numerator by multiplying RajahCart's expected future sales of INR8 billion by the price-to-sales ratio of 3 times:

$$\text{INR}24,000 \text{ million} = \text{INR}8,000 \text{ million} \times 3$$

Substitute the numerator and denominator (ROI) into Equation 11 to

$$IV_t = \text{INR}1,600 \text{ million} = \frac{\text{INR}24,000 \text{ million}}{15}.$$

As illustrated above, an early-stage model is comprised of a firm's future market potential discounted at an expected rate of return, which compensates investors for the high likelihood of failure among start-ups. Using Equation 10, we can show the IRR on the RajahCart investment:

$$15 = (1 + \text{IRR})^6; \text{IRR} = 57.04\%.$$

A final point to consider is the perspective of the analyst valuing an early-stage firm. The investor is typically a private investor considering a new equity investment over the development phase. In contrast to a non-controlling investor, as is commonly assumed in equity valuation, here we identify the new investor's post-financing ownership stake by distinguishing between the company's value before and after a new financing. Equation 12 and Equation 13 demonstrate the relationship between firm value prior to the new investment (or **pre-money valuation**), the intrinsic value following the new equity investment (**post-money valuation**), and the new investor's ownership stake in the firm (or fractional ownership):

$$\text{Pre-money valuation} = \text{Post-money valuation} - \text{New equity} \quad (13)$$

$$\text{Fractional ownership} = \frac{\text{New equity}}{\text{Post-money valuation}} \quad (14)$$

We now revisit the prior example to explore these relationships.

KNOWLEDGE CHECK: RAJAH CART START-UP COMPANY VALUATION



In conjunction with the earlier post-money valuation of RajahCart at INR1,600 million, a private investor takes an INR500 million equity stake in a new financing transaction.

1. Calculate the new investor's fractional ownership and the pre-money valuation of RajahCart.

Solution

Using Equation 13, we solve for the investor's fractional ownership as 31.25%:

$$\text{Fractional ownership} = 31.25\% = \frac{\text{INR}500 \text{ million}}{\text{INR}1,600 \text{ million}}$$

We may solve for the pre-money valuation using Equation 11:

$$\text{Pre-money valuation} = \text{INR}1,100 \text{ million} = 1,600 \text{ million} - 500 \text{ million}.$$

Private equity investments in general and venture capital in particular will be addressed in greater detail later in the curriculum.

Model Construction During the Growth Phase

Unlike early-stage firms with uncertain prospects and a high failure rate, companies entering an expansion phase generate revenue—and in some cases, earnings—with a proven product and seek to rapidly capitalize on existing market opportunities. The growth strategy typically involves attracting new equity capital to increase assets and quickly grow sales to achieve sustainable profitability over a specific horizon. In contrast to start-up valuations with a singular focus on future sales potential, key equity

value drivers in the growth phase include detailed revenue and profit projections to achieve targeted financial results. Firms in this phase typically retain and reinvest earnings given the higher expected rate of return over the above-trend growth period.

As in the case of early-stage companies, the valuation over the growth phase aims to meet required investor returns by targeting a specific terminal valuation based on a multiple. A key difference is that growth companies focus on earnings-based multiples rather than those based on sales used for start-ups. Also, in contrast to start-ups, growth firm valuations take advantage of company financial statements as a basis for future projections. Despite a limited history, the use of differing growth rate assumptions for financial statement components from a base year can offer advantages over the simpler two-stage growth model. The following example illustrates the construction of a valuation model during the growth phase.

KNOWLEDGE CHECK: PREMVOL S.A. GROWTH COMPANY VALUATION



PremVol is a private French company that offers commercial drone services for geological and environmental applications. It was founded five years ago. To meet its ambitious five-year plan to more than quadruple revenue and achieve sustainable profitability, PremVol is considering an equity capital injection to fund an asset expansion to support its plan. PremVol's base year (Year 5) income statement is summarized below:

Position	Year 5 (EUR)
Sales	29,610,000
COGS	25,380,000
Gross profit	4,230,000
SG&A	4,230,000
Profit before tax	---
Net income	---

PremVol has set a target of 35% annual sales growth over five years, with COGS and SG&A both expected to rise at a rate of 25%. For computational convenience, depreciation expense has been included as part of COGS. PremVol's tax rate is assumed to be 30%.

1. Calculate PremVol's Year 10 sales and net income based on the above assumptions.

Solution

Solve for sales in Year 10 as follows:

$$\begin{aligned}\text{Sales(Year 10)} &= \text{Sales (Year 5)} \times (1 + \text{Sales growth rate})^5 = \\ \text{EUR}29,610,000 \times (1.35)^5 &= \text{EUR}132,772,230.\end{aligned}$$

Earlier lessons showed net income to be a function of sales less expenses (COGS plus SG&A) after taxes:

$$\text{Net income} = [\text{Sales} - \text{COGS} - \text{SGA}](1 - t)$$

Solve for COGS and SG&A in Year 10 as above for sales:

$$\text{COGS(Year 10)} = \text{EUR}77,453,613 = 25,380,000 \times (1.25)^5$$

$$SG\&A(\text{Year } 10) = \text{EUR}12,908,936 = 4,230,000 \times (1.25)^5$$

Substitute into the above equation for net income:

$$\begin{aligned} \text{Net income}(\text{Year } 10) &= [132,772,230 - 77,453,613 - 12,908,936](1 - 0.3) \\ &= \text{EUR}29,686,777. \end{aligned}$$

2. A growth equity analyst determines that mature service businesses like PremVol typically exhibit a price-to-earnings ratio of 15 times. The analyst assumes that the present value of PremVol's cash flows during the next five years is approximately zero. Assuming the analyst's firm is targeting a 5 times ROI on a PremVol investment over the next five years, solve for PremVol's estimated intrinsic value following the new investment.

Solution

Use a revised version of Equation 11 to find PremVol's intrinsic value. As a first step, solve for the numerator, PremVol's terminal value based on expected Year 10 net income of EUR29,686,777 from Question 1 and the P/E ratio of 15:

$$\text{Terminal value (Year } 10) = \text{EUR}445,301,655 = \text{EUR}29,686,777 \times 15$$

Next, solve for the present value using 5 times ROI:

$$IV_t = \text{EUR}89,060,331 = \frac{\text{EUR}445,301,655}{5}.$$

As shown above, a growth model retains features of early-stage companies, seeking sustainable profitability over an investment horizon discounted at an expected rate of return that compensates investors for the elevated risk over the growth phase versus mature, stable companies. For example, we may use Equation 10 to show PremVol's implied IRR as close to 38%:

$$5 = (1 + \text{IRR})^5 ; \text{IRR} = 37.97\%.$$

However, it also allows for differing growth rates among financial variables to establish a more-detailed valuation narrative over a period of high growth.

Model Construction for Firms in Decline

Many mature firms enter a stage of decline due to competitive forces, changing customer preferences and/or consumer tastes, technological change, or poor management, among other factors. Common features among these firms beyond falling sales, loss of market share, and lower profitability is a greater likelihood of **financial distress**, defined as the inability to meet future financial obligations to debtholders. Financial distress may result in the sale of the company to another firm or private investor, the restructuring of assets and liabilities under new management, or liquidation of the firm's assets to pay its creditors. If asset sale proceeds are not sufficient to cover debts, shareholders face a total loss of their investment.

Despite such companies often having a long operating history over the economic cycle, use of either constant growth assumptions or financial models based on projecting existing financial statement relationships into the future may fail to fully capture the effects of potential financial distress. One approach is to adjust key areas of focus highlighted in Exhibit 19 when valuing firms in decline as follows:

- *Cash flow adjustments.* Analysts must identify and adjust key balance sheet and income statement items affected by a firm's decline. For example, key customers or suppliers may switch to competitors or assets that are obsolete or substandard may face impairment. The timing and magnitude of these event-driven changes are uncertain.
- *Denominator issues.* Discount rates are difficult to estimate given the possibility of financial distress. Specifically, while the cost of debt is highly sensitive to the likelihood of default, the required return on equity will also vary widely as a company faces insolvency.
- *Terminal value and forecast horizon issues.* Future cash flow estimates imply that a firm will remain a going concern over the forecast horizon. Also, terminal value estimates based on multiples calculated using ratios from non-distressed industry peers will introduce distortion if a firm faces restructuring or liquidation.

For a firm in decline, it may be preferable to incorporate the likelihood of financial distress more explicitly into valuation model construction, as described in Equation 14:

$$IV_t = \text{Going-concern value} \times (1 - \text{Prob}(\text{Distress})) + \text{Distress sale value} \times \text{Prob}(\text{Distress}). \quad (15)$$

While Equation 14 is quite useful for firms in decline, the framework is reasonable to use for any firm for which future probability of business failure is a distinct possibility, as in the following example.

KNOWLEDGE CHECK: PREMVOL VALUATION WITH PROBABILITY OF DISTRESS



In an earlier example, an analyst values PremVol, a private French company offering commercial drone services, at EUR89,060,331 as a going concern. However, the analyst believes that there is a 20% probability that PremVol's business may face the danger of prohibitive regulatory constraints that will limit the commercial use of drones during the next five years, causing the firm's management to liquidate its assets for EUR50,000,000 in five years. The analyst continues to assume an ROI of 5 times.

1. Which of the following value estimates is closest to the analyst's updated valuation incorporating the concerns about PremVol's future liquidation?

- A. EUR71,250,000
- B. EUR73,250,000
- C. EUR81,250,000

Solution

B is correct. Choice B is correct because PremVol's value in the updated scenario reflects the weighted average of its going-concern value and its distress sale value as shown below:

$$\text{Distress sale value} = \frac{\text{EUR}50,000,000}{5} = \text{EUR}10,000,000.$$

$$IV_t = \text{EUR}89,060,331 \times 0.80 + \text{EUR}10,000,000 \times 0.20 = \text{EUR}73,248,265.$$

Choice A is incorrect because this response implies a distress sale value of zero.

Choice C is incorrect because this response fails to discount the future distress sale value to a present value by applying the ROI.

The Effect of Shareholder Control on Valuation Model Construction

Equity investors may have a non-controlling, financial interest in a company or hold a larger stake seeking to control its activities. A **financial investor** is typically one who has no influence over a firm's operations. When a company's shares are widely held by financial investors or management, an analyst typically bases their valuation on how the firm is operated by its existing management. Changes in the composition of a company's ownership, such as the emergence of a controlling investor, however, may signal significant changes that if anticipated may be incorporated into the disaggregated equity valuation approach.

Controlling investors are often classified as **strategic investors** or **activist investors**. They may hold either a significant minority or majority stake and seek to initiate changes to increase shareholder value via seats on the board of directors, a majority shareholder vote, or by assuming full control of a firm and taking it private. These changes may include, but are not limited to, the sale of company assets, business combinations, and expense reduction or restructuring. When activist investors spell out their intentions to existing public shareholders seeking their vote in support of a restructuring plan, analysts can translate the proposed operating, investing, and financing changes into a detailed forecast for valuation purposes under a scenario in which these efforts are successful. The topic of corporate restructuring will be addressed later in the CFA® Program curriculum.

QUESTION SET 2



1. A valuation model for a company in the start-up phase of the company life cycle is most likely based on:
 - A. a P/E multiple using expected earnings at the end of the forecast horizon.
 - B. a P/S multiple using expected sales in the next year.
 - C. a P/S multiple using expected sales at the end of the forecast horizon.

Solution

C is correct. Choice C is correct because an analyst valuing a start-up should apply a terminal value based on sales, not earnings, at the end of an expected high growth phase rather than in a year.

Choice A is incorrect because the valuation model for a start-up is less likely to include earnings in an early phase than sales.

Choice B is incorrect because the next year's sales do not provide a complete picture of a start-up company's full revenue growth potential.

2. Equity valuation models for start-up or growth companies are most likely to include terminal value estimates using:
 - A. sales or earnings multiples.
 - B. constant growth.

C. asset multiples.

Solution

A is correct. Choice A is correct because start-up and growth companies derive their value from revenue growth. As a result, terminal value estimates are most likely to be based on multiples of sales or earnings multiples.

Choice B is incorrect because constant growth is a more likely assumption for mature companies rather than start-ups or growth firms.

Choice C is incorrect because the use of an asset multiple would require a balance sheet forecast, which is less applicable to start-up or growth firms.

3. An analyst seeking to value the equity of a company that faces the possibility of financial distress is most likely to:

A. increase the discount rate for future cash flows.

B. incorporate the probability and value associated with distress.

C. estimate a terminal value using non-distressed comparables.

Solution

B is correct. Choice B is correct because this approach captures the value of the firm as a going concern while incorporating the value if it experiences distress in the future.

Choice A is incorrect because of the challenges associated with estimating the effects of distress on the costs of debt and equity.

Choice C is incorrect because the terminal value estimation assumes the company is a going concern, distorting the analyst's valuation.

EQUITY VALUATION AND FINANCIAL STATEMENT MODEL OUTPUT

4



estimate and interpret the value of an equity security based on the outputs of a financial statement forecast model

The prior two lessons focused on financial model construction within an equity valuation process. In this lesson, we shift our focus to present a completed model in a step-by-step process. While the illustrations used in this lesson reflect numbers from a hypothetical company, the process shown generalizes to valuing any real-world company where historical financial statements are available and when an analyst invests time to build a narrative about possible outcomes.

Illustration of Equity Valuation

To value equity using a disaggregated model, we follow a basic process described in the second lesson of this learning module.

1. Gather historical financial statements:
2. Construct an income statement forecast.
3. Construct a balance sheet forecast.

4. Use the forecasted financial statements to compute forecasted free cash flow metrics.
5. Estimate a terminal value associated with model outputs.
6. Compute a present value estimate from the free cash flows and terminal value.

The essence of the valuation process occurs in the second and third steps above. The construction of income statement and balance sheet forecasts provides the inputs required to compute forecasted free cash flow metrics. Furthermore, the financial statement forecast narrative is likely to define how an analyst might estimate terminal value.

Analyst's Initial Valuation Scenario of Folium AG

For the initial illustration, a detailed initial scenario presents the process. We then present two additional model scenarios and assign probabilities to each scenario to derive an equity value for a company.

Folium AG is a hypothetical European company that has a liquid market for its publicly listed equity securities. The company operates in a mature, cyclical industry and has identified an attractive niche that allows for growth opportunities.

We first consider the company's historical financial reports.

Historical Financial Statements

Folium's sales have grown from EUR100 million to EUR140 million over the past five years. However, the company's profitability has struggled, as net income has declined from EUR12 million to EUR7 million over the same period. Exhibit 20 shows its historical income statements over the last six years.

Exhibit 20: Folium AG Historical Income Statements (in EUR millions)

Income Statement	Years					
Accounts	t-5	t-4	t-3	t-2	t-1	t
Revenue	100	107	121	128	130	140
COGS	40	45	55	60	58	65
Gross profit	60	62	66	68	72	75
SG&A	30	33	38	40	44	45
D&A	10	11	12	14	15	15
EBIT	20	18	16	14	13	15
Interest expense	5	5	6	7	6	6
Interest income	1	1	1	1	1	1
EBT	16	14	11	8	8	10
Taxes	4	4	3	2	2	3
Net income	12	10	8	6	6	7

The company is asset intensive, with significant investments in both working capital and long-term assets. Folium also maintains significant and growing cash balances as it has not paid dividends to its shareholders over the last six years. Exhibit 21 shows six years of historical balance sheets for the company.

Exhibit 21: Folium AG Historical Balance Sheets (in EUR millions)

Balance Sheet	Years					
	t-5	t-4	t-3	t-2	t-1	t
Assets:						
Cash & ST investments	40	45	52	49	60	68
Accounts receivable	20	31	40	41	30	33
Inventory	30	30	35	37	38	37
Other current assets	5	7	10	7	5	7
Total current assets	95	113	137	134	133	145
Gross PP&E	100	110	120	135	150	160
Accumulated depreciation	70	81	93	107	122	137
Net PP&E	30	29	27	28	28	23
Intangibles	10	15	20	15	14	22
Total assets	135	157	184	177	175	190
Liabilities & Equity:						
Accounts payable	20	25	30	28	27	25
Other current liabilities	5	5	6	4	3	4
Total current liabilities	25	30	36	32	30	29
Total debt	80	85	95	90	85	90
Other long-term liabilities	10	12	15	11	10	14
Total liabilities	115	127	146	133	125	133
Preferred stock	0	0	0	0	0	0
Common stock	5	5	5	5	5	5
Retained earnings	15	25	33	39	45	52
Treasury stock	0	0	0	0	0	0
Total equity	20	30	38	44	50	57
Total liabilities & equity	135	157	184	177	175	190

Note: ST = short-term.

Financial Statement Forecast Model

An analyst has been following Folium AG's stock during recent months and believes that the market is overly pessimistic about Folium management's efforts to continue its niche-oriented growth strategy while improving profitability. The company's stock price has declined recently and has been trading at a market capitalization of about EUR200 million in recent weeks. The analyst believes that the stock may be undervalued, thus providing an attractive investment opportunity at its current price.

After performing other analyses of the company's financial statements, the analyst builds a simple disaggregated model with the following assumptions, shown in Exhibit 22, over a five-year forecast horizon. The following assumptions create the foundation for the base-case valuation scenario that the analyst views as the most likely. Later, the analyst plans to re-estimate the model with at least two additional scenarios to account for more optimistic and pessimistic assumptions.

Exhibit 22: Assumptions for Folium AG Base Case Scenario

Forecasting Items	t+1	t+2	t+3	t+4	t+5	Notes
Revenue	5%	6%	7%	7%	6%	Growth rates
Expenses	6%	6%	4%	4%	3%	Growth rates
Tax rate	30%	30%	30%	30%	30%	% of EBT
Assets & liabilities	7%	6%	5%	5%	6%	Growth rates
Financing needs	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Financing needs are negative in all forecast years; zero dividends in forecast

KNOWLEDGE CHECK: FOLIUM AG PROFIT MARGIN AND INVESTMENTS

- Which one of the following best summarizes the analyst's future assumption about Folium's profit margins?
 - Improving profit margins in the early years of the forecast followed by declining profit margins
 - Declining profit margins in the first year of the forecast and improving profit margins in the final three years
 - Stable profit margins throughout all years of the forecast

Solution

B is correct. Choice B is correct because the analyst has forecasted the expense growth rate as greater than the revenue growth rate during the first year of the forecast and revenue growth rates greater than expense growth rates during the final three years of the forecast.

Choice A is incorrect because this statement implies that revenue growth rates are greater than expense growth rates early in the forecast, while the reverse is true in later years. This implication is the opposite of what is observed from the analyst's forecast assumptions.

Choice C is incorrect because this statement would imply that revenue growth rates and expense growth rates would be equal throughout all years of the forecast assumptions.

- Which one of the following best summarizes the analyst's future assumptions about Folium AG's investment spending relative to revenue forecast?
 - Higher investment spending in the first year of the forecast with investment declining relative to revenues until the last year of the forecast
 - Lower investment spending in the first year of the forecast with investment increasing as revenues increase in the later years of the forecast

- C. Higher investment spending in the first two years of the forecast with declining investment in the final three years of the forecast

Solution

A is correct. Choice A is correct because asset growth rate is greater than revenue growth rate in the first year of the forecast. The asset growth rate is equal to the revenue growth rate in the second year of the forecast and falls below revenue growth in the following two years.

Choice B is incorrect because this statement most closely describes the assumed revenue growth rate over time.

Choice C is incorrect because this statement most closely describes the assumed expense growth rates over time.

After completing a thorough analysis of Folium AG, including its strategic plan, competitive environment, and macroeconomic fundamentals, the analyst creates a forecast based on a narrative that Folium is likely to begin a significant turnaround in its profitability beginning in year $t+3$. The analyst assumes that Folium management will invest disproportionately in assets in year $t+1$ to support this turnaround.

From the set of assumptions outlined above, the analyst creates the following financial statement model shown in Exhibit 23 and Exhibit 24. As an additional component to these exhibits, an analyst is likely to create other exhibits (tables or graphs of key financial ratios) from this information.

Exhibit 23: Folium AG Income Statement Forecast (in EUR millions)

Income Statement Forecast	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$
Revenue	147.00	155.82	166.73	178.40	189.10
COGS	68.90	73.03	75.96	78.99	81.36
Gross profit	78.10	82.79	90.77	99.40	107.74
SG&A	47.70	50.56	52.58	54.69	56.33
D&A	15.90	16.85	17.53	18.23	18.78
EBIT	14.50	15.37	20.66	26.49	32.63
Interest expense	6.36	6.74	7.01	7.29	7.51
Interest income	1.06	1.12	1.17	1.22	1.25
EBT	9.20	9.75	14.82	20.41	26.38
Taxes	2.76	2.93	4.45	6.12	7.91
Net income	6.44	6.83	10.37	14.29	18.46

Exhibit 24: Folium AG Balance Sheet Forecast (in EUR millions)

Balance Sheet Forecast	t	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$
Cash & ST investments	68.00	68.91	70.66	76.56	86.14	98.68
Accounts receivable	33.00	35.31	37.43	39.30	41.27	43.74
Inventory	37.00	39.59	41.97	44.06	46.27	49.04
Other current assets	7.00	7.49	7.94	8.34	8.75	9.28
Total current assets	145.00	151.30	158.00	168.26	182.42	200.74
Net PP&E	23.00	24.61	26.09	27.39	28.76	30.49

Balance Sheet Forecast	t	t+1	t+2	t+3	t+4	t+5
Intangibles	22.00	23.54	24.95	26.20	27.51	29.16
Total assets	190.00	199.45	209.04	221.85	238.70	260.38
---	---	---	---	---	---	---
Accounts payable	25.00	26.75	28.36	29.77	31.26	33.14
Other current liabilities	4.00	4.28	4.54	4.76	5.00	5.30
Total current liabilities	29.00	31.03	32.89	34.54	36.26	38.44
Total debt	90.00	90.00	90.00	90.00	90.00	90.00
Other long-term liabilities	14.00	14.98	15.88	16.67	17.51	18.56
Total liabilities	133.00	136.01	138.77	141.21	143.77	147.00
---	---	---	---	---	---	---
Preferred stock	0.00	0.00	0.00	0.00	0.00	0.00
Common stock	5.00	5.00	5.00	5.00	5.00	5.00
Retained earnings	52.00	58.44	65.27	75.64	89.93	108.39
Treasury stock	0.00	0.00	0.00	0.00	0.00	0.00
Total equity	57.00	63.44	70.27	80.64	94.93	113.39
Total liabilities & equity	190.00	199.45	209.04	221.85	238.70	260.38

Before moving on to presenting the estimated free cash flows, we note that the analyst could have combined some line items in both the income statement and balance sheet forecasts given the simplistic structure of the initial assumptions. However, in this case, the analyst chooses to maintain more line items to accommodate a more comprehensive set of assumptions in future valuations of Folium's equity.

Free Cash Flow Forecast Estimates from the Model

The analyst uses the income statement and balance sheet information presented in Exhibit 23 and Exhibit 24 to generate five-year forecasts of FCFF and FCFE, as shown in Exhibit 25. Recall from Equation 3 that NOPAT equals revenue less COGS, SG&A, and D&A after taxes.

Exhibit 25: Folium AG's FCFF and FCFE Forecast (in EUR millions)

	t+1	t+2	t+3	t+4	t+5
NOPAT	10.15	10.76	14.46	18.54	22.84
Investment in LT assets:					
Net PP&E	1.61	1.48	1.30	1.37	1.73
Intangibles	1.54	1.41	1.25	1.31	1.65
Other LT liabilities	-0.98	-0.90	-0.79	-0.83	-1.05
Change in net WC	3.36	3.08	2.72	2.86	3.60
Total net investment	5.53	5.07	4.48	4.70	5.93
FCFF	4.62	5.69	9.98	13.84	16.92
---	---	---	---	---	---
	t+1	t+2	t+3	t+4	t+5
Net income	6.44	6.83	10.37	14.29	18.46
Net investment	5.53	5.07	4.48	4.70	5.93

	t+1	t+2	t+3	t+4	t+5
Net borrowing	0.00	0.00	0.00	0.00	0.00
FCFE	0.91	1.75	5.89	9.58	12.54

Notes: LT = long-term; WC = working capital.

As a contrast to the forecasted FCFF and FCFE results shown above, the analyst notes the FCFF and FCFE computed from Folium AG's historical financial statements, as shown in Exhibit 26:

Exhibit 26: Folium AG's Historical FCFF and FCFE (in EUR millions)

	t-4	t-3	t-2	t-1	t
FCFF	2.86	0.64	6.50	19.75	6.50
FCFE	5.00	7.00	-3.00	11.00	8.00

KNOWLEDGE CHECK



1. Explain why the analyst's assumptions about Folium's investment and profitability create the observed pattern for forecasted FCFF and FCFE.

Solution

High investment in the early years of the forecast creates low free cash flows for Folium while improving profitability later in the forecast.

The analyst's FCFF forecast for Folium shows no significant improvement in cash flows available to investors over the first two years of the five-year forecast horizon. In the third year of the forecast, Folium's forecasted efforts are expected to begin generating significant increases in FCFF. From the base year (*t*) FCFF of EUR6.5 million, Folium is expected to grow FCFF to EUR16.92 million by the final year of the forecast horizon.

Terminal Value Estimate from the Model

After considering both FCFF and FCFE forecasts, the analyst decides to value Folium based on the FCFF forecast. The analyst considers the forecast assumption of zero debt to be an issue in the reliability of the FCFE model. In the next learning module, we consider this issue further as we assess differences in valuation results across models.

The analyst considers two possible approaches for estimating Folium's terminal value:

1. Constant growth
2. Relative valuation

The analyst has some concerns about the ability to define an appropriate peer group for Folium given the company's specialized niche market within the industry. As a result, the analyst decides to continue the valuation process by estimating terminal value using a constant growth model. To do so, the analyst employs Equation 16 and Equation 17 to estimate Folium's terminal firm value at the end of year *t*+5 by using the sustainable growth rate of its forecasted FCFF at the end of year *t*+5:

$$\text{Firm value}_{t+n} = \frac{\text{FCFF}_{t+n} \times (1 + \text{FCFF SGR}_{t+n})}{\text{WACC} - \text{FCFF SGR}_{t+n}}, \quad (16)$$

where

$$\begin{aligned} & \text{FCFF SGR}_{t+n} \\ &= \left(1 - \frac{\text{FCFF}_{t+n}}{\text{NOPAT}_{t+n}}\right) \left(1 - \frac{\text{FCFF}_{t+n}}{\text{NOPAT}_{t+n}}\right) \times \left(\frac{\text{NOPAT}_{t+n}}{\text{Book value of total capital}_{t+n-1}}\right). \end{aligned} \quad (17)$$

KNOWLEDGE CHECK: ESTIMATING FOLIUM AG'S TERMINAL VALUE



The analyst valuing Folium assumes that the forecasted FCFF in year $t+5$ is EUR16.92 million (see Exhibit 25). Forecasted NOPAT for Folium in year $t+5$ is EUR22.84 million (using Exhibit 23 and Exhibit 24). The analyst has forecasted the company's book value of capital to be EUR184.93 million at the end of year $t+4$. The analyst uses a weighted average cost of capital (WACC) of 7.33% for valuing Folium.

1. Calculate the analyst's estimate of the constant growth rate used to estimate Folium's firm value.

Solution

The analyst estimates a constant growth rate of 3.2% for use in estimating Folium's terminal firm value as demonstrated below.

$$\begin{aligned} \text{FCFF SGR}_{t+n} &= \left(1 - \frac{\text{FCFF}_{t+n}}{\text{NOPAT}_{t+n}}\right) \times \left(\frac{\text{NOPAT}_{t+n}}{\text{Book value of total capital}_{t+n-1}}\right) \\ &= \left(1 - \frac{16.92}{22.84}\right) \times \left(\frac{22.84}{184.93}\right) = 0.2592 \times 0.1235 = 0.032. \end{aligned}$$

The 25.92% represents the analyst's forecasted reinvestment rate, and the 12.35% represents the analyst's forecasted return on capital.

2. Calculate the analyst's estimate of Folium's terminal firm value as of the end of year $t+5$ using the constant growth model.

Solution

The analyst's estimate of terminal value is approximately EUR423 million, as shown below.

$$\begin{aligned} \text{Firm value}_{t+n} &= \frac{\text{FCFF}_{t+n} \times (1 + \text{FCFF SGR}_{t+n})}{\text{WACC} - \text{FCFF SGR}_{t+n}} \\ &= \frac{16.92 \times 1.032}{0.0733 - 0.032} = \frac{17.46}{0.0413} = 422.76. \end{aligned}$$

While the terminal value estimate in the example above is calculated using a constant growth model, the analyst may prefer to communicate the terminal value estimate in terms of a multiple. To do so, the analyst must choose an appropriate multiple. In the example above, terminal value is computed as a firm value. Thus, the denominator must reflect a fundamental variable consistent with firm value, such as revenue, EBITDA, or EBIT. As described in an earlier learning module, the analyst can assess whether the terminal value multiple is reasonable compared to an appropriate peer group or the company's own historical multiples by converting the terminal value computed using a constant growth model.

Present Value Estimate

Finally, the analyst determines the present values of the FCFF forecasts as well as of the terminal value, as shown in Exhibit 27 for Folium.

Exhibit 27: Folium FCFF and Terminal Value Forecasts (in EUR millions)

	t+1	t+2	t+3	t+4	t+5
FCFF	4.62	5.69	9.98	13.84	16.92
Terminal value	---	---	---	---	422.76

An earlier learning module provides the following equity valuation equation when using FCFF as the cash flow metric, as shown in Equation 18:

$$IV_t = \left[\sum_{i=1}^n \frac{FCFF_{t+i}}{(1+WACC)^i} + \frac{E(P_{t+n})}{(1+WACC)^n} \right] - MV_{Debt} \quad (18)$$

KNOWLEDGE CHECK

The analyst notes that Folium has EUR90 million of debt on its balance sheet as of the end of year t and estimates that the market value of this debt is equal to its book value. The analyst estimates that Folium's cost of equity is 10% and that its WACC is 7.33%.

- Which one of the following is closest to the analyst's estimate of the correct equity value of Folium as of the end of year t ?

- A. EUR246 million
- B. EUR336 million
- C. EUR372 million

Solution

A is correct. Choice A is correct as shown below.

$$IV_t = \left[\sum_{i=1}^n \frac{FCFF_{t+i}}{(1+WACC)^i} + \frac{E(P_{t+n})}{(1+WACC)^n} \right] - MV_{Debt}$$

$$IV_0 = \left[\frac{4.62}{(1.0733)^1} + \frac{5.69}{(1.0733)^2} + \frac{9.98}{(1.0733)^3} + \frac{13.84}{(1.0733)^4} + \frac{16.92 + 422.76}{(1.0733)^5} \right] - 90$$

$$= [4.30 + 4.94 + 8.07 + 10.43 + 308.70] - 90 = 246.44.$$

Choice B is incorrect because EUR336 million reflects firm value, not equity value from the calculation process shown above.

Choice C is incorrect because a result of EUR372 million results if the analyst fails to calculate the present value of the terminal value in the calculation process shown above.

Analyst's Alternative Valuation Scenarios for Folium AG

The analyst's first valuation scenario represented a most likely scenario for Folium based on a thorough analysis of the company and its competitive environment. However, the analyst recognizes that the valuation exercise should account for additional potential outcomes. To begin this process, the analyst creates two alternative scenarios to the initial set of assumptions: an optimistic scenario and a pessimistic scenario. The analyst creates these scenarios such that the financial results are reasonable approximations of potential outcomes for Folium over the five-year forecasting horizon. Both scenarios create significantly different valuation results as compared to the base case equity valuation of EUR246 million presented earlier.

Alternative Assumptions

The analyst's optimistic case includes Folium management making greater investment expenditures over the forecast horizon combined with expected higher revenue growth and a greater profit margin expansion. These assumptions are summarized in Exhibit 28:

Exhibit 28: Folium AG—Optimistic Scenario

Forecasting Items	t+1	t+2	t+3	t+4	t+5	Notes
Revenue	5.0%	8.0%	9.0%	7.5%	6.0%	Growth rates
Expenses	6.0%	6.0%	4.0%	4.0%	3.0%	Growth rates
Tax rate	30%	30%	30%	30%	30%	% of EBT
Assets & liabilities	8.0%	8.0%	7.0%	6.0%	6.0%	Growth rates
Financing needs	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Financing needs are negative in all forecast years; zero dividends in forecast

The analyst's downside scenario includes Folium revenue growth stagnating to relatively low levels compared to historical growth. Profit margins continue their recent decline, and investment expenditures are assumed to decline over time to maintain free cash flow. These assumptions are summarized in Exhibit 29.

Exhibit 29: Folium AG—Downside Scenario

Forecasting Items	t+1	t+2	t+3	t+4	t+5	Notes
Revenue	5%	4%	3%	2%	3%	Growth rates
Expenses	6%	5%	4%	3%	3%	Growth rates
Tax rate	30%	30%	30%	30%	30%	% of EBT
Assets & liabilities	8%	6%	4%	4%	3%	Growth rates
Financing needs	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Financing needs are negative in all forecast years; zero dividends in forecast

After the analyst submitted the three valuation scenarios, a question was raised about the valuation associated with a fourth scenario. Specifically, the analyst's base case scenario was questioned as reflecting significant optimism. In response, an alternative

scenario was prepared in which expenses and assets are assumed to grow at the same rate as revenues. This scenario is referred to as the stable scenario and is shown in Exhibit 30:

Exhibit 30: Folium AG—Stable Scenario

Forecasting Items	t+1	t+2	t+3	t+4	t+5	Notes
Revenue	5%	6%	7%	7%	6%	Growth rates
Expenses	5%	6%	7%	7%	6%	Growth rates
Tax rate	30%	30%	30%	30%	30%	% of EBT
Assets & liabilities	5%	6%	7%	7%	6%	Growth rates
Financing needs	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Cash: 100% Debt: 0% Equity: 0%	Financing needs are negative in all forecast years; zero dividends in forecast

Selected Valuation Results Across Scenarios

The scenarios yielded significant variation in Folium's equity value estimation. A summary of results are shown in Exhibit 31:

Exhibit 31: Selected Results Across Folium Valuation Scenarios

Scenario	Optimistic	Base	Stable	Downside
Firm value (in EUR millions)	436.56	336.61	177.41	88.16
Equity value (in EUR millions)	346.56	246.61	87.41	−1.84
Revenue CAGR (5-year)	7.09%	6.20%	6.20%	3.39%
EBIT margin (Year t+5)	20.65%	17.26%	10.71%	7.21%
FCFF (Year t+5)	22.23	16.92	8.14	5.42
FCFF SGR	3.19%	3.21%	3.37%	1.75%

The analyst's base case and optimistic scenarios both produce Folium equity value estimates that suggest significant equity price upside for the company's stock from the current market capitalization of approximately EUR200 million. A key driver of the valuations in these two scenarios is profit margin improvement. Comparison of the base case and stable scenarios highlights this difference. Specifically, if Folium's management is unable to improve EBIT margins significantly from 10.7% under the stable scenario, then the analyst's assumed value improvements are questionable. Finally, the downside scenario implies that slowing revenue growth and declining

margins may cause Folium's firm value to fall below the face value of its debt. In this scenario, the computed value of equity is below zero. However, in this case, the market value of the firm's debt would fall to a degree such that equity value remains positive (although it should be close to zero). These financial distress scenarios are addressed in detail later in the curriculum.

Interpretation of Scenario Valuation Results

The analyst believes strongly that the base case scenario components reflect Folium's revenue growth, profitability, and investment characteristics on a forward-looking basis. To signal strong confidence, the analyst applies a subjective 60% probability to this scenario. As the stable environment is believed to be the second most likely outcome, the analyst assigns a 20% probability to this valuation. Finally, the analyst assigns 10% probabilities to both the optimistic and pessimistic scenarios and their associated valuations.

KNOWLEDGE CHECK: INTERPRETING FOLIUM AG'S FORECAST RESULTS



1. If Folium's equity is currently trading at EUR200 million, which one of the following is a likely conclusion about the analyst's assessment of the company's equity value?

- A. Folium's equity is clearly undervalued.
- B. Folium's equity is clearly overvalued.
- C. Folium's equity is approximately fairly valued.

Solution

C is correct. Choice C is correct, as shown by the probability-weighted average of Folium's equity value across the four scenarios:

$$\begin{aligned}\text{Folium equity value} &= 0.1 \times 346.56 + 0.6 \times 246.61 + 0.2 \times 87.41 + 0.1 \times 0 \\ &= 200.10.\end{aligned}$$

To account for the possibility that the downside scenario implies a Folium bankruptcy, an equity value of zero is assumed. Nevertheless, the probability-weighted average across the scenarios implies that Folium's current equity value is in line with the analyst's valuation scenarios and the subjective probabilities.

Choice A is incorrect given the probabilities assumed by the analyst.

Choice B is incorrect given the probabilities assumed by the analyst.

The exercise above illustrates that a company's equity value reflects a range of possible future outcomes. The analyst assigns subjective probabilities to the discrete values associated with each scenario. However, the analyst's subjective estimates of each scenario's probability are unlikely to reflect the true, but unknowable, probabilities. In hindsight, only one scenario is realized, and market participants constantly reassess future possibilities as new information becomes available. As a result, financial statement models are updated as companies file quarterly or semiannual financial reports or other economic, market, or industry changes occur.

A More Detailed Set of Assumptions

A second analyst assessing the original Folium equity valuation suggests that the assumption structure is too simplistic to properly reflect the company's valuation. The analyst creates a new scenario to address these concerns. Exhibit 32 includes a more detailed set of assumptions as follows:

1. Separately consider COGS, SG&A, and D&A;
2. Separately consider working capital and net PP&E growth;
3. Incorporate a target debt ratio of 46%;
4. Allocate the financing surplus across cash, debt, and equity; and
5. Provide interest rates associated with borrowing and holding cash.

Exhibit 32: Folium AG—Expanded Assumptions

	t+1	t+2	t+3	t+4	t+5	Notes
Revenue	5%	6%	7%	7%	6%	Growth rates
COGS	6%	6%	4%	4%	3%	Growth rates
SG&A	4%	5%	4%	4%	3%	Growth rates
Depreciation	7%	6%	5%	5%	6%	PP&E growth rate
Tax rate	30%	30%	30%	30%	30%	% of EBT
Dividend pay-out ratio	0%	0%	0%	0%	0%	% of NI
WC assets	6%	6%	4%	4%	3%	Growth rates
WC liabilities	6%	6%	4%	4%	3%	Growth rates
Net PP&E growth	7%	6%	5%	5%	6%	Growth rates
Debt ratio	46%	46%	46%	46%	46%	% of con-current total assets
Balancing figures:						
Cash	20%	20%	20%	20%	20%	% of financ-ing needs
Debt	0%	0%	0%	0%	0%	% of financ-ing needs
Equity	80%	80%	80%	80%	80%	% of financ-ing needs
Sum	100%	100%	100%	100%	100%	---
---	---	---	---	---	---	---
Interest expense	6%	6%	6%	6%	6%	% of prior year total debt
Interest income	2.5%	2.5%	2.5%	2.5%	2.5%	% of prior year cash

Note: NI = net income.

Using a cost of equity of 10%, the analyst finds that this updated valuation of Folium's equity using forecasted FCFF and a terminal value estimate from a constant growth model (using sustainable growth rate as in prior examples) generates a present value estimate of approximately EUR219 million. This result is approximately 11% lower than the similar model presented earlier.

The hypothetical company featured in the preceding valuation example exhibits characteristics of a relatively mature company. Specifically, Folium generated positive free cash flows over the course of the financial statement history shown. As a result, an analyst may have considered valuing Folium's equity by assuming growth rates of its base year (t) free cash flows.

A less mature company, however, is likely to generate higher growth rates and negative free cash flows. In such a situation, a disaggregated model becomes the only possible choice for an analyst choosing to estimate value using a present value model, as demonstrated in the following case study.

CASE STUDY



Valuing Growth Companies

Consider an analyst who is valuing the equity of Veranta Inc., a high-growth publicly listed company that went IPO recently. Veranta has grown its revenue from USD70 million to USD190 million over the last two years. Industry observers expect the company to continue its strong growth trajectory, with a potential for USD700 million in revenues in five years. However, Veranta is currently generating negative profits and free cash flow. As a result of the negative cash flow, an analyst is unable to value Veranta's equity using base year cash flow estimates. Exhibit 33 shows Veranta's last three years of income statements and balance sheets.

Exhibit 33: Veranta Inc.'s Historical Financial Statements (in USD millions)

	$t-2$	$t-1$	t
Revenue	70	130	190
COGS	30	58	96
Gross profit	40	72	94
SG&A	50	80	140
D&A	3	3	3
EBIT	-13	-11	-49
Interest expense	4	4	4
Interest income	2	2	2
EBT	-15	-13	-51
Taxes	-2	-2	-6
Net income	-13	-11	-45
Cash & ST investments	75	98	60
Accounts receivable	12	19	33
Inventory	5	8	17
Other current assets	3	6	9
Total current assets	95	131	119
Gross PP&E	30	30	30

Cash & ST investments	75	98	60
Accumulated depreciation	12	15	18
Net PP&E	18	15	12
Intangibles	10	13	23
Total assets	123	159	154
---	---	---	---
Accounts payable	15	27	42
Other current liabilities	4	3	4
Total current liabilities	19	30	46
Total debt	60	70	80
Other long-term liabilities	4	10	14
Total liabilities	83	110	140
Preferred stock	0	0	0
Common stock	80	100	110
Retained earnings	-40	-51	-96
Treasury stock	0	0	0
Total equity	40	49	14
Total liabilities & equity	123	159	154

Assumptions Used to Value Veranta

The analyst following Veranta creates an initial valuation scenario in which the company achieves USD700 million in revenue in year $t+5$. Additionally, the analyst assumes that the company can achieve scale such that its operating and net profitability are able to generate positive free cash flow before the end of the five-year forecast horizon. Exhibit 34 shows the assumptions used to generate the financial model for valuing Veranta's equity.

Exhibit 34: Initial Assumptions for Veranta Base Case Scenario

	t+1	t+2	t+3	t+4	t+5	Notes
Revenue	40%	35%	30%	25%	20%	Growth rates
COGS	35%	30%	25%	20%	15%	Growth rates
SG&A	30%	25%	20%	15%	10%	Growth rates
Depreciation	35%	30%	25%	20%	15%	PP&E growth rate
Tax rate	30%	30%	30%	30%	30%	% of EBT
Dividend payout ratio	0%	0%	0%	0%	0%	% of NI
WC assets	38%	33%	28%	23%	18%	Growth rates
WC liabilities	38%	33%	28%	23%	18%	Growth rates
Net PP&E growth	35%	30%	25%	20%	15%	Growth rates
Debt ratio	30%	30%	30%	30%	30%	% of con-current total assets
Balancing figures:						
Cash	10%	10%	10%	90%	90%	

	t+1	t+2	t+3	t+4	t+5	Notes
Debt	10%	10%	10%	10%	10%	
Equity	80%	80%	80%	0%	0%	
Sum	100%	100%	100%	100%	100%	
Interest expense	6%	6%	6%	6%	6%	
Interest income	2.5%	2.5%	2.5%	2.5%	2.5%	

Using the assumptions shown in Exhibit 34, the resulting model is partially summarized by the forecasted ratios shown in Exhibit 35.

Exhibit 35: Veranta Inc. Historical and Forecasted Metrics

	t-2	t-1	t	t+1	t+2	t+3	t+4	t+5
Revenue growth		85.7%	46.2%	40.0%	35.0%	30.0%	25.0%	20.0%
Gross margin	57.1%	55.4%	49.5%	51.3%	53.1%	54.9%	56.7%	58.5%
EBIT margin	-18.6%	-8.5%	-25.8%	-18.7%	-11.7%	-5.0%	1.5%	7.9%
Net margin	-18.6%	-8.5%	-23.7%	-13.9%	-8.6%	-3.9%	0.7%	5.2%
A/R to sales	17.1%	14.6%	17.4%	17.1%	16.9%	16.6%	16.3%	16.1%
Inventory to COGS	16.7%	13.8%	17.7%	18.1%	18.5%	19.0%	19.4%	19.9%
Net PP&E to sales	25.7%	11.5%	6.3%	6.1%	5.9%	5.6%	5.4%	5.2%
Sales to total assets	0.569	0.818	1.234	1.531	1.777	1.978	2.053	1.875
Debt to total assets	48.8%	44.0%	51.9%	34.0%	31.7%	30.7%	29.6%	28.4%
ROE	---	-27.5%	-91.8%	-264.8%	-97.2%	-65.0%	18.0%	135.1%
ROC	---	-9.3%	-36.3%	-37.0%	-32.4%	-17.8%	6.6%	34.9%
FCFE	---	3.00	-48.00	-61.89	-30.44	-14.49	10.96	53.96
FCFF	---	-5.31	-56.24	-38.58	-33.91	-21.10	1.55	34.41

- The analyst's next step in valuing Veranta's equity is to estimate a terminal value. Which one of the following best describes an analyst's approach to terminal value estimation for Veranta?
 - The analyst is likely to estimate Veranta's terminal value using a constant growth model.
 - The analyst is likely to estimate Veranta's terminal value using a multiple applied to the company's forecasted year $t+5$ sales or earnings.
 - The analyst is likely to estimate Veranta's terminal value using a multiple applied to the company's forecasted year $t+1$ earnings.

Solution

B is correct. Choice B is correct because a relative valuation approach will be most appropriate for Veranta given the company's continued growth trajectory with high revenue growth and improving margins.

Choice A is incorrect because the constant growth model is most appropriate for estimating terminal value when the company has reached stability in its financial characteristics. That is not the case for Veranta as it is still expe-

riencing growth of 20% per year and its profit margins are likely to improve. Additionally, other characteristics also appear to lack stability. Overall, Veranta is likely not a good candidate for employing a constant growth model to estimate terminal value at year $t+5$ of its forecast.

Choice C is incorrect because of two issues. First, Veranta has negative expected earnings in year $t+1$, so an earnings-based multiple will yield a negative result for the company's value. Second, applying a multiple to a year $t+1$ forecast of earnings yields a current value estimate, not a terminal value.

QUESTION SET 3



1. Which one of the following statements most correctly describes equity valuation models using financial statement forecasts?
 - A. Equity value is estimated using a free cash flow forecast calculated based on growth of a base year free cash flow.
 - B. Equity value is estimated using a free cash flow forecast calculated from assumptions made about growth of financial statement items.
 - C. Equity value is estimated including a terminal value estimated from base year earnings or revenues.

Solution

B is correct. Choice B is correct because the equity value reflects free cash flows calculated from the future financial statements constructed from an analyst's assumptions about different components of financial statements, such as revenue growth, profit margins, investment, and financing needs.

Choice A is incorrect because this statement implies an equity valuation in which an analyst does not construct a financial statement forecast. Rather, a base year free cash flow is constructed from historical financial statements, and forecasted free cash flows are simply computed from an analyst's growth assumption.

Choice C is incorrect because this statement incorrectly implies that an analyst estimates a terminal value from base year information. If, instead, terminal value is estimated from earnings or revenues from the final year of the financial statement forecast, this would be a more correct statement.

2. An analyst calculates a terminal value estimate using a constant growth model and uses this estimate to create a multiple of earnings. Which one of the following most correctly describes an advantage of converting this terminal value estimate into a multiple?
 - A. The multiple may provide a way to assess the reasonableness of the terminal value estimate.
 - B. The multiple ensures that the analyst has properly defined an appropriate peer group.
 - C. The multiple provides a more appropriate methodology for calculating terminal value compared to a constant growth model.

Solution

A is correct. Choice A is correct because the analyst can compare the terminal value estimate multiple to a peer group multiple or to the company's own multiple history to assess whether the multiple is economically reason-

able. As an example, a terminal value estimate that yields a P/E much higher than the P/E multiples of a group of comparables or the firm's own historical P/E multiple would suggest that the constant growth model estimate should be re-estimated with differing assumptions.

Choice B is incorrect because the analyst still needs to properly define an appropriate peer group. Simply converting the terminal value estimate into a multiple does not ensure that the analyst also identifies an appropriate peer group.

Choice C is incorrect because the statement implies that the terminal value is initially calculated using a multiple (either from a peer group or from the company's own history). However, this contradicts that the terminal value is initially computed using a constant growth model.

3. Vibria Company is a firm in a high growth phase of its life cycle, with revenue growth rates near 30% and negative profitability and cash flow. Analysts expect continued strong revenue growth and improving profitability for Vibria to continue for at least five years, by which time the company's earnings are expected to be positive. Which of the following statements best describes a present value model for valuing Vibria's equity?
- A. A five-year financial statement model to forecast Vibria's free cash flows and a terminal value estimate generated by a constant growth model
 - B. A five-year financial statement model to forecast Vibria's revenues and a terminal value estimate generated by an assumed multiple of Year 5 revenues
 - C. A five-year financial statement model to forecast Vibria's free cash flows and a terminal value estimate generated by a multiple of forecasted Year 5 revenues or earnings

Solution

C is correct. Choice C is correct because the model described allows for the calculation of a positive terminal value estimate and also reflects the effect of negative free cash flows occurring during the forecast period of five years.

Choice A is incorrect because a constant growth model requires both positive free cash flow and stability of the company's financial fundamentals. In this case, insufficient information is provided as to whether the company is expected to be cash flow positive by the end of the five-year forecast horizon. If the company is still investing heavily to finance continued growth, it is plausible that investment may be greater than the company's profitability.

Choice B is incorrect because this describes a valuation derived solely from a revenue forecast. While such a process might be reasonable for a company in the start-up phase of the life cycle, this process is inadequate for a company in the growth phase.

SUMMARY: FINANCIAL STATEMENT FORECASTING IN EQUITY VALUATION

- Disaggregated valuation models allow an analyst to communicate a narrative associated with separate assumptions across financial statement forecasts of revenue, profitability, investment, and financing.
- Disaggregated models of revenue, expenses, and assets often rely on historical information and may involve bottom-up or top-down approaches to establishing a company's financial statement forecast.
- The level of disaggregation used in constructing financial statement forecast models changes over a company's life cycle. Companies at various stages are valued based on different forecast metrics.
- Analysts may incorporate the possibility that a company may face future financial distress using a probability-weighted average of a going-concern valuation and a bankruptcy or liquidation valuation.
- A basic process of company valuation involves the following steps: 1) gather historical financial statements; 2) construct financial statement forecasts over a future time horizon; 3) use the forecasted financial statements to compute forecasted free cash flow metrics; 4) estimate a forecasted terminal value associated with model outputs; and 5) compute a present value estimate from the forecasted free cash flows and terminal value.

PRACTICE PROBLEMS

1. An analyst is constructing a dividend valuation of a company and has assumed the following earnings per share (EPS) and dividend payout ratio for the next four years.

Year	EPS	Payout Ratio (%)
0	4.00	40
1	4.60	40
2	5.29	40
3	5.819	60
4	6.10995	80

The analyst's table above reflects a disaggregated model in which of the following years?

- A. Years 1 and 2
 - B. Years 2 and 3
 - C. Years 3 and 4
2. An analyst is forecasting selling, general, and administrative (SG&A) expenses for a company for next year. She estimates a regression of the annual percentage change in SG&A (y -variable) on annual percentage change in sales (x -variable) using the last 20 years of the company's income statement information. The analyst's regression produces the following output:

	Coefficients	Standard Error	t-Statistic
Intercept	-0.008	0.011	2.00
% Δ Sales	1.15	0.115	10.00

Which one of the following statements is most correct about the analyst's forecast of the company's SG&A expense for the next year if she utilizes the regression results?

- A. The analyst forecasts that SG&A expense will grow by more than sales in the coming year regardless of the sales forecast.
 - B. The analyst forecasts that SG&A expense will grow by less than sales in the coming year in the event of a 10% increase in sales.
 - C. The analyst forecasts that SG&A expense will grow by less than sales in the coming year in the event of a 5% increase in sales.
3. An analyst has created an income statement and balance sheet forecast for PT Indorinjani, an Indonesian manufacturer. The following table shows a simplified

version of the company's balance sheet (in IDR thousands).

	Year t	Year $t+1$
Non-cash assets	122,000	130,540
Non-debt liabilities	133,000	136,010

The analyst forecasts that PT Indorinjani's net income (in thousands) for the next year ($t+1$) is 6,440 and that the company is expected to pay 50% of its net income as dividends.

Which one of the following most closely states PT Indorinjani's financing needs for next year and its FCFE if the financing needs are met entirely by using surplus cash?

- A. Financing needs are 2,310, and FCFE is 910.
 - B. Financing needs are -910, and FCFE is 910.
 - C. Financing needs are 2,310, and FCFE is 3,220.
4. Which of the following statements best describes the characteristics of a start-up company valuation?
- A. Start-up company valuation models focus on the first-year cash flow forecast, with less importance assigned to the expected terminal value.
 - B. Start-up company valuation models forego a near-term cash flow forecast and primarily focus on the firm's expected future terminal value.
 - C. Start-up company valuation models are usually based on a constant cash flow growth assumption plus a terminal value.
5. Which of the following statements best describes the key features of a disaggregated model for a growth company?
- A. Key equity value drivers in the growth phase include detailed revenue and profit projections to achieve targeted financial results.
 - B. Key equity value drivers in the growth phase include the level of earnings and the stability of earnings.
 - C. Key equity value drivers in the growth phase include the assumption of a constant growth rate and its use in estimating terminal value.
6. Which of the following statements best represents the process of how free cash flow metrics are estimated in a disaggregated forecast model for equity valuation purposes?
- A. Income statement and balance sheet forecasts provide the necessary inputs required to compute forecasted free cash flow metrics.
 - B. Assumed cash flow growth rates provide the necessary inputs required to compute forecasted free cash flow metrics.
 - C. Historical financial statements provide the inputs required to compute forecasted free cash flow metrics.

SOLUTIONS

1. C is correct. Choice C is correct because the increase in assumed dividend payout ratio for Years 3 and 4 causes assumed dividends per share to grow more rapidly than EPS during those two years.

Choice A is incorrect because the dividend payout ratio remains at 40% for both years, so EPS and dividends per share are assumed to grow at the same rate in those two years.

Choice B is incorrect because the dividend payout ratio remains at 40% in Year 2 before increasing in Year 3. As a result, EPS and dividends per share grow at the same rate in Year 2.

2. C is correct. Choice C is correct because the regression coefficient of 1.15 implies that the SG&A growth rate is faster than sales while the negative intercept term offsets the effect of growth. Thus, SG&A expense will be lower than sales growth only at lower sales growth rates. Confirming that this is true at 5% sales growth is shown below:

$$\text{SG\&A growth rate} = -0.008 + 1.15 \times 5\% = 4.95\%.$$

Choice A is incorrect because the forecasted SG&A expense growth rate will only be higher than the sales growth rate at higher sales growth rates. As shown above, at a low growth rate of 5%, the forecasted SG&A expense growth rate is lower than the forecasted sales growth rate.

Choice B is incorrect because the forecasted SG&A expense growth rate will be higher than the forecasted sales growth rate when it is 10%. Applying the same calculation as above yields a forecasted SG&A expense growth rate of 10.7% when sales growth is forecasted at 10%.

3. A is correct. Choice A is correct as shown in the calculations shown below.

To answer the question, we use the following two equations:

Financing needs

$$= \Delta \text{Operating assets} - \Delta \text{Non-debt liabilities} - \Delta \text{Retained earnings}$$

$$\text{FCFE} = \Delta \text{Cash} + \text{Dividends} + \text{Stock repurchases} - \text{Stock issuance}$$

$$\begin{aligned} \text{Financing needs} &= (130,540 - 122,000) - (136,010 - 133,000) - (6,440 - 3,220) \\ &= 2,310 \end{aligned}$$

$$\text{FCFE} = -2,310 + 3,220 + 0 - 0 = 910.$$

Choice B is incorrect because the effect of the dividends are not considered in the financing needs calculation. However, the FCFE is correct because both dividends and change in cash are part of FCFE, and the payment of dividends is simply financed by drawing down the company's cash balance in the correct solution above.

Choice C is incorrect because FCFE does not incorporate the drawdown of the company's cash balance to account for the financing needs.

4. B is correct. B is correct because start-up firms generate little to no near-term cash flow, with the full value often placed on the terminal value.
A is incorrect, as start-up firms have little to no cash flow in their initial years of existence.
C is incorrect, as constant growth assumptions are more appropriate for mature

firms.

5. A is correct. A is correct because growth companies focus on achieving scale and profitability over time via rapid growth.
B is incorrect because growth firms often have low to no earnings and are seeking growth to achieve scale and targeted earnings over time.
C is incorrect because growth firms typically are not expected to expand at a constant rate.
6. A is correct. A is correct because income statement and balance sheet forecasts provide the necessary inputs required to compute forecasted free cash flow metrics.
B is incorrect because assumed cash flow growth rate inputs are used in present value models, not disaggregated forecast models.
C is incorrect because while historical financial statements may form the basis for revenue, expense, and asset forecasts, they do not provide the inputs to compute free cash flow forecasts.

LEARNING MODULE

9

Industry and Competitive Analysis

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	describe the process of industry and competitive analysis
☐	describe industry classification methods and compare methods by which companies can be grouped
☐	determine an industry's size, growth characteristics, profitability, and market share trends
☐	analyze an industry's structure and external influences using Porter's Five Forces and the PESTLE framework

INTRODUCTION: INDUSTRY AND COMPETITIVE ANALYSIS

1

The preceding Equities learning modules estimated intrinsic share value based on discounted cash flow, relative value, and disaggregated valuation modeling approaches. These quantitative tools based on forecasted income statement or balance sheet items should be substantiated by a detailed analysis of the business environment within which a firm operates and the risks it faces. In the next two learning modules, we turn our attention to industry and company analysis. This process should provide a consistent narrative supporting model results, completing the elements necessary for a thorough equity research report. This is likely to require iteration as an analyst refines the quantitative model to incorporate the most critical information obtained from industry and company analysis.

The second lesson provides a basic overview of the steps involved in the process of industry and competitive analysis. Next, we address the classification of a firm within an industry and the associated challenges. We then highlight the process of surveying an industry's size, growth, profitability, and composition to anticipate future industry and company trends. Finally, we revisit Porter's Five Forces to assess the impact of industry structure and introduce PESTLE analysis to assess external factors affecting industry and company performance. The competitive strategy of a firm and evaluation of its past, present, and expected future performance will be the focus of the following learning module.

PRE-TEST: INDUSTRY AND COMPETITIVE ANALYSIS

1. Which of the following represents a limitation of using third-party industry classification schemes?

- A. Currency differences
- B. Single-product companies
- C. Strict hierarchical taxonomies

Solution:

C is correct because commercial classification schemes, such as the Global Industry Classification Standard (GICS), are examples of strict hierarchical taxonomies that classify companies into a single group, regardless of whether a firm offers products or services in different industries.

A is incorrect because currencies have no impact on industry classification.

B is incorrect because single-product companies are easily classified.

2. Which of the following statements best characterizes the difference between defensive and growth companies?

- A. Growth companies tend to be profitable across all phases of the business cycle, while defensive companies are not.
- B. Growth companies tend to expand revenue at above-trend rates across all phases of the business cycle, while defensive companies do not.
- C. Growth companies tend to acquire competitors across all phases of the business cycle, while defensive companies do not.

Solution:

B is correct because higher revenue expansion among growth companies tends to occur independent of the business cycle.

A is incorrect because growth companies may or may not exhibit a high level of profitability depending on their stage of development, and their profitability is independent of the business cycle.

C is incorrect because the distinction of growth versus defensive companies is related to business cycle sensitivity, not the acquisition of competitors.

3. Which of the following statements best characterizes the measurement of industry concentration using the Herfindahl–Hirschman Index (HHI)?

- A. The Herfindahl–Hirschman Index is calculated as the sum of the squares of competitor market shares.
- B. The Herfindahl–Hirschman Index is calculated as the sum of squares of the number of competitors within an industry.
- C. The Herfindahl–Hirschman Index is calculated as the sum of the competitor market shares of the top four competitors in an industry.

Solution:

A is correct because the HHI is based on the sum of squares of market shares. A higher HHI indicates greater concentration.

B is incorrect because the HHI is based on the sum of squares of market shares, not the number of industry competitors.

C is incorrect because the HHI is based on the sum of squares of market shares for all industry participants, not just the top four companies.

4. Which of the following best characterizes the impact of profitability on an industry's value creation?

- A. As a measure of excess return, the difference between an industry's ROIC and its cost of equity is the best measure of profitability.
- B. As a measure of excess return, the difference between an industry's ROIC and its WACC is the best measure of profitability.
- C. As a measure of excess return, the difference between an industry's FCFE and its WACC is the best measure of profitability.

Solution:

B is correct because the ROIC reflects the return on capital and the WACC measures the cost of capital. Thus, the difference (or ratio) between ROIC and WACC reflects the excess return in an industry and is an effective measure of profitability.

A is incorrect because the ROIC measures both debt and equity holder returns, while the cost of equity measures only shareholder return.

C is incorrect because the FCFE measures shareholder return, while the WACC combines the cost of debt and equity.

5. Which of the following statements best describes the potential effect of an industry's workforce as suppliers of labor under Porter's Five Forces framework?

- A. The use of well-organized workers increases the bargaining power of suppliers within an industry.
- B. The use of well-organized workers increases the threat of substitutes within an industry.
- C. The features of an industry's workforce is not a factor that is considered under Porter's Five Forces framework.

Solution:

A is correct because the bargaining power of well-organized workers may force firms to pay higher wages or face competition among rivals in hiring and retaining these employees as suppliers of labor to firms in an industry.

B is incorrect because the features of workers do not increase the threat of substitutes, as they are suppliers of labor to firms in an industry.

C is incorrect because workers are considered to be suppliers with bargaining power.

6. Which of the following statements best describes the role of PESTLE analysis in industry and competitive analysis?

- A. PESTLE analysis seeks to better understand the threat of entrants in an industry or substitutes for its products and services.
- B. PESTLE analysis focuses on the bargaining power of suppliers and customers within an industry.
- C. PESTLE analysis focuses on the effect of external influences on an industry's growth rate and market share dynamics.

Solution:

C is correct because the PESTLE framework considers external political, economic, social, technological, legal, and environmental influences on industry growth and market share changes.

A is incorrect because it describes two distinct elements included in Porter's Five Forces analysis related to an industry's competitive rivalry.

B is incorrect because it lists two separate elements of Porter's Five Forces analysis related to competitive forces within an industry.

2

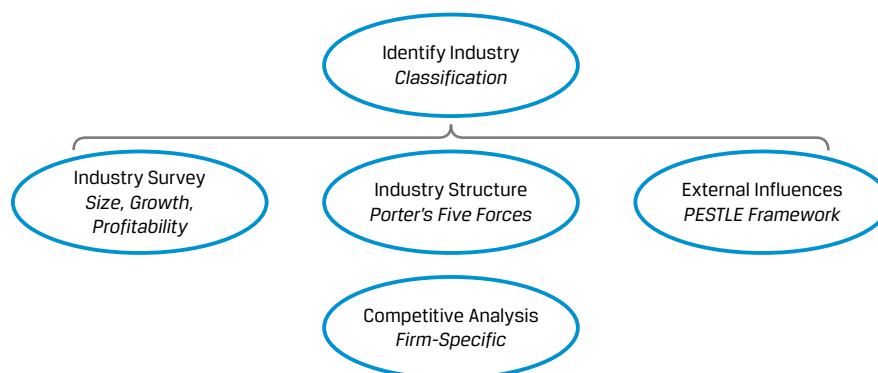
INDUSTRY AND COMPETITIVE ANALYSIS PROCESS



describe the process of industry and competitive analysis

Industry and competitive analysis provides the necessary context to both support the numerical inputs in an equity valuation model and establish a valuation narrative consistent with these inputs. The competitive opportunities, threats, and underlying value drivers faced by a firm are often shared by companies offering similar goods or services to customers that are grouped within an industry. Exhibit 1 outlines the steps associated with the industry and competitive analysis process.

Exhibit 1: Industry and Competitive Analysis Process



This lesson provides a descriptive overview of the process of industry and competitive analysis, with details of the various components of the process provided in subsequent lessons. Exhibit 1 lists the following five steps in the industry and competitive analysis process.

Define a Firm's Industry

Understanding a firm's market position versus key competitors and the overall market for related goods and services is predicated on first establishing the scope of an industry. This may be achieved using third-party industry classification schemes, among other approaches, as described in the following example and addressed in detail in the following lesson.

EXAMPLE 1**Defining Novo Nordisk A/S's Industry Classification**

As described in an earlier learning module, Novo Nordisk A/S is a Danish multinational company whose activities are concentrated in the areas of diabetes and obesity. Exhibit 2 shows the company's classification within the health care sector and the pharmaceutical industry.

Exhibit 2: Pharmaceutical Industry Classification (GICS)

Export			Settings		Classification Browser				
Classification		GICS	Currency	USD	Selected Industry	Pharmaceuticals			
Classification Hierarchy		Count	Industry Summary						
Search for an industry			Filter By	None					
Energy		2368	Max Cap >		Value	Count			
Materials		7660	Aggregate Statistics						
Industrials		10995	Total Market Cap		4,777	1201			
Consumer Discretionary		7755	Poverty Rankings		41.3%	1448			
Consumer Staples		3990	Total Revenue		1,181	1497			
Health Care		5536	Total Revenue 1YR Growth		-0.58%	1418			
Health Care Equipment & Services		2283	Member Companies					Min Cap	Revenue
Pharmaceuticals, Biotechnology & Life Sciences		3303	1. ELLI LILLY & CO					777,428	34,128
Biotechnology		1508	2. NOVO NORDISK B					201,246	23,716
Pharmaceuticals		1527	3. JOHNSON & JOHNSON					385,538	85,168
Life Sciences Tools & Services		268	4. ROCHE HLDG-GENUS					258,708	65,388
Financials		6225	5. MERCK & CO					238,220	60,120
Information Technology		7294	6. NOVARTIS AG REG					242,138	46,668
Communication Services		2950	7. ASTRAZENECA PLC					224,078	45,818
Utilities		1277	8. PFIZER INC					199,188	58,508
Real Estate		3516	9. SANDOZ					136,138	50,238
			10. BMS TOI MYER SQB					115,178	45,018
			11. COVIS INC					81,438	8,548
			12. CHUGAN PHARMA CO					79,818	7,938
			13. GSK PLC					76,468	37,738
			14. MERCK KGAA					72,868	32,738
			15. DAIICHI SANKYO					62,488	11,098
			16. SUN PHARMA INDU					52,048	5,778
			17. TAIKANG PHARMACEU					44,778	20,538
			18. JANSSEN PHARMACEU					42,798	3,228
			19. JANSSEN					39,248	5,608
			20. JANSSEN					31,208	14,938

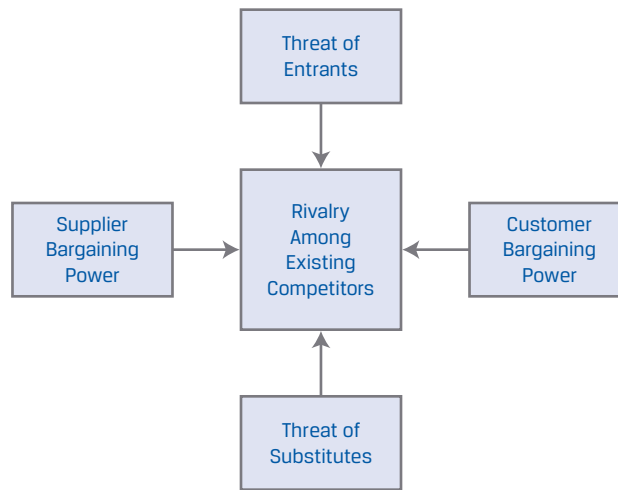
Source: Bloomberg.

Survey the Industry

Once an analyst has determined the relevant scope of an industry, an important next step in gauging a firm's potential growth and profitability is to gather data related to size, growth, and profitability and assess the existing composition of the firms included.

Evaluate Industry Structure

Porter's Five Forces analysis addresses the level of competitive rivalry within an industry by incorporating key factors including the threat of new entrants and product substitutes, as well as customer and supplier bargaining power. Porter's Five Forces framework helps analysts understand influences affecting industry profitability. This framework was introduced in an earlier Economics learning module and is shown in Exhibit 3.

Exhibit 3: Porter's Five Forces

Porter's Five Forces framework provides context and possible scenarios supporting a narrative on how an industry's structure might change in the future. For example, consider the threat of entrants and substitutes in the Novo Nordisk example introduced earlier.

EXAMPLE 2**Porter's Five Forces and Novo Nordisk A/S**

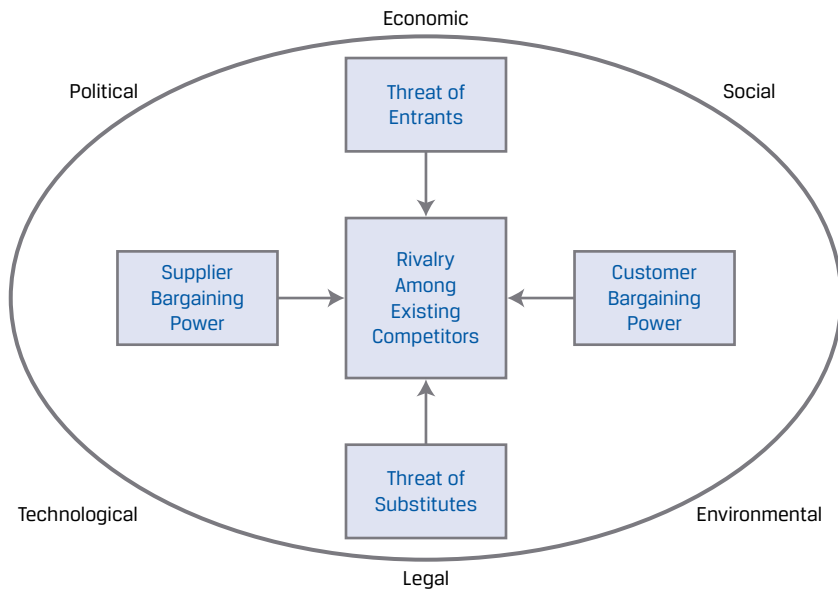
As described in an earlier learning module, the Danish pharmaceutical company Novo Nordisk A/S established itself as a dominant player in diabetes and weight-loss solutions due largely to the success of Ozempic, a patented drug with approximately USD14 billion of revenue, which represented over 40% of its total 2023 revenue. Following its initial release in 2017 to treat diabetes, the observed weight loss in clinical trials led to the release of a higher-dose solution in 2021 to treat obesity.

Although Novo Nordisk's patented solution provided the company with an initial advantage over its rivals, challenges to its market dominance rapidly emerged. Given soaring demand, the company faced production constraints that limited its ability to fulfill customer orders. In the United States, the cost per patient was over USD900 per month before insurance reimbursement in 2023. Limitations on coverage of these costs by health insurance companies limited demand and created opportunities for lower-cost but less effective solutions. Key industry competitors, such as Eli Lilly, Pfizer, and Amgen, accelerated development of products similar to Ozempic, including Pfizer's more attractive oral alternative to Novo Nordisk's injectable therapy. Once Novo Nordisk's patent expires in 2032, generic manufacturers can enter the market at lower prices. These forces are likely to erode Novo Nordisk's market dominance, reducing its revenue and profitability over time.

Assess External Influences

In addition to factors affecting industry dynamics, external forces such as political, economic, social, technological, legal, and environmental influences on an industry are captured in what is referred to as **PESTLE analysis**, as shown in Exhibit 4 and addressed in a later lesson.

Exhibit 4: Porter's Five Forces and External Influences



PESTLE analysis provides a framework allowing analysts to assess industry dynamics, including the growth rate. We revert to the Novo Nordisk example to illustrate how one of these influences, political forces in the form of government involvement, affects both the pharmaceutical industry and individual company revenue.

EXAMPLE 3

Government Health Care Policies and the Pharmaceutical Industry

Because many governments regulate the practice of medicine, pharmaceutical firms producing prescription drugs depend on government approval for use, as well as reimbursement policies that make new therapies affordable for patients. For example, the US Food and Drug Administration (FDA) requires a lengthy preclinical and clinical research process to determine the efficacy and possible side effects of new drugs prior to sale.

In the case of diabetes and weight loss solutions, Eli Lilly first applied for a patent for its antidiabetic medication tirzepatide in 2016, which was followed by three phases of clinical trials completed in 2021. The company received FDA approval for the use of tirzepatide as an antidiabetic therapy (sold as Mounjaro) in 2022 and as a weight loss medication under the name Zepbound in late 2023.

The external influence of government regulation in health care is an important determinant of the timing and magnitude of challenges to Novo Nordisk's market dominance in this market.

Evaluate Competitive Strategy

The final step in the industry and competitive analysis process considers the strategy of an individual firm in the context of its industry and its competitive advantage. This stage of the analysis is addressed in detail in the next learning module.

QUESTION SET 1



1. Which of the following represents the best approach in defining an industry?

- A. Use of Porter's Five Forces analysis
- B. Use of a third-party classification scheme
- C. Use of the PESTLE framework

Solution:

B is correct because third-party classification schemes apply strict hierarchical taxonomies to categorize firms within a single industry group, regardless of whether a firm offers products and services in different industries.

A is incorrect because Porter's Five Forces analysis is used to explain industry profitability.

C is incorrect because the PESTLE framework is used to explain the industry's growth rate.

2. Which stage of the industry and competitive analysis process addresses the impact of environmental regulations on an industry?

- A. Porter's Five Forces analysis stage
- B. Industry survey stage
- C. PESTLE framework stage

Solution:

C is correct because the PESTLE framework addresses external forces affecting an industry, such as political, economic, social, technological, legal, and environmental factors.

A is incorrect because Porter's Five Forces analysis addresses competitive forces within an industry, such as intercompany rivalry.

B is incorrect because the industry survey stage involves estimating industry size, historical growth rate, and profitability and identifying company market share trends over time.

3. Which stage of the industry and competitive analysis process incorporates the threat of new entrants into the industry?

- A. Porter's Five Forces analysis stage
- B. Industry survey stage

C. PESTLE framework stage**Solution:**

A is correct because one of the five forces included in this stage is to analyze the threat of new entrants into the industry.

B is incorrect because the industry survey stage involves estimating industry size, historical growth rate, and profitability and identifying company market share trends over time.

C is incorrect because the PESTLE framework addresses external forces affecting an industry, such as political, economic, social, technological, legal, and environmental factors.

INDUSTRY CLASSIFICATION

3



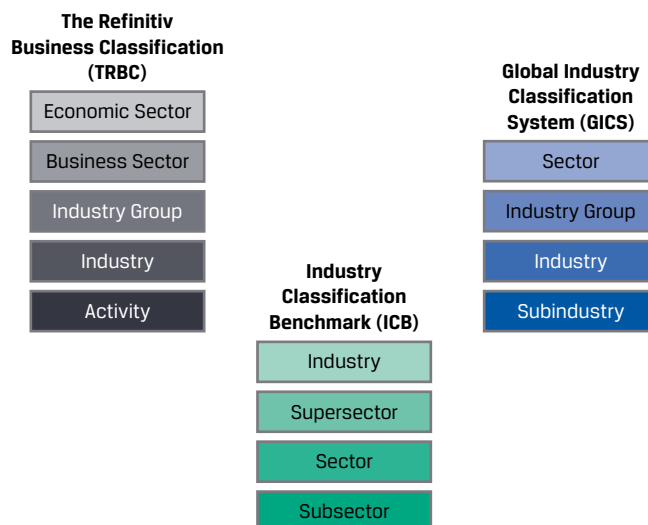
describe industry classification methods and compare methods by which companies can be grouped

An industry is commonly defined as a group of firms that offer comparable products or services from the perspective of a customer. The importance of such groupings for financial analysts is that such firms typically face similar competitive forces and value drivers. While this explains their use in establishing peer multiples for relative valuation purposes, described in an earlier learning module, industry analysis is an important element for all equity valuation approaches. The use of industry classifications provided by third parties enables analysts to consistently compare companies in the same sector. While the use of third-party classification aims to achieve uniformity in analysis and benchmarking, analyst discretion remains important, as described later in this lesson.

Third-Party Industry Classification Schemes

Commercial classification schemes include the Global Industry Classification Standard, offered by MSCI and S&P Dow Jones Indices; the Industry Classification Benchmark (ICB), from FTSE Russell; and the Refinitiv Business Classification (TRBC), created by Refinitiv. These schemes group companies by the similarity of products and services offered, are global in nature, and are reviewed and updated at least annually, with new companies added more frequently. Schemes such as the North American Industry Classification System (NAICS) and the European Nomenclature of Economic Activities (NACE) are also used within countries or regions. Exhibit 5 revisits the three major global classification schemes introduced in an earlier learning module.

Exhibit 5: Third-Party Industry Classification Schemes



GICS and ICB cover public companies, while TRBC also covers private companies, non-profits, and government entities. Data aggregators (such as Bloomberg, FactSet, and S&P Capital IQ) and stock exchange operators (such as the London Stock Exchange) either use the industry classification schemes as shown in Exhibit 5 or have their own that are substantially similar.

While each of the three classification schemes shown in Exhibit 5 has slightly different groupings, all are hierarchical taxonomies, with a company classified within a single group in the lowest tier, and the groups themselves making up the higher tiers. GICS, ICB, and TRBC have slightly different groupings, tiers, and names.

Limitations of Third-Party Classification Schemes

While these classification schemes are a useful starting place for research and other uses, analysts face several key limitations when using such schemes to conduct industry research:

- Company groupings with significant business model variations or that sell substitute products
- Classification of companies operating in multiple industries
- Geographical considerations
- Changes in groupings over time affecting the comparability of statistics

Analysts may find that industry groupings in the third-party schemes are either too narrow or too broad, as scheme providers use discretion in constructing groups based on product similarity and business model. In these cases, analysts may either apply higher or lower classification tiers or modify standardized groupings based on discretion and purposes of analysis. Industry classifications may in some cases involve significant variations in business models among firms, such as their ownership of real estate, as described in the following example.

EXAMPLE 4**Real Estate Use at McDonald's Inc. vs. Yum! Brands**

McDonald's Corporation (MCD) and Yum! Brands, Inc. (YUM), are constituents within the Restaurant subindustry under the GICS industry classification within the Consumer Discretionary sector and the Hotels, Restaurants & Leisure industry. YUM has around 60,000 fast-food restaurants worldwide, as compared to over 40,000 for McDonald's. That said, the two firms take different approaches to real estate in their business models.

While both MCD and YUM have a very high percentage of franchised versus company-owned locations, McDonald's owns a much higher proportion of the underlying real estate used by its restaurant operators, as evidenced by its balance sheet and income statement figures.

For example, MCD has more than double the proportion of net property, plant, and equipment (PP&E) to total assets relative to YUM (68.4% versus 31.4% as of 2023). In addition, 63.7% of MCD's gross PP&E is attributable to land and buildings and improvements on owned land, as opposed to only 3.5% for YUM.

While both companies generated franchise fees as a significant proportion of their revenue, a far larger proportion of McDonald's revenue comes from franchisee rent payments for company-owned property, as illustrated in Exhibit 6.

Exhibit 6: Selected Financial Statement Items for MCD and YUM, 2023

Item	MCD (USD millions)	%	YUM (USD millions)	%
Total Revenue	25,494	100.0	7,076	100.0
Of Which:				
Company Sales	9,742	38.2	2,142	30.3
Franchise-Based and Other Revenue	15,752	61.8	4,934	69.7
Total Franchise Revenue	15,437	100.0	3,247	100.0
Of Which:				
Property Revenue	9,840	63.7	115	3.5
Other Franchise Revenue	5,597	36.3	3,132	96.5

Sources: MCD and YUM 2023 10-K reports.

The generation of rental payments on company-owned property is a contributing factor in McDonald's Corporation's higher 2023 operating margin (45.7%, versus 32.8% for YUM). Also, an analyst may choose to separately consider the value of MCD's real estate portfolio in an overall company valuation.

Adherence to strict taxonomies results in the assignment of companies operating in multiple industries to a grouping within GICS, ICB, and TRBC based on a firm's largest product or service by revenue. This treatment creates challenges for analysts seeking to identify industry participants if a competitor is a division of a company classified in a different industry or sector. Alternative industry classification providers may classify companies operating in multiple industries differently, as described in the following example.

EXAMPLE 5

Amazon.com’s Industry Classification

Amazon.com started as a book seller and is now one of the world’s largest companies. Amazon.com’s business model has continued to evolve since the company went public in 1997. As over 60% of the company’s global revenue remains in its online retail segment, Amazon continues to be considered a Consumer Discretionary company under the GICS classification, along with such companies as Starbucks Corporation.

Like other technology or e-commerce firms, however, such as Microsoft, IBM, and Alibaba, growth in Amazon’s cloud infrastructure and platform services business within the Amazon Web Services (AWS) business segment was a strong driver of the company’s performance for several years. The cloud computing industry is difficult to isolate under industry classification schemes given its strict hierarchical taxonomy, but the Bloomberg Industry Classification Standard (BICS) offers analysts the ability to break down a company’s key segments across classifications, as shown in Exhibit 7.

Exhibit 7: Amazon.com’s Industry Classification (BICS)



Source: Bloomberg.

Exhibit 7 also shows Amazon’s classification as a participant in the traditional supermarket industry via its ownership of Whole Foods, Inc., on the right as well as a breakdown of its revenue across other industries including its e-commerce, cloud computing, and internet advertising businesses.

Given their global reach, the GICS, ICB, and TRBC classification schemes are more appropriate for the largest firms with recognizable brands that produce goods and services distributed worldwide. Firms operating in segmented national or local markets, however, may be less comparable under these industry classification schemes. This segmentation can arise due to customer behavior, transportation or other costs, or regulations that restrict competition. Differences in economic conditions, the degree of competition, or other factors across segmented markets reduce the comparability of companies, although they may offer the same products or services. For example, governments often regulate the practice of medicine through the licensing of facilities and physicians, as well as the provision of state-sponsored health care in certain countries. As a result, analysts typically consider the health care services industry on a national or regional rather than global basis.

Finally, industry classifications may change over time. For example, GICS conducted a significant reclassification in 2023, including the shift of several firms, including Visa and Mastercard, from the Data Processing industry categorized within the Information Technology sector to Transaction and Payments Processing Services within the Financial sector. This is an example of classification changes affecting

industry groupings and statistics. Public offerings and delistings from acquisitions and bankruptcies also change the composition of the groupings over time, raising potential issues with historical return statistics. Analysts should use restated historical statistics that conform to the latest groupings and adjust for changes in the composition of groupings when possible.

Alternative Methods of Grouping Companies

Industry or product-based approaches to grouping companies are not the only grouping methods and may be less useful in contexts outside industry analysis. Other approaches used include geography and business cycle sensitivity.

Geography

Under this approach, companies may be classified by country, which are then aggregated by their stage of development, such as developed, emerging, and frontier markets. Country-based classification typically involves the country where the issuer is incorporated, the country of the primary listing of its equity securities, or the location of its headquarters. Companies operating within a specific jurisdiction may have more common value drivers because of political, economic, and social policies and conditions than companies offering similar products in other countries, as illustrated in the following example.

EXAMPLE 6

The Nifty India Consumption Index

An earlier Equities learning module described the Nifty India Consumption Index based on firms whose activities are primarily focused on domestic sales in India. This grouping of 30 companies publicly listed on the National Stock Exchange of India spans such industries as Consumer Non-Durable Goods, Health Care, Autos, and Media and Entertainment.

While operating in different sectors, firms such as Hindustan Unilever Limited, a consumer goods manufacturer and majority-owned subsidiary of Unilever PLC, and Bajaj Auto Limited, a leading manufacturer of two- and three-wheel motorcycles for consumer transportation and delivery, may be more affected by the Indian consumer market than to global value drivers affecting their respective sectors.

It is important to note that a geographic classification typically does not consider the composition of a firm's revenue or location of its assets, although these factors may be of importance to financial analysts for valuation purposes. For example, although Toyota Motor Company may fall under Japan using a geographic classification, its largest market by revenue is North America, which is also the location of most of its assets.

Business Cycle Sensitivity

Firms are often classified as either **cyclical companies** or **defensive companies** based on the impact of economic fluctuations on their revenue and profitability. Cyclical companies are firms that experience greater fluctuations in revenue and profitability over the business cycle, while defensive companies are those whose sales growth, profitability, and valuations are less affected by changes in macroeconomic activity.

Industries that supply, produce, and distribute discretionary goods and services or durable goods, such as housing, automobiles, appliances, and capital equipment, tend to be more cyclical. These products are generally long-lived, giving the buyer

flexibility as to when to purchase or replace them. Inputs for cyclical industries, including commodities, such as lumber, steel, and copper, and certain business services (e.g., heavy equipment rentals) also face demand fluctuations. For cyclical industries, revenue fluctuations result in greater operating profit fluctuations in the presence of high fixed costs. Defensive industries often include essential goods and services subject to inelastic demand, resulting in less revenue variation over the business cycle.

This approach usually involves grouping entire sectors from the industry classification schemes discussed earlier. Defensive sectors include consumer staples, health care, and utilities; demand for their underlying products tends to remain stable over the business cycle. Cyclical firms involve the financial, basic materials, consumer discretionary, and industrial sectors. **Growth companies** are the third category of firms, which are relatively insensitive to the economic cycle. Firms within growth industries, which include those in such areas as emerging technology or biotechnology, generally exhibit rapid revenue growth regardless of the underlying economic environment, which may or may not be profitable, depending on their stage of growth. The key question for analysts considering investment strategies related to growth industries is how long the above-trend growth will persist and the level of profitability as the industry matures. Categorization based on business cycle sensitivity combined with a geographic approach may be advantageous when countries vary with respect to their position in the business cycle.

QUESTION SET 2



1. Which of the following statements best characterizes the treatment of companies operating in multiple industries under a strict hierarchical taxonomy?
 - A. Strict hierarchical taxonomies classify these types of companies under more than one industry as reflected by their most important products and services by revenue.
 - B. Strict hierarchical taxonomies classify these types of companies by the single largest product or service they offer in terms of revenue.
 - C. Strict hierarchical taxonomies classify these types of companies in a separate category to reflect that they offer more than one product or service.

Solution:

B is correct because strict hierarchical taxonomies lead to the classification of companies in a particular group, regardless of whether a firm operates in multiple industries.

A is incorrect because strict hierarchical taxonomies do not allow for inclusion of firms in more than one industry.

C is incorrect because strict hierarchical taxonomies do not include a separate multiple industry company category.

2. Which of the following statements best characterizes the criterion used to group companies using a geographic approach?
 - A. The location of most of their assets
 - B. The country in which the company is incorporated

C. The country where the company generates most of its revenue

Solution:

B is correct because an issuer's domicile is typically used to classify companies by geography, which may differ from the location of their assets or where they generate most of their revenue.

A is incorrect because asset location is not a determinant of geographic classification.

C is incorrect because the country that is the major source of a company's revenue is not a determinant of geographic classification.

3. Which of the following statements best describes company groupings based on their sensitivity to the business cycle?

A. In an economic downturn, defensive companies experience increases in revenue and profitability, while cyclical companies experience declines in revenue and profitability.

B. In an economic upswing, defensive companies experience decreases in revenue and profitability, while cyclical companies experience increases in revenue and profitability.

C. During periods of economic fluctuation, defensive companies experience more stability in revenue and profitability, while cyclical companies experience less stable revenue and profitability.

Solution:

C is correct because defensive companies experience more stability of revenue and profits, while cyclical company revenue and profitability fluctuate more over the economic cycle.

A is incorrect since defensive companies experience more stable revenue and profitability during an economic downturn, not necessarily increases.

B is incorrect because defensive companies experience more stable revenue and profitability during an economic upswing, not decreases.

INDUSTRY SURVEY

4



determine an industry's size, growth characteristics, profitability, and market share trends

Once an analyst has classified a company into an industry, the next step in the process is to survey the industry by estimating its overall size, determining its historical growth rate, evaluating the nature of that growth, measuring its profitability, and identifying major industry participants and **market share** trends over time. A firm's industry market share is the percentage of total industry sales generated by the company over a specific period. This information on the current state of an industry establishes a basis for analysts to evaluate key issues and opportunities and project future trends using the frameworks on industry structure and external influences described in the following lesson.

Industry Size

Two basic approaches exist in estimating the size of an industry: a top-down approach and a bottom-up approach. A top-down approach involves estimating total revenue across an entire industry over a given period. These data are typically compiled by a reputable third-party data provider or government statistical agency. For example, many governments track and publish periodic retail sales data. One shortcoming of this approach is that these data releases may not be granular enough to distinguish between retail industries supplying distinct types of goods and services.

In contrast, a bottom-up approach involves the sum of the revenue generated by industry participants for specific goods or services for a given period. However, this does not necessarily correspond to the sum of revenue disclosures for all industry constituents. For example, an earlier example illustrated that Amazon.com is a participant in the supermarket industry. Simply adding Amazon.com's total revenue, however, would include both its Amazon Web Services cloud computing business and its Whole Foods supermarket business, therefore overstating the size of the supermarket industry. An advantage of bottom-up analysis for industry groupings dominated by large, publicly traded companies with clear product and/or customer-based sales delineations is the ability to directly measure industry size. However, challenges may arise when an industry includes significant numbers of private companies with meaningful market share. Specifically, data for these industry participants may be limited or unavailable. Also, when companies operating in multiple industries are involved, differences in segment breakdowns and how companies disclose operating segments may require analysts to make estimates based on forms of disclosure outside of a firm's audited financial statements.

When information challenges make industry-wide figures impractical or impossible to calculate using firm-specific data, it is common practice to estimate industry size using a top-down approach, whether this includes economic indicators published by government agencies, data from third-party consultancies that conduct surveys, or industry data from issuers' investor presentations based on proprietary sources. For example, it is best to estimate the size of an industry that includes thousands of small businesses using figures published by a government agency, as described in the following example.

KNOWLEDGE CHECK: MARKET SHARE OF TOP AUSTRALIAN FOOD RETAILERS



Australia's food retail industry comprises several thousand supermarket, grocery store, and convenience store outlets across the country. According to the Australian Bureau of Statistics, Australian food retailing sales for the 12-month period ending June 2024 totaled AUD170.45 billion. Two publicly listed companies—Woolworths Group Limited (WOW) and Coles Group Limited (COL)—dominate the industry, while most of the remaining outlets are either owned and operated independently (Independent Grocers of Australia) or owned by a private company (ALDI).

In order to estimate the food retail market share of the two largest participants, a financial analyst compiles the information in Exhibit 8 for the year ending 30 June 2024.

Exhibit 8: Selected Financial Information for Woolworths and Coles

Woolworths Group	AUD millions	%
Total Revenue	67,922	100
Of Which:		
Australian Food	50,741	75
New Zealand Food	7,551	11
Big W	4,685	7
Australian B2B	4,589	7
Other	356	1

Coles Group	AUD millions	%
Total Revenue	43,684	100
Of Which:		
Supermarkets	39,042	89
Liquor	3,692	8
Express	1,190	3
Other	950	2

Source: Bloomberg.

The analyst notes that Woolworths' business lines include its grocery stores under the Australian Food segment, a wholesale business unit (B2B), New Zealand-based food outlets, and discount department store chain Big W, while the Supermarkets segment makes up Coles Group's food retail operations.

1. Estimate the Australian food retail market share of both Woolworths Group and Coles Group based on the information provided.

Solution:

Woolworths' food retailing activities are in its Australian Food segment, so we consider this revenue only as a percentage of total food retailing:

$$\text{WOW market share} = 29.8\% = \frac{\text{AUD } 50.741 \text{ billion}}{\text{AUD } 170.45 \text{ billion}}$$

Similarly, Coles Group's Supermarkets revenue fully captures its food retailing activities, which may be used to determine its food retail market share:

$$\text{COL market share} = 22.9\% = \frac{\text{AUD } 39.042 \text{ billion}}{\text{AUD } 170.45 \text{ billion}}$$

Note that the use of either company's total revenue would overstate its market share as defined within the context of the government statistic used to measure the overall industry size.

Industry Growth Characteristics

An industry's pattern of historical growth is usually compared to broader economic measures of activity over time and should be evaluated over the cycle to gauge an industry's sensitivity to economic fluctuations.

Since growth in all industries in aggregate should technically equal macroeconomic growth, we would expect more mature industries that have reached their addressable market and potential customers to grow at a rate near or below that of

the economy. Key issues for investors in a mature industry include monitoring for disruptive threats, changes in competitive intensity, and the potential onset and speed of decline. Sensitivity to the business cycle tends to be higher for industries involving discretionary sales, consumer durable or capital goods, or companies whose revenue is exposed to the interest rate cycle.

Expansion rates in growth industries usually exceed that of the economy and are usually not sensitive to business cycles, because they have not yet reached full penetration of their total addressable market. Key questions for analysts related to growth industries include the length of time over which above-trend growth will persist and the expected peak penetration rate, which are more difficult to estimate in earlier stages of growth.

Industry growth rates may be affected by structural changes affecting the demand for certain products and services or by more severe economic shocks. For example, the effect of the COVID-19 pandemic on industry growth is described in the following example.

EXAMPLE 7

Industry Growth Rates during the COVID-19 Pandemic

The unprecedented restrictions on economic activity and free movement in public spaces that began in early 2020 as governments sought to limit the spread of COVID-19 had a profound negative effect on industry growth in many sectors, while having a positive impact on others.

For example, sectors dependent on travel and the in-person delivery of goods and services faced severe setbacks due to travel bans, economic lockdowns, and quarantines. Global passenger air travel fell by 60% in 2020, while the sharp growth in e-commerce during the same period increased global air freight yields by 30% during the same year.

The decline in both road traffic and passenger air travel had an adverse effect on the oil and gas industry amid rising inventory and falling prices. According to the International Energy Agency, primary energy demand fell by 4% in 2020. Half of the decline was attributable to decreased automobile traffic, and over a third was due to the drop in passenger air travel.

Industries and firms within industries that benefited from the pandemic because of the same economic forces included cloud computing, videoconferencing applications, and industries with internet-based business models, such as streaming services and online shopping. For example, while overall US retail sales fell slightly in 2020, e-commerce sales jumped by over 30% over the same period.

Industry Profitability Measures

Given the potential earnings distortions associated with gains, losses, and non-operating income, industry profitability is commonly measured using either operating profit margins $\left(\frac{\text{Operating income}}{\text{Total revenue}}\right)$ or return on invested capital (ROIC). Recall from an earlier

Corporate Finance learning module that ROIC is an aggregate measure of value creation across all investments, which can be compared to the weighted average cost of capital (WACC) as a measure of investors' required rate of return:

$$\text{ROIC} = \frac{\text{NOPAT}_t}{\text{Average invested capital}} \quad (1)$$

where the numerator, NOPAT_t , is net operating profit after taxes and the denominator is equal to total long-term liabilities and equity averaged between the beginning and end of a reporting period. A positive difference between an industry's ROIC and its WACC indicates that the industry generates excess accounting return to investors. Note that this measure includes returns and costs to both debtholders and shareholders and is analogous to the excess return measure equal to the difference in return to shareholders (ROE) and the required equity rate of return (r_e) from the residual income valuation approach introduced earlier in the curriculum. ROIC is often measured using a time series and in some cases is shown as a ratio $\left(\frac{\text{ROIC}}{\text{WACC}}\right)$, as illustrated in the following example.

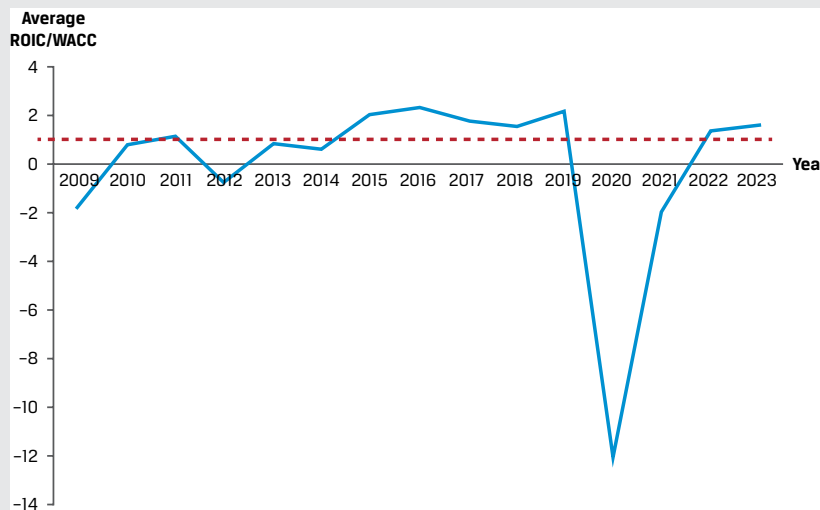
EXAMPLE 8

Measuring Value Creation in the Passenger Airline Industry

The prior example illustrated global passenger airline industry exposure to economic shocks. As shown in the prior learning module using the case of Deutsche Lufthansa, airlines exhibit significant cyclicalities in revenues and expenses, with costs that are driven in part by volatile jet fuel prices and revenue dependent on both business and premium travelers, which can fluctuate over the business cycle, as well as leisure travel, which is dependent on disposable income.

Given these industry features, it is important to consider whether passenger airlines are able to create value over the business cycle. Exhibit 9 uses data compiled from Bloomberg on the ratio of ROIC to WACC from a sample of passenger airlines, including Delta Airlines, United Airlines, IAG Group (the parent company of British Airways, Iberia, and Aer Lingus, among others), Deutsche Lufthansa, and Air France, over a 15-year period to estimate excess accounting returns in the airline industry over this time period. The red dashed line is set at a ratio of exactly 1, and years above (below) this benchmark indicate positive (negative) excess returns.

Exhibit 9: Average ROIC/WACC for Selected Airlines, 2009–2023



Source: Bloomberg.

As shown in Exhibit 9, the airline industry generated returns that were below its cost of capital following the 2008 Global Financial Crisis, reaching positive excess returns for a single year (2011) before oil price spikes over the following

years eroded profit margins despite a recovery in passenger demand. While the industry as a whole enjoyed positive excess returns over a five-year period thereafter, the COVID-19 pandemic had a severe negative effect on ROIC, leading to negative returns until 2022.

While the earlier example shows how an industry as a whole may or may not create value over time, variations among industry participants indicate the dispersion of value creation across companies in the same sector. For example, in 2014 as global oil prices fluctuated near USD100 per barrel, US-based carriers Delta and United generated ROIC well in excess of WACC, while their European-based industry competitors did not.

Challenges in obtaining and evaluating these profitability measures arise if there is a substantial number of private companies in an industry. The following are two common methods to overcome this challenge in practice to estimate industry profitability:

- Measure the profitability of publicly traded industry constituents and assume that the private company's profitability is similar.
- Obtain data from a government agency or third-party consultancy.

For example, the same government agency that publishes retail sales data also conducts an annual survey of retailers' gross margins and operating expenses, which may be used to calculate retail industry operating margins.

As in the case of industry revenue, changes in profitability over time as compared to growth in the overall economy shed light on an industry's sensitivity to the overall business cycle.

Market Share Trends and Major Industry Participants

Industry market shares are measured by dividing each participant's annual revenue by total industry revenue. Given the challenges in measuring industry size mentioned earlier, market shares should be interpreted as a range rather than a point estimate. Significant changes in market share or emerging new industry participants are indicative of the potential for intensive competitive rivalry, as described in the following lesson.

The degree of industry concentration is an essential element of an industry survey. The lower the industry concentration, the greater the competitive intensity, unless an industry is segmented or has a high level of product differentiation. Rising industry concentration or consolidation is generally associated with falling competitive intensity and higher profitability. A common measure of industry concentration is the **Herfindahl–Hirschman Index**, calculated as the sum of the squares of competitor market shares:

$$HHI = \sum_{i=1}^n s_i^2, \quad (2)$$

where s is the market share of market participant i stated as a whole number (i.e., 50% share = 50, not 0.50).

KNOWLEDGE CHECK: THE EFFECT OF MERGERS ON THE HERFINDAHL–HIRSCHMAN INDEX



An industry consists of seven firms, with market shares of 30%, 20%, and five firms with 10% each.

1. Calculate the Herfindahl–Hirschman Index for the industry.

Solution:

Use Equation 2 to solve for the HHI:

$$HHI = \sum_{i=1}^n s_i^2$$

$$= 30^2 + 20^2 + 5 \times (10^2) = 1,800.$$

2. Calculate the new HHI if two of the smallest firms in the industry were to merge with one another.

Solution:

Use Equation 2 to calculate the new HHI:

$$HHI = \sum_{i=1}^n s_i^2$$

$$= 30^2 + 20^2 + 20^2 + 3 \times (10^2) = 2,000.$$

As shown in the example, a higher HHI is associated with greater industry concentration, with the maximum HHI being for a monopoly with a value of 100^2 , or 10,000. Regulators in some countries consider industries with an HHI greater than 1,800 to be highly concentrated, with proposed transactions increasing the HHI by over 100 points in such markets receiving closer scrutiny and possible regulatory challenges to maintain a level of competition within an industry.

Market share changes over time provide insight into the degree of competitive rivalry among existing industry participants, as well as new industry entrants. It is important to consider the effect of mergers and acquisitions on market share, as well as whether a company has a rising or falling market share on an organic (non-acquired) basis, as discussed in the following example.

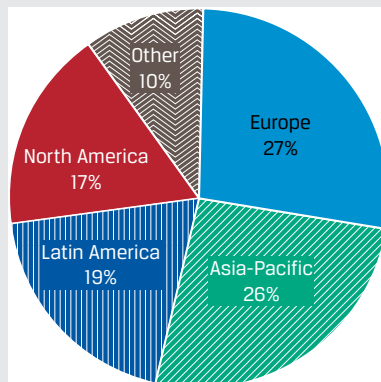
EXAMPLE 9

The Global Beer Industry

Given the large number of privately held craft breweries, total sales in the global beer industry are best estimated using a top-down approach. Because brewery sales are not generally tracked by government statistical agencies, third-party industry providers are the best source for such estimates.

According to data analytics and research firm Euromonitor International, 2023 global beer sales totaled USD750 billion, with the estimated geographic breakdown shown in Exhibit 10.

Exhibit 10: 2023 Global Beer Sales by Region (total = USD750 billion)



Source: Bloomberg.

A major drawback associated with the top-down approach to the beer market when assessing the competitive landscape among global brewers is that the vast majority of firms in the industry are small microbreweries that serve local markets. We can gauge the degree of market segmentation by comparing top-down industry figures with a bottom-up analysis.

Using a bottom-up approach to the beer industry based on major publicly listed brewers, the Bloomberg Industry Classification Standard includes nearly 100 companies with total 2023 beer-related revenue of USD250.87 billion, and the largest 10, representing 64% of total sales, are shown in Exhibit 11.

Exhibit 11: Major Global Brewers and Selected 2023 Data

Company	2023 Revenue (USD bn)	% of BICS Total Revenue
AB Inbev	52.65	21.0%
Heineken Holding	32.83	13.1%
Ambev SA	14.49	5.8%
Asahi Group Holding	14.20	5.7%
Molson Coors	11.70	4.7%
Carlsberg	10.68	4.3%
Constellation Brands	8.16	3.3%
Budweiser Brewing	6.86	2.7%
China Resources	5.20	2.1%
Kirin Holdings	4.87	1.9%

Source: Bloomberg.

A comparison of the top-down and bottom-up approaches to the beer market in this example suggests that roughly two-thirds of global beer sales take place among small, mostly privately held brewers in segmented local markets.

As the previous example suggests, gauging overall market size and distinguishing the degree of market segmentation are important elements of industry and competitive analysis. The following example shows how an analyst might quantify the competitive intensity of the beer industry using this information.

KNOWLEDGE CHECK: CALCULATING THE HERFINDAHL-HIRSCHMAN INDEX FOR THE GLOBAL BEER INDUSTRY



An equity analyst compiling an industry report based on the information in the prior example seeks to measure the competitive intensity of the global beer industry.

1. Estimate the Herfindahl–Hirschman Index for the global beer industry using the information in Exhibit 11, assuming that each of the remaining 90 industry participants has a market share of 0.4%.

Solution:

Use Equation 2 to solve for the Herfindahl–Hirschman Index:

$$\begin{aligned} \text{HHI} &= \sum_{i=1}^n s_i^2 \\ &= 21^2 + 13.1^2 + 5.8^2 + 5.7^2 + 4.7^2 + 4.3^2 + 3.3^2 + 2.7^2 + 2.1^2 + 1.9^2 + 90 \times (0.4^2) \end{aligned}$$

$$= 759.92.$$

2. Calculate a “revised” Herfindahl–Hirschman Index for the global beer industry using revenue figures from Exhibit 11, the top-down estimate of a USD750 billion industry size, and the assumption that the remainder of the industry is composed of firms with a 0.4% market share.

Solution:

First, we must calculate each of the top 10 firms’ market share using a denominator equal to the USD750 billion top-down estimate. For example, AB Inbev’s revised market share is equal to 7.0% (or $\frac{\text{USD}52.65 \text{ billion}}{\text{USD}750 \text{ billion}}$). The full revised Exhibit 11 figures are as follows:

Company	2023 Revenue (USD bn)	% of Top-Down Revenue
AB Inbev	52.65	7.0%
Heineken Holding	32.83	4.4%
Ambev SA	14.49	1.9%
Asahi Group Holding	14.20	1.9%
Molson Coors	11.70	1.6%
Carlsberg	10.68	1.4%
Constellation Brands	8.16	1.1%
Budweiser Brewing	6.86	0.9%
China Resources	5.20	0.7%
Kirin Holdings	4.87	0.6%

Since the sum of the top 10 firms’ market shares has fallen from 64% to 21.6% of the total market, we assume the remaining 78.4% of the market is made up of 196 firms (or $\frac{78.4\%}{0.4\%}$) with 0.4% market share each.

Use Equation 2 to solve for the “revised” Herfindahl–Hirschman Index:

$$\begin{aligned}
 \text{HHI} &= \sum_{i=1}^n s_i^2 \\
 &= 7^2 + 4.4^2 + 2 \times (1.9^2) + 1.6^2 + 1.4^2 + 1.1^2 + 0.9^2 + 0.7^2 + 0.6^2 + 196 \times (0.4^2) \\
 &= 114.33.
 \end{aligned}$$

Given the degree of estimation required to gauge an industry’s size, growth characteristics, profitability, and/or market share trends, it is often important to substantiate measures of industry and competitive analysis using more than one approach. As the prior knowledge check shows, even quantitative measures of competitive intensity, such as HHI calculations, are sensitive to analyst assumptions.

QUESTION SET 3



- Which one of the following statements best characterizes the measurement of an industry’s size?
 - An appropriate bottom-up approach to measuring industry size is to simply add up the total revenue figures of all firms in an industry.

- B.** Economic indicators published by government agencies, data from third-party consultants, or industry data may be used in a top-down approach to estimating industry size.
- C.** Estimated consumer expenditure on specific goods or services over a specific period may be used in a top-down approach to estimating industry size.

Solution:

B is correct because government economic indicators, consultant data, or industry data may be used to estimate industry size in a top-down approach.

A is incorrect because it fails to account for firms that operate in multiple industries and private companies that may not release revenue information.

C is incorrect because the basis for measuring industry size is total revenue, not customer spending.

2. Which of the following approaches best characterizes the relationship between an industry's growth rate and the overall economy?

- A.** Industries growing more slowly than the overall economy are in decline.
- B.** Industries with less cyclical growth than the overall economy are mature.
- C.** Industries that are relatively insensitive to business cycles are growth industries.

Solution:

C is correct because growth industries are insensitive to business cycles, since they have not yet reached full saturation of customer demand or penetration of their total addressable market.

A is incorrect because mature industries may also grow but at a slower rate than the economy since the aggregate of all industries constitutes the overall growth rate.

B is incorrect because mature industries may be cyclical or defensive.

3. Which of the following statements best addresses the impact of market share concentration in an industry?

- A.** The fewer firms competing in an industry, the higher the Herfindahl–Hirschman Index will be for that industry.
- B.** The fewer firms competing in an industry, the higher the competitive intensity will be in that industry.
- C.** The fewer firms in an industry, the less likely that regulatory authorities will prevent a merger that consolidates industry participants.

Solution:

A is correct because the Herfindahl–Hirschman Index is equal to the sum of squares of industry participant market share, which increases as the number of firms competing in an industry declines.

B is incorrect because fewer firms in an industry reduces the competitive intensity of that industry.

C is incorrect because mergers that consolidate market share and reduce competition in an industry are more likely to be scrutinized and challenged by regulatory authorities.

PORTER'S FIVE FORCES AND THE PESTLE FRAMEWORK

5



analyze an industry's structure and external influences using Porter's Five Forces and the PESTLE framework

The prior lesson established the size, growth, profitability, and composition of companies grouped in an industry based on historical information. As a next step in completing a thorough forward-looking analysis of an industry and the companies in an industry, it is critical to evaluate the dynamic forces operating within an industry, as well as external factors that influence its growth and profitability.

The Use of Porter's Five Forces in Equity Analysis

Recall from an earlier Economics learning module that Porter's Five Forces framework was used to assess industry structure. As shown previously in Exhibit 3, these forces are as follows:

- Threat of entrants
- Threat of substitutes
- Bargaining power of customers
- Bargaining power of suppliers
- Rivalry among existing competitors

If some or all of these five industry forces are intense, industry return on capital is unlikely to exceed the cost of capital. This is consistent with earlier Economics learning modules that described firms in perfectly competitive markets in which the price of products and services is equal to marginal cost. Less intense industry pressures, in contrast, often lead to highly profitable industries. In the previous lesson, we considered an industry's profitability on a historical basis, whether it is a source of value creation, and how this value creation is distributed among its participants. For new industries with little or no data or that are still in an early, unprofitable stage, this industry structure analysis can help point to the likelihood, timing, and level of potential future profitability.

Using the analogy of a water-filled trench surrounding a fortified castle, such investors as Warren Buffett have described a long-standing competitive advantage as an **economic moat**. The wider the moat, the greater the potential to sustain profitability in excess of capital costs over longer periods. A wide moat may exist for all the firms in an industry with fewer constituents, such as an oligopoly. Alternatively, economic moats may be present for just one or two profitable firms among many in a highly competitive industry whose returns as a whole fall short of the WACC. While factors affecting the moats vary widely across industries and companies, analysts and data providers have sought to distill and apply this information in the investment selection process as described in the following example.

EXAMPLE 10**Morningstar's Economic Moat Rating**

The US-based investment research and ratings firm Morningstar, Inc., created a proprietary rating system designed to gauge the relative size of a company's economic moat using the following criteria:

- Intangible assets
- Switching costs
- Network effect
- Cost advantages
- Efficient scale

For example, VanEck, a US-based investment management firm that operates globally, has established exchange-traded funds (ETFs) using security selection based on Morningstar's economic moat rating of a firm.

In what follows, we align factors such as those affecting the relative width of an economic moat with Porter's Five Forces to address how analysts evaluate the competitive rivalry within an industry.

Assessing the Threat of New Entrants

The evaluation of market share trends from the prior section is an important first step in determining how new competitors will affect an industry. Beyond the *number* of entrants over a specific period, their *impact* on industry incumbents is an important determinant of future competitive rivalry. For example, the acquisition of startups by technology firms, such as Alphabet's acquisition of YouTube or Meta's purchase of WhatsApp, serves to neutralize the threat of new industry entrants on industry incumbents. In other cases, entrants such as Tesla in the automotive industry pose an enduring threat to existing automotive manufacturers. Incumbent responses to these threats include partnerships with new entrants to combat the competitive threat, as in the case of electric vehicles described next.

EXAMPLE 11**Volkswagen AG's Joint Venture with Rivian Automotive, Inc.**

Volkswagen AG, the world's second-largest auto manufacturer, based in Germany, and US-based electric vehicle (EV) producer Rivian Automotive announced a USD5.8 billion joint venture in November 2024.

Volkswagen built a strong engineering reputation over many decades with its traditional internal combustion engine technology, but the auto industry's shift to EVs created competitive challenges for the company. While Rivian proved to be an effective innovator in computing and software technology in the EV space, it lacked the cash to capitalize on these developments by increasing its production scale. The joint venture allows Volkswagen to gain direct access to the technology necessary to upgrade its vehicles to the digital age.

While Rivian and VW will each have a 50% stake in the new venture, Volkswagen will contribute capital and fund 75% of the shared platform costs, while Rivian will fund the remainder.

Government regulation may limit or prohibit entry into an industry in some cases, such as for investor-owned regulated utilities granted the right to supply power, gas, or water in a specific region. In other cases, regulation may function as a barrier to entry given the cost of complying with regulations that only larger companies can afford. Other forms of entry prevention include government subsidies for domestic firms or restrictive licensing requirements. The case of patents involves a *temporary* barrier to market entry, granting firms the exclusive right to develop new products or technologies, while preventing competitors from providing the same offering until the patent expires. In this case, financial analysts must assess the value of a competitive advantage over a finite patent protection period, as described next.

EXAMPLE 12

Pfizer's Competitive Advantage in Lowering Cholesterol

Atorvastatin, branded as Lipitor, is a cholesterol-lowering drug developed by Pfizer Inc., a US-based multinational pharmaceutical firm. First developed in 1985, clinical trials of the drug progressed until the company received exclusive patent protection from 1997 until 2011. Lipitor generated substantial revenue for Pfizer, peaking at USD12.9 billion in 2006 annual sales and resulting in over USD125 billion in cumulative sales for Pfizer over the period of patent protection.

While atorvastatin remains among the most highly prescribed medications in developed countries, Pfizer experienced a revenue drop of over 50% from the drug once patent protection expired as generic drug manufacturers entered the market and eliminated the company's competitive advantage.

The previous case highlights an important feature of the non-generic pharmaceutical industry—specifically, the importance of continuous research and development of a new generation of patent-protected drugs. Analysts researching an industry must develop a strong understanding of these industry dynamics and their effect on company fundamentals.

Preventing the threat of entry over an indefinite period may include some or all of the factors described in an earlier example related to the width of an economic moat, including the following:

- *Intangible assets:* In addition to the value of patents, brands that command customer loyalty represent a barrier to entry to firms that may vary widely across industries. While the value of the Lipitor brand failed to protect Pfizer's revenue when generic drugmakers entered the market, consumer luxury brands and brands in the food and beverage industry benefit from faithful customers.
- *Switching costs:* Costs or expenses faced by a customer when changing from one product or service can serve as a substantial barrier to entry unless the perceived long-term benefits of switching far outweigh the near-term costs. These costs may include opportunity costs, risks associated with new firms or products, and expectations of upgrades from existing providers. For example, switching business software providers (such as from SAP to Salesforce) often requires lengthy implementation processes led by third-party consulting firms and establishing connectivity to other internal software applications, which is often a costly and lengthy process. In contrast, switching service providers who use a smartphone app presents no limitations due to switching costs.

- *Network effect:* As described in an earlier Economics learning module, a **network effect** involves a situation in which the value of a product or service grows as adoption by households and/or businesses increases. The large demand-side economies of scale associated with the network effect of an established firm in an industry limits the ability of new entrants to compete, despite the potential benefits of a new product or service. The economic benefits to industry incumbents with strong network effects depends on the degree to which a provider can monetize the network, which often increases as network usage grows. Key examples include social media platforms, such as Meta, and payment systems, including Visa and Mastercard. While advertising revenue supports most social media platforms, volume-based transaction and other fees for data processing generate revenue for payment platforms, which grows in value to end users as the network expands.
- *Cost advantage:* Firms that are able to provide goods or services to customers at a lower cost have a pricing advantage over new entrants. Cost advantages usually arise in combination with one or more additional sources of competitive advantage, such as proprietary technology, brand loyalty, or scale. In other cases, incumbents benefit from **economies of scope** across multiple products. Economies of scope arise when the cost of producing two or more distinct goods or services is less than that of producing each separately. Examples include the bundling of cable television, wi-fi connectivity, and telephone service with one provider.
- *Efficient scale:* Efficient scale markets are industries that typically face mature demand for a commoditized product with inelastic demand and high entry costs. In these cases, entrants lack the incentive to enter a market in which additional capacity would drive returns below the cost of capital. Examples include the freight railway industry, in which the cost of maintaining rail infrastructure is very high, while the market attains efficiency through supporting a few operators on a shared rail network.

Assessing the Threat of Substitutes

The threat of substitutes rises when another product or service becomes available that fulfills the same or similar customer need as an existing product, particularly when offered at a lower price. In some cases, the threat of substitutes may arise across seemingly different industries. The following example describes the competitive rivalry among small and large videoconferencing application providers.

EXAMPLE 13

Videoconferencing Application Substitutes

While the use of videoconferencing applications soared more than thirtyfold during 2020, the competitive rivalry among firms offering this service, including Slack Technologies, Zoom Video Communications Inc., Microsoft Corporation, and Alphabet Inc., began prior to the COVID-19 pandemic.

Slack and Zoom represented small, innovative players in the market. As a collaboration pioneer, Slack positioned itself as a customizable platform with a focus on integration with third-party applications. Salesforce Inc. acquired Slack in 2021, allowing its integration with customer relationship management software. Despite the advantage of integration with thousands of third-party apps, Slack is text focused, with weaker video capabilities. Zoom launched its meeting video software in 2013 and went public in 2019. It is known for superior

video and audio quality, a user-friendly interface, and scalability. Zoom was a dominant video conferencing provider during the pandemic but has since faced competition from more established rivals.

As larger market participants, Microsoft and Google have added video conferencing features to existing platforms. Microsoft Teams was launched in 2017 and became a top choice for many organizations given its integration within the Microsoft 365 offering, combining chat and file sharing collaboration tools with videoconferencing. Google Meet was also launched in 2017 and is integrated with the Google Workspace ecosystem, such as Gmail and Google Drive.

Although competition among videoconferencing providers continued as the COVID-19 pandemic subsided, these applications also substitute for in-person meetings and office work, reducing the need for business travel and commercial office space capacity, while expanding opportunities in such areas as remote education and telemedicine.

Assessing the Bargaining Power of Suppliers and Customers

Beyond the threat of industry entrants and substitutes for the goods and services produced in an industry, a thorough analysis must also consider the impact of an industry's suppliers and customers on competitive rivalry.

The bargaining power of suppliers can vary widely across industries. This power is in general positively correlated with the concentration of available suppliers to an industry, the degree to which the materials or components required are unique or tailored to a specific firm, and the cost of switching providers.

When few concentrated providers supply inputs to a specific firm or industry, their bargaining power can be very high. As an example, Apple Inc., the world's largest purchaser of semiconductor chips for its smartphones and other devices, buys all of its advanced chips from TSMC, the leading multinational semiconductor contract manufacturing and design firm. While Apple is beholden to TSMC for its advanced semiconductor chips, it has retained control of the designs and intellectual property for these components in its devices.

The effect of supplier concentration is not limited to material inputs, because the licensing of proprietary technology or intellectual property can have similar effects on bargaining power and the competitive rivalry among industry participants. Examples include the licensing for certain chip designs in the semiconductor industry and streaming content such as live sports or popular entertainment among media content providers.

The presence of switching costs is an additional factor affecting the bargaining power of suppliers. In the case of both semiconductors and index providers, the typically high cost of changing suppliers contributes to their market power. In the case of semiconductors, switching the source of this critical input represents a strategic change that may require product changes and significant investments and time, while ETF issuers that commonly feature the name of a benchmark index in their product offering and compare their funds against others using a similar index might jeopardize the comparability of their offering if switching to an alternative index.

The supply of specialized workers in an industry may also affect the competitive rivalry within the industry if the necessary labor is highly skilled, requires licensing, or is well organized. For example, the bargaining power of unionized factory workers may force firms to negotiate wage increases that are favorable to workers. Labor strikes are a disruptive force among manufacturing companies and transportation companies, such as commercial airlines.

In contrast to industries with few specialized suppliers and high switching costs, supplier bargaining power is at its lowest in industries with many small-scale providers of commoditized inputs that may be easily substituted. Examples include retail

industry segments, such as clothing or housewares, that sell private-label products; low-tech manufacturing with a high interchangeability of components; or restaurants or food service businesses. In cases where industry participants are large buyers of inputs, they have the power to exert considerable influence over suppliers in terms of pricing, quality standards, delivery times, and payment terms.

The relative bargaining power of an industry's customers depends on the features of the underlying product or service and the nature of the industry's customer base.

Highly commoditized products and services, or those for which switching costs are extremely low, are common among industries where customers have a high degree of bargaining power. This power drives competition among industry participants in the form of price competition, product differentiation, bundling of services, or other strategies to increase customer loyalty. By rewarding more frequent customers with incentives, such as discounts, frequent flyer miles, or other exclusive offers, these firms attempt to overcome the low switching costs related to their products and services. The following example describes these strategies in the case of convenience stores.

EXAMPLE 14

Competitive Strategies in the Convenience Store Industry

Convenience stores occupy a distinct space in the food retailing industry given their smaller store size in high traffic areas designed to meet the more frequent, everyday needs of consumers making small, ready-to-eat purchases at premium prices compared to standard grocery stores. Consumers also often have a wide range of substitutes when weighing a decision to shop at these stores.

This segmented market is composed of both small domestic and large global private and public firms across sectors, including food retail and oil and gas. For example, major global competitors range from the privately held Japanese Seven & I Holdings to the publicly held French food retailer Carrefour S.A., British multinational oil and gas giant British Petroleum, and Alimentation Couche-Tard (ATD), a publicly held Canadian operator of convenience stores in over a dozen jurisdictions worldwide.

Whether located in central business districts, congested suburban areas, or along highways, these stores use a range of strategies to bundle offerings and grow their numbers of repeat customers. For example, ATD's Circle K stores offer a loyalty program in which customers earn points for purchases and special rewards for fuel purchases that are redeemable for free discounted items. Subscription-based programs to save on fuel costs or offer unlimited coffee or tea for a monthly fee promote returning customers. Use of the company's mobile app allows users to track reward points, search for company locations, and access personalized offers, as well as pre-order items and pay via the company's app.

In other cases, a large and concentrated customer base for an industry's products or services often significantly increases customers' bargaining power over suppliers within an industry. As described earlier, when few large buyers use an industry's output, they have a greater ability to dictate pricing, quality standards, delivery times, and payment terms.

While outsourcing remains a common strategy in the manufacturing sector, a key competitive strategy often used in industries that face a high degree of supplier or customer bargaining power is **vertical integration**. Vertical integration involves the expansion of operations into an earlier or later stage of the production or supply chain associated with the delivery of a product or service to end users.

Expansion of a firm's activities to an earlier stage of the production process is commonly referred to as upstream or backward integration, in which a firm or group of firms within an industry seek to take control of suppliers or raw material production. This control can be in the form of exclusive supplier agreements, joint ventures as described in the earlier example of Volkswagen and Rivian, or outright ownership of resources or production facilities. The example of rare mineral inputs and battery production for electric vehicles is illustrated next.

EXAMPLE 15

Electric Batteries and the Automaker Supply Chain

The shift from internal combustion technology to battery-powered electric vehicles has had a significant impact on the automotive industry supply chain. The most valuable component of pure EVs, electric batteries, represents over a third of total vehicle value.

Critical minerals, such as lithium, cobalt, manganese, nickel, and graphite, are refined and processed to create the electrodes that form the basic building block of EV batteries. Given that the supply chain of minerals for EV batteries is concentrated in the Chinese Mainland, global automakers have taken steps to diversify their sources of this critical production input.

For example, when US-based EV automaker Tesla, Inc., began its mass production of battery-powered passenger cars in 2014, it was solely reliant on batteries supplied by Panasonic Corporation. As the company reached a higher scale of production, it pursued upstream integration by directly partnering with mining companies to secure lithium, nickel, and other battery-grade minerals and partnering with Panasonic in research and technology to initiate its own battery production at its Gigafactory in 2016. While Tesla continues to expand its battery production capacity at Gigafactory facilities worldwide, it continues to partner with Panasonic, as well as South Korea-based LG Energy Solutions and Chinese Mainland battery manufacturer Contemporary Amperex Technology Co., Limited (CATL), to meet its global battery needs.

Traditional US automakers, such as General Motors Company and Ford Motor Company, have taken similar steps to ensure access to key EV battery raw materials and entered joint ventures to secure EV battery production capacity. For example, in 2019, GM and LG Energy Solutions entered into a joint venture, Ultium LLC, to *jointly* build EV batteries in the United States. Ford Motor Company, in contrast, entered an agreement to partner with CATL in which it retains *full* ownership of new US-based battery manufacturing facilities while licensing CATL's technology.

In the prior example and other cases of backward integration, firms in an industry must weigh the reduced dependence on external parties and scarce resources with significant upfront investments and reduced flexibility resulting from expanding operations across the supply chain.

The alternative to upstream integration involves a shift towards eliminating intermediaries, such as distributors, retailers, or other providers that interact directly with end users of a firm's products or services. As described in the following example, this allows a firm to extract a higher proportion of the value chain associated with the final purchase price of goods and services, improve efficiency, or take advantage of scale in order to offer a more integrated customer experience.

EXAMPLE 16**Amazon Prime and Downstream Integration**

Amazon.com initiated a USD79 annual membership in 2005 for US customers that offered two-day free shipping of the company's products. As the company broadened its member services and added customers, it made major investments in fulfillment centers, vehicles, and air cargo services in order to establish its own logistics and delivery network, competing directly with shipping industry incumbents, such as Federal Express (FDX) and United Parcel Service (UPS).

Amazon expanded its Prime membership services to include music and film streaming while increasing its global reach. In 2022, the company surpassed both UPS and FDX in US delivery volume, and it generated over USD20 billion in membership fees alone in the first half of 2024, with over 200 million members in 25 countries.

Assessing Rivalry among Existing Competitors

In addition to the ease of industry entry, substitutes, and the bargaining power of suppliers and customers, analysts consider a number of other factors when assessing industry rivalry, including the following:

- Price competition
- Product differentiation
- Relative size of major competitors and market share changes
- Industry growth rates and profitability

As an analyst completes the evaluation of industry structure, it is important to highlight the relative influence of these factors on competitive rivalry. While no definitive assessment exists by which to gauge their relative importance, a thorough analysis allows for the incorporation of scenarios in which one or more factors may dominate the others under different circumstances. Consider the following example based on the passenger airline industry from an earlier example.

EXAMPLE 17**Porter's Five Forces Analysis for the Passenger Airline Industry****Threat of New Entrants: Low to Moderate**

Potential entrants in the passenger airline industry face significant capital investment requirements, with industry incumbents benefiting from economies of scale. Such regulatory hurdles as strict aviation regulations and licensing requirements in some markets serve as deterrents to new industry entrants.

While established global airlines benefit from extensive networks and frequent flyer programs, in some regions or markets, low-cost carriers (LCCs) have a lower entry barrier due to flexible leasing options, limited schedules, and the use of secondary airports. As a result, the threat of entry is low for major global carriers but moderate for LCCs in many regional and underserved markets.

Threat of Substitutes: Moderate

While driving and high-speed rail travel represent substitutes for short-haul passenger air travel, long-haul flights have few alternatives. Business-related travel remains affected by remote work trends and virtual meetings, as described

in an earlier example. However, few alternatives to air transportation exist for leisure travel and tourism. As a result, substitutes for passenger air travel are a moderate threat and are more pronounced for shorter trips and business-related travel, while longer flights and leisure travel are less susceptible to substitution.

Bargaining Power of Suppliers: High

Passenger airline companies are dependent on aircraft manufacturers, fuel deliveries, and unionized employees as key suppliers. The global dominance of jet aircraft manufacturers Boeing and Airbus leave airlines with little bargaining power. As a major expense item for passenger airlines, oil prices are subject to significant volatility, which affects operating margins. Airline pilots, engineers, and maintenance workers are often well organized, affording them the ability to negotiate wage and benefit increases, driving up airline operating expenses.

Airlines are heavily dependent on these suppliers, which have strong bargaining power due to their critical role in operations and the lack of substitutes.

Bargaining Power of Customers: High

Passenger airline customers are very sensitive to price differences, which are highly transparent given the widespread practice of online travel booking. While airlines use such strategies as branded credit cards, frequent flier programs, premium seating, and travel lounge access, customer loyalty is limited given low switching costs, particularly for short-haul trips and popular destinations.

Demand for non-essential air travel is also sensitive to the business cycle. Business travelers may face constraints due to restrictive travel policies, limit travel in economic downturns, and may forgo travel altogether in favor of remote meetings as described earlier. Leisure travelers may delay discretionary holiday travel or substitute ground travel for flights when seeking to limit expenses. As a result, customer bargaining power in the passenger airline industry is considered high.

Intensity of Competitive Rivalry: High

A high degree of industry rivalry results from numerous global and regional airlines and low-cost carriers competing on price, routes, and service quality while facing high operating expenses and suppliers with significant bargaining power.

Regulatory constraints limit the degree of industry consolidation, while barriers to exit may lead to greater price competition during economic downturns. As illustrated in an earlier example, the airline industry faces cyclical demand and returns below the cost of capital over the cycle, intensifying the rivalry among competitors.

The relative importance of these industry factors may also be influenced by *external* factors that play a role in the valuation process, such as the likelihood of an economic downturn or higher and more volatile oil prices. We now turn our attention to these elements and their effect on industry and competitive analysis.

PESTLE Framework

While Porter's Five Forces analysis seeks to better explain the competitive rivalry among firms *within* an industry, factors driven by forces outside an industry often affect future growth prospects and the relative market positions of individual firms within the industry based on their ability to anticipate and adapt to changing conditions.

The PESTLE framework, which analyzes political, economic, social, technological, legal, and environmental influences on an industry, is a common approach. While Porter's Five Forces pertain to the determinants of industry profitability, PESTLE

analysis is more concerned with an industry's growth rate and market share dynamics based on these influences. Analysis of these external forces helps analysts identify themes or narratives that may be incorporated into the equity valuation process with a positive or negative influence on an industry or companies within an industry.

For example, the COVID-19 pandemic had a profound, immediate adverse effect on such sectors as transportation and physical retail, while accelerating the digitalization of payments. Long-lasting effects of COVID-19 include the trend toward remote work, which affected air travel and commercial real estate.

Political Influences

Political influences include those that *indirectly* affect sector constituents, such as changes in trade, fiscal and monetary policies, regulations, and geopolitical factors, as well as *direct* government purchase or sale of an industry's goods and services. In some industries, government-owned or controlled industry participants are common, which increases the potential effect of political factors on competitive analysis. For example, basic commodity producers or utilities are often owned or controlled by governments, such as oil producers Saudi Aramco and PetroChina; electricity producers, including Électricité de France SA (EDF); or mining companies, including National Copper Corporation of Chile (Codelco).

In many cases, although industries may be dominated by privately held or publicly listed companies, the existence or potential threat of changes in government regulations, restrictions, or taxes may adversely impact future revenue and profitability. In contrast, subsidies or reimbursement of costs or fees associated with a product or service can improve an industry's prospects or cause customers to favor one product over another. For example, governments in the United Kingdom, the United States, and the European Union offer consumers *direct* tax incentives for purchasing new electric vehicles and *indirect* tax incentives for driving EVs given that their use of electricity is not subject to national gasoline taxes, as in the case of cars with traditional internal combustion engines.

As described earlier, governments often closely regulate the health care industry through the licensing of medical facilities and physicians but also through the reimbursement of medication costs under state-sponsored programs. The degree to which pharmaceutical drug costs are reimbursed by government agencies can be a key driver of product revenues, as described in the following example.

EXAMPLE 18

Government Regulations and Pharmaceutical Industry Revenue

The timing of release and the revenue and profitability potential of new drugs often hinge on government regulations, as described in the previous examples concerning Pfizer Inc. and Novo Nordisk A/S.

Pfizer's cholesterol-lowering statin drug, Lipitor, was developed in a multi-phase government-sanctioned trial before receiving official approval for use in combating coronary heart disease. While under exclusive patent protection, US government health care agencies, such as Medicaid, covered the cost of Lipitor given its proven efficacy in lowering cholesterol and continue to reimburse the expense of low-cost generic versions of the drug.

Novo Nordisk A/S developed its patented drug Ozempic and received official approval for its use to treat type 2 diabetes in 2017. Its subsequent approval at higher doses as a weight loss drug in 2021 held promise for many non-diabetic patients struggling with obesity. However, Medicaid and other US health care

agencies limited their Ozempic reimbursement to the treatment of diabetes only for the first several years of patent protection, constraining Novo Nordisk's revenue and earnings potential for the drug.

Economic Influences

Economic influences include short-term effects of economic fluctuations and long-term growth trends and structural changes that affect industry dynamics. Industry and competitive analysis must take the expected level of growth and inflation into consideration. The industry's expected future performance should be consistent with these expectations, with adjustments based on whether an industry is considered to be defensive, cyclical, or in an above-trend growth phase.

The influence of economic factors extends to longer-term trends, such as an economy's stage of development and its competitiveness. As described in an earlier Economics learning module, developing countries attract capital investment, which increases labor productivity, leading to growth rates that exceed those of developed countries. Shifts in productive capacity from developed to developing markets lead many goods-producing industries to transfer mass production to low-wage countries, while service industries often take advantage of business process outsourcing (BPO), as described in the following example.

EXAMPLE 19

BPO in the Philippines

As one of the fastest growing economies in Southeast Asia, the Philippines has rapidly become a large provider of business process outsourcing activities, which exceeded USD35 billion in 2023 revenue and employed 1.7 million Filipinos. Key industries that take advantage of BPO include financial services, health care, and e-commerce firms.

For example, US-based companies, such as JPMorgan Chase & Co., United Healthcare Group Inc., and Amazon.com, use BPO operations in the Philippines for customer service and back-office processing, often cutting the operational costs of such tasks in half by relocating tasks from domestic workers to this low-cost jurisdiction.

Exchange rates also influence industry growth and profitability by affecting both costs and the competitiveness of exports versus imports. When a domestic currency depreciates relative to other currencies, the cost of inputs, such as raw materials and energy, rises, reducing industry profit margins unless firms have a high degree of pricing power and are able to pass these costs on to consumers. Currency depreciation also boosts export competitiveness by making domestically produced goods and services less expensive for foreign buyers.

Social Influences

Cultural and consumer trends, demographic changes, and changes in lifestyles or attitudes in a population are among the most important social influences that can affect industry growth and profitability.

In some cases, social influences arise as a result of economic changes. For example, developing countries with rising levels of per capita GDP typically experience significant changes in consumer demand, shifting from a focus on low-cost necessities among consumers with little discretionary income to greater demand for higher-quality,

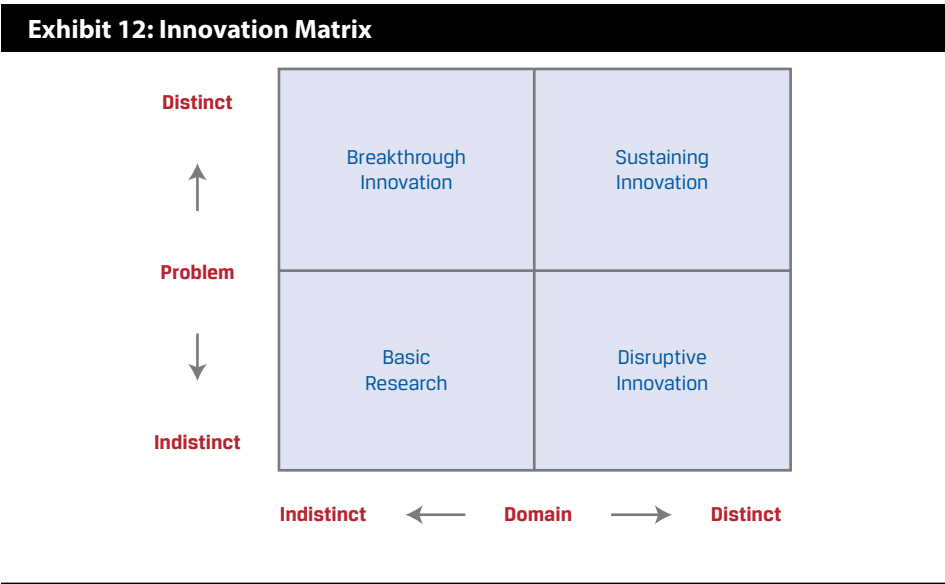
brand-name products. Rising per capita income levels drive increased demand for durable goods, such as appliances, furniture, and automobiles, as well as growth in spending on leisure, entertainment, and travel.

Product differentiation in developed economies often focuses on quality, brand image, and features that tend to cater to diverse consumer preferences rather than solely competing on price. These features may include convenience, flexibility, or health consciousness but may also include transparency and disclosures surrounding product origin, including the local sourcing of inputs or the socially responsible procurement of raw materials using fair labor practices and safe working conditions.

Industries and firms that are able to cater to local tastes, social trends, and cultural differences across markets are best able to combine steady growth in developed markets with more rapid expansion in developing markets. For example, restaurant franchising chains, such as McDonald’s Corporation and Subway IP LLC, adapt their menus to local tastes and locally source ingredients and dishes in order to broaden their appeal.

Technological Influences

Technological change can result in the creation of new products or services, transform business models, and provide incremental improvements for existing industries or render existing offerings obsolete. As a key driver of total factor productivity for the overall economy and within an industry, analysts should consider when and how technological change will affect competition within a sector. It is useful to classify different forms of technological advancement based on their impact on a specific **domain** or sector and the problem or issue they are designed to address. A domain refers to an area in which an innovation is applied, which may include a product, business process, or specific market segment. While an innovation’s scope is defined by its domain, it may address a problem that can be directly translated into a new product or service or offer a less distinct solution without immediate commercial application. These types of technological advancements are often categorized on the basis of a so-called innovation matrix, as shown in Exhibit 12.



Starting from the bottom left quadrant of Exhibit 12, **basic research** involves developing an understanding of fundamental scientific principles that typically have no immediate commercial application. Basic research often builds a foundation for future innovation, and while governments and universities are often the largest sources of

basic science funding, companies in information technology, manufacturing, and the pharmaceutical industry commonly fund such research through grants or directly conduct research to advance product development. For example, in the field of medicine, laboratory experiments or animal testing may be used to determine the efficacy of a specific compound, which must be replicated and further assessed for harmful side effects before experimental use on humans. Pharmaceutical companies, such as Pfizer and Novo Nordisk, seek to capitalize on successful basic research by initiating early-stage clinical trials of medications that may receive approval for widespread use. Despite the large scale of basic research, commercial application of the results to generate revenue is typically years in the making and far from certain.

Breakthrough innovation involves technology that establishes a new product, service, or business model or fundamentally changes existing businesses with a significant effect across industries. A prominent example of breakthrough innovation is the development of the rechargeable lithium-ion battery. Following years of basic materials research, the first commercialization of the technology took place in the early 1990s. Continued refinement of the technology, including improved energy density, battery life, and safety features, paved the way for its widespread adoption in smartphones, computers, power tools, and automobiles.

Innovations with the most immediate financial impact on industries, products, and business models include **sustaining innovations** and **disruptive innovations**. Sustaining innovations are improvements in product or service performance and the addition of marginal features without a fundamental change in functionality or operation. These innovations are often led by industry incumbents, making improvements to better serve existing or adjacent customers' needs. Successive improvements to reduce rechargeable lithium-ion battery size and prolong battery life in smartphones and automobiles represent sustaining innovations.

Disruptive innovations are changes that create a new market or redefine an existing one with a different value proposition that may not immediately appeal to existing customers who are satisfied with the existing offering. While industry incumbents may lead disruptive innovation, it also provides an opportunity for entrants to rapidly capture market share. For example, the existing cable television industry was disrupted by streaming video delivered over the internet. While streaming technology was at first a rudimentary offering that appealed to consumers without cable subscriptions, it eventually grew to overtake cable in the number of subscribers. New entrants have relatively little at stake when pursuing a disruptive innovation because they do not have an existing profitable business to protect. As for incumbents, they can either invest in the disruptive innovation themselves—accelerating the decline of their existing business to maintain or grow market share—or ignore or more slowly adopt the innovative technology and lose market share, while continuing to generate strong profits in the near or medium term.

Legal Influences

Legal influences on an industry include changes in laws and regulations from courts and policymakers that impose constraints or costs on industry participants, alter an industry's business practices or economic outcomes, or fundamentally change an industry's growth and profitability prospects by either banning or allowing certain activities. These influences may either promote or constrain business activity in one jurisdiction versus another, causing firms to move or outsource certain operations or target growth in areas with fewer legal restrictions, as described in the following example.

EXAMPLE 20**Regulatory Constraints and the Global Pharmaceutical Industry**

As described in earlier Pfizer Inc. and Novo Nordisk A/S examples, the global pharmaceutical industry is tightly regulated. Firms introducing new medications must go through a series of clinical trial protocols and extensive new drug approval processes from agencies including the US Food and Drug Administration and the European Medicines Agency (EMA) in the European Union. While the FDA and EMA approval processes differ, both are focused on the highest health and safety standards for new medications when granting intellectual property protection, such as patents.

Given its competitive advantage in terms of relatively low production costs, a highly skilled workforce, and local government policies that promote generic drug use, the Indian pharmaceutical industry rapidly developed by specializing in reducing the prices of drugs that are no longer under patent protection through the local manufacture of active pharmaceutical ingredients. As a result, India supplies over 20% of the world's generic drugs, with a US generic drug market share of over 40%.

Differing pharmaceutical industry regulatory and legal regimes have played a key role in the development of the industry in India. For example, Dr. Reddy's Laboratories, a multinational drug company based in Hyderabad, India, began as a domestic supplier to Indian drug manufacturers, later expanding operations to export medications to unregulated markets. As the company grew, it established FDA-approved facilities in India that manufactured active pharmaceutical ingredients. In 2001, Dr. Reddy's became the first Indian company to launch a generic drug in the United States, and it listed its public shares in the United States during the same year. Since that time, the company has continued to expand in Europe and the United States through acquisitions and partnerships in order to conduct clinical trials and develop new drugs.

Examples of industries strongly affected by legal influences include tobacco and cannabis. Developed economies have imposed wide-ranging laws and restrictions on the manufacturing, distribution, marketing, sale, and use of tobacco products to discourage consumption and the associated adverse health consequences. Because most of the legal constraints on consumption focus on high cigarette taxes and smoking bans in public places, tobacco companies have adapted by increasing their offerings of smokeless tobacco products, with over a third of industry revenue generated by these alternative offerings.

In contrast to the tobacco industry, the legalization of cannabis represents an evolving legal environment in which some markets, such as Canada and Germany, have legalized its production, sale, and use. Other markets, such as the United States, have varying laws at the local level, while it remains a controlled and illegal substance in most Asian jurisdictions. Given both the social stigma and the well-publicized negative health effects of smoking in many developed markets, cannabis industry offerings have rapidly evolved to include edible products, which represent a growing proportion of overall cannabis industry revenues. The examples of the pharmaceutical, tobacco, and cannabis industries demonstrate how industries evolve and adapt to legal constraints and a changing legal environment over time.

Legal factors are intertwined with technological influences in cases where innovations occur with no established legal framework or precedent.

For example, a key area of sustained innovation in the automotive industry involves the introduction of autonomous driver assistance features. These include speed control, autonomous steering and acceleration, and environmental detection capabilities using a combination of cameras, radar, and lidar technology. However, the introduction of fully autonomous vehicles with no human intervention remains experimental in many countries and is subject to legal limitations, as described next.

EXAMPLE 21

Self-Driving Cars and the Legal Landscape

Although many automakers have developed various levels of driver-assisted technology, the use of fully automated vehicles (AVs), where all driving tasks are performed without human intervention, remains at an experimental stage. While no widely accepted global regulatory AV framework exists, as in the case of the Geneva Convention on Road Traffic, signed by over 100 countries, many governments seek to establish uniform rules surrounding operational responsibility, safety standards, software reliability, environmental performance, and cybersecurity, as well as the protection of people and property outside the AV.

While no formal nationwide AV laws exist in the Chinese Mainland as of 2024, the Ministry of Public Security has established 32,000 km of roads for AV testing and 16,000 test licenses, with 20 cities testing autonomous taxis and buses. In the case of Australia, existing legal barriers prohibited full AV operation on public roads in 2024, although updated regulations are under discussion.

The EU passed legislation in 2022 establishing a legal framework for the approval of automated and fully driverless vehicles, with each country slated to create its own legislation. Germany's first trial of highly automated self-driving taxis began on public roads in mid-2024. While no uniform US federal law on AVs was in place as of 2024, several US states, including Nevada and Arizona, initiated local legislation allowing the early adoption of autonomous vehicles on a limited scale.

Environmental Influences

Environmental influences on industry and competitive analysis are often closely associated with political, social, and/or legal influences. Energy producers and heavy consumers of energy are among the industries whose growth and profitability are most heavily impacted by environmental factors.

A key question for energy producers involves the risks and opportunities associated with the so-called **energy transition**, or structural shift mandated by many global governments to reduce the emission of greenhouse gases in overall energy use. As national governments implement energy policy changes at varying speeds and with different combinations of taxes and subsidies, industry participants take different approaches in adapting to incentives and expected policy changes, as described in the following example.

EXAMPLE 22**RWE AG and Enel S.p.A. Approaches to the Energy Transition**

Germany-based RWE AG and Italian utility Enel S.p.A. are multinational firms that represent the largest electric utilities in their domestic markets. While both firms are focused on increasing renewable energy capacity and reducing fossil fuel emissions in alignment with European Union policies, their strategies reflect the unique regulatory, tax, and subsidy policies of their respective countries.

RWE AG has committed to invest over EUR50 billion in renewable technologies through 2030, 40% of which is slated for onshore wind and solar projects, 35% for offshore wind, and the remainder for hydrogen and battery technology. Germany's *Energiewende* transition policy prioritizes the phaseout of both coal and nuclear power in part by subsidizing investment in power plants that may be converted to hydrogen.

In contrast, around the same time, Enel S.p.A. committed to investing EUR43 billion over a three-year period in its grid networks and renewable projects. Around 60% of Enel's total investment is to be allocated to the improvement and expansion of its electricity grids to accommodate sources of renewable energy. Enel's investment strategy seeks to promote infrastructure digitalization to make electricity distribution more efficient, with the goal of equipping 80% of its customers with smart meters by 2025.

Transition risks and opportunities are of similar importance for industries representing energy consumers, such as airlines, global shipping, the automotive industry, and other manufacturing industries that are users of fossil fuels responsible for carbon emissions. Changes in regulations, taxes, and/or consumer preferences will continue to lead companies in these sectors to evolve their business operations or face market share losses to competing solutions with lower carbon emissions.

QUESTION SET 4

1. Which of the following represents the prevention of the threat to industry entrants for an indefinite period?

- A. Patent protection
- B. Licensing requirements
- C. Proprietary technology

Solution:

C is correct because the existence of proprietary technology can prevent the threat of entry only as long as other potential market participants fail to replicate a company's technology.

A is incorrect because patent protection provides a competitive advantage by preventing industry entry for a finite period of time.

B is incorrect because licensing requirements provide a competitive advantage for a specific period of time.

2. Which of the following set of factors is most likely to lead to greater bargaining power among an industry's suppliers?

- A. Higher concentration of available suppliers, more tailored inputs, and high switching costs

- B.** Lower concentration of suppliers, less tailored inputs, and higher switching costs
- C.** Higher concentration of suppliers, more tailored inputs, and lower switching costs

Solution:

A is correct because more concentrated suppliers, more tailored inputs, and high switching costs lead to higher supplier bargaining power.

B is incorrect because less concentrated suppliers and more standard inputs reduce supplier bargaining power.

C is incorrect because low switching costs reduce supplier bargaining power.

3. Which of the following forms of innovation represents technology that establishes a new product or changes businesses with a significant effect across industries?

- A.** Breakthrough innovation
- B.** Sustaining innovation
- C.** Disruptive innovation

Solution:

A is correct because breakthrough innovation involves a new product, service, or business model that may fundamentally change existing businesses across multiple industries.

B is incorrect because sustaining innovation involves incremental technological improvements often led by industry incumbents that improves existing products or services.

C is incorrect because disruptive innovation can create a new market or enter an existing one with a different value proposition.

SUMMARY: INDUSTRY AND COMPETITIVE ANALYSIS

- Analysts apply industry and competitive analysis to define and identify value drivers, competitive forces, and external factors affecting groups of companies that offer similar products and services to their customers.
- The first step in conducting industry and competitive analysis is to classify an industry and its constituents, whether using third-party classification schemes or other criteria, such as geography.
- Once the scope of an industry is established, an industry survey, including overall size, growth, and profitability and covering the competitive landscape, provides a basis for further assessment. Industry dynamics may be compared to economic trends to determine the industry's stage in the life cycle, business cycle sensitivity, and level of competitive rivalry, including market shares and trends of key industry participants.
- Porter's Five Forces provide a framework to analyze an industry's structure as defined by the level of competitive rivalry. Key factors include the threat of new entrants, the threat of substitutes, and customer and supplier bargaining power. These forces may be evaluated based on the width of an economic moat, which is affected by intangible assets, switching costs, network effects, cost advantages, or efficient scale within an industry or among firms.
- PESTLE analysis is conducted to assess the effect of political, economic, social, technological, legal, and environmental factors on an industry's prospective growth rate and market share dynamics.

PRACTICE PROBLEMS

1. Which of the following statements best describes the characteristics of growth companies?
 - A. Unlike defensive or cyclical companies, growth companies tend to exhibit smaller revenue declines in an economic downturn.
 - B. Unlike defensive or cyclical companies, growth companies tend to exhibit higher profitability during economic expansion.
 - C. Unlike defensive or cyclical companies, growth companies exhibit higher revenue growth independent of economic fluctuations.
2. Which of the following statements provides the best justification for grouping companies using a geographic rather than industry approach?
 - A. Firms that face similar economic conditions in the same country may have more common value drivers than those in the same industry.
 - B. Firms listed on the same exchange may have more common value drivers than those in the same industry.
 - C. Firms that report financial statements in the same currency may have more common value drivers than those in the same industry.
3. Which of the following statements best describes the challenges of using a bottom-up approach to measure industry size?
 - A. A bottom-up approach is difficult to apply when government statistics for a specific sector are unavailable.
 - B. A bottom-up approach is difficult to apply when an industry classification involves a strict hierarchical taxonomy.
 - C. A bottom-up approach is difficult to apply when an industry includes a large proportion of private companies or small businesses.
4. Which of the following statements best describes third-party industry classification schemes?
 - A. Third-party classification schemes combine companies with a similar business model into the same industry.
 - B. Third-party classification schemes align companies to a single industry that represents their primary operating activity.
 - C. Third-party classification schemes assign the different activities of companies operating across industries to each of their different industries.
5. Which of the following statements best describes the use of vertical integration as a competitive strategy?
 - A. Vertical integration may be used to expand operations into an earlier stage of production to offset supplier bargaining power.

- B.** Vertical integration may be used to outsource production to offset consumer bargaining power.
 - C.** Vertical integration may be used to expand operations in order to reduce the threat of substitutes for an industry's products or services.
- 6. Which of the following statements best describes the contrast between the impacts of basic research and sustaining innovation on an industry?
 - A.** Basic research involves developing an understanding of scientific principles, while sustaining innovation aims to minimize a firm's environmental impact.
 - B.** Basic research creates new industries, while sustaining innovation fundamentally changes existing businesses across multiple industries.
 - C.** Basic research generally has no immediate commercial application, while sustaining innovation involves incremental improvements to existing products or services.
- 7. Which of the following sets of industry circumstances minimizes the threat of entry?
 - A.** A greater proportion of intangible assets, lower network effects, and lower switching costs
 - B.** A lower proportion of intangible assets, greater cost advantages, and less efficient scale
 - C.** A greater proportion of intangible assets, higher network effects, and higher switching costs
- 8. Which of the following statements is consistent with lower industry concentration?
 - A.** Lower industry concentration is consistent with a higher Herfindahl–Hirschman Index.
 - B.** Lower industry concentration is generally consistent with greater competitive intensity.
 - C.** Lower industry concentration is generally consistent with lower product differentiation.

SOLUTIONS

1. C is correct because growth company performance is relatively insensitive to the business cycle.
A is incorrect because the revenue of growth companies is typically less related to economic cycles.
B is incorrect because growth companies exhibit higher revenue growth, which may or may not be accompanied by high profitability.
2. A is correct because companies operating within a jurisdiction may have more common value drivers due to political, economic, and social policies and conditions than those operating in various jurisdictions within an industry.
B is incorrect because operating within the same industry typically results in more common value drivers relative to listing on a common exchange.
C is incorrect because operating within the same industry results in more common value drivers relative to having the same reporting currency.
3. C is correct because it is more difficult to estimate revenue for private companies or small businesses than to access disclosures for large, publicly listed companies.
B is incorrect because strict hierarchical taxonomies are not a challenge specific to a bottom-up approach to estimating industry size.
A is incorrect because the lack of government statistics for a sector is a challenge for a top-down—not a bottom-up—approach to estimating industry size.
4. B is correct because third-party classification schemes are based on a strict hierarchical taxonomy that assigns each company to a single industry.
A is incorrect because companies within the same industry may have different business models.
C is incorrect because companies operating across industries are assigned to a single industry that represents a firm's largest business activity.
5. A is correct because vertical integration may be used to offset supplier bargaining power by expanding production to an earlier stage.
B is incorrect because vertical integration involves the insourcing rather than the outsourcing of production processes.
C is incorrect because vertical integration does not address the threat of substitutes for a company's products or services.
6. C is correct because basic research involves discoveries with no immediate commercial impact, while sustaining innovation makes small improvements to existing products or services.
A is incorrect because sustaining innovation does not necessarily address a product's environmental impact.
B is incorrect because basic research generally requires longer periods until commercialization, while breakthrough innovation fundamentally changes existing businesses across multiple industries.
7. C is correct because more intangible assets, higher network effects, and higher switching costs all represent barriers to entry.
A is incorrect because higher, not lower, network effects and switching costs deter industry entrants.
B is incorrect because higher, not lower, intangible assets and more efficient scale are factors limiting industry entrants.

8. B is correct because lower industry concentration involves many industry participants supplying a good or service, which generally leads to higher competitive intensity.

A is incorrect because lower industry concentration is consistent with a lower, not higher, Herfindahl–Hirschman Index.

C is incorrect because product differentiation is not determined by the degree of industry concentration.

LEARNING MODULE

10

Company Analysis: Past, Present, and Future

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	explain a company's competitive strategy and position within an industry
☐	evaluate a company's revenue drivers, operating profitability, and pricing power
☐	evaluate a company's working capital, capital investments, and capital structure

INTRODUCTION: COMPANY ANALYSIS: PAST, PRESENT, AND FUTURE

1

This learning module builds on prior industry and competitive analysis by focusing on an individual company's strategy and positioning. Financial analysts must gain a thorough understanding of these elements in order to explain the key **drivers** of a company's operating, investing, and financing activities in the past, as well as the present and future. These factors support the evaluation of company performance using income statement and balance sheet metrics. A thorough company analysis provides critical context for company valuation and serves as a key component of a comprehensive equity research report.

In the second lesson, we address a firm's market and pricing strategy and its competitive position within an industry. The firm's business model and its addressable market, its current stage in the company life cycle, and the size of its economic moat as described in the prior learning module are important elements in this analysis. Next, we turn our focus to how these factors affect a firm's income statement, including revenue drivers, **pricing power**, and operating profitability and its balance sheet dynamics, including investments in assets and debt and equity financing. This analysis provides a narrative to substantiate revenue and earnings forecasts as introduced in an earlier learning module, allowing analysts to support an investment recommendation and intrinsic value estimate for a specific company in an equity research report.

PRE-TEST: COMPANY ANALYSIS: PAST, PRESENT, AND FUTURE

1. Compared to a firm's serviceable addressable market, its obtainable addressable market is further constrained by the company's:

- A. focus.
- B. scale.
- C. geographic presence.

Solution:

B is correct. B is correct because a firm's serviceable addressable market depends only upon a firm's focus and geographic presence, while its obtainable addressable market is further constrained by the firm's current scale and strategy.

A is incorrect because a firm's serviceable market also depends on its focus.

C is incorrect because a firm's serviceable market also depends on its geographic presence.

2. Cost leadership strategies are *most* common for:

- A. startup companies.
- B. differentiated products.
- C. commoditized products.

Solution:

C is correct. C is correct because cost leadership strategies can offer companies an attractive competitive strategy when little to no differentiation exists among competitor products.

A is incorrect because startup companies are commonly establishing a business model with little to no revenue and negative cash flow and are usually not in the position to feasibly establish cost leadership.

B is incorrect because a lack of commoditization within an industry is more likely to lead to differentiation strategies than cost leadership.

3. A firm is *most likely* to have pricing power if it:

- A. supplies price-elastic goods.
- B. supplies price-inelastic goods.
- C. uses price-elastic goods as inputs.

Solution:

B is correct. B is correct because price-inelastic goods are those for which changes in price have little effect on demand, increasing the pricing power of firms who supply them.

A is incorrect because price-elastic goods are those for which changes in price have a greater effect on demand, lowering the pricing power of firms who supply them.

C is incorrect because using price-elastic goods as inputs has an indeterminate effect on a firm's pricing power.

4. Which of the following is *most likely* an example of a top-down approach to company revenue analysis?

- A. The firm's geographic sales distribution.
- B. The firm's product mix.
- C. The firm's market share within an industry.

Solution:

C is correct. C is correct because the economy, the firm's industry, and the firm's market share reflect top-down approaches to analyzing company revenue.

A is incorrect, as a firm's geographic sales distribution represents a bottom-up, not a top-down, approach to company revenue analysis.

B is incorrect, as a firm's product mix represents a bottom-up, not a top-down, approach to company revenue analysis.

5. A shorter cash conversion cycle implies that a company:

- A. has more working capital.
- B. pays its suppliers more quickly.
- C. efficiently manages inventory and receivables while obtaining favorable payment terms from suppliers.

Solution:

C is correct. C is correct because the cash conversion cycle is equal to days of inventory on hand plus days sales outstanding less days payable outstanding. Shorter cash conversion cycles therefore involve minimizing inventory and collection periods while maximizing the time to meet payable obligations.

A is incorrect because a shorter cash conversion cycle implies less, not more, working capital.

B is incorrect, as more quickly meeting payment obligations to suppliers does not shorten but rather lengthens the cash conversion cycle.

6. In a given year, a company can increase its free cash flow to equity (FCFE) by increasing its:

- A. debt.
- B. working capital.
- C. capital expenditures.

Solution:

A is correct. A is correct because an increase in debt is considered a cash flow to shareholders. A company's FCFE is equal to cash flow from operations minus capital expenditures plus net debt issuance.

B is incorrect, as an increase in working capital is a use of funds that lowers cash flow from operations and reduces FCFE.

C is incorrect, as increasing capital expenditures reduces FCFE.

2

COMPANY COMPETITIVE STRATEGY AND INDUSTRY POSITION

explain a company's competitive strategy and position within an industry

An earlier Corporate Finance learning module described how companies create value via their business model, which includes the firm's existing and prospective customers, products or services offered to those customers, key resources and assets used by the business, as well as its key partners and suppliers.

A firm's competitive strategy is a key determinant of its business model. All firms have a competitive strategy, whether intentional or not. An *intentional* strategy is one which informs the company's business model, sets measurable goals communicated internally and externally to investors, tracks performance and incorporates a feedback loop to refine and improve the company's future strategy. An *unintentional* strategy results from managers pursuing their own incentives, repeating whatever was done in the past, or simply copying industry leaders or professional norms.

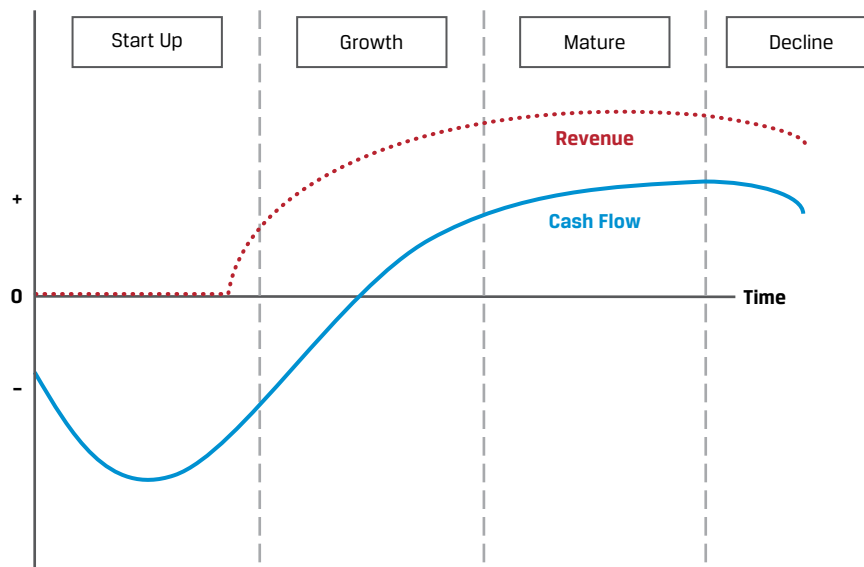
Companies with effective strategies are more likely to create value for their investors. While public stock prices may reflect perceptions about value creation, it is difficult to correctly judge value creation until time has passed. To evaluate competitive strategies on a forward-looking basis, analysts should consider three dimensions:

- How well does the strategy defend the firm against Porter's Five Forces?
- Is the strategy aligned with, or at least not in conflict with, external industry factors identified in a PESTLE (political, economic, social, technological, legal, and environmental) analysis?
- Does the firm have the resources and capabilities to execute the strategy?

The ability of individual firms to create value on a sustained basis is influenced by the company and industry's stage of development, a company's reaction to the competitive dynamics within its industry, as well as the external influences that shape its operating environment.

Competitive Strategies for Startups, Growth Companies, and Firms in Decline

A company's competitive strategy and positioning varies significantly over the company life cycle. As described in earlier Corporate Finance learning modules, firms are commonly categorized by their life-cycle stage as shown in Exhibit 1. We first address strategic implications for startups, growth firms, and firms in decline in this section of the lesson and devote the remainder of the lesson to mature firms.

Exhibit 1: Company Life-Cycle Stages

Startup companies usually involve firms with little or no revenue, negative cash flow, and few assets, while growth companies often transition to profitability and exhibit above-trend revenue and earnings growth. Given the limited operating history of these firms, analysts must rely on factors beyond historical performance when conducting company analysis.

In the earliest stage of development, a company's primary focus is often to *establish* a business model. When engaged in new product development, a startup firm must create a prototype or otherwise establish the ability to address an unmet customer need. These firms face a very high level of risk and likelihood of failure as illustrated in the following example.

EXAMPLE 1

BioAge Labs' Obesity Drug Innovation

An earlier Equities learning module described the application of Porter's Five Forces in the health care industry's product development of diabetes and obesity solutions due to the success of Ozempic, Novo Nordisk's patented weight-loss drug that generated nearly USD17 billion of revenue in 2024.

One example of an early-stage company is BioAge Labs Inc. (BIOA), a US-based biotechnology firm. Unlike large health care firms such as Eli Lilly or Pfizer with existing products, the company had no existing products or revenue, focusing solely on clinical development of new therapies for obesity and metabolic diseases. In early trials, BioAge Labs' leading oral compound azelaprag showed promise in enhancing muscle metabolism, reducing muscle atrophy, and maintaining fitness among sedentary subjects. This initial success led to several rounds of private equity funding followed by an IPO in September 2024.

Exhibit 2: BioAge Labs Inc.'s Share Price, September 2024–March 2025

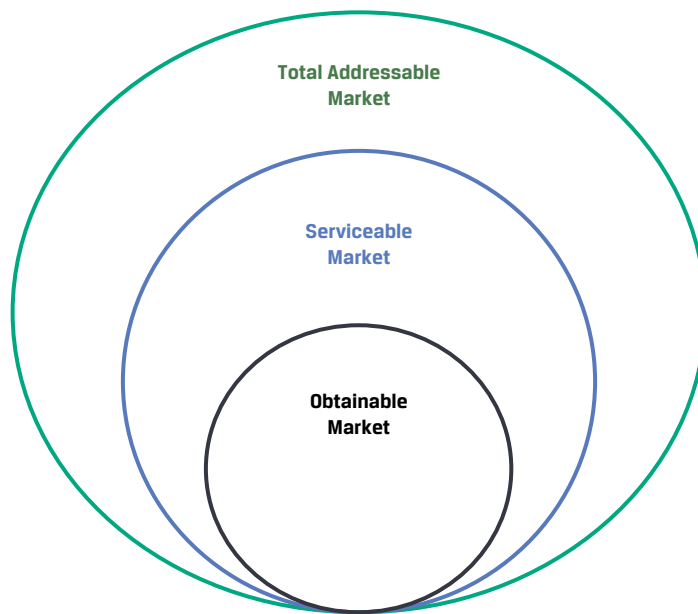


Source: Bloomberg.

Shortly before the IPO, the company initiated a Phase 2 trial with human subjects and combining azelaprag with Eli Lilly's existing obesity drug to target superior results. BioAge Labs' market capitalization reached a high of nearly USD900 million as the trial progressed. However, in December 2024, the firm announced the discontinuation of the trial as some patients faced elevated levels of liver enzymes, causing the company's share price to plunge by 80%. In March 2025, BioAge Labs officially abandoned its development of azelaprag due to the safety concerns raised and elected to focus on other therapies.

The strategy of startup firms developing new products is often centered on innovation, partnerships, and raising equity capital to fund research and development. Despite the presence of promising elements as in Example 1, a startup company's success is likely to face many challenges and setbacks.

Existing industries with low barriers to entry may face disruptive new market entrants that engage in business model innovation. Unlike those firms offering new products, these companies may deliver the same product to those sold by industry incumbents in a new way to capture market share. When an early-stage or growth company enters a more established market, company analysis is often focused on sizing the firm's market potential and its ability to reach or sustain above-trend growth. The size of a firm's revenue potential is often analyzed by estimating its **total addressable market (TAM)**, **serviceable market**, and **obtainable market** for each of the company's goods or services. The definition of TAM reflects the largest potential market for a good or service. The estimation of a serviceable market refines a company's TAM by incorporating constraints associated with the company's focus and geographic presence. Finally, defining an obtainable market integrates the company's current scale and strategy into the size of the serviceable market. Exhibit 3 illustrates these measures of potential market size.

Exhibit 3: Addressable Market Measures

For instance, consider IndiGo Air, a high-growth Indian firm that is a relatively new entrant in the established airline industry. By following a low-cost, no-frills strategy that offers affordable fares and a high degree of operational efficiency, the company has captured significant market share since it was founded. A forward-looking evaluation of IndiGo Air's strategy, keeping in mind IndiGo's growth company status, must take the potential market size into account as shown in the following example.

EXAMPLE 2**Estimating IndiGo Air's Addressable Market**

The ultra-low-cost Indian passenger airline, IndiGo Air, operated by InterGlobe Aviation Limited, initiated service in 2006. The airline expanded to international flights after five years and became India's largest airline by market share in the following year. IndiGo Air's revenue grew at a compound average growth rate (CAGR) over the decade through 2023 of 17% per year.

Analysts estimating IndiGo Air's future growth trajectory must consider the company's addressable market. In an earlier Equities learning module, available seat kilometers (ASK) was introduced as a measure of airline capacity equal to the number of seats available for scheduled airline passengers multiplied by kilometers flown. Given COVID-19-related flight disruptions during the period, Exhibit 4 shows that total ASK declined across the selected Asian airlines between 2019 and 2023, and the total ASKs likely are a reasonable estimate of IndiGo's TAM. However, IndiGo Air's ASK grew substantially, and its relative market share approximately doubled versus its Asian airline competitors.

Exhibit 4: Available Seat Kilometers (ASK) for Selected Asian Airlines

ASK (millions)	2023	%	Change	2019	%
Total	1,673,065	100.0%	---	1,734,294	100.0%
China Southern	316,217	18.9%	−1.7%	344,062	20.6%
Air China	292,513	17.5%	0.3%	287,788	17.2%
China Eastern	244,960	14.6%	−1.5%	270,254	16.2%
Singapore Airlines	163,468	9.8%	−0.5%	171,211	10.2%
Qantas	141,357	8.4%	1.8%	111,870	6.7%
IndiGo Air	139,300	8.3%	3.8%	75,051	4.5%
All Nippon Airways	120,968	7.2%	−1.0%	138,515	8.3%
Japan Airlines	82,513	4.9%	−0.5%	90,110	5.4%
Korea Air	80,777	4.8%	−1.2%	101,108	6.0%
Cathay Pacific	48,924	2.9%	−3.5%	107,991	6.5%
Air New Zealand	42,067	2.5%	0.3%	36,335	2.2%

Source: Bloomberg.

This total addressable market should be further adjusted to reflect IndiGo Air's *serviceable* addressable market or the portion of the TAM that may be reached based upon IndiGo's geographic presence as described below.

An important constraint in the transportation industry that also applies to air travel in Asia is **cabotage restrictions**. Cabotage restrictions limit or prevent airlines from one jurisdiction from carrying passengers or cargo within the domestic borders of another jurisdiction. Since foreign airlines are generally prevented from operating domestic flights elsewhere, IndiGo Air's serviceable market within Asia is limited to cross-border flights and those within India, which constitute approximately 55% of the totals in Exhibit 4. This also implies that IndiGo faces a different set of competitors for domestic travel, which composed 60% of its flights in 2023, compared to those it competes with on cross-border routes within Asia.

Finally, IndiGo Air's *obtainable* addressable market is constrained by the full capacity utilization of its current fleet, which reached 300 Airbus aircraft in early 2023, as well as its ability in the near term to acquire or lease additional aircraft.

As illustrated in the previous example, a growth company's revenue potential is closely linked to its ability to expand beyond its obtainable addressable market while considering near-term operational constraints. Successful expansion requires identifying demand drivers and overcoming industry-specific regulatory or competitive challenges. Asian passenger airlines are restricted from offering domestic travel within foreign jurisdictions but face varying degrees of domestic competition. For example, rail links between Japanese or Chinese Mainland destinations offer a close substitute for air travel, while the lack of a domestic high-speed rail network benefits carriers such as IndiGo Air and enhances their cross-border competitiveness. The following knowledge check underscores the importance of these parameters when assessing competitive strategies and addressable markets for growth firms.

KNOWLEDGE CHECK: SERVO AND TEA, LTD.

The prior learning module described Australia's food retail industry as composed of several thousand supermarket, grocery, and convenience store outlets with annual revenue of over AUD170 billion.

Servo and Tea is a fictional convenience store outlet with six outlets along Australia's Highway 1 in Western Australia. The chain attracts travelers as a fuel stop and convenience store, combined with a specialty restaurant catering to tourists. Given its success in the region, Servo and Tea plans two additional outlets in Western Australia.

1. Which of the following opportunities best describes Servo and Tea's serviceable market?

- A. Sales potential for the company's existing outlets at full capacity
- B. Same-store sales growth and the rollout of its two new outlets
- C. Existing and planned outlets as well as expansion to other regions

Solution:

C is the correct answer because Servo and Tea's existing outlets as well as potential expansion are serviceable given its focus and geographic presence.

A is incorrect because it represents Servo and Tea's existing market, not its serviceable market.

B is incorrect as it represents the company's obtainable, not serviceable, market.

2. AUD170 billion is *best* described as Servo and Tea's:

- A. obtainable market.
- B. serviceable market.
- C. total addressable market.

Solution:

C is correct because Servo and Tea's total addressable market reflects a large-scale market for Servo and Tea's products and services. In this case, the Australian retail food industry as a whole serves as a reasonable proxy.

A is incorrect because Servo and Tea's obtainable market includes same-store sales growth and its two new outlets.

B is incorrect because the serviceable market includes the potential for Servo and Tea to expand to other regions but does not amount to the entire Australian retail food industry.

Companies are considered to be in decline if their revenue is either falling or growing at a pace well below that of nominal GDP or well below that of industry competitors. Weak revenue performance is often accompanied by declining profitability and results from changing tastes and lower consumer demand, the emergence of new and attractive substitutes, or technological obsolescence. For example, many firms whose revenue was historically dominated by print media or fixed line telecommunications either are no longer in existence, consolidated with other businesses, or adapted to embrace internet or wireless technology.

Firms in decline often operate in industries with excess capacity, a factor that intensifies competitive rivalry among industry participants. Common competitive strategies for companies in this life-cycle phase include cost management, restructuring

operations, and downsizing or exiting businesses. However, in some instances, companies may pursue strategies to reinvent weak business lines. An example of one firm's strategy to address stagnating lines of business is outlined in Example 3.

EXAMPLE 3

LG Corporation's Strategic Renewal

Founded in 1947 by Koo In-hwoi as a chemical company, the South Korean conglomerate LG Corporation subsequently expanded to the electronics, household appliance, information technology, power generation, and telecommunications industries.

In the home appliance and electronics industries, LG initially focused on low-priced appliances and cellular phones referred to as “feature phones” with LCD screens and keypads. The firm's established market position was challenged in the late 1990s and early 2000s amid growing competition and accelerating technology. For example, new home appliance producers such as Chinese Mainland-based Haier Group increased industry rivalry in the affordable appliance segment, while mobile phone competitors such as Apple and Samsung introduced innovations such as touchscreens and operating systems.

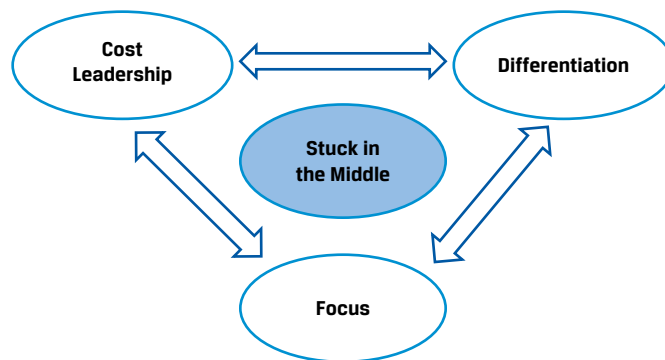
LG's competitive response to these changes was to heavily invest in research and development, leading to breakthroughs in display technology and smart appliances. For example, LG pioneered the use of artificial intelligence and internet of things (IoT) technology to allow smartphones or voice assistants to control and monitor devices. As a result, LG became a dominant provider in the growing premium home appliance market.

Conversely, LG's attempt to compete with industry leaders Apple and Samsung in the smartphone business was less successful. LG made extensive investments in a model upgrade based on a Windows operating system, and this model was poorly received by the market. Although LG eventually shifted to an Android-based operating system, it was unable to capture sufficient market share and eventually decided to exit mobile phone production in 2021.

Example 3 highlights the dynamic nature of competitive strategies. The forces of creative destruction in which new innovations render existing products obsolete is a constant threat to most businesses. Catalysts such as consumer demand changes or technological developments are often at work, whether firms are in an early phase seeking to scale above-trend growth or in a phase of contraction where downsizing and cost efficiency gains are more prevalent. We now turn our attention to strategies common among mature, established businesses.

Competitive Strategies among Established Companies

Firms with an established business model and operating history offer a basis for comparison when considering competitive strategy on a forward-looking basis. Exhibit 5 illustrates three common competitive strategies identified by Professor Michael Porter, whose Five Forces analysis was introduced in earlier learning modules.

Exhibit 5: Competitive Strategies of Established Firms

Variations of these generic strategies shown in Exhibit 5 are prevalent across companies and industries, while an unsatisfactory outcome is for a company to be “stuck in the middle,” that is, to participate in an industry where it is neither a cost leader, nor differentiated, nor focused.

Cost Leadership

A **cost leadership** strategy involves becoming the lowest-cost provider of goods or services among all competitors within an industry. The prevalence of cost leadership strategies is greater in industries where products are relatively commoditized or those that offer little to no differentiation among competitor products. As these goods and services, as well as the firms producing them, are usually in a mature phase with limited industry innovation, customers tend to be more cost conscious and less focused on distinguishing among offerings. Examples of such industries include firms involved in processing or refining commodities as well as consumer staples.

Cost leadership strategies may take various forms across different companies and industries. For instance, firms may seek to reduce average cost per unit by increasing the scale of production or capitalize on economies of scope by reducing the average cost of supplying multiple products. As described in the prior learning module, the global beer industry consists of many small microbreweries serving local markets and relatively few large publicly listed brewers such as AB Inbev and Heineken Holding. These large global beer producers have followed a strategy of acquiring small craft breweries and then diversifying their product portfolios while lowering costs by expanding the use of existing storage and distribution networks.

Firms may also pursue a cost leadership strategy by controlling costs. For firms with high fixed costs, this often involves optimizing capacity utilization as described with the case of IndiGo Air in the following example.

EXAMPLE 4

IndiGo Air's Cost Leadership Strategy and Load Factor Results

IndiGo Air's rapid revenue growth was achieved in part due to its pursuit of a cost leadership strategy among its Asian airline industry peers. As described in the prior learning module, airlines often exhibit high cyclicalities in revenues

and expenses, with costs driven in part by jet fuel price volatility and fluctuating revenue over the business cycle. COVID-19-related shutdowns had a severe adverse impact on travel, from which many airlines were slow to recover.

Key elements of IndiGo Air's cost leadership strategy include the primary use of a single family of Airbus 320 aircraft to minimize maintenance and pilot and crew training costs, focusing on point-to-point or direct flights between destinations with high demand, and maintaining rapid turnaround times to minimize aircraft down time. The company further reduced its operational burden and increased flexibility to meet demand by entering a high proportion of so-called wet leases, or agreements where a third-party lessor or owner provides the crew, maintenance, and insurance in addition to the aircraft.

The results of IndiGo Air's cost leadership may be demonstrated by considering the airline's load factor, or proportion of revenue passenger kilometers (RPKs) to its available seat kilometers (ASK), versus industry competitors:

$$\text{Load factor} = \frac{\text{Revenue passenger kilometers}}{\text{Available seat kilometers}}$$

RPKs involve the total distance traveled by fare-paying passengers, and the ratio of RPK to ASK is a common measure of airline capacity utilization. Exhibit 6 shows the load factor for IndiGo Air as well as selected Asian airline competitors as of 2023.

Exhibit 6: Load Factors for Selected Asian Airlines, 2023

	RPK (millions)	ASK (millions)	Load Factor
Total	1,323,403	1,673,065	79.1%
China Southern	246,947	316,217	78.1%
Air China	214,173	292,513	73.2%
China Eastern	182,299	244,960	74.4%
Singapore Airlines	143,890	163,468	88.0%
Qantas	116,895	141,357	82.7%
IndiGo Air	119,700	139,300	85.9%
All Nippon Airways	90,930	120,968	75.2%
Japan Airlines	63,973	82,513	77.5%
Korea Air	68,053	80,777	84.2%
Cathay Pacific	42,258	48,924	86.4%
Air New Zealand	34,285	42,067	81.5%

Source: Bloomberg.

As shown in Exhibit 6, IndiGo Air's 2023 load factor of 85.9% is exceeded only by Singapore Airlines and Cathay Pacific, both carriers based in major global transit hubs (Singapore and Hong Kong SAR) and focused almost exclusively on long-haul intercontinental routes with limited competition.

Over the cycle including the COVID-19 downturn, Exhibit 7 illustrates that IndiGo Air's load factor performance shows a high degree of resiliency as compared to both the Asian total and the top 2023 load factor performers.

Exhibit 7: Load Factors for Selected Asian Airlines, 2019–2023

	2023	2022	2021	2020	2019
Total	79.1%	73.8%	61.4%	64.0%	81.4%
Singapore Airlines	88.0%	85.4%	30.1%	13.3%	82.4%
Cathay Pacific	86.4%	73.2%	27.4%	57.3%	81.8%
IndiGo Air	85.9%	82.1%	73.6%	68.7%	87.1%

Source: Bloomberg.

IndiGo Air's cost leadership among low-cost air carriers is demonstrated by its cost per air seat kilometer, or CASK. According to its 2025 investor presentation, IndiGo Air's CASK excluding fuel and foreign exchange effects was USD0.0337 versus an average USD0.0579 among selected low-cost carriers.

Framing a company's past results in the context of its competitive strategy as shown in the example provides analysts with a basis for projections and forward-looking analysis.

Differentiation

Differentiation as a competitive strategy can take many forms, from the quality or features of a product or service to its brand positioning through advertising to the customer experience a firm offers. The degree to which a differentiation strategy provides a competitive moat protecting a firm's market share or sustained profits depends upon whether these elements may be replicated by competitors. While patents or copyrights offer legal barriers to competition over the long term, other attempts to offer products and services that are distinct from those of competitors offer a short-lived advantage.

For example, passenger airlines have expanded well beyond delivering travelers to their destinations, with major carriers offering co-branded credit cards, in-flight meals and entertainment, frequent flyer programs that reward customer loyalty, and premium cabin seating, among other features. Although airlines generally face no restrictions when providing ancillary services, their conversion into higher revenue and greater profitability can vary widely across competitors. In other cases, competitors such as IndiGo Air differentiate their services by forgoing such offerings to cater to budget-conscious travelers.

In contrast to frequent flyer programs designed to create loyalty over a series of purchases, subscription-based business models can substantially increase a firm's ability to differentiate itself through customer engagement. Companies can closely monitor customer activity and offer incentives tailored to customer preferences while generating a steady, predictable revenue stream. The following example compares the differentiated strategies of two firms applying this strategy in the retail sector.

EXAMPLE 5

Costco Wholesale versus Amazon Prime

Costco Wholesale Corporation is a US-based firm that operates large membership-based warehouse retail outlets in the United States, Canada, and Mexico as well as a dozen other countries. While the company originally catered only to businesses, it has grown to nearly 900 locations and over 130 million individual and business subscribers. In 2024, Costco's nearly USD5 billion in membership

fees constituted 2% of total revenue and roughly two-thirds of its net income. While the company has relied on traditional physical retail stores, it has a small but growing e-commerce presence.

Costco Wholesale's business model is centered on a membership-based club format that offers discounted prices on a narrow selection of consumer products supplied in bulk. By generating recurring revenue through membership fees and operating warehouse-like stores with sparse layouts, Costco can offer goods at prices that are usually below those of non-subscription competitors.

Amazon Prime is a subscription-driven outgrowth of Amazon.com's successful e-commerce platform. Initiated in 2005 as a free two-day and discounted one-day shipping service in the United States, the Prime offering rapidly expanded across countries and services, reaching over 200 million subscribers in 25 countries as of 2025. In 2024, Amazon.com generated over USD44 billion, or nearly 7% of its total revenue, in the form of subscription fees.

What began as a faster delivery service, Amazon Prime has expanded across the company's ecosystem to offer an integrated combination of retail sales, third-party marketplace fees, and cloud services. By expanding its Prime offering to make video streaming services, Amazon Music, groceries, books, and medical prescriptions available at discounted prices, Amazon Prime represents both a differentiated offering and a formidable competitive moat.

Focus

Competitive strategies anchored around a focus-based approach usually incorporate elements of both cost leadership and differentiation that are applied to specific areas of a firm's business. Companies with intentional strategies frequently assess their relative performance across products and services, regions, or stages of the product cycle and adjust their business model. This may result in a firm redirecting its efforts to concentrate on a specific region, a narrower mix of products and services, or a specific area of value creation for existing lines of business. When a firm's focus is centered around a differentiation strategy, it typically seeks to specialize in a niche area with a unique product for which it can command premium pricing. Drug companies may choose to focus their business models on research and development activities as shown in the following example.

EXAMPLE 6

AstraZeneca's Ambition 2030 and Beyond

AstraZeneca plc, a British and Swedish pharmaceutical and biotech company, successfully averted a hostile takeover bid from US-based Pfizer Inc. in 2014, remaining independent in part by promising to grow annual revenue from USD26 billion to over USD45 billion in a decade. The company achieved this goal by narrowing its business model to focus on research, development, and commercialization of therapies in oncology, cardiovascular, renal, and metabolism (CVRM) as well as respiratory and immunology. At the same time, AstraZeneca divested stakes in over-the-counter medicines and formed partnerships to develop new therapies.

For example, in 2020 AstraZeneca reached an agreement with Japanese drug-maker Daiichi Sankyo Limited to co-develop and co-commercialize Enhertu, a patented oncology therapy for breast and lung cancer. AstraZeneca committed to making contingent payments in exchange for a 50–50 split in profits from the drug outside of Japan.

Building on its success, AstraZeneca announced new Ambition 2030 goals at its 2024 Investor Day, targeting USD80 billion revenue and an operating margin in the mid-30% range. To achieve this goal, AstraZeneca set out to further sharpen its research and development focus across five global R&D sites, highlighting a dozen new molecular entities, which are drug molecules not previously approved for use in humans, each of which was already in a trial phase with the potential to generate at least USD5 billion in revenue per year once approved. The company also added high-value new technologies such as more targeted therapies and gene editing to its deepening pipeline and partnerships.

By combining a competitive strategy with measurable financial goals, this example underscores the link between strategy and specific revenue drivers in generating financial projections used in equity valuation.

In other cases, a firm's focus may be primarily driven by cost considerations. Companies employing this strategy seek a competitive advantage by offering products or services at the lowest price within a specific, narrow market segment as described in the case of Ryanair (also the subject of relative valuation examples in an earlier Equities learning module) in the following example.

EXAMPLE 7

Ryanair's Cost-Based Focus Strategy

Ryanair Holdings was founded in 1984 as a budget airline flying between Ireland and London's Gatwick Airport. The company grew rapidly amid European airline deregulation in the late 1980s and 1990s as cabotage restrictions for intra-EU flights by European carriers were lifted. As of 2024, Ryanair Group operated 590 aircraft with over 3,500 daily flights to 235 destinations in 37 countries.

Ryanair's competitive strategy is focused on short- and medium-haul point-to-point routes within Europe. The carrier avoids major hubs in favor of secondary airports such as London Stansted to reduce costs and turnaround times, often operating multiple daily flights between key destinations. Ryanair reduces maintenance and training costs via its standardized fleet, over 95% of which is composed of company-owned Boeing 737 aircraft. The company caters to budget-conscious travelers by maintaining low base fares while charging extra for baggage, seat selection, and food. The company also reduces overhead and advertising costs by maintaining direct digital sales to customers via its website and app.

In the same way that an analyst might track product development milestones or actual revenue associated with new products to estimate a firm's future performance, cost-based competitive strategies provide a basis for analysts to project key drivers of performance using income statement and balance sheet measures as shown in the following lessons.

Exhibit 8 summarizes the forms, impact, applications, and risks of the competitive strategies outlined earlier.

Exhibit 8: Forms, Impact, Applications, and Risks of Common Competitive Strategies

	Cost Leadership	Differentiation	Focus
Forms	• Economies of scale • Access to materials or supply chain • Cost controls • Aggressive pricing • Low-cost distribution • Economies of scope	• Quality/features • Advertising/brand • Service/distribution • Intellectual property • Customer experience • Premium pricing • Product or service may be bundled with other services	• Elements of both cost leadership and differentiation focused on specific group or product
Five Forces Impact	• Threat of new entrants • Customer bargaining power • Industry rivalry	• Threat of new entrants • Threat of substitutes • Customer bargaining power • Industry rivalry	• Threat of new entrants • Threat of substitutes • Customer bargaining power
Industry Applicability	• Capital intensity • Price-conscious customers • Lack of product differentiation • Limited industry innovation	• Less price-conscious customers • Customers value product uniqueness • Industry innovation with changing product features	• Competitors face challenges in servicing specific customers, regions or delivering specific products
Risks	• Cost inflation, lack of discipline • Technological change affecting costs or market share • Customer desire to elevate product quality (premiumization)	• Imitation • Increased buyer sophistication • Pricing premium becomes excessive • Exclusivity prevents market share growth	• Large competitors outcompete on price • Narrowing of differences among groups • Increased buyer sophistication

QUESTION SET 1



1. A growth company’s revenue potential depends on its ability to:

 - A. grow its scale within its obtainable market.
 - B. increase its market share within today’s total addressable market.
 - C. expand in its serviceable market within operational constraints.

Solution:

C is correct because the company’s serviceable market represents the opportunity available for a growth firm’s product and geographic focus.

A is incorrect as a company’s obtainable market is one that is based upon a firm’s current scale and strategy, not growth in its scale.

B is incorrect because a growth company’s potential may or may not involve capturing market share and the total addressable market for its product may not be serviceable for a given company.
2. A firm that specializes in a narrow group of customers with a specific product with top-tier pricing is considered to follow which competitive strategy?

 - A. A differentiation strategy
 - B. A cost-based focus strategy
 - C. A differentiation-based focus strategy

Solution:

C is correct because the firm’s strategy is aimed at a market niche based upon a unique product offering.

A is incorrect as the firm's strategy involves a focused approach.

B is incorrect because even though the firm operates in a narrow market segment, its strategy does not reflect a low-cost provider approach.

3. Firms engaged in managing costs, restructuring operations, and downsizing or exiting businesses are commonly associated with which life-cycle phase?

A. Startup phase

B. Decline phase

C. Growth phase

Solution:

B is correct because firms in decline often face intense industry rivalry, resulting in cost-cutting, restructuring, and exiting businesses to maintain profitability.

A is incorrect because startup firms focus on establishing a business model to address an unmet customer need.

C is incorrect because growth firms seek to scale operations in a serviceable addressable market under operational constraints rather than downsizing.

4. Firms seeking an advantage by selling at the lowest price within a specific market segment are considered to follow:

A. a cost leadership strategy.

B. a differentiation strategy.

C. a cost-based focus strategy.

Solution:

C is correct because the firm's strategy is to pursue the lowest cost within a specialized market niche.

A is incorrect as a cost leadership strategy pursues a broad market approach rather than focusing on a consumer segment.

B is incorrect because even though a differentiation strategy involves operating in a narrow market segment, it does not reflect a low-cost approach.

COMPANY REVENUE, PRICING POWER, AND OPERATING PROFITABILITY

3



evaluate a company's revenue drivers, operating profitability, and pricing power

The following lessons build on the competitive strategies outlined in Lesson 2 by connecting a firm's business model, competitive strategy, market position, and operating environment to its performance. The income statement and balance sheet from the three-statement model introduced in earlier Level I learning modules and shown in Exhibit 9 provide a useful reference in these lessons for mapping company strategies to financial performance. We begin with a focus on a firm's income statement.

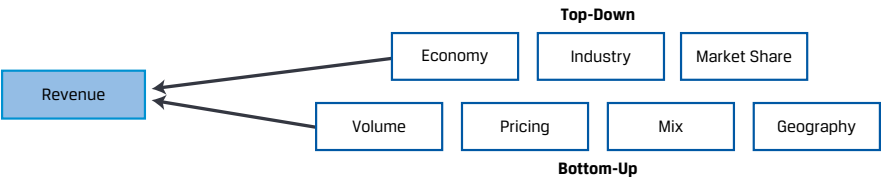
Exhibit 9: Income Statement and Balance Sheet

Income Statement	Balance Sheet ($t = 0$)	
Revenue	Cash ₀	Current Liabilities ₀
Operating Costs	Current Assets ₀	Debt ₀
Depreciation and Amortization	LT Assets ₀	Equity ₀
Operating Income	Balance Sheet ($t = 1$)	
Non-Op Income	Cash ₁	Current Liabilities ₁
Non-Op Expense	Current Assets ₁	Debt ₁
Debt Service	LT Assets ₁	Equity ₁
Taxes		
Net Income		
Dividends and Distributions		
Retained Earnings		

Revenue Analysis and Forecasts: Top-Down and Bottom-Up Approaches

Analysts often estimate a company’s revenue potential within the context of its industry growth characteristics as described in the prior learning module. Best practices in analyzing and forecasting revenue involve a combination of top-down analysis that incorporates insights from the overall economy and industry or industries within which a firm operates and a bottom-up approach that considers firm-specific elements. Exhibit 10 summarizes the drivers considered in these approaches to revenue analysis.

Exhibit 10: Top-Down versus Bottom-Up Revenue Drivers

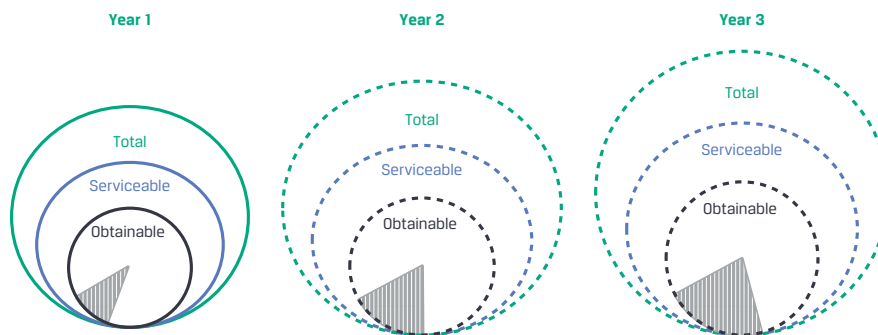


As in the case of other financial statement figures, this analysis involves identifying drivers or causative factors that explain both the level of and changes in a variable such as sales and understanding the evolution of such drivers over time. It is also important to consider expected future changes in company performance *relative* to other key industry participants. For example:

- Does the company lead or lag in adopting new technologies, business channels, or other factors driving industry trends?
- Has the company adequately planned for anticipated market size changes by initiating capital investments and research and development?
- How do the company's announced plans compare to those of major competitors facing similar market forces and external factors?

A top-down approach involves sizing a firm's future market opportunity and its prospective market share. A specific revenue forecast based on a firm's future share of its obtainable market, as depicted earlier in Exhibit 3, establishes consistency with industry and economic trends as well as competitor activity.

Exhibit 11: Firm Revenue Projections in an Industry Context



As shown in Exhibit 11, this revenue projection involves forecasting the total addressable, serviceable, and obtainable markets as well as a firm's market share within its *obtainable* addressable market. For example, in the case of IndiGo Air described earlier (see Example 2), a top-down revenue forecast might consider the company's prospective share of the future domestic Indian and cross-border Asian passenger air travel markets. The company's obtainable market is limited to flights using its existing fleet as well as any anticipated new aircraft deliveries.

A bottom-up approach to company revenue forecasting usually involves decomposing revenue into drivers such as sales volume and price, product lines, and geographies, while anticipating future changes in revenue composition. Supporting evidence for firm-specific revenue projections can vary widely across industries, depending in part upon whether revenue is recurring or episodic. In some cases, growth depends on increasing sales to the same customers, while in others, it may rely on new products, new channels, and/or new customers.

For example, retail industry analysts commonly distinguish between same-store sales and revenue from new outlets to compare repeat sales in existing locations to those of new markets. An earlier Equities learning module addressed the case of Starbucks Inc. over a period of declining same-store sales growth rates, which had an adverse effect on the company's share price. Subscription-based business models are often assessed by separating subscription renewals from new subscribers, which

are often attracted by discounted memberships or other promotions not extended to existing customers. The following example highlights changes in a company’s product mix due to changing tastes and technology within a bottom-up approach.

EXAMPLE 8

Toyota Motor Company’s Changing Product Mix

An earlier Equities learning module considered a top-down forecast of global automaker Toyota Motor Company’s 2025 revenue using 2024 as a base year. Using the company’s audited financial statements, footnotes, and financial data provider information, the earlier module showed that Toyota generated JPY45,095 billion in revenue in FY 2024 with 9.44 million in global vehicle sales, 39.2% of which were in North America.

A closer analysis of Toyota’s product mix based on US sales figures reveals significant changes in the composition of vehicles sold, as shown in Exhibit 12.

Exhibit 12: Toyota Motor North America US Sales, FY 2024 versus 2023

Vehicle Category	2024	2023	Daily Sales Rate Growth (%)	Volume Unit Growth (%)
Total Units	2,332,623	2,248,477	3.1%	3.7%
Cars	621,696	596,656	3.1%	4.2%
Trucks	1,643,340	1,584,433	3.0%	3.7%
Non-Electrified	1,326,162	1,591,143	---	−16.7%
Electrified	1,006,461	657,334	55.1%	53.1%
% Electrified	43.1%	29.2%	---	---

Source: Toyota Motor North America, FY 2024 and 2023 ending 31 March.

While electric and hybrid vehicles accounted for 20% of total US automobiles sold in calendar year 2024, Toyota’s US proportional sales of electrified vehicles of 43.1% reached over double the industry average, led by large increases in hybrid sales among its Camry and RAV4 SUV models. The shift to electrified vehicles under Toyota’s so-called “multi-pathway strategy” involved offering a wider range of carbon-neutral powertrains to compete with electric vehicle producers, such as Tesla and BYD, as well as traditional automakers. While hybrid models typically sell for a few thousand dollars more, on average, than their gasoline-powered equivalents, tax incentives often make such vehicles less costly to buyers.

To strengthen its US supply chain, Toyota also initiated battery manufacturing outside of Japan for the first time, with a USD14 billion capital investment in a new North Carolina battery plant starting production in 2025 and a wholly owned battery plant in Shanghai in 2027 with expected annual production capacity of 100,000 units.

As suggested in the example, significant product mix changes may also affect a firm’s profitability. In this case, hybrid vehicles commanding higher prices than gasoline-powered cars may have the potential to generate higher profits for auto-makers if costs of manufacturing hybrid cars are similar to those of manufacturing gas-powered vehicles.

Operating Profitability

Beyond revenue analysis and forecasting, assessing the past, present, and expected future profitability of a firm is an important next step in the process. Company analysts have three key choices when measuring profitability as illustrated in Exhibit 13.

Exhibit 13: Key Profitability Measures

Net Income	Operating Income	EBITDA
Revenue	Revenue	Revenue
Operating Costs	Operating Costs	Operating Costs
Depreciation and Amortization	Depreciation and Amortization	
Operating Income		
Non-Op Income		
Non-Op Expense		
Debt Service		
Taxes		

These three levels of profitability are commonly expressed as margins, or income as percentages of revenue. While often used interchangeably in company presentations and financial press releases, a thorough company analysis should distinguish between these income and margin measures, taking their strengths and weaknesses into account, as described in earlier Equities learning modules, as follows:

- *Net income and net margin:* As a widely reported measure of earnings available to shareholders employed in many metrics, net income provides the most comprehensive measure of profitability as it incorporates operating, financing, and investing activities as well as taxes. The comprehensive nature of this measure makes it difficult to compare across companies for the same period or the same company over time. For example, changes in capital structure, capital expenditure, or non-recurring items may distort this measure of firm profitability, with significant deviations from a company's true returns from ongoing operations in its primary area of business. Also, for early-stage companies or those in highly cyclical industries with periods of net losses, net income is a less useful indicator. For most companies in the growth, mature, or decline phases of the life cycle, however, net margin is a useful metric of analysis across companies and time.
- *Operating income and operating margin:* The exclusion of financing costs, taxes, and non-recurring items increases the comparability of this core measure of profitability when converted into operating margins across firms in an industry and the same company over time. Firms are also less

likely to experience negative operating margins versus net margins over the economic cycle. However, the inclusion of net investment expenditures, reflecting depreciation and amortization, may misrepresent true operating profitability for firms with different investment policies. For example, firms increasing capital expenditures today to expand and upgrade production capacity may experience lower operating margins in the near future due to higher depreciation, while those failing to invest will show higher operating margins in the near future, which may overestimate operating margins because of today's underinvestment.

- **EBITDA and EBITDA margin:** Although not a recognized earnings measure under GAAP or IFRS accounting standards, analysts will commonly encounter EBITDA as a profitability gauge and find it useful under specific circumstances. As a measure of operating profitability before discretionary expenditures, EBITDA allows the comparison across firms with different leverage, taxes, and investment expenditure policies. That said, no universal agreement exists on how this non-standardized earnings measure may be adjusted, so a degree of caution is warranted when EBITDA is used for comparative purposes.

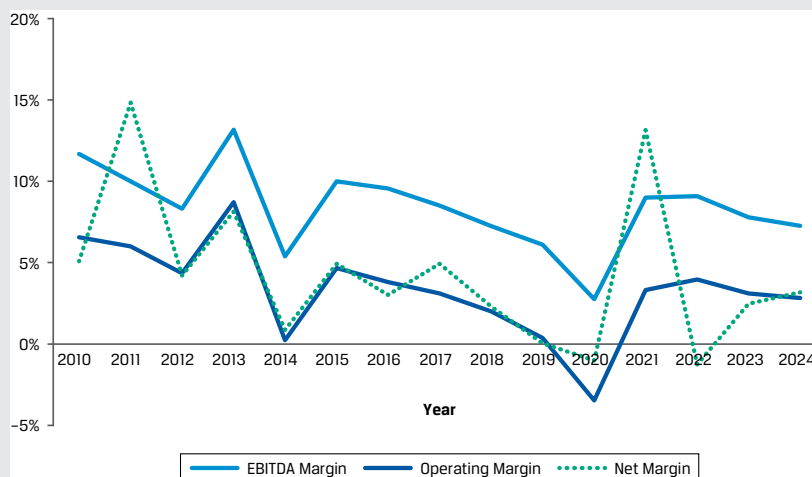
The following example contrasts these profitability measures for Ford Motor Company, introduced in earlier Equities learning modules.

EXAMPLE 9

Ford Motor Company's Profit Margins over the Economic Cycle

The highly discretionary nature of automobile purchases contributes to the cyclical nature of the global automotive industry. As the world's sixth largest manufacturer of passenger cars and commercial vehicles, Ford Motor Company's production and financial performance fluctuates over the economic cycle. Ford's EBITDA, operating, and net margins are shown in Exhibit 14.

Exhibit 14: Ford's EBITDA, Operating, and Net Margins, 2010–2024



Source: Bloomberg.

Although Ford's EBITDA margin and operating margin are closely aligned over the period, its net margin shows greater volatility, with positive spikes in 2011 and 2021 due to factors beyond its operations. For example, Ford's net margin soared to 15% in 2011 due to the firm's strong recovery from the severe

economic downturn following the Global Financial Crisis. Unlike its two largest US-based competitors, Ford avoided bankruptcy but increased its debt burden, the reduction of which improved its net margin. From 2010 to 2011, Ford's net interest expense fell by over USD1 billion.

In the aftermath of the COVID-19 pandemic in 2020, Ford's operating income recovered from losses in 2021, but its net margin soared to over 13%. This was due to an extraordinary USD8.2 billion gain on the IPO of electric vehicle startup Rivian, Inc. in which it held an ownership stake.

An alternative to the profitability ratios described in Example 9 is to analyze the components of a company's cost structure in greater detail over time as well as in comparison with its peers. This approach introduces more granularity into the company analysis process, allowing an analyst to measure a firm's financial progress in comparison to its competitive strategy and stated goals, to sharpen the valuation narrative, and to compare a firm's operating performance based on specific cost components to its peers.

EXAMPLE 10

Deconstructing Ryanair's Margin and Expenses versus Competitors

An analyst seeking to better understand European budget passenger airline Ryanair Holdings PLC's competitive strategy described earlier may consider a closer look at its operating margin as well as its underlying expenses over time versus European airline competitors. Exhibit 15 summarizes key income statement information from Ryanair and its peers in the three years following the COVID-19 pandemic-related shutdowns of 2020.

Exhibit 15: Selected Data for Ryanair Holdings and Competitors, 2021–2023

Statistic	2021	2022	2023
Operating Margin (% of sales)			
Industry Peers	–25.6%	4.3%	7.7%
Ryanair Holdings	–7.1%	13.4%	15.3%
Wages & Salaries (% of sales)			
Industry Peers	34.9%	16.0%	15.7%
Ryanair Holdings	14.4%	11.1%	11.2%
Fuel Costs (% of sales)			
Industry Peers	25.3%	32.0%	27.7%
Ryanair Holdings	35.4%	37.4%	38.3%
Maintenance Costs (% of sales)			
Industry Peers	10.3%	6.0%	5.2%
Ryanair Holdings	5.3%	3.5%	3.1%

Source: Bloomberg.

European airline competitors include Air France-KLM, Deutsche Lufthansa AG, easyJet PLC, Finnair Oyj, Icelandair Group HF, International Consolidated Air, Norwegian Air Shuttle ASA, Pegasus Hava Tasimaciligi AS, SAS AB, Turk Hava Yollari AO, and Wizz Air Holdings Plc.

Recall that Ryanair's focus is on short- and medium-haul routes for no-frills, budget-conscious travel within Europe, with a standardized fleet to minimize maintenance costs and low overhead and advertising costs achieved through direct digital sales to customers via its website and app.

While Ryanair's labor expenses and maintenance costs compare favorably to its European competitors as a percentage of sales, note that Ryanair's fuel costs constitute a much larger proportion of revenues than for its peers. While Ryanair has a reputation for cost efficiency, the greater-than-average impact of fuel on its cost structure stems from its lower average revenue per passenger as a low-cost carrier as well as the diminished fuel efficiency of short-haul, high-frequency flights. An analyst considering a future scenario involving higher jet fuel prices might therefore consider the greater potential adverse impact on Ryanair's profitability versus its competitors. In addition, the analyst should confirm that lower labor and maintenance costs result from greater efficiency rather than underinvestment in upkeep and personnel versus peers.

Pricing Power

A firm's pricing power reflects its ability to freely set prices and other economic terms for its products without adversely affecting demand. The more pricing power a company has, the greater its ability to improve margins in a stable market environment or offset rising costs under an inflationary scenario.

The features of goods and services supplied by a firm are one determinant of its pricing power. **Price elasticity**, or the change in quantity demanded relative to a price change, provides one measure of pricing power. Goods that are considered necessities, such as health care, tend to be price inelastic, with small changes in price having little effect on demand, while luxury goods or leisure travel tend to be price elastic, with price changes having a greater impact on demand. Firms offering price inelastic goods typically have greater pricing power than those supplying price elastic goods.

Other determinants of a firm's pricing power include the market structure of a firm's industry as described in earlier Economics learning modules, a company's competitive position in the markets in which it operates, and the sales channels used by firms, such as e-commerce- or subscription-based models that may increase a firm's ability to control its margins for products and services. Consider the following example of Netflix.

EXAMPLE 11

Netflix, Inc.'s, Revenue Drivers and Pricing Power

An earlier Equities learning module described how Netflix, Inc., the US-based video streaming service provider of films and television programming, greatly expanded its subscriber base during the COVID-19 pandemic and was able to retain and grow its business thereafter using tiered pricing and other strategies. Exhibit 16 provides a snapshot of Netflix's revenue and subscriber growth over the five years after the pandemic began in 2020.

Exhibit 16: Selected Financial Information for Netflix, Inc., 2020–2024

	2020	2021	2022	2023	2024
Total Revenue (USD mm)	24,996	29,698	31,616	33,723	39,001
Percentage Change	24%	19%	6%	7%	16%
% Domestic	43%	40%	37%	39%	38%
% International	57%	60%	63%	61%	62%
Total Subscribers (mm)	203.7	221.8	230.7	260.3	301.6
% Domestic	33%	31%	28%	29%	27%
% International	67%	69%	72%	71%	73%
Average Monthly Revenue per User (USD)	10.91	11.67	11.76	11.64	11.70
Domestic	13.61	14.50	15.78	16.27	17.33
International	9.67	10.11	10.38	10.14	9.19

Note: mm = millions.

Source: Bloomberg.

Note that this exhibit excludes Netflix's declining DVD business over the period, which constituted less than 1% of revenue. Following rapid growth during the COVID-19 pandemic when in-person entertainment options were severely restricted, the company faced increased competition from streaming providers such as Disney+ and Hulu.

Netflix's response included efforts to crack down on password sharing and a strategy to increase customer engagement and loyalty by creating and streaming original content. The company was also able to protect its leading market share position by introducing tiered pricing, with ad-free premium subscriptions and less expensive subscriptions that include advertisements. The company's growth in both subscribership and revenue per subscriber represents a shift from a "price taker" position streaming third-party content to a provider with unique offerings and greater pricing power.

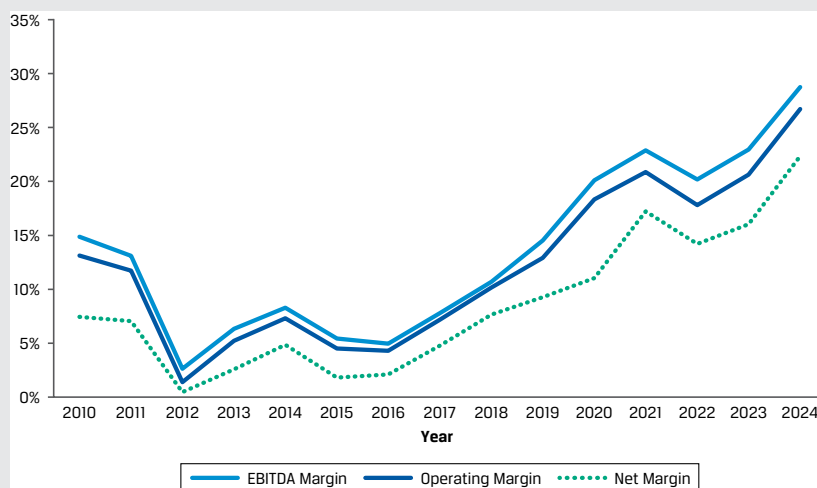
Firms operating in less competitive markets, such as monopolistic competition, oligopoly, or monopoly structures, have a higher degree of pricing power. They can raise prices or change other economic terms such as contract length without materially affecting sales volumes. A firm's *relative* market share may be more important than its absolute share in terms of its ability to affect pricing dynamics. For example, a firm with 10%–15% market share may be a market leader in supplying goods where many competitors are present or a market follower if it is the third largest among fewer suppliers. Markets with stronger pricing power are often characterized by significant product differentiation, high barriers to entry, a lack of substitutes, high customer switching costs, and/or a high degree of customer loyalty or retention. We revisit the example of Netflix to consider how the company's evolution has affected its measures of profitability over time.

EXAMPLE 12**Netflix's Evolving Business and Profitability Margins**

Founded as a DVD mail-order rental business in the late 1990s, Netflix soon shifted to a subscription-based service, initiating its streaming business over a decade later. Starting in 2010, the company aggressively ramped up spending to license copyrighted content from major film studios, and in 2013 the company released its first original series, *House of Cards*.

Netflix's annual content expenditures rose from nearly USD5 billion in 2016 to more than triple that amount in 2024, a large majority of which was spent on creating original content. Exhibit 17 shows how Netflix's profitability margins have evolved over this transformational period.

Exhibit 17: Netflix's EBITDA, Operating, and Net Margins, 2010–2024



Source: Bloomberg.

As shown in Exhibit 17, Netflix's margins reached a low in 2012 as the company rapidly increased its spending on external content, invested in its streaming delivery, and expanded internationally. The company's rapid expansion of content expenditure coincided with a sharp rise in its margins, underscoring its ability to attract and retain subscribers while maintaining its strong pricing power. While Netflix's EBITDA and operating margins are closely correlated, the company's increased use of debt financing to fund content is largely responsible for the widening gap between operating and net margins over time.

In the most competitive markets in which competitors sell very similar products, firms are price takers. That is, prices are dictated by the forces of supply and demand, with all firms generally selling at the same price. If a firm offers its products at a price above the prevailing market price, it should face declining sales as customers will buy from other firms. Offering products below the market price, conversely, would not only reduce profits but also would cause competitors to do the same, leading to a price war. Although capital returns in highly competitive markets can vary, over the long run they tend to approximate firms' cost of capital, as competition drives prices to equal marginal cost. In a competitive industry, low-cost producers may earn returns that exceed the cost of capital if they have a sustained cost advantage over others who supply the same goods at their marginal cost. For example, while the global price

of petroleum is determined by supply and demand, the majority state-owned Saudi Arabian oil producer Saudi Aramco has by far the lowest production costs among global competitors of less than USD10 per barrel.

Operating Leverage

An important distinction affecting the stability of a firm's profit margins is the proportion of fixed versus variable costs defined in earlier learning modules as operating leverage. We may define operating profit by using this distinction as shown in Equation 1.

$$\text{Operating profit} = [Q \times (P - VC)] - FC, \quad (1)$$

where Q is the quantity of goods sold in a period, P is the unit price of goods sold, VC equals variable operating cost *per unit*, and FC represents total fixed operating costs.

Examples of variable operating costs include inventory for retail firms or raw materials and hourly wages for manufacturing companies, which fluctuate with the level of output. Fixed operating costs are stated on a total, rather than per unit, basis, representing those costs that do not change within a given output range in the short run, including employee salaries, depreciation and amortization, software expenses, insurance, and some utility costs.

While standard financial disclosures do not classify expenses as fixed or variable costs, an earlier learning module introduced the use of regression analysis to estimate how a specific expense line item on the income statement changes in response to a given change in revenue. The greater an expense's sensitivity to revenue changes, the more likely it is to be categorized as a variable cost. Note that this use of historical data requires an assumption that the firm's cost structure has remained unchanged over the period.

In general, companies with a higher proportion of fixed costs, and therefore higher operating leverage, experience greater changes in cash flow and profitability for a given change in revenue than firms with low operating leverage. For instance, the greater margin fluctuations faced by Ford Motor Company over the business cycle, as shown in Exhibit 14, results in part from higher operating leverage as compared with Netflix in Exhibit 17.

QUESTION SET 2



1. A top-down approach to estimating company revenue is *most likely* to emphasize the company's:

- A. market share.
- B. product mix.
- C. geographic exposure.

Solution:

A is correct because gauging a company's obtainable market share represents a top-down approach to estimating the revenue level that may be reached given the company's current scale and strategy.

B is incorrect because an emphasis on product mix is more consistent with a bottom-up approach.

C is incorrect because an emphasis on geographic exposure is more consistent with a bottom-up approach.

2. When comparing companies with significant differences in leverage, taxes, and investment expenditure, it is *most* appropriate to focus on:

- A. net margin.
- B. EBITDA margin.
- C. operating margin.

Solution:

B is correct because EBITDA is a measure of operating profitability before discretionary expenditures, allowing for more direct comparison among companies with different financing structures, taxes, and capital expenditure policies.

A is incorrect as net margin levels may be distorted by differences in interest expense and taxes among other factors.

C is incorrect as operating margins may be affected by differences in capital expenditure policies.

3. For a company with a high degree of operating leverage, a 10% increase in revenue should increase cash flow and profitability by:

- A. less than 10%.
- B. 10%.
- C. more than 10%.

Solution:

C is correct because companies with high operating leverage, or significant fixed versus variable costs, experience a greater change in cash flow and profitability for a given change in revenue than companies with low operating leverage.

A is incorrect because operating leverage increases the sensitivity of profit and cash flow to sales growth.

B is incorrect because it characterizes a company that has only variable costs.

4. A company is *most likely* to enjoy pricing power when:

- A. barriers to entry are low.
- B. it sells a price-elastic product.
- C. it has a high relative share of the market.

Solution:

C is correct. Companies operating in less competitive markets, such as monopolistic competition, oligopoly, or monopoly structures, have a higher degree of pricing power. They can raise prices or change other economic terms such as contract length without materially affecting sales volumes. A company's relative market share may be more important than its absolute share in terms of its ability to affect pricing dynamics.

A is incorrect because low barriers to entry are associated with less pricing power.

B is incorrect because pricing power is associated with price inelasticity.

WORKING CAPITAL, CAPITAL INVESTMENTS, AND CAPITAL STRUCTURE

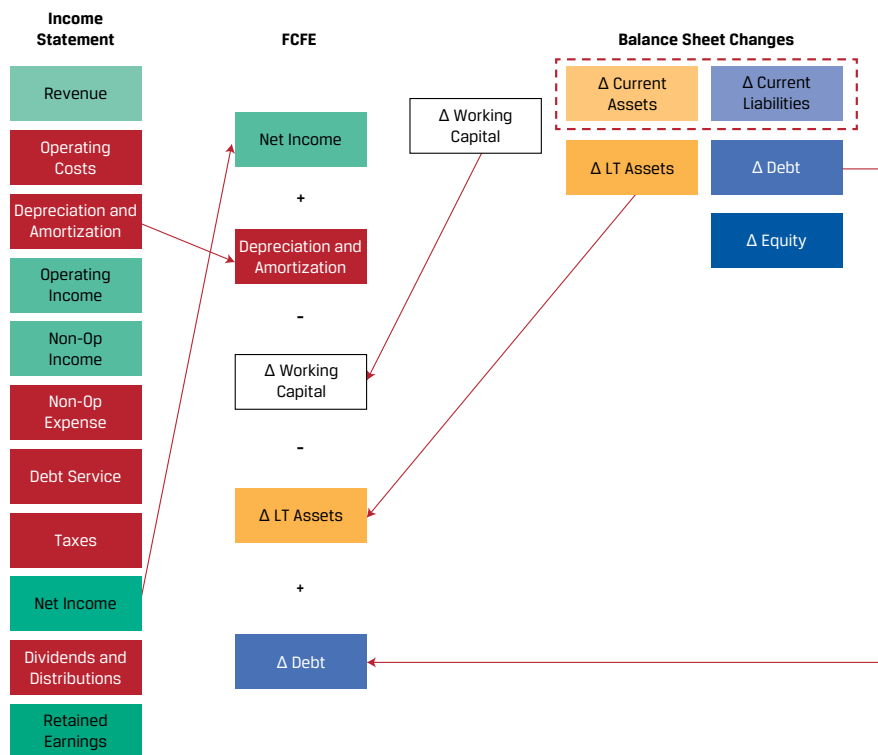
4

- evaluate a company's working capital, capital investments, and capital structure

A detailed company analysis should not only consider a firm's strategy and performance in the context of its income statement but should also incorporate key balance sheet elements. Key drivers of a company's performance reflected in its balance sheet include investments in working capital, long-term (LT) fixed assets, and incremental financing.

Free cash flow to equity (FCFE) was introduced in earlier Equities learning modules as the remaining cash flow available to shareholders once operating expenses, including taxes, are covered and all necessary investments for continuing operations and growth are made. The composition of FCFE from income statement and balance sheet measures is shown in Exhibit 18 and defined in Equation 2.

Exhibit 18: FCFE from Income Statement and Balance Sheet Changes



$$\text{FCFE} = \text{Net income} + \text{Depreciation \& amortization} - \Delta \text{ Working capital} - \Delta \text{ LT Assets} + \Delta \text{ Debt},$$

(2)

where

$$\Delta \text{ Working capital} = \Delta \text{ Current assets} - \Delta \text{ Current liabilities}$$

and

Net debt issuance (Δ Debt) = Additional borrowing – Debt repayment.

As shown in Exhibit 18 and Equation 2, FCFE captures the sources and uses of cash flow that summarize firm activities. Cash flow sources include net income and non-cash expenses such as depreciation and amortization from the income statement and incremental debt on the balance sheet, while cash flow uses include growth in working capital and long-term assets.

The level and composition of FCFE varies widely among companies and industries, as well as over the company life cycle. For example, startup firms with little revenue and negative net income typically require significant capital investments to scale operations. Early-stage firms in a startup or growth phase often have negative FCFE, as they often lack the stable cash flows necessary to support debt in the capital structure. As firms grow and mature, the incremental need to make capital investments declines, and the firm may be able to increase the proportion of debt financing in its capital structure.

Working Capital

Earlier Financial Statement Analysis and Corporate Finance learning modules defined and measured the length of a company's operating cycle. The speed and efficiency at which firms convert investments in raw materials, inventory, and other resources into cash can improve a firm's competitive positioning and increase returns to investors.

The nature of the operating cycle varies widely across industries, with manufacturing firms often focused on sourcing raw materials, components, and other inputs from suppliers to meet customer orders. Firms in the retail sector primarily focus on sourcing and selling finished inventory and minimizing the period over which cash is collected from customers. Alternatively, business software or telecommunications firms have relatively low working capital as they are not engaged in the production or distribution of physical goods.

Activity ratios such as inventory turnover, days of inventory on hand, receivables turnover, days of sales outstanding, and days of payables outstanding are used to measure operating cycle efficiency. For example, the cash conversion cycle, in days, is expressed in Equation 3 as follows:

Cash conversion cycle

= Days of inventory on hand + Days sales outstanding – Days payable outstanding.

(3)

Consider the following example of how Costco Wholesale's working capital management is aligned with its competitive strategy.

EXAMPLE 13

Costco Wholesale's Working Capital Management

Example 5 described Costco Wholesale's business model as a membership-based club offering reduced prices on a narrow set of products supplied in bulk. Costco's warehouse stores stock an average of fewer than 4,000 stock keeping units (SKUs) each, as compared to 120,000 SKUs at a Walmart, Inc. store. By stocking far fewer items, and often a single brand per item rather than multiple brands of the same product, Costco is able to streamline its inventory management and increase its bargaining power with suppliers to obtain favorable terms.

One approach to assessing the relative financial impact of Costco's approach versus that of Walmart is to break down the cash conversion cycle components from Equation 3 as shown in Exhibit 19.

Exhibit 19: Cash Conversion Cycle for Costco and Walmart, 2019–2024

Days	2019	2020	2021	2022	2023	2024
Costco Wholesale						
Inventory on Hand	30.73	29.68	28.21	29.32	30.15	28.89
Sales Outstanding	3.82	3.37	3.11	3.24	3.47	3.58
Payables Outstanding	31.30	32.27	32.10	30.58	31.01	29.94
Cash Conversion Cycle	3.24	0.78	−0.77	1.98	2.61	2.53
Walmart, Inc.						
Inventory on Hand	41.71	41.02	38.92	43.16	44.51	41.50
Sales Outstanding	4.22	4.38	4.19	4.71	4.84	4.71
Payables Outstanding	44.07	43.47	41.80	43.25	42.89	41.31
Cash Conversion Cycle	1.86	1.93	1.31	4.63	6.45	4.91

Source: Bloomberg.

As shown in Exhibit 19, despite Walmart's longer average period of payables outstanding, its relatively high inventory on hand and slower payment collection from customers contributed to a cash conversion cycle that is twice as long on average (3.52 versus 1.73) than that of Costco, underscoring the contribution of the company's working capital management to its overall performance.

In contrast to retail industry participants managing finished goods inventories, payables, and receivables, working capital management for manufacturing companies usually involve longer, more capital-intensive, and more complex production cycles. Production efficiency as measured by working capital may be a major source of competitive advantage for some firms, while in others, increasing working capital for a given level of sales may reflect operational challenges.

For example, while most global automakers have standardized the manufacture of major components such as transmissions and drivetrains as well as processes, Japan-based Toyota Motor Company mentioned earlier took the additional step under its multi-pathway strategy to align diverse powertrain technologies such as battery electric vehicles and plug-in hybrids within a unified production framework. The following example describes a case in which design and production issues had significant adverse effects on a company's management of working capital.

EXAMPLE 14

Boeing Company's Max 737 Safety Issues and the Impact on Inventory

The Boeing Company, a US-based aerospace manufacturer and defense contractor, first certified its new 737 MAX 8 aircraft in 2017 as a fuel-efficient version of previous models. While the new product launch initially spurred hundreds of new aircraft orders from global airlines, two subsequent fatal air crashes led to the grounding of 737 MAX aircraft by most civil aviation authorities in March 2019 until late 2020. A faulty flight stabilizer was identified as the likely cause of the accidents, and further investigations uncovered unrelated flight computer and cockpit display design flaws.

In addition to an estimated USD20 billion in direct costs related to fines, compensation, and legal costs due to the accidents and grounding, Boeing incurred significant additional costs related to the inventory backlog and requirements for repairing defective aircraft as well as related financing costs.

While aircraft orders typically extend well over a year, the grounding and subsequent delays due to unrelated flight computer and cockpit display design issues forced Boeing to stockpile undeliverable aircraft as inventory, a situation that was slow to resolve due to canceled orders, supply chain challenges, and another safety incident involving a Boeing MAX aircraft in early 2024. Exhibit 20 shows Boeing's cash conversion cycle from 2016 through 2024.

Exhibit 20: Boeing's Cash Conversion Cycle, 2016–2024

Days	2016	2017	2018	2019	2020	2021	2022	2023	2024
Inventory on Hand	147.7	214.2	268.7	344.0	448.1	489.8	450.0	406.7	441.2
Sales Outstanding	35.6	24.2	13.6	18.5	17.3	14.1	14.9	12.7	15.4
Payables Outstanding	52.9	40.4	55.6	59.7	75.2	71.6	57.1	56.6	56.1
Cash Conversion Cycle	130.4	198.1	226.7	302.8	390.1	432.3	407.8	362.9	400.5

Source: Bloomberg.

As shown in Exhibit 20, Boeing's cash conversion cycle more than tripled following the Boeing MAX incidents and grounding as well as the COVID-19 pandemic, with the increase in inventory as the primary cause.

Unlike the assignment of inventory to individual units as in the retail industry, so-called **program accounting** is used to measure inventory under GAAP for specific long-term programs or projects, which takes overall project profitability into account rather than unit costs. In the case of Boeing, a significant proportion of the rise in inventory balances is related to deferred production costs, which may result in costs that could exceed the eventual recoverable value of aircraft.

As shown in the example, the longer asset conversion cycle of manufacturing companies can magnify the impact of production challenges on working capital via changes in inventory. In other sectors, such as software services, firms hold no inventory and working capital consists simply of managing payables and receivables.

Under some business models, declining or negative working capital may provide a significant *source* of funding for firms as described in the following example.

EXAMPLE 15

Anheuser-Busch InBev SA's Working Capital Management

Anheuser-Busch InBev SA/NV (ABI), the world's largest brewer, was formed by combining three firms, namely Brazil-based AmBev, Belgium-based InBev, and US-based Anheuser Busch. In 2016, ABI purchased global beer industry competitor SABMiller for a price that exceeded USD100 billion.

The SABMiller acquisition created the first global beer company, with prospective benefits in the form of greater scale, the potential for cost savings and efficiencies, as well as increased bargaining power with its suppliers.

Exhibit 21 shows how ABI's cash conversion cycle evolved before and after the acquisition but prior to the COVID-19 pandemic that began in 2020.

Exhibit 21: ABI's Cash Conversion Cycle, 2014–2019

Days	2014	2015	2016	2017	2018	2019
Inventory on Hand	57.64	62.15	69.64	68.54	76.48	77.63
Sales Outstanding	24.81	27.99	31.50	30.17	31.62	29.64
Payables Outstanding	201.61	241.50	249.32	247.74	279.94	278.68
Cash Conversion Cycle	-119.16	-151.36	-148.18	-149.03	-171.85	-171.42

Source: Bloomberg.

As Exhibit 21 shows, ABI was able to steadily reduce its cash conversion cycle over the period due in large part to improved payable terms from its suppliers. This improvement in working capital management reduced the firm's funding needs and supported its ability to repay acquisition-related debt.

Capital Investments

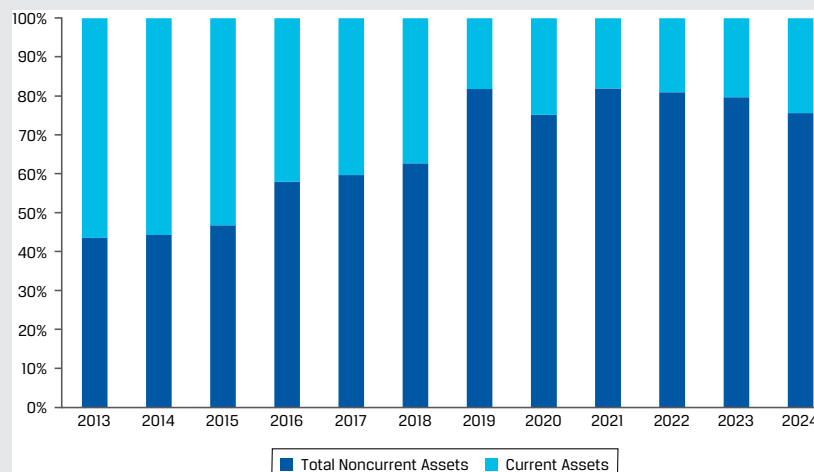
In contrast to the short-term nature of working capital, the acquisition of long-term tangible and intangible assets is capitalized, with those assets with a finite useful life expensed over time through the income statement as non-cash depreciation and amortization. As a result, company analysts often consider **net capital expenditure**, or total capital expenditure less depreciation and amortization, as a measure of a firm's increase in productive capacity.

As demonstrated earlier in the case of working capital, a firm's current and expected future business model is an important determinant of its need for long-term fixed assets. For example, manufacturers of automobiles or aircraft described in earlier examples invest heavily in *tangible* assets to sustain and grow productive capacity. In other cases, firms in service- or technology-based industries are able to generate revenue and earnings with a relatively small, efficient asset base with lower financing requirements. Firms that generate revenue from licensing fees for software or other intellectual property have a high proportion of *intangible* assets that are amortized over their useful life. Changes in a company's business model may necessitate changes in capital investments as described in the following example.

EXAMPLE 16

Netflix's Content Creation Strategy and Capital Expenditures

Example 12 described Netflix, Inc.'s, evolution from DVD mail-order movie rentals to streaming acquired films and television shows to a major developer of original content. The impact of this shift on Netflix's current versus non-current asset composition since 2013 is shown in Exhibit 22.

Exhibit 22: Netflix's Balance Sheet Composition, 2013–2024

Source: Bloomberg.

The shift from largely current to non-current assets coincided with the company's extensive investments in content creation. Netflix's net content assets rose from just over USD2 billion of non-current assets in 2013 to more than USD32 billion in 2024.

According to the company's annual report, Netflix recognizes both licensed and produced content as "net content assets" on its balance sheet. While the fee per title is capitalized for licensed content, Netflix capitalizes production and development costs for original content. Content amortization is based on past and expected future viewing patterns, with asset amortization based on the shorter of each title's contractual availability or estimated use over 10 years. Given the expectation of more viewership upon release, amortization is on an accelerated basis, with an average of over 90% of a content asset's value amortized within four years.

While the *level* of capital investment is an important metric to support future earnings growth, the *return* on invested capital is a major focus of analysts when assessing a firm's relative performance. Earlier Equities learning modules described the return on invested capital (ROIC) as an aggregate measure of value creation across all investments, which is often compared to the weighted average cost of capital (WACC) as a measure of excess return:

$$\text{ROIC} = \frac{\text{NOPAT}_t}{\text{Average invested capital}}, \quad (4)$$

where the numerator, NOPAT, is net operating profit after taxes and the denominator is equal to the average of total long-term liabilities and total equity between the beginning and end of a reporting period. A positive difference between a firm's ROIC and its WACC indicates that the company generates excess accounting return for its investors. We use this measure in Example 17 to compare individual airlines from among those included in an industrywide analysis in the prior learning module.

EXAMPLE 17**Comparison of Global Airlines Based on Excess Returns to Capital**

The prior learning module showed that the global airline industry on average generated a ROIC that fell short of the average WACC ($\text{ROIC}/\text{WACC} < 1$) after the 2008 Global Financial Crisis. The overall industry earned excess returns in a single year before oil price rises eroded profit margins in the early 2010s. Lower oil prices beginning in 2014 contributed to positive excess returns in 2015–2019. COVID-19-related border closures and flight cancellations in 2020 caused air travel to come to a near standstill, with US and European governments offering financial assistance to airlines. As a result, a company analysis might consider two distinct periods, namely the decade prior to COVID-19 (2009–2019) as well as the years after 2020 (2021–2023). Exhibit 23 breaks down ROIC/WACC performance for selected global airlines over these two periods.

Exhibit 23: ROIC/WACC Performance for Selected Global Airlines

Airline	ROIC/WACC Average 2009–2019	ROIC/WACC Average 2021–2023
Air France	0.747	−0.816
American Airlines	N/A	−0.206
Delta Airlines	1.304	1.042
IAG Group	1.160	0.772
Lufthansa	0.770	0.238
United Airlines	0.878	0.519

Source: Bloomberg.

Delta Airlines emerges as the strongest performer over these two periods based upon these measures, while Air France appears to be the weakest, with the exception of American Airlines, which filed for bankruptcy protection in 2011.

Delta Airlines' competitive strengths after the Global Financial Crisis was a result of its 2008 merger with Northwest Airlines, positioning it as the world's largest airline at the time and providing greater scale to bolster its domestic and international networks. With a strong focus on reliability and on-time performance and selected investments in new and used aircraft, Delta kept costs low while maintaining efficiency. The company's post-COVID recovery began with growth in domestic leisure travel, while the company invested in premium services such as preferred seating and lounge options in anticipation of stronger business travel.

As a partially government-owned enterprise, Air France lacked the flexibility to restructure operations following the Global Financial Crisis. The company's focus on long-haul flights and recurring labor challenges resulted in higher costs, less operational efficiency, and lower profits. The company benefited from continued government support following the COVID-19 pandemic and initiated a modernization program to upgrade its fleet and restructure its costs. Despite improvements, its performance lags behind that of other global airline competitors.

Capital Structure

A firm's optimal capital structure depends upon its business model, the nature of its operations and asset conversion cycle, and its stage in the company life cycle.

As described earlier, early-stage firms that have greater business risk, little to no revenue, and negative cash flow are usually financed with equity capital, while more mature businesses with stable cash flows are able to support debt financing. A company's relative reliance on debt financing, the type and tenor of bonds, and loans outstanding as well as the firm's access to additional debt financing are important factors to consider in company analysis.

The impact of a firm's debt on its financial performance is reflected in the income statement by the **degree of financial leverage**, or the sensitivity of net income to changes in operating income as shown in Equation 5.

$$\text{Degree of financial leverage} = \frac{\% \Delta \text{ Net income}}{\% \Delta \text{ Operating income}} \quad (5)$$

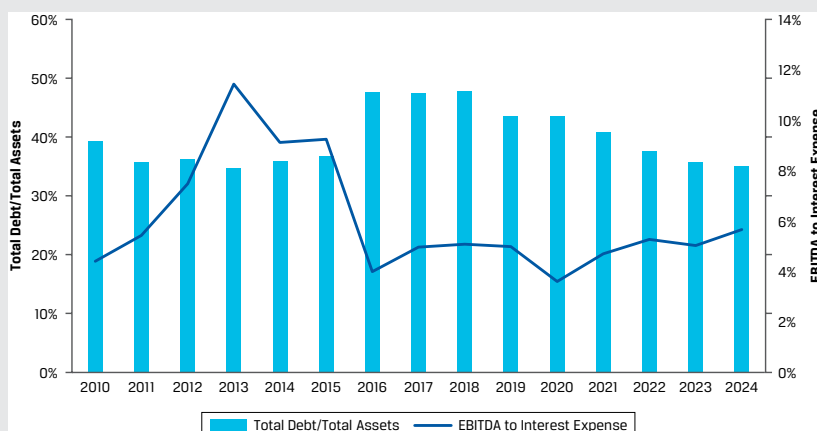
The degree of financial leverage increases as interest expense rises relative to net income. Similar to operating leverage, financial leverage can augment profits but also magnify losses and increase the likelihood of financial distress. Financial distress involves a high degree of uncertainty regarding a firm's ability to meet debt obligations due to weaker earnings or net losses.

The relative ability to support long-term debt on the balance sheet depends upon the underlying business model and the types of assets being financed. For example, capital-intensive businesses such as manufacturing companies or public utilities have relatively long-lived assets that are often financed by long-term bonds, while retail and service-based businesses tend to fund operations on a shorter-term basis. Changes in the relative use of debt when firms ramp up capital investments, restructure operations, or acquire other companies should be incorporated in disaggregated financial models to ensure that use of cash for debt service and repayment is properly captured. We return to the example of Anheuser Busch InBev to examine the company's management of its capital structure following the acquisition of SABMiller.

EXAMPLE 18

Anheuser Busch InBev's Capital Structure

An earlier Equities learning module described the Anheuser-Busch InBev's acquisition of SABMiller in 2016. The deal was financed almost entirely with debt. Exhibit 24 shows the significant increase in ABI's debt ratio between 2015 and 2016 and the associated decline in its interest coverage ratio.

Exhibit 24: Evolution of AB InBev Debt and Interest Coverage Ratios

Source: Bloomberg.

AB InBev was unable to reduce its debt load in the first few years following the SABMiller acquisition as the company was forced to navigate both weak beer demand and asset sales requested by international regulators. However, starting in 2022, the company was able to reduce its debt ratio to levels in place prior to the acquisition, with interest coverage ratios also showing signs of improvement.

The prior example illustrates a common situation in which a significant corporate acquisition leads to a substantial increase in debt ratio and a decrease in interest coverage. Analysts should not only explain the increase in financial leverage but also assess the viability of a company's plan for subsequent reductions in leverage following an acquisition financed with debt.

QUESTION SET 3



- Which of the following conditions is *most* consistent with a firm that has a high degree of financial leverage?
 - The firm's earnings before interest and taxes (EBIT) are negative.
 - The firm's net income is highly sensitive to changes in operating income.
 - The firm relies primarily on equity financing to fund its capital expenditures.

Solution:

B is correct because financial leverage refers to the use of debt financing and reflects how net income changes with fluctuations in operating income. High sensitivity of net income to operating income changes indicates a high degree of financial leverage.

A is incorrect because negative EBIT reflects operational performance, not necessarily high leverage.

C is incorrect because equity reliance implies lower financial leverage, not higher.

- A decline in a firm's net capital expenditure is *most* consistent with:
 - increasing depreciation charges.
 - reduced investments in long-term assets.

C. an aggressive expansion strategy.

Solution:

B is correct because net capital expenditure is calculated as capital expenditures minus depreciation and amortization. A reduction in long-term asset investments directly leads to a decline in net capital expenditures.

A is incorrect because higher depreciation would typically follow new investments, not a reduction.

C is incorrect because expansion strategies usually involve increased capital expenditures.

3. Does a lengthening of the cash conversion cycle affect a company's financing needs?

A. No.

B. Yes, the financing needs are reduced.

C. Yes, the financing needs are increased.

Solution:

C is correct because the cash conversion cycle is the time a company requires to convert net claims to cash. A longer cash conversion cycle is therefore a use of cash that must be financed.

A is incorrect because a longer cash conversion cycle is a use of cash that requires financing.

B is incorrect because a longer cash conversion cycle is a use of cash that requires financing.

4. Which of the following trends is *most* consistent with a company deleveraging after a major acquisition?

A. Declining interest coverage and rising debt-to-equity ratio

B. Rising return on equity (ROE) driven by increased borrowing

C. Decreasing debt levels and improving debt-to-equity ratio

Solution:

C is correct because deleveraging refers to the process of reducing debt. This results in lower debt levels and a lower (i.e., improved) debt-to-equity ratio.

A is incorrect because declining interest coverage and a higher debt ratio suggest rising leverage, not deleveraging.

B is incorrect because rising ROE due to increased debt implies leverage is increasing, not decreasing.

SUMMARY: COMPANY ANALYSIS: PAST, PRESENT, AND FUTURE

- A company's competitive strategy should be evaluated based on its effectiveness in defending a firm against Porter's Five Forces while considering external industry influences as well as the resources and capabilities at its disposal.
- While early-stage firms in startup and growth phases of the life cycle concentrate on developing a product and sizing the addressable market, established firms typically focus on strategies involving cost leadership, differentiation, or focus.
- Revenue analysis and forecasting usually combines top-down analysis, which incorporates an overall economic and industry outlook with a firm's expected market share, and bottom-up factors such as sales volume, pricing, product mix, and geographic sales distribution.
- Profitability measures include net income, a comprehensive but difficult metric to compare across firms and over time; operating income, a core measure that is challenging to compare when investment policies differ; and EBITDA, a non-standardized operating profit measure before discretionary spending that is subject to adjustment. These profitability measures are often expressed as percentages of revenue, or margins.
- Additional factors that affect changes in a company's revenue and profitability include pricing power and operating leverage.
- Working capital analysis assesses the relative speed and efficiency with which firms can convert inputs and inventory into cash over an operating cycle to generate returns. The operating cycle varies widely across firms and industries.
- Capital investment analysis assesses the size, timing, and return on a firm's investments in long-term tangible and intangible assets, with the return on invested capital often measured relative to the weighted cost of capital.

PRACTICE PROBLEMS

1. Which of the following strategies is *most likely* to apply to economies of both scale and scope?
 - A. Cost leadership
 - B. Cost-based focus strategy
 - C. Differentiation-based focus
2. Focus strategies typically focus on:
 - A. differentiation only.
 - B. cost leadership only.
 - C. either cost leadership or differentiation.
3. In a declining industry, which of the following factors is *most likely* to intensify competitive rivalry?
 - A. Excess capacity
 - B. Product differentiation
 - C. Research and development
4. Higher working capital efficiency is *most likely* to be associated with an increase in:
 - A. days sales outstanding.
 - B. days inventory on hand.
 - C. days payables outstanding.
5. Return on invested capital (ROIC) is *best* described as a measure of:
 - A. pre-tax returns to shareholders.
 - B. value created across all assets.
 - C. the amount needed for a company to earn excess returns.
6. A firm raises debt to finance a new production facility. Prior to the facility opening, the firm will *most likely* experience:
 - A. an increased operating margin.
 - B. an increase in financial leverage.
 - C. a reduction in net working capital.

SOLUTIONS

1. A is correct because economies of scale and scope are most applicable under a cost leadership strategy, since each aims to reduce the average cost of supplying goods and services.
B is incorrect because it is difficult to achieve economies of scope in a focus strategy.
C is incorrect because differentiation focus strategies are unlikely to benefit from economies of scale or scope.
2. C is correct because a focused strategy may employ either cost leadership or differentiation aimed at a specific group or product.
A is incorrect because either cost leadership or differentiation can be the basis for a focus strategy.
B is incorrect because either cost leadership or differentiation can be the basis for a focus strategy.
3. A is correct because declining industries often face excess capacity due to declining demand, a factor that intensifies competitive rivalry among industry participants.
B is incorrect as product differentiation may occur at any life-cycle stage and is not necessarily a competitive driver in declining industries.
C is incorrect as research and development is more likely to be a factor affecting competitive rivalry among startup or growth companies.
4. C is correct because an increase in a firm's payables outstanding results in a shorter cash conversion cycle, which is equal to days of sales outstanding plus days of inventory on hand minus days of payables outstanding.
A is incorrect as an increase in days sales outstanding will lengthen, not shorten, the cash conversion cycle.
B is incorrect as an increase in days inventory on hand will lengthen, not shorten, the cash conversion cycle.
5. B is correct because the ROIC measures return to all investors, as it includes both long-term liabilities such as debt and average equity in the calculation.
A is incorrect because ROIC measures after-tax return to all investors.
C is incorrect because it describes the weighted average cost of capital (WACC), not ROIC.
6. B is correct because issuing debt increases the firm's liabilities and associated interest expenses, thereby increasing financial leverage.
A is incorrect because operating margin excludes interest and would not be affected by financing decisions.
C is incorrect because raising debt typically affects long-term liabilities, not current assets or liabilities directly associated with net working capital.

LEARNING MODULE

11

Equity Analyst Research Reports

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	describe the elements that should be covered in a detailed company research report
☐	compare sell-side and buy-side equity analyst reports and their conclusions
☐	explain why value estimates from the same modeling approach may differ based on analyst assumptions regarding revenue, profitability, investment, and financing

INTRODUCTION: EQUITY ANALYST RESEARCH REPORTS

1

This learning module explores equity research reports as a primary means through which analysts communicate their assessment of a firm's investment potential. These reports integrate concepts and methodologies from CFA® Program topic areas, such as Quantitative Methods, Economics, Corporate Finance, and Financial Statement Analysis, as well as earlier Equities learning modules. Reports **initiating coverage** usually include a detailed narrative that analyzes the company's business model, industry, and competitive position with the goal of translating financial analysis into a **price target** and investment recommendation.

The second lesson of this learning module addresses the key components of an **equity research report**. The differing perspectives of sell-side versus buy-side analysts and its impact on research reports prepared by their respective firms is the subject of the third lesson. Sell-side reports are widely distributed to clients, whereas buy-side reports are confidential and are designed to support specific internal investment decisions. The fourth and final lesson considers the reconciliation of varying views among analysts as expressed in research reports. Differing assumptions about revenue growth, risk factors, and other elements can lead to divergent recommendations and distinct target prices which affect underlying market prices.

The skills necessary to prepare an equity research report are important, but analysts in many other roles must also be able to critically evaluate equity research reports. Although the number of equity analysts may be declining as a result of structural

changes to the industry, the volume of financial analysis produced is increasing rapidly as a result of technological innovations, including artificial intelligence. Thus, equity research skills are more important than ever.

PRE-TEST: EQUITY ANALYST RESEARCH REPORTS



1. An equity research report typically begins with:

- A. a comprehensive environmental, social, and governance (ESG) performance assessment.
- B. the analyst's investment thesis and target price.
- C. a discussion of the company's accounting policies.

Solution:

B is correct. The beginning of a research report typically includes a summary of the analyst's recommendation, including the target price and investment thesis, to quickly convey the rationale behind the recommendation.

A is incorrect. ESG assessments, if included, are typically found later in the report and are not a standard part of the opening summary.

C is incorrect. Detailed discussions of accounting policies are more relevant to the financial analysis section and are not emphasized upfront.

2. The primary purpose of an equity analyst writing a company research report is to:

- A. fulfill regulatory disclosure requirements.
- B. communicate an unbiased summary of recent company events.
- C. provide a persuasive investment recommendation based on analysis.

Solution:

C is correct. The goal of an equity research report is to provide a recommendation—such as “buy,” “hold,” or “sell”—that is supported by detailed analysis of the company and its stock.

A is incorrect. Regulatory disclosures are the responsibility of the company, not the analyst, and are not the goal of the report.

B is incorrect. Although recent company events should be discussed, the analyst's report uses this information only in the broader context of conducting the analysis used to inform the recommendation.

3. Which type of equity research report is most likely to remain confidential and tailored to internal investment strategy?

- A. Sell-side report
- B. Buy-side report
- C. Third-party research analyst report

Solution:

B is correct. Buy-side reports are written for internal decision makers at investment firms and are not made public. They often align closely with the firm's specific strategy and holdings.

A is incorrect. Sell-side reports are published and distributed widely to clients and the public.

C is incorrect. Third-party analyst reports are generally developed for a larger audience than one firm.

4. Which of the following statements regarding equity analyst recommendations based on a benchmark is most accurate?
- A. Companies whose shares are expected to appreciate must receive an overweight rating.
 - B. Companies whose shares are expected to appreciate cannot receive an underweight rating.
 - C. Companies whose shares are expected to appreciate may receive an overweight, underweight, or neutral rating.

Solution:

C is correct. Equity analyst recommendations based on a benchmark are relative in nature, with ratings based on expected equity return in relation to an index or analyst coverage universe.

A is incorrect. Companies whose shares are expected to have a positive rate of return well below a benchmark usually receive a neutral rating.

B is incorrect. In some cases of expected strong benchmark performance, it is possible for a company whose shares appreciate modestly to receive an underweight rating relative to peers.

5. Assuming two analysts use the same valuation model to estimate equity value for the same company, they are *least* likely to estimate significantly different values because of significant differences in:

- A. terminal value assumptions.
- B. profit margin assumptions.
- C. financing assumptions.

Solution:

C is correct. Analysts should take care when building valuation models to ensure that they do not inadvertently assume significant value creation from their underlying financing assumptions. Although the form of financing used may have some positive or negative effect on a company's value, it is important to remember that value creation by firms is predominantly a function of the company's operating strategies and results. Thus, financing differences should not lead to significant valuation differences between two analysts.

A is incorrect. Terminal value differences often lead to different valuation results.

B is incorrect. Differing profit margin assumptions are likely to lead to significant value differences between two analysts.

6. Analysts should *most likely* be wary of valuations featuring projections in which:

- A. revenue and profits do not grow similarly.
- B. revenue and investment do not grow similarly.
- C. revenue and investment are highly uncertain.

Solution:

B is correct. The cash flow benefits of additional income from growing revenue are not likely to occur without proportional investment expenditures.

A is incorrect. Revenue and profit growth may diverge when a valuation assumes increasing or decreasing profit margins over time.

C is incorrect. Highly uncertain revenue and investments are likely when a company is hard to value. Although this uncertainty is important to assess, it should not be a cause for analyst concern.

2

EQUITY ANALYST RESEARCH REPORT CONTENTS



describe the elements that should be covered in a detailed company research report

Company research reports serve different purposes for different audiences, with those estimating a company's intrinsic value versus its current market price using qualitative and quantitative analysis of its business model, industry dynamics, and financial performance, which are referred to as equity research reports. Exhibit 1 provides an illustration of the primary elements included in such a report.

Exhibit 1: Basic Structure of Equity Research Report



The perspective of an equity analyst as well as the context within which the analysis is conducted are important features that distinguish equity analyst reports. For example, when a report is created for wider distribution among external users, it is commonly referred to as a **sell-side research report**. These reports are often issued by underwriters of debt and equity securities for their clients, which include both the underlying issuers seeking to publicize their security offerings as well as prospective investors in these securities. In contrast, a company research report created for internal purposes by an asset manager to substantiate specific investment recommendations is commonly referred to as a **buy-side research report**. These distinctions will be addressed further in the next learning module.

The context of a research report is an important factor that influences the contents, as shown in Exhibit 1. For example, a report initiating coverage typically includes a comprehensive discussion and thorough analysis of a company's industry, competitive position, and outlook, which serves to introduce a company to readers for the first time, often following an IPO. In contrast, ongoing or periodic research reports

are typically issued in response to episodic events, such as mergers or acquisitions, as well as widely anticipated announcements, such as periodic earnings releases as described in the following example.

EXAMPLE 1

InterGlobe Aviation Ltd. Research Update by Bullish Analyst

InterGlobe Aviation Ltd. operates the largest passenger airline in India, IndiGo, as its primary operating business. The company generated INR674 billion in the fiscal year ended in March 2024 with net income of INR82 billion. In early 2025, a sell-side firm issued an analyst report for InterGlobe Aviation Ltd. following the release of the company's quarterly financial results. The report provided an overweight recommendation with an expected price appreciation for the stock of almost 20% within the next year.

Despite earnings per share (EPS) falling 16% short of expectations in the latest quarter, the analyst raised their price target by 3%. According to the analyst report, the earnings shortfall was attributed to unfavorable foreign exchange movements that are not expected to recur. The report also cited expected revenue gains from higher load factors as justification for the positive stock recommendation.

As shown in this example of a sell-side research report, an analyst considers both transitory and long-term factors affecting a company's outlook, which are distilled into a specific recommendation.

Analysts provide recommendations on a company's stock, such as "buy," "sell," or "hold," to summarize their views about expected share price performance, which may be defined in absolute price return terms or on a relative basis versus an index, a relevant industry or regional benchmark, or the universe of companies covered by a specific analyst. These absolute or relative returns are measured based on the relationship between the current share price and a price target or expected future share price, which reflects the results of an analyst's valuation. This performance is expected over a specified time horizon, which is usually one or two years. Some analysts may utilize alternative terminology, such as "overweight," "underweight," or "neutral," to express their buy, sell, or hold recommendations in the form of a relative portfolio weighting.

No uniform standard exists to classify buy, sell, or hold recommendations. As a general rule, however, shares that tend to be rated a "buy" are expected to appreciate at least 10%–15% on an absolute basis or in excess of a benchmark, stocks with a "sell" rating are expected to decline or fall short of a benchmark by at least 10%–15%, and a "hold" or neutral rating falls between these two ratings.

KNOWLEDGE CHECK: ANALYST RECOMMENDATIONS



A junior financial analyst covering the European beverage industry compiles the following table of current share prices and target prices based on her equity valuation results over a one-year time horizon for four industry peers (see Exhibit 2).

Exhibit 2: Comparison of Current and Target Share Prices

Company	Current Share Price	Target Price (1 year)
Carlsberg AS	DKK910	DKK930
Anheuser-Busch InBev SA/NV	EUR62	EUR77
Diageo PLC	GBP1,840	GBP1,900
Pernod Ricard SA	EUR88	EUR83

Overall, the relevant equity benchmark is expected to increase by 12% over the next twelve months.

1. Which of the following conclusions is most likely to result from using a relative recommendation approach to rating the four companies?
 - A. Overweight for Anheuser-Busch InBev, underweight for Pernod Ricard, and neutral for Carlsberg and Diageo
 - B. Overweight for Anheuser-Busch InBev, Carlsberg, and Diageo and underweight for Pernod Ricard
 - C. Not enough information is provided to draw conclusions about relative rating recommendations

Solution:

A is correct. Anheuser-Busch is expected to outperform on a relative basis, Pernod Ricard is expected to underperform, and Carlsberg and Diageo are expected to fall in between as neutral based on the relationship between current share prices and the respective company target prices.

Calculate expected price appreciation based on the target price as follows:

$$\text{Expected One Year Appreciation} = \frac{\text{Target Price} - \text{Current Share Price}}{\text{Current Share Price}}$$

$$\text{Carlsberg: } 2.20\% = \frac{930 - 910}{910}$$

$$\text{Anheuser-Busch InBev: } 24.19\% = \frac{77 - 62}{62}$$

$$\text{Diageo: } 3.26\% = \frac{1,900 - 1,840}{1,840}$$

$$\text{Pernod Ricard: } -5.68\% = \frac{83 - 88}{88}$$

As a result, Anheuser-Busch is expected to outperform the benchmark by more than 10%, Carlsberg and Diageo are expected to underperform the benchmark by less than 10%, and Pernod Ricard is expected to underperform the benchmark by more than 10%.

B is incorrect. It assumes shares should be overweighted if they appreciate on an absolute rather than relative basis.

C is incorrect. All current and target share prices are comparable on a relative basis, allowing an analyst to draw an appropriate conclusion.

In some cases, even a stock that was expected to have a modest return in absolute terms might receive a “sell” or underweight rating if strong outperformance is expected among its peers. The focus of a specific company research report may vary based on the analyst’s perspective and the context within which it was written. An equity research report, however, usually contains a set of common elements, as described next.

Common Elements of an Equity Research Report

Analysts write equity research reports to persuade readers to follow and act on their recommendation. As a result, an equity report must provide convincing evidence substantiating the analyst’s opinions and recommendation. The analyst, however, cannot make a compelling case without first providing readers with a thorough understanding of the company, industry, and the competitive environment, including relevant risk factors. This qualitative information is complemented by the analyst’s quantitative analysis and a complete valuation analysis, integrating the analyst’s research into a consistent and coherent narrative.

Analyst Opinion and Investment Summary

An effective report should begin by clearly summarizing the analyst’s recommendation. A key aspect of a recommendation is the price target, which may be communicated as a point estimate or as a range. Although analyst reports often consider a one-year time horizon, this period may vary based on company-specific milestones or events, such as successful research trials, regulatory approvals, or patent expiry. As a result, the analyst’s price target can be interpreted as implying an expected return on the stock, as illustrated in the following example.

EXAMPLE 2

Changes in BioAge Labs Inc.’s Analyst Recommendations

An earlier Equities learning module provided a description of an early-stage US biotechnology company, BioAge Labs Inc. (BIOA). In December 2024, shortly after its IPO, BIOA announced the discontinuation of the trial of its obesity drug. Exhibit 3 shows a Bloomberg screenshot of analyst recommendations, price targets, and the company’s stock price around the trial discontinuation using Bloomberg’s ANR function.

Before the trial discontinuation, the consensus analyst expectation was for BIOA’s stock price to increase significantly to levels above USD40 per share. The new information coincided with not only substantial declines in BIOA’s stock price but also with a sharp decline in price targets to approximately USD5–USD7 per share.

Exhibit 3: BioAge Labs Analyst Recommendations



Source: Bloomberg.

Following the summary recommendation, an analyst should aim to condense key aspects of their investment rationale into a section referred to as an **investment thesis**. An effective investment thesis succinctly articulates the analyst’s findings and conclusions. It often includes company-specific drivers of performance as well as the underlying economic and industry environment that will contribute to the analyst’s predicted outcome for the company. The thesis should be directly aligned to achieving the price target within the anticipated time horizon. The investment thesis articulates

why an analyst believes the stock is mispriced, while specific **catalysts**, or expected events or developments related to a company's performance, are listed and described that could prompt a revaluation and move the market price toward its intrinsic value. Catalysts are an important component of a research report, especially when the investment thesis depends on such specific events or developments.

The first page of an equity research report usually includes basic company and share information, such as the following:

- the stock's ticker symbol,
- the company's primary sector and industry,
- the stock's market capitalization,
- the stock's market float,
- recent developments, and
- a summary of recent and forecasted financial data, especially earnings.

This summary information is discussed in more detail in the remainder of the report. In some cases, as shown in Exhibit 1, tables, graphs, and charts, including key evidence supporting the analyst's conclusions and assumptions, are provided in the margin of the document. Although these elements appear at the beginning of an equity research report, this portion of an equity research report should be written last to ensure its proper integration with subsequent content.

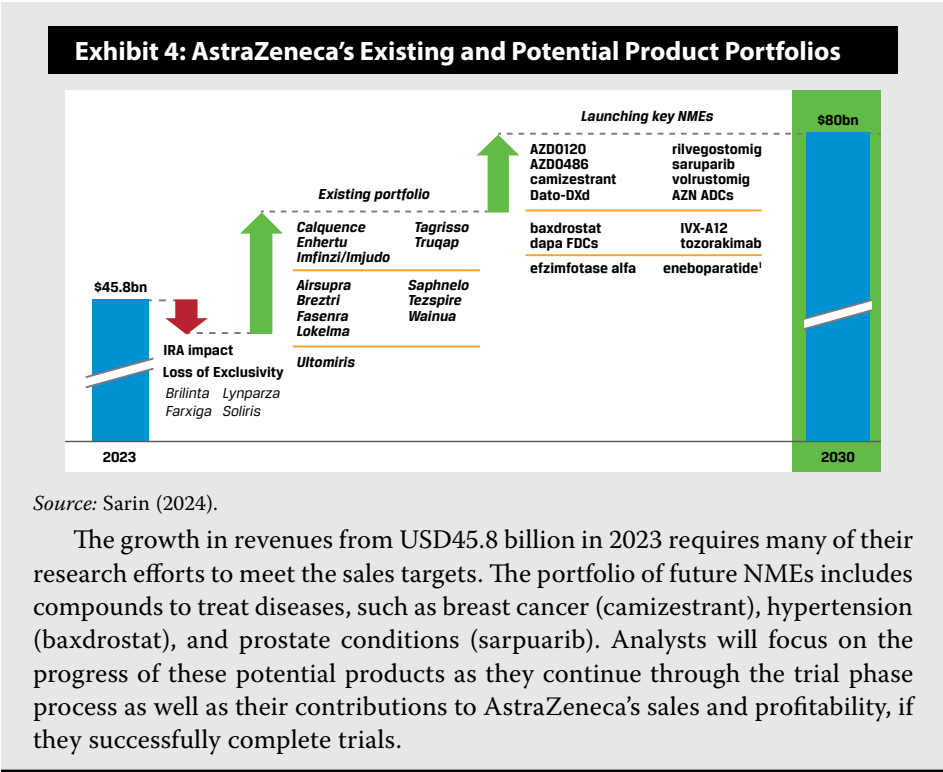
Business Description, Industry Overview, and Competitive Positioning

An equity research report in which an analyst initiates company coverage should include a detailed description of the firm and its business model, including its products, services, major suppliers, and customers. It should convey a clear understanding of how the company generates value and its strategies, including a discussion of the key drivers of revenues and expenses. The management discussion and analysis of a firm's annual report, additional regulatory filings, industry publications, and financial press releases are the best starting point for this discussion. Although financial analysts covering industries involving advanced technology such as biotechnology or semiconductors often bring industry experience or specialized knowledge to their roles, it is important that company and business model descriptions be tailored to suit a broad population of potential investors rather than an industry specialist audience. This communication approach is addressed in the next example.

EXAMPLE 3

AstraZeneca's Research and Development Focus

An earlier Equities learning module described AstraZeneca's Ambition 2030 goals and its portfolio of new molecular entities (NMEs) with enough market potential to achieve the company's goal of USD80 billion in revenue by 2030. Exhibit 4 shows AstraZeneca's existing product portfolio as well as the portfolio of NMEs in trial phases.



In addition to distinct company-specific issues, such as those highlighted in the previous example, a firm’s pricing, distribution, and the stability of its market share are other factors that might be discussed within this portion of an equity research report.

As equity investors commonly consider a firm’s competitors to be the nearest substitutes for owning its shares, an effective report should contain an overview of the firm’s competitive environment, including a detailed analysis of the industry or industries within which it operates. An analyst should develop a group of peer companies for competitive analysis. Although an analyst may be able to utilize some firms’ annual reports to identify specific competitors, many firms avoid naming specific competitors in their investor filings. In other cases, peer groups of companies are explicitly identified in filing information, such as proxy statements in the United States. Although these peers may extend beyond direct competitors, this information may also be valuable for an analyst building a peer group. Analysts should consider building peer groups using a variety of resources, including from data providers as shown in the following example.

EXAMPLE 4

Finding a Peer Group for Birkenstock Holding PLC

The October 2023 IPO of Birkenstock Holding PLC (BIRK) was introduced in an earlier Equities learning module. As a shoe manufacturing company with a global brand reach, defining a peer group for BIRK may seem like a straightforward task. Birkenstock Holding, however, does not provide a discussion of specific competitor companies in its investor filings; thus, analysts must rely on other sources to define the peer group.

Data providers such as Bloomberg and FactSet provide tools for identifying comparable companies based on industry classification systems. For example, Bloomberg includes a relative value function (RV); see the output example shown in Exhibit 5.

Exhibit 5: Bloomberg RV Function Screenshot for BIRK

Source: Bloomberg.

The introduction of various criteria used to identify comparable companies using both Bloomberg and FactSet yields quite different results. For example, ON Holding AG (ONON) is the most commonly identified company comparable to BIRK based on several searches using a Bloomberg terminal, whereas Crocs Inc. (CROX) is the most commonly cited company considered to be comparable to BIRK using FactSet. These searches also identified a number of other companies focused heavily on shoe brands, such as Skechers USA Inc. (SKX) and Dr. Martens PLC (DOCS), firms with broader offerings of outdoor clothing as well as companies focused solely on footwear manufacturing. A comparison of all of these companies, however, may not adequately account for Birkenstock's shift to luxury shoe brands following an investment in the company by a group backed by the LVMH (Louis Vuitton Moët Hennessy) luxury goods maker.

The previous example highlights the challenges faced in identifying an appropriate peer group, a task that requires an analyst to exercise significant judgment about the choice of comparable companies. A thorough analysis of the firm's industry structure and external influences often implicitly or explicitly incorporates elements of Porter's Five Forces and the political, economic, social, technological, legal, and environmental (PESTLE) framework introduced earlier in the curriculum.

While providing a useful checklist, analysts should avoid a superficial exercise of simply listing and addressing each of Porter's Five Forces and the six elements that make up the PESTLE framework. A preferable approach involves focusing primary attention on those factors that have the greatest substance and impact on a firm's performance.

Financial Analysis and Valuation

The most effective equity research reports combine the qualitative aspects of a company's business model, industry overview, and competitive positioning with quantitative analysis that supports a robust valuation of the firm.

Financial Analysis

A first step involves conducting a thorough analysis of the company's financial statements as illustrated in earlier Financial Statement Analysis learning modules. When compiling this section of the report, the analyst's goal should not be to provide an exhaustive financial statement analysis, but rather to focus on key financial measures

and ratios that reflect their economic analysis of the company and support the investment thesis and valuation narrative. It is important to directly review the firm's financial disclosures and footnotes rather than simply rely on information sourced from financial data providers. This more detailed approach to financial statement analysis helps analysts to identify material risk factors, such as poor earnings quality, or questionable revenue or expense recognition policies. The financial analysis portion of an equity research report often includes industry-specific ratios, such as proven reserves for oil companies, revenue/available rooms for the hotel industry, or airline-specific metrics as illustrated in the following example.

EXAMPLE 5

InterGlobe Aviation Research Report and Airline-Specific Performance Measures

The equity research report on InterGlobe Aviation includes a variety of airline-specific metrics. An important capacity metric in the airline industry is available seat kilometers (ASK), which multiplies the number of available seats on all of the company's flights by the total kilometers flown by these airlines. By framing the company's revenues and costs per ASK, the research report provides readers with an understanding of the unit economics of the airline, which can be directly compared with those of other airlines. These metrics are reported by airlines in their filings, and research analysts often build ASK and similar metrics into their financial statement models.

InterGlobe Aviation management reported 22% ASK growth as passenger traffic continued to improve following the extreme negative effects of the COVID-19 pandemic. The research analyst forecast continued capacity growth for the company with double-digit percentage increases in ASK within the model. InterGlobe Aviation *revenue* per ASK (RASK) is forecasted by the analyst to decline slightly relative to fiscal 2024 RASK. The analyst's *costs* per ASK (CASK) forecast implies significant unit cost improvements in non-fuel-related expenses for the company. Overall, the analyst's model implies improvements in the spread between the company's RASK and CASK during the forecast years of fiscal years 2025–2027, although this spread is somewhat smaller than in fiscal year 2024.

Valuation

A comprehensive analysis of a firm's financial statements is an essential ingredient for the valuation exercise used to substantiate an analyst recommendation. Analysts routinely employ more than one valuation approach to determine a firm's price target based on intrinsic value. Analysts may use absolute or relative valuation models. For example, the level of detail used to construct a cash flow projection model based on income statement and balance sheet positions will vary based on a firm's distinct value drivers. Use of relative value measures requires an analyst to identify the appropriate ratios for comparison and ensure that these multiples are consistently defined and calculated for the firm and a group of carefully selected peers.

EXAMPLE 6**Initiating Coverage of Birkenstock Holding PLC**

Shortly after its IPO in October 2023, analysts began releasing reports initiating coverage on Birkenstock Holding PLC (BIRK) stock. One such report issued a hold recommendation on the stock with a price target of approximately 5% greater than BIRK's traded price shortly before the report's release.

The report's price target was based on a discounted cash flow (DCF) valuation. The analyst's narrative underlying the valuation relies on BIRK's dominant position within a niche market. The company's unique sandals are made in Europe, rather than being outsourced to Asia, as is commonplace in footwear manufacturing. The company's customers tend to be loyal, and the company can maintain this loyalty without incurring significant marketing expenses. The report's underlying themes translated to strong forecasted revenue growth with a double-digit growth in earnings before interest and taxes (EBIT) over the first 10 years of the forecast.

Given that BIRK is listed in the United States while operating in Europe, the report benchmarked BIRK's price target against 10 comparable companies trading in both the United States and Europe, including peers mentioned earlier, such as ON Holding, Crocs, and Dr. Martens, as well as global brands such as Nike and Adidas. Specifically, the report compared several different valuation multiples along with margin and growth characteristics.

Investment Risks

An analyst's price target should incorporate all potential opportunities and risks faced by a firm over the forecast time horizon. This price target point estimate, however, should be complemented by a discussion of potential events or outcomes that might adversely affect the firm's performance. This information may be sourced from risk factors listed in the management discussion and analysis section of the firm's annual report, earnings call transcripts, investor presentations, or news releases. These risks may be company-specific or affect an industry or the economy as a whole. Company-specific risks include revenue shortfalls related to a specific product or client, operational risks such as delayed product launches or recalls, or challenges in executing on a stated business strategy. Industry-wide risks include cyclicality, disruptive market entrants, or regulatory pressures. Macroeconomic risks, such as significant changes in inflation, interest rates, or currency levels, may have important effects on an analyst's valuation and recommendation on a company's stock. In other cases, structural changes in customer behavior or tastes affecting one industry or company may spill over to another, as described in the following example.

EXAMPLE 7**Effects of Ozempic on Food Industry Companies**

Although the introduction of weight-loss drugs, such as Ozempic, is considered a potential contributor to the expected revenue growth of some drug companies, analysts also had to consider potential risks to food companies associated with these new medications. In a 2023 study conducted by Walmart on consumer behavior based on its pharmacy and grocery shopping data, the retailer found that those customers using GLP-1 drugs bought less food than those that were not. Other research shows that those individuals on GLP-1 drugs consume less ultraprocessed foods and less food in general. Given that the growth in GLP-1

weight loss drug use worldwide is expected to continue, larger changes in food consumption patterns should be expected in the future. This “Ozempic effect” revealed changes in consumer eating habits that were likely to have meaningful effects on the revenues of some food companies. To mitigate these risks, some companies adopted new strategies, such as creating single-portion meals, smaller snack sizes, and healthier alternatives to offset the impact of lower consumption.

This example highlights a case of a *transitory* risk factor, or one in which companies adapt and adjust to changing consumer tastes and preferences. Companies failing to adapt will likely face adverse competitive consequences if these changes in preferences become more permanent. A compelling assessment of company risks should consider the firm’s relative adaptability to changing market conditions versus its peers as well as mitigants of the risks associated with the company’s business, the competitive environment, and the economic outlook. The most important opportunities and downside risks should be explicitly translated into a valuation framework. For example, airline analysts might incorporate different jet fuel price scenarios into their calculations when considering a stock price range, whereas biotechnology company research reports often focus on the expected timing and success of drug trials as well as the status of competitor research and development of similar compounds.

Analysts should translate key risks into their valuation framework explicitly. For example, analysts covering airline stocks may consider different jet fuel price scenarios when constructing alternative valuations, whereas health care analysts might focus on the timing and likelihood of new drug approvals, as well as research and development progress among direct competitors.

Sustainability

A thorough equity research report often includes a section specifically devoted to sustainability and other relevant factors important to investors. The focus of this section of an equity research report may vary across jurisdictions and industries. For example, in markets such as the European Union where institutional equity investors commonly adhere to ESG principles when selecting equity investments, this section of the report may address a company’s strengths, vulnerabilities, and potential controversies it faces in these areas. In the case of specific firms, those in industries in which sustainability factors are expected to have a material financial impact on revenue and earnings warrant a separate, more detailed analysis of these factors within this section. For example, energy producers; significant energy consumers, such as electric utilities; or companies whose products rely heavily on energy consumption often have a major focus on energy transition plans as described in the following example.

EXAMPLE 8

Toyota’s Multipathway Strategy and Sustainability Goals

Toyota Motor Company describes its sustainability strategy as consisting of multiple pathways in its 2024 North American Environmental Sustainability Report. Rather than focusing solely on a rapid transition to all-electric vehicles, Toyota is pursuing a variety of initiatives to meet its goal of carbon neutrality in North American operations by 2035.

While Toyota Motor Company is increasing the electrification of its new vehicle sales, the company is also pursuing collaborations to work toward its environmental sustainability goals. A specific partnership with Chevron involves pilot-testing a renewable gasoline blend that will reduce a gasoline-powered vehicle’s life-cycle carbon emissions. This project reflects a recognition that gasoline-powered vehicles are likely to remain in operation well into the 2030s.

CFA Institute and Equity Research

As part of its mission, CFA Institute provides experiential learning opportunities for students related to equity research as described in the following example.

EXAMPLE 9

CFA Institute Research Challenge

The CFA Institute Research Challenge provides a distinct opportunity for university students around the world to develop valuable career skills by taking on a role within an equity research team. Teams of three to five students analyze publicly traded companies selected by their local CFA Society® and present their research and recommendations in both written and oral form. The winning team in each local competition is chosen based on the thoroughness of its analysis and knowledge of a single company, rather than whether their predictions of the stock price are correct. Winners then move on to compete in broader-based regional, national, and global competitions.

As of 2025, participation in the CFA Institute Research Challenge had grown to include well over 1,000 universities from almost 100 countries and regions globally. Across the 18 years of competitions, the global champions have come from 15 different universities and 12 countries and regions. Students can learn more about the CFA Institute Research Challenge through their local CFA Societies or by visiting the CFA Institute website.

QUESTION SET 1



1. Which of the following is *most likely* a key element included in an equity research report when initiating coverage on a company?
 - A. A general summary of economic conditions affecting global equity markets
 - B. A listing of all regulatory filings made by the company in the past fiscal year
 - C. A comprehensive analysis of the company's business model, competitive environment, and valuation

Solution:

C is correct. Initiating coverage reports typically contain a thorough analysis of a company's operations, industry setting, competitive position, and valuation. These elements provide a comprehensive view of the investment opportunity and support the analyst's recommendation.

A is incorrect. A general summary of global economic conditions may be relevant context but is not a core element of an equity research report focused on a specific company.

B is incorrect. Although regulatory filings may be referenced as sources of data, a full listing of past filings is not a standard feature of an equity research report and does not directly support valuation or recommendations.

2. Which section of an equity research report is *most likely* to address potential supply chain disruptions and increasing crude oil prices for an airline company?

- A. Investment risks
- B. Valuation and financial analysis
- C. Business description, industry overview, and competitive positioning

Solution:

A is correct. The risk section is where the analyst discusses potential events or outcomes that might adversely affect the firm's performance.

B is incorrect. The financial analysis section is meant to focus on key financial ratios and measures that reflect the analyst's economic analysis and support their investment thesis. It is not typically used to delve into specific scenarios around these drivers.

C is incorrect. The business description section may talk about the supply chain or oil pricing as general financial drivers but is unlikely to elaborate on specific scenarios.

3. The financial analysis section of an equity research report should:

- A. focus on key financial measures and ratios.
- B. provide an exhaustive financial statement analysis.
- C. use standardized measures sourced from financial data providers.

Solution:

A is correct. The analyst's goal should not be to provide an exhaustive financial statement analysis, but rather to focus on key financial measures and ratios that reflect their economic analysis of the company and support the investment thesis and valuation narrative.

B is incorrect. The analyst's goal should not be to provide an exhaustive financial statement analysis, but rather to focus on key financial measures and ratios.

C is incorrect. The analyst should directly review the firm's financial disclosures and footnotes rather than simply relying on information sourced from financial data providers.

4. The first page of an equity research report is expected to include:

- A. a list of potential merger and acquisition (M&A) targets for the company under review.
- B. a target price with associated time horizon and expected return.
- C. an ESG impact score benchmarked against global sustainability indices.

Solution:

B is correct. A target price, along with a time horizon and implied return, is a key component of the analyst's recommendation and is typically highlighted on the first page of an equity research report.

A is incorrect. Speculative M&A targets are rarely listed in the early sections of a report and would typically be part of a strategic discussion buried deeper in the analysis, if included at all.

C is incorrect. Although ESG factors may be discussed, they are not a standard item on the first page of the report unless material to the investment thesis.

SELL-SIDE VERSUS BUY-SIDE EQUITY ANALYST REPORTS

3

- ☐ compare sell-side and buy-side equity analyst reports and their conclusions

Although the elements of equity research reports are similar, the perspective of a research analyst is an important distinction. Sell-side equity analysts are commonly employed by investment banks creating reports for their clients, and buy-side analysts typically create internal reports used to support the investment decision making process. Although each of these analyst groups seeks to provide insight and analysis critical to more informed investor decision-making, a variety of important differences exist between these report types. Exhibit 6 provides a comparison of sell-side and buy-side equity research report features.

Exhibit 6: Comparison of Sell-Side vs. Buy-Side Equity Research Reports		
Feature	Sell-Side Research	Buy-Side Research
Primary audience	External investors, traders, institutional and retail clients	Internal investment teams and portfolio managers
Objective	Provide research and recommendations to firm clients	Support internal investment decisions for portfolio managers and fund managers
Depth of analysis	Broad coverage across multiple stocks and industries, with standardized formats	In-depth and focused on specific stocks or sectors that align with the firm's investment strategy
Valuation	Commonly accepted models, such as DCF and multiples with emphasis on earnings forecasts	May incorporate proprietary models with focus on type of firm being valued
Level of detail	Often more general to cater to a broad audience	Customized for internal decision-making
Regulatory constraints	Restrictions imposed on analyst research because of the analyst firm's relationships with covered companies or investment banking clients	Largely unconstrained as reports are used for internal portfolio management rather than public dissemination

Subsequent sections of this lesson provide additional discussion of each of these two types of equity research reports.

Sell-Side Equity Analyst Research Reports

Investment banks incorporate equity research into their business model to support their capabilities in underwriting and issuing securities, managing corporate advisory relationships, and enhancing broker/dealer trading for clients. Equity research teams in sell-side firms focus on specific industry sectors by conducting research on selected companies within a sector. The sector analyst typically has deep industry knowledge, possibly gained from experience working within the targeted industry.

Sell-side firms operate within a framework of regulatory constraints that separates research from underwriting and trading activities to ensure the objectivity of research recommendations. This separation of research from investment banking operations limits potential conflicts of interest between the incentives for sell-side research analysts to conduct independent research and those of a sell-side firm’s investment banking unit to solicit underwriting and other business from companies being analyzed. For example, although underwriters may have access to material non-public information, such as pending mergers, acquisitions, debt, or equity offerings, sell-side analysts cannot receive such material information from companies unless it is made public, limiting the information that may be used in research reports. Also, analysts are restricted from issuing reports for a period around an IPO or secondary equity offering. The second European Markets in Financial Instruments Directive (known as MiFID II) aimed to further separate trading and research by requiring asset managers to explicitly pay for sell-side investment research separately from trade execution costs.

Although the details of research reports in many cases are subject to permissioning, financial data providers such as Bloomberg often summarize information on these fundamentals as shown in the following example.

EXAMPLE 10

Analyst Information Summarized in Bloomberg EEO Function

Bloomberg’s Consensus Overview (EEO) function provides summary information for a variety of different financial data items of interest for investors and other analysts following specific stocks. Exhibit 7 shows a screenshot of the EEO function for Tesla Inc. (TSLA) on 24 April 2025.

Exhibit 7: Analyst Estimates of Tesla Inc. Financial Data

TSLA US Equity		Actions	Export	Settings	Consensus Overview				
Tesla Inc		Periodicity	Quarters	Source	Standard	Cur	USD		
More estimates are available		MODI		Broker	Guidance				
1. Headline 2. Headline Growth 3. Company-Specific									
		Q1 2025 Act	Q2 2025 Est	Q3 2025 Est	Q4 2025 Est	Q1 2026 Est			
		03/31/2025	06/30/2025	09/30/2025	12/31/2025	03/31/2026			
3 Months Ending									
(1) EPS, Adj+	0.270	0.494 ↓	34	0.564 ↓	33	0.649 ↓	33	0.619 ↓	17
(2) EPS, GAAP	0.120	0.398 ↓	26	0.472 ↓	25	0.574 ↓	25	0.555 ↓	12
(3) Revenue	19.335B	24.525B ↓	37	26.015B ↓	36	28.263B ↓	36	25.786B ↓	20
(4) Gross Margin %	16.300	16.860 ↓	30	17.033 ↓	29	17.170 ↓	29	17.848 ↓	18
(5) Operating Profit	399.000M	1.560B ↓	36	1.846B ↓	35	2.199B ↓	35	2.060B ↓	19
(6) EBIT	399.000M	1.545B ↓	32	1.857B ↓	31	2.198B ↓	31	2.109B ↓	17
(7) EBITDA	2.814B	3.520B ↓	23	3.820B ↓	23	4.233B ↓	23	4.228B ↓	13
(8) Pre-Tax Profit	1.162B	1.847B ↓	22	2.021B ↓	22	2.305B ↓	22	2.162B ↓	11
(9) Net Income Adj+	934.000M	1.805B ↓	31	2.041B ↓	31	2.328B ↓	31	2.324B ↓	17
Current Multiples		5. Hide Multiples							
		Last 4 Qtrs Act	Next 4 Qtrs Est	FY 2025	FY 2026	FY 2027			
Price/EPS, Adj+	114.25		110.08	114.81	79.43	57.26			
Price/Book	11.04		11.42	10.66	9.49	8.16			
Price/Cash Flow	48.69		30.08	64.49	49.94	43.81			
EV/Revenue	8.37		7.66	7.80	6.47	5.31			
EV/EBITDA	67.48		50.35	55.10	41.69	31.15			
EV/EBIT	127.09		103.92	120.16	73.03	50.70			
EV/OPP	127.09		104.51	115.53	70.34	47.94			
Dividend Yield	0.00			0.00	0.00				

Source: Bloomberg.

As one example of data shown in Exhibit 7, Tesla's estimated adjusted EPS for its quarter ending on 30 June 2025 is USD0.494. This amount incorporates the estimates of 34 different analysts and reflects a decrease from prior estimates. The EPS figure cited earlier is based on non-GAAP EPS, and 26 analysts forecast GAAP EPS. Individuals using this data must decide which EPS format best fits their decision-making framework. Tesla's estimated revenue incorporates forecasts from 37 different analysts. The tab labeled "Headline Growth" shows the estimates as growth rates. Overall, the EEO function provides a helpful tool for analysts benchmarking their forecasts against the market consensus. These sell-side forecasts provide a useful consensus discovery function for buy-side analysts and investors.

Except in cases in which users explicitly pay for sell-side equity research, research is a cost center rather than a source of revenue. As a result, firms are strategic about their choices regarding their research activities. Specifically, firms must decide the following:

- which industries are covered by their research teams,
- how many companies are covered within each industry, and
- the desired characteristics of companies covered.

One source of investment bank revenue is generated by trading commissions, so whether or not a company is covered is likely to depend on the expected trading volume of its stock. For example, hedge funds are likely to be consumers of research and bank broker/dealers may use their firms' equity research reports on stocks of particular interest to these clients as a selling point to influence trading volumes and commissions. Sophisticated investors who value in-depth, quality research tend to reward those firms producing it.

In contrast, large public issuers also place value on the quality of an underwriter's equity research reports. For example, strong ongoing equity research coverage helps to attract institutional investors by increasing available information to investors, thus reducing information asymmetry. Equity coverage of a firm by one or more major underwriters is a signal to market participants of credibility and enhances a firm's standing in the capital markets.

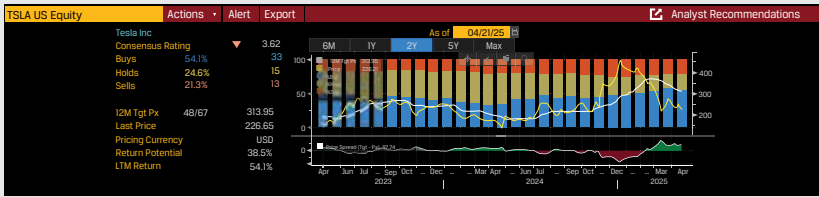
Changes in equity research price targets are a common focus of the financial press given the broad market following of a number of sell-side analysts. Despite the many constraints faced by sell-side analysts, they are usually reluctant to issue sell recommendations that might result in negative trading sentiment by market participants and potentially harm relations with management of the company. In general, taking a **contrarian view** on a stock as a sell-side analyst requires a high degree of conviction, as recommendations are in the public domain and involve a high degree of risk to one's reputation when rejecting the consensus and being wrong. Analysts, however, often change recommendations between buy and hold and sometimes may issue sell recommendations. A rapid increase in a company's stock price may create the impetus for analysts to change their ratings on a stock as shown in the following example.

EXAMPLE 11

Analyst Recommendations and Tesla Stock’s Rise and Fall in Late 2024 Through Early 2025

As the demand for electric vehicles grew in the early 2020s, Tesla stock was a primary beneficiary. During the second half of 2023 and early 2024, however, investor excitement for Tesla stock had calmed. Exhibit 8 shows a Bloomberg screenshot summarizing Tesla’s price and analyst recommendations from April 2023 to April 2025.

Exhibit 8: Analyst Equity Investment Recommendations for Tesla Inc.



Source: Bloomberg.

Analyst buy recommendations reached a low point in March 2024, as many switched to hold recommendations on Tesla. These recommendations coincided with prices that, in hindsight, were at the lowest levels observed during the time period under consideration. Analyst buy recommendations gradually become a greater percentage than holds through the middle of 2024. During the last few months of 2024, Tesla’s stock price exploded as analyst recommendations shifted more toward buys. During this time, analyst price targets for Tesla were lower than the stock’s traded price, but this situation was quickly reversed in early 2025 as both the US stock market, and Tesla stock in particular, suffered significant declines. Nevertheless, analysts were not equally quick to reverse their buy ratings on Tesla, as 54% of analysts still had buy ratings on the stock as of late April 2025.

Buy-side analysts write their own research reports, as described in the following section, but the buy side regularly follows sell-side research to gain the detailed knowledge it contains, although these buy-side firms may not necessarily follow sell-side analyst recommendations.

Buy-Side Equity Analyst Research Reports

Research conducted within buy-side firms is typically confidential and reported only to investment decision makers within a firm. These reports are frequently used by buy-side firm investment committees to both support investment decisions and monitor existing positions. An individual company stock for which a report is created has likely passed a rigorous **stock screening** process, which features the following general steps:

- identify an appropriate initial set of stocks, such as all firms within a specified index;
- construct and apply quantitative metrics and criteria that are consistent with the decision maker’s investment strategy, such as a minimum required earnings growth rate over a specified threshold for an investor with a growth focus;

- construct and apply secondary metrics that ensure companies continuing to pass the screen fit well with the criteria sought by the investment strategy; and
- refine screening criteria such that the number of stocks passing all screens is sufficiently limited so that each company is able to be researched at an appropriate level of detail to distinguish between the investment potential of each stock.

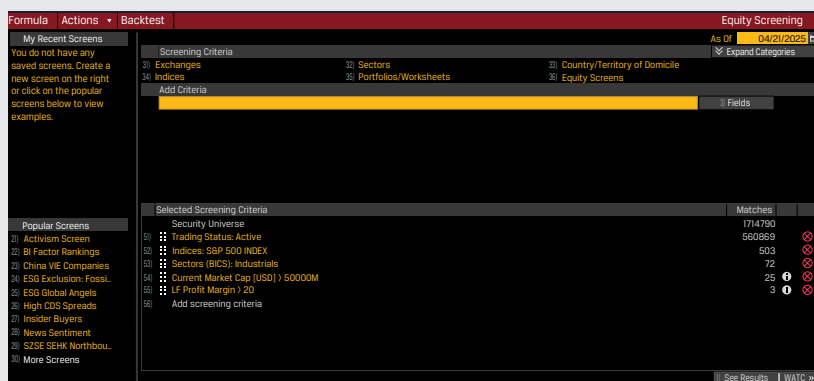
A variety of finance applications provide stock screening tools, including applications powered by artificial intelligence. More advanced screening tools are available through higher-priced data providers, such as Bloomberg, as shown in the following example of a simple screen.

EXAMPLE 12

Illustration of Bloomberg Equity Screening Tool

Analysts can create equity screens within the Bloomberg terminal using the equity screening (EQS) function. Exhibit 9 shows a Bloomberg screenshot of a simple screen for large-cap S&P 500 industrial sector stocks with high profit margins in their latest fiscal year.

Exhibit 9: Bloomberg EQS Illustration



Source: Bloomberg.

The screening process begins with the universe of almost 561,000 securities. The example screen then selects only the stocks included in the S&P 500 index. The next screen selects only stocks classified within the industrials sector, as defined by the Bloomberg Industrial Classification System (BICS). The second-to-last screen limits the search for stocks with a current market capitalization of more than USD50 billion. Finally, the last screen narrows the field to those stocks with profit margins greater than 20%. Information about the three companies that met the criteria are found by clicking on the “See Results” indicator.

A drawback of this screening processes is that the use of arbitrary threshold values may exclude potentially attractive stock candidates, causing them to drop out of an analyst’s consideration, as demonstrated in the following knowledge check.

KNOWLEDGE CHECK: A SIMPLIFIED EQUITY SCREEN ILLUSTRATION



An analyst has 10 stocks in the Consumer Staples sector under consideration and wants to identify the top 2 or 3 stocks on which to conduct in-depth research. The analyst is screening for firms with high return on capital (ROC), low debt, and low equity beta. The statistics for the population of firms are shown in Exhibit 10.

Exhibit 10: Equity Screen Illustration

Stock	ROC	Rank order	Debt-to-assets	Rank order	Equity beta	Rank order
1	8%	7	49%	7	0.49	3
2	23%	1	60%	10	0.44	1
3	8%	7	45%	6	0.48	2
4	12%	4	52%	9	0.64	6
5	9%	6	37%	5	0.81	10
6	7%	9	30%	3	0.67	7
7	5%	10	26%	2	0.77	9
8	11%	5	49%	7	0.58	5
9	16%	3	35%	4	0.54	4
10	20%	2	22%	1	0.71	8
Average	11.9%	---	40.5%	---	0.613	---
Median	10.0%	---	41.0%	---	0.61	---

1. The analyst decides to set the averages of each screening variable as the threshold values. Identify all stocks that meet or exceed the analyst's screening criteria.

Solution:

Stock 9 is the only one that is above average for return on capital and below average for debt ratio and equity beta. Its return on capital of 16% is greater than the average of 11.9%. Its debt ratio of 35% is below the average of 40.5% and its equity beta of 0.54 is below the average of 0.613.

2. The analyst seeks to add one or two more stocks for possible consideration in the research process. Identify the stock that is most likely to be attractive given the analyst's screening preferences.

Solution:

Stock 10 is a likely stock to be considered along with Stock 9. Stock 10 has a high return on capital of 20% and a low debt ratio of 22%. On these two characteristics, Stock 10 appears to be superior to Stock 9. Stock 10, however, did not initially pass the screening criteria associated with equity beta. Its beta of 0.71 is above the average beta of 0.613.

Stock 2 may also be worth considering because its return on capital of 23% is the highest in the population with below average equity beta of 0.44. Its debt ratio of 60%, however, is the highest among the population of firms.

A rank ordering of each of the three screening characteristics across all the stocks is an alternative way to screen the stocks. For example, an analyst using equal weights across the rank orders of ROC, debt ratio, and beta would narrow the set of 10 to Stocks 10, 9, and 2.

Regardless of the screening process employed, buy-side firms must selectively decide where to invest their time and effort in conducting in-depth research reports. Unlike sell-side firm research which tends to focus on larger companies with significant trading volume, buy-side research may have greater flexibility to pursue other priorities, such as smaller stocks that receive less institutional attention.

Research by Activist Short Sellers

Although the name may imply a focus on buying stocks, buy-side research actually supports a range of investment decisions, including buying, holding, or selling securities. **Activist short sellers** conduct research on stocks that are likely to decline significantly in value. In contrast to typical buy-side reports that are kept confidential, activist short sellers often circulate their research publicly following the release of research to their limited partner investors. The broad circulation of these reports is used to influence market consensus and the price of a stock as well as to attract new clients. The research done by these types of firms is commonly focused on identification of misinformation or fraud that is not currently being recognized in the price of the target company stocks as discussed in the following example.

EXAMPLE 13

Hindenburg Research: An Example of a Successful Activist Short Seller

Hindenburg Research began as Nathan Anderson's pseudonym for posting his short-selling ideas on a crowdsourced finance research website, Seeking Alpha, in 2017. Hindenburg Research gained notoriety in late 2017 with research on a biotech-turned-crypto-currency firm's stock in which Anderson suspected manipulative trading activity by the company's largest shareholder.

In September 2020, Hindenburg's profile grew with its research on electric-vehicle start-up Nikola, in which Hindenburg argued of fraud being perpetuated by the company's founder and CEO. The US Securities and Exchange Commission (SEC) found the research sufficiently credible to warrant prompt investigation of the claims. The founder of Nikola was ultimately convicted criminally for securities fraud for his false claims about the company's business.

In January 2025, Nathan Anderson announced the closure of Hindenburg Research. The firm had completed analysis on its pipeline of outstanding cases. As is typical for activist short sellers, the firm remained small with a team of only 11 staffers. In Anderson's closing message on the firm's website, he discussed plans to eventually share materials and videos about Hindenburg Research's business model.

Importance of Communication in Equity Research Reports

An important commonality is that all equity research should reflect the highest analytical and ethical standards. A well-written report cannot compensate for poor analysis, although a poorly written report can detract from the credibility of an excellent analysis. An effective research report:

- contains timely information;
- is written in clear, incisive language;
- is objective and well researched, with key assumptions clearly identified;
- distinguishes clearly between facts and opinions;
- contains analysis, forecasts, valuation, and a recommendation that are internally consistent;
- presents sufficient information to allow a reader to critique the valuation;
- states the key risk factors involved in an investment in the company; and
- discloses any potential conflicts of interests faced by the analyst.

QUESTION SET 2



1. Compared with a sell-side report, a buy-side equity research report is *most likely* targeted to:

- A. external clients.
- B. internal investment teams.
- C. high-net-worth individuals.

Solution:

B is correct. Buy-side reports are typically confidential and intended for internal use by portfolio managers and investment teams.

A is incorrect. It describes sell-side research reports. Buy-side reports are typically confidential and intended for internal use by portfolio managers and investment teams.

C is incorrect. High-net-worth individuals are one type of client a sell-side research report might be intended for. Buy-side reports are typically confidential and intended for internal use by portfolio managers and investment teams.

2. Activist short-seller research reports are typically released to the public:

- A. as soon as they are produced.
- B. after being released to limited partners.
- C. only after the misinformation or fraud described within them is exposed.

Solution:

B is correct. Activist short-seller reports are created for internal purposes and to serve the needs of limited partner investors, whereas wider public distribution occurs later to influence market consensus and attract new clients.

A is incorrect. This describes sell-side reports, which are disseminated broadly.

C is incorrect. Activist short sellers seek to publicize their views on a stock to influence market consensus and drive price changes.

3. Why might a sell-side analyst be reluctant to issue a negative rating on a stock?

- A. Issuing a negative rating may damage the firm's investment banking relationships with the company.
- B. Negative ratings are prohibited by most regulatory bodies unless justified by audited financial statements.
- C. Sell-side analysts are ethically obligated to maintain positive ratings on all covered securities.

Solution:

A is correct. Sell-side firms often have investment banking relationships with companies they cover. Analysts may be reluctant to issue negative recommendations if doing so could jeopardize those relationships or reduce client trading interest.

B is incorrect. Regulatory bodies do not prohibit negative ratings, although such ratings must be supported by reasonable analysis.

C is incorrect. Ethical standards do not require positive ratings; analysts must issue objective recommendations based on their analysis.

4. Which of the following is *most likely* a feature of a sell-side equity research report?

- A. It excludes financial models to maintain client confidentiality.
- B. It includes proprietary investment strategies and is distributed only within the firm.
- C. It includes a price target and is updated regularly in response to company earnings releases.

Solution:

C is correct. Sell-side reports typically include a target price and are regularly updated following material events, such as earnings announcements.

A is incorrect. Financial models are often included in sell-side reports to support recommendations and are not withheld for confidentiality reasons.

B is incorrect. Proprietary strategies used internally are characteristic of buy-side, not sell-side, research.

5. Sell-side analysts tend to be subject to greater regulation than buy-side analysts because:

- A. they deal directly with the general public.
- B. they face more potential conflicts of interest.
- C. they control a large proportion of investable assets.

Solution:

B is correct. Sell-side firms receive their revenue from investment banking and trading operations. These parts of the organization may have interests that are not aligned with those of external investors. For example, a negative rating on a stock may make that company unwilling to send the firm investment banking deals, so the analysts could feel pressure to maintain positive ratings even if the analyst does not fully support them.

A is incorrect. Sell-side firms deal primarily with buy-side clients. Both types of analyst deal with the general public only indirectly.

C is incorrect. The sell-side does not directly control investable assets.

4

EQUITY RESEARCH RECONCILIATION

- ☐
- explain why value estimates from the same modeling approach may differ based on analyst assumptions regarding revenue, profitability, investment, and financing

A critical aspect of the investment landscape is that market participants disagree about the values of securities. Different participants express their opinions about the value of securities through their trading actions, and these opinions may be informed by analyst research reports. In this learning module, we focus on discussing why equity valuation models used by analysts produce different value estimates for the same stock. We first focus on growth implications associated with firms facing highly uncertain growth prospects.

“Hard-to-Value” Equities and Model Growth Implications

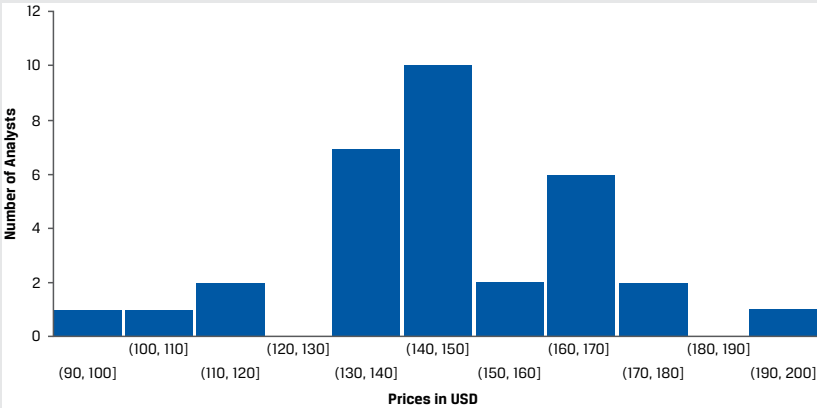
In the prior Equities learning modules, the focus of numeric examples typically was on illustrating principles of valuing stocks using fictional examples. Although these examples have occasionally included scenarios, we have not yet highlighted the issue of valuations of the same stock across different analysts. Some stocks are labeled as “**hard to value**” because of significant dispersion across well-informed market participants about the company’s intrinsic value. In the following example, we examine a real-world scenario in which analysts had starkly different opinions about the company’s value.

EXAMPLE 14

Analyst Target Prices of Nvidia Stock in Late August 2024

On 29 and 30 August 2024, 32 analysts posted target prices for Nvidia, the dominant provider of chips used in artificial intelligence (AI) applications at the time. As a result of the market’s exuberance for the potential of AI applications, Nvidia’s stock price had risen by almost 8× over the prior two-year period. At the end of August 2024, Nvidia’s stock price was approximately USD120 per share. Exhibit 11 shows a histogram of the target prices across 32 analysts at that time.

Exhibit 11: Analyst Target Prices on Nvidia Stock at End of August 2024



Source: FactSet.

Although more than half of the analysts had price targets between USD130 and USD150 per share, several had expectations of the stock rising to USD170 and higher, with one posting a target of USD200. Very few analysts were bearish on Nvidia's price, and several analysts posted target prices below the company's recent price with one analyst setting a price as low as USD90 per share.

This example shows an instance in which one analyst sets a target price at more than 50% below that of another analyst. What factors drive such differences among professional analysts following the same company? Three of the most important distinctions are:

- differences in opinion about company revenue growth potential over a forecast horizon,
- differences in opinion about company profitability changes over a forecast horizon, and
- differences in terminal value estimates.

This example highlights a straightforward application that fundamental company variables may be reasonable sources of differences in valuation. Although this example assumed that different analysts might consider similar multiples for a company's stock regardless of the level of optimism (or pessimism), candidates should recall from the learning module on relative valuation that fair multiples may differ across stocks based on differences in underlying firm fundamentals. The same principle should likely apply across analysts looking at the same stock when these analysts have differing assumptions about fundamentals, such as expected growth rates.

A useful approach to assess a stock's price is to assume the price (P_t) as equal to intrinsic value (IV_t) within a present value model, and then solve for an underlying assumption that forces the intrinsic value estimate to be equal to the price shown in Equation 1 for a two-stage growth model valuing a company's forecasted free cash flow to equity (FCFE).

$$P_t = IV_t = \sum_{i=1}^n \frac{FCFE_{t+i}}{(1+r_e)^i} + \frac{FCFE_{t+n+1}}{(1+r_e)^n} \frac{1}{r_e - g_l} \quad (1)$$

Each analyst forecasts the following variables for a company:

- FCFE over a forecast horizon, and FCFE for the year beyond the horizon;
- length of the forecast horizon in years, n ;
- cost of equity, r_e and
- long-run growth rate of FCFE for years beyond the forecast horizon, g_l .

If the only unknown variable is the long-run growth rate, Equation 1 may be algebraically rearranged, as shown in Equation 2:

$$g_l = r_e - \frac{FCFE_{t+n+1}}{(1+r_e)^n \left[P_t - \sum_{i=1}^n \frac{FCFE_{t+i}}{(1+r_e)^i} \right]} \quad (2)$$

Although the second term in Equation 2 appears to be complex, the intuition of the equation may be stated as follows: The long-run growth rate implied by a stock's current equity market capitalization, P_t , is equal to the difference between the assumed cost of equity r_e and the forward FCFE yield at the end of the forecast horizon. Specifically, the denominator of the second term shows the implied terminal value necessary for a current value, IV_t , to be equal to its current price, P_t . In the following knowledge check, we illustrate how two analysts employing similar near-term FCFE figures but differing forecast horizons may arrive at differing conclusions about the long-run growth rate of FCFE for the same fictional company.

KNOWLEDGE CHECK: LONG-RUN GROWTH RATES IMPLIED BY CURRENT PRICE



Merlion Microtech is a maker of semiconductor chips with a growing presence in the graphics processing unit (GPU) industry space. The company is publicly listed on the Singapore Exchange. Merlion Microtech's current market cap of equity is SGD30,000 million. Two analysts covering the stock have created FCFE forecasts for Merlion over the next four years with the same expectations. One of the analysts, however, extends their FCFE forecast two years longer than the other analyst. Both analysts use 12% as the cost of equity. Exhibit 12 shows each analyst's FCFE forecast, as well as the present value (PV) calculations and sums of the PVs for each analyst forecast.

Exhibit 12: FCFE Forecasts for Merlion Microtech (in SGD millions)

Year	Analyst A FCFE	Analyst B FCFE	Analyst A FCFE PV	Analyst B FCFE PV
1	500	500	446.4286	446.4286
2	750	750	597.8954	597.8954
3	1,500	1,500	1,067.6704	1,067.6704
4	2,000	2,000	---	1,271.0362
5	---	2,500	---	1,418.5671
6	---	3,000	---	---
Sum of PVs	---	---	2,111.9944	4,801.5976

1. Calculate the implied long-run growth rate for Merlion Microtech stock based on Analyst A's forecast of FCFE.

Solution:

Analyst A's forecast implies a 6.90% long-run growth rate of FCFE beginning after the end of the three-year forecast horizon. The calculation process is shown using Equation 2:

$$g_l = r_e - \frac{\text{FCFE}_{t+n+1}}{(1+r_e)^n \left[P_t - \sum_{i=1}^n \frac{\text{FCFE}_{t+i}}{(1+r_e)^i} \right]}$$

Given Analyst A's forecast, the fourth-year FCFE of SGD2,000 million is the numerator of the second term in the equation. The first term in the denominator of the equation is the future value factor given a discount rate of 12% and a horizon of three years which is equal to approximately 1.4049.

$$(1+r_e)^n = (1.12)^3 = 1.4049.$$

The bracketed second term in the denominator reflects the present value difference between Merlion Microtech stock's current traded value of SGD30,000 million and the sum of the FCFE PVs given in Exhibit 12 as SGD2,111.9944. Thus, the bracketed term equals SGD27,888.0056. The computation of the equation's second term is shown as follows:

$$\frac{2,000}{(1.4049)[27,888.0056]} = 0.051.$$

Thus, the long-run growth rate of FCFE for Merlion Microtech stock is 6.90%, as follows:

$$g_I = 12\% - 5.1\% = 6.9\%.$$

2. Compared with the long-term growth rate implied by Analyst A's forecast, the long-term growth rate implied by Analyst B's forecast will be:

- A. lower.
- B. the same.
- C. higher.

Solution:

A is correct. The second term in Equation 2 is higher based on Analyst B's forecast compared with that of Analyst A's forecast, as follows:

$$\text{Analyst A second term of Equation 2} = \frac{2,000}{(1.4049)[27,888.0056]} = 0.051.$$

$$\text{Analyst B second term of Equation 2} = \frac{3,000}{(1.7623)[25,198.4024]} = 0.0676.$$

Because the second term is subtracted from the cost of equity to obtain the long-run growth rate, Analyst B's forecast implies a lower long-run growth rate. Effectively, Analyst B's forecast of continued strong growth of FCFE for two additional years reduces the long-run growth expectations beyond the end of the explicit FCFE forecast horizon. Furthermore, from the comparison of Analyst B's forecasted annual growth rates in Year 5 (25%) and Year 6 (20%) with the long-term growth rate estimated in Step 1 (6.9%), we can clearly conclude that Analyst B's implied long-term growth rate is lower than that implied by Analyst A's forecast.

B and C are incorrect. See the solution process for correct answer A.

This knowledge check shows an example in which an analyst's longer forecast horizon of underlying company cash flows implies less aggressive assumptions about long-run growth of cash flow. Note also, however, that if Analyst B had assumed sufficiently lower growth in FCFE for Years 5 and 6 of the forecast, the implied long-run growth rate would be higher. Thus, the longer forecast horizon is not sufficient by itself to induce lower implied long-run growth expectations.

An additional lesson from the computation of implied long-run growth expectations across different analyst forecasts may be related to analyst interpretations regarding these results. Analyst A may consider the 6.9% growth rate expectation implied by the stock's current price and believe that the market is too optimistic about the stock. This inference may lead Analyst A to avoid recommending this stock for purchase. In contrast, Analyst B may be more inclined to view the implied long-run growth expectations as reasonable (or perhaps "too low"); thus, this analyst may view the more modest growth expectations implied by the stock's current price as a good argument to recommend the stock for purchase.

Financial Statement Valuation Drivers

Prior material in the Equities learning modules has established that the intrinsic value of a company's stock is driven by expectations of future free cash flow or may be expressed as a multiple of a fundamental variable, such as earnings. Candidates should recall the following three FCFE-related equations from prior Equities content.

$$FCFE = \text{Net income} - \text{Net investment} + \text{Net debt issued}, \quad (3)$$

$$\text{Equity reinvestment rate} = \frac{\text{Net investment} - \text{Net debt issued}}{\text{Net income}} = 1 - \frac{FCFE}{\text{Net income}}, \text{ and} \quad (4)$$

$$\text{Long-run FCFE growth} = \text{Equity reinvestment rate} \times \text{Return on equity}. \quad (5)$$

Embedded in any analyst's free cash flow forecasts are expectations of future revenue potential, profitability, investment needs, and possibly financing choices for the company being valued as well as expectations for long-run growth expectations. In this lesson, we will focus on the valuation effects of each of these components.

Effects of Revenue and Profitability Changes on Value

Companies that exhibit significant revenue growth are often considered to be generating value as market participants may extrapolate large revenue improvements into higher free cash flow in both the near-term and potentially longer horizons. Nevertheless, analysts need to exercise caution in constructing their own valuation models as well as in evaluating others' models. Revenue changes may affect other valuation drivers of a model.

An observed revenue change for a company could be assumed to have positive or negative effects on net income, depending on the direction and size of the revenue change and concurrent changes in the company's cost structure.

Exhibit 13 provides a conceptual guide in terms of considering likely effects of differing revenue and profit margin changes on company equity value within an equity valuation model.

Exhibit 13: Likely Value Changes from Revenue and Net Margin Changes

Changes (relative to market expectations)	Net margin +	Net margin –	No net margin change
Revenue +	Positive value change > Positive revenue change	Value change < Revenue change; Value change may be negative	Positive value change = Positive revenue change
Revenue –	Value change > Negative revenue change; Value change may be positive	Negative value change < Negative revenue change	Negative value change = Negative revenue change
No revenue change	Positive value change associated with profit margin improvement	Negative value change associated with profit margin decline	No value change

Generally, positive revenue changes compared with market expectations are likely to be associated with positive equity value changes. The middle column (Net margin –) highlights that if positive revenue changes coincide with cost pressures, then negative value changes may result.

Analysts should consider the possibility that a company's revenue improvements may be accompanied by disproportionately large expense increases. In such a situation, company profit margins will decline, and that combination of analyst assumptions may create a potentially negative effect on the company's estimated equity value as explained in the following knowledge check.

KNOWLEDGE CHECK: MERLION MICROTECH EQUITY VALUE WITH REVENUE INCREASE AND NET MARGIN DECREASE



Analyst A follows up the work on valuing Merlion Microtech with an explicit assumption regarding the forecast of Merlion's revenue and net income over the next four years, along with expected FCFE. The analyst assumes 12% cost of equity and a 6% long-run growth rate to compute terminal value at the end of Year 3. The analyst's forecast is shown in Exhibit 14.

Exhibit 14: Analyst's Forecast

Year	Revenue	Net income	Net investment	Net debt issued	FCFE
1	8,000	800	2,500	2,200	500
2	12,000	1,200	2,200	1,750	750
3	24,000	2,400	1,900	1,000	1,500
4	32,000	3,200	1,600	400	2,000

Using the FCFE forecast, Analyst A estimates that Merlion's market capitalization of equity equals SGD25,838 million using this initial set of assumptions.

Analyst A then considers that Merlion's management is making aggressive strides in its GPU business lines and believes that a 10% per year improvement in revenues may be possible over the next four years. Analyst A, however, also has uncovered supply chain issues that may create challenges to Merlion's cost structure in this scenario. Specifically, Analyst A assumes that the company's net margin (currently assumed at 10%) declines by 25 basis points in each of the following four years, implying that by Year 4, the company's net margin is only 9%. The analyst further assumes no changes to the forecast of net investment or net debt issued for Merlion over the four years.

1. Calculate the analyst's updated net profit and FCFE forecast for Merlion Microtech under this scenario of higher revenue with lower profit margins.

Solution:

In all four years of the forecast, the updated revenue may be calculated by multiplying the prior forecast numbers by 10%. Net profit for each year reflects the product of the revised net margins and the updated revenue forecasts.

To convert the forecasted net income to forecasted FCFE, Analyst A uses the same difference shown between net income and FCFE in the original scenario. By doing so, the analyst effectively assumes no change in Merlion's net investment and net debt issued. The overall effect of these assumptions is shown in Exhibit 15.

Exhibit 15: Effect of Assumptions

Year	Revenue forecast	Net margin forecast	Net income forecast	FCFE forecast
1	8,800	9.75%	858	558
2	13,200	9.50%	1,254	804
3	26,400	9.25%	2,442	1,542
4	35,200	9.00%	3,168	1,968

In this knowledge check, the increase in revenue is accompanied by a decline in profit margins over the forecast horizon. Analysts should be aware of the possibility that higher revenue might be associated with changes in a company's profitability potential. For example, although current period sales may be rising, industrial or manufacturing firms may face declining profits if they need to retool or make supply chain or production changes that result in higher short-term costs. These linkages are an important part of an analyst's narrative that should be supported by detailed company and industry analysis.

Effects of Investment Changes on Value

A key principle introduced in Corporate Finance learning modules is that corporations make investment expenditures that add value to the company. Thus, the "net investment" component of a company's FCFE should be viewed as possibly foreshadowing future revenue, profitability, and cash flow improvements for a company.

An effective valuation narrative articulates how company strategy influences the interaction between revenue and costs. However, it should also consider potential relationships among revenues, expenses, and investment expenditures as suggested in the following knowledge check.

KNOWLEDGE CHECK: MERLION MICROTECH EQUITY VALUE WITH REVENUE AND NET INVESTMENT INCREASE

Analyst A continues to explore assumptions surrounding Merlion Microtech's valuation. The analyst is concerned that assuming revenue growth without considering additional investment needs may create an upward bias in the valuation. To address this concern, Analyst A increases both revenue and net investment by 10% relative to the base case scenario, while reverting back to the original assumption of a 10% net margin. If Merlion Microtech's net debt issued is not altered from the prior assumptions, Analyst A's assumptions lead to the revised forecast shown in Exhibit 16.

Exhibit 16: Revised Forecast

Year	Revenue	Net income	Net investment	Net debt issued	FCFE
1	8,800	880	2,750	2,200	330
2	13,200	1,320	2,420	1,750	650
3	26,400	2,640	2,090	1,000	1,550
4	35,200	3,520	1,760	400	2,160

Analyst A continues to assume that Merlion Microtech's cost of equity equals 12% and that the company's long-run growth rate is 6%.

1. Compared with the original scenario, under the revised scenario the equity value will increase by:

- A. less than 10%.
- B. 10%.
- C. more than 10%.

Solution:

A is correct. The FCFE forecast is less than 10% between the two scenarios as shown in Exhibit 17.

Exhibit 17: Two FCFE Scenarios

Year	Original forecast FCFE	Updated forecast FCFE	Percentage change between forecasts
1	500	330	-34%
2	750	650	-13.3%
3	1,500	1,550	3.3%
4	2,000	2,160	8%

From the percentage changes in Exhibit 17, it is clear that the value change between scenarios will be less than 10%. The terminal value estimate will be 8% higher in the new scenario, whereas the FCFE during the forecast horizon shows negative changes in Years 1 and 2 and only a small positive change in Year 3.

B is incorrect. Value will only change by exactly 10% if forecasted FCFEs change by 10% between the two scenarios. A 10% change in the equity value estimate would occur if Analyst A also assumed a 10% increase in the net debt issued amounts between the two scenarios.

C is incorrect. A change greater than 10% would require additional assumptions to create larger percentage changes in the FCFE forecast between the two scenarios.

The knowledge check highlights a valuation scenario in which an analyst assumes that revenue, income, and investment change similarly. In general, analysts should be wary of valuations in which revenue and income changes are assumed without proportional changes in investment. The cash flow benefits of additional income from growing revenue are not likely to occur without proportional growth in investment expenditures.

Overall, an analyst's assumptions about a company's investment behavior is likely to have a direct effect on the analyst's free cash flow forecast and may have implications for the factors associated with long-run growth of the company's free cash flow. Exhibit 18 provides a brief summary.

Exhibit 18: Summary of Likely Forecasting Assumptions from Investment Assumptions

Investment assumption	Free cash flow outcomes	Possible long-run free cash flow growth rate implications
Investment growth rate > Income growth rate	Reduction in free cash flow	Higher long-run reinvestment rate; Lower long-run return on capital
Investment growth rate = Income growth rate	No upward or downward pressure on free cash flow	Consistent with long-run stable growth of free cash flow
Investment growth rate < income growth rate	Increase in free cash flow	Lower long-run reinvestment; Higher long-run return on capital

The third column of Exhibit 18 highlights that one of the factors associated with long-run free cash flow growth is a company's return on capital. An analyst forecast should be clear regarding assumptions about a company's forecasted return on capital, given that this factor has a direct effect on whether growth is valuable or not.

Analysts also apply financing assumptions as part of their valuation models. Although the form of financing used may have some positive or negative effect on a company's value, value creation by firms is predominantly a function of the company's operating strategies and results. Thus, analysts should take care when building valuation models to ensure that they do not inadvertently assume significant value creation from their underlying financing assumptions.

Divergence of Opinion Among Equity Analysts

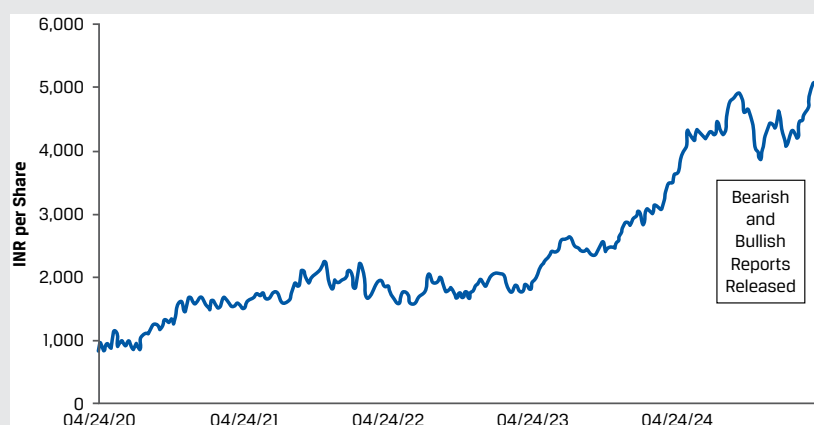
A common misconception among junior analysts is that equity research reports based on similar historical financial data can and should usually converge to a market consensus view associated with a particular share price. In fact, differing views of a company's future prospects lead to a wider range of investor expectations, encouraging price discovery and supporting efficient markets. As in the extreme case of activist short sellers described earlier, contrarian views among equity analysts, or those that run counter to market consensus, often provide valuable insight to market participants in challenging the status quo and thus contribute to the proper functioning of equity markets.

As a summary of this lesson, we revisit the case of InterGlobe Aviation by introducing a bearish report in addition to the bullish report discussed in the earlier lessons to demonstrate how two analysts working with the same firm's data may arrive at very different conclusions regarding the stock's intrinsic value.

EXAMPLE 15

Significant Differences in Analyst Opinions of InterGlobe Aviation Stock

Analyst research about InterGlobe Aviation stock was featured earlier in this learning module. The stock has shown excellent performance during the five years following the outbreak of COVID-19. In particular, the company's stock showed large gains during 2023 and early 2024. Exhibit 19 shows the price of InterGlobe Aviation stock from April 2020 through April 2025.

Exhibit 19: InterGlobe Aviation Stock Price, 2020 to 2025

Source: FactSet.

IndiGo's parent company showed strong stock price performance from 2023 to early 2025; two separate research reports published during this period highlight a considerable difference in opinion about the expected prospects of the stock through 2025 and 2026. In mid-2024, one analyst from a firm that we label "Bearish" expected a significant decline of more than 40% in InterGlobe Aviation's price. In early 2025, an analyst from another firm that we label "Bullish" completed a report with a price target well over 2× that of the bearish analyst's price target.

Although the two analysts employed different relative valuation methodologies to estimate their price targets for InterGlobe Aviation stock, the bearish forecast demonstrated a pessimistic view of the company's expected financial performance over the next couple of years. Exhibit 20 shows ratios of the bearish analyst's revenue and EBITDA forecasts to those of the bullish analyst for the fiscal year ending in March 2026.

Exhibit 20: 2026 Comparisons of Analyst Forecasts

Forecasts	Bearish forecast to Bullish forecast
Fiscal year 2026 revenue	0.854
Fiscal year 2026 EBITDA	0.651

The bearish analyst forecasts only about 85% of the revenue forecasted by the bullish analyst for InterGlobe Aviation. More important, the company's forecasted EBITDA from the bearish analyst is only 65% of that forecasted by the bullish analyst. Although other differences are certainly driving the wide gap between the analysts' price targets for InterGlobe Aviation, the bearish analyst's pessimism about the company's ability to generate revenues and profits in the near term is a primary driver.

An interesting contrast with the previous example is how analyst perspectives can differ based on their location. Some global research analysts may view InterGlobe Aviation as a broader opportunity to gain exposure to the Indian stock market, whereas analysts based in India may focus more closely on region-specific competitive issues that are less visible to international observers. To some extent, analysts will always disagree about company valuation, but the degree of such disagreements often reflects disparities in analyst assumptions about a company's growth potential, profitability, or both.

QUESTION SET 3

1. Analyst A and Analyst B are valuing a company using a FCFE model. Analyst A has a bullish opinion but Analyst B is bearish. The difference in opinion is *best* explained if Analyst B's expectations include greater:
- A. deleveraging.
 - B. revenue growth.
 - C. positive net present value (NPV) investment opportunities.

Solution:

A is correct. In an FCFE model, delevering reduces the cash available to equity. This reduced cash, in turn, would reduce the valuation applied to equity.

B is incorrect. Greater revenue growth is more likely to result in higher FCFE and an increased valuation.

C is incorrect. Greater opportunities for positive NPV investments should result in a higher valuation.

2. Increasing the size of an investment assumption while keeping net income and debt issuance constant in a FCFE model will:
- A. have no effect on FCFE unless discount rates change.
 - B. increase FCFE and raise the estimated value of equity.
 - C. reduce FCFE and may lower the estimated value of equity.

Solution:

C is correct. Increasing net investment while keeping net income and debt issuance constant reduces FCFE, which leads to a lower present value of expected cash flows and potentially a lower valuation for the stock.

A is incorrect. FCFE is directly affected by investment assumptions, independent of discount rates.

B is incorrect. Higher investment increases capital outlays, reducing FCFE, not increasing it.

3. An analyst is valuing two companies and believes that both have virtually identical prospects. The current market price implies a lower long-term growth rate for Company 1 than for Company 2. The *most* appropriate conclusion is that:
- A. Company 1 is a better value.
 - B. Company 2 is a better value.
 - C. both companies are fairly valued.

Solution:

A is correct. Its market price implies lower growth, but the analyst believes its growth will be similar. On a relative basis, then, Company 1 offers similar growth at a lower price and is thus the better value.

B is incorrect. Company 1 is the better value.

C is incorrect. Company 1 is the better value.

SUMMARY: EQUITY ANALYST RESEARCH REPORTS

- Equity research reports typically begin with an analyst's investment recommendation which is supported by a price target. An investment thesis, consisting of key narrative points supporting the recommendation, is also shown early in a report. Detailed company and industry as well as financial analysis and valuation provide background support for the recommendation.
- Reports are structured to persuade readers through coherent qualitative and quantitative analysis. The details contained within an equity research report vary depending on whether the report is generated by a sell-side or buy-side firm.
- Sell-side reports are prepared by investment banks or brokerages for external distribution and often follow standardized formats. Sell-side analysts face constraints and risks when issuing negative recommendations and thus often limit themselves to buy or hold recommendations.
- Buy-side reports are prepared for internal investment decision-making and typically remain confidential within the firm. Buy-side analysts may use more proprietary models than sell-side analysts to value companies.
- Companies with significant dispersion across analysts about future growth potential are often considered "hard to value" because of difficulties in reconciling the intrinsic value of their equities.
- Analysts may produce differing price targets because of differences in assumptions about revenue growth, profit margins, investment levels, or financing choices.

REFERENCES

Sarin, Aradhana. 2024. "Delivering Shareholder Value." Investor Day Presentation. https://www.astrazeneca.com/content/dam/az/Investor_Relations/events/ID/Delivering-Shareholder-Value-presentation.pdf.

PRACTICE PROBLEMS

1. An effective investment thesis:
 - A. includes a detailed analysis of the company's financial statements.
 - B. articulates why the analyst believes the stock is mispriced.
 - C. includes a detailed analysis of the industry or industries within which it operates.
2. An equity research report that focuses on misinformation or fraud not currently reflected in the stock price is *most likely* written by:
 - A. a buy-side analyst.
 - B. a sell-side analyst.
 - C. an activist short seller.
3. Compared with sell-side reports, buy-side equity research reports are typically:
 - A. more customized.
 - B. more constrained by regulatory restrictions.
 - C. produced to provide recommendations to firm clients.
4. The purpose of a stock screening process is to:
 - A. expand the investable universe.
 - B. provide more accurate valuations.
 - C. focus resources on opportunities that are consistent with the investment strategy.
5. An increased revenue forecast is *most likely* to result in an increased valuation estimate if it is accompanied by an increase in:
 - A. net margin only.
 - B. capital investment only.
 - C. both net margin and capital investment.
6. An analyst valuing a company's stock revises income statement assumptions to increase revenue growth while decreasing margins. The *most likely* implication is that:
 - A. the lower margins will result in a lower equity value.
 - B. the higher revenue growth will result in higher equity value.
 - C. the equity value change depends on the magnitude of the offsetting assumptions.

SOLUTIONS

1. B is correct. An effective investment thesis incorporates information as to why the stock's price should converge to the price target. The thesis succinctly articulates the analyst's findings and conclusions. It often includes not only company-specific drivers of performance but also the underlying economic and industry environment that will contribute to the analyst's predicted outcome for the company. The thesis should be directly aligned to achieving the price target within the anticipated time horizon.
A is incorrect. An effective investment thesis reflects an analyst's abilities to condense key aspects of their investment rationale into a short summary. Detailed financial statement analysis is not part of the investment thesis.
C is incorrect. An effective investment thesis does not include detailed information from the overall research. Rather, the analyst condenses this information into the key points driving why the stock's current mispricing should correct within the anticipated time horizon.
2. C is correct. Activist short sellers focus on identifying misinformation or fraud that is not currently reflected in a company's share price.
A is incorrect. Buy-side reports are done within, or on behalf of, a buy-side firm that is typically confidential and not focused on misinformation or fraud.
B is incorrect. Sell-side reports are written for external clients and typically do not focus on misinformation or fraud.
3. A is correct. Buy-side reports tend to be more customized for internal decision making. In contrast, sell-side reports are more general because they cater to a broad audience.
B is incorrect. Buy-side reports are largely unconstrained.
C is incorrect. Buy-side reports are intended for internal use by firm decision makers.
4. C is correct. Screening reduces the number of potential stocks to those that fit criteria that are consistent with the strategy.
A is incorrect. Screening reduces the size of the investable universe.
B is incorrect. Equity screening applies before valuation models are used, and it does not ensure accuracy in valuation outcomes.
5. A is correct. Increased revenue plus increased margin will result in higher forecasted free cash flows and a positive value change.
B is incorrect. Increased capital investment will result in lower free cash flows and may result in decreased value.
C is incorrect. The positive valuation effect of the net margin increase may be offset by the effect of increased capital investment.
6. C is correct. The offsetting effects of higher revenue growth and lower margins will result in profits that may be higher, lower, or possibly equal to the profits assumed in the original scenario. For example, small declines in margins with large revenue growth improvements may result in higher FCFE and higher equity value. In contrast, large declines in margins combined with relatively small revenue growth improvements may result in lower FCFE and lower equity value.
A is incorrect. Lower margins are only likely to result in a revised lower equity value if they more than offset increased revenue growth. Specifically, lower margins may still result in higher net income because of higher revenue growth

assumptions.

B is incorrect. Higher revenue growth is not a guarantee of higher equity value if net income assumptions are revised downward because of lower margins.

LEARNING MODULE

12

The Capital Asset Pricing Model, Market Model, and Other Factor-Based Equity Models

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	explain the use of the CAPM and the market model to estimate a company's required return on equity
☐	describe the arbitrage pricing theory and the importance of multi-factor models used in equity investing

INTRODUCTION: THE CAPITAL ASSET PRICING MODEL, MARKET MODEL, AND OTHER FACTOR-BASED EQUITY MODELS

1

A critical component of equity valuation is the estimation of a discount rate used to calculate the present value of a company's expected future cash flows. In earlier Equities learning modules, we have simply assumed a discount rate. This learning module addresses the source of this discount rate by estimating a company's required return on equity. Analyst estimates of a company's required return on equity are critical inputs for equity valuation.

In this learning module, we first discuss the **capital asset pricing model (CAPM)**, a primary tool used by analysts to estimate required returns on equity. The second lesson focuses on the estimation of a company risk factor, known as beta, and the expected equity **market risk premium (MRP)**. This estimation is extended to the use of comparable companies to estimate an industry beta as well as the consideration of a **country risk premium (CRP)** for equities outside of the most highly developed markets. While the CAPM is a **single-factor model**, the third lesson addresses models incorporating multiple risk factors for explaining equity returns and illustrates their use in factor-based investing.

PRE-TEST: THE CAPITAL ASSET PRICING MODEL, MARKET MODEL, AND OTHER FACTOR-BASED EQUITY MODELS


1. When the CAPM is used to estimate the required rates of return of two individual stocks that operate in the same country at the same point in time, differences in the estimates are *best* attributed to differences in:

- A. risk-free rates.
- B. beta estimates.
- C. market risk premium estimates.

Solution:

B is correct because differences in beta estimates are the only source of differences in estimated costs of equity when using the CAPM at a single point in time. The beta estimates reflect the relative risk for stocks of different companies.

A is incorrect because the risk-free rate reflects macroeconomic conditions, not company-specific difference. Thus, this variable does not change, as required returns are estimated across different securities or at the same point in time.

C is incorrect because the expected market risk premium reflects macroeconomic conditions, not company-specific differences. Thus, this variable does not change as required returns are estimated across different securities at the same point in time.

2. An adjusted beta is typically used:

- A. to leverage an industry beta.
- B. as part of an unlevering process.
- C. to account for mean reversion of beta towards 1.0.

Solution:

C is correct. Choice C is correct because an adjusted beta estimate incorporates a weighted average approach, as shown below.

$$\text{Adjusted BETA} = w_1 \times \text{Raw BETA} + w_2 \times 1.0.$$

As long as the weight on the raw beta is less than 1.0, the adjusted beta calculation causes the adjusted beta to be closer to 1.0.

A is incorrect because this statement describes part of an industry comparable approach to estimating beta.

B is incorrect because this statement describes the process of calculating an unlevered beta from an industry beta.

3. An implied market risk premium is:

- A. a forward-looking estimate that incorporates a current stock market index level, a current government rate, and assumptions about future index growth.
- B. a historical estimate that incorporates the historical average rate of return on a market index and the historical average rate of return on a government security.

- C.** a forward-looking estimate that incorporates a current stock market index level, historical average rate of return on a government security, and assumptions about future index growth.

Solution:

A is correct because an implied market risk premium is forward-looking. It utilizes only current information, such as the stock market index level and the current government rate, along with assumptions about future growth in the stock market index.

B is incorrect because this statement describes a historical approach to estimating a market risk premium.

C is incorrect because this description uses a historical average rate of return for the government security instead of the current rate of return.

4. Which of the following statements best describes the use of an industry beta to estimate a company beta?

- A.** An industry equity beta may be used to compute a company equity beta to reflect differences between the financial leverage of the company relative to that of the industry.
- B.** An industry beta cannot be used for private company valuation given the lack of historical price data available for the private company's equity.
- C.** Use of an industry beta is inappropriate for companies who have experienced a recent change in financial leverage given the lack of comparability to past data.

Solution:

A is correct because once an analyst has chosen a comparable set of firms and summarized individual company levered betas across these firms, the industry levered beta may be converted into an unlevered beta, then relevered and applied as a company beta estimate.

B is incorrect because industry betas are applicable to private companies whose shares do not trade; a comparable set of firm betas can provide a reasonable company beta estimate in the absence of company-specific data.

C is incorrect because industry betas can provide a better estimate for firm risk when company-specific data do not capture changes in equity risk associated with recent significant changes in company debt.

5. Which one of the following most correctly describes the relation between risk and expected return for the CAPM and arbitrage pricing theory (APT)?

- A.** CAPM relates systematic risk to expected return, while APT relates total risk to expected return.
- B.** CAPM relates a single risk factor to the expected return, while APT relates multiple risk factors to expected return.
- C.** CAPM relates market risk to expected return, while APT relates four empirically defined risk factors to expected return.

Solution:

B is correct. Choice B is correct because CAPM is a single-factor model in which investors are exposed to only market risk, while APT is a general multi-factor model. In the APT, the factors are theoretical and therefore not defined either in terms of the identity or quantity of factors.

A is incorrect because like the CAPM, the APT also relates systematic risk (not total risk) to expected return. However, the APT assumes that systematic risk is likely to be reflected by multiple factors, rather than only one factor.

C is incorrect because APT is a theoretical model that does not identify specific factors.

6. Assuming that the market is one of the factors in the multi-factor model, which one of the following statements is *most correct* about market beta estimates from a single-factor model versus the market beta estimate from a multi-factor model?
- A. The single-factor market beta is always greater than the market beta in a multi-factor model.
 - B. The single-factor market beta is always less than the market beta in a multi-factor model.
 - C. The single-factor market beta is unlikely to be equal to the market beta in a multi-factor model.

Solution:

C is correct. Choice C is correct because the inclusion of additional factors beyond the market factor is likely to change the market beta between the two models. However, the direction of the change is uncertain after the inclusion of the other factors into the model.

A is incorrect because the inclusion of the other factors may cause market beta to increase or decrease from its estimate in the single-factor model.

B is incorrect because the inclusion of the other factors may cause market beta to increase or decrease from its estimate in the single-factor model.

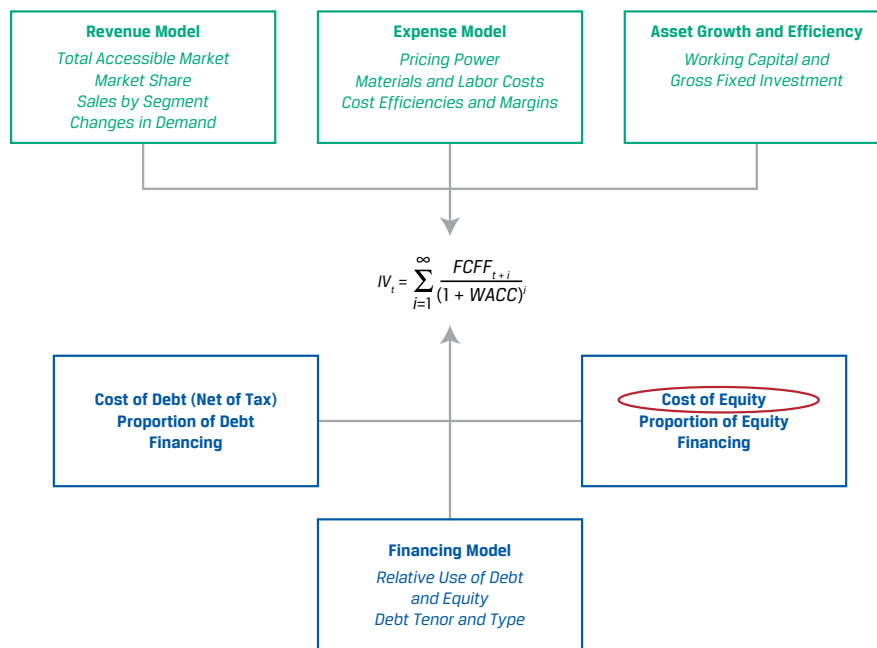
2

USE OF THE MARKET MODEL AND CAPM TO ESTIMATE COST OF EQUITY



explain the use of the CAPM and the market model to estimate a company's required return on equity

Earlier Equities learning modules on valuation have primarily focused on estimating a company's free cash flows. In this learning module, we explicitly address the estimation of a company's cost of equity, sometimes referred to as the required return on equity, as an important input in the company valuation process, as shown in Exhibit 1:

Exhibit 1: Illustration of Company Valuation Framework

Note: IV = intrinsic value.

A company's cost of equity is an important component of its weighted average cost of capital (WACC), the discount rate in the free cash flow to the firm (FCFF) valuation framework. Alternatively, equity valuation based on free cash flow to equity (FCFE) uses a company's cost of equity as the discount rate.

An earlier Quantitative Methods learning module introduced the capital asset pricing model. The CAPM is often used to estimate the required return on a company's stock. This single-factor model states that the expected return of a security or portfolio is determined by:

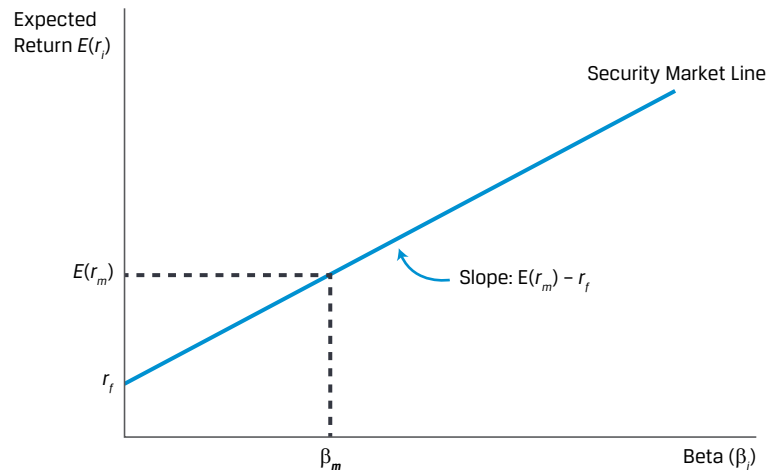
- the risk-free rate, r_f
- the expected market risk premium (MRP), $E[r_m] - r_f$ and
- the security or portfolio **beta estimate**, β_i .

Equation 1 shows the CAPM equation:

$$E[r_i] = r_f + \beta_i (E[r_m] - r_f). \quad (1)$$

Equation 1 is also known as the **security market line (SML)** equation. While the variable solved in Equation 1 is the expected return of a stock or a portfolio, it may also be interpreted as the required return on equity for investors, as illustrated in Exhibit 2.

Exhibit 2: Security Market Line



In this learning module, the focus is primarily on individual stocks, although the portfolio aspect is discussed later in this lesson with an application to industry betas. The required rate of return for a firm's equity is equivalent to the cost of its equity.

The beta estimate of a single stock reflects the ratio of the covariance with the market and the market's variance. This ratio may be decomposed, as shown in Equation 2:

$$\beta_i = \frac{\sigma_{i,m}}{\sigma_m^2} = \frac{\rho_{i,m} \times \sigma_i \times \sigma_m}{\sigma_m^2} = \frac{\rho_{i,m} \times \sigma_i}{\sigma_m}, \quad (2)$$

where

$\sigma_{i,m}$ = the covariance of the returns of security i and the return on the market portfolio

σ_m^2 = the variance of the market portfolio's returns

$\rho_{i,m}$ = the correlation coefficient of the returns of security i and the market

σ_i = the standard deviation of the returns of security i

σ_m = the standard deviation of the market returns

KNOWLEDGE CHECK: ESTIMATING BETA AND REQUIRED RETURN



- Suppose that the risk-free rate is 4% and the expected return on the market portfolio is 10%, with a standard deviation of 22%. A German company, Müller Metals, has a standard deviation on its stock returns of 44%. The correlation between Müller's returns and market returns is 0.60.

Calculate the beta and required return for Müller Metals' equity.

Solution:

Using Equation 2, we first calculate β_i :

$$\beta_i = \frac{\rho_{i,m} \times \sigma_i}{\sigma_m} = \frac{0.60 \times 0.44}{0.22} = 1.20.$$

Using Equation 1, we then calculate the required return using the beta estimate:

$$E[r_i] = r_f + \beta_i (E[r_m] - r_f) = 4\% + 1.20(10\% - 4\%) = 11.20\%.$$

The risk-free rate, r_f , and the expected MRP, $E[r_m] - r_f$ reflect macroeconomic conditions and will not change as required returns are estimated across different securities at the same point in time. In this instance, differences in beta estimates across firms are the only source of variation in estimated costs of equity when using the CAPM. The beta estimates reflect the relative risk for stocks of different companies. However, risk-free rates and MRPs are likely to be different when an analyst is estimating the required returns for securities in different countries. For example, the cost of equity for a foreign competitor of Müller Metals (from the example above) may likely incorporate a different risk-free rate and/or MRP.

As time passes, all of the variables in Equation 1 and Equation 2 are likely to change, as interest rates change and risks affecting the overall stock market and individual firms change. The combination of these factors may result in changes in the cost of equity, which may affect stock prices even in the absence of new information affecting cash flows.

KNOWLEDGE CHECK: CHANGES IN MÜLLER METALS' COST OF EQUITY



1. Recall our earlier example in which the cost of equity for Müller Metals is shown to equal 11.20%. In the initial example, the risk-free rate is 4%, the expected return on the market portfolio is 10%, the standard deviation of the market returns is 22%, Müller's return standard deviation is 44%, and the correlation between Müller's returns and the market returns is 0.60.

Which one of the following changes (while holding the other two variables constant) over a short period of time would most likely cause an increase in Müller's cost of equity?

- A. Risk-free rate increases
- B. Standard deviation of market returns increases
- C. Correlation between Müller's and the market returns increases

Solution:

C is correct. Choice C is correct because an increase in correlation, holding the standard deviations of both the market and the firm returns constant, is consistent with an increase in the beta estimate using Equation 2:

$$\beta_i = \frac{\sigma_{i,m}}{\sigma_m^2} = \frac{\rho_{i,m} \times \sigma_i \times \sigma_m}{\sigma_m^2} = \frac{\rho_{i,m} \times \sigma_i}{\sigma_m}.$$

Because correlation is in the numerator of the above equation, an increase in correlation is associated with an increase in beta. Holding risk-free rate and expected market return constant, an increase in beta is consistent with an increase in the firm's cost of equity using Equation 1.

A is incorrect because an increase in risk-free rate may not result in an increase in cost of equity. Holding the expected market return constant implies that the expected MRP declines with an increase in the risk-free rate. Given that beta is greater than one for Müller, the effect of a decline in the

MRP more than offsets the increase in the risk-free rate. For example, the solution below illustrates that an increase in risk-free rate to 5% is associated with a decline in Müller's cost of equity to 11%:

$$E[r_i] = r_f + \beta_i (E[r_m] - r_f) = 5\% + 1.20(10\% - 5\%) = 11.00\%.$$

B is incorrect because an increase in the standard deviation of market returns is consistent with a decline in the beta estimate because this variable is in the denominator of the equation. Thus, this change would more likely result in a lower cost of equity.

A beta estimate consists of two factors reflecting the relation between the stock and market returns:

- the correlation coefficient of the stock's and the market returns, and
- the ratio of the stock's standard deviation of returns to the market's standard deviation of returns.

For most individual stock beta estimates, the stock's returns will be more volatile than the market's returns, reflecting firm-specific risk of an individual stock. Thus, the second factor will most likely reflect a ratio greater than 1.0. The beta estimate reflects the interaction of the higher volatility of an individual stock relative to the market while incorporating the correlation of the stock with the market.

In practice, analysts are unlikely to estimate stock betas directly using Equation 2. While Equation 2 is mathematically correct, beta estimates are typically estimated through regression, although reported beta estimates may also reflect proprietary adjustments made by financial data providers, such as Bloomberg.

Beta Estimation

The beta estimate is a crucial measure in the CAPM and reflects an asset's estimated sensitivity to market movements. On the one hand, a company with "average" exposure to stock market movements will exhibit a beta of approximately 1.0. On the other hand, a **low-beta stock** is one with a beta below 1.0 and fluctuates less than market index movements. A **high-beta stock** is one with a beta greater than 1.0 and fluctuates more than market index movements.

Analysts may estimate betas for individual equities from historical data using linear regression as illustrated earlier in the Quantitative Methods curriculum. In practice, third-party data sources are widely used for obtaining beta estimates as discussed in the following example.

EXAMPLE 1

Bloomberg BETA Function

While beta estimates for individual company stocks are available from a variety of online sources, the Bloomberg BETA function provides a robust tool for gathering beta estimates. Exhibit 3 shows a screenshot of the Bloomberg BETA function output for a beta estimation of Amazon.com (AMZN) equity as of 14 January 2025.

Exhibit 3: Bloomberg BETA Function for Amazon.com



Source: Bloomberg.

Exhibit 3 shows AMZN's beta estimate derived from weekly returns over the prior two years regressed on the weekly returns of the S&P 500 Index (S&P 500). The resulting raw beta coefficient from this regression is shown as 1.241, while adjusted beta is shown as 1.161. Additionally, the output column includes regression statistics, such as a correlation between AMZN and S&P 500 returns of 0.636 and the regression beta's standard error of 0.15.

The Bloomberg BETA function screenshot shown in Exhibit 3 illustrates a common methodology for reporting a stock's beta estimate. An **adjusted beta** estimate incorporates the potential for measurement error from the regression analysis using a weighted average approach, as shown below:

$$\text{Adjusted BETA} = w_1 \times \text{Raw BETA} + w_2 \times 1.0, \quad (3)$$

where $w_2 = (1 - w_1)$.

Adjusted beta seeks to account for expected mean reversion when estimated beta is used for forward-looking valuations. Specifically, the betas of high-beta stocks are likely to be lower in the future, and betas of low-beta stocks are likely to be higher in the future. The weighted average approach that considers the likely mean reversion toward an average beta of 1.0 may be more accurate when an analyst uses historical data to predict future stock price risk.

KNOWLEDGE CHECK: BLOOMBERG ADJUSTED BETA FOR AMAZON.COM



1. Recall from the Example 1 and Exhibit 3 that the raw beta for Amazon.com stock was shown as 1.241, while the adjusted beta was 1.161.

Solve for the weighting scheme used to determine Amazon.com's adjusted beta in the above example.

Solution:

Substitute the raw beta and adjusted betas into Equation 3:

$$1.161 = w_1 \times 1.241 + w_2 \times 1.0$$

$$1.161 = w_1 \times 1.241 + (1 - w_1) \times 1.0$$

$$w_1 = 0.67, w_2 = 0.33 (= 1 - 0.67).$$

Note that this weighting scheme reflects a commonly used approach for adjusted beta, which is incorporated into Bloomberg's BETA function.

Calculating beta estimates for the CAPM typically starts by collecting weekly or monthly returns for the asset and the market index over a one- to five-year period. The risk-free rate is then subtracted from both the asset ($r_i - r_f$) and market ($r_m - r_f$) returns to arrive at excess returns. The beta regression equation takes the form shown in Equation 4:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i(r_{m,t} - r_{f,t}) + \epsilon_t \quad (4)$$

where α_i is the intercept and ϵ_t is the error term.

The slope coefficient, β_i , from this regression is an estimate of the asset's historical beta during the time horizon of the regression. This beta estimate may be used in Equation 1 to estimate the required return of the asset. While this form of regression is theoretically correct, beta estimates in practice, such as in the Bloomberg BETA function example shown earlier, are often obtained by regressing the asset's returns on the market's returns, as shown in the following **market model**:

$$r_{i,t} = \alpha_i + \beta_i \times r_{m,t} + \epsilon_{i,t} \quad (5)$$

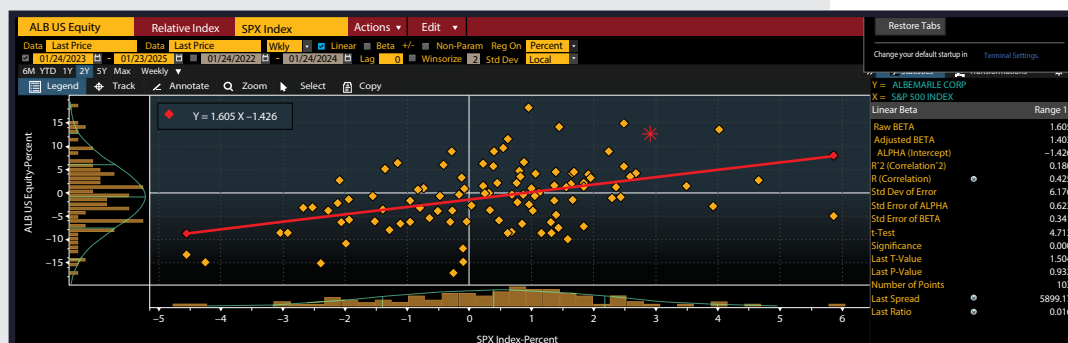
In principle, Equation 4 and Equation 5 will produce similar beta estimates. An important difference is that the market model is easier to implement because it omits the need to gather data for the realized rate of return of the risk-free asset proxy. As discussed earlier, beta estimates reflect the relative risk of an individual stock compared to the overall market. Any specific company's beta estimate likely reflects both industry-specific characteristics and company-specific characteristics as highlighted in the following example.

EXAMPLE 2

Company and Industry Characteristics and Beta Estimates

Company beta estimates typically reflect both industry- and company-specific characteristics. Some firms compete in industries with highly predictable demand for products or services and relatively stable costs and earnings, resulting in company stocks that are more likely to have low betas (i.e., below 1). Mature firms within the consumer staples industry are an example.

On the other hand, companies in industries with conditions resulting in unpredictable product revenue are more likely to exhibit high beta estimates (i.e., well above 1). Commodity-focused firms may be especially prone to market forces due to volatile underlying commodity prices. As an example, Albemarle Corporation, a US-based manufacturer of lithium compounds, is a major supplier to makers of batteries for electric vehicles (EVs). The price of lithium rose rapidly in 2021–22, sharply declining thereafter. This lithium price volatility may lead investors to view firms such as Albemarle as having a higher risk relative to the overall equity market. Exhibit 4 shows the Bloomberg BETA function applied to Albemarle's stock.

Exhibit 4: Bloomberg BETA for Albemarle Corporation

Source: Bloomberg.

Exhibit 4 shows a market model raw beta estimate for Albemarle Corporation of approximately 1.6. This relatively high beta estimate is consistent with observations of declining revenue growth and profitability for Albemarle as volatile lithium prices weakened.

Although Example 2 highlights that a specific stock's beta estimate is affected by industry and company-specific factors, it is important to note that market participants' views of these factors can change over time. As a result, analysts following specific stocks and/or industries must understand changes in beta estimates and factors driving these changes in market assessments of risk.

Practical Issues in Beta Estimation

Analysts using the market model and CAPM to estimate the required rate of return typically must address several practical issues in the process of estimating betas for individual securities. Key elements for analysts to consider include the following:

- Definition of "market" portfolio
- Potential noise of a single beta estimate
- Time horizons and periodicity of returns used

Market portfolio: The CAPM is a theoretical model in which the market portfolio consists of *all* risky assets, regardless of whether the asset trades (such as public equities) or does not trade (such as human capital gained through education or training programs). In practice, CAPM use has gravitated towards the most liquid asset markets, specifically public equities. In this context, the definition of a market portfolio is typically guided by features of the stock(s) being analyzed. As an example, it may be worthwhile to compare risk for two stocks of firms operating in different geographies with beta estimates computed using a broad market index that is common to both geographies. In general, caution may be warranted when the local market index is potentially too narrowly defined, as in the following example.

EXAMPLE 3

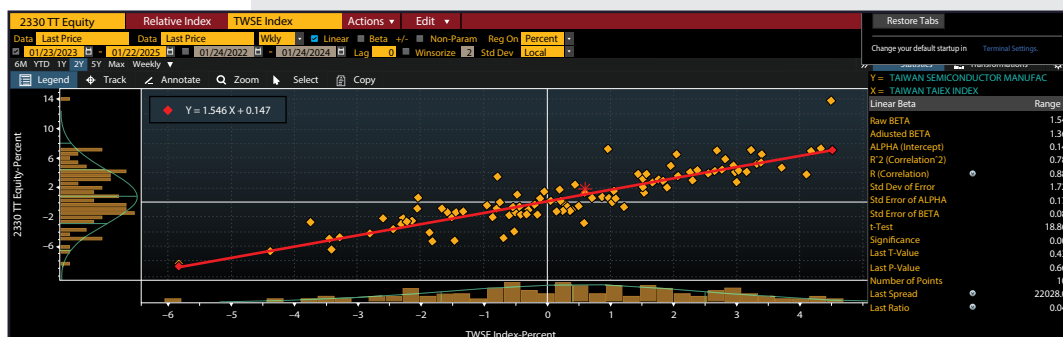
Taiwan Semiconductor Manufacturing Co Ltd (TSMC) and Market Portfolio Definition to Estimate Beta

TSMC, a global leader in semiconductor manufacturing, is based in Taiwan. A significant majority of trading in its stock occurs on the Taiwan Stock Exchange (TWSE). However, the company is global, with nearly two-thirds of its revenues

coming from the United States. Investors in TSMC may also take positions in the company’s stock by trading in the sponsored American Depositary Receipt (ADR), and this trading in US markets constitutes a portion of trading in TSMC.

While cost of equity estimation for US-based multinational companies often utilizes the S&P 500 as the market portfolio, the choice of index for a multinational such as TSMC is less clear. Exhibit 5 shows a screenshot of the Bloomberg BETA function for TSMC with the market portfolio specified as the Taiwan Stock Exchange Weighted Index (TAIEX). This is a market-capitalization-weighted index of the 800+ stocks listed on the TWSE.

Exhibit 5: TSMC BETA Estimate (TAIEX)



Source: Bloomberg.

Exhibit 5 shows that TSMC’s Bloomberg adjusted beta is equal to 1.364, estimated from weekly returns from January 2023 to January 2025. Notably, TSMC is, by far, the largest constituent stock in the TAIEX with a weight of over 17% as of February 2025. Given its relatively large impact on TAIEX performance, a high correlation of 88.4% is observed between TSMC and TAIEX returns during the estimation period.

An analyst may change the market portfolio used by selecting a different Relative Index in the top row of the Bloomberg screen. Possible alternatives for TSMC are the S&P 500 or the MSCI World Index. For example, TSMC’s Bloomberg adjusted beta falls to 1.281 when using the MSCI World as the market portfolio for the same estimation period.

Noise: As in the case of any estimate, beta estimates face the potential for error or noise. Two primary measures may be used to gauge the estimate’s noise. First, the R squared of a market model regression reflects the percentage of the stock’s variation explained by the market’s variation, and in a single-factor market model, the correlation coefficient between the stock and market returns equals the square root of the R squared statistic. Thus, lower R squared and correlation measures imply beta estimates that are subject to greater statistical noise. A second, more direct, metric is the beta’s standard error. A relatively small standard error of the beta estimate implies a more precise estimate.

Time horizons and return periodicities: In most CAPM applications, a stock’s beta estimate should reflect a forward-looking view of its market risk. For example, the cost of equity used in stock valuation is used to discount future expected cash flows. In practice, analysts rely on backward-looking techniques, such as regression, to estimate beta coefficients for stocks. However, analysts must use their judgment to select appropriate time periods and time increments for regressions to estimate beta. While the use of approximately 100 observations based on weekly data over the past two years is a common default approach, Bloomberg’s BETA function allows users to alter these time parameters, as shown in the following example.

EXAMPLE 4**Amazon.com Beta Estimates Using Alternative Time Parameters**

In Example 1, a Bloomberg screenshot illustrated the measurement of a beta estimate for Amazon.com stock of 1.241 based on a two-year regression of weekly returns. Using the Bloomberg adjusted beta equation, this regression beta was transformed to a value of 1.161. While this beta estimation reflects the default settings in Bloomberg, the BETA function can be set to different historical time horizons as well as differing return frequencies. Exhibit 6 shows a screenshot of the Bloomberg BETA function after changing the historical time horizon to five years with returns measured in monthly time increments.

Exhibit 6: Bloomberg BETA for Amazon.com (Monthly Returns)

Source: Bloomberg.

The raw beta estimate of 1.152 is slightly lower than the prior example's estimate of 1.241. The regression's R squared is similar to the example shown earlier, while the beta estimate's standard error of 0.183 is somewhat higher than the earlier example's standard error of 0.15. This change is likely due to the smaller number of return observations.

While any time horizon can be chosen within the BETA function, the return increments are limited to fairly common time frames, such as daily, weekly, and monthly. Using a one-year time horizon and daily returns shows a raw beta estimate of 1.526 for AMZN as well as a higher R squared and a lower standard error of the beta estimate.

The prior example highlights that analysts must use professional judgment in determining the parameters of a beta estimate using the CAPM. An analyst might choose one specific estimate or combine information from multiple beta estimates through a simple or weighted average process. Communication of underlying methodologies and assumptions is an important part of an analyst's reporting process.

Using Industry-Level Betas to Estimate a Single-Security Beta

As an alternative to relying on beta estimates using single security return data, analysts may consider estimating the beta for a company's stock based on an industry beta estimate. An industry equity beta may be used as part of the computation of a company equity beta to reflect differences between the financial leverage of the company

relative to that of the industry. This alternative approach may help alleviate possible concerns about the statistical noise associated with beta estimates from single-security regressions discussed above.

In earlier Equities learning modules on both relative valuation and industry analysis, we addressed the creation of a set of comparable companies formed to compare company valuations or assess their competitive environment. If the comparable companies have equivalent, or very similar, levels of business risk, then the **unlevered beta**, β_u , of the comparable companies should either be equal to or very close to that of the firm being analyzed. An unlevered beta is a typically unobserved risk measure that reflects a firm's or industry's equity beta estimate in the absence of financial leverage. Because an unlevered beta is unlikely to be observed, analysts gather information on a company's **levered beta**, β_L . After summarizing individual company levered betas across all comparable firms, an analyst converts the industry levered beta into an unlevered beta, as shown in Equation 6.

$$\beta_u = \frac{\beta_{L,Comps}}{1 + (1 - t) \times \frac{D}{MVE}}, \quad (6)$$

where

$\beta_{L,Comps}$ = the average levered beta of the set of comparable firms

t = the average tax rate for the set of comparable firms

$\frac{D}{MVE}$ = the average debt-to-market value of equity ratio for comparable firms

While Equation 6 uses the average industry debt-to-market value of equity ratio to estimate unlevered beta, note that this measure is also sensitive to the choice of comparable companies.

CASE STUDY



Converting Levered Betas into an Unlevered Beta

An analyst covering five biopharmaceutical companies gathers beta estimates, debt, market capitalization, and tax rate data for these firms, which the analyst believes face similar business risk, as follows:

Company	Beta	Debt	Market Cap	Tax Rate
Amavit	1.58	20,000	24,000	18%
Bonheur	1.35	18,000	82,000	22%
Calabana	1.52	21,000	33,000	15%
Dahlia	1.29	5,300	17,000	20%
Evitas	1.41	5,990	12,000	25%

- Which one of the following is closest to the correct estimate of the unlevered beta for this group of comparable companies?

- A. 1.02
- B. 1.14
- C. 1.43

Solution:

A is correct. Choice A is correct, as shown below. First, compute the average of the five observed beta estimates:

$$1.43 = \frac{1.58 + 1.35 + 1.52 + 1.29 + 1.41}{5}.$$

Next, calculate the ratio of debt to market value of equity for each firm as well as the average:

$$\text{Amavit: } \frac{20,000}{24,000} = 83.33\%$$

$$\text{Bonheur: } \frac{18,000}{82,000} = 21.95\%$$

$$\text{Calabana: } \frac{21,000}{33,000} = 63.64\%$$

$$\text{Dahlia: } \frac{5,300}{17,000} = 31.18\%$$

$$\text{Evitas: } \frac{5,990}{12,000} = 49.92\%$$

Average debt to market cap = 50 %

$$= \frac{83.33\% + 21.95\% + 63.64\% + 31.18\% + 49.92\%}{5}.$$

Finally, compute the average tax rate:

$$\text{Average tax rate} = 20\% = \frac{18\% + 22\% + 15\% + 20\% + 25\%}{5}.$$

Substituting these values into Equation 6 gives us an unlevered beta of 1.02:

$$\beta_u = \frac{\beta_{L,Comps}}{1 + (1 - t) \times \frac{D}{MVE}} = \frac{1.43}{1 + (1 - 20\%) \times 50\%} = 1.02.$$

B is incorrect because 1.14 is obtained as the unlevered beta estimate if the analyst incorrectly uses debt divided by the sum of debt and market capitalization, instead of debt divided by market capitalization.

C is incorrect because 1.43 is obtained as the average of the five observed betas. This answer would only be appropriate if all five of the firms had no debt. In this case, the observed beta estimates would reflect unlevered betas for each firm, and the average of these would reflect an average unlevered beta.

The most common argument for using the industry comparables approach to estimating beta is related to the statistical noise associated with a beta estimate obtained directly from the market model regression. Since much of the variation of a single company's stock returns is likely to include diversifiable risk, the beta estimate reflects significant amounts of statistical noise. This noise is typically illustrated by a relatively low R squared and high standard errors from a regression. When a single-stock regression produces a beta estimate deemed as being statistically imprecise, the analyst may consider employing the comparable company approach.

If an analyst can identify a portfolio of companies that face similar business risks, then combining the returns of such a portfolio into a market model regression should produce a regression with stronger explanatory power. Specifically, the correlation with market returns should be noticeably greater when using portfolio returns rather than a single security, as shown in the following example.

EXAMPLE 5

Regression Correlations: Single Security versus Comparable Company Portfolio

A US airline industry analyst requires beta estimates for individual airlines and for a portfolio of the largest airlines. As of the end of 2023, the five largest airlines by enterprise value, market capitalization, or revenue in the United States included Delta Air Lines (DAL), United Airlines Holdings (UAL), American Airlines Group (AAL), Southwest Airlines (LUV), and Alaska Air Group (ALK). The analyst runs market model regressions of each stock's daily returns between the last trading days of December 2022 and December 2023 using the S&P 500 as the market index and conducts the same regression on the returns of an equally weighted portfolio of these five stocks. Exhibit 7 shows a table summarizing the beta estimates, their standard errors, and the correlation of firm returns with S&P 500 returns.

Exhibit 7: Summary of Airline Beta Regression Results

	Beta Estimate	Standard Error	Correlation with S&P 500
LUV	1.021	0.134	0.434
ALK	1.204	0.147	0.461
UAL	1.283	0.156	0.462
AAL	1.341	0.148	0.499
DAL	1.174	0.124	0.516
Portfolio	1.205	0.118	0.542

The analyst estimates betas of the individual airlines ranging from 1.021 for Southwest Airlines to 1.341 for American Airlines Group. The beta estimate for the five-stock portfolio from a regression analysis is 1.205, which also represents the average of the beta estimates for the five individual stocks. While the beta estimate for the portfolio reflects an average, the portfolio beta estimate's standard error of 0.118 is lower than those estimated in the individual stock beta estimations. Furthermore, the correlation of returns with market index returns of 0.542 is highest for the portfolio.

Beta estimates from regressions reflect the market's assessment of relative equity risk and therefore should also reflect leverage differences. From the prior example, Southwest Airlines has the lowest estimated beta, while American Airlines Group shows the highest beta estimate. As it turns out, Southwest Airlines also had the lowest ratio of total debt to market value of equity of the five airlines, while American Airlines Group had the highest ratio.

A comparable company approach to beta estimation explicitly incorporates leverage differences between the comparable set of companies relative to an individual firm being analyzed. Once the beta estimate from the industry comparables has been converted into an unlevered beta that is applicable to the firm being analyzed, then the firm's beta estimate may be relevered, as shown in Equation 7:

$$\beta_{L,firm} = \beta_u \times \left[1 + (1 - t) \times \frac{D_{firm}}{MVE_{firm}} \right]. \quad (7)$$

Equation 7 reflects a transformation of Equation 6 in which the levered beta becomes the unknown variable. Application of Equation 7 incorporates the firm-specific financial leverage, as shown below in the extension of the earlier example.

CASE STUDY



Estimating Santevie's Equity Beta from Unlevered Beta

The analyst covering biopharmaceutical companies is analyzing an additional French firm, Santevie SE, that recently completed its IPO and has been publicly traded for a very short time frame. As a result of the stock's short price history, there are no published beta estimates because of data limitations for the purpose of running a regression on its stock returns against a market index. The analyst observes that Santevie's market capitalization is currently EUR5 billion, and its total debt is EUR3 billion. Santevie's tax rate is 25%.

The analyst views the set of five biopharmaceutical companies analyzed earlier as an appropriate comparable set of companies because Santevie is developing very similar types of drugs to these companies. The average levered beta of the comparable company set is 1.43, and their unlevered beta is 1.02. The average ratio of debt to market capitalization for the comparable companies is 50%, and their average tax rate is 20%.

1. Which one of the following statements most correctly describes the analyst's estimation of beta for Santevie?

- A. Santevie's beta is greater than 1.02 but less than 1.43.
- B. Santevie's beta is less than 1.02.
- C. Santevie's beta is greater than 1.43.

Solution:

C is correct. Choice C is correct because the resulting levered beta for Santevie is greater than 1.43, as shown below:

$$\beta_{L,firm} = \beta_u \times \left[1 + (1 - t) \times \frac{D_{firm}}{MVE_{firm}} \right] = 1.02 \times \left[1 + (1 - 25\%) \times \frac{3}{5} \right] = 1.48.$$

The bracketed term reflects the leverage adjustment factor in the levering process. In this instance, the leverage adjustment for computing Santevie's levered beta is greater than the leverage adjustment factor used to unlever the comparable company set's observed betas in the earlier example.

A is incorrect because Santevie's beta estimate would only be less than the observed average beta of the comparable companies if its leverage adjustment factor is less than that of the comparable company set's leverage adjustment factor.

B is incorrect because a levered beta will never be less than its associated unlevered beta.

The transformation of betas results in the following set of inequalities expressed in Equation 8:

$$\beta_{L,Comps} \geq \beta_u \leq \beta_{L,firm} \quad (8)$$

The magnitude of the inequalities implied by Equation 8 reflect average industry leverage and company leverage. For example, the unlevering and levering adjustments made using Equation 6 and Equation 7 have relatively small effects when both an industry

and the firm employ low financial leverage. This adjustment process is most evident when financial leverage ratios differ significantly between the comparable company set and the firm being analyzed.

The use of industry betas has other applications. For example, private company valuation is a case in which analysts may use an industry beta as a starting point in estimating a company beta given the lack of historical stock price data necessary to conduct market model regressions. Alternatively, companies that have recently altered their financial leverage may also be better suited to beta estimation through an industry approach because of the explicit adjustments based on current levels of debt ratios. A historical regression may not adequately capture changes in equity risk associated with recent significant changes in company debt.

A final caveat to this comparable company approach to estimating beta: An analyst using this approach must remember that the unlevered beta being estimated reflects the estimated asset risk of the set of comparable companies and is being assumed as the asset risk of the firm being analyzed. If the comparable company set and the firm being analyzed have different levels of asset risk, this approach creates a flawed beta estimate.

Estimation of Expected Equity MRP

The expected equity MRP represents the expected incremental return that investors demand as compensation for holding risky equity securities rather than a risk-free asset. In typical practice, an MRP is measured as the difference between the expected return on a broad-based and diversified stock index and the return on a government security issued in a country closely aligned with the stock index. As such, MRPs differ across geographies and are subject to change over time. The size of the MRP, in conjunction with interest rates on government securities, defines the required return on equity for securities deemed to be of “average risk” (i.e., beta estimate equal to 1.0).

Even for developed markets, the size of an MRP is often a source of disagreement across analysts. Similar to beta estimates for individual companies, MRPs are subject to estimation error. Furthermore, analysts may employ differing methodologies to estimate MRPs, resulting in additional possible sources of disagreement. For markets outside of the most developed countries, a MRP for a specific country may include a country risk premium (CRP) added to the MRP for a benchmark developed country, as shown in Equation 9:

$$E[r_i] = r_f + \beta_i (\text{MRP}_{\text{developed}} + \text{CRP}), \quad (9)$$

where

$$\text{MRP}_{\text{developed}} = (E[r_m] - r_f) \text{ for a given developed market}$$

$$\begin{aligned} \text{CRP} = & \text{Sovereign Bond Yield Spread} \\ & \times \frac{\text{Annualized Equity Market Standard Deviation}}{\text{Annualized Bond Market Standard Deviation}} \end{aligned}$$

The CRP is the additional risk premium required by investors to assume the risk of a stock in a given market. It is equal to the product of the target country's observed or estimated sovereign bond yield spread denominated in a reserve currency, such as USD or EUR, and the ratio of standard deviations in the country's equity versus debt market. CRP estimates are provided by financial data vendors, such as Bloomberg, and compiled annually by Aswath Damodaran, a finance professor at the Stern School of Business at New York University.

By adding CRP to a developed market MRP before multiplying by beta, Equation 9 implicitly incorporates country risk into a particular stock's risk premium. The following knowledge check demonstrates an application of this approach.

KNOWLEDGE CHECK: COST OF EQUITY FOR A DEVELOPING COUNTRY STOCK

1. Ahmet Trading S.A. is a publicly traded Turkish conglomerate firm listed on the Borsa Istanbul. An analyst uses regression analysis of Ahmet's historical share price versus the Borsa Istanbul 100 Index to derive an adjusted beta of 1.15 for the company's shares. She notes that Turkish sovereign EUR-denominated bonds trade at a yield spread of 275 basis points (2.75%) over a EUR benchmark rate of 4.50%. She also selects the FTSE Euro 100 as a developed market equity benchmark, observing an expected for this index of 5.00%.

$$(\text{MRPE}[r_m] - r_f)$$

If annualized Turkish equity market volatility (i.e., standard deviation) is double that of the Turkish bond market, calculate the expected rate of return for Ahmet Trading.

Solution:

Use Equation 9 to calculate the expected rate of return on Ahmet Trading shares as follows:

$$E[r_i] = r_f + \beta_i (\text{ERP}_{\text{developed}} + \text{CRP}).$$

First, we may calculate the CRP as 5.50% ($= 2.75\% \times 2$), adding this to the FTSE Euro 100 MRP ($\text{MRP}_{\text{developed}}$) to arrive at an overall MRP of 10.5%. We may multiply this by the adjusted beta, β_i , and add the risk-free rate (r_f), as shown in Equation 9:

$$16.58\% = 4.50\% + 1.15 \times (10.5\%).$$

Analysts typically estimate MRPs using either 1) historical market and government security returns or 2) current market and government security prices along with forward-looking assumptions. Details of these two general approaches are covered later in the curriculum. Our focus in this lesson is to provide basic explanations of the approaches.

Historical Approach to MRP Estimation

An analyst may estimate an MRP by using the difference between historical averages of the total returns on a broad-based stock market index and of the total returns on a government-issued security from the same country or region. This approach presumes availability of return data for a sufficiently long period such that the sample mean of the return represents an unbiased expectation of future returns.

When using a historical approach to estimate an MRP, analysts are faced with methodological choices. First, should the averages reflect arithmetic means or geometric means? Second, what maturity of government security should proxy for the risk-free asset?

HISTORICAL MRP: AN EXAMPLE OF US DATA

A variety of authors and studies provide historical return data from stock market indexes and government securities, especially in developed markets. For a wide array of reasons, the US data are often viewed as a benchmark when discussing

expected MRPs. Although no two studies are alike, the data shown in Exhibit 8 are reasonable representations of MRP estimations in the United States from historical data.

Exhibit 8: Estimates of US MRPs Through 2024 Using Historical Data

	1928–2024	1975–2024
Stocks – Bills (arithmetic average)	8.44%	9.25%
Stocks – Bills (geometric average)	6.63%	8.02%
Stocks – Bonds (arithmetic average)	7.00%	7.03%
Stocks – Bonds (geometric average)	5.44%	6.22%

Source: Data from https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html.

The first two rows subtract average US T-bill rates during the year from the total return on large stocks in the United States. The final two rows instead use average 10-year US Treasury bond rates, instead of T-bills, as the risk-free security. Historical MRPs using short-term government securities range from 6.63% to 9.25%. When using long-term government securities, historical MRPs range from 5.44% to 7.03%.

An MRP estimated using the geometric average (i.e., compound returns) will always be smaller than an MRP using the arithmetic average of the same data. This occurs specifically because volatility reduces the compounding effect of returns on riskier assets.

Short-term government securities should provide lower returns, on average, relative to long-term government securities because of maturity risk premiums for longer-horizon securities. Thus, an MRP using a long-term government security as the risk-free asset should be lower relative to an MRP using a short-term security. In principle, investing in equities reflects a long-term investment because the underlying company cash flows being valued occur over very long time horizons. As such, equity valuation should generally utilize MRPs calculated with long-term government security returns.

Forward-Looking Approaches to MRP Estimation

Given that the MRP should reflect expected future returns, a more intuitive approach is to incorporate forward-looking assumptions about investor expectations. An analyst may utilize several different methodologies to generate a forward-looking MRP estimate. One methodology relies on fundamental valuation principles already covered in prior Equities learning modules, and the resulting MRP estimate is sometimes referred to as an **implied market risk premium**.

An implied MRP is estimated from a current stock market index level and a forecast of future cash flows, including a terminal value assumption. At its most fundamental level, this methodology can be stated using a dividend constant growth model. If the index price level is assumed to equal the value of the index, we can solve for the expected return of an index using a constant growth assumption, as shown in Equation 10:

$$E[r_m] = \frac{\text{Div}_{1,m}}{P_{0,m}} + g_{l,m}, \tag{10}$$

where

$\text{Div}_{1,m}$ = the next year's expected dividends on the stock market index

$P_{0,m}$ = the current price level of the stock market index

$g_{l,m}$ = the long-run constant growth rate of the dividends on the stock market index

After calculating the expected return for a market index, an analyst simply subtracts a current government security rate to complete the computation of an implied MRP, as shown in Equation 11:

$$\text{Implied MRP} = E[r_m] - r_f \quad (11)$$

Equation 10 may be modified to incorporate a more comprehensive cash flow measure, such as free cash flow to equity (FCFE), and/or to utilize a more complicated forecast of future expected cash flows. For example, Damodaran has published annual and monthly measures of an implied MRP for the United States based on the S&P 500 over many years using a two-stage valuation model of assumptions about the index's future expected FCFE (see <https://pages.stern.nyu.edu/~adamodar/>).

KNOWLEDGE CHECK: CALCULATING AN IMPLIED MRP



1. The S&P 500 generated dividends of USD71.50 per share in 2024.

The index value at the end of 2024 was USD5,881.63. The compound annual growth rate of the S&P 500's dividends per share over the prior 10 years was 6.33%. The market yield on constant-maturity 10-year US Treasury bonds at the end of 2024 was 4.58%.

If the recent historical growth rate on S&P 500 dividends is a reasonable estimate of long-run dividend growth on the S&P 500, which of the following is closest to the most correct estimate of an implied MRP?

- A. 1.75%
- B. 2.97%
- C. 3.04%

Solution:

C is correct. Choice C is correct as shown below.

$$E[r_m] = \frac{\text{Div}_{1,m}}{P_{0,m}} + g_{l,m} = \frac{71.50 \times (1 + 0.0633)}{5881.63} + 0.0633 = 7.62\%.$$

$$\text{Implied MRP} = E[r_m] - r_f = 7.62\% - 4.58\% = 3.04\%.$$

The dividend per share in the numerator needs to incorporate an expected growth rate to estimate 2025 expectations, instead of using 2024 dividends.

A is incorrect because 1.75% reflects a subtraction of the Treasury rate from the long-run growth rate in dividends but omits the dividend yield portion of the calculation.

B is incorrect because 2.97% is the result if 2024 dividends per share were used instead of estimating 2025 dividends per share by multiplying by 1 plus the growth rate as in the correct method.

Issues Related to MRP Estimation

To estimate a company's cost of equity, analysts must decide on an appropriate MRP point estimate or consider a range associated with the potential estimation error for purposes of conducting sensitivity analyses on their valuations. As mentioned earlier, MRPs vary across geographies. While the MRPs in the most developed economies are likely in the low-to-middle digit percentages (i.e., 3%–6%), analysts valuing stocks in developing countries may utilize MRPs in the high single digits or even into the double digits.

MRP estimates also vary over time. This point highlights an important difference between historical versus forward-looking estimates of MRPs. Higher stock market returns over time cause average stock market index returns to grow relative to average returns on government securities. This change in average returns suggests an increase in historical MRP, which further suggests a higher discount rate on equities. However, a higher discount rate on equities is consistent with lower valuations as opposed to the higher valuations created by a rising stock market. Thus, in such instances, present value approaches to valuing stocks become more likely to suggest overvaluation. On the other hand, forward-looking approaches to estimating MRP, using current prices, are more likely to interpret rising markets as an indicator that investors require less compensation for bearing risk. Thus, rising stock market prices often are associated with lower MRPs over time when using forward-looking MRP estimates.

Ultimately, analysts must decide upon specific numbers for their estimates of MRPs. A variety of data sources are available from which analysts can assess the underlying methodologies used in making their decisions. On the one hand, forward-looking methodologies are more consistent with the use of MRPs to estimate company costs of equity, but analysts must be comfortable with the underlying assumptions. On the other hand, historical approaches provide concrete evidence of the excess returns on stock market indexes over time, but these historical returns may not adequately reflect forward-looking return potential.

QUESTION SET 1



1. An analyst estimates a stock's beta to be low at 0.80, and the standard deviation of its returns is 25%. The standard deviation of market returns is 20%. The correlation between the stock and the market is *closest* to:

- A. 64%.
- B. 80%.
- C. 100%.

Solution:

A is correct. Choice A is correct because we may rearrange Equation 2 to solve for the correlation coefficient when the beta estimate is known.

$$\beta_i = \frac{\rho_{i,m} \times \sigma_i}{\sigma_m}.$$

$$\rho_{i,m} = \frac{\sigma_m}{\sigma_i} \times \beta_i = \frac{0.20}{0.25} \times 0.80 = 0.64.$$

B is incorrect because 80% equals the square root of the correct correlation, not the correlation itself.

C is incorrect because it reverses the numerator and denominator in the equation.

2. Adjusted beta estimates are typically:

- A. lower than unadjusted betas.
- B. higher than unadjusted betas.
- C. closer to 1.0 than unadjusted betas.

Solution:

C is correct. Choice C is correct because adjusted beta is calculated based upon a weighted average:

$$\text{Adjusted beta} = w_1 \times \text{Raw beta} + w_2 \times 1.0,$$

where $w_2 = 1 - w_1$. For companies with raw (i.e., regression) betas greater than 1.0, this weighting causes the adjusted beta to decline relative to the regression beta. For companies with raw (i.e., regression) betas less than 1.0, this weighting causes the adjusted beta to increase relative to the regression beta. Thus, adjusted betas are closer to 1.0 than the regression (i.e., raw or unadjusted) betas.

A is incorrect because this statement is only true for stocks with raw betas greater than 1.0.

B is incorrect because this statement is only true for stocks with raw betas less than 1.0.

3. Which one of the following provides the most likely rationale for using an industry-based beta estimate to compute a company beta estimate?

- A. The company's regression-based beta estimate has a low *R*-squared value.
- B. Financial leverage of the company is similar to that of its industry.
- C. Asset risk of the company is not similar to that of its industry peers.

Solution:

A is correct because the low *R* squared from the company's regression-based beta estimate implies that the estimate's error is large. The industry-based approach should produce a beta estimate with lower standard error.

B is incorrect because a lack of difference in financial leverage is not necessary to make the industry-based approach useful. The effect of leverage differences on a company's beta is captured by the unlevering and relevering processes.

C is incorrect because establishing a set of comparable companies with similar asset risk to that of the single company under consideration is a prerequisite for using the industry-based approach.

4. The size of an MRP estimate is:

- A. a consensus estimate agreed upon across analysts.
- B. an estimate that uses the same methodology across analysts.
- C. a noisy estimate subject to error.

Solution:

C is correct. Choice C is correct because, regardless of the method employed, MRPs are either based on a sample of historical data or assumptions about the future. As a result, MRPs are estimates and subject to error.

A is incorrect because the size of MRPs is often a source of disagreement across analysts.

B is incorrect because MRPs may be estimated using a historical methodology or a forward-looking methodology. Furthermore, there are likely to be methodological differences within both historical and forward-looking approaches to estimating MRPs.

5. A developed country's market index is priced at USD20,000. Dividends on the index are expected to be USD250 during the next year and are expected to grow at a rate of 6% per year. The country's risk-free rate is 3.25%. The implied MRP for the country is:

- A. 3.25%.
- B. 4.00%.
- C. 8.00%.

Solution:

B is correct. Choice B is correct because the implied MRP from the given information is as follows:

$$E[r_m] = \frac{\text{Div}_{1,m}}{P_{0,m}} + g_{l,m} = \frac{250}{20,000} + 6\% = 7.25\%.$$

$$\text{Implied MRP} = E[r_m] - r_f = 7.25\% - 3.25\% = 4\%.$$

A is incorrect because 3.25% is simply the country's risk-free rate, not an implied MRP.

C is incorrect because 8.00% is incorrectly computed if the analyst adds the growth rate of 6% to the risk-free rate of 3.25% and then subtracts the dividend yield of 1.25%.

3

ARBITRAGE PRICING THEORY



describe the arbitrage pricing theory and the importance of multi-factor models used in equity investing

The prior lesson focuses on using the CAPM to estimate a company's cost of equity. The CAPM reflects a single source of risk: variation in stock market returns. More specifically, investors demand compensation for the risk that stock market returns sometimes are below those available from very low-risk government securities. In the CAPM, this market risk is the only form of systematic, or non-diversifiable, risk, and beta estimates from Equations 3 or 4 are the only relevant risk metrics in estimating a company's cost of equity.

Empirical research of the CAPM has consistently shown that market risk is not the only source of risk recognized by investors in equities. Thus, a wide array of both academic and practitioner research of stock markets has searched for additional sources of risk. In other words, what are other possible risk factors associated with equity investing for which investors expect to earn compensation for bearing risk? In this lesson, we describe the foundation of multi-factor models and their importance in building stock portfolios.

Arbitrage Pricing Theory

In the CAPM, the beta estimate reflects the exposure of a single stock or a portfolio to a market index return. **Arbitrage pricing theory (APT)** provides a general framework for a **multi-factor model**. Rather than a single risk factor, APT allows for multiple sources of systematic risk to equity investors as implied by Equation 12.

$$E[r_i] = r_f + \beta_{i,1} \times E(f_1) + \beta_{i,2} \times E(f_2) + \dots + \beta_{i,k} \times E(f_k), \quad (12)$$

where

$\beta_{i,k}$ = the beta estimate (or risk exposure) of asset i to factor k

$E(f_k)$ = the expected risk premium of factor k

The APT does not define what the factors are, nor does the model state the number of factors. Rather, each factor reflects a theoretical risk to equity investing that can be reduced through portfolio diversification. In other words, just as in the CAPM, the average investor holds the market portfolio of stocks. Because this portfolio is fully diversified, the only risk remaining is systematic. A significant difference between the APT framework versus the CAPM is that systematic risk in the APT reflects more than one risk factor. Furthermore, the APT does not necessarily imply that excess stock market returns are one of the factors (although it certainly may be one of the factors). Ultimately, researchers conduct experiments to identify relevant factors that reflect systematic risks in equity investing.

Multi-Factor Models in Practice

Research into building multi-factor models is a constantly evolving process. Factors may be built using observed data from sources, including 1) company stock market data, 2) macroeconomic data, and 3) company accounting data. By the early 2020s, published research has proposed, tested, and found positive results for hundreds of different factors. Many of these factors may be products of data mining or spurious correlation, so most investment practitioners and scholars believe that the population of factors can be distilled to a small number that effectively reflect all relevant systematic risks. One practical example is a four-factor model using the following firm variables as factors to explain returns for stocks:

- Excess stock market returns (market factor)
- Firm size (size factor, or SMB, small minus big)
- Firm value (value factor, or HML, high minus low)
- Firm stock price momentum (momentum factor, or UMD, up minus down)

While the market factor is the same as described in the CAPM, the definition of the other three factors is not obvious. To use these three factors to explain returns, researchers must estimate portfolio returns resulting from an investment strategy that captures each factor. Equation 13 shows the four-factor model in functional form.

$$E(r_i) = r_f + \beta_{i,\text{Market}} \times E(r_m - r_f) + \beta_{i,\text{Size}} \times E(\text{SMB}) + \beta_{i,\text{Value}} \times E(\text{HML}) + \beta_{i,\text{Momentum}} \times E(\text{UMD}), \quad (13)$$

where

$\beta_{i,Size}$ = the exposure of stock (or portfolio) i to the size factor, $E(SMB)$

$E(SMB)$ = the expected return on a zero-investment portfolio consisting of a long position in small market-capitalization stocks and an equal-sized short position in large-capitalization stocks

$\beta_{i,Value}$ = the exposure of stock (or portfolio) i to the value factor, $E(HML)$

$E(HML)$ = the expected return on a zero-investment portfolio consisting of a long position in stocks with relatively high ratios of book value of equity to market value of equity (corresponding to a low price-to-book ratio), commonly referred to as **value stocks**, and an equal-sized short position in stocks with relatively low ratios of book value of equity to market value of equity (or a high price-to-book ratio), typically categorized as **growth stocks**

$\beta_{i,Momentum}$ = the exposure of stock (or portfolio) i to the momentum factor, $E(UMD)$

$E(UMD)$ = the expected return on a zero-investment portfolio consisting of a long position in stocks with the highest returns and an equal-sized short position in stocks with the lowest returns

The general factor definitions shown above are addressed in greater detail later in the CFA Program curriculum.

For any individual stock or portfolio, betas may be estimated for each of the factors by gathering historical data for the factors described and regressing the stock returns on the variables listed. As opposed to market beta estimates for which an average value is 1.0, the average value for the other three factor betas should be zero. In other words, negative beta estimates are quite common for the other three factors, as shown in the following example of a four-factor model estimation done on several US stocks as of December 2023.

CASE STUDY



Estimation of the Four-Factor Model: Application to Technology Stocks

Wharton Research Data Services (WRDS) provides a user interface for accessing a wide array of finance and business research databases. WRDS includes an interface called Beta Suite, which allows users to estimate the single-factor CAPM as well as two specific multi-factor models.

In 1993, finance professors Eugene Fama and Kenneth French (Fama and French 1993) published a three-factor model in which they added size and value factors to the market factor. In 1997, Mark Carhart (1997) added an additional momentum factor to Fama and French's three-factor model. Beta Suite offers WRDS users the ability to estimate these two multi-factor models.

To illustrate the implications of using the four-factor model on estimation of corporate cost of equity, betas are estimated using this model for three well-known technology firms: Apple (AAPL), Microsoft (MSFT), and Nvidia (NVDA). Beta estimates are also derived for MicroStrategy (MSTR), a technology company that later shifted its focus to raising capital to purchase bitcoin.

The four-factor model is estimated for each of the four stocks using 60 months of data ending in December 2023 for each of the risk factors. Exhibit 9 shows the beta estimates.

Exhibit 9: Beta Estimates from Four-Factor Model

Stocks	$\beta_{i,Market}$	$\beta_{i,Size}$	$\beta_{i,Value}$	$\beta_{i,Momentum}$
AAPL	1.36	-0.40	-0.46	0.08
MSFT	0.96	-0.58	-0.48	0.03
NVDA	1.58	-0.34	-1.17	-0.27
MSTR	1.86	3.21	-1.01	-0.59

1. In the context of the four-factor model, NVDA is *best* described as:

- A. a small-cap value stock.
- B. a growth stock with high market risk.
- C. a momentum stock with high value exposure.

Solution:

B is correct because market beta being significantly greater than 1.0 implies high market risk, while a large negative coefficient on the value factor implies a growth orientation.

A is incorrect because NVDA's exposure to the size and value factors are negative, which indicates that it is a large-cap growth stock.

C is incorrect because NVDA's exposure to both the value and momentum factors are negative, not positive.

Exhibit 9 shows contrasts in terms of the risk exposures faced by these stocks. Three of the four stocks (AAPL, NVDA, and MSTR) show significantly above average market beta estimates, while MSFT has roughly average market risk.

MSTR exhibits very large positive exposure to the size factor. Specifically, MSTR is expected to perform well when small stocks are outperforming large stocks. The other three stocks show negative exposure to the small stock premium.

All four stocks behave as "growth" stocks due to negative exposure to the value premium. In other words, all four are likely to perform better when stocks with higher price-to-book multiples outperform stocks with low price multiples.

MSTR and NVDA both have negative exposures to the momentum factor. Specifically, during periods when equity markets favor stocks with the highest recent returns, stocks with negative exposure to the momentum factor are expected to earn lower returns.

As with the CAPM, analysts may utilize the estimation of a factor model to obtain a cost of equity estimate for a stock. The following case study continues the examination of the four technology stocks to illustrate the estimation of their costs of equity from the four-factor model with a contrast against cost of equity estimates from the single-factor CAPM model.

CASE STUDY

Estimated Factor Risk Premiums and Nvidia Estimated Cost of Equity

As in the application of the CAPM model to estimate the cost of equity, risk premiums for each factor are required. Kenneth French provides historical US data back to 1927 for each of the factors on his publicly available website. Estimates based on these data result in the following approximate monthly risk premiums (rounded to two basis points) for each factor (see Exhibit 10).

Exhibit 10: Monthly Average Risk Premiums of Four-Factor Model

Factor	Historical Average Monthly Risk Premium
Market	0.69%
Size	0.18%
Value	0.34%
Momentum	0.63%

Source: Data from https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

1. Recall from the earlier case study in which NVDA’s beta estimates are re-ported as follows:

- Market beta = 1.58
- Size beta = −0.34
- Value beta = −1.17
- Momentum beta = −0.27

Furthermore, the four-factor model utilizes the US one-month Treasury bill as the risk-free security. As of the end of 2023, the annualized one-month US Treasury rate is 5.60%.

Calculate an estimate of NVDA’s cost of equity using the four-factor model results.

Solution:

Applying Equation 13, we multiply each beta estimate by its corresponding risk premium and add the risk-free rate:

$$E[r_i] = r_f + \beta_{i,\text{Market}} \times E(r_m - r_f) + \beta_{i,\text{Size}} \times E(\text{SMB}) + \beta_{i,\text{Value}} \times E(\text{HML}) + \beta_{i,\text{Momentum}} \times E(\text{UMD}).$$

Since the risk premiums shown above are monthly, first we must multiply each premium by 12 to annualize, as shown below:

Factor	Monthly Premium	Annual Premium
Market	0.69%	8.28%
Size	0.18%	2.16%
Value	0.34%	4.08%
Momentum	0.63%	7.56%

$$E[r_i] = 5.60\% + 1.58 \times 8.28\% - 0.34 \times 2.16\% - 1.17 \times 4.08\% - 0.27 \times 7.56\% = 11.13\%.$$

The cost of equity estimate for NVDA stock based on the four-factor model as of the end of 2023 is 11.13%.

As discussed earlier, beta estimates of multi-factor models are sensitive to the time horizon as well as return increments used. Furthermore, the factor risk premiums are based on historical averages, which may also change based upon the time horizon under consideration.

In a multi-factor model, a beta estimate associated with market risk is unlikely to equal the beta estimate from a single-factor CAPM or market model regression. Even in uncommon cases when these are the same, required return estimates are still likely to differ, as shown in the following continuation of Nvidia's beta exposures.

CASE STUDY



Contrasting Nvidia's Cost of Equity Using CAPM versus the Four-Factor Model

In the earlier case study in which the four-factor model is estimated, NVDA's beta estimate on the market factor is 1.58. Using the CAPM as the underlying model in Beta Suite instead of the four-factor model, NVDA's beta on the single market factor is also estimated to equal 1.58.

1. Calculate an estimate of NVDA's cost of equity based on the CAPM under the assumption that expected MRP is 8.28% per year and the risk-free rate is 5.60% on an annualized basis.

Solution:

Nvidia's cost of equity using the CAPM is 18.68%, as shown below.

$$E[r_i] = r_f + \beta_{i,Market} \times E(r_m - r_f) = 5.60\% + 1.58 \times 0.0828 = 18.68\%.$$

2. From the prior case study using the four-factor model to estimate NVDA's cost of equity, the estimate is 11.13%, as compared to 18.68% using the CAPM. Which one of the following statements best describes the source(s) of disagreement between the two models?

- A. The 18.68% estimate reflects higher market exposure for NVDA compared to the 11.13% estimate.
- B. The 18.68% estimate reflects additional positive exposure for NVDA compared to the 11.13% estimate.
- C. The 18.68% estimate reflects omitted negative exposures for NVDA compared to the 11.13% estimate.

Solution:

C is correct. Choice C is correct because the 11.13% cost of equity estimate from the four-factor model incorporates three additional factors beyond the market factor, and these three factors all exhibit negative exposures for NVDA. The single-factor CAPM only reflects market exposure, so this model omits the negative exposures of the other three factors while estimating the same market beta as shown for the four-factor model.

A is incorrect because both models estimate the same market exposure (i.e., beta) of 1.58 for NVDA. Thus, this statement cannot be correct in explaining the difference in these two cost of equity estimates for NVDA.

B is incorrect because the four-factor model's estimate is lower because the other three risk factors show negative, not positive, exposure for NVDA.

As shown in the case study above, the conclusions from a factor model may differ from those from the CAPM about a company’s cost of equity (or interpreted as investors’ required return on the company’s equity) because of the risk factors chosen in a model. While the example above shows a situation in which the CAPM produces an overestimate relative to the four-factor model, the relationship between estimates will vary across firms depending upon the nature of risk exposures.

KNOWLEDGE CHECK: CONTRASTING COST OF EQUITY ESTIMATES



1. An analyst estimates the costs of equity for three different companies. The risk-free rate is 4.50%, and the risk premium on the market factor is 6%. The analyst first estimates the CAPM with the following results:

Company	Single-Factor Beta	Cost of Equity
Drysdale Inc.	1.10	11.1%
Eggerton Corp.	0.95	10.2%
Flatcher Inc.	1.80	15.3%

The analyst decides to contrast the cost of equity estimates from the single-factor model with estimates from a four-factor model with the following estimates:

Company	$\beta(\text{mkt})$	$\beta(\text{SMB})$	$\beta(\text{HML})$	$\beta(\text{UMD})$
Drysdale	0.95	0.40	0.30	−0.10
Eggerton	1.05	−0.20	0.50	−0.05
Flatcher	1.50	1.20	−0.30	0.20
Annual Factor Risk Premiums	6.0%	1.5%	4.0%	6.5%

Which one of the following companies will show the most significant change in the cost of equity estimate when using the four-factor model?

- A. Drysdale
- B. Eggerton
- C. Flatcher

Solution:

B is correct. Choice B is correct because the cost of equity estimate for Eggerton increases by 200 basis points, from 10.2% to 12.2%, when switching the estimation model from the single-factor CAPM to the four-factor model. The calculation of the four-factor model for Eggerton is shown below:

$$E[r_i] = r_f + \beta_{i,Market} \times E(r_m - r_f) + \beta_{i,Size} \times E(\text{SMB}) + \beta_{i,Value} \times E(\text{HML}) + \beta_{i,Momentum} \times E(\text{UMD}).$$

$$E[r_i] = 4.5\% + [1.05 \times 6.0\% - 0.20 \times 1.5\% + 0.5 \times 4.0\% - 0.05 \times 6.5\%]$$

$$= 12.2\%.$$

The increase in Eggerton's market beta from the single-factor model to the four-factor model and the positive value beta are primary factors associated with increasing Eggerton's cost of equity estimate between the two models.

A is incorrect because the cost of equity estimate only increases by 30 basis points, from 11.1% to 11.4%. Positive size and value beta estimates are effectively offset by a decline in the market beta estimates between the two models.

C is incorrect because the cost of equity estimate only increases by 10 basis points, from 15.3% to 15.4%. The decline in the market beta estimate effectively offsets the positive exposures on size and momentum.

This lesson introduces multi-factor models within the context of estimating discount rates for purposes of equity valuation. These models can serve as important tools in the development of an investor's investment strategy as well as their performance evaluation, topics addressed later in the curriculum.

The examples in this lesson have focused on one specific type of multi-factor model: the four-factor model. Important aspects of factor models involve both clearly defining each risk factor as well as determining the frequency of observation and portfolio rebalancing, the details of which are shown later in the curriculum.

Finally, the search for the best set of factors is an ongoing exercise among both investment practitioners and academic researchers. As such, analysts should strive to follow developments with respect to newly discovered factors as well as those that may have been proven to not explain returns.

QUESTION SET 2



1. Which one of the following is the best description of the value factor utilized in a four-factor model?
 - A. The expected return on a zero-investment portfolio consisting of a long position in stocks with high ratios of book value of equity to market value of equity and an equal-sized short position in stocks with low ratios of book value of equity to market value of equity
 - B. The expected return on a zero-investment portfolio consisting of a long position in stocks that have been appreciating in value and an equal-sized short position in stocks that have been depreciating in value
 - C. The expected return on a zero-investment portfolio consisting of a long position in small market-capitalization stocks and an equal-sized short position in large-capitalization stocks

Solution:

A is correct. Choice A is correct because the value factor is constructed based upon the ratios of book value of equity to market value of equity across companies. The expected return on a zero-investment portfolio that buys value stocks (i.e., those with high book-to-market ratios) and sells growth stocks (i.e., those with low book-to-market ratios) reflects the "value premium."

B is incorrect as this response describes the momentum factor used in the four-factor model.

C is incorrect as this response describes the size factor used in the four-factor model.

2. An average risk stock has which of the following beta estimates for each factor?

- A. $\beta_{i,Market} = 1.0$, $\beta_{i,Size} = 1.0$, $\beta_{i,Value} = 1.0$, $\beta_{i,Momentum} = 1.0$
- B. $\beta_{i,Market} = 1.0$, $\beta_{i,Size} = 0.0$, $\beta_{i,Value} = 0.0$, $\beta_{i,Momentum} = 1.0$
- C. $\beta_{i,Market} = 1.0$, $\beta_{i,Size} = 0.0$, $\beta_{i,Value} = 0.0$, $\beta_{i,Momentum} = 0.0$

Solution:

C is correct. Choice C is correct because an average risk stock in the four-factor model has a beta estimate of 1.0 on the market factor as in the CAPM. An average risk stock should have zero exposure to the other three factors because the beta estimates for these other factors are centered on a value of 0.0 (as the average stock return is that of the market and is neither small nor large, neither value nor growth, and neither appreciating nor depreciating in price relative to the market).

A is incorrect because this pattern of beta estimates suggests average market beta while also indicating strong positive exposure to size, value, and momentum factors. Specifically, this stock performs well when small stocks outperform large stocks, performs well when value stocks outperform growth stocks, and performs well when winning stocks outperform losing stocks.

B is incorrect because this pattern of beta estimates suggests a strong positive exposure to the momentum factor, implying that the stock performs better in an environment when stocks with high prior returns continue to produce high returns in subsequent time periods.

3. Which one of the following is a correct statement about multi-factor models in practice?

- A. Investment practitioners and researchers have reasonably decided upon the set of factors that explain the expected returns of stocks.
- B. Investment practitioners and researchers have developed models suggesting that there are hundreds of different factors that explain expected stock returns.
- C. Investment practitioners and researchers agree that factors should be defined using only macroeconomic data or company accounting data.

Solution:

B is correct. Choice B is correct because as of the early 2020s, published research has proposed, tested, and found positive results for hundreds of different factors.

A is incorrect because research into building multi-factor models is a constantly evolving process. Many of the hundreds of factors “discovered” may be products of data mining or spurious correlation, so most investment practitioners and scholars believe that the population of factors can be distilled to a small number that effectively reflect all relevant systematic risks. The search for the best set of factors is an ongoing exercise among both investment practitioners and academic researchers.

C is incorrect because, in addition to macroeconomic data and company accounting data, factors may be built using company stock market data. For example, the four-factor model includes market capitalization of equity to

define firm size as well as for the denominator of the book value of equity-to-market value of equity ratio used to define the value factor.

SUMMARY: THE CAPITAL ASSET PRICING MODEL, MARKET MODEL, AND OTHER FACTOR-BASED EQUITY MODELS

- Models of expected returns may be used to estimate a firm's required return on equity, also referred to as its cost of equity.
- The capital asset pricing model (CAPM) is a single-factor model in which an expected individual stock or portfolio return is explained by a risk-free rate plus an expected market risk premium (MRP) multiplied by a beta estimate that reflects risk exposure to market movements.
- In developing markets, a country risk premium (CRP) is added to an MRP and is derived using an observed or estimated sovereign bond yield spread and the ratio of standard deviations in the country's equity versus bond markets.
- Both beta estimates and expected MRPs may be measured using different estimation approaches, each with their own advantages and shortcomings.
- Multi-factor models provide for multiple sources of systematic risk to explain expected returns, and multi-factor models may use equity market risk, as in the CAPM, as one of the factors. Other factors may incorporate macroeconomic and company data.
- Multi-factor models may imply higher or lower expected rates of return on individual stocks as compared to the single-factor CAPM model depending on the stock's exposures to the additional risk factors.

REFERENCES

- Carhart, Mark M. 1997. "On Persistence in Mutual Fund Performance." *Journal of Finance* 52, no. 1 (March): 57–82. <https://doi.org/10.1111/j.1540-6261.1997.tb03808.x>.
- Fama, Eugene F. and Kenneth R. French. 1993. "Common Risk Factors in the Returns on Stocks and Bonds." *Journal of Financial Economics* 33 (1): 3–56. doi:10.1016/0304-405X(93)90023-5

PRACTICE PROBLEMS

1. An analyst observes that a stock's standard deviation of returns is 30%, and its returns show a correlation of 75% with those of the market. The market returns have a standard deviation of 18%. The stock's beta is *closest* to:
 - A. 0.45.
 - B. 0.80.
 - C. 1.25.
2. Assuming a risk-free rate of 4%, compared to a stock with a beta (β_a) of 0.70, the cost of equity for a stock with a beta (β_b) of 1.40 will be:
 - A. twice as high.
 - B. less than twice as high.
 - C. more than twice as high.
3. Gableton is a privately held mining company based in Australia with properties producing gold, silver, and copper. An analyst is assessing the potential for Gableton in the IPO market and has created a set of 10 comparable publicly traded mining companies in Australia. The beta estimate for an equally weighted portfolio of these 10 companies is 0.65. The average ratio of debt to market capitalization across the 10 companies is 40%, and the average tax rate is 30%. Gableton's ratio of debt to estimated market value of equity is 60%, and its tax rate is 25%. Which of the following is closest to the analyst's estimate of Gableton's beta?
 - A. 0.51
 - B. 0.74
 - C. 0.94
4. An analyst seeks to estimate the expected rate of return for Air Astana, a publicly traded company based in Kazakhstan. She observes EUR-denominated Kazakh sovereign bonds at 125 bps over the EUR government yield of 3.5%. She estimates an adjusted beta of 1.1 and observes a 4.5% ERP for the STOXX Europe 600 Index used as a developed market reference. If the annualized Kazakh equity market volatility is 1.5 that of the bond market, which of the following is the best estimate of Air Astana's rate of equity return?
 - A. 9.83%
 - B. 10.06%
 - C. 10.51%
5. A stock exhibits the following beta estimates from the four-factor model:
$$\beta_{i,Market} = 0.75, \beta_{i,Size} = 0.40, \beta_{i,Value} = 0.35, \beta_{i,Momentum} = -0.10$$
The stock is *best* described as having below-average exposure to:
 - A. all four factors.
 - B. the momentum factor only.

C. the market and momentum factors only.

6. The risk-free rate is 4%, and a stock has the following factor exposures:

Factor	Stock Beta	Factor Risk Premium
Market	1.2	5%
Size	-0.5	2%
Value	-0.4	4%

Based on the three-factor model, the company's cost of equity is *closest* to:

- A. 3.4%.
- B. 7.4%.
- C. 10.0%.

SOLUTIONS

1. C is correct. Choice C is correct as shown below:

$$\beta_i = \frac{\sigma_{i,m}}{\sigma_m^2} = \frac{\rho_{i,m} \times \sigma_i \times \sigma_m}{\sigma_m^2} = \frac{\rho_{i,m} \times \sigma_i}{\sigma_m} = \frac{0.75 \times 0.30}{0.18} = 1.25.$$

A is incorrect because an estimate of 0.45 is computed from the above inputs if the market's volatility of 18% is incorrectly used in the numerator of the above equation and the stock's volatility of 30% is incorrectly used in the denominator.

B is incorrect because an estimate of 0.80 is the result if the calculation is incorrectly computed as the market's volatility of 18% divided by a denominator specified as the correlation of 0.75 multiplied by the stock's volatility of 30%.

2. B is correct. Choice B is correct because the CAPM implies that the cost of equity across the two companies is as follows:

$$E[r_a] = r_f + \beta_a (E[r_m] - r_f) = 4\% + 0.70(E[r_m] - 4\%) = 1.2\% + 0.70 \times E[r_m]$$

$$E[r_b] = r_f + \beta_b (E[r_m] - r_f) = 4\% + 1.40(E[r_m] - 4\%) = -1.6\% + 1.40 \times E[r_m]$$

Taking the ratio of these two expressions is simplified below:

$$\frac{E[r_a] = 1.2\% + 0.70 \times E[r_m]}{E[r_b] = -1.6\% + 1.40 \times E[r_m]}$$

The expression above implies that the low-beta company's cost of equity is greater than 0.7 multiplied by the expected return on the market (as evidenced by the positive addition of 1.2% in the numerator), while the high-beta company's cost of equity is less than 1.4 multiplied by the expected return on the market. Thus, the ratio will always be greater than 0.5, regardless of the value of the expected market return.

A is incorrect because this statement would only be true in the expression above if the risk-free rate were zero.

C is incorrect because this statement would be true if the risk-free rate were negative.

3. B is correct. Choice B is correct as shown below.

The unlevered beta is first computed from the comparable company data:

$$\beta_u = \frac{\beta_{L,Comps}}{1 + (1 - t) \times \frac{D}{MVE}} = \frac{0.65}{1 + (1 - 30\%) \times 40\%} = 0.5078.$$

Then, use the unlevered beta along with Gableton's debt and tax rate ratios to relever the beta:

$$\begin{aligned} \beta_{L,firm} &= \beta_u \times \left[1 + (1 - t) \times \frac{D_{firm}}{MVE_{firm}} \right] = 0.5078 \times [1 + (1 - 25\%) \times 60\%] \\ &= 0.7363. \end{aligned}$$

A is incorrect because 0.51 is the unlevered beta estimated from the observed beta estimate of the comparable company portfolio after adjusting for their financial leverage factor.

C is incorrect because 0.94 is computed from levering the observed portfolio beta by Gableton's financial leverage factor.

4. C is correct. C is correct because it accurately determines the CRP by multiply-

ing the 125 bp spread by 1.5, which is added to the MRP and multiplied by beta, which combined with the risk-free rate gives us an estimated return on equity of 10.5125% using Equation 9 as follows:

$$E[r_i] = r_f + \beta_i (\text{ERP}_{\text{developed}} + \text{CRP})$$

$$10.5125\% = 3.50\% + 1.1 [4.50\% + (1.25\% \times 1.5)].$$

A is incorrect because it uses the country credit spread in Equation 9 rather than calculating the CRP by multiplying it by the ratio of equity to bond market volatility.

B is incorrect because it is calculated by multiplying only the CRP by beta rather than the CRP plus the MRP.

5. C is correct. Choice C is correct because this stock has a market beta less than 1.0, size and value factor betas that are greater than zero, and a negative momentum factor beta.

A is incorrect because the positive beta estimate on size and value reflect above-average exposure.

B is incorrect because a beta below 1.0 for the market factor also indicates below-average exposure.

6. B is correct. Choice B is correct as shown below for the three-factor model.

$$E[r_i] = r_f + \beta_{i,\text{Market}} \times E(r_m - r_f) + \beta_{i,\text{Size}} \times E(\text{SMB}) + \beta_{i,\text{Value}} \times E(\text{HML}).$$

$$E[r_i] = 4.0\% + [1.20 \times 5.0\% - 0.50 \times 2.0\% - 0.40 \times 4.0\%] = 7.4\%.$$

A is incorrect because a result of 3.4% reflects the sum of the stock's risk premiums, but it neglects to add the risk-free rate of 4%.

C is incorrect because the result of 10.0% reflects only the risk premium associated with the stock's market risk added to the risk-free rate. The two other factors with negative risk exposures serve to effectively reduce the cost of equity estimate.

Glossary

- Activist investors** Investor who may hold either a significant minority or majority stake and seek to initiate changes to increase shareholder value via seats on the board of directors, a majority shareholder vote, or by assuming full control of a firm and taking it private. Also see *strategic investors*.
- Activist short sellers** Investment firms that conduct research on stocks that are expected to decline significantly in value with a focus on identification of misinformation or fraud that is not currently being recognized in the price of the target company stocks.
- Adjusted beta** A beta estimate that reflects a weighted average of a stock's beta estimate from a market model regression and the market average beta estimate of 1.0. For example, adjusted betas reported by Bloomberg use weights of two-thirds on the regression beta estimate and one-third on the beta of 1.0.
- Adjusted EBITDA** Earnings measure defined as EBITDA plus one-time, non-recurring expenses or minus non-operating or non-recurring income for which no universally accepted standards of calculation exist.
- Arbitrage pricing theory (APT)** A theoretical framework allowing for multiple sources of systematic risk to equity investors.
- Asset-based valuation model** Equity valuation model that estimates share value by seeking to determine the market value of a firm's net assets.
- Available seat kilometers (ASKs)** Measure of capacity used by airline industry analysts equal to the product of the number of seats available and the total number of kilometers those seats were flown over a period.
- Average daily volume** Measure of liquidity that reflects the average number of shares traded each business day over a pre-determined period.
- Back door listing** Acquisition of an existing publicly listed company by a private company with the primary purpose of avoiding an IPO process.
- Base effect** A statistical effect in which a variable's calculated growth rate is sensitive to the base year referenced because of the variable's volatility.
- Basic research** Development of an understanding of fundamental scientific principles that generally have no immediate commercial application.
- Beneficial owner** True underlying owner of shares; it often differs from that of the custodian or broker in whose name they may be held at a securities depository.
- Beta** A measure of the sensitivity of price changes in a particular asset relative to changes in the market portfolio. This is a measure of systematic risk, risk that cannot be diversified away.
- Beta estimate** A measure of stock or portfolio sensitivity to movements in a broad-based market index defined as the ratio of the covariance of stock or portfolio returns with market return to the market's variance of returns.
- Bid-ask spread** The lowest ask price (price at which a trader will *sell* a specified quantity of a security) minus the highest bid price (price at which a trader will *buy* a specified quantity of a security).
- Block trades** Very large stock trades typically executed off exchange usually defined as an order of 10,000 shares or more for a specific issuer.
- Book value of equity** The difference between the carrying value of assets and liabilities in a firm's most recent financial statements.
- Breakthrough innovation** A technological change involving a new product, service, or business model that may fundamentally change existing businesses across multiple industries.
- Broker-dealers** Financial intermediaries that either act on behalf of clients as an agent or serve as principal in a financial transaction with a client.
- Buy-side research report** A company research report done within, or on behalf of, a buy-side firm that is typically confidential and reported only to investment decision makers within a firm.
- Cabotage restrictions** Barriers to trade that limit firms from one jurisdiction from carrying passengers or cargo within the domestic borders of another jurisdiction.
- Callable preference shares** A form of preferred stock that grants an issuer the right, but not the obligation, to repurchase preferred shares at a predetermined price.
- Callable preferred shares** Shares that may be redeemed by an *issuer* prior to maturity, often either at par value or at a predetermined premium to par, for which an investor expects to receive a premium in the form of a higher dividend versus non-callable preferred shares.
- Capital asset pricing model (CAPM)** A simple market equilibrium framework that relates an asset's expected return to its systematic (market) risk, measured by beta.
- Catalysts** Expected events or developments related to a company's performance that are listed and described as part of an equity research report that may influence the market price.
- Common shares** An ownership interest that entitles an investor to both economic and voting rights.
- Composite equity index** Comprehensive equity market index that includes securities representing more than 90% of a selected market's value.
- Contrarian view** An equity analyst research report recommendation, price target, or investment thesis that runs counter to market consensus.
- Conversion ratio** The fixed number of common shares available for conversion per preferred share under a convertible preferred share structure.
- Convertible preference shares** A form of preferred stock that may pay a dividend and be converted into common shares at a fixed conversion ratio, or number of common shares received for each preferred share.

- Convertible preferred shares** A form of preferred stock that often pays a dividend and offers the right to be converted into a fixed number of common shares for each preferred share after a specific period. Investors typically receive a lower dividend in exchange for conversion rights.
- Core earnings** Earnings adjusted to exclude non-recurring items used to calculate sustainable earnings in the price-to-earnings ratio.
- Corporate restructuring** Transformative phase in which a firm may acquire another company, sell a division or significant assets, or undertake measures to improve performance.
- Cost leadership** A competitive strategy among firms within an industry that involves becoming the lowest-cost provider of goods or services.
- Country risk premium (CRP)** The additional equity risk premium required by investors to assume the risk of a stock in a given market equal to the product of an observed or estimated sovereign bond yield spread and the ratio of standard deviations in the country's equity versus debt market.
- Cult companies** Firms for which investor decisions may be based upon a loyal or fanatical product following rather than expected company performance.
- Cumulative preferred shares** A form of preferred stock for which unpaid dividends accrue and must be paid in full before dividends on common shares can be paid.
- Custodian** Financial intermediary that maintains securities ownership records and manages the safekeeping, settlement, and reporting for securities held by investors.
- Cyclical companies** Companies that experience high fluctuations in revenue, profitability, and valuation over the business cycle.
- Declaration date** The day that an equity issuer makes a public statement declaring a specific dividend.
- Defensive companies** Companies whose sales growth, profitability, and valuation are less affected by fluctuations in macroeconomic activity.
- Degree of financial leverage** The sensitivity of net income to changes in operating income.
- Delay costs** Costs that arise from the inability to execute a desired trade immediately.
- Depository receipts (DRs)** An equity investment created when the equity shares of a foreign company are deposited with a domestic bank (i.e., a depository or depository) that issues receipts representing the deposited shares.
- Designated market maker** Dealer selected by an exchange to ensure orderly, continuous trading by buying, selling, or holding when the market opens, closes, and at times of market imbalance or high volatility.
- De-SPAC transaction** Completion of a business combination between a public special purpose acquisition company (SPAC) and a private company.
- Direct listing** Primary market issuance that involves the listing of already issued and existing shares on an exchange at a price determined by the market without the use of an underwriter.
- Disaggregated valuation models** Models in which expected future cash flows are estimated from financial statement forecasts instead of by directly growing base year cash flows.
- Disruptive innovations** Technological or other changes that either create a new market or enter an existing one with a different value proposition that may not immediately appeal to existing customers who are satisfied with an existing offering.
- Dividend discount model** Present value equity valuation model which assumes that a shareholder's sole access to firm value is through capital income in the form of cash dividends.
- Dividend rate** Annualized amount of the most recent periodic (quarterly or annual) dividend used to solve for the dividend yield.
- Dividends** Discretionary net income distributions to shareholders.
- Dividend yield** The annual per-share dividends for a company divided by the share price.
- Domain** An area in which an innovation is applied, which may include a product, a business process, or a specific market segment.
- Drivers** Causative factors that explain the past, present, and expected future levels changes in a financial variable such as sales.
- Dual class share structure** Share structure in which differences in the distribution of economic and/or voting rights exist among classes that are held by different shareholder groups.
- Dual listing** Shares issued for trading in two or more markets globally.
- Earnings before interest, tax, depreciation, and amortization (EBITDA)** A broad measure of operating performance that includes revenues less operating expenses but excludes payments to debtholders, non-cash expenses, and taxation.
- Earnings yield** The inverse of the price-to-earnings ratio, or earnings divided by the current price (E/P), commonly used as a measure of equity return.
- Economic moat** A qualitative measure of a firm's tenable competitive advantage in an industry based on factors that defend a firm's business against entrants, substitutes, and other competitive pressures.
- Economic profit** Measure of return based on the profit generated by a firm once it covers the opportunity cost of both debt and equity capital.
- Economies of scope** Cost savings arising for firms that offer multiple products when the cost of producing two or more distinct goods or services is less than that of producing each separately.
- Energy transition** Structural shift mandated by many global governments to reduce the emission of greenhouse gases in overall energy use.
- Enterprise value** A measure of firm value that incorporates both debtholder and shareholder claims—namely, the sum of the market value of a firm's equity, preferred shares, and debt less its cash and short-term investments.
- Equity exchange** A rules-based venue where market participants regularly meet to trade equity securities.
- Equity prospectus** The memorandum distributed to prospective investors at the time of primary share issuance. It includes a description of the firm's business, financial information, and key features of equity securities, including shareholder voting and economic rights.
- Equity research report** A company research report incorporating both quantitative and qualitative information and analysis of the company, its industry, the competitive environment, and financial and valuation analyses, including assessment of relevant risks (both financial and non-financial).
- Ex-dividend date** The first business day that a share trades without (or "ex") the dividend.

- Explicit costs of trading** Direct, out-of-pocket expenses incurred during the execution of a trade, such as brokerage fees, commissions, taxes, and regulatory charges.
- Factor-based investing** Investment selection based upon fundamental factors believed to be underlying drivers of equity performance, such as earnings growth and well-capitalized balance sheets.
- Factor investing** Investment selection process that seeks to identify statistical characteristics important to risk and return used to group companies and make investment decisions.
- Financial distress** The inability of a firm to meet future financial obligations to debtholders, suppliers, or customers, which may result in the sale of the company to another firm or private investor, the restructuring of assets and liabilities under new management, or liquidation of the firm's assets to pay its creditors.
- Financial investor** An investor that typically has no influence over a firm's operations.
- Firm value** A measure of a company's value used in the denominator of multiples equal to the sum of the market value of debt and the market value of equity.
- Floating bar valuation chart** A side-by-side comparison of a stock's actual trading range and one or more estimated value ranges from various valuation approaches; it is sometimes referred to as a football field valuation chart.
- Forward multiple** A forecast-based multiple calculated using values for the next 12-month period.
- Fractional share ownership** Situation in which a shareholder owns a fraction of a share with a similar proportional claim to distributions and net assets.
- Free cash flow** The excess of operating cash flow over capital expenditures.
- Free cash flow to equity (FCFE)** Cash flows available to shareholders once debt service payments are made and other capital expenditure and working capital needs are met, which are discounted at an equity rate of return to determine a share's intrinsic value.
- Free cash flow to the firm (FCFF)** Cash flows available to all capital providers that are discounted at a firm's weighted average cost of capital (WACC) to determine a share's intrinsic value.
- Fundamental factors** Measures other than market capitalization used to weight constituents in an index that may incorporate an issuer's revenue, earnings, dividends, or book value.
- Goodwill** Purchase price of companies that exceeds the fair value of identifiable assets and liabilities recorded as a balance sheet asset.
- Growth companies** Firms that exhibit rapid revenue growth that outweighs the significance of broad economic or other factors over a period.
- Growth equity** Private equity investment in a young company with a business model and growing revenue.
- Growth stocks** Category of stocks that exhibit relatively low ratios of book value of equity to market value of equity, which correspond to high price-to-book ratios.
- Hard to value** A description of certain stocks because of significant dispersion across well-informed market participants about the company's intrinsic value.
- Herfindahl-Hirschman index** A common measure of industry concentration which is calculated as the sum of the squares of competitor market shares.
- High-beta stock** A stock with a beta estimate greater than 1.0 that fluctuates more than movements in the market index, on average.
- Home asset bias** A situation in which investors demonstrate a preference for assets that are issued in domestic versus foreign markets.
- Implicit costs of trading** Indirect costs that result from the market impact of trading, such as the bid-ask spread.
- Implied constant growth rate** The hypothetical rate of steady growth into the indefinite future necessary for intrinsic value to equal current market value.
- Implied market risk premium** A forward-looking estimate of a market risk premium that is estimated from a current stock market index level and a forecast of future cash flows and a terminal value.
- Initiating coverage** A specific type of sell-side equity research in which an analyst writes a comprehensive report for a new publicly traded company following its IPO or on a company that the sell-side firm has not previously provided research coverage.
- Intangible assets** Non-monetary, non-physical items of value owned or controlled by a company that provide economic benefits.
- Intrinsic value (or fundamental value)** Estimated value of a security based upon an analysis of key characteristics believed to be important factors affecting a firm's future cash flows.
- Investment thesis** The key aspects of an analyst's rationale underlying a stock recommendation communicated clearly in a summary early in an equity research report.
- Justified multiple** The estimated fair value of a specific multiple that can be supported using the method of comparables or the method of forecasted fundamentals.
- Justified price multiple** The estimated fair value of a multiple used for relative valuation, which can be supported using the method of comparables or the method of forecasted fundamentals.
- Leading dividend yield** Forward-looking rate of return on equity based on cash dividends calculated using forecasted dividends per share over the next year divided by the current share price.
- Leveraged buyout (LBO)** Situation in which one or more private investors takes control of and restructures a mature underperforming public company using significant debt financing, with the intention of selling it at a higher future valuation.
- Levered beta** A beta estimate that reflects the financial leverage of an individual firm's stock or a portfolio of stocks.
- Limit orders** Trade instructions that impose restrictions on the sale or purchase of shares. A buy limit order specifies a maximum share price to be paid, while a sell limit order indicates a minimum price to be received.
- Liquidating dividend** An extraordinary distribution to shareholders in the form of a return of capital rather than an earnings distribution.
- Liquidation preference payout** Predefined distribution of cash proceeds to preferred shareholders made in full before any distributions are made to common shareholders.
- Low-beta stock** A stock with a beta estimate below 1.0 that fluctuates less than movements in the market index, on average.
- Management buyout (MBO)** Situation in which the group of investors taking a company private consists primarily of the firm's existing management.

- Market breadth** Gauge of market liquidity that reflects the number of orders at different price levels, both above and below, but very close to the best bid and offer prices, which may indicate overall market direction.
- Market-clearing price** Price level at which all buy and sell orders for a security can be executed, leaving no excess demand or supply.
- Market depth** Measure of liquidity based on the relative size of orders posted that are at or very near the best currently available bid and offer prices, reflecting a market's capacity to sustain large orders without significant adverse price movements.
- Market float** The proportion of the constituent security that is available to the investing public.
- Market impact** The effect of immediately executing an equity trade on transaction prices.
- Market model** A variation of the CAPM model in which an asset's returns are regressed on the market's returns.
- Market orders** Trade instructions to a broker to buy or sell a specific number of shares at the best available current market price.
- Market resiliency** Gauge of market liquidity that considers the relative ability of an issuer's shares to maintain a stable market-clearing price and quickly revert to such a price following a market shock.
- Market risk premium (MRP)** The extra return investors expect from holding a broad market portfolio, such as the S&P 500, compared to a risk-free asset like a government bond.
- Market share** The percentage of total industry sales generated by a single company over a specific period.
- Meme stocks** Equity issuers whose shares are traded by individual investors who are heavily influenced by social media posts in making their investment decisions. These stocks often exhibit high share price volatility and high trading volumes.
- Method of comparables approach** The use of an appropriate multiple as a basis for comparing a company's value to other companies considered to have similar characteristics that relies on the economic rationale of the law of one price.
- Method of forecasted fundamentals** Multiples based on projected fundamental characteristics of the business used to compare a company's value to its peers.
- Mid-market price** The average of the lowest ask and highest bid price.
- Multi-factor model** Any model designed to explain returns with multiple variables and which may be theoretical, such as APT, or which may be empirically defined with specific variables measuring different risk factors.
- Multi-market indexes** Comprised of indexes from different countries/markets, designed to represent multiple security markets.
- Multiple** A key metric used in relative valuation approaches involving the ratio of a market-based price with a measure of sales, earnings, or book value of a firm used for peer comparisons.
- Net capital expenditure** A measure of a firm's change in productive capacity equal to its total capital expenditure minus depreciation and amortization.
- Net debt** Debt minus cash and short-term investments included as a component of a firm's enterprise value.
- Net investment** Measure of asset increases to support growth consisting of both long-term assets net of D&A and working capital changes.
- Net operating profit after taxes (NOPAT)** Earnings measure that equals revenue less COGS, SG&A, and D&A after taxes.
- Net operating profit after tax (NOPAT)** A measure of income to both debt and equity holders as represented by operating income multiplied by 1 minus the tax rate.
- Network effect** A situation in which the value of a product or service grows as adoption by households and/or businesses increases.
- Non-systematic risk** (also called unsystematic or idiosyncratic risk) The portion of total risk that is unique to a specific company, industry, or asset and can be reduced or eliminated through diversification.
- Normalized EPS** An estimate of EPS that represents the company's true long-term earning power irrespective of changes in the business or economic cycle.
- Obtainable market** The largest potential market that may be reached by a specific firm given the company's current scale and strategy.
- Opportunity costs** Unrealized profit or loss arising from the failure to execute a trade promptly.
- Participating preferred shares** A form of preferred stock that entitles shareholders to receive the standard preferred dividend plus the opportunity to receive an additional dividend if the company's profits exceed a prespecified level.
- Payment date** Date that the company electronically transfers a dividend payment to shareholders.
- Peer group** Set of comparable companies selected as a benchmark for relative value comparisons that have cash flow, risk, and growth characteristics similar to those of the target firm being valued.
- Perceived value** An individual assessment of how a security *should* be priced, which may vary widely across individuals and may differ significantly from the observed market price.
- PESTLE analysis** Refers to the assessment of external influences on an industry or company due to political, economic, social, technological, legal, and environmental factors.
- PIPE (private investment in public equity) transaction** A private offering of publicly listed equity securities to a select group of investors.
- Porter's Five Forces** A framework first described by Michael Porter to analyze an industry's structure and potential returns as defined by the level of competitive rivalry, which incorporates key factors including the threat of new entrants and product substitutes, as well as customer and supplier bargaining power.
- Post-money valuation** An early-stage firm's value following a new equity investment.
- Preference share par value** Nominal face value of a preferred share.
- Preference shares** Also referred to as preferred stock, these shares have characteristics of both debt securities in that they pay dividends and common shares, given that they are often perpetual.
- Pre-money valuation** An early-stage firm's value prior to a new equity investment.
- Present value models** Valuation models that estimate the intrinsic value to a firm's investors of its expected future capital income and capital appreciation as of today using an appropriate discount rate.
- Present value of growth opportunities** The difference between a company's current stock price and the present value of its forward earnings per share.

- Price/earnings-to-growth (PEG) ratio** The ratio of P/E to expected earnings growth; used to compare growth-adjusted P/Es across comparable firms.
- Price elasticity** A measure of pricing power that involves the change in quantity of goods demanded relative to a price change.
- Price target** An expected price or price range within a defined time horizon for a stock communicated within an equity research report that reflects the results of an analyst's valuation of the stock.
- Price-to-book ratio (P/B)** Valuation measure that is calculated as a ratio of the current share price to the book value per share.
- Price-to-earnings ratio (P/E)** The ratio of share price to earnings per share.
- Price-to-sales ratio** The market price per share divided by annual net sales per share, where net sales are total sales minus returns and customer discounts.
- Price to tangible book value** A multiple with share price in the numerator and tangible book value in the denominator; intangible items, such as goodwill, are excluded.
- Pricing power** A firm's relative ability to freely set prices and other economic terms for its products and services without adversely affecting demand.
- Primary equity markets** Markets in which an issuer sells a new equity security to investors to raise ownership capital or transfer ownership from private to public investors.
- Primary follow-on offering** A dilutive offering of new shares in the primary market for a company that currently has a public share listing.
- Private placement** Restricted sale of securities in which only a selected investor, or group of investors, may purchase the securities.
- Privatization** The transfer of ownership of a government-owned corporation to private or institutional investors either directly or through an IPO.
- Program accounting** An accounting approach used to measure inventory under GAAP for specific long-term programs or projects, which takes overall project profitability into account rather than unit costs.
- Proxy advisors** Third parties engaged to assist institutional investors with services, including information gathering and assessment, voting recommendations, and recordkeeping associated with exercising shareholder votes.
- Proxy statement** Document distributed by a company to shareholders with timing and voting details for the company's annual shareholder meeting along with proposals slated to be voted upon and associated background information.
- Proxy vote** Election process where a designated third party votes on a shareholder's behalf.
- Public offering** Sale of securities in which any member of the public may participate.
- Putable preference shares** A form of preferred shares that grants an investor the right, but not the obligation, to sell preferred shares back to the issuer at a predetermined price.
- Putable preferred shares** Shares that may be sold by an investor back to an *issuer* prior to maturity, often either at par value or at a predetermined price, for which an investor expects to receive a lower dividend versus non-putable preferred shares.
- Put contingency price** Price at which investors may sell putable preference shares back to the issuer.
- Real estate investment trusts (REITs)** A specific type of company that owns a portfolio of income-producing properties that must distribute at least 90% of its annual taxable income to shareholders.
- Record date** Date that a shareholder registered with a securities depository is deemed to own shares for purposes of receiving an upcoming dividend.
- Regular cash dividends** Cash distributions to shareholders paid at known intervals.
- Reinvested dividends** Cash dividends that a shareholder receives and uses to purchase additional shares issued by the same company.
- Residual income model** An equity valuation approach that seeks to estimate the periodic return to shareholders above their required rate of return. The present value of residual income represents the difference between the share price (P_t) and the book value of equity per share ($BVE/Share_t$).
- Restricted stock units (RSUs)** Shares rewarded to employees as part of annual compensation that are restricted from sale during a vesting period that may last several years.
- Return on capital (ROC)** Measure of the return to both equityholders and debtholders equal to the ratio of after-tax EBIT to total capital.
- Reverse stock split** Corporate action involving a decrease in the number of shares outstanding and a proportionate increase in the price per share, which leaves the total market value of equity and an investor's proportional ownership claim unchanged.
- Scenario analysis** Probability-weighted valuation that combines a base case with alternative outcomes, allowing the use of more favorable or adverse forecasts to arrive at a single expected value.
- Seasoned equity offering** Sale of additional units of a previously issued equity security, an accelerated process given market familiarity with the issuer.
- Secondary equity markets** Markets in which existing stocks are subsequently traded among investors.
- Security market line (SML)** A linear relationship, derived from the capital asset pricing model (CAPM), that shows an asset's expected return as a function of its systematic (market) risk, measured by beta.
- Sell-side research report** A company research report produced by a sell-side firm for distribution to the firm's clients to support its securities underwriting and corporate advisory business lines.
- Serviceable market** The maximum addressable market by a specific firm given its current focus and geographic presence as well as growth opportunities.
- Settlement** Official transfer of securities to the buyer's account and cash to the seller's account.
- Shareholder proposal** An equity investor recommendation or requirement that the company or its board take a specific action submitted for a shareholder vote.
- Shareholders of record** Investors whose identity is confirmed by a securities depository or custodian as the owner of a security as of the record date.
- Share repurchase** Also known as a share buyback, this is a transaction in which a company uses cash to buy back its own shares.
- Single-factor model** A model designed to explain the expected return of an asset or portfolio that uses only one factor. A standard example of this is the CAPM, which bases equity return solely on stock market risk.

- Special purpose acquisition company (SPAC)** Shell or blank check company created for the purpose of raising equity capital to acquire an unspecified private company within a pre-agreed time frame.
- Special situations** An area of private capital investment that targets return by investing in stressed, distressed, or event-driven opportunities.
- Stock dividend** Distribution to shareholders of additional stock, which divides the firm's market value of shareholders' equity into smaller pieces without a change in the value or a shareholder's proportional ownership.
- Stock screening** A multistep process used by buy-side analysts to identify a relatively small number of stocks that meet the firm's investment strategy and are worthy of being researched at an appropriate level of detail.
- Stock split** An increase in the number of shares outstanding and a proportionate decrease in the price per share, with no change to the total market value of equity and an investor's proportional ownership claim.
- Strategic investors** Investors who may hold either a significant minority or majority stake and seek to initiate changes to increase shareholder value via seats on the board of directors, a majority shareholder vote, or by assuming full control of a firm and taking it private. Also see *activist investors*.
- Sustainable growth rate (SGR)** A growth rate that reflects a company's level of investment and its productivity of investment, often estimated by multiplying a company's reinvestment rate by its return on capital.
- Sustaining innovations** Incremental product or service improvements that do not fundamentally change the functionality of existing goods or services. These innovations are often led by industry incumbents making improvements to better serve customer needs.
- Systematic risk** The inherent overall market risk that affects all assets.
- Take-private transaction** This involves the purchase of a publicly listed company by private investors in which the target company's shares are no longer publicly traded.
- Terminal value** The expected value of a share at the end of the investment horizon—in effect, the expected selling price.
- Total addressable market (TAM)** The largest potential market that may be reached by a specific good or service.
- Trailing dividend yield** Measure of the rate of return on equity based on cash dividends calculated by dividing the dividend rate by the current market price per share.
- Trailing multiple** Multiple calculated based on financial statement variables from the most recent 12-month reporting period.
- Turnover ratio** The average daily volume divided by market float used as a measure of equity market liquidity.
- Unlevered beta** A beta estimate derived from a levered beta estimate. It is typically an unobserved risk measure that reflects a firm's or industry's equity beta estimate in the absence of financial leverage.
- Value stocks** Category of stocks that exhibit relatively high ratios of book value of equity to market value of equity corresponding to low price-to-book ratios.
- Value-to-book multiple** A relative valuation that compares firm value to the book value of capital using the ratio of market value to book value of both debt and equity.
- Venture capital (VC)** Private equity investment in a start-up or early-stage company involving high risk and a high rate of failure.
- Vertical integration** A strategy adopted by industries facing a high degree of supplier or customer bargaining power that involves the expansion of operations into an earlier or later stage of the production or supply chain to deliver a product or service to end users.
- VIX index** A financial measure that estimates the expected volatility of the S&P 500 Index.
- Volume-weighted average price (VWAP)** A measure of intraday share price performance calculated by weighting the traded price by the volume of shares traded at that price.
- Vote by proxy** Process in which shareholders delegate their voting rights to a third party.